TENSE IN ENGLISH

Its structure and use in discourse

Renaat Declerck

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Renaat Declerck



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To my wife and children

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Contents

Preface viii
List of abbreviations x

1 Introduction
1 Aim and scope of the work 1
2 Some theoretical and methodological preliminaries 7

2 Tense in discourse 1: general principles
1 Situation 14
2 Temporal zero-point (t_O) 14
3 The two time-spheres 16
4 Time of orientation (TO) 17
5 Sectors 19
6 Present sector, pre-present, post-present 19
7 Absolute sectors 19
8 Temporal domain 20
9 Binding TO 20
10 Central TO 21
11 Ancillary TO 21
12 Shift of temporal domain 21
13 Shift of temporal perspective 24
14 Temporal subordination 25
15 The expression of domain-internal relations 26
15.1 The relations in a past time-sphere domain 26
15.2 The relations in a pre-present sector domain 28
15.3 The present sector 34
15.4 Relations in the post-present sector 36
16 Sloppy simultaneity 41
17 Alternatives to temporal subordination 45
17.1 Shifting the domain 46
17.2 Establishing a simultaneous domain 49
17.3 Re-establishing the domain 52
18 Different ways of establishing a domain 59

19 Direct and indirect binding 62
20 Shift of temporal perspective 66
21 Temporal perspective and modality 76
21.1 Remoteness from reality, i.e. nonfactuality 77
21.2 Tentativeness 78
22 Temporal focus 80
23 Point of view 89
24 Implicit TO 92
25 Adverbial time clauses 97
25.1 *When*-clauses 98
25.2 *Before*-clauses 101
25.3 *After*-clauses 107
26 Conclusion 113

3 Tense in discourse 2: restrictions and possibilities
1 The choice and interpretation of the temporal domain 116
1.1 Unembedded clauses 116
1.2 The choice of temporal domain in subclauses 132
1.3 Conclusion 139
2 The choice of binding TO 141
2.1 Binding rules for subclauses 142
2.2 Binding rules for unembedded clauses 150
3 The choice between the Present Perspective System and the Future Perspective System 153

4 Tense in discourse 3: two test cases
1 Tense in indirect speech 157
1.1 Theories of tense in indirect speech 158
1.2 Restrictions on the use of absolute tenses in complement clauses 182
1.3 Further remarks 191
2 The Present Perspective System versus the Future Perspective System in conditional clauses 192
2.1 Introductory remarks 192
2.2 Types of conditionals that use the FPS 198
2.3 Conclusion 221

5 The structure of tense 1: previous analyses
1 Reichenbach (1947) 224
1.1 Reichenbach's analysis 224
1.2 Objections to Reichenbach's system 225
2 Comrie (1985) 232
2.1 Comrie's analysis 232
2.2 Objections to Comrie's theory 234
3 Conclusion 248

6 The structure of tense 2: general principles
1 Primitives and relations 249
1.1 The primitives 249
1.2 Relations 253
1.3 Time adverbials 253
1.4 Time of orientation 255
1.5 Chains of relations 256
1.6 Further preliminaries 256
1.7 Temporal vagueness 269
2 Progressive sentences 273
3 Repetitive and habitual sentences 277
4 Time adverbials 284
5 Conclusion 292

7 The structure of tense 3: analysing the tenses
1 The preterit 295
2 The present tense 313
3 The present perfect 319
4 The past perfect 354
5 The future tense 369
6 The future perfect 375
7 The conditional tense 379
8 The conditional perfect 383

8 Conclusion
1 Summary 387
2 Further remarks 393

References 397
Index 416

Preface

I have been interested in the English tense system for quite a long time. However, it was not until I read Comrie's (1985) book on tense that I decided to investigate that system in detail myself. The reason was that, although I found the book excellent in many respects, I could not agree with some of Comrie's basic claims. I therefore put pen to paper and wrote an article (Declerck 1986a) in which I refuted the claims in question and suggested an alternative theory. However, it soon became clear to me that my article had been a bit premature. I found that the theory I had suggested was not quite satisfactory. I also found that there were other interesting problems in connection with tense which were worth considering besides the ones I had treated. And last but not least I found that the linguistic literature contained a lot of work on tense which I had not read while writing my article. So I decided to do the job over again, and to do it differently this time. The present book is the result.

One of the things that struck me while reading the (innumerable) books and articles on tense that have been written over the last few decades is the proliferation of theoretical claims that have been made in connection with tense. As is well known to anybody familiar with the field, these claims are divergent and often conflicting, although the data from which they are derived are relatively uncontroversial. The problem, apparently, is that there has been no coherent theoretical framework within which to interpret the data. I realized this was a gap which had to be filled before any progress could be made, and I decided to undertake the task of doing so. Offering a coherent theoretical framework is therefore one of the major purposes of the present book.

This being said, I hasten to add that this is the only respect in which the work is theoretical. It is not theoretical in the sense that it is written within the framework of one of the current linguistic theories (transformational grammar, Montague grammar, functional grammar, etc.), nor is it meant to adduce evidence either for or against any of these theories. As noted by Comrie (1985:viii), none of the current linguistic theories

seems to have much to say about tense. I have therefore preferred to build a theory of tense in isolation rather than squeeze it into the bodice of some linguistic theory that is not particularly well suited to deal with tense.

Another feature of the book which some linguists may find surprising is that it is written without any formalism. Though I agree that formalization may sometimes be useful when a very precise semantic description is called for, I personally prefer to avoid formalism unless it is really necessary. For one thing, formalizing is just a way of describing meaning (i.e. it is a descriptive tool; it does not explain anything). For another, not everybody finds the formalized version of a statement more illuminating and readable than its nonformalized counterpart. In my opinion, the theory of tense I have developed can be presented without any formalism whatever. I have therefore abstained from using any.

Finally, I am very grateful to Bernard Comrie for the numerous comments he made on an earlier version of this book, and to Shigeki Seki for the assistance he gave me in proofreading the manuscript.

R.D.
Catholic University of Leuven
14 March 1990

Abbreviations

FPS	Future Perspective System
MidS	middle of the situation
NP	noun phrase
PPS	Present Perspective System
RST	Represented Speech and Thought
SoT	Sequence of Tenses
SS	situation span
TE	established time
t_0	temporal zero-point
TO	time of orientation
TOE	time which is both TE and TO
TO_{sit}	situation-TO
TS	time of the situation
TU	time of utterance
VP	verb phrase

1 Introduction

1 AIM AND SCOPE OF THE WORK

The number of books and articles that have been written on the subject of tense in recent years is quite impressive.[1] This shows both that the interest in the phenomenon of tense is great and that the problems in connection with it are numerous. It is also striking how many different 'theories' of tense have been proposed. Clearly, tense is a subject on which the last word has not yet been said.

Of course, this is not in itself a justification for writing yet another book on tense. The reason why I have deemed it necessary to write this book is that the vast literature on tense still shows a couple of conspicuous gaps – gaps which need to be filled if the theory is to make any progress whatsoever.

The first gap is the absence of what could be called a 'descriptive theory' of tense. What I mean by this is a theory which functions as an intermediate level between the concrete data and the abstract, hypothetical treatments of tense in theoretical frameworks. It is my conviction that the reason why there has been a proliferation of widely divergent and often conflicting theoretical approaches to tense is the lack of such a descriptive theory. As I see it, a descriptive theory is a coherent set of generalizations, which function as ordering principles and render it possible to classify the observational data and hence to interpret them correctly. At present there is apparently no complete descriptive theory of this kind (though there are bits and pieces scattered around), and this is the reason why the data are so often interpreted differently. The recent literature shows too many examples of intricate theoretical constructs that are inadequate because they are based on faulty interpretations of otherwise sound observations.

To illustrate the need for a descriptive theory, I will review a couple of sets of data, which have given rise to a variety of claims. Since the claims

1. Schulze (1985) offers a bibliography which is twenty-five pages long in spite of the fact that it covers only a period of (roughly) ten years and does not aim at being complete.

are usually contradictory, a large part of them must of necessity be mistaken. Yet all of them are familiar in the linguistic literature,[2] and all of them are claimed to be supported by the data.

1.1 As is well known, English has various means of referring to the future. The following examples illustrate some possibilities:

(1) (a) I *will do* it tomorrow.
(b) I *am leaving* tomorrow.
(c) I *am going to leave* tomorrow.
(d) I will do it tomorrow if I *have* time. (*will have)

Data like these have led linguists to formulate widely divergent claims. The fact that four different kinds of verb form are used to refer to the future is sometimes interpreted as meaning that English does not really have a future tense. Some linguists consider the form with *will* to represent the future tense in English and claim that there is '*will*-deletion' in the *if*-clause of (1,d). Some treat the use of the present tense in (1,b) and that in the *if*-clause of (1,d) as instances of one and the same phenomenon. Some of those that hold this view claim that there is *will*-deletion in both cases. Others argue that (1,d) is different from (1,b) and that the use of the present tense in (1,d) is due to linguistic economy, because futurity is sufficiently expressed by the head clause. In contrast, some linguists claim that the present tense in (1,d) is due to the fact that the speaker treats the future fulfilment of the condition as a present fact. Others say that the present tense is used in the *if*-clause because the *if*-clause is temporally subordinated to the head clause. In sum, the few data available in (1,a–d) are interpreted as evidence for widely divergent claims. Each of these claims has some *prima facie* plausibility and would seem to be in accordance with the data of (1,a–d). It is my belief that in order to judge which of them is/are correct and which of them are not, we must stop considering (1,a–d) in isolation. What we must do is try and build a descriptive theory of tense which is consonant not only with (1,a–d) but also with the hundreds of other observations that can be made. If we succeed in doing this, we can judge the soundness of the theoretical claims simply by checking whether or not they are in keeping with the descriptive theory as a whole.

1.2 A second illustration concerns sentences like the following:

(2) (a) I *am* hungry.
(b) London *lies* on the Thames.

2. The fact that all of the claims that are presented below are widespread is one of the reasons that I will abstain from adding references to them. The other reason is that all these claims will be discussed at greater length at different places in the book and that all necessary references will be given there.

(c) John's dog *chases* cars.
(d) The exhibition *opens* next month.
(e) I *hear* you have been promoted.
(f) He just *walks* into the room and *sits* down in front of the fire without saying a word to anyone. (Palmer 1988:39)

In these six examples the present tense is used with reference to different times. In (2,a) the reference is to a relatively short interval which includes the moment of speaking. Example (2,b) refers to an interval which is indefinitely long and encompasses not only the present but also the past and the future. Example (2,c) refers to a habitual activity consisting of a repeated number of occurrences; the habit includes the moment of speech and extends into the past and possibly also into the future; it is not necessary that any of the occurrences should coincide with the moment of speech itself. In (2,d) the reference is to a future event. In (2,e) *hear* refers to an event that took place in the past and is still relevant at the present moment; it is thus roughly equivalent to *have heard.* In (2,f), finally, we have the so-called historical present, which refers to events that took place in the past.

Again, these data have given rise to various interpretations. Some linguists claim that there is no reference to the present (the moment of speech) in generic or habitual sentences like (2,b) and (2,c). Others hold that both sentences refer to a state which is represented as holding at the moment of speech. The latter group usually claim that the basic meaning of the present tense is reference to present time and that the uses in (2,d–f) are special uses. Others conclude from (2,a–f) that there is no relation between the present tense and the expression of present time. In fact, they claim that the present tense does not refer to time at all, but simply functions as the unmarked tense in the English tense system, and is therefore compatible with any time reference.

1.3 Like the present tense, the past tense (preterit) can be used in different ways:

(3) (a) Yesterday I *met* John in the street.
(b) The two girls *were* usually known by their surnames, Banford and March.
(c) I *wanted* to have a word with you.
(d) I would appreciate if you *came* on time.

Sentence (3,a) might be part of any everyday conversation. Example (3,b) is the first sentence of a literary work (D.H. Lawrence's novella *The fox*). Examples (3,c) and (3,d) illustrate so-called 'modal' uses of the preterit: in (3,c) *wanted* is an 'attitudinal' preterit, which is used for tentativeness and has present time reference; in (3,d) *came* refers to an action which could take place in the future.

Most linguists hold that the basic meaning of the preterit is reference to past time, and that the modal uses are special ones. Others, however, point out that what all the sentences in (3) have in common is the idea of 'remoteness' or 'distance': in (3,a) the preterit represents the event as temporally distant from the moment of speech; in (3,b) it represents the event as distant from present reality, i.e. as belonging to a fictional work; in (3,c) the preterit conveys the idea of psychological or social distance, i.e. modesty, tact; in (3,d) the preterit again expresses some distance from reality, since the future event is not represented as factual but as one whose actualization is only possible, and not even very likely. For linguists who set store by these considerations, the basic meaning of the preterit is not reference to past time but the expression of distance from present reality. If this interpretation is combined with the view that the present tense does not refer to present time (see above), it leads to the conclusion that there is no basic connection between tense and time. However, even this view can lead to quite different theories. Most people that believe that there is no fundamental relation between tense and time accept the view that the basic distinction between the past tense and the present tense is that between representing situations as non-actual (remote) and representing them as actual (see e.g. Janssen 1990). However, Weinrich (1964, 1970) holds that the essential difference is that the past tense is used in 'Erzählsituationen' (narrative contexts), whereas the present tense is used in'Besprechungssituationen' (contexts of discussion). And there are still other (slightly different) views besides these.

1.4 When the contents of a sentence in the present tense are reported in the form of an indirect speech clause embedded under a verb in the past tense, the embedded clause usually makes use of the preterit:

(4) (a) John said: 'I am hungry.'
(b) John said that he *was* hungry. (*is)

The observation that we use *was*, not *is*, in (4,b) has been interpreted in at least three different ways. A small group of linguists argue that the use of the preterit in the subclause of (4,b) is in no way different from its use in the head clause: in both cases the preterit is used because the event or state in question is to be located in the past (i.e. before the moment of speech). Others argue that *was* is not an 'absolute' tense form (like *said*) but a 'relative' one. That is, the preterit is used because the state referred to in the *that*-clause is represented as simultaneous with the action referred to in the head clause. According to these linguists, the past tense is the tense form to be used for the expression of simultaneity if the head clause is also in the past tense. A third group of linguists (probably the majority) believe that there is a 'sequence of tenses' rule which mechanically backshifts the present tense to the past tense in clauses that

are embedded under a head clause in the past tense. Some of these linguists still accept that this rule is semantically motivated, but others (e.g. Comrie 1986) claim that it has no semantic motivation whatever, i.e. that it is a purely formal rule.

1.5 The above four illustrations concern data which have given rise to a variety of different interpretations and explanations, even in theories of tense that have been developed independently of any particular theoretical framework (such as transformational grammar, Montague grammar, etc.). In treatments of tense that do make use of a theoretical framework the danger of deriving theoretical conclusions from faulty interpretations of data is even greater. One example may be sufficient here. Ladusaw (1977) observes that the sentence

(5) Mary saw the unicorn that walked.

may be true not only if the event of the unicorn walking was simultaneous with the event of Mary seeing the unicorn but also if the unicorn walked before or after Mary saw it. This observation is correct, and I will argue below that it can be explained from the fact that both clauses in (5) locate an event before the time of speech independently of each other. That is, each of the tense forms in (5) expresses no more than that the event referred to took place in the past; neither of them conveys any information as to the temporal order in which the events took place. Since the tense forms leave the temporal relation between the events unspecified (vague), and since there is no other (e.g. adverbial) indication of this relation, the sentence may be true irrespective of whether the seeing is prior to, simultaneous with, or posterior to the walking. However, Ladusaw interprets the data differently. He claims that (5) is three-ways ambiguous, and that this ambiguity is due to the fact that the sentence may be derived from three different sources: a higher past tense sentence lowered to complement position, a base past tense complement, and a base present tense complement to which the sequence of tenses rule has applied. In his opinion, these three sources correspond with the interpretations that the time of walking is before, after, or simultaneous with the time of seeing, respectively.

I think that the above illustrations are sufficient to substantiate the claim that there is not much point in developing 'higher order' theories of tense as long as there is no basic descriptive theory to help us to classify and interpret the data.[3] It is therefore one of the aims of the present book to

3. The numerous theories developed in tense logic are clear examples of theories that suffer from the defect of being based on scanty, and moreover often misinterpreted, data. It is therefore not surprising that these theories have little to offer that is useful for a linguistic theory of tense in English.

fill this gap and offer such a basic theory. This is done in the first half of the book. In chapter 2 I develop a full descriptive theory of tense, in which all the data (as far as I am aware of them) are given their proper places. This theory consists of a number of rules and principles which (in my opinion) provide for the different uses of the tenses in English and predict which tenses can be used in particular environments and contexts.[4] Since, obviously, not all principles are operative in all contexts, ample attention is devoted to the necessary restrictions and conditions. For some of the major principles this is not done in chapter 2, but in the next chapter, where the possibilities and restrictions can be explored at greater length. Chapter 4 then closes off the first part of the book with a scrutiny of two areas of English grammar which have proved to be particularly troublesome and have therefore frequently been discussed in recent times: the use of tenses in indirect speech, and the use of the present tense vs. *will* in conditional clauses referring to the future. These two subjects are seen as test cases for the descriptive theory developed in the previous chapters.

The second part of the book concerns matters of a different kind. It no longer deals with the use of the tenses in simple and complex sentences and in stretches of text and discourse, but rather with the internal structure of the tenses, i.e. with the temporal schemata which they realize. Here, I think, there is a second gap to be filled in the linguistic literature. This time it is not a gap in the sense that there is no theory – the theories abound – but in the sense that there is no *satisfactory* theory.

My quest for temporal schemata begins in chapter 5, where I discuss the theory that is by far the most influential in the recent literature, viz. the analysis proposed by Reichenbach (1947). After summing up the various objections that can be made to this theory I conclude by separating the elements that are useful from those that should be given up in favour of a better alternative. I then proceed to discuss the theory more recently advanced by Comrie (1985), which departs from Reichenbach (1947) in a number of ways. Here again I argue that the theory is valuable in some respects but needs revising on other points. The conclusion is that, if we want to build a theory which produces the correct temporal schemata for the different tenses, we must adopt the valuable elements from the theories of both Reichenbach and Comrie and remedy their defects.

In chapter 6 a theory is developed which I think comes up to these requirements. Here I single out the 'primitives' that are to be the corner-stones of the temporal schemata and examine the temporal relations that can hold between them. I also discuss the role of temporal

4. The rules in question concern the use of tenses not only in unembedded clauses but also in embedded clauses and in discourse.

adverbials and examine how the theory can account for generic and progressive sentences. Chapter 7, finally, applies the theory to each of the English tenses. That is, it examines what is the basic temporal structure that is realized by the present tense, the preterit, the present perfect, etc.

2 SOME THEORETICAL AND METHODOLOGICAL PRELIMINARIES

There are a few theoretical and methodological questions that must be briefly gone into before I can conclude this introduction.

2.1 I should like to make clear from the outset that I do not subscribe to such theories as claim that there is no connection between tense and time. I accept Lyons' (1977:68) definition of tense, which says that tense 'grammaticalizes the relationship which holds between the time of the situation that is being described and the temporal zero-point of the deictic context'.[5] As stressed by Lyons (1977:682), 'the crucial fact about tense, whether we are talking about sentences or propositions, is that it is a deictic category. A tensed proposition, therefore, will be, not merely time-bound, or even temporally restricted: it will contain a reference to some point or period of time which cannot be identified except in terms of the zero-point of utterance.' (see also Rescher and Urquhart 1971:27).

According to this definition, only finite verb forms can be said to express tense. Nonfinite verb forms (i.e. infinitives, gerunds and participles) express a single temporal relation, viz. they relate the situation to some other time, which may or may not be the temporal zero-point (the time of speaking).[6] (In fact, the reference time in question is usually the time of the head clause situation.) A tense, by contrast, expresses the various relations that hold between the situation and the zero-point. If the tense is an 'absolute tense' (such as the preterit), it relates the situation directly to the zero-point. If it is a 'relative tense' (like the past perfect), it relates the situation to a reference time which is itself related to the zero-point (either directly or via other reference times). In other words, tenses realize temporal schemata which necessarily involve the zero-point, whereas nonfinite verb forms do not. Nonfinite clauses will therefore be treated as tenseless (untensed). (This accords with the fact that nonfinite clauses are by definition dependent clauses: they cannot be

5. We find the same definition in other words in Fleischman (1982:10): 'tense will be defined as a deictic category of grammar, marked formally by an affix, particle, auxiliary, etc., whose primary function is to mark sequence of events in direct or indirect relation to a temporal zero-point, which is the moment of the speech event'.
6. For the time being I will disregard the exceptional possibility that the temporal zero-point is a time other than the coding time (time of speaking) (see chapter 2, section 2).

used as statements and cannot, therefore, be assigned a truth value.)[7]

Clauses whose verb form is in the subjunctive mood are also untensed (see Heny 1982:112). Like nonfinite forms, subjunctive forms merely express a single temporal relation. Thus, the 'present subjunctive' just expresses simultaneity. For this reason it can be used after any tense (see Er-Rayyan 1986:72):

(6) Bill suggests/suggested/will suggest/had suggested/would suggest that we leave.

It should be noted that the term 'tense' is not always defined so restrictively as is done in Lyons (1977) and here. For example, Comrie (1985) also treats tense as a deictic category (p. 14), but (in spite of the fact that he refers to Lyons' treatment of deixis in a footnote) he accepts a much broader definition of deixis: according to him, a verb form is deictic if it relates a situation to a reference point, even if this referent point is not the speech time.[8] Comrie therefore uses the term 'relative tenses' to refer to nonfinite verb forms that have 'relative time reference, i.e. time reference relative to a deictic centre other than the present moment' (p. 16), as in *Those **sitting** on the benches were forced to move.* In this book the term 'relative tense' will be defined differently. As far as I can see, the issue is largely a terminological one. However, it is important to see the precise meaning that is going to be ascribed to the term 'tense'.

2.2 The linguistic literature reveals a total lack of consensus as to the question of how many tenses there are in English. A great many linguists hold that we can speak of different tenses only if we have to do with morphologically differentiated verb forms. This leads to the conclusion that there are only two tenses in English: the past tense and the present (or 'nonpast') tense.[9] In this theory, such verb forms involving different auxiliaries as *will do, has done, will have done* all belong to the same (in this case: present) tense. However, other linguists hold that tenses may be

7. It should be clear from these remarks that we cannot subscribe to McCawley's (1973) claim that *have* in English is always the realization of past tense. Our position is that *have* means no more than anteriority. (As noted by Lyons (1977:89), anteriority is not a deictic notion and should therefore be distinguished from 'past'.) In tenseless constructions (e.g. infinitives, gerunds, participles) *have* just expresses this temporal relation; when used as a finite form (i.e. in one of the perfect tenses) the anteriority relation in question is part of a temporal schema that ultimately relates the time of the situation to the coding time. For example, in the past perfect tense *had* expresses a relation of anteriority between the time of the situation and some other time, which is itself 'past' with respect to the coding time.
8. In his review article on Comrie (1985), Dahl (1987:491) also draws attention to this difference between Lyons' definition of deixis and Comrie's.
9. See e.g. Jespersen (1931:3), Trager and Smith (1951:77), Twaddell (1960), Ota (1963:2), Joos (1964:120), Chomsky (1965), Jacobs and Rosenbaum (1968), Streng (1972), Enç (1987:634), etc.

formally marked by auxiliaries as well as by inflectional morphemes, and therefore consider the above verb forms as realizations of different tenses. In their opinion, English has a whole array of tenses, for which we can use such traditional labels as *present tense*, *preterit* (or *past tense*), *present perfect*, *past perfect*, *conditional tense*, *conditional perfect*, *future tense*, *future perfect*.

My own position in this book will be in accordance with the latter tradition. Though the issue is not of crucial importance, I wish to point out that this is not simply an arbitrary decision in favour of one of the existing traditions but a well-motivated choice. In my opinion, there are several reasons why the view that English has only two tenses should not be adopted:

2.2.1. The claim that there are only two tenses in English is mainly based on two observations: viz. the observation that the present tense and the preterit are the only tenses that are morphologically marked (the other tenses being formed with the help of auxiliaries); and the observation that there are only two temporal perspectives in English (corresponding, roughly, to the cases in which the so-called 'sequence of tenses' rule applies or does not apply). However, it is doubtful whether either of these observations necessarily leads to the claim that there are only two tenses in English:

2.2.1.1 The observation that there are basically two temporal perspectives can lead us to conclude that there are two sets or kinds of tenses, but does not warrant the conclusion that there are only two tenses. (In French, where the future is morphologically marked, so that it is difficult to deny the existence of a future tense, we also find that 'there exist basically two temporal perspectives: Past and Non-Past' (Rohrer 1986: 79). The nonpast perspective is therefore typical of a *set of tenses*, one of which is the future tense.) In other words, the claim that English and French (and most languages that have tense) 'treat the distinction of past and non-past as being of greater importance than the distinction of present and non-present or future and non-future' (Lyons 1977:809) is a claim about how these languages handle distinctions of *time*. We may conclude from this that the tenses in these languages are better subclassified into 'past tenses' vs. 'non-past tenses' than into 'present tenses' vs. 'non-present tenses' or 'future tenses' vs. 'non-future tenses'; but there is no basis for concluding that these languages make use of only two tenses.

2.2.1.2 As to the observation that only the preterit is morphologically distinct from the present: what reason is there for taking precisely this characteristic as criterial? It is well known that it is often the case that

one and the same idea is expressed morphologically in one language and by means of an independent morpheme in another. The definite article is a free morpheme in English, but it may be a suffix in Swedish. Many languages use suffixes to express relations which are expressed by means of prepositions in English. Why should we not accept that some tenses can be expressed morphologically while others make use of free morphemes (auxiliaries)?[10] This question becomes particularly pertinent when we observe that in some cases a morphological indication of tense has developed out of an auxiliary. The modern French future tense suffix is a case in point: Latin originally had a synthetic future (e.g. *cantabo* 'I will sing'); later on this was replaced by the 'periphrastic or analytic structure *cantare habeo*, which itself ultimately agglutinated and evolved into the actual Romance synthetic – or, as they are often called, 'simple' – futures *chanterai, cantaré, canterò'*, etc.' (Fleischman 1982:1). The French present perfect (*j'ai chanté*) has similarly originated as a periphrastic verb phrase which adopted the function of marking perfect, but unlike the future, it has not been subject to resynthesis but has remained analytic. This example from French shows that whether or not a tense is morphologically marked is the result of a historical development determined by arbitrary factors that have nothing to do with the essence of the phenomenon of tense itself. I therefore think it is unwarranted to claim that tenses can only be marked by an inflectional morpheme and not by a free morpheme such as an auxiliary.[11]

2.2.2 Apart from the fact that they do not involve a particular inflectional tense morpheme, the present perfect and the future are often discarded from the English tense system on the plea that instead of expressing tense they express aspect and modality, respectively. Again, I do not wish to subscribe to this view. In my opinion, the claim that 'the future is not a tense at all, but a mode' (Cygan 1972:9) is unwarranted, because it is an overstatement. All that we can say is (1) that the future tense has modal connotations (since an utterance about a situation that has not yet held is, of necessity 'a subjectively modalized utterance: a prediction rather than a statement' (Lyons 1977:815),[12] and (2) that in all

10. Dahl (1985:22) makes the same point when he argues that limiting oneself to inflectionally marked categories is too narrow an approach seeing that syntactically and morphologically expressed categories frequently have the same function in different languages. Moreover, it is rightly pointed out by Canavan (1983:46) that 'in semantic terms, past, present, and future are contrastive sense-relations which are independent of morphological considerations'.
11. We can also refer here to Lyons' (1977:678) remark that 'semantically ... tense is a category of the sentence' and that 'the view that tense is necessarily an inflectional category of the verb' is therefore incorrect.
12. In fact, the future tense is intrinsically connected not only with nonfactive (irrealis) modality but also with obligative and desiderative modality (see Kress 1977; Fleischman 1982:14-16; Comrie 1989:59).

Indo-European languages future tenses have developed out of modal forms and are 'rarely, if ever, used solely for making statements or predictions, or posing and asking factual questions, about the future' (Lyons 1977:816). To my mind, it does not follow from this that English has no future tense. There is no *a priori* reason why a verb form should be unable to express a temporal relation and a modal connotation at the same time (in the same way as a verb form often expresses both a temporal relation and an aspectual meaning). Besides, the French future tense (which, as noted above, has developed from a structure expressing obligation) is as modal as its English counterpart, but its existence as a tense is less controversial because it involves an inflectional morpheme.[13] Furthermore, it should be pointed out that at least one of the future tenses, viz. the conditional tense (expressing future-in-the-past), has a use which does not have the connotation of nonfactive (irrealis) modality, since it asserts the occurrence of a situation at a time which is past with respect to the present moment (but future with respect to the past reference time):

(7) (a) Fifteen years later Bill would be the richest man in town.
(b) Berlin would not fall before 1945.

And finally, Dahl (1985:105ff.) and Comrie (1989:53-6) adduce cross-linguistic and diachronic evidence that the future in modern English (and in many other languages) is a tense that has certain secondary modal uses, rather than the other way round. (For further arguments against the view that there is no future tense in English see Matthiesen (1983: 407–11).)

Similar arguments can be adduced in connection with the present perfect. Granted that this verb form may express perfect aspect (though it is not easy to define what exactly is meant by this), this is no reason for claiming it is no tense.[14] Those who do make the claim[15] must argue that the present perfect does not express a temporal schema of its own,

13. Diachronic observations also provide no basis for calling the preterit a tense while denying the future this status. According to Kuryłowicz (1972:181), 'it must be stressed that in all languages – as far as we are able to infer from history or from reconstruction – the forms of both the pret. and the fut. result from a revaluation (semantic change) of old presents'. Thus, 'forms like *I wrote*, G *ich schrieb* go finally back to the IE perfect, which according to the IIran. and Gr. testimony denoted originally a pres. state or result of action. . . . The IE fut. was originally a desiderative, a pres. form expressing the desire of action.'
14. That tense and aspect are not incompatible is of course clear from the fact that all tenses easily allow the use of *be. . .-ing* to signal progressive aspect. But even apart from this, it has often been noted that it is not possible to draw a neat distinction between tense and aspect. 'Thus, for example, a "completive" marker inherently signals anteriority of a situation with respect to some temporal axis, and in some contexts (in so-called tenseless languages) may function specifically to indicate temporal ordering' (Botne 1981:16).
15. See, for instance, Visser (1970:2192): 'The term "tense" for the perfect is apparently a misnomer.'

but realizes the temporal schema of a 'true' tense plus the expression of perfect aspect. It is striking, however, that there is no unanimity as to which of the tenses this 'true tense' is: whereas some linguists (e.g. Palmer 1988:35) claim that the present perfect is a combination of the present tense with perfect aspect,[16] others (e.g. Comrie 1985) claim that the present perfect realizes the same temporal schema as the preterit and differs from the latter only in that it also expresses perfect aspect: 'however perfect differs from past, it is not in terms of time location' (Comrie 1985:78). This is a curious situation, indeed. Does the present perfect minus perfect aspect yield the temporal schema of the present tense or that of the preterit? In my opinion it yields neither; the present perfect realizes a temporal schema of its own, and should therefore be considered a tense of its own. (I will come back to this point at greater length in chapters 5 and 7.)

2.2.3 Apart from the above two arguments, there are some minor pieces of evidence that are often adduced for claiming that the *shall/will*-future and the present perfect are not tenses. For example, it is pointed out that *shall/will* is only one of several means to express future time, as we can also use the present tense (progressive or nonprogressive) or a construction with *be going to* to this effect. Once again, I do not think this argument (which is advanced e.g. by Smith (1978b:49)) is valid, for several reasons:

1 The same thing can be pointed out in connection with languages that contain an inflectionally marked (and hence unimpeachable) future tense: in French too it is possible to use the present tense or a construction with *aller* ('go' or 'be going') with future time reference: e.g. *Il vient demain* ('He comes tomorrow'), *Il va le faire* ('He is going to do it').
2 A great many verbs (e.g. the statives) do not allow the use of a present tense or *be going to* to refer to the future: **Tomorrow I know/*am knowing/??am going to know why he did it*. The *shall/will*-future, in contrast, is possible with any verb.
3 The *shall/will*-future is a verb form which is capable of establishing future time reference by itself. If we use the present tense, a future time interpretation will only offer itself if it is invited by a time adverbial, by the context or by pragmatic factors. Taken by itself, the present tense does not suggest future time.
4 *Shall* and *will* are the only forms that can combine with a perfect infinitive to form the future perfect. This suggests that the typical realization of the future tense also involves one of these auxiliaries (but then with a present infinitive).

16. Palmer actually speaks of 'perfect phase' rather than 'perfect aspect'.

5 *Shall* and *will* are also the forms that combine with a progressive infinitive to form the future progressive tense.

The above arguments, I think, warrant the conclusion that it is the construction with *shall/will* that is the unmarked means of expressing future time in English and which may therefore be given the label of 'future tense'.

The overall conclusion of section 2.2 is that, even if it is correct that English speakers basically distinguish between a past and a present 'perspective',[17] there is no reason to claim that English has only two tenses.[18] I will therefore work with the full set of eight tenses referred to above.

17. In chapter 2 I will introduce the notions 'past time-sphere' and 'present time-sphere' to account for this.
18. Sørensen (1978) and Davidsen-Nielsen (1987) adduce further arguments supporting this conclusion.

2 Tense in discourse 1: general principles

The following terms and principles will prove indispensable when we describe the various ways in which tenses are used in discourse.

1 SITUATION

Like Comrie (1976, 1985), Lyons (1977) and many others, I will use the term 'situation' as a cover term for the various 'things' that can be expressed in a clause (viz. actions, processes, states, events . . .).

2 TEMPORAL ZERO-POINT (t_0)

I will adopt Lyons' (1977:278) term 'temporal zero-point' (t_0) to refer to the time which is the ultimate 'origin' of all the temporal relations expressed in the sentence, i.e. the time to which all the situations referred to in the sentence are directly or indirectly related, and which is not itself represented as dependent on any other (more basic) time. (Bull (1963) calls this time the 'prime axis of orientation'; Chung and Timberlake (1985:204) speak of the 'tense locus'.) In English, two kinds of times can serve as t_0: the time of utterance (coding time) (i.e. the time of speaking or writing) and, if it does not coincide with the former, the time of decodification (reception).[1] The unmarked situation is for the two times to coincide. This is the case in 'face-to-face conversation' (Fillmore

1. This double possibility concerning the choice of t_0 (temporal deictic centre) runs parallel with the double choice that is sometimes possible in connection with the spatial deictic centre. For example, verbs like *come* and *bring* allow the spatial deictic centre to lie either with the speaker (as in (i) or with the addressee (as in (ii)) – see Fillmore (1966, 1975):
   (i) *Come* and stay for the weekend and *bring* the children.
   (ii) Thanks, we'd love to *come*. Can we *bring* the dog too?
   It should also be noted that the time of coding and the time of decodification are not the only times that can function as t_0 in natural language. Comrie (1985:90ff.) observes that there are African languages in which the 'deictic center' for the tense system (i.e. t_0) is 'yesterday' (so that there are different tenses to locate situations at, before or after yesterday).

1976:92), which is the 'standard situation of communication' (Sennholz 1985:193) or 'canonical situation of utterance' (Lyons 1977:637). In some cases, however, the time of decodification is later than the coding time, and the speaker may choose the former as t_0. The following are some typical examples:

1 Fillmore (1976:93) records the case of a note found on an office door which read *I am in room 2114*. In this case the time referred to by the present tense 'is not the time at which the message was written but that time at which the message was expected to be interpreted'. (If the note had said *I have gone to room 2114* the same remark would have been applicable.) Road signs of the kind *You are now leaving West Berlin* constitute similar examples (see Comrie 1985:16).
2 Some newspapers are printed the day before they are issued. In that case the editors may use a tense system in which t_0 is not the time when the articles are written but the time of publication. Thus, if someone dies on 23 July this may be reported (in a text written that same day) as *Yesterday X died* (see Latzel 1974:285).[2]
3 The writer of a letter may take the time of decodification as t_0. Thus, in a letter which is written and sent before Christmas but which is expected to reach its destination only after Christmas, we may read such sentences as *I hope you had a nice Christmas* (see Fillmore 1972:167).[3]
4 Pre-recorded programmes may take the hearing time as t_0 (e.g. *The programme you are now listening to was recorded last Tuesday*).
5 The author addressing the reader of a book may take as t_0 the stage in the book where the reader has arrived. For example:

   (1) You have already read a good thirty pages and you begin to take to the story. At a certain moment you will notice ... (Lo Cascio 1986:195)

6 Descriptions of travel itineraries may also take the time of reading as t_0:

   (2) Today the train takes us to London. We visit the British Museum and Tate Gallery ...

2. Note that in this case the journalist selects as t_0 the official date of publication, since it is expected that most of the public are going to read the paper that day. Whoever reads the paper after that date must therefore compute all deictic indications of time (i.e. deictic tenses and adverbs) from the temporal standpoint of this publication time.
3. In Latin it was customary to use 'epistolary tenses' when writing letters. In this tense system, t_0 is the time of reception, and the past tense is used for any situation preceding this time, i.e. even for situations holding at the time of writing the letter. Thus we read sentences of the type *I was writing to you to tell you that ...* (where English would use *I am writing to you to tell you ...*).

7 Stage directions are typically written in the present tense. Here t_0 is the time when the play is actualized (i.e. read, rehearsed, acted, recalled in memory . . .)

In examples like these, deictic temporal adverbs (i.e. adverbs like *now*, *yesterday*, *today*, *tomorrow* which are interpreted as defining a time interval relative to t_0) are also related to the time of decodification.

3 THE TWO TIME-SPHERES

As has often been noted (see e.g. Bull 1963), there is a difference between 'objective (physical) time' and 'linguistic time'. From the point of view of physics, time is a unidirectional continuum, which may be represented by a line (the 'time line') and which consists of two parts, the past and the future, separated by the present. The present is no more than a point which divides the past from the future; it has no extension, and it continuously moves from left to right on the time line. Whereas physical time exists 'in abstraction from any given language' (Quirk *et al.* 1985:175), 'linguistic time' is time as it is perceived and talked about by language users. The most important difference between physical time and linguistic time (or, in the terminology of Lewis (1986:49), between 'real, objective time' and 'psychological time') is that the tense system of a language does not normally treat the past, the present and the future as equally important. It usually divides time into two (rather than three) 'time-spheres'. This is the case in English, where the tense system divides time into two time-spheres: the past time-sphere and the nonpast or present time-sphere.[4] The past time-sphere is an indefinite length of time which lies completely before t_0 (and hence does not include t_0). The nonpast (or present) time-sphere is an indefinite timespan including t_0. Some of the English tenses are used to locate situations in the past time-sphere. These 'past time-sphere tenses' (or 'past tenses') are the preterit, the past perfect, the conditional and the conditional perfect. The other tenses (viz. the present tense, present perfect, future and future perfect) are 'present (time-sphere) tenses'. They represent situations as lying in the nonpast time-sphere, where they may be anterior, simultaneous or posterior to t_0.

The claim that English divides time into two, not three, time-spheres is in keeping with the fact that only the past and the present tenses are morphologically marked in English. In my opinion, however, this is not the primary reason why one should subscribe to the claim. The main

4. Most languages that have tense distinguish these two time-spheres. That is, they 'treat the distinction of past and non-past as being of greater importance than the distinction of present and non-present or future and non-future' (Lyons 1977:809). There are exceptions, though. For example, in languages like Hopi and Dyirbal the basic temporal distinction is between the future and the non-future (see Whorf 1956; Ultan 1978:89; Chung and Timberlake 1985:205).

reason is that the hypothesis that English works with only two time-spheres is the only hypothesis that neatly accounts for all the data. This is something that I will attempt to make clear in the first part of this book. The fact that English has only two morphological tenses may be additional evidence, but I do not think we should set great store by this observation. As argued in chapter 1, there is no *a priori* reason why the number of morphologically marked tenses should run parallel to the number of time-spheres distinguished by the language. In French, where there are more than two morphologically marked tenses, we also find that 'there exist basically two temporal perspectives: Past and Non-Past' (Rohrer 1986:79). Moreover, it may be a rather arbitrary matter whether a particular tense is morphologically marked or not. In chapter 1 examples have been given which show that whether or not a tense is morphologically marked is the result of a historical development possibly determined by arbitrary factors that have nothing to do with the essence of the phenomenon of tense itself. This means (1) that there need not be a one-to-one relation between the number of morphological tenses and the number of time-spheres, and (2) that there is no reason for assuming that only morphological tenses count as real tenses. I will therefore join the tradition which distinguishes eight English tenses: the present, present perfect, future, future perfect, preterit, past perfect, conditional and conditional perfect. The first four of these are present time-sphere tenses, the others are typical of the past time-sphere.[5]

4 TIME OF ORIENTATION (TO)

This is one of a number of terms which will be explained at great length in chapter 6. At this point I will restrict myself to a brief definition, based on claims which will be substantiated only in chapter 6.

Let us take the following example as a starting-point:

(3) John was in the kitchen this afternoon.

In this sentence several times are referred to or implied:

1 A first time interval is the timespan indicated by *this afternoon*. I will refer to such a time (established by a time adverbial) as the 'Established Time'.
2 Secondly, there is the time interval at which the situation is located. This may be the Established Time as a whole (i.e. John may have been in the kitchen throughout the period indicated by *this afternoon*), but

5. Some linguists use the terms 'present time-sphere' and 'past time-sphere' in a way which is different from ours. For example, Poutsma (1922:14) distinguishes between three time-spheres (viz. the past, present and future time-spheres) and claims that the present perfect locates a situation in the past (rather than the present) time-sphere. (The claim that English recognizes three time-spheres is also made by Charleston (1955:265), Vasudeva (1971:164), and many others.)

it may also be a subpart of this (i.e. John may have been in the kitchen for some time in the course of the afternoon). This 'time interval at which the situation is located' will be called the 'situation-TO' (situation time of orientation). It is by definition so that the situation-TO either coincides with the Established Time or is included in it.

3 The length of time that John actually was in the kitchen will be called the 'time of the full situation'.

4 In a sentence like (3) the reference need not be to the full situation. The sentence is true not only if the time of the full situation coincides with the situation-TO but also if only a subpart of it does. That is, even if it was the case that John was in the kitchen all day, sentence (3) only refers to that part of the full situation that was simultaneous with the situation-TO (which either coincides with or is included in the Established Time *this afternoon*). The notion 'time of the situation' (TS) will therefore be defined as that subinterval of the full situation (possibly the whole full situation) that is located in time by the sentence, i.e. that is represented as simultaneous with the situation-TO.

In sum, the following claims can be made about sentence (3):

1 The adverbial identifies the Established Time.
2 The time that John actually was in the kitchen is the time occupied by the full situation.
3 The sentence locates a situation at a certain time, thus making a claim about a situation and a particular time interval. The claim in question is that the time of the situation and the time interval in question are simultaneous (i.e. coincide).
4 The time of the situation in question (TS) may be the time of the full situation or a subinterval of this.
5 The time interval in question (which we have called the situation-TO) may be the Established Time or a subpart of it.
6 The use of the past tense makes clear that the situation-TO is anterior to t_0.

Of these various terms, only the notion 'situation-TO' will be needed in this chapter. Roughly speaking, it is the time with which the situation is represented as simultaneous. The reason why I call this time the situation-TO is that I will be using the term TO for any time that functions as the origin of a temporal relation. Thus, t_0 is a TO because the situation-TO is anterior to it. The situation-TO is a TO because the situation is simultaneous with it. It may also be the TO of another temporal relation. For example, in *John remained in the kitchen after he had finished his dinner* the situation-TO of the first clause is the TO to which the situation-TO of the second clause is anterior.

5 SECTORS

When a situation is located in the past or present time-sphere, its situation-TO divides the time-sphere into three sectors: (1) the portion of the time-sphere up to the TO (the 'anterior sector'); (2) the portion of the time-sphere that begins at TO and stretches into the future (the 'posterior sector'); (3) a portion of the time-sphere that is centred around the TO (the 'simultaneous sector'). Thus, in *John said he was tired because he had worked hard and that he would go to sleep early* the TO of the situation denoted by *said* divides the past time-sphere into three sectors; the situations indicated by *was tired*, *had worked hard* and *would go to sleep* are located in the simultaneous sector, anterior sector and posterior sector, respectively.

6 PRESENT SECTOR, PRE-PRESENT, POST-PRESENT

The fact that t_0 lies in the present time-sphere entails that the present time-sphere is automatically divided into three sectors: the sector anterior to t_0, the sector simultaneous with t_0, and the sector posterior to t_0. For ease of reference I will use the terms 'pre-present sector', 'present sector' and 'post-present sector' to refer to these.

7 ABSOLUTE SECTORS

Whereas the present time-sphere is automatically divided into three sectors, the past time-sphere is not subdivided into sectors as long as no situation has been located in it. This means that the past time-sphere can be treated as a single sector, which is defined as lying completely before t_0. Together with the three present time-sphere sectors, the past sector then constitutes the set of 'absolute' sectors, i.e. the four sectors that are defined in direct relation to t_0. The tense system provides special means of locating a situation in each of these sectors. English uses the preterit, present perfect, present tense and future tense to locate a situation in the past, pre-present, present and post-present, respectively.[6] These are therefore the four tenses that can be used as 'absolute' tenses.[7] The other

6. Grammatical handbooks state that the English speaker's choice between the preterit and the present perfect depends on whether or not he is thinking of a time interval which is completely over at t_0. Translated into our present terminology this means that the choice depends on whether he is conceiving the timespan in which he is going to locate the situation as the past time-sphere (past sector) or as the pre-present sector (i.e. as that portion of the present time-sphere that is anterior to t_0).
7. The claim that there are four absolute tenses is also made by Curme (1931:354). Still, it is a rather unfamiliar claim in the linguistic literature. As noted before, most linguists recognize only two absolute tenses, viz. the preterit and the present tense; some also include the future tense (e.g. Huddleston (1969), Wunderlich (1970), Vasudeva (1971:164), Halliday (1976), Matthiesen (1983:371), King (1983), Comrie (1985), Enç (1986)). However, I hope to show that we can build a much neater theory of tense if we treat the present perfect as an absolute tense too (i.e. if we treat the pre-present sector as different from both the past and the present sectors).

four tenses can only relate a situation to a TO which is already established in one of the sectors. That is, they can only be used as 'relative' tenses.[8]

It should be noted that the absolute sectors cannot be defined in terms of objective time. They depend on how the speaker conceptualizes time. As has often been noted, the present (in our terminology: the present sector) can be conceived as a point (as in *I declare the meeting closed*) or as a period of indefinite length (e.g. *London lies on the Thames*). Similarly, the pre-present can be conceived either as very short or as stretching indefinitely far back, while the past time-sphere may be conceived as distant from t_0 or as almost reaching up to it. It follows that the same situation can often be conceived as lying in the past sector or as lying in the pre-present (compare *I met Ann just now* with *I have just met Ann*).

8 TEMPORAL DOMAIN

By temporal domain we mean the time interval taken up by a situation or by a number of situations which are temporally related to each other by means of special tense forms. In

(4) John said he was tired because he had worked hard and that he would go to sleep early.

the reference is to a 'past domain', i.e. a time interval which lies in the past time-sphere and which comprises the times of the four situations referred to. These four situations are related to each other in terms of simultaneity, anteriority or posteriority – relations which are expressed by the verb forms themselves.

9 BINDING TO

In a domain, each situation-TO is related to another TO. We will say that the latter is the 'binding TO' of the situation-TO in question.[9] In (4), the

8. Our use of the terms 'absolute' and 'relative' tenses is in accordance with Sgall *et al.* (1986:165) and many other linguistic treatments of tense. Alternative terms are 'deictic' vs. 'anaphoric' (see e.g. Lo Cascio (1986) and Adelaar and Lo Cascio (1986)), 'primary' vs. 'secondary' (Matthiesen 1983:381), 'primary' vs. 'relational' (King 1983), and 'exophoric' vs. 'endophoric' (Brecht 1974:495). (It should be noted, however, that these linguists do not normally include the present perfect in the set of absolute tenses.)
9. The term 'binding time' is used in much the same sense in Lo Cascio (1986), Adelaar and Lo Cascio (1986), and several other studies. Linguists that are oriented towards tense logic or formal semantics often speak of 'evaluation time'. (See e.g. Enç 1987). However, the notion 'evaluation time' does not always correspond with what we call the 'binding TO'. In fact, it is a rather hybrid term (see also below) and I will therefore avoid using it in this book.

situation-TO of *said* acts as binding TO for the situations denoted by *was tired* and *would go to sleep.* The situation-TO of *was tired* is the binding TO of *had worked hard.*

10 CENTRAL TO

The central TO of a domain is the TO of the situation which establishes the domain and which acts as binding TO for the next situation that is introduced into the domain. The central TO in example (4) is the situation-TO of *said.* The situation-TO's corresponding with *was tired* and *would go to sleep early* are related to this TO in terms of simultaneity and posteriority, respectively. The choice of tenses is in accordance with this.

11 ANCILLARY TO

For ease of reference we will use the term 'ancillary TO' for any situation-TO that is temporally subordinated, i.e. which is not the central TO of the domain in which it occurs.

12 SHIFT OF TEMPORAL DOMAIN

When clauses follow each other in a stretch of discourse, the situations which they describe may or may not be located in the same temporal domain.[10] Consider:

(5) (a) I noticed that Bill was talking with Emma and that they seemed excited about something.
(b) Suddenly the bell rang. Bill stood up from his chair, went to the front door and opened it.

In (5,a) the three situation-TOs are simultaneous with each other. That is, the past tense *noticed* is an absolute tense form establishing the central TO of a past domain, and the other two preterit verb forms are relative tense forms which represent situations as simultaneous with this central TO. The three situations are thus located within the same domain. In (5,b) the situations follow each other, but the tense forms do not relate any of them to one of the others in terms of anteriority or posteriority. (As is clear from (5,a), the preterit can be used either as an absolute tense establishing the central TO of a past domain or as a relative tense representing a situation-TO as simultaneous with some TO of a past domain. But the preterit cannot be used as a relative tense expressing

10. In some cases the choice of temporal domain remains free, in others it is determined by the context and the kind of clause. Such restrictions will be investigated in chapter 3.

anteriority or posteriority in a past domain. The relative tenses used for this purpose are the past perfect and the conditional tense, respectively – see below.) Since all the past tenses in (5,b) are thus absolute tense forms, we must conclude that each clause in (5,b) locates a situation in a different past domain. I will use the term 'shift of temporal domain' to refer to this phenomenon. The fact that such shifts are possible means that it is a general principle that, in a stretch of discourse, the temporal domain may or may not be kept constant as the discourse progresses.

Examples like (5,b) illustrate one possibility of shifting the temporal domain in one and the same time-sphere: a situation-TO which follows the central TO of the current domain in time is not related to the latter TO but is represented as the central TO of a new domain. Another possibility is for an ancillary TO which is prior to the central TO to develop into the central TO of a new domain. This is the case, for instance, in

(6) John had been in Venice before. He went there when he was only fourteen.

In this example the situation-TO of the first clause is an ancillary TO anterior to the central TO. The same time (viz. the time of John visiting Venice) is referred to in the second clause, but this time the situation-TO is no longer treated as an ancillary TO: the situation-TO of *he went there* is the central TO of a new past domain. This shift of domain is marked by the shift of tense (from the past perfect to the preterit). It should be noted that such a shift of domain within the same time-sphere is not normally obligatory. Instead of (6), the speaker might also have uttered (7):

(7) John had been in Venice before. He had been there when he was only fourteen.

In this sequence the situation-TO of the second clause is not represented as a new central TO but as an ancillary TO anterior to the central TO of the current domain.

The above examples have revealed two possibilities of shifting the domain in one and the same time-sphere. However, it goes without saying that a shift of domain may also be a shift from one time-sphere or sector to another. For example:

(8) (a) I saw him yesterday, I have seen him today, and I will see him again tomorrow.
(b) He said that Betty is a very clever girl.
(c) Bill said that the meeting will take place tomorrow.

In (8,a) we notice a shift from a past time-sphere domain to a pre-present sector domain, and then to a post-present sector domain. In (8,b) the

shift is from a past to a present domain. In the latter example, the shift is optional, since the speaker could also have chosen to keep the domain constant, as in

(9) (a) He said that Betty was a very clever girl.[11]
(b) Bill said that the meeting would take place tomorrow.

As we will see in chapter 3, the decision to shift the domain rather than keep it constant will generally be based on pragmatic considerations.

When there is a shift of temporal domain, the temporal relation between the old domain and the new one can be signalled by temporal adverbs (e.g. *later*, *afterwards*), but not by the tenses that are used. Tenses can only express temporal relations that hold between TO's that belong to the same domain. This is clear from examples like the following:

(10) (a) John became ill. He died two weeks later.
(b) John became ill. He would die two weeks later.

In (10,a), where there is a shift of temporal domain, the time adverb *later* situates the second domain relative to the first. However, the tense forms (*became*, *died*) do not provide the information that the second domain is posterior to the first; the only information they furnish is that the two domains belong to the past time-sphere. In (10,b) the two clauses belong to the same domain. The information that the situation-TO of the second clause is posterior to that of the first clause is now furnished both by the temporal adverb and by the tense form of the second clause. A similar illustration is provided by the following sentences:

(11) (a) Two weeks ago the chairman of the board unexpectedly died. He had been a wise man, whose contribution to the firm had been invaluable.
(b) Two weeks ago the chairman of the board unexpectedly died. He had been a wise man, whose contribution to the firm was invaluable.
(c) Two weeks ago the chairman of the board unexpectedly died. He was a wise man, whose contribution to the firm was invaluable.

These three sentences show three different ways of locating the same three situations in time. In (11,a) the first preterit (*died*) establishes the central TO of a past domain, and the two situations referred to in the second sentence are related as anterior to this central TO. In doing so, they are not related to each other, but it is pragmatically clear that the two situations can only be understood as being roughly simultaneous

11. In section 25 a test will be discussed that distinguishes between preterits that shift the domain and preterits that do not.

with each other. Sentence (11,b) differs from (11,a) only in that the third situation (*was invaluable*) is no longer represented as anterior to the first. However, this anteriority relation remains recoverable, since for pragmatic reasons the third situation is interpreted as simultaneous with the second, which is itself represented as anterior to the first. In (11,c), finally, the second clause shifts the domain. The information that the second and third situations are anterior to the first is no longer linguistically expressed. It is, however, still recoverable from our pragmatic knowledge of the world.

It should be noted that the third past tense form in (11,b–c) (viz. *was invaluable*) can in principle be interpreted either as a relative tense form (expressing the domain-internal relation of simultaneity) or as an absolute tense form (establishing its own domain).[12] In (11,c) this does not lead to ambiguity because it is pragmatically clear that even if a new domain is created, it must be interpreted as (roughly) simultaneous with the domain established by *was a wise man*.[13] In some cases, however, the double possibility does entail ambiguity. Consider:

(12) He asked whether he might give me a piece of advice, which he assumed would help me to solve my problems.

As pointed out by Hajicová, Panenová and Sgall (1973:199), a sentence like this is ambiguous because the nonrestrictive relative clause may or may not be interpreted as forming part of the (direct speech) utterance which is being reported. In my opinion, these two interpretations correlate with the two readings that can be assigned to the preterit *assumed*: if *assumed* incorporates the situation into the existing domain, it represents the assumption as simultaneous with the situation of the matrix (*might give me a piece of advice*) and therefore suggests that both situations were referred to in the original direct speech utterance; if *assumed* establishes a new domain, the suggestion is that there is no such direct link between the reported utterance and the assumption. (See also section 2.1 of chapter 3 and section 1.3 of chapter 4.)

13 SHIFT OF TEMPORAL PERSPECTIVE

The notion of 'shift of temporal perspective' will only be briefly introduced here. It will be discussed at greater length in section 20 below. What I mean by it is that the speaker sometimes treats a situation-TO from a particular sector as if it belonged to another sector. This means that the tense used to refer to the situation-TO in question or to relate

12. As we will see below, the shift of domain interpretation is the marked possibility in subclauses.
13. The possibility of interpreting two independently established domains as simultaneous will be discussed more extensively in section 17.

another situation-TO to it is not a tense typical of the sector to which it belongs, but is a tense that is characteristic of another sector. A typical example of this is the use of the so-called 'historical present': situations from the past time-sphere are represented as if they belonged to the present sector. The same thing can be done in connection with situations belonging to the future: in some contexts the speaker may also represent a situation from the post-present sector as if it belonged to the present sector:

(13) (I can well imagine what will happen. I can see it happening before me:) John *comes* home, a little tipsy, as always; he *finds* his bed occupied by a stranger, and he *makes* the most awful scene imaginable.

As we will see, shifting the temporal perspective is a possibility which is very often made use of in English. However, it would be premature to discuss other examples here. The examples given should be sufficient to create an intuitive understanding of what shifting the perspective means.[14]

14 TEMPORAL SUBORDINATION

As we have seen, each of the four absolute sectors has its own tense to establish a domain in the sector. A situation whose TO is to become the central TO of a past domain is referred to by the preterit. The pre-present, present and post-present sectors use the present perfect, present tense and future tense, respectively, for the same purpose. Once a domain has been established, other situation-TOs can be introduced into it. In that case each new situation-TO is related (bound) to one of the TOs that are already part of the domain. I will refer to this process as 'temporal subordination' (see Allen 1966; Tregidgo 1979; Wekker 1980). It is important to stress, as Tregidgo (1979) does, that temporal subordination is not the same thing as syntactic subordination (of clauses). The two differ in several respects: (1) subordinate clauses may or may not show tense subordination (see *He said he would/will come*); (2) 'subordinate tenses' (i.e. relative tenses) may sometimes occur in unembedded clauses (e.g. when a sentence like *He had worked hard all day* is the opening sentence of a novel); and (3) in a sentence like *The boy who told me the news had witnessed the accident* the subordinate clause uses an absolute tense form and the superordinate clause a subordinate tense form.

14. In order to avoid any misunderstanding I wish to state explicitly that the way in which I will be using the term 'shift of temporal perspective' is not the same as the way in which the term is used by Kamp and Rohrer (1983).

15 THE EXPRESSION OF DOMAIN-INTERNAL RELATIONS

Each absolute sector has its own system to express the (domain-internal) relations that are created by the process of temporal subordination. The purpose of this section is to investigate these systems in detail.

15.1 The relations in a past time-sphere domain

The system of tenses expressing relations in a past domain is relatively simple:

15.1.1 Simultaneity in a past domain

To represent a situation-TO as simultaneous with some TO in the domain we use the preterit, irrespective of whether the binding TO is the central TO (as in (14,a)) or an ancillary TO (as in (14,b–d)):

(14) (a) He said that he *was feeling* hungry.
(b) He said he had panicked when the milk *boiled* over.
(c) He said he would do it when he *had* time.
(d) He had been talking to someone while he *was waiting* for Mary.
(e) It's a pity Alice had to pass away when she *did*. (E. Caldwell, *This very earth*)

However, it should be noted that two situations that are simultaneous (in terms of real time) are not necessarily *represented* as simultaneous, i.e. temporally related to each other. Since the speaker has a large measure of freedom in describing situations, he may also make use of other possibilities that are available in the English tense system (and which will be discussed in later sections of this chapter). Thus, the speaker of (14,d) might also have said *He had been talking to someone while he had been waiting for Mary*. In that case he would not have represented the second situation as simultaneous with the first, but rather as anterior to the (implicit) TO which also binds the first situation. (In section 19 this type of tense subordination will be referred to as 'indirect binding'.)

15.1.2 Anteriority in a past domain

To represent a situation-TO as anterior to some other (central or ancillary) TO in the past domain, the past perfect is used:

(15) (a) He thought I *had been living* there for some time.
(b) He said he would do it after the others *had left*.
(c) He had said he *had done* it all by himself.
(d) The lawyer consulted Mary after she *had left* John after she

> *had thrown* a pot of hot coffee at him when he yelled at her after they *had* just *come* home from the party. (Smith 1975:73)

Here again it should be noted that the temporal relation which exists between two situations in real time is not necessarily expressed by the speaker reporting the situations. This explains why sentences of the type *I spoke to the boy whose father had died a week earlier* can alternate with sentences like *I spoke to the boy whose father died a week earlier.* In the former case the subclause situation is temporally subordinated to the head clause situation. That is, *had died* is a relative tense form expressing anteriority in a past domain. However, there is no temporal subordination when the preterit is used: *died* is an absolute preterit establishing a new domain (i.e. shifting the domain).

It is also worth noting that apart from the past perfect there is another tense that expresses anteriority in a past domain, viz. the conditional perfect (*would* + perfect infinitive). However, the conditional perfect does more than just express anteriority. It also makes clear that the binding TO to which the situation is anterior is itself posterior to another TO in the past domain. The conditional perfect thus expresses two relations within the past domain. For the time being I will restrict myself to the relative tenses that just express one relation. The conditional perfect will be discussed in section 17.

15.1.3 Posteriority in a past domain

To represent a situation as posterior to some TO in the domain we use the conditional tense:

(16) (a) I thought he *would help* me.
(b) He had promised that he *would* henceforth *behave* himself.
(c) He promised that he *would* soon *tell* me when he *would make* his decision.

It should be noted that the conditional tense (*would* + infinitive) is not the only verbal means of expressing posteriority in a past domain. There are other possibilities, such as the use of *was/were going to* or *was/were to*:

(17) (a) He said there *was going to be* a storm in a minute.
(b) The King arrived at 11 a.m. The Queen *was* not *to arrive* until the day after.

As is well known, these alternatives have special connotations and therefore typically appear in different kinds of context. Together with *would* they constitute a set of forms which are more or less in complementary distribution. For ease of reference I will only mention the

conditional tense when speaking about the expression of posteriority in a past domain. It should be borne in mind, however, that there are other forms besides this.[15]

15.2 The relations in a pre-present sector domain

A pre-present domain is always established by a present perfect. As is well known, a present perfect locates a situation in a period that starts before t_0 and leads up to it. The situation located in this period can either lie entirely before t_0 or lead up to (and include) t_0. In the former case the present perfect is said to have an 'indefinite' ('existential') meaning, in the latter it is 'continuative'. Now, for the expression of temporal relations it will make a great difference whether t_0 is or is not included in the situation-TO that functions as binding TO. We therefore have to treat the two possibilities separately.

15.2.1 Domain established by an indefinite perfect

As is well known, an indefinite perfect is normally only used to introduce a situation (i.e. to establish a domain in the pre-present sector), not to go on speaking about it after this has happened. In order to do that we normally switch to the preterit. (The reason for this will be explained in chapter 7.) This means that the indefinite perfect will only be used to establish the central TO of the domain. If we wish to relate other situation-TOs to this central TO, we shift the temporal perspective. That is, what is initially a pre-present domain is reinterpreted as a past domain, and the system used to express domain-internal relations is that typical of a past domain. For example:

(18) I have tried doing it differently, but *I didn't like* the result.

The clause *I have tried doing it differently* locates a situation (which does not reach up to t_0) at some indefinite TO within a period reaching up to t_0 (i.e. within the pre-present sector). However, once the speaker has done this, and wishes to say more about the situation in question, his next sentence shows a shift of temporal perspective: the speaker now concentrates on the situation itself, which lies wholly before t_0, and disregards the link with t_0. This link with t_0, which was expressed in the first clause (where the speaker is not thinking of the past time-sphere but is concerned with the state of the world at t_0) is no longer relevant in the next part of the discourse. The speaker now concentrates on the past situation itself. It follows that the domain, which was initially established

15. In some cases futurity in a past domain can also be expressed by the preterit, as in *We had to hurry, because the ship left at 4.30*, or *He told me he was leaving the next day*. In section 20 this use of the preterit will be analysed as a shift of temporal perspective.

in the pre-present sector, is further developed as if it were a past time-sphere domain. There is thus a shift of temporal perspective from the pre-present sector to the past time-sphere. The situation functioning as central TO is referred to in the present perfect tense, but the other situations introduced into the domain are referred to by means of past time-sphere tenses. This is clear from (18) and from similar examples like the following:

(19) (a) I have eaten lobster once, but I can't say I *enjoyed* it.
(b) He's been here once or twice, while his wife *was* on holiday.
(c) They have eaten all the fudge while you *were* out. (Smith 1978b:53)

Sentences like these clearly show the shift of interest from the *fact that* the situation has taken place to the *occasion when* it took place, and the corresponding shift of temporal perspective from the present time-sphere to the past time-sphere.

It follows from this shift that not only the preterit but also the other tenses typical of the past time-sphere can be found in subclauses relating a situation to the TO of a situation described by means of an indefinite perfect:

(20) (a) I have never promised that I *would help* you.
(b) I have never denied that I *had used* that money.
(c) I have seen her a couple of times, when I *left* the office early.
(d) Has she ever told you anything after she *had quarrelled* with her husband?
(e) I have just met your brother. He *was coming* out of the bank.
(f) The Egyptian leader has died at a most awkward moment for the Russians, who *were trying* to consolidate their position in the area. (McCoard 1978:82)

It goes without saying that this shift of temporal perspective can only occur if the situation introduced in the present perfect sentence lies wholly before t_0, i.e. if the perfect is an indefinite perfect. If it is a continuative perfect, the temporal perspective remains within the present time-sphere (see below):

(21) (a) I have been watching this programme since it started and I *have enjoyed/am enjoying* it very much.
(b) I have been digging the garden, while you *have been doing* nothing.

There are two further things to be noted:

1 Sometimes the indefinite perfect establishing the pre-present domain does not refer to a semelfactive situation. It refers to a repetitive situation that consists of an indefinite number of subsituations. In that case the

domain is felt to be a domain that reaches up to t_0, even if the individual subsituations belonging to it lie before t_0. When adding a new clause, the speaker may be guided by either of these factors, so that there are two possibilities as far as the tense is concerned:

a. In most cases the speaker makes use of the possibility of shifting the temporal perspective and uses one of the tenses typical of the past time-sphere:

(22) (a) I have often cried when I *was feeling* lonely.
(b) I have often cried when I *had been laughed* at by the others.
(c) I've fed all three, . . . as much as I *wanted.* (Allen 1966:168)
(d) I flatter myself that when prattle *was needed* I have never been found wanting. (Fenn 1987:78)
(e) Well, father, I've often tried to faint when I *wanted* something that you *wouldn't give* me. (Fenn 1987:93)

b. If the two situations are simultaneous, both can be referred to by means of the present perfect:

(23) (a) I have often cried when I *have felt* lonely.
(b) John has batted well when he *has played.* (Hornstein 1975:92)
(c) The right honourable gentleman, once or twice, when I *have inquired* into these things, has sheltered himself behind the plea that this is a self-governing colony. (Poutsma 1926:259)

We will see below that in such sentences both the head clause and the subclause establish their own pre-present domain. (These domains happen to be simultaneous, as is often the case – see section 17.2.)

2 When the shift of temporal perspective takes place (as in (19)–(20)), the speaker just relates the subclause situation to the pre-present binding TO. In this way no relation is expressed between the bound situation-TO and t_0. However, in some cases the speaker may wish to represent a relation of some kind between the subclause situation and t_0. He may, for example, wish to represent the subclause situation as still relevant to the structure of the world at t_0. In that case he does not relate the subclause situation to the head clause situation-TO but rather relates it to t_0. That is, instead of applying the temporal subordination process (involving a shift of temporal perspective) he shifts the domain, i.e. he has the subclause establish its own domain relative to t_0 (i.e. its own pre-present domain). This is the case in examples like the following:

(24) (a) The doctor has already confirmed that Bill *has sprained* his ankle.
(b) The news agency has reported that the astronauts *have landed.*

The present perfects in the subclauses of these examples are absolute present perfects.[16] Other absolute tense forms, such as the preterit or the future tense, can be found too:

(25) (a) The doctor has already confirmed that Bill *will be* unable to walk to school for some time because he *has sprained* his ankle.
(b) The doctor has already confirmed that Bill *sprained* his ankle yesterday.
(c) There *have been* times in my life when I required soothing, and then I *have felt* that a whiff of tobacco *stills* and *softens* one like a kiss of a little child. (Poutsma 1926:259)

The last example illustrates both temporal subordination and domain shifting. In the first conjunct, the perfect *have been* establishes a pre-present domain and the preterit *required* expresses simultaneity within this domain. Then there is a shift to a new pre-present domain, established by *have felt*.[17] This domain is not further expanded, for the present tense forms *stills* and *softens* again shift the domain, this time to the present sector.

It should also be noted that the fact that the present perfects in the subclauses of (24,a–b) establish their own pre-present domains means that these forms do not express a direct temporal relation between the situation of the head clause and that of the subclause. This is in keeping with our claim that only past time-sphere tenses can be used to express relations within a domain established by an indefinite present perfect.

15.2.2 Domain established by a continuative perfect

When the present perfect has continuative meaning, the domain which it establishes in the pre-present sector includes t_0. In that case there is no special system of tense forms to represent domain-internal relations. Since the domain includes t_0, all relations must start from t_0, i.e. they must be 'absolute relations' of the kind that establish domains in different sectors. This gives the following results:

16. The same is true in examples such as Berezovsky's (1978:87) *Someone has taken my book while I have been out* and Smith's (1975:72) *Alice has left while the red queen has been talking*. However, there is something unusual about this domain-establishing use of the present perfect in *while*-clauses. In clauses of this type it appears much more natural to use a past tense expressing simultaneity, i.e. to use a tense form that is in keeping with the temporal meaning of the conjunction.
17. The presence of the adverb *then* entails that the domain established by *have felt* is actually interpreted as simultaneous with the domain established by the first clause. As we will see below, there is nothing unusual about this. It often happens that two independently established domains are felt to cover the same time interval. What I am arguing here is that *have felt* does not express simultaneity within the domain established by *have been*; that the new domain which it establishes happens to be simultaneous with the other domain is immaterial to this.

15.2.2.1 Reference to an anterior situation

A situation that is anterior to a situation that is located in the pre-present sector necessarily lies wholly before t_0. To refer to it the speaker can therefore make use of the two possibilities which he generally has to refer to situations that are over at t_0: he can use a preterit to locate the situation in the past time-sphere or he can use an indefinite perfect to locate it (as lying wholly before t_0) in the pre-present sector.[18] For example:

(26) (a) I have known for some time that Jim *has sprained* his ankle.
(b) I have known for some time that he *did* not *do* it.

In such sentences the subclause establishes a domain of its own. That is, there is a shift of domain rather than temporal subordination.

15.2.2.2 Reference to a simultaneous situation

It goes without saying that a situation can only be conceived as simultaneous with a situation reaching up to t_0 if that situation itself also includes t_0. Because of this, two tenses can be used to refer to the situation in question, viz. the present perfect (as a continuative perfect) or the present tense. The former will be used if the speaker wants explicitly to represent the situation as starting in the past and continuing into t_0;[19] if this is not the case, the present tense is more natural. For example:

(27) (a) Ever since this morning I have been working while you *have been doing* nothing.
(b) Jim has known for some time that Joy *has been/is* in Reno. (Berezovsky 1978:91)
(c) Up to now he has always maintained that he *works* in a bank.
(d) Since then I've felt I'*ve missed* some vital experience. (Vermant 1983:73)
(e) You know, ever since we started this treasure thing, I've had the strangest feeling we'*re being* followed. (ibid)
(f) All the time I'*ve been* away, it's been shrinking and shrinking. (Fenn 1987:7)

18. The question of why the speaker may choose a past tense rather than a present perfect, or vice versa, need not concern us here. We will go into it in chapter 7.
19. In order to avoid any misunderstanding I wish to stress that in such cases the present perfect or present tense is not a relative tense. (There is only one relative tense that can express simultaneity in a pre-present domain, viz. the preterit.) Rather the present perfect and present tense forms in the subclauses of (27,a–g) are absolute tense forms, each of which establishes a domain which is (roughly) simultaneous with the head clause domain (since both domains cover a timespan which reaches up to t_0). (As noted before, the possibility of interpreting two independently established domains as simultaneous will be discussed more extensively in section 17.)

(g) Have all these things been fizzling away in here while we'*ve been* at the inquest? Suppose they'd been all burnt up. (A. Christie, *4.50 from Paddington*, London: Fontana, p. 63.)

In some cases the present tense is the only possibility because the present perfect would have indefinite rather than continuative meaning:

(28) (a) I have felt for weeks that you *want* me. (Rigter 1982:119)
(b) Ever since that day I have been telling them that John *lives* in London.

If the perfect refers to a repetitive situation, there are two possibilities. Compare:

(29) (a) He has always told me that he *works* in a bank.
(b) He has always told me that he *worked* in a bank.

In (29,a) *has told* is treated as a continuative perfect, whereas in (29,b) it is treated as an indefinite perfect. It follows that the subclause situation (which is simultaneous with the head clause situation) is referred to in the present tense in (29,a) and in the past tense in (29,b). This double possibility follows from the fact that the head clause situation consists of a number of subsituations. The situation as a whole reaches up to t_0 (hence the possibility of interpreting *has told* as a continuative perfect), but the various subsituations lie completely before t_0 (hence the possibility of treating *has told* as an indefinite perfect). The same two possibilities are also to be observed in the following examples:

(30) (a) 'I have always known that you *cared*', she sighed as he closed her in his arms. (Rigter 1982:119)
(b) I have always known that you *care* for me.

15.2.2.3 Reference to a posterior situation

Since a continuative perfect establishing the central TO of a domain represents this TO as including t_0, any situation that is posterior to the central TO must also be posterior to t_0. To refer to such a situation we therefore use the future tense (i.e. the absolute tense used for establishing a post-present domain):

(31) Ever since this morning he has repeated that he *will move* to London.

However, if the perfect establishing the domain represents the situation as repetitive, there is again the possibility of treating the perfect as an indefinite perfect (referring to subsituations that lie wholly before t_0). In that case there is a switch of temporal perspective to the past time-sphere, so that the tense used for posteriority is the conditional tense (or another form expressing posteriority to a past TO, e.g. *was going to* + infinitive):

(32) He has been telling me for months that he *was going to invite* me, but he has not done it yet.

This example (in which the *that*-clause is temporally subordinated) should be compared with (33), where the speaker treats *has been telling* as a purely continuative perfect and has the *that*-clause establish its own post-present domain:

(33) He has been telling me for months that he *is going to invite* me.

15.3 The present sector

If the present tense is used as an absolute tense, it establishes a present domain, i.e. a domain whose central TO is t_0 or is centred around t_0. The expression of temporal relations is then as follows:

15.3.1 Simultaneity

The present tense is used to represent a situation as simultaneous with the central TO, or with any other TO that includes t_0.

(34) (a) I am working, while he *is doing* nothing.
(b) I am just saying to the others that *I think* John *does* not *mean* what he *is saying.*

It should be noted that, in a present domain, simultaneity is the only temporal relation that can be expressed by means of a finite verb form. Since the central TO of a present domain is or includes t_0, reference to an anterior or posterior situation by means of a finite verb form will automatically mean that the situation is located in one of the other absolute sectors. Needless to say, the verb form in question is then an absolute tense form (which creates its own domain). (However, things are different when a nonfinite verb form is used, as in *He seems to have been ill.* Nonfinite verb forms cannot establish domains; they can only express temporal relations. In the above example the perfect infinitive thus expresses anteriority in the present domain.)

15.3.2 Reference to an anterior situation

What is anterior to a TO including t_0 will be referred to by either the past tense or the present perfect, depending on whether the speaker wishes to locate the situation in the past time-sphere or in the pre-present sector:

(35) (a) I am just explaining that I *did* it last night.
(b) I am just explaining that I *have* not *been* able to do it.

As noted above, these *that*-clauses involve an absolute tense form establishing a new domain. If this domain is expanded (i.e. if the

that-clause situation-TO serves as binding TO for another situation-TO), the tense to express this relation will be in accordance with the systems typical of past and pre-present domains, respectively:

(36) (a) I am just explaining that I have not been able to do it because I *have been feeling* ill.
(b) I am just explaining that I have not been able to do it because I *was feeling* ill (when I had to do it).
(c) I am just explaining that I did it last night because I *had* time then.

(37) (a) You know that I have always said that I *will/would help* you.
(b) I admit that I doubted if that *would be* possible.

(38) (a) I know that you confessed to John that you *had gone* out with Mary.
(b) I know that you have always denied that you *went/had gone* out with Mary.

In these examples some of the subclauses shift the domain, whereas others exhibit temporal subordination. For example, in (36,a) the head clause establishes a present domain, the *that*-clause shifts the domain (by establishing a pre-present domain) and the *because*-clause again shifts the domain (by establishing another, but simultaneous, pre-present domain). Sentence (36,b) is like (36,a) except for the fact that the *because*-clause now exhibits temporal subordination. In (36,c) there is a shift from a present domain (*am explaining*) to a past one (*did*), and then temporal subordination (*had*).

15.3.3 Reference to a posterior situation

In order to represent a situation as posterior to a TO including t_0 we will make use of the future tense (which is the absolute tense that is normally used to refer to the post-present) or some other expression establishing a post-present domain:[20]

20. In spite of the fact that *be going to* and *will/shall* are not always interchangeable, they do have the same basic temporal meaning. After investigating the use of the future tense and the *go*-future in various languages, Fleischman (1982:97) concludes that her findings 'militate strongly in favor of regarding the *go*-paradigm as a legitimate future-tense form'. Haegeman (1989:291) similarly argues 'that at the level of sentence meaning *be going to* and *shall/will* are equivalent, and that the difference between them is to be found in the constraints they impose on the processing in context of the utterance in which they occur'.

As for the use of the present tense with future time reference (e.g. *I'm leaving tomorrow*), we will see in section 20 that this is an instance of a 'shift of temporal perspective', i.e. the speaker represents the future as if it were the present. However, the present tense does refer to the future and therefore establishes a post-present domain.

(39) (a) I know John *will be* in London tomorrow.
(b) Why don't you admit you *are leaving* the country?
(c) I think it's *going to rain.*

To relate another situation to the bound post-present situation-TO we will use the tenses that are typically used to represent relations in a post-present domain (see below).

15.4 Relations in the post-present sector

When the speaker wishes to express a domain-internal relation in the post-present sector (i.e. when he wishes to relate a situation-TO to a TO in a post-present domain rather than to t_0), there is normally a shift of temporal perspective: the binding TO in question starts behaving as if it were t_0. This opens up all sorts of possibilities, for t_0 is the time from which four different sectors (viz. the four absolute sectors) are defined. The same thing now becomes possible in the post-present domain: the central TO of the domain can become the point from which four different sectors are defined, and the systems of expressing domain-internal relations in these sectors are exactly the same as those observed above in connection with the absolute sectors (which are directly related to t_0). This gives the following result:

15.4.1 Anteriority in a post-present domain

As we have seen, a situation that is anterior to t_0 (in the sense of 'lying completely before t_0') is located either in the past time-sphere or in the pre-present sector. Since a post-present TO behaves as if it were t_0, the same two possibilities are available when a situation is to be represented as anterior to it. That is, the expression of anteriority to a post-present TO involves a shift of temporal perspective to either the past time-sphere or the pre-present sector.

15.4.1.1 When there is a shift of temporal perspective to the past time-sphere, the bound situation is represented as lying completely before the binding TO and is not represented as lying in a period leading up to that TO. This is the case in (40,a–d). (Note that the situation in question is anterior to t_0 in (40,a), simultaneous with t_0 in (40,b), and posterior to t_0 in (40,c–d). This means that the position of the situation relative to t_0 is irrelevant to this system.)

(40) (a) The police will believe that he *was killed* yesterday.
(b) The police will find out that you *were staying* here today, and not in London.
(c) (*said while planning someone's murder*) The police will believe that he *was killed* tonight.

(d) From now on any new employees that *didn't work* hard enough will be fired at the end of the month.

It should be noted that the use of the past tense here does not entail a shift of domain because the past tense does not relate the subclause situation to t_0. It relates it to a post-present TO which merely behaves as if it were t_0. In other words, because the binding TO is a pseudo-t_0, the past tense is a pseudo-absolute tense, which actually serves to express anteriority in the post-present domain.

The pseudo-past subclause situation can serve as binding TO for other situations that are introduced into the domain. The latter situations are then referred to by means of the tense system that is typical of the past time-sphere. That is, the preterit is used for simultaneity, the past perfect for anteriority, and the conditional tense for posteriority:

(41) (a) (*said when planning someone's murder*) The police will think that he was killed when he *came* home after he *had attended* the meeting at his club. They will believe that he was murdered by the syndicate because he *had said* he *would* soon *leave* America.
(b) I will lie and say that when I reached the station the train *had left.* (Schopf 1984:299)

15.4.1.2 When a situation is located in the time-interval leading up to the central TO of the post-present domain, this time-interval is treated as if it were the pre-present sector. This means that the present perfect is used to locate situations in it:

(42) (a) He will soon find out that you *have been following* him.
(b) What will happen when the others *have left*?
(c) John is coming when he *has eaten.* (Wachtel 1982:340)

If the present perfect is of the 'indefinite' type, any situation that is related to the situation-TO established by the present perfect will again be represented as if it belonged to a past domain:

(43) (a) He will wonder if she has ever said that she *was* unhappy.
(b) The police will want to know if anybody has ever heard her say that her brother *had told* her that he *would change* his will.

If the present perfect represents its situation as continuing up to the post-present central TO, we will use the same system for relating a situation to this continuative situation as we use for relating a situation to a TO established by an absolute continuative perfect (i.e. a present perfect which represents its situation as continuing up to t_0):

1 *Simultaneity:* If not only the binding situation but also the bound one continues into the relevant post-present TO, the bound situation is expressed by either the present perfect or the present tense:

(44) (a) When Mary asks me what I've been doing in the library, I will tell her that I have been studying hard since nine o'clock while she *has been doing* nothing.
(b) She will whisper to you that she has felt for a long time that you *want* her.

2 *Anteriority:* If the new situation is to be represented as anterior to the situation that reaches up to the relevant post-present TO, there will be a shift of perspective to either the past time-sphere or the pre-present sector:

(45) (a) Tomorrow the police will tell him that they have known for some time that he *killed* his wife.
(b) Tomorrow the police will tell him that they have known for some time that he *has lied* to them.

3 *Posteriority:* If the new situation is posterior to the situation which reaches up to the post-present binding TO, it is represented as if it belonged to the post-present sector:

(46) He will tell you that he has been hoping that you *will help* him.

15.4.2 Simultaneity in a post-present domain

A situation that is simultaneous with the post-present binding TO is represented as if it were lying in the present sector. This means that the present tense is used.[21]

(47) They will believe that Jack *is* back in town.

It should be noted that sentences like these are potentially ambiguous, because the present tense in the complement clause can often also beinterpreted as an absolute tense (shifting the domain to t_0).[22] This is not surprising, because it is a general characteristic of such complement

21. This accords with the following statement by Allen (1966:168):
It appears, then, that the use of present verb forms (to show 'same time') and of present perfect verb forms (to show 'earlier time') in clauses included in sentences (or other clauses) referring to future time is actually what one should expect and not a violation of the rules of logic. The use of *will* in an included clause in such sentences would not be 'more correct', as Reichenbach suggests, nor are present verb forms used in such included clauses because 'futurity is sufficiently indicated in the main verb', as Jespersen states.
22. See Ejerhed (1981:239), who offers evidence that this is a question of true ambiguity, not a matter of the complement clause being unspecified or vague with respect to temporal index.

clauses that they can easily shift the domain to a sector which is different from that in which the head clause domain is located (e.g. *I believe that he was/has been/is/will be here*).

As will be noted below, the use of a present tense to express simultaneity in a post-present domain is restricted to some kinds of subclause (e.g. complement clauses, conditional clauses, restrictive relative clauses). In unembedded clauses and in the other types of subclause (e.g. nonrestrictive relative clauses) the post-present domain must be 're-established' by the use of another future tense form. Hence the difference between (48,a) and (48,b):

(48) (a) You will be met by a man who is wearing a red tie.
(b) You will be met by a man. He will be wearing a red tie.

I will go further into this in section 17.

Since, in sentences like (47), the binding post-present TO behaves as if it were t_0, the bound situation-TO which is represented as simultaneous with it also behaves as if it were t_0. This means that any of the four absolute tenses can be used to relate another situation to it:

(49) (a) Everybody will see that you are doing nothing while your wife *is doing* all the work.
(b) They will find out that Jack thinks that he *was* born in London.
(c) (Do not inform Jack, or) he will go and tell the police that he knows why the murder *has been committed.*
(d) (Do not inform Jack, or) he will go and tell the police that he knows that a murder *will be committed.*

15.4.3 Posteriority in a post-present domain

To represent a situation as posterior to a post-present TO the speaker will use the future tense, thus representing the situation as if lying in the post-present sector (i.e. in the future of t_0).

(50) He will say that he *will* never *leave* her.

This TO which is posterior to the central TO of the post-present domain may become the binding TO of another situation that is introduced into the domain. In that case it also behaves as if it were t_0, so that all the shifts of temporal perspective noted above may again take place. That is, the situations related to this TO may be represented as if they belonged to the past time-sphere, the pre-present, the present or the post-present:

(51) (a) She will think that John will tell the others that he *was* here today.
(b) She will think that John will tell the others that he *has* not *told* them the truth yet.

(c) She will think that John will tell the others that he *is* dissatisfied with their behaviour.
(d) She will think that John will tell the others that he *will tell* them the truth later.

And, of course, the TO of the situation of the most deeply embedded clause of (51,d) again behaves like t_0 if other situations are related to it, etc.

To close off this section on temporal subordination, I would like to point out that the above theory neatly accounts for an interesting difference that we may observe between (52,a–c) and (53,a–b):[23]

(52) (a) John had left at five o'clock.
(b) John had left when Bill arrived.
(c) John will have left at five o'clock.
(53) (a) John will have left when Bill arrives.
(b) (John will no longer be there at midnight.) He will have left when Bill arrived.

As has often been pointed out in the linguistic literature, (52,a) is ambiguous between two readings, which we can paraphrase as 'It was the case that John had left at five o'clock' and 'At five o'clock it was the case that John had already left'. (The latter interpretation is strongly invited if we give *at five o'clock* initial position in (52,a), or when we add *already*.) Sentence (52,b) is ambiguous in exactly the same way. Sentence (52,c) is ambiguous too: *at five o'clock* again indicates either the time of John's leaving or the TO to which John's leaving is anterior. Sentences (53,a–b), by contrast, are not ambiguous: (53,a) can only mean that John's leaving will be anterior to Bill's arrival, whereas (53,b) can only mean that the two situations will coincide. The theory of temporal subordination we have developed accounts for this difference between (52,b) and (53,a–b). In (52,b) both situations are located within the same past time-sphere domain. The past perfect *had left* represents John's leaving as anterior to some TO. Since the past tense can be used to express simultaneity with any TO in the past domain, it is not clear from the verb form *arrived* whether Bill's arrival is to be interpreted as simultaneous with the time of John's leaving or as simultaneous with the TO to which John's leaving is anterior. Hence the ambiguity of (52,b). In (53,a–b) there is no such ambiguity because the reference is now to a post-present domain. In a post-present domain different verb forms are used to express simultaneity. To express simultaneity with the central

23. Some of these sentences use a future perfect in the head clause. This is a tense to which I have not yet accorded a place in the system of tenses that I am developing. It will be investigated in section 17. However, this presents no problem for the observation that is made here, since the point that is made concerns the use of the tenses in the *when*-clauses of these examples.

TO, the present tense is used. This is the case in (53,a). To express simultaneity with a situation that is anterior to the central TO, the past tense is used. This is the case in (53,b), where the future perfect *will have left* represents John's leaving as anterior to a post-present TO and the past tense *arrived* represents Bill's arrival as simultaneous with John's leaving. (That the past tense must be used for simultaneity follows from the fact that in a post-present domain the expression of anteriority involves a shift of temporal perspective to the past time-sphere.) So, both the ambiguity of (52,b) and the non-ambiguity of (53,a–b) are predictable from our theory.

16 SLOPPY SIMULTANEITY

When the tense of a clause represents a situation as simultaneous with a binding TO, this does not necessarily mean that the binding situation and the bound one must be strictly simultaneous. When we say that two situations are simultaneous, this normally means that they either coincide or overlap:

(54) (a) He felt unhappy when he *said* that.
(b) We will be in London when he *arrives*.

However, there is only 'sloppy' simultaneity in sentences like the following:

(55) (a) When John *came* home, Mary told him about the accident.
(b) (Be careful in your speech tonight.) The newspapers will print everything you *say*.
(c) When the bell *rings*, everybody goes home.
(d) If Mary *dies*, John will come into a fortune.

Each of these examples refers to two situations which do not overlap but follow each other. Still, the tense used in the subordinate clause is each time the tense that is typically used to express simultaneity, not anteriority. Anteriority would be expressed as follows:

(56) (a) When John *had come* home, Mary told him about the accident.
(b) The newspapers will print everything that you *have said*.
(c) When the bell *has rung*, everybody goes home.
(d) If (by the end of the year) Mary *has died*, John will come into a fortune.

It appears, then, that the tense forms expressing simultaneity can be used in cases where there is 'sloppy' rather than strict simultaneity. In these cases the speaker disregards the fact that the two situations do not really overlap but concentrates on the fact that they follow each other closely and that there is some logical relation between them. The use of a tense

form expressing simultaneity then means that the speaker treats the two situations as belonging to the same 'occasion' and hence as falling within the same time interval.[24] Another clear illustration of such a use is provided by examples like the following, where the relevant verb form is a form whose basic meaning is the expression of simultaneity, viz. a present participle:

(57) (a) Opening the drawer he took out a revolver. (Thomson and Martinet 1969:166)
(b) Raising the trapdoor she pointed to a flight of steps. (ibid.)
(c) The locomotive jumped the rails, narrowly missing a couple of trees near the railway.

The fact that verb forms expressing simultaneity can be used in a sloppy way illustrates the fact that simultaneity is the unmarked relation of the three that can be expressed by verb forms.[25] This is especially clear in conditional sentences referring to the future, where the expression of a temporal relation between the two clauses is less important than the expression of the logical link ('if p then q'). In such conditional sentences the *if*-clause therefore seldom needs to express anteriority (though it does in examples like *You will not get a full pension if you haven't worked for forty years*). It follows that most *if*-clauses involve a tense form expressing simultaneity. This may even be the case when the two clauses contain adverbials referring to different times, as in *If John doesn't do it today, I will do it myself tomorrow*. In such a sentence the speaker represents the two situations as if simultaneous if he is just thinking of them as two post-present situations that are logically connected with each other.[26] It is only when he wants to draw explicit attention to the fact that one situation is over when the other takes place that he selects the verb form expressing anteriority (*If John hasn't done it today, I will do it myself tomorrow*).[27]

24. In fact, this also holds for the absolute use of the present tense, which represents a situation as simultaneous with (i.e. as including) t_0. Leech (1971:3) points out that the 'instantaneous' use of the present tense does not require that the situation is strictly simultaneous with t_0. A sentence like *Now I open the envelope* may be uttered seconds before the action takes place. As Leech points out, 'it is subjective rather than objective simultaneity that is conveyed'.
25. Simultaneity forms are unmarked in two respects: they are formally less complex than the verb forms expressing anteriority or posteriority, and they are also semantically unmarked in that they are the forms that are chosen when the use of a marked form seems unnecessary (i.e. they represent what Dahl (1985:19) calls the 'default category').
26. The fact that conditional sentences represent the contents of the two clauses as simultaneous is also apparent from the fact that the head clause can be introduced by *then*. This use of *then* may not be purely temporal, but it is reminiscent of the temporal meaning of the adverb, which is 'at that time', i.e. 'at the time referred to in the *if*-clause'.
27. The fact that the two clauses of a conditional sentence need not be strictly simultaneous induces Huddleston (1969:804) to reject Allen's (1966) claim that *if*-clauses in the present tense are temporally subordinated to their head clauses. What I am arguing is

When-clauses constitute another type of subclause in which the expression of sloppy simultaneity is particularly frequent. It is therefore not surprising that it has sometimes been claimed (e.g. by Richards (1987:368–9)) that *when*-clauses basically define 'cases' (in the sense of Lewis (1975)) rather than expressing a temporal relation, and that they can therefore appear in the same tense irrespective of whether their situation precedes, follows or overlaps the head clause situation. This is what happens in the following examples, where it is only our pragmatic knowledge of the world that suggests an interpretation in terms of simultaneity, anteriority and posteriority, respectively:

(58) (a) When he lived at Abbotsford, Scott always dined at home.
(b) When Jocelyn received the letter, she wrote a reply.
(c) When Jocelyn went on holiday, she notified the police.

In each of these sentences *when* means something like 'when it was the case that' and refers to a situation that is represented as (sloppily) simultaneous with the main clause situation.[28]

When we look for further instances of tense forms expressing sloppy simultaneity, our attention is inevitably caught by the observation that in adverbial time clauses the present tense is used to refer to situations that actually lie in the post-present sector. However, we should not jump to the conclusion that this use of the present tense is also an instance of sloppy simultaneity. There is an important difference between this use of the present tense and the cases of sloppy simultaneity referred to above. Apart from conditional clauses, the use of a verb form expressing sloppy simultaneity never appears to be obligatory in the latter. The speaker can also use a verb form expressing anteriority or posteriority (as the case may be). This is not true as concerns the use of the present tense in adverbial time clauses referring to the future. As is clear from (59,a–b), using the present tense is not just an option here: it is the rule. The same is true of the use of the past tense in examples like (59,c–d). Here too the use of a verb form expressing posteriority renders the sentence ungrammatical:

(59) (a) John will leave before Mary *arrives*. (*will arrive)
(b) I will stay here until she *comes* back. (*will come)
(c) Bill intended to wait until the shop *opened*. (*would open)
(d) It would not be long now before the shop *opened*. (*would open)

that Allen's analysis is correct, and that Huddleston's criticism is based on too narrow a conception of simultaneity.

28. The conclusion that *when* is temporally vague is confirmed by the fact that it can collocate not only with verb forms expressing simultaneity but also with verb forms expressing anteriority (in which case it is interpreted as 'after'). The conclusion is also confirmed by the fact that *when* can replace both *while* (which requires reference to a durative situation) and *as* (which often refers to a punctual situation).

The use of present and past tense forms in time clauses like these is usually treated as a kind of anomaly. The observation that the situation referred to in the time clause is posterior to that of the head clause has induced most scholars to claim that a verb form expressing posteriority would be a more logical choice. (This claim is corroborated by the fact that cognate languages like German and Dutch actually allow the use of the future and conditional tenses in adverbial time clauses.) For that reason some transformationalists have treated the present tense forms in time clauses as derived from future tense forms by means of so-called '*will*-deletion'.[29] However, the use of simultaneity forms does not seem so illogical if one considers the fact that the conjunctions *when*, *after*, *before*, etc., are semantically equivalent to prepositional phrases of the form 'at/after/before the time when/that'.[30] In section 25 diachronic evidence will be adduced to show that conjunctions like *after* and *before* have actually developed from Old English prepositional phrases of that kind. This observation helps us to see why adverbial time clauses referring to a posterior situation use the tenses that normally express simultaneity. The reason why we use *arrives* rather than *will arrive* in (59,a) is that Mary's arrival is not represented as posterior to John's leaving but rather as simultaneous with some time later than John's leaving. In other words, English does not treat the time clause situation as posterior to the head clause situation. It treats it as simultaneous with a post-present TO that is implied by the semantic structure of the conjunction. (This simultaneity is signalled by *at* in the paraphrase 'before the time at which'.)[31]

Once it is understood that English treats *when*, *before*, *until*, etc. as equivalent to 'at/before/until the time at which', it becomes clear that it is only seemingly the case that sloppy simultaneity is expressed in (59,a–d). What is expressed is genuine simultaneity: the time clause situation is really simultaneous with the TO that is implicitly referred to by the temporal conjunction. The idea that there is something special about this expression of simultaneity (i.e. that the simultaneity is sloppy) arises only if one assumes that the verb form of the time clause relates the time clause situation directly to the head clause situation. However, this assumption is not correct. (I will go further into this question in section 25.)

29. This analysis is accepted by, amongst others, Fodor (1968:13), Kittredge (1969:15), Ross (1970), Sampson (1971:588), Vasudeva (1971:167) and Wekker (1980:99). Ejerhed Braroe (1974) points out some problems for this *will*-deletion hypothesis.
30. See Clark 1973:50: 'It can be shown ... that *John left before Mary arrived* is derived from *John left before the time at which Mary arrived*, in which *before* is explicitly used as a temporal preposition.' Ritchie (1980:209) similarly argues that the conjunction *when* is a reduction of 'at the time when' (in which *when* = 'at which').
31. In most other European languages (e.g. French, Dutch) it has become possible to use the verb forms expressing posteriority instead of the forms expressing simultaneity. This means that the loss of the noun or pronoun that originally accompanied *before*, *until*, etc. has obscured the simultaneity relation, so that it has become possible to see the time clause situation as related to the head clause situation in terms of posteriority.

A case in which we do have sloppy simultaneity is when a complement clause depending on *hope* or *bet* represents its posterior situation as if it were simultaneous. This possibility exists provided it is clear from an adverbial or from the pragmatics of the situation that the reference is really to the future: *I hope/bet I (will) get there before the others.* (However, sloppy simultaneity cannot be expressed after other, similar verbs: **I expect/am sure/suppose/am afraid I get there before the others.*)

The fact that simultaneity is the unmarked one of the three temporal relations is also clear from infinitive and gerund clauses. Because these clauses are tenseless, and hence incapable of establishing a domain, they always express a domain-internal relation. In doing so, they can make use of three forms: a 'perfect' form expressing anteriority; a 'present' form expressing simultaneity; and a form with *be going to* expressing posteriority. For example:

(60) (a) John seems/seemed/had seemed to *be* ill.
(b) John seems/seemed/had seemed to *have been* ill.
(c) John seems/seemed/had seemed to *be going to be* ill.

However, there are numerous cases in which the form used for expressing simultaneity may or must be substituted for either of the other two forms, provided anteriority or posteriority is clear from the context or from the pragmatics of the sentence:

(61) (a) He admits *being/having been* there yesterday.
(b) After *locking/having locked* the safe, he left the room.
(c) I expect to *see* him tomorrow.

All this makes clear that verb forms expressing simultaneity are often used even if there is no simultaneity in the strict sense of the word.

17 ALTERNATIVES TO TEMPORAL SUBORDINATION

When the speaker utters more than one clause, the result is what we may call a text or a stretch of discourse. When combining clauses, the speaker has to decide, for each clause that is added, whether he is going to use a relative tense form (incorporating the new situation into an existing domain) or an absolute tense form (creating a new domain). In section 15 we have explored the former possibility (which we have called temporal subordination); let us now have a look at the latter.

When a new situation is referred to, it is not always incorporated into the current domain (i.e. temporally subordinated), even if it belongs to the same absolute sector. Instead, it may be related again to t_0, i.e. it may be located again in the sector in question (which is defined relative to t_0). There appear to be three typical ways in which this may be done: by shifting the domain; by establishing a simultaneous domain; and by re-establishing the domain. As we will see, all three operations are

possible in some types of clause only. They are not normally possible in dependent clauses whose situation-TO is automatically related to the TO of the head clause (e.g. time clauses introduced by *when* or *while*, conditional clauses),[32] but they can occur in unembedded clauses and in subclauses whose situation-TO is not automatically felt to be related to the TO of the head clause (e.g. relative clauses, subclauses of reason, etc.). In such clauses the situation-TO may often be directly related to t_0.

17.1 Shifting the domain

We are not talking here about such shifts of domain as are due to a shift of absolute sector (as in *He says that he* ***was*** *ill*), but about shifting the domain within the same sector. Such a shift occurs when a domain has been established and a new situation is introduced in such a way that it is not incorporated into the existing domain but rather becomes the central TO of a new domain. The following (a) sentences are examples of this. They are contrasted with the (b) sentences, in which there is no shift of domain because the new situation (described in the second clause) is incorporated into the domain established by the first clause:

(62) (a) The man left the town and *was* never *heard* of again.
(b) The man left the town and *would* never *be heard* of again.
(63) (a) John said that Mary *witnessed* the accident.
(b) John said that Mary *had witnessed* the accident.
(64) (a) Henry knew Mary. He *met* her three years before.
(b) Henry knew Mary. He *had met* her three years before.

Sentences like these (a) examples are sometimes accounted for in terms of 'tense simplification', because they use an absolute tense instead of a more complex relative tense.[33] In fact, the phenomenon can be explained as a case of shift of temporal domain. Since the shift in question is presumably motivated by a desire to use less complex tense structures wherever possible, the label 'tense simplification' is not quite a misnomer, as long as it is understood that the phenomenon is due to a shift of temporal domain and not to some kind of idiosyncratic transformational rule.

The desire for simplicity is not the only reason why a shift of domain (within the same sector) may occur. In some cases the shift follows simply from the fact that two different situations are located relative to t_0 rather than relative to one another (see above):

32. However, we will see below that there are exceptional cases in which even *if*/*when*/*while*-clauses are not incorporated into the head clause domain.
33. According to Smith (1980:369), the possibility of using the past tense in the *that*-clause of (63,a) shows that the past tense can be 'relative in embedded contexts'. It goes without saying that I disagree with this view. I claim that the preterit in (63,a) is an absolute tense form, not a relative one.

(65) (a) The news agency has spread the news that the astronauts *have landed.*

(b) The police have confirmed that the hijackers *have released* the passengers.

(66) After the party I will tell you all about the people we *will meet* there.

In such examples the speaker just locates the two situations in the pre-present or post-present sector without stating the temporal relationship between them.[34] (Of course, we know from pragmatic considerations that one situation is anterior to the other, but this anteriority is not expressed by the tense form.)[35]

Because it follows from a shift of domain that no temporal relation is expressed between the relevant situations, a co-operative speaker will only make use of the possibility of shifting the domain if the shift does not obscure the temporal relations between the situations or if these relations are deemed irrelevant (see Grice's (1975) Maxims of Quantity and Relation (Relevance)).[36] Thus, we may say either (67,a) or (67,b) (because the temporal relation is anyhow expressed by *after*), but we cannot replace (68,a) with (68,b) without suggesting a different meaning (because *when* is temporally vague):

34. In my opinion, sentences like the following also illustrate this kind of use of an absolute tense form in a subordinate clause:
    (i) She sewed a whole dress for Alice before they *rowed* her across the river.
    (ii) Percy got there before I *did.*
    (iii) They were considered unsafe for passenger use until Elisha Graves Otis *invented* a protective ratchet device which prevented falls.
    These examples come from Allen (1966:170), who wrongly analyses the italicized verbs as referring 'to a time later than the time of the main verb', i.e. as relative tense forms. In my opinion, the verb forms in question shift the domain and are therefore absolute tense forms (which represent the situations as lying in the past time-sphere). The temporal relation between the situations of the head clause and the subclause is therefore not expressed by the tense of the subclause, but rather by the temporal conjunction.

    This does not mean, of course, that a past tense in a *before/until*-clause cannot be a relative tense. As we have seen, such clauses frequently represent their situation as simultaneous with an implicit TO (rather than as posterior to the head clause situation), and the preterit is the relative tense that is used for this in a past domain. The following is an example:
    (iv) He hoped to be able to speak to her before she *left.*
    Here the past tense does not represent the situation as lying in the past of t_0 but relates it to an implicit TO that is posterior to the matrix situation. (In languages like Dutch, where it is possible for an adverbial time clause to relate its situation directly to the head clause situation, the conditional tense can substitute for the subclause preterit in (iv), but not in (i)–(iii). For further discussion, see section 25.)
35. The temporal relation may also be expressed by a relative adverb, as in
    (i) (He arrived at two o'clock.) An hour later he *was* dead.
    In this example the tense form is an absolute deictic, whereas the adverb is a relative one.
36. The Maxim of Quantity says that a co-operative speaker will make his contribution neither more nor less informative than is required. The Maxim of Relation (Relevance) stipulates that he should only say what is relevant.

(67) (a) I always felt hungry after I *had worked* in the garden.
(b) I always felt hungry after I *worked* in the garden.
(68) (a) I always felt hungry when I *had worked* in the garden.
(b) I always felt hungry when I *worked* in the garden.

In most cases, however, our knowledge of the world will guide us to the correct interpretation, so that the shift can be applied without any problem. This is clearly the case in the (a) sentences of (62)–(64).[37] It is also the case in (69,a), but not in (69,c):

(69) (a) I will never forget that you *are treating* me like this.
(b) I will never forget that you *have treated* me like this.
(c) Next time I will tell your father that you *are* not at school.
(d) Next time I will tell your father that you *have* not *been* at school.

In (69,a) the present tense is naturally interpreted as referring to t_0. This means that the tense is interpreted as a shift of domain from the post-present to the present. This shift is not obligatory. Since what happens at t_0 is anterior to the post-present head clause TO, we can also use the present perfect to incorporate the subclause situation into the existing domain, see (69,b). There is no great semantic difference between these two ways of locating the subclause situation: in both cases it is understood as being anterior to the head clause situation. This is not the case in (69,c–d). In (69,c) the present tense can hardly be interpreted as shifting the domain (i.e. as referring to t_0). It is naturally interpreted as a relative tense expressing simultaneity. Replacing it by a present perfect (expressing anteriority) therefore entails a drastic change of meaning, see (69,d). The fact that the shift of domain interpretation is possible in (69,a), and not in (69,c), apparently follows from the fact that only the former contains such lexical items as *forget* (which inherently implies anteriority) and *like this* (with a demonstrative that is orientated towards t_0). Moreover, the use of *next time* in (69,c) makes clear that the statement concerns a time which is later than t_0.

One important conclusion to be drawn from the above observation is that a shift of domain is marked with respect to temporal subordination. There are serious restrictions on the possibility of shifting the domain

37. Richards (1982:86) remarks that a sentence like *Sam changed the tyre and Bill filled the tank* is ambiguous between the readings 'at the same time' and 'at different times'. Similarly, Ladusaw (1977) observes that the sentence *Mary saw the unicorn that walked* may be true irrespective of whether the walking took place after, before, or at the same time as the seeing. To account for this 'ambiguity' Ladusaw argues that the sentence may be derived from three different sources. My explanation is that in examples like these the temporal relation between the two situations is left vague because there is no tense form which relates them to each other: both tense forms are absolute tense forms establishing a past domain. In other words, the location of the two situations relative to each other is not expressed, because both situations are directly related to t_0.

(within the same absolute sector), whereas there are no such restrictions on temporal subordination (apart from the fact that there are clause types that do not allow it – see below).[38]

17.2 Establishing a simultaneous domain

As we have seen, the expression of domain-internal relations in the pre-present or post-present sector as a rule involves a shift of temporal perspective to another sector:

1 We use the tenses typical of the past time-sphere to relate a situation to a pre-present sector situation that does not include t_0:

(70) (a) I have never been drunk while I *was* on duty.
(b) I have never been drunk after *I had drunk* five glasses.
(c) I have never said that *I would come* back before the end of the week.

2 When a new situation is related to a post-present one, there is a shift of temporal perspective entailing that the tenses used are those typical of the past or pre-present sector (for anteriority), the present sector (for simultaneity) or the post-present sector (for posteriority):

(71) (a) He will say that he *was* ill after he *had drunk* ten glasses tonight.
(b) He will say that he *has* never *seen* me before.
(c) He will say that he *is* ill because he *has drunk* too much.
(d) He will think that he *will* soon *be promoted.*

In examples like (70,a–c) and (71,a–d) the shift of perspective to another absolute sector creates no problem of interpretation, because the subclauses are of the suitable type: the situation expressed in a temporal clause or complement clause is most naturally interpreted as temporally related to the TO of the head clause (although exceptions remain possible, see below). However, in clauses which do not evince this temporal dependency, the shift of perspective to another absolute sector can result in ambiguity:

38. The observation that a shift of domain within the same absolute sector is a marked possibility concurs with the fact that languages may differ in the extent to which they allow it. English appears to be fairly liberal in this respect. For example, in a sequence of clauses in the past perfect, English easily allows the substitution of an (absolute) past tense for the (relative) past perfect in the later clauses:
(i) His thoughts went back to the past. He had gone to school in his home village and *had gone/went* to university at the age of 19. There he (*had*) *met* a girl called Sandy . . .
As noted by Salkie (1989:25), French consistently uses the past perfect in such sequences. It does not have the 'stylistic option' of switching to the preterit.

(72) (a) Some day your father will come back, and he will be surprised by what *is going on.*
(b) The newspapers will publish everything that *is said.*
(c) You will be met by a man who *is wearing* a dark coat.
(d) I'll travel faster than you *do.*

Sentences like these are ambiguous because the situation in the subclause can be interpreted either as simultaneous with the situation of the matrix or as holding at t_0. In order to rule out the second interpretation, the speaker may prefer to use the future tense:

(73) (a) ... and he will be surprised by what *will be going on.*
(b) The newspapers will publish everything that *will be said.*
(c) You will be met by a man who *will be wearing* a dark coat.
(d) I'll travel faster than you *will.*

The use of the future tense is especially recommendable when the subclause precedes the matrix. Compare:

(74) (a) What *is going on* will surprise your father.
(b) What *will be going on* will surprise your father.

Unless (74,a) is part of a context referring to the post-present sector, its first clause will not be interpreted as referring to that sector. To ensure that interpretation we normally have to use (74,b).[39]

It appears from these examples that there are two systems to refer to a situation in a post-present domain. Let us call them the 'Present Perspective System' (PPS) and the 'Future Perspective System' (FPS), respectively. The Present Perspective System consists of relative tenses and is used to relate a situation to a post-present TO which behaves as if it were t_0. This means that the Present Perspective System involves a shift of temporal perspective and uses either the preterit or the present perfect for anteriority, the present tense for simultaneity, and the future tense (or an equivalent expression) for posteriority. The Future Perspective System, in contrast, involves tenses that establish post-present domains, viz. the future tense (as absolute tense) and the future perfect. (As we will see below, the latter combines the function of absolute tense (establishing a domain) with that of a relative tense (expressing a relation

39. As noted by Enç (1986:68), the use of *will* in a relative clause can also create ambiguity: the sentence *Mary will see the unicorn that will walk* 'can be true if the walking is after Mary's seeing the unicorn or after the moment of utterance'. In our system this ambiguity follows from the principle that the tense of a relative clause may be either a relative tense (in which case *will* expresses the domain-internal relation of posteriority) or an absolute tense (in which case *will* establishes a new domain, which, as we will see presently, may be simultaneous with the head clause domain). Sentence (74,b) in principle also yields the two interpretations, but pragmatic considerations bring out the latter of the two.

within the domain).) Since unembedded clauses referring to the post-present normally require absolute tenses, it is the Future Perspective System that is used in them:

(75) The whole family will assemble here next year. Even John and Susan *will come*. They *will have been married* then for forty years.

The use of the Future Perspective System in the second clause of (75) is an example of what we could call 'establishing a simultaneous domain'. In the first clause of (75) a situation is located in the post-present sector. In doing so it becomes the central TO of a post-present domain. The second clause introduces a new situation, but it is not incorporated into the domain. That is, it is not related (in terms of simultaneity) to the central TO of the current domain. Because no domain-internal relation is expressed, there is no shift of perspective, hence no use of the Present Perspective System. Instead, the situation of the second clause is again located in the post-present sector by the use of the future tense. It thus becomes the central TO of a new post-present domain. However, it is clear from the pragmatics of (75) that this new domain is in fact not temporally different from the old: it coincides with it. For this reason we will not speak of a shift of domain but of the establishment of a simultaneous domain.[40] We have a shift of domain when the use of an absolute tense creates a new domain (in the same sector) which precedes or follows the previously mentioned domain (e.g. *He will find the key, open the box and find the will*). There is no such shift when the two domains are simultaneous.

Shifting the domain and establishing a simultaneous domain are operations which can apply to the various absolute sectors.[41] The following examples offer some illustrations:

(76) (a) He got up from his bed and *opened* the door.
(b) He has visited Greece, and he *has been* in Italy too.

40. Allen (1966:178) observes that in
(i) He *won't speak* to her when she *comes* tomorrow although he *will look* at her from a distance.
'all three verb-clusters refer to tomorrow'. He concludes that English has two tense systems to express simultaneity with a head clause in the future tense: subclauses that are 'bound' (i.e. closely temporally connected with the matrix) use the present tense, while those that are 'free' use the future tense. My analysis differs from this in that I consider only the present tense as a relative tense expressing simultaneity in a future domain. Future tense forms like *will look* in (i) are treated as absolute tense forms that establish a domain which (for pragmatic reasons) is interpreted as simultaneous with the previously established future domain.

41. Since a situation that belongs to the present sector necessarily includes t_0, there cannot be a shift of domain in the present sector. If two present sector situations are felt to establish different present domains, these domains must be simultaneous, because both must include t_0. (As noted in section 16, for two situations or domains to be simultaneous it is not necessary that they coincide completely; it is sufficient that they overlap.)

(c) The chairman has admitted that the decision *has* already *been made.*
(d) She will be caught and she *will be brought* to trial.

(77) (a) He lay on his bed and *smoked* a cigarette.
(b) Someone has used my typewriter and *has torn* the ribbon while doing so.
(c) She will look at the photograph and (she *will*) *think* of you.

The examples in (76) illustrate domain shifting; the examples in (77) refer to simultaneous domains. As should be clear from these examples, it largely depends on pragmatic factors (the nature of the situations etc.) whether or not the two domains are interpreted as simultaneous in such sentences.

17.3 Re-establishing the domain

Let us consider example (75) (repeated here) again:

(75) The whole family will assemble here next year. Even John and Susan will come. They will have been married then for forty years.

The third clause of (75) refers to a situation which is anterior to the central TO of the current domain. However, this relation of anteriority cannot be expressed by the use of the present perfect (which expresses anteriority in the Present Perspective System) because the third clause is an unembedded clause. In unembedded clauses we cannot use the Present Perspective System to refer to the post-present. (The reason may be that, if we did, the situations referred to would be interpreted as temporally related to t_0 rather than to some post-present TO.) In an unembedded clause, locating a situation in the post-present sector requires the use of a tense form which makes clear explicitly that the domain belongs to the post-present sector. The tense forms of the Future Perspective System come up to this requirement:[42] the future tense forms in the first two clauses of (75) establish two simultaneous post-present domains; the future perfect in the third clause both re-establishes the domain (by the use of *will*) and expresses anteriority in it (by means of *have*). The future perfect thus combines the functions of absolute and relative tense; we can call it an 'absolute–relative tense'.

42. When the post-present location of the domain is clear from the (linguistic or extralinguistic) context, the present tense can also be used (as in *The train leaves at 6*). In that case the speaker establishes a post-present domain, but treats it as if it were a present one. This kind of shift of temporal perspective, which is semantically motivated, will be discussed in section 20.

Re-establishing a domain is not the same thing as establishing a simultaneous domain. A simultaneous domain is established by representing a new situation as the central TO of a new domain. The future perfect in (75) does not do this. It represents a new situation as anterior to the central TO of an already existing domain. However, in doing so it also re-establishes the domain, because it reaffirms (by the presence of *will* in the tense form itself) that the domain is located in the post-present sector. This reaffirmation is necessary because the kind of clause in which the future perfect can be used is the kind in which the domain can easily be shifted, viz. unembedded clauses and subclauses whose situation-TO need not be temporally dependent on the situation-TO of the matrix (e.g. relative clauses, subclauses of reason, concession, comparison, place, etc.).[43] The future perfect cannot be used to represent a situation as anterior to a post-present TO if the latter TO is the situation-TO of a syntactically higher clause. Thus, we can say *John will have left when you arrive* but not **You will arrive when John will have left.*[44]

Apart from the future perfect, there are a couple of phrasal forms that also establish a post-present domain and indicate a relation in it (and which therefore also belong to the Future Perspective System), viz. the phrases '*will be going to* + infinitive' and '*will be about to* + infinitive':

(78) (a) Call on me at lunchtime on Monday – *I'll be going to speak to the boss about it that afternoon.* (Leech 1971:57)
(b) Henry will be going to leave town. (Matthiesen 1983:387)
(c) She will be about to leave when we get there.

Like the future perfect, these phrases are absolute–relative forms because they both establish a post-present domain (by the use of *will*) and indicate a relation (in this case: posteriority) in it. However, they cannot always be used to express future-in-the-future, because both *be going to* and *be about to* have particular connotations and are not normally used to express unqualified posteriority.[45]

The fact that the Future Perspective System is used to establish or re-establish a domain, whereas the Present Perspective System is used to

43. As we will see below, these are the same types of clause as the ones in which the 'Sequence of Tenses' rule is often not observed. The reason is that 'what these clauses indicate can be semantically independent of the time indicated by the verb of the main clause' (Ota 1963:113), so that the tense used can easily be absolute rather than relative. (Allen (1966:177) speaks of 'free' (rather than 'bound') clauses.)
44. This does not mean that the future perfect cannot appear in a subclause. Sentences like *John will say that he will have left* are impeccable, but they form no exception to the rule, because the situation-TO of the head clause is not the TO to which the subclause situation is represented as anterior: the time before which John will have left is not the time referred to by *will say* but some other time, which is not specified in the sentence itself.
45. *Be going* to represents a future situation as having its roots in the present (see Binnick 1971, 1972; Haegeman 1983b, 1989), whereas *be about to* represents it as lying in the immediate future.

express domain-internal relations, entails that they have the following distribution:

1 In unembedded clauses the Future Perspective System is used:

(79) (a) You *will know* the answer tomorrow. (*know)
(b) You *will have heard* the answer by tomorrow. (*have heard)

(Note that the use of the present tense in sentences like *I am leaving tomorrow* or *The ship sails in an hour* does not run counter to this claim. As we will see below, one of the uses of the present tense is to represent post-present situations as if they were present ones. In this use the present tense is an absolute tense, not one of the relative tenses of the Present Perspective System. (I have already noted that the future tense is not the only means available to the speaker for establishing a post-present domain. Other means (which can only be employed under certain conditions) are the use of *be going to* or *be to*, or the present tense (which represents a shift of temporal perspective from the post-present to the present – see section 20 below).)

2 Only the Present Perspective System can be used in subclauses whose situation is represented as temporally related to that of the matrix. With few exceptions,[46] conditional clauses and adverbial time clauses are always of this type:[47]

(80) (a) If John *is* there, I will speak to him. (*will be)
(b) I will help you when you *leave*. (*will leave)
(c) I will tell you when/after they *have left*. (*will have left)
(d) I will stay here until she *comes* back. (*will come)
(e) John will arrive before Mary *has left*. (*will have left)

As noted above, adverbial time clauses use verb forms expressing simultaneity with an implicit posterior TO rather than verb forms expressing posteriority. This explains the use of the present tense in (80,d–e). (The Present Perspective System uses the present tense for simultaneity and the future tense for posteriority.)

46. There are a few exceptional types of conditional sentences in which the *if*-clause is not temporally subordinated to the head clause and therefore uses the Future Perspective System. These types will be investigated in detail in chapter 4.
47. Leech (1971:60) explains the use of the present tense after *if*/*when* as follows: 'the happening referred to is not a prediction but a fact that is taken as given. . . . [Because of this] it is appropriate to use the Present Tense, with its assumption of factual certainty.' However, this claim (also made by Rigter (1980a:436)) is refuted by the observation that one of the exceptional cases in which *will* has to be used in *if*-clauses is precisely when the future fulfilment of the condition is treated as a present fact, as in *If, as you say, John will be here tomorrow, there is no point in sending him the parcel by post* (see chapter 4). I claim that the present tense after *if*/*when* is not an absolute tense (as Leech and Rigter consider it to be) but a relative tense.

3 In subclauses whose situation is not represented as temporally dependent on that of the matrix, the Future Perspective System is used. Nonrestrictive relative clauses are always of this type:

(81) (a) I will ask John, who *will know* the answer tomorrow. (*knows)
(b) I will ask John, who *will have heard* the answer by then. (*has heard)

Note that *when*-clauses too can be used as nonrestrictive relative clauses:[48]

(82) (a) I . . . would like to look at the Preface more closely on my return, when I *will be* a little less rushed. (Hirtle 1981:221)
(b) Wait till Tuesday, when you *will see*. (ibid.)[49]

4 In some types of subclause (e.g. restrictive relative clauses) the situation may or may not be represented as temporally dependent on the situation of the matrix:

(83) (a) When she gets paid next month she will tell us that she will soon buy a new hat with the money she *has saved/will have saved* from that paycheck.
(b) You will live to see the day when there *will* not *be* an English soldier on the soil of France. (*or*: is) (Hirtle 1981:221)

Whether such a relative clause uses the Present Perspective System or the Future Perspective System depends on whether or not the speaker finds it advisable to express explicitly in the subclause that the subclause situation belongs to a post-present domain. And this in its turn may depend on how much is clear from the use of adverbial indications of time and from the pragmatics of the situations referred to. This appears from examples like the following:

48. See Ejerhed Braroe (1974:20): 'in any nonrestrictive relative clause' the tense marker 'retains the meaning it has in an unembedded S'.
49. The Future Perspective System appears to be obligatory in nonrestrictive relative time clauses. This means that the *when*-clause of *I will do it tomorrow, when I have more time* is not a relative clause but an adverbial one (which is used as a kind of apposition to *tomorrow*). (Hirtle (1981:222) makes the same claim when he compares the following two sentences:
(i) I will work in the garden till the evening, and then, when it *will be* cooler, I will walk to Blooms-End.
(ii) I will work in the garden till the evening, and then, when it *is* cooler, I will walk to Blooms-End.
According to Hirtle, the *when*-clause in (i) 'is incident not to *will walk* but to *then*: it tells of the temperature at that time of the day', but in (ii) it is 'incident to' *will walk*: 'the walking is to take place only after it cools off, whenever that may be in the evening'.)

(84) (a) In five minutes' time a bomb will explode that *is hidden* in the building.
(b) It is not impossible that some time or other a bomb will explode which *is hidden* in our building.

In both sentences the present tense form *is hidden* can be interpreted as expressing simultaneity with its head clause. (Let us disregard the possibility of interpreting *is hidden* as shifting the domain to t_0.) In (84,a), where the head clause situation is very close to t_0, this entails that the subclause situation (which is a state of indefinite length) is easily interpreted as already holding at t_0. If this is not the case (i.e. if the bomb has not been hidden yet), it is advisable for the speaker to re-establish the reference to the post-present by using the future tense (*In five minutes' time a bomb will explode that will be hidden in this building*). In (84,b), on the other hand, the head clause situation is not represented as close to t_0 and is moreover represented as uncertain. There is therefore no suggestion that the simultaneous subclause situation already holds at t_0. In consequence, there is no need to emphasize the post-present nature of the domain by using the future tense (although it remains possible to do so).

Because-clauses may also make use of either of the systems:

(85) (a) John will be caught because he *will set off* the alarm.
(b) John will be caught because he *sets off* the alarm.

According to Sampson (1971:588), these two sentences are not equivalent: in (85,a) the *because*-clause gives the reason why I state that John will be caught; in (85,b) it gives the reason why John will be caught.[50] In my opinion, this explanation is dubious. It does not seem to be applicable to examples like the following, where *will* has to be used if the reference is to the future, even if the *because*-clause is meant to express the reason for the head clause situation:

(86) (a) You will have to give John a hand tomorrow because he *will be* alone in the shop. (?*is)
(b) You will have to give John a hand tomorrow because the others *will have left*. (*have left)

Sentences like these make clear that *because*-clauses will anyhow recur to the Future Perspective System if the forms from the Present Perspective System would naturally be interpreted as relating the situation to t_0 rather than to a post-present TO.

50. Sampson, who accepts Ross' (1970) 'hyperclause hypothesis', even claims that (85,a–b) are derived from different underlying structures:
(i) I assert [John will be caught] because he will set off the alarm.
(ii) I assert [John will be caught because he will set off the alarm].
According to Sampson, (85,b) is derived from (ii) via *will*-deletion.

The statement that the future perfect is an 'absolute–relative tense' should not be misunderstood. It must be stressed that the future perfect is only an absolute tense in the sense that it re-establishes the domain (i.e. reaffirms the post-present location of the central TO of the domain) into which it introduces the new situation. It is not an absolute tense in the sense that it makes explicit the temporal relation that holds between the new situation and t_0. In fact, the future perfect does not tell us anything about this temporal relation at all, and can therefore be used irrespective of whether the new situation holds before, at or after t_0:

(87) (a) If it rains tomorrow, we'*ll have worked* in vain yesterday. (Comrie 1985:73)
(b) If it rains tomorrow, we'*ll have worked* in vain today.
(c) If it rains the day after tomorrow, we'*ll have worked* in vain tomorrow.

(88) By next week I'll be sick of exams: I'*ll have had* four exams last week and another four in the coming week. (Leech 1971:54)

(89) John *will have come* home tomorrow. (Hornstein 1975:76)[51]

(Sentences like (87,a–b) may seem a bit odd in isolation because a situation which already holds at t_0 or before t_0 is more easily related to t_0 than to some post-present TO. However, there are contexts in which they are impeccable.)[52]

As a final illustration, consider the following sentence:

(90) It will no doubt rain the day after tomorrow, and we'll have worked in vain yesterday/today/tomorrow, while the others were having a good time.

The first clause establishes a post-present domain (by using the future tense as absolute tense). The second clause, which is syntactically independent, both re-establishes the domain and represents a new situation as anterior to its central TO. In order to do this it uses the future perfect. The situation-TO indicated by the future perfect then acts as binding TO for the *while*-clause. In doing so it behaves as if it were a past sector TO (see above), so that the tense that is selected to express simultaneity in the *while*-clause is the preterit (*were having*).

51. Note that sentences like these become ungrammatical if we substitute an epistemic modal for *will*: **John must/may have come home tomorrow*. This casts doubt on the (fairly widespread) belief that there is no future tense in English, i.e. that *will* expresses epistemic modality rather than future time. (See also Rauh 1984:14–15.)
52. Comrie (1985:73) offers the following setting for (87.a): 'Suppose that a group of people worked yesterday on building a new dam. Their work is not yet completed, so it is crucial that it should not rain before they complete the work, otherwise what they have built so far may be washed away. Unfortunately, rain has been forecast for tomorrow. Under these circumstances, one of the workers might remark *If it rains tomorrow, we'll have worked in vain yesterday*'. Similar settings can easily be imagined for (87.b–c).

The dual function of the future perfect is shared by the tense which is the past time-sphere counterpart of the future perfect, viz. the conditional perfect. Consider:

(91) John said he *would have left* (before Bill arrived).

The conditional perfect here represents the situation-TO (the time of John's leaving) as anterior to a past TO (the time of Bill's arrival), which is itself posterior to the central TO of the domain (the time of saying). The conditional perfect thus has a double function: it locates the situation-TO as anterior to some other TO in the past domain, and at the same time it makes clear that the latter TO is posterior to the central TO or to some other TO in the domain. This double function is similar to what we have observed in connection with the future perfect. There is one difference, however. The future perfect locates its situation-TO as anterior to a TO that is directly related to t_0 (viz. the central TO of the post-present domain). For this reason we have called the future perfect an 'absolute–relative tense' and have said that it (re-)establishes a post-present domain. We cannot make similar remarks in connection with the conditional perfect. The binding TO to which the situation-TO is represented as anterior is here no longer a time that is directly related to t_0 (i.e. the central TO of the domain). It is a TO which is itself temporally subordinated. For this reason the conditional perfect cannot be said to re-establish the domain – though it does 'reaffirm' it in the sense that it refers to TOs that must necessarily belong to a past domain. The conditional perfect is therefore a 'doubly relative' tense rather than an 'absolute–relative' one.[53]

For the rest, however, the conditional perfect is very similar to the future perfect. This becomes clear when we consider its distribution. The conditional perfect alternates with the past perfect in exactly the same way as the future perfect does with the present perfect. This means that the conditional perfect is the appropriate tense in unembedded clauses and in subclauses whose situation is not represented as temporally dependent on that of the matrix (e.g. nonrestrictive relative clauses), whereas the past perfect is used in subclauses whose situation is

53. Comrie (1986:272) applies the label 'absolute–relative tense' to the past perfect, because the meaning of this tense 'combines absolute time reference (one time point is located prior to the here-and-now) with relative time reference (the action referred to by the verb in the pluperfect is located prior to this contextually established reference point)'. I do not think this is a felicitous use of the term 'absolute–relative tense'. If we accept this criterion, any tense that is not a (pure) absolute tense is an absolute–relative one. Since all those tenses which I treat as (pure) relative tenses relate a situation to a TO which is itself related (directly or indirectly) to t_0, all of them must be considered absolute–relative tenses if we adopt Comrie's criterion. This is unfortunate, for it blurs the difference between tenses like the future perfect which both establish a domain and indicate a relation in it and tenses like the past perfect which just express a temporal relation.

represented as temporally dependent on that of the matrix (e.g. adverbial time clauses, conditional clauses):[54]

(92) (a) He *would have heard* the answer before the end of the day. (*had heard* is not ungrammatical, but it does not express 'anteriority in the future')
(b) I said I would go to John, who *would have returned* by then. (*had returned)
(c) He said he would help me when the others *had left*. (*would have left)

A final remark to be made is that the future perfect can sometimes be used as a 'doubly relative' tense (like the conditional perfect) rather than as an 'absolute–relative' one. This is the case when the origin of the posteriority relation is not t_0 but a post-present TO, as in *He will say that he will have finished before five o'clock.* In this example, *will say* creates a post-present domain and *will have finished* expresses a double relation within this domain: it represents the finishing as anterior to a TO which is itself posterior to the time indicated by *will say*. (As noted above, the latter time behaves as if it were t_0.) In other words, in the same way as the four absolute tenses can be used as 'pseudo-absolute' tenses to express relations in a post-present domain, the future perfect can be used as a 'pseudo-absolute–relative tense'. In that case the origin of the posteriority relation is not t_0 but a post-present TO.

18 DIFFERENT WAYS OF ESTABLISHING A DOMAIN

Up to now we have pointed out two ways in which a domain can be established. The first, which is by far the commoner of the two, is the establishment of a domain by the use of an absolute tense form; the second is the re-establishment of a domain by the use of an absolute–relative tense like the future perfect. We will now consider two further possibilities. The first is illustrated by the following sentence:

(93) I hope John had left the house by five o'clock.

In this sentence the head clause establishes a present domain, which is not expanded by the subclause. The subclause shifts the domain, i.e. it establishes a new domain in the past sector. However, it does not do so in the usual way, i.e. by the use of an (absolute) past tense setting the central TO of the new domain. Rather, the tense that is used is a relative tense (*had left*), describing a situation which is represented as anterior to

54. We will see below (footnote 113 and chapter 4) that the conditional perfect can occasionally appear in conditional clauses and adverbial time clauses. In that case the use of the conditional perfect instead of the past perfect is semantically motivated. This is also true of the exceptional use of the conditional perfect in nonfactual *before*-clauses that is noted in footnote 105.

the central TO. The central TO itself is not represented as a situation-TO. That is, it is not established by a verb form (referring to a situation). Instead, it is indicated by the adverbial *by five o'clock*. However, it is clear that the adverbial does not do this on its own (since *five o'clock* could also indicate a time that lies in the future of t_0). The tense form *had left* also plays a part in the identification of the central TO: it not only expresses anteriority but also conveys the information that the binding TO (which is the central TO here) belongs to the past time-sphere. The adverbial and the tense thus co-operate to establish the central TO: the tense makes clear that it belongs to the past time-sphere, the adverbial specifies where exactly it is located in that time-sphere.

A further possibility of establishing a domain is illustrated by the following sentences:

(94) (a) Next week John and Mary will come to our party. Two days before they *will have been married* for ten years.
(b) I expect that the others *will have left* soon/before six o'clock.

These examples differ from our previous examples involving a future perfect in that the future perfect can no longer be said to 're-establish' the domain here. Clearly, the post-present domain is established for the first time. Again, the tense and the adverb co-operate to specify the central TO: the future perfect reveals that the central TO is a time-interval that lies in the post-present sector, the adverb further specifies its location in that sector.

It should be noted that it would not be correct to say that the adverbial establishes the domain on its own, while the tense form just expresses a domain-internal relation. This point is worth stressing, for analyses of this kind have actually been proposed in the linguistic literature. For example, Allen (1966:165) makes the following claim:

> the verb-cluster in the following sentence, for example, could be analyzed as expressing time-relationship – that is, as indicating 'same time as the time shown by the identifying time-expression *six months ago*':
> (2) I gave Charlie a check for those apple trees six months ago.

Partee (1984) also suggests this kind of analysis. According to her, tenses are 'anaphoric' (like pronouns), and in sentences like *John arrived at three* the 'antecedent' is the time indicated by the adverbial. This analysis again treats *arrived* as a relative tense form relating a situation to a TO established by a time adverbial. In my opinion, analyses of this kind are mistaken. I claim that in the above example *John arrived at three* it is the tense form *arrived* that establishes the domain, not the time adverbial. In other words, the central TO of the domain is the situation-TO. (In this example the time indicated by the adverbial (which, in section 4, we have called the 'Established Time') happens to coincide with the situation-TO,

but this is not necessarily the case. In *John arrived yesterday* the Established Time is a much longer interval than the situation-TO.)[55]

There are several pieces of evidence supporting this analysis:

1 The claim that, in an example like *John arrived at three*, the adverb establishes the central TO of the past domain and the tense just represents the situation as simultaneous with this leads to the undesirable conclusion that *arrived* is a relative tense form in this example, whereas it is an absolute tense form in the corresponding sentence without an adverbial (*John arrived*). (The only way of avoiding this problem would be to claim that the preterit is always a relative tense form, i.e. that in *John arrived* the preterit expresses simultaneity with a contextually identifiable TO. However, there are many instances of the use of a past tense without there being a contextually identifiable TO. Consider, for example, the use of a sentence like *John arrived by train* as the first sentence of a novel, or the use of the preterit in *John is no longer the outstanding scientist that he was.*)
2 How is it possible for *at three* to establish a particular kind of domain if the adverb does not even express whether the time referred to lies before or after t_0? (When a sentence like *John arrived at three* is the opening sentence of a novel, it is the past tense that informs the reader that *at three* must be referring to some past time.)
3 Consider:

(95) (a) Everything will be all right tomorrow.
(b) *Everything is all right tomorrow.

If *tomorrow* established the central TO of the post-present domain by itself and the tense just represented the situation as simultaneous with this, it would be (95,b) (with a present tense form expressing simultaneity) that was grammatical, not (95,a) (whose verb form does not express simultaneity). In actual fact, it is just the other way round.[56] Moreover, the future tense in (95,a) would be interpreted as

55. In chapter 6 we will see that the Established Time either includes the situation-TO or coincides with it. However, this relationship is not one of the temporal relations expressed by the tense form (except in the perfect tenses – see below).
56. As noted above, the use of the present tense in combination with an adverbial of future time is not always ungrammatical:
(i) We are leaving tomorrow.
However, the present tense is not a relative tense form in such examples. In section 20 we will analyse the use of the present tense as the result of a shift of temporal perspective from the post-present to the present sector. It is therefore similar in function to an (absolute) future tense.
Incidentally, examples like (i) form another type of sentence in which the adverbial may co-operate with the tense in establishing a domain. Unless the sentence is couched in a context referring to the post-present, the presence of the adverbial is essential (see Crystal 1966:5). Without it the present tense would naturally be understood as referring to the present.

a relative tense form relating the situation as posterior to the time established by the adverb. That is, (95,a) would be interpreted as meaning 'tomorrow it will be the case that everything will be all right at some time later than tomorrow'. Needless to say, (95,a) cannot be interpreted this way.

4 A further argument is to be derived from examples like the following:

(96) Yesterday morning John left after he had had a quarrel with Jim.

In this example, the subclause situation is temporally subordinated to the central TO of the domain established by the head clause. Clearly, this central TO is the situation-TO of the head clause, not the time indicated by the adverbial. That is, (96) represents the quarrel as anterior to John's leaving, not as anterior to yesterday morning.

It appears from these arguments that, when a domain is established by a clause containing an adverbial, it is the situation-TO of the clause and not the adverbial that establishes the central TO of the domain. The only exception concerns clauses like (93) and (94), where a relative tense form is used in a context which does not contain a situation-TO that can be interpreted as the central TO of the domain. In that case the verb form and the adverb co-operate in establishing the domain. The relative tense form makes clear what kind of domain (past, pre-present or post-present) it is, while the adverb indirectly specifies the location of the central TO (by specifying the Established Time that either includes or coincides with the central TO).

19 DIRECT AND INDIRECT BINDING

We have noticed above that, when two clauses follow each other, the second may or may not be bound by (i.e. temporally subordinated to) the first. If it is not bound by it, it is related to t_0 and hence establishes a domain (i.e. it either shifts the domain or establishes a simultaneous domain or re-establishes the domain). The two possibilities (binding or no binding) are illustrated by the following sentences:

(97) (a) Bill has often told us that he *was* unhappy.
(b) John has been suspended because he *has behaved* badly.

In a similar way, there are often two possibilities when the speaker develops a past domain. Compare:

(98) (a) Bill knew that John *had* often *said* that he *was* unhappy.
(b) Mary knew that John *had left* because he *had felt* unhappy.

Sentence (98,a) illustrates what I will call 'direct binding': each subclause is temporally subordinated to its matrix (superordinate clause). This

means that the situation-TO of the second *that*-clause is represented as simultaneous with that of the first, which is itself anterior to the situation-TO of the head clause. The fact that the situation-TO of the second *that*-clause is also anterior to that of the head clause is not expressed by its tense form. In (98,b) temporal subordination happens differently: here not only the *that*-clause but also the (more deeply embedded) *because*-clause represents its situation-TO as anterior to that of the head clause. The temporal relation between the situation-TOs of the two subclauses is therefore left unexpressed. This means that only the *that*-clause is bound directly (i.e. bound by its matrix). The *because*-clause is bound indirectly (i.e. temporally subordinated to a clause which is not its matrix).

A further illustration of the difference between direct and indirect binding is offered by the following sentences:

(99) (a) He knew that John would be absent when the parcel *arrived.* (= direct binding) (*would arrive)
(b) I said I would go to John, who *would be* back by then. (= indirect binding) (*was back)
(c) She had been to Brussels for a flying visit to a nephew who *was/had been staying* at the Hilton. (= direct or indirect binding)

(100) (a) I hoped that John would go away after he *had got* his way. (= direct binding) (*would have got)
(b) What we hoped was that we would receive aid from John, who *would have been informed* by then. (= indirect binding) (*had been informed)

Whereas the time clauses of (99,a) and (100,a) must be bound directly (i.e. temporally subordinated to the *that*-clause), the nonrestrictive relative clauses in (99,b) and (100,b) must be bound indirectly, while the restrictive relative clause in (99,c) allows the two possibilities. This means that the speaker cannot always choose freely between direct and indirect binding: the choice depends in the first place on the kind of clause that is used to describe the bound situation. As a matter of fact, only few of the types of subclause that allow direct binding also allow indirect binding.[57] Indirect binding is clearly the marked possibility.[58]

57. Temporal clauses do not normally allow indirect binding. Still, occasional examples can be found:
(i) He said that John had waited outside while Mary *had locked* up. (Huddleston 1969:792)
(ii) Chris had been reading while he *had been eating.* (Berezovsky 1978:104)
Huddleston (1969:793) remarks that this use of the past perfect with *while* is 'comparatively rare' and that the preterit is normally used instead. With other temporal conjunctions, however, indirect binding by means of a past perfect is not really exceptional:
(iii) As he *had walked* up the avenue, he had, for two reasons, dreaded his visit more than usual. (Poutsma 1926:283)

It is also worth noting that the choice between the preterit/past perfect and the conditional/conditional perfect in sentences like (99)–(100) is the past time-sphere counterpart of the choice between the Present Perspective System and the Future Perspective System in a post-present domain. Thus, (99)–(100) run completely parallel to (101)–(102):

(101) (a) I know that John will be absent when the parcel *arrives*. (PPS) (*will arrive)
(b) I think that I will go to John, who *will be* back by then. (FPS) (*is)
(c) She will go to Brussels for a flying visit to a nephew who *is staying/will be staying* at the Hilton. (PPS or FPS)

(102) (a) I hope that John will go away after he *has got* his way. (PPS) (*will have got)
(b) What we hope is that we will receive aid from John, who *will have been informed* by then. (FPS) (*has been)

As we have seen, the use of the Future Perspective System in such sentences means that the speaker establishes a simultaneous domain or re-establishes the existing post-present domain. In (99,b) and (100,b) the conditional and the conditional perfect do not (re-)establish a domain but effect what we may call 'rebinding': the relevant subclause is not bound by the TO of its matrix but by the TO that already binds that of its matrix. Rebinding is thus a special case of indirect binding.[59]

Direct and indirect binding are also to be observed in post-present domains:

Notice also that the past perfect *had thrown* in
(iv) John had left the room when Mary *had thrown* the ashtray. (McCoard 1978:201)
can be accounted for in two ways. We can say that *had thrown* represents the throwing as anterior to the head clause situation and is therefore an instance of direct binding. But we can also say that both *had left* and *had thrown* represent their situations as anterior to the same (unspecified) binding TO. In that case *had thrown* is an instance of indirect binding. (The second explanation is the only possible one if we wish to interpret the leaving and the throwing as simultaneous – an interpretation which is pragmatically less plausible but not impossible.)

58. Salkie (1989:27) notes that English speakers can choose between the preterit and the past perfect in the relative clause of (i), but that only the preterit can be used in the French equivalent of (i). In our terminology this means that a French relative clause must be bound directly. This, of course, is in keeping with our claim that indirect binding is a marked possibility.
(i) One of them had given her once a bun from the corner bakery which (*had*) *contained* a sticky sweet filling.

59. In the same way as adverbial time clauses do not normally allow the Future Perspective System, they do not normally allow rebinding by means of the conditional or conditional perfect. Still, occasional exceptions can be found:
(i) . . . so I suggested she should come down the week-end after Christmas while the family *would* still *be* there. (A. Christie, *4.50 from Paddington*, London: Fontana, p. 92)
Other exceptional uses of *would* (*have*) will be referred to in sections 24 and 25, and in chapter 4 (footnote 64).

(103) (a) After your journey you will admit that you have often felt unhappy when you *were* alone in your cabin. (PPS, direct binding)

(b) After your journey you will admit that you have often felt unhappy when you *have been* alone in your cabin. (PPS, indirect binding)

There is direct binding of the *when*-clause in (103,a) and indirect binding (more specifically, rebinding) in (103,b) (where the *when*-clause is represented as anterior to the time denoted by *you will admit* (i.e. the time which behaves as if it were t_0) rather than as simultaneous with that of *you have felt unhappy*).

In some cases the indirectly bound situation does not belong to the same temporal domain as the head clause situation:

(104) She is sad because John *had left*.

There is a shift of domain here from the present to the past. The past perfect *had left* does not establish the central TO of the past domain but refers to a situation-TO which is anterior to that (and which is therefore a bound TO). The central TO itself (which binds *had left*) is not the situation-TO of the head clause. (It must therefore be a TO which is somehow retrievable from the preceding context.) This means that there is indirect binding (but no rebinding).

In the preceding paragraphs we have defined direct and indirect binding in connection with subordinate clauses. We have said that a subclause is bound directly if its binding TO is to be found in its matrix; otherwise it is bound indirectly. A similar distinction can now be made for unembedded clauses that use a relative tense (i.e. whose situation is bound). We will say that such a clause is bound directly if its binding TO is to be found in the immediately preceding or following clause. There is indirect binding if the bound clause and the binding one are separated by one or more other clauses. For example:

(105) (a) I *had lost* my wallet and could not pay for the meal.

(b) I could not pay for the meal. I *had lost* my wallet.

(c) It was midnight when we finally reached the house. It was clear at once that John *had* already *arrived*. He *had* simply *walked* in. He *had searched* all the cupboards for whisky and *had eaten* all the bread in the house. My wife was so angry with him that she told him to leave the house and spend the night in the shed.

In both (105,a) and (105,b) *I had lost my wallet* is bound directly. In (105,c) the clause *John had already arrived* is bound directly by its matrix. The other (unembedded) past perfect clauses are also bound by the latter, so the binding is now indirect. (The sentence *My wife was so*

angry . . . finally shifts the domain.) Since the four past perfect clauses are all bound by (the TO established by) *it was clear* and not by each other, the tense forms do not express the temporal order of the situations. However, this temporal order is to a certain extent recoverable from the order in which the situations are mentioned.[60]

20 SHIFT OF TEMPORAL PERSPECTIVE

We speak of a shift of temporal perspective when the tense system that is characteristic of a particular sector is used in referring to another sector, i.e. if a TO from one sector behaves as if it belonged to a different sector. In section 15 some instances of this have been dealt with which form an integral part of the English tense system. One such instance was the use of past sector tenses to relate situations to a TO that has been established in the pre-present sector (e.g. *I have met him once or twice when I was jogging*). In this case we can speak of a shift of temporal perspective from the pre-present sector to the past sector. Another instance was the switch from the post-present sector to one of the other sectors in some types of subclause relating a situation to a post-present TO (e.g. *They will discover that you were here tonight*; *What will you say you are doing?*; *They will think you have betrayed them*). In such examples the expression of a domain-internal relation typically involves shifting the perspective to another sector.

There are other cases of shift of perspective that are inherent in the English tense system itself. One example is the use of the present tense to refer to the future, as in *The train arrives at 5.47* or *John is leaving tomorrow*. Such sentences involve a shift of temporal perspective because they represent a post-present situation as if it were holding at t_0.[61] The same kind of shift can also occur in subclauses. Compare:

(106) (a) John says that the train *will arrive* at 5.47.
(b) John says that the train *arrives* at 5.47.

(107) (a) John says that he *will leave* tomorrow.
(b) John says that he *is leaving* tomorrow.

60. In chapter 3 we will see that the order in which (bounded) situations are mentioned usually reflects the order in which they have taken place. However, there are exceptions to this rule.

61. The use of the present tense in such clauses referring to the future has often been treated as resulting from '*will*-deletion' (see e.g. G. Lakoff 1971:339). (It has already been noted that the same hypothesis has been advanced in connection with the use of the present tense in temporal and conditional clauses referring to the future – see footnote 29.) Arguments against such a rule have been advanced by Jenkins (1972), Vetter (1973), Ejerhed Braroe (1974). (The most telling of these arguments is that the *will*-deletion hypothesis wrongly predicts that *John will say that he loves you* is synonymous with (because derived from) *John will say that he will love you.*)

In (106,a) and (107,a) the head clause establishes a present domain and the *that*-clause establishes a post-present one. (As noted above, a finite verb form cannot express posteriority within a present domain. What is posterior to t_0 must necessarily be located in the post-present sector. This automatically produces a shift of domain from the present to the post-present.) In (106,b) and (107,b), on the other hand, the *that*-clause represents its situation as if holding at t_0 instead of as posterior to it. This is a shift of perspective: a situation that is actually posterior to t_0 is represented as if it were simultaneous with it.

In examples like the above ones, the shift of temporal perspective from the post-present to the present is effected in order to achieve a particular semantic effect: semantically too, the situation is dragged into the present. That is, whereas (106,a) and (107,a) express a prediction or expectation (i.e. represent their situations as hypothetical), (106,b) and (107,b) represent their situations as if they had already become a fact at t_0.[62] This is possible because in such examples the future situation is logically entailed by the structure of the world that holds at t_0: the speaker can represent the future arrival of the train at 5.47 as a fact because this time schedule is part of the structure of the world that holds at t_0; in (107,b) the future situation is part of the world structure that holds at t_0 because the situation has already been decided on (planned, arranged) at t_0. (See also Rigter 1983:175.)[63]

The shift of perspective which takes place in (106,b) and (107,b) has a counterpart in the past time-sphere:

(108) (a) John said he *would be leaving* for France the next day.
(b) John said he *was leaving* for France the next day.

(109) (a) John said that the train *would leave* at 4.50.
(b) John said that the train *left* at 4.50.

In each of these sentences the subclause refers to a situation which is posterior to the head clause situation. In the (a) sentences this posteriority is expressed by the conditional tense, which is the normal tense to express posteriority in a past domain. In the (b) sentences this

62. In some languages the shift may even be from the future to the pre-present. In Greek, a sentence like *I am coming!* may be in the aorist (perfective preterit). In using this tense the speaker stresses the fact that he has already made the decision to come soon (see Dorfmüller-Karpusa 1988:143).

63. As we will see below, the basic function of the progressive form of the present tense is to represent a situation as in progress at t_0 or at a time including t_0. The use of the progressive present to refer to the future is only a metaphorical extension of this: since the future situation depends on a present arrangement (intention, decision), the situation is felt to have a present initiation. In other words, t_0 is felt to be included in the reference. This is the reason why the temporal perspective is shifted, i.e. why the situation is represented as if it were lying in the present rather than in the post-present sector. (Essentially the same analysis is offered by Allen (1966:215), Zydatiss (1976:148), Botne (1981:116), Johnson (1981:156), Lewis (1986:95).)

posteriority is not expressed by the verb form. In fact, the subclause situation is represented as if it were simultaneous with the head clause situation rather than posterior to it. However, in this case we cannot speak of a shift of perspective: a shift of perspective always means that a situation which belongs to a particular absolute sector is represented as if it belonged to another. This is not the case in (108,b) and (109,b), since all the clauses of (108)–(109) locate their situations in the past time-sphere. Still, (108,b) and (109,b) resemble (106,b) and (107,b) in that they represent the *that*-clause situation, which is actually posterior to the head clause situation, as simultaneous with the latter.

Apart from sentences like *The train leaves at 5*, a shift of perspective from the post-present to the present can also occur in the head clause of a conditional sentence, as in *If you behave like that tomorrow, you* ***lose*** *your pocket money.* Huddleston (1979:337) comments on this: 'The absence of predictive modality ... makes the threat that it would typically be used to make somewhat more forceful ...: the occurrence of the event under the stated condition is guaranteed in the here-and-now.'[64] Another instance of the shift of perspective is the type *Here we are and here we* ***stay*** *till they throw us out* (see also Nehls 1988:300).

There is also a shift of temporal perspective when the present tense is used to report a situation which is in fact anterior to t_0:

(110) (a) I *hear* you have been promoted.
(b) They *tell* me she is pregnant again.
(c) I *gather/understand* there's been trouble again.
(d) The ten o'clock news *says* that there's going to be a bad storm. (Quirk *et al.* 1985:181)
(e) As Aristotle *tells* us, ... (Saurer 1981:65)

64. The following are some further examples:
(i) (a) One more step and you *are* a dead man.
(b) But you know if your son, when of age, refuses to marry his cousin, her whole fortune *is* then at her own disposal. (Poutsma 1926:250)
(c) Laugh and the world *laughs* with you; frown and you'll wrinkle your face. (ibid.) (There is a shift of perspective in the first sentence, not in the second.)
This kind of shift of perspective can entail the use of the present perfect instead of the future perfect:
(ii) (a) One more word and you *have sealed* your fate.
(b) If they make another goal we *have lost* the game.
In English this use of the present perfect is rare and typically dependent on the expression of a condition. It is not so heavily restricted, however, in cognate languages like Dutch and German, where it need not occur in a conditional context:
(iii) (a) Morgen *ben* ik *vertrokken.* ('I have left tomorrow')
(b) Falls jemand nach mir fragt: Ich *bin* morgens *verreist.* (Latzel 1974:312) ('In case someone asks for me: I have left tomorrow')
(iv) In einer halbe Stunde *habe* ich meinen Vortrag *beendet.* (Grewendorf 1982:77) ('In half an hour I have finished my speech')

This kind of shift of perspective from the pre-present to the present is possible only with verbs referring to the productive or receptive end of the process of communication, i.e. with verbs of saying, hearing and understanding. According to Crystal (1966:271), its function is to dramatize the communication act. In my opinion, however, its main function is to heighten the relevance of the past situation. As noted by Comrie (1985:41), 'statements about the present moment are more relevant than those about other times', so that representing past situations as if they were present ones is a means of highlighting their relevance. A somewhat similar kind of shift of perspective to the present is to be observed in examples like *Someone is asking for you* or *What is she saying?*. The situations referred to in such sentences may have taken place just before t_0 but they are represented as if they were going on at t_0. Here again the motivation for the shift appears to be present relevance: although the speech act itself is over, it is still a matter of actual concern.

Quirk *et al.* (1985:181–2) point out several other cases in which the past is treated as if it were the present. In their opinion, this possibility follows from the fact that 'the past can remain alive in the present':

(111) (a) The Book of Genesis *speaks* of the terrible fate of Sodom and Gomorrah.
(b) In *The Brothers Karamazov*, Dostoevsky *draws* his characters from sources deep in the Russian soil.
(c) Brahms *is* the last great representative of German classicism.

In sentences like these the speaker is not thinking of the time interval in history when something was created or when the creator lived, but rather of the work that was created, which continues to exist (and hence to be relevant) in the present. The shift of perspective is thus again based on semantic considerations.

In some cases the shift of temporal perspective to the present sector is motivated by the fact that the present tense is the tense that can function as 'default tense' in the English tense system (see Papp 1985:98; Dahl 1985, 1987:493). By 'default tense' I mean the tense that is used when the speaker wishes to describe a situation without bothering to locate it in time.[65] That it is the present tense that is used in this way follows from the fact that the present tense is the tense with the simplest temporal

65. The default tense is also the only tense that is used by non-native speakers who have only a very elementary knowledge of the language and have not begun to master the tense system yet. Experiments conducted by Godfrey (1986:204–5) have revealed that the English present tense is, indeed, used in this way. Moreover, Godfrey's experiments also revealed that native speakers switch from the past tense to the present 'when overloads in production processing or other sources distract the speaker from consistent use of an ongoing past tense' (p. 136). This too is consistent with the hypothesis that the present tense represents the default category.

structure: it just represents the situation as holding at t_0 and expresses no other temporal relation besides this. (As we have seen, simultaneity (i.e. the relation 'holding at') is the unmarked of the temporal relations. This explains why nonsimultaneous situations are often represented by verb forms expressing simultaneity – see section 16.)[66] A typical example of a shift of perspective that is motivated by the desire to use the default tense is the substitution of the present tense for any other tense in newspaper headlines:[67]

(112) (a) Trade unions *seek* assurances.
(b) US *send* warships to Gulf of Oman.

Other examples where the default tense is used because time reference seems irrelevant are sentences that figure in photographic captions, descriptions of drawings, etc.:[68]

(113) (a) (*photographic caption*) The Queen *arrives* for the State Opening of Parliament. (Quirk *et al.* 1985:182)
(b) (*description of a painting*) Roman soldiers *nail* Jesus onto the cross.

A shift of temporal perspective to the present is also very frequent in historical summaries, tables of date, etc. In these cases the shift is probably motivated by the fact that, like a fictional work, a historical text creates its own universe of discourse, which is represented as having its own temporal zero-point (see below):

(114) (a) 1876 – Brahms *completes* his first symphony. (Quirk *et al.* 1985:182)
(b) Napoleon *is exiled* to St Helena, where he *will die* a couple of years later.

66. It should be clear from this definition of 'default tense' that I do not consider this use of the present tense to be its basic use. It follows that I cannot endorse the view (held by e.g. Joos (1964), Weinrich (1964), Lyons (1977), Hirtle (1975)) that the present tense is basically an 'unmarked' or 'neutral' tense, i.e. a tense with no definite time reference, which can therefore be used to refer to present, past and future. At the end of this section I will adduce several pieces of evidence against such a view.
67. In newspaper headlines grammatical markers that are not essential for the interpretation (e.g. articles, titles, the *-ing* form as marker of progressive aspect) are omitted (*as in Britain seeks peace with Soviets*). I suggest that tenses (grammatical markers of temporal relations) are also subject to this, i.e. that the tense marking on the verb is normally effected by the default tense, viz. the present tense. This explanation would seem to be in keeping with the fact that sometimes there is no tense marking at all (e.g. *British economy looking up*), and with the fact that 'the present tense forms are not used when it is essential that an unambiguous past time reference is given' (Mårdhi (1980:173). The reason why unneccessary words and morphemes are dropped is presumably the aim for brevity (see Rot 1988:35–6).
68. An alternative explanation for the use of the present tense in such sentences is that t_0 is the time of decodification. That is, the present tense is used to describe what can be seen in the photograph or painting at the time of reading the caption.

In the above paragraphs I have pointed out several types of sentences in which a past situation is treated as if it were present. A shift of temporal perspective in the opposite direction appears to be possible too. The conventional use of the past tense in narrative appears to be an instance of this. As has often been pointed out, a fictional work creates its own time. The 'present' of a novel has nothing to do with the present of the author (the time of writing). It is a present which comes alive whenever the work is read. That is, it is a present that coincides with the time of reading (t_0). And since a literary work can be read an indefinite number of times, the present of the work continues to coincide with t_0. In this way it is a timeless present, i.e. a present which remains present, unaffected by the flow of time which, in the real world, constantly turns the present into the past. Because of this, the most suitable tense to represent the present of the novel would seem to be the present tense. And indeed, the present of the novel can be represented as lying in the present sector (i.e. as simultaneous with t_0), while what is past or future in the novel is represented as lying in the pre-present and post-present sectors, respectively. This system of tenses, which is in principle the most plausible, is that which we actually use when summarizing a novel, a story, the previous episode of a serial, etc.:

(115) (a) That night Jenny *meets* Lester in the woods and *tells* him about her brother's plan.
(b) John and Joy Jennings, who *have been fighting* a gang led by Red Regan, *have followed* the sinister goatherd Khari to a mountain hide-out, where they *stumble* upon a coded message from Red's lieutenant, Hercule Judd. (Leech 1971:38)

However, this unmarked system of tenses is not the one normally used in the fictional work itself. Since writers aim to create the illusion that the story is authentic, the convention has grown to represent the fictional situations as if they had really held in the past. This entails a shift of temporal perspective: the present of the novel is situated in the past sector. The ruling tense is now the preterit; the past perfect is used for anteriority, the conditional tense for posteriority. Even in science fiction, where the happenings are claimed to take place in the post-present sector (e.g. the twenty-second century), the temporal perspective usually remains that of the past sector:

(116) In the year 2123 man finally *reached* Mars. The commander of spaceship N-145 *set* foot on the planet on 4 July. He *stayed* there for three weeks, and then *continued* his voyage to Jupiter.

However, a writer may choose to deviate from the 'narrative past' convention in order to achieve a particular effect. There exist a few novels that are written in the present tense, e.g. Joyce Cary's *Mister*

Johnson, Camus' *L'Étranger*, Werfel's *Das Lied von Bernadette*. Dieling and Kempter (1983:63) note that in Thomas Mann's novel *Doktor Faustus* part of the fictional situations are described in the present tense. This is because Mann uses two different tense systems, in order to establish two different time levels. The present tense system is used to describe the hero's life, the past tense system is used to refer to the time when the fictitious narrator records this life story. The use of the present tense here means that the author wants a particular fictitious timespan to be interpreted as 'the present'. (Because this is an unusual thing to do, Dieling and Kempter call this use of the tense system 'metaphorical'.)[69] Similarly, the author of a work of science fiction can deviate from the tradition and represent the events (which are claimed to lie in the future of t_0) as belonging to the post-present sector. An example of this (noted in Bronzwaer (1970:70)) is Michael Frayn's novel *A very private life*, which opens with the sentence 'Once upon a time there will be a little girl called Uncumber.' In the opening pages of this novel the author uses the future tense to describe the situations that push forward the action, the past tense for situations that are anterior to these (and which may be anterior or posterior to t_0), and the present tense to give background information which does not belong to the story itself.

Another clear example of a shift of temporal perspective is the use of the historical present in oral narrative: situations that belong to the past time-sphere (or form part of a narrative that is situated in the past) are represented as if they belonged to the present sector.[70] For example:

(117) Do you know what John did to me the other day? Well, just as I was leaving for the office, he *comes in* and *tells* me I'*ve won* on the pools and some big chap *will come* and pay me over £1,000. I nearly fainted on the spot. And two minutes later he *tells* me it's all a hoax and I *haven't won* anything at all.

Although linguists usually speak only of the 'historical present', examples like this make clear that the various present time-sphere tenses are involved in this shift of temporal perspective.

The reason why the shift is applied is often claimed to be that 'the historic present describes the past as if it is happening now: it conveys something of the dramatic immediacy of an eye-witness account' (Quirk *et al.* 1985:181). Kiparsky (1968) has argued that, although this explanation may hold for modern English, it does not apply to older Indo-European languages like Classical Greek, Old Irish, Old Norse, etc.,

69. Dieling and Kempter rightly point out that this use of the present tense is different from that of the 'historical present': 'Das Präsens soll hier nicht dramatisieren, sondern "Gegenwart" suggerieren.' ('The present tense is not used here to dramatize but to suggest "present time".')
70. According to Matthiesen (1983:401), the historical present is 'best interpreted as a *tense metaphor*: we choose to speak of the past *as if* it were the present'.

where the historical present is 'a replacement of the moribund injunctive' (p. 47).[71] According to Spielman and Gardner (1979:316ff.) and Wolfson (1982:13), the explanation does not satisfactorily account for the use of the historical present in conversational narrative either. (More particularly, it does not account for the fact that there is always a switching back and forth between the historical present and the preterit.) Since for our present purpose the question of what effect the shift of perspective has (and hence why it is applied) is less important than the mere ascertainment that this type of shift of temporal perspective is possible, I will not pursue this question here. For further discussion the reader is referred to Spielman and Gardner (1979:315–27), Schiffrin (1981), Wolfson (1982), Bache (1985:107), Fleischman (1985), Godfrey (1986), Koops (1986), Wright (1987) and Johnstone (1987).

It should be noted that, in narrative, the shift of temporal perspective may be not only from the past to the present but also from the post-present to the present:[72]

(118) (a) I can well imagine what will happen. I can see it happen before me: John *sets out* his plans, Mary *disagrees*, they *start* shouting at each other and in no time there is a terrible row. I've seen it happen often enough to know that it is going to end like this.

(b) Here's what you have to do tomorrow morning. You *walk* to the station, and *get* on the first train. (Crystal 1966:5)

The conclusion from this section can be that there are quite a few possibilities of shifting the temporal perspective,[73] and that in most cases

71. If this is correct, it suggests that the historical present in those languages is an instance of the use of the present as default tense, for it was typical of the injunctive that it was a verb form that was indifferent to time (and modality). However, it should be pointed out that Kiparsky's (1968) theory has been severely criticized by McKay (1974) and Thomas (1974).

72. I speak of a 'shift from the past to the present' because the speaker using the historical present as it were steps out of the past time-sphere to select a tense from the present time-sphere. Other people (e.g. Cooper 1986:31–2) speak of a 'shift from the present to the past' because the tense that is normally used to refer to the present is now used to describe the past. This difference in nomenclature need not reflect a different interpretation of the phenomenon. It is a purely terminological difference, which is quite immaterial to the analysis that is suggested here.

73. Some of these possibilities (viz. the historical present, the present tense referring to the future) are also recognized as special by Randriamasimanana (1987), who treats them as instances of 'displacement'. This concept is defined as follows:

> Displacement describes two types of situation: The first is one in which Speaker S refers to an Entity/Event present within his/her own Here and Now, and does so as if it was not there; the second is one where Speaker S refers to an Entity/event absent from his/her own Here and Now, and yet does so as though it was present there. (p. 193)

Although this definition clearly resembles the way we have defined our term 'shift of temporal perspective', only some of Randriamasimanana's examples of 'displacement' actually qualify as cases of a shift of perspective. Some of his examples are illustrations of what we call a shift of domain (e.g. the use of the present tense in *John said that the*

the shift is to the present sector. It has become clear that the present tense need not always locate a situation in the present sector or express simultaneity with a present TO. In many cases it is used to speak of a past or future situation *as if* it belonged to the present sector or *as if* it were simultaneous with a present TO.[74]

This double use of the present tense has often been misinterpreted as an argument for claiming that the present tense does not basically refer to present time. Thus, Lewis (1986:64) claims that 'it is *not* characteristic of the present simple that it is associated with Present Time; the feature which all uses of the present simple share is that temporal reference is unnecessary'. Langacker (1978:867) similarly argues that 'the present-tense morpheme does not really mark present tense. Its use with future sense, the historical present, and its habitual interpretation with perfective verbs have made this clear to virtually all investigators.'[75] And analogous claims are made by Bolinger (1947:436), Twaddell (1960:5), Joos (1964), Weinrich (1964), Hirtle (1975), Waugh (1975), Lyons (1977), Smith (1981:214), Harris (1982:273), Wolfson (1982:33–4), King (1983:106), Koops (1986:125–6), Janssen (1989:71, 1990) and many others.

The analysis I have been arguing does not accord with this. I hold that the present tense does basically refer to t_0, and that the other uses are 'metaphorical', i.e. result from a shift of temporal perspective.[76] In my opinion, there are several pieces of evidence against the theory that the present tense does not refer to present time but is a neutral tense which is compatible with present, past and future time reference. The following are the most important:

1 One unfortunate consequence of the 'neutral tense' theory is that it does not allow a unitary treatment of the simple present and the progressive present. It is, indeed, a fact (admitted by some of the above linguists), that 'present continuous forms do contain reference to time or temporal interpretation' (Lewis 1986:64).

earth is flat), others exemplify our notion of shift of temporal focus (which will be explained in section 22 below).

74. Some linguists (e.g. Rauh 1984:21–3) interpret these observations differently. They claim that the present tense in these uses continues to represent a situation as simultaneous with t_0, but that t_0 is no longer the time of utterance or the time of reception. I do not find this view attractive, because the notion of t_0 as speech/reception time continues to be relevant in such cases of 'displaced t_0'. For example, the historical present is clearly interpreted as referring to a situation that lies before t_0 (speech/reception time). The claim that the situation in question is represented as simultaneous with (a displaced) t_0 therefore obliges us to recognize two t_0s of a different kind. This is not as neat a solution as the one I have argued.
75. However, in a later study, Langacker (1982:291) gives up this view and also treats the historical present and future reference uses of the simple present as metaphors.
76. This kind of analysis is also argued by e.g. Matthiesen (1983:401ff.), Windfuhr (1985:451).

2 Another problem for the theory is that it is hard to see how a sentence like *John is in the kitchen* should not involve temporal reference. Lewis (1986:65) claims that all the uses of the simple present have in common that 'questions of restriction' like *When?* or *On what occasion?* are not appropriate. It may be true that a sentence like *When is John in the kitchen?* is not very felicitous (on a single-event reading and with reference to present time), but this is for pragmatic reasons which have nothing to do with the claim Lewis is making: *When is John in the kitchen?* is infelicitous for the same reason as *When did John leave two minutes ago?* is, viz. because the answer to the question is given in the question itself (which runs counter to Grice's Maxims of Quantity and Relation (Relevance)). The unacceptability of the question is thus evidence for, not against, the claim that the present tense intrinsically refers to present time. That sentences like *John is in the kitchen* do involve temporal reference is furthermore clear from the fact that a sequence like the following is impeccable:

(119) John is in the kitchen. – What! Right now? He was not to arrive before teatime.

Obviously, the second speaker is reacting here to the temporal reference that is implicit in the use of the present tense in the first sentence.

3 The observation that the present tense often turns out to be used with past or future time reference does not warrant the conclusion that it is a neutral tense which is compatible with any time reference. If this conclusion were correct, the present tense could always be substituted for the preterit and for the future tense. However, this is not the case. In a complex sentence like *John left when we arrived* we cannot substitute the present tense for the preterit in the head clause (**John leaves when we arrived*), nor in the time clause (**John left when we arrive*). Moreover, such uses as are grammatical (e.g. the historical present) are highly restricted and felt to be 'special'.[77] Obviously, the present tense is not the normal tense to refer to the past. A use like that of the historical present is clearly a marked use, which is therefore better analysed as metaphorical (i.e. as a shift of temporal perspective) (see also Nehls 1975:276). Similarly, the use of the present tense to refer to the future is subject to restrictions. Apart from the fact that it is not quite semantically equivalent to the future tense, it is possible with some kinds of verbs only. (For example, it is not normally possible with stative verbs.)

77. See Chung and Timberlake (1985:213): 'The historical present is commonly explained in terms of markedness (as the unmarked tense, the present can supposedly substitute for other, more marked tenses), but this interpretation does not account for the specialized narrative value of the historical present.'

4 As noted by Nehls (1975:276) and Wolfson (1982), the widespread view that the historical present renders the past more vivid by representing past situations as if they were happening in the present is incompatible with the view that the present tense is timeless or unmarked for temporal reference.

5 If one interprets the fact that the present tense can be used with future time reference (e.g. *He is coming tomorrow*) as evidence that there is no basic relation between the present tense and reference to present time, then one must draw an analogous conclusion in connection with the past tense, since this can also occur with present or future time reference (e.g. *John* ***was coming*** *here today/tomorrow, but the strike makes it impossible*). However, it seems difficult to see how a language can have two different neutral tenses which can refer to either present, past or future.

6 If a present tense form is interpreted as referring to the past or future, this time reference actually follows from the presence of a suitable time adverbial, from the context or from the pragmatics of the sentence. Thus, a sentence such as *They are having a party* is not interpreted as having future time reference if there is no adverbial or contextual indication of future time. Similarly, a present tense is not interpreted as a historical present, unless it is couched in a past context.[78] On the other hand, we do interpret a present tense as referring to the present if it is used in isolation.[79] All this substantiates the claim that the present tense does basically refer to t_0 and that the other uses are metaphorical, i.e. instances of a shift of temporal perspective from one of the non-present sectors to the present one.

21 TEMPORAL PERSPECTIVE AND MODALITY

It is not impossible that the so-called 'modal' uses of the past time-sphere tenses can also be traced back to a shift of temporal perspective. These modal uses have in common that they express some kind of 'remoteness'. The following subtypes can be distinguished:

78. It is a general feature of sentences in the historical present occurring in conversational narrative that they must be intermixed with sentences in the past tense (see Spielman and Gardner 1979:315; Wolfson 1982:3). Another indication that the historical present is a marked choice is that it cannot occur in any context where a past tense can occur. Wolfson (1982) points out several restrictions on the use of the conversational historical present. For example, in her corpus it never occurs in time clauses with *when* (p. 16); it is 'limited to performed stories as opposed to reports' (p. 29); and even there it can be used 'only when the time orientation is related to the time of the story' and not to give background information (p. 30).
79. I will argue in chapter 6 that even generic sentences in the present tense should be analysed as referring to the present: such sentences describe a world structure state and represent it as holding at t_0.

21.1 Remoteness from reality, i.e. nonfactuality

21.1.1 In some types of subclause the preterit and the past perfect are used to represent a situation as contrary to fact. In doing so they behave as they always do when used as relative tenses, i.e. the preterit expresses simultaneity, the past perfect anteriority. This is the case irrespective of the tense of the head clause. The following exemplify the typical cases:

(120) (a) I wish you *had been* here yesterday.
(b) I wished John *was* there too.
(c) It's high time we *left*.[80]
(d) Your hair looks as if you *had* never *combed* it in your life.
(e) I'd rather you *told* me the truth.

21.1.2 Conditional sentences use past time-sphere tenses (in both clauses) to represent the fulfilment of the condition and the actualization of the consequent as hypothetical (i.e. not impossible, but rather unlikely) or counterfactual:

(121) (a) If I *told* you the truth, you *wouldn't believe* me.
(b) If we *hadn't missed* the train, we *wouldn't be waiting/ wouldn't have been waiting* on this cold platform now.

Relative clauses with a conditional connotation use the same verb forms:

(122) (a) He *would* never vote for a candidate who *did* not *condemn* abortion.
(b) Any person who *had refused* to obey these orders *would have been thrown* in prison.

21.1.3 In unembedded clauses, the use of a past tense to represent a present situation as imaginary (nonfactual) is exceptional. Still, occasional examples can be found:

(123) What a warm evening! We *could/might be* in Spain.

In this example the past tense is selected to represent a present possibility as nonfactual.

A quite different use of past tense forms in head clauses that can probably also be explained in terms of nonfactuality is the conventional use of preterits in children's games in some European languages. When casting the parts to be played in the game, French, Spanish, Italian, etc., children will typically use past tenses: *I was a rich king and you were a*

80. In this example too the past tense has counterfactual meaning: it suggests that we should already have left (see King 1983:111). This implication is absent in *It's time for us to leave.*

poor boy, and you came to me to ask me for money. In this way the children make clear that what they are doing is just a game, i.e. something imaginary rather than factual. (For further details see Warnant (1966), Rodari (1973).)

21.2 Tentativeness

With some verbs (*wonder, expect, hope, want, think*, etc.) the past tense is used to render a statement less direct, more tentative, hence more tactful, polite:

(124) (a) I *wanted* to ask you a question.
(b) I *wondered* if you heard anything on the grapevine about this new building Harrison's having put up. (Fenn 1987:187)
(c) I *was hoping* that you would be able to help me tomorrow.
(d) I *thought* you might be mistaken.[81]

The use of past time-sphere tenses to express nonfactuality or tentativeness constitutes a shift of temporal perspective from the present to the past:

1 What is represented as holding at t_0 is necessarily represented as factual. To express nonfactuality, many languages therefore recur to the tenses of the past time-sphere.[82] In this way the same tenses are used for what is remote from present time and what is remote from present reality.[83]
2 The use of the past tense for tentativeness is directly related to its temporal use. The mechanism is simply that the speaker locates his hope, wonder, thinking, etc., in the past in order to suggest that he is not necessarily committed to them at the time of speech. This is a means of making a request less pressing (as in (124,a–c)) or of forestalling severe criticism in case the statement turns out to be incorrect (as in 124,d)).

This explanation means that we treat these modal uses of past time-sphere tenses as metaphorical: the fact that the past time-sphere expresses remoteness from t_0 entails that situations which belong to the

81. *I thought* can also be used when the object clause is counterfactual:
(i) You don't know that man? How odd! I thought he was a good friend of yours.
In that case *thought* can be argued to have its normal temporal meaning: 'I thought so until now, when you disabused me of the notion' (see Erades 1975:2).
82. James (1982:376) comes to the conclusion that the use of the past tense for 'irrealis' 'occurs in a number of different language families'.
83. In the case of counterfactual conditionals we use a tense which normally expresses posteriority (viz. the conditional tense or the conditional perfect). According to Tedeschi (1981:256) this is because 'we evaluate counterfactual conditional sentences as if we returned to the past and looked at possible futures with respect to that past'.

present time-sphere can be represented as if they belonged to the past time-sphere if they are to be interpreted as 'remote' in some way or other.[84]

The observation that the past tenses are used to express different kinds of remoteness has led many linguists to hypothesize that the basic distinction between past tenses and present tenses is the distinction between 'remote' and 'non-remote' (or 'non-actual' versus 'actual') rather than the distinction between past and present time. Past time, i.e. temporal remoteness, is then just seen as a special case of remoteness. This hypothesis is suggested by Lyons (1977:819–20) and unconditionally accepted by Twaddell (1963:8), Joos (1964:121), Steele (1975), Nehls (1975:289), Leuschner (1977:99), Kress (1977), Langacker (1978), Hutchinson (1985), Lewis (1986), Janssen (1989:71–2, 1990) and many others, who consequently claim that 'tense and time have in principle nothing to do with each other' (Leuschner 1977:99). On this view, tense is a kind of modality, and it simply depends on the context whether a past tense form is interpreted as indicating past time or as indicating nonfactuality or tentativeness. It will be clear from what I have argued that I do not subscribe to this view. I hold that the basic distinction is that between the two time-spheres, i.e. the past tenses primarily express past time, and their modal uses are metaphorical extensions of this. Everything that I have argued up to now supports the view that the use of tenses is basically a question of time-spheres, TOs and temporal relations, and not a question of modality. Many observations that have been made would not fall into place if we accepted the alternative hypothesis. For this reason I do not think it is necessary to devote a lot of space to refuting it, but perhaps it is worth making the following remarks. First, if the past tense had as its basic meaning the feature 'distant from present reality and time', we would expect it to be used not only to indicate the distant past but also to refer to the distant future.[85]

However, this expectation is not borne out. (As a matter of fact, there appear to be no languages known that grammaticalize not-now as a single

84. Fleischman (1989) offers a more detailed study of the use of past tenses as 'the extension of temporal distance to modal distance' (p. 5) in various languages. Her interpretation of the data is exactly the same as ours: 'distance from the speaker's "now" along a temporal axis is metaphorically converted into distance conceptualized in terms of other grammatical and pragmatic notions that languages find it appropriate to mark.' (p. 3).

85. A comparison is in order here with the adverbs *now* and *then* and with the temporal use of the demonstratives *this/these* and *that/those*. As is well known, *then* is used in the sense of 'not now' and can refer both to the past and to the future. The demonstratives divide time in a similar way: *this* and *these*, which imply absence of distance, refer to the present (*this week, these days*), whereas *that* and *those*, which imply distance, refer either to the past or to the future (*at that time, during those days, that afternoon*).

tense – see Comrie (1985:15).) Secondly, the claim that there is no fundamental relation between the past tense and the expression of past time naturally leads the proponents of this theory to the more general conclusion that there is no relation between tense and time (see above). This means that these linguists also assume that there is no relation between present tense and present time – an assumption that was rejected in the previous section. Thirdly, as will be stressed in chapters 5 and 6, the use of the preterit does not deny that the situation referred to includes the present moment. All it does is express that the situation held at some past time. Thus, *Five minutes ago John was in the garden* does not imply that John is not in the garden at t_0. It just states a simultaneity relation between the past time indicated by *five minutes ago* and the situation of John being in the garden. This observation proves that the (nonmodal) use of the preterit is not a question of remoteness from the present, but a question of how a situation is located in time. (For further arguments against the 'past tense = remoteness' hypothesis, see James (1982:398), Matthiesen (1983:405–6), Huddleston (1984: 148–50), Comrie (1985:20).)

22 TEMPORAL FOCUS

As we have seen, a head clause referring to a situation that includes t_0 will normally use a tense that represents the situation as including t_0 (usually the present tense, sometimes the present perfect). Thus, since it is presently the case that the Eiffel Tower stands in Paris, this fact will normally be reported as (125,a) rather than as (125,b):

(125) (a) The Eiffel Tower stands in Paris.
(b) The Eiffel Tower stood/had stood/will stand/etc. in Paris.

Similarly, when a clause is temporally subordinated to another and the bound situation-TO is simultaneous with the binding one, then the relation expressed will normally be that of simultaneity, not anteriority or posteriority. Thus, (126,a) is a quite normal sentence, but (126,b) would wrongly suggest that the Eiffel Tower did not stand in Paris when Bill told his children about it:[86]

(126) (a) Bill told his children yesterday that the Eiffel Tower *stood* in Paris.
(b) Bill told his children yesterday that the Eiffel Tower *had stood/would stand* in Paris.

86. Since not only the time of the head clause situation but also t_0 is included in the time of the subclause situation, the speaker may also say *Bill told his children yesterday that the Eiffel Tower stands in Paris.* In that case there is a shift of domain from the past to the present sector (see chapter 4).

To capture these observations I will introduce the notion of 'temporal focus'. In (125,a) and (126,a) the temporal focus is 'unmarked', i.e. it is in accordance with the following principle:

(127) The temporal focus is unmarked in the following cases:[87]
  (a) in clauses that are not temporally subordinated: if the way in which the tense locates the situation in a particular absolute sector corresponds with the situation's actual location in time (i.e. if a past situation is located in the past or pre-present, a present situation in the present and a future situation in the post-present);
  (b) in clauses that are temporally subordinated: if the temporal relation expressed corresponds to the actual location of the situation relative to the binding TO (i.e. if a situation that held before the binding TO is represented as anterior to it, etc.).

In sentences like (125,b) and (126,b) the temporal focus is 'shifted'. That is, the way the situation is located in time is not the unmarked way described in (127). It follows that the sentences suggest a reading which is actually false: (125,b) suggests that the assertion 'The Eiffel Tower stands in Paris' is not true at t_0, and (126,b) suggests that the same assertion was not true at the time when Bill spoke to his children.

The possibility of shifting the temporal focus in order to suggest that a situation no longer holds (or may no longer hold) at t_0 is explicitly made use of in sentences like the following:

(128) (a) I *was going to do* it tomorrow (but I probably won't have time).
  (b) He *was coming* here himself tomorrow.
  (c) Yesterday they *left* tomorrow. (Huddleston 1969:787)

These sentences refer to an intention or arrangement whose actualization lies in the post-present. By locating the intention/arrangement in the past rather than in the present, the speaker suggests that the intention/arrangement is, or at least may be, no longer valid at t_0. In this way he avoids having to commit himself to a positive assertion concerning the future actualization of the situation referred to.

It is important to see that what we observe in (128,a–c) is a shift of temporal focus, not a shift of temporal perspective. A shift of perspective would mean that the past tenses were understood as referring to a

87. 'Unmarked' is to be understood here as 'temporally unmarked'. I am not claiming that the tense that is unmarked in this sense is also the tense that is preferably used in a particular sentence. There are sentence types in which the shift of focus has become conventional to a point that using the unmarked tense is distinctly odd. Thus, Riddle (1978:82) points out that we are much more likely to say *Wasn't he weird!* than *Isn't he weird!* when someone with strange clothes has just gone past us.

situation which actually belongs to a nonpast sector. This is not what we observe in (128,a–c). What we observe is that the focus is placed on the pastness of the intention rather than on its present relevance.

In (128,a–c) a marked temporal focus is used in order to suggest that the intention may no longer be valid. This kind of effect is not always present. Some kinds of shift of temporal focus are conventionalized and are not felt to have any bearing on the truth value of the sentence. For example, when my secretary comes into my office just as a visitor is leaving it, I can use either (129,a) or (129,b) to inform her (a minute later) of the identity of the person who has just left (see Rigter 1980c:855):

(129) (a) That *is* my brother.
(b) That *was* my brother.

Since the man in question is still my brother at t_0, (129,a) is the sentence with the unmarked temporal focus. In (129,b) the temporal focus is shifted to the moment when the secretary saw my brother and perhaps wondered who he was. This kind of shift is quite normal in present-day English. The following is a similar example:

(130) (a) I'm going to take the dog up to my room.
(b) You can't do that! Don't you remember that notice on the receptionist's desk that *stated* that dogs *weren't* allowed in the hotel rooms?

If the speaker of (130,b) had said . . . *that states that dogs aren't allowed in the hotel rooms*, he would have placed the temporal focus at t_0 (the time of his utterance). Instead, he has preferred to shift the temporal focus to the time when he and the addressee checked in at the hotel and read the notice on the receptionist's desk. In this way the situations are located in the past time-sphere rather than in the present one.

The above examples illustrate the possibility of a marked temporal focus in clauses that are not temporally subordinated. The following is an example in a clause that is temporally subordinated:

(131) When dinner was over, Bill said that he was going to fetch the dog from the car and take it up to his room. His wife protested that he could not do that. She pointed out that there *had been* a notice on the receptionist's desk which *had stated* that dogs weren't allowed in the hotel rooms.

In the same way as the preterit *stated* in (130,b) locates the situation (which still holds at t_0) at some time before t_0, *had stated* in (131) represents the situation as anterior to the relevant binding TO rather than as simultaneous with it. The same is true of *had been.*

In many cases the reason why the speaker shifts the temporal focus is that he wishes to represent the situation from the point of view of

someone else. (As noted by Bronzwaer (1970:48), the reader more easily identifies and empathizes with a character if the events are told from that character's point of view.) Thus, in the following examples the subclause situations are described as they were perceived or experienced by the people who participated in them:

(132) (a) They went to Florence, which *lay* in a wide valley. (translated from Wunderlich (1970:139))
(b) They had difficulty in climbing, for the northern slope of the Matterhorn *was* steep. (ibid.)
(c) My parents talked to the visitors in French because they *didn't understand* English.

In each of these sentences the preterit in the subclause could be replaced by a present tense. That the speaker prefers to use a past tense means that he wants to assume the temporal standpoint of the relevant participants in the events: he wants to express that it was *they* who observed *there* and *at that time* that the situation referred to in the subclause was holding. For this reason Wunderlich speaks of 'subjective preterit'. (If the speaker had substituted present tense forms for the preterits, he would have represented the situations from his own temporal standpoint, i.e. as simultaneous with t_0).[88]

The observation that the speaker is often free to choose the temporal focus accounts for many of the 'abnormal' uses of past and future tenses pointed out by R. Lakoff (1970). Lakoff claims that the existing theories of tense (which somehow relate the time of the situation to the time of utterance) are unsatisfactory because they cannot account for these special uses. In her opinion, an adequate theory of tense (which, as she herself admits, she cannot offer) 'must be able to incorporate in its underlying structures concepts such as the interrelationship of the actual time of occurrence of events with the involvement of the speaker, hearer or other participants in these events' (p. 838). As far as I can see, Lakoff's observations also have to do with the speaker's freedom to choose a marked temporal focus (such as the temporal standpoint of the hearer, or of one of the participants in the situation) and are therefore neatly accounted for in our theory. Let us, for instance, consider the following sentences (R. Lakoff 1970:839):

88. Jespersen (1931:155) refers to the same phenomenon when he observes:
> There is a curious use of the preterit dependent on a not expressed main verb in cases like this: Someone has told me 'I am going to Bristol on Thursday'. A little later I remember that he is going, but forget the date, and ask: 'What day were you going to Bristol?' or 'Was it on Thursday or Friday you were going to Bristol?' (= What day did you say you were . . .) cf. 'what was it again?'

A sentence like *What name was it please?* can be used similarly, but it can also be used in a context where the name in question has not been mentioned or asked yet. In that case *was* is a modal preterit which is used for tentativeness (see Latzel 1974:293; Lewis 1986:14).

(133) (a) The animal you saw *was* a chipmunk: see, there he is running up a tree.
(b) That thing rustling in the bushes over there *will* no doubt *be* a chipmunk: let's wait till it comes out.

Lakoff notes that the use of the tenses is special here because the past tense does not 'describe something that is no longer true at present' and because the future tense does not 'describe something that has not yet occurred' (p. 839).[89] I claim that what is special about these tense forms is that they are the result of a shift of temporal focus from t_0 to the time sector in which the situation (of the relevant participant observing the chipmunk) is located. This is the past sector in (133,a) (hence the use of the preterit) and the post-present sector in (133,b) (hence the use of the future tense).[90] Since the shift in question forms an alternative to locating the situation in the present sector, Lakoff's observation that the tense form does not imply that the situation is not holding at t_0 is automatically accounted for.[91]

Another illustration we can borrow from Lakoff (1970:848) concerns sentences that make a statement that is valid at all times. Although such generic (gnomic) sentences are usually in the present tense, there are a couple of (rather proverbial) examples in which the preterit or the future tense is used:

89. These descriptions of the meanings of the preterit and the future tense are anyhow inaccurate. As noted above, the past tense does not exclude that the situation continues into the present (see *John was in the kitchen a minute ago. So, I presume he is still there*). And neither is it inherent in the meaning of the future tense that the situation begins only after t_0 (see chapters 5 and 7).
90. One might suggest that in (133,b) *will* is not an instance of the future tense but the modal *will* signalling probability, expectation. However, R. Lakoff (1970:840) points out that there are examples in which this explanation clearly does not work:
(i) Those mangos will be three for a dollar.
(ii) !Those mangos are probably three for a dollar.
The shopkeeper using (i) to a customer knows perfectly well that the fruit are three for a dollar. It would be ridiculous for him to use (ii). In my opinion, this proves that the *will* in (i) is a true future tense form. It results from the speaker adopting the standpoint of the hearer paying for the mangos at some time in the future of t_0. As usual, this shift of temporal focus is performed to achieve a particular effect. By locating the act of paying in the future rather than in the present, the speaker avoids putting pressure on the person who has to pay. The shift of temporal focus thus renders the sentence less direct, and therefore more polite.
91. R. Lakoff (1970:840) observes that we cannot substitute *used to be* for *was* in (133,a). In my opinion, the reason for this is that (133,a) is a 'specificational' (identifying) sentence, i.e. a sentence which specifies a value (*a chipmunk*) for a variable (*the animal you saw*). As I have argued elsewhere (Declerck 1988a:85), a specificational sentence cannot involve *used to be*, unless the variable is such that different values can be assigned to it at different times, as in (i,a–b). Clearly, the variable in (133,a) is such that only one value can be assigned to it at any time. The same is true of (ii).
(i) (a) The colour that she preferred used to be blue.
(b) It used to be John who did most of the work.
(ii) The murderer of Tom is/*used to be John.

(134) (a) Boys will be boys.
(b) The course of true love never did run smooth.

According to Lakoff, the use of these tenses is special, because the reference is really to 'all time'. Our explanation is that examples like these make use of the possibility of shifting the temporal focus. In (134,a) the speaker exploits the fact that, if a situation is true at all times, future instances of this situation are predictable. He can therefore shift the temporal focus to these future instances. Similarly, the speaker of (134,b) can locate the situation in the past time-sphere because he knows that the hearer will himself 'draw the conclusion that what has hitherto been true is so still and will remain so to the end of time' (Jespersen 1924:259).

The speaker's freedom to choose the temporal focus is also clear from sentences like the following, which involve a modal auxiliary or semi-auxiliary:

(135) (a) The treasure *had to* be there. It *could not be* in any other place.
(b) The treasure *must have been* there. It *cannot have been* in any other place.

Each of the clauses in (135,a–b) expresses epistemic necessity. In (135,a), this necessity concerns the truth of a conclusion that was arrived at in the past and concerned a situation that was then holding; the speaker therefore assumes the temporal standpoint of the person drawing the conclusion, i.e. he locates the situation of drawing a conclusion in the past sector (*had to, could not*) and represents the situation that is being interpreted as simultaneous with this (*be there, be in any other place*). In (135,b), in contrast, the speaker expresses his own present conviction that a situation must have held in the past; he therefore locates the act of inference in the present sector (*must, cannot*) and represents the situation interpreted as anterior to it (*have been there, have been in any other place*). In other words, in both (135,a) and (135,b) the temporal focus resides with what we might call the 'evaluation time' (i.e. the time when the inference is made), but this time is a past interval in (135,a) and a present one in (135,b).

There is a similar difference between a past focus (past evaluation of a situation that was then holding) and a present focus (present evaluation of a past situation) in pairs like the following:

(136) (a) He *seemed/appeared/happened to be* a good scientist.
(b) He *seems/appears/happens to have been* a good scientist.
(137) (a) What John did *was* study.
(b) What John did *is* study.

(138) (a) It *was* Bill who left the room.
(b) It *is* Bill who left the room.[92]
(139) (a) It *was* true/a fact that John was there.
(b) It *is* true/a fact that John was there.
(140) (a) It *was* interesting/puzzling that Betty was absent.
(b) It *is* interesting/puzzling that Betty was absent.

In each of these examples the speaker uses the present tense to express his own current evaluation of the subclause situation and the preterit to express how the situation was appreciated in the past. The function of the tense of the head clause is thus to place the temporal focus on the time that is to be interpreted as the time at which the subclause situation is evaluated.

Examples involving an alternation of a present focus and a post-present focus are also available:

(141) (a) It *is* John who *will win* the prize.
(b) It *will be* John who *wins* the prize.
(142) (a) It *is* a fact that John *will* soon *be* the chairman.
(b) It *will* soon *be* a fact that John *is* the chairman.

In (141,b) and (142,b) there is reference to a post-present evaluation time. The subclauses refer to situations that will be evaluated at those times and are therefore simultaneous with them. In the following example the situation to be evaluated is anterior to the post-present evaluation time:

(143) If it rains tomorrow, we *will have worked* in vain yesterday. (Comrie 1985:73)[93]

This sentence involves a shift of temporal focus, because the conclusion expressed in the head clause is represented as a future conclusion, although it is actually reached at t_0. That is, the head clause situation is represented as anterior to a post-present TO, although it could also have been expressed as anterior to t_0. (In the latter case the sentence would have been *If it rains tomorrow, we worked in vain yesterday.*) This kind of shift of temporal focus from t_0 to some future evaluation time is not very common in English, but it is very often observed in French. French speakers very easily choose to use the 'futur antérieur' (future perfect) instead of the 'passé composé' (present perfect/preterit). In English too, it is not impossible to express (144,a) in the form of (144,b), but this type of shift of focus is by no means as usual as it is in French.

92. Self-evidently it is excluded that a present situation should be evaluated (verified) in the past. Hence the unacceptability of examples like **It was John who is in the room*, **It was a fact that Bill is 25 today* (see Berezovsky 1978:17).
93. See footnote 52 for a suitable context for this sentence.

(144) (a) Ronald Reagan *has been* the first president to defend this policy.
(b) Ronald Reagan *will have been* the first president to defend this policy.

For French examples and a useful discussion, see Steinmeyer (1987).

Some special uses of *will*, which are traditionally treated as 'modal' (see e.g. Palmer 1988:136ff.) can be accounted for in a similar way. The uses in question are those in which *will* is used in spite of the fact that the statement made would seem to be valid at t_0, so that there is no *a priori* reason for not using the present tense. In that case *will* expresses one of the following ideas:

1 belief or conjecture:[94]

(145) (a) They *will be* across the border by now.
(b) (*when the bell rings*) That *will be* the milkman.
(c) It *will have cost* a good deal, I suppose. (Kruisinga 1931:497)

2 Inherent capacity:

(146) (a) This jug *will hold* two litres.
(b) This auditorium *will seat* 320 people.

3 Characteristic (predictable) behaviour:

(147) (a) Oil *will float* on water.
(b) He *will talk* for hours if you give him a chance.

These uses of *will* can be explained as cases in which the temporal focus is shifted from the present to the post-present. Instead of simply claiming that a situation is holding at t_0, the speaker suggests that it will become apparent in the future that the situation is holding. That is, the speaker takes a post-present TO (rather than t_0) as the time at which the truth of the statement is evaluated.[95]

94. This 'modal' meaning is usually mentioned in connection with *will*, not *shall*. However, Kruisinga (1931:495) gives the following example involving *shall*. This supports our suggestion that the 'belief' interpretation results from the use of a future tense with an implicit TO, rather than being a separate modal meaning of the auxiliary.
(i) The missionary was not indeed unknown even in early times; but we *shall* probably *be* right in saying that when the system was most powerful, organization of missions was far more commonly in the interest of some sect within Islam than in the interest of Islam itself.

95. Smith (1978b:49) claims that sentences like *The store will have your book by now* and *The documents will have arrived last week* show that '*will* can occur with Present and Past, as well as Future, sentences'. In her opinion, this is a 'strong argument' for not treating *will* as a tense, for 'if *will* were to be treated as a future, it would be necessary to set up at least one other *will* to account for sentences such as [(145)–(147) above]'. The theory I have argued runs counter to these claims. I hold that *will* does refer to the future in such sentences and that there is consequently no reason for claiming that there is no future tense with *will* in English.

Needless to say, the idea of a future evaluation time is essential to these uses of *will*. Sentences like (145,a–c) are not interpreted as asserting a fact, but rather as expressing a belief which is expected to be confirmed in the future.[96] The same is true of (146,a–b). And (147,a–b) also imply future evaluation, since it is characteristic of such sentences that they imply that the truth of the statement is verifiable by experimentation. If this is not the case, *will* cannot be used (as in *Liverpool lies*/**will lie on the River Mersey*).[97]

In sentences like (145)–(147) *will* is used in spite of the fact that the situations referred to do not lie in the post-present. (This is clearest in (145,c), where the situation referred to by *will have cost* actually lies in the past.) For this reason *will* is often treated as a modal auxiliary expressing belief (conjecture, probability) rather than as the future tense auxiliary (see e.g. Canavan 1983:151). However, it may be argued that what we observe in (145,c) is a quite normal future perfect, and that the 'modal' interpretation results from the particular kind of time reference effected and a Gricean implicature. As we have seen, the future perfect tense locates a situation as anterior to a post-present binding TO but does not tell us anything about the temporal location of the situation relative to t_0. The situation in question may therefore be a situation that lies before t_0. However, if this is the case, as in (145,c), the speaker referring to the past situation will normally represent it as anterior to t_0, and not as anterior to some other TO (especially if the latter remains unidentified). This follows from the Gricean Maxim of Relation (Relevance), since 'other things being equal, statements about the present are more relevant than those about other times' (Comrie 1985:41–2). This means that the speaker evaluating a past situation will relate that situation to t_0 and not to some post-present TO, unless he wishes to suggest that evaluation of the past situation is not possible at t_0 but will become possible in the future. In other words, the use of the future perfect suggests that the speaker is not quite certain about what he is saying: the evaluation expressed is not presented as a present fact but as a prediction, i.e. as a belief which may or may not be confirmed in the future. (The main reason why a speaker may choose to represent his evaluation in this way is that it renders his statement more tentative and hence politer.)

96. Legault (1976:74) notes that *That'll be the milkman* implies something like 'as you will see when you open the door'. Bennett and Partee (1978:43) paraphrase *John will have eaten the fish by now* as 'It will be known that John has eaten the fish now.' Such a temporal, rather than modal, analysis of *will* accords with the fact that the French future tense can be used in exactly the same way: *Ce sera le laitier* ('It will be the milkman').

97. King's (1983:112) comment on (147,a) is that 'the use of the future ... looks to the future as prediction in a "wait and see" fashion'. Similarly, Bybee and Dahl (1989:93) note: 'This use is also a prediction, though it is a prediction about a state of affairs in present time. It is typically used when the verification of the present prediction lies in the future.'

It may be noted, finally, that the above analysis (according to which the use of *will* to express belief (inference) is not a separate modal use which is unrelated to the temporal use of *will*) accords perfectly with Dahl's (1985:105) conclusion that it is prediction that is the dominant component of meaning of the future tense in the sixty-four languages he has investigated, i.e. the future is a tense that has certain secondary modal uses, rather than the other way round. Moreover, this analysis also appears to be supported by diachronic evidence:

> While there is widespread attestation of the development of future tenses from such modal values as desiderativity and obligation and also from aspectual or aktionsart values (e.g. German *werden* 'become'?), in particular among the Germanic and Romance languages, I am not aware of any examples from these language families where future time reference develops diachronically out of inferential modality. My impression is that the future time reference develops out of other modalities and only with the establishment of the category's future time reference does the possibility of inferential modal interpretation arise. In at least most instances, the modal or aspectual origin of the future tense, while suitable as a candidate for reanalysis as future time reference, is not suitable as a candidate for direct reanalysis as an inferential. In other words, at least diachronically it may well be that inferentiality is parasitic on future time reference, rather than vice versa.
>
> (Comrie 1989:62)

23 POINT OF VIEW

It was noted in the previous section that a shift of temporal focus often entails a shift of point of view. If a situation is located in the past instead of in the present, this is often because the speaker wants to represent it as it was perceived or experienced by a participant in the situation at the time that the situation held. In that case assuming someone else's temporal standpoint is a means of representing the situation from that person's point of view.

However, the speaker can also adopt someone else's point of view in sentences where there is no shift of temporal focus. The use of past time-sphere tenses in 'represented speech and thought' (free indirect speech)[98] is a good example of this.[99] In such sentences the speaker

98. The label 'represented speech and thought' comes from Banfield (1982). 'Free indirect speech' is a translation of the French term 'style indirect libre' (coined by Bally 1912). Other terms are 'erlebte Rede' (Lorck 1921), 'represented speech' (Jespersen 1924: 290), 'quasi-direct discourse' (Coulmas 1986:8) and 'narrated monologue' (Cohn 1978). Van Gunsteren-Viersen (1980) offers a good overview of its historical development and use.

99. I will restrict the discussion to represented speech and thought, but the same remarks

represents things from the point of view of someone else, but there is no shift of temporal focus because the situations that are located in the past time-sphere could not be located in any other sector. That is, even if the speaker represented the situations from his own temporal standpoint, he would have to locate them in the past time-sphere rather than in the present. This is clear when we compare the following sentences:

(148) (a) (One day Mary's father asked her about her plans for the future.) He wanted to know what she *intended* to do after the summer holidays. He asked if she *would be going* to university. He added that he and Mother *had* always *hoped* she *would go* to Oxford.

(b) (One day Mary's father asked her about her plans for the future.) What *did* she *intend* to do after the summer holidays? *Would* she *be going* to university? He and Mother *had* always *hoped* she *would go* to Oxford.

In (148,a), which is a typical example of (normal) indirect speech, the speaker represents things from his own temporal standpoint and point of view.[100] He reports a number of interrogative speech acts performed by Mary's father in the past, and quite logically locates these speech acts in the past time-sphere. In (148,b) we have a stretch of RST (Represented Speech and Thought). This means that the acts of communication are now described without apparent mediation by the speaker. The speaker has withdrawn from the picture, so that the situations are no longer represented from his point of view, but from the point of view of a participant in the situations. However, this shift of point of view has no effect whatever on the tense forms. The tense forms are still the relative tense forms of (148,a). The only difference is that the situations functioning as central TOs in (148,a) (*he wanted to know/asked/added*) are no longer expressed in (148,b), so that the clauses (which are still temporally subordinated) are no longer syntactically subordinated.[101]

apply to other types of narrative in which the speaker represents things from someone else's point of view, such as 'represented perception' (see e.g. Fehr 1938; Steinberg 1971; Brinton 1980; Banfield 1982:200) and 'figural narration' (Ruin 1986:65).

100. This point is stressed by Halliday (1985:231): 'Hypotactic projection preserves the deictic orientation of the projecting clause, which is that of the speaker.' Banfield (1973:20–1) adduces a number of syntactico-semantic arguments confirming this. See also Coulmas (1986:2–3), Li (1986:38–9).

101. In RST, the tense forms (like the personal pronouns) continue to be related (directly or indirectly) to t_0. But the other deictics (i.e. the place adverbials and the time adverbials that are 'shifters' in the sense of Jakobson (1957) (e.g. *now, tomorrow, yesterday, an hour ago*)) are related to the deictic centre of the relevant character, i.e. to the 'fictional NOW' (Ruin 1986:66). (For a detailed study of the use of *now* in RST and other narrative past contexts, see Dry (1979). Schiffrin (1987:228–66) discusses both the temporal use of *now* and its use as a discourse marker.)

When a sentence does not contain any place or time adverbial, it is not always possible to know whether it is an example of RST or not. For example:

(i) John quickly made his way through the forest. It was bitterly cold.

The use of relative past tenses in RST clauses like those of (148,b) follows logically from the way in which RST works. The mechanism underlying RST is that someone else's temporal standpoint is adopted in order to suggest that the situation is expressed from his point of view. This means that all situations are related to a particular past time (the time when the relevant character said, thought or observed something) rather than to t_0. Tense forms that relate past situations directly to t_0 are not interpreted in terms of RST. For example, clauses that involve a shift of temporal focus from the past time-sphere to the present are not interpreted as RST clauses. Compare:

(149) (a) (When I met John, he immediately started asking me questions about Bill.) He wanted to know where Bill *was*, and when he *would come* back.
(b) (When I met John, he immediately started asking me questions about Bill.) He wanted to know where Bill *is*, and when he *will come* back.
(c) (When I met John, he immediately started asking me questions about Bill.) Where *was* Bill, and when *would* he *come* back?
(d) (When I met John, he immediately started asking me questions about Bill.) Where *is* Bill, and when *will* he come back?

Sentences (149,a–b) are instances of normal indirect speech. In (149,a) the complement sentences are temporally subordinated to the matrix; in (149,b) they show a shift of domain (and hence of temporal focus) to t_0. (This is possible because the questions are still relevant at t_0.) In (149,c–d) the interrogative clauses are no longer embedded under a verb of inquiry. This results in an RST reading in (149,c), where the clauses are still temporally subordinated, but not in (149,d), where the temporal focus is shifted to the present. (In the latter sentence the questions are attributed to the present speaker. That is, they may originally have been John's questions, but the speaker has made them his own.)

The fact that clauses expressing RST have the syntactic status of unembedded clauses entails that they differ from the corresponding indirect speech clauses when the reference is to the post-present. As we have seen, there is a double system of tenses to refer to situations in the

There is no indication here to tell us whether *it was bitterly cold* is an auctorial statement or represents the cold as perceived by John. (This concurs with Lyons' (1977:821) remark that *It was raining* can be interpreted either as 'It is a fact that it was raining' or 'as meaning, as it were, "It was a fact that it is raining" '. In the latter case there is a 'deictic projection into a past ... intensional world'. Lyons also notes that this difference 'would come out quite strikingly in a language like Latin, which would use a so-called indirect discourse construction' for the latter interpretation.)

post-present sector. In unembedded clauses we use the Future Perspective System (i.e. the future tense or the future perfect); in complement clauses we normally use the Present Perspective System (which inherently involves a shift of temporal perspective to one of the absolute sectors). Since indirect speech clauses are complement clauses, whereas RST clauses are unembedded clauses, it is not the same system that is used in the two cases:

(150) (a) (It's no use asking him for details.) He will say that he *has* not *seen* anything, that he *has* not *heard* anything, and that he *does* not *know* anything at all.
(b) (It's no use asking him for details.) He *won't have seen* anything, he *won't have heard* anything, and he *won't know* anything at all.

Whereas the Present Perspective System is used in the indirect speech subclauses of (150,a), the Future Perspective System is used in the (unembedded) RST clauses of (150,b).

24 IMPLICIT TO

When a tense relates a situation to a TO, this TO is normally expressed either by a clause (in which case it is a situation-TO) or by an adverb. When this is not the case, the TO in question is usually recoverable from the context. This is the case in examples like the following:

(151) (a) We must be quick or he *will have left* (before we get there).
(b) Mary is sad because Jim *had beaten* John. (prior to a TO referred to in the preceding context)

However, there are cases in which the relevant TO is not recoverable from the preceding context. Consider:

(152) (a) We will attack the city before the sun *has gone* down.
(b) The letter was stolen before we *had read* it.

In sentences like these the head clause situation is located, not before the subclause situation, but before some (unspecified) time to which the subclause situation is anterior. That is, the temporal relations expressed in (152,a–b) are the same as in the following paraphrases:

(153) (a) We will attack the city before (a time at which) it is the case that the sun has already gone down.
(b) The letter was stolen before (a time at which) it was the case that we had already read it.

These paraphrases logically entail the following:

(154) (a) We will attack the city at a time at which it is not yet the case that the sun has gone down.
(b) The letter was stolen at a time when it was not yet the case that we had read it.

And these, in their turn, entail

(155) (a) The sun will not have gone down yet when we attack the city.
(b) We had not yet read the letter when it was stolen.

As is clear from (154,a–b) and (155,a–b), *before*-clauses involving a perfect tense are 'nonfactual': they emphasize that the subclause situation has not held yet (i.e. is still nonfactual) at the time of the head clause situation.[102] However, they are not 'counterfactual' (even though they are often called so – see e.g. Lutzeier 1976), as they do not deny the actualization of the subclause situation at any time. It is clear that, even if it is true that the sun will not have gone under before we attack the city, it will go under afterwards. And (152,b) does not rule out the possibility that we ultimately retrieved the letter and read it.[103] (See also Heinämäki 1972, 1978:53ff.; Wachtel 1982:339.)

102. This explains Hornstein's (1975:103) observation that we will normally use (i), not (ii), to express the idea 'All year John has been telephoning Mary in order to prevent her from coming':
(i) All year John has been telephoning Mary before she comes.
(ii) All year John has been telephoning Mary before she has come.
The non-factuality of *before*-clauses in the present perfect or past perfect is also clear from the use of non-assertive words like *any*:
(iii) The police intervened before anything serious had happened.

103. A counterfactual interpretation can of course be enforced by the pragmatics of the sentence or by the context. This is the case in *The letter was destroyed before I had read it*, since it is pragmatically impossible to read a letter after its destruction. In such *before*-clauses (which can only be interpreted as counterfactual) the past tense can be substituted for the past perfect with no apparent difference: *The letter was destroyed before I read it*. However, when the head clause situation does not pragmatically exclude the subclause situation, the different tenses suggest different interpretations. Compare:
(i) (a) John read the letter before I had read it.
(b) John read the letter before I read it.
(c) John will read the letter before I have read it.
(d) John will read the letter before I read it.
Sentence (i,a) states that I had not (yet) read the letter when John read it. It does not say anything about whether or not I did read it afterwards. Sentence (i,b) suggests that I read the letter, but only after John had read it first. Similarly, (i,c) expresses no more than that John will read the letter at a time when I have not read it yet; (i,d) suggests that I will read the letter too, but only after John. In sum, the use of a non-perfect tense in a *before*-clause suggests actualization. (Since the suggestion is cancellable, as in *The letter was destroyed before I read it*, we should consider it an implicature, not an entailment – see Tedeschi (1981:264).)

Sentences like (152,a–b) make use of an 'implicit TO': the time referred to in the paraphrases (153,a–b) and (154,a–b) is not directly indicated by an adverb or by the fact that a situation is located at it. Still, there is implicit reference to that time, since the subclause situation is located as anterior to it.[104] The fact that this reference is only implicit means that we know nothing about the temporal location of this TO except for the fact that the situation referred to in the past perfect is anterior to it. However, in those cases in which it is clear from the context or from pragmatic factors that the past perfect situation eventually took place, the implicit TO tends to be interpreted as lying right after the situation in question (so that it is practically felt as coinciding with the final point of the situation). In that case the distance between the head clause situation and the implicit TO can be adverbially indicated:[105]

104. In Declerck (1979a) I tried to account for the nonfactual reading of *before*-clauses with a past perfect by hypothesizing that the past perfect was the result of backshifting (what we would now call a shift of temporal perspective) in order to express nonfactual modality. (As we have seen, this type of shift of temporal perspective is often used to express nonfactuality (e.g. in conditional clauses) or tentativeness.) I claimed that the past perfect could not be explained in terms of temporal relations, since the past perfect expresses anteriority, whereas in actual fact the subclause situation is not anterior to the head clause situation. I now think this explanation was mistaken. The notion of implicit TO renders it possible to treat the past perfect in the usual way, viz. as a verb form expressing anteriority. There are several pieces of evidence supporting this analysis. First, the use of a past perfect in a *before*-clause produces a reading which is nonfactual but not necessarily counterfactual. A modal past perfect, by contrast, always expresses counterfactuality. Second, a present perfect in a *before*-clause equally triggers a nonfactual interpretation; yet there are no modal present perfects. Finally, if the use of the past perfect were the result of a shift of temporal perspective in order to express modality, we would expect the past perfect to be possible in nonpast environments (in the same way as *I wanted to ask you something* can be used instead of *I want to ask you something*). However, the past perfect does not appear to be possible in *before*-clauses depending on a head clause in the present or future tense.

105. What is indicated by such an adverbial is the distance between the actualizations of the two situations. The presence of such an adverbial thus makes clear that the past perfect situation did eventually take place. Because of this, examples like (156,a–b) look better with the past tense in the *before*-clause. In fact, the past perfect (which emphasizes nonfactuality) appears to be possible only because the distance indicated by the measure phrase is very long. If it is a short timespan that is indicated, the past perfect is hardly acceptable:

    (i) I took the pill just before I ate/?had eaten dinner.

    In sentences like (156,a–b) the measure phrase indicates the temporal distance between the head clause situation and the end (completion) of the *before*-clause situation. This system becomes unnatural when that distance is very short. In that case we normally indicate the distance between the head clause situation and the beginning of the *before*-clause situation, using the preterit in the *before*-clause instead of the past perfect.

    Another consequence of the fact that the use of a measure phrase indicating the temporal distance between two past situations implicates that both situations are factual (i.e. took place in the past) is that a measure phrase cannot normally collocate with a past perfect in a *before*-clause which is not only nonfactual but also counterfactual (i.e. which refers to a purely imaginary situation which never held).

(156) (a) John left the house long before the others had arrived.
(b) The Vikings sailed to America centuries before Columbus had discovered it.

Not only *before*-clauses may make use of an implicit TO. There is also an implicit TO in sentences like the following:

(157) It *will have cost* a good deal, I suppose. (Kruisinga 1931:497)

In section 22 it was noted that such sentences show a shift of temporal focus from the present to the post-present sector: a situation which lies before t_0, and which could therefore be represented as temporally related to (and hence as evaluated at) t_0, is actually related to some post-present TO. Since the TO in question is not identified, sentences like (157) also illustrate the use of an implicit TO.

Sentences like (157) provide us with a first argument for claiming that the future perfect involves an implicit TO: since the element *have* in *will have* expresses anteriority (see also Smith 1976a, 1976c), there must be a TO functioning as binding TO for this anteriority relation. A second argument can be added to this, which is derived from sentences like the following:

(158) (a) If it rains tomorrow, we will have worked in vain yesterday/now/this afternoon, while the others *were doing* nothing.
(b) In six months' time we will perhaps have lost everything we *had saved* in that period.
(c) In six months' time we will perhaps have lost all the money that we *were going to use* for buying a house.
(159) (a) In six months' time we will perhaps have lost everything we *have saved* in that period.
(b) I will have finished my homework when John *comes* back.

Sentences (158,a–c) show that past time-sphere tenses are used to relate a situation to the situation-TO of a clause in the future perfect. Our theory that such a situation-TO is anterior to an implicit post-present TO accounts for this, for we have seen that a TO that is anterior to a post-present TO always functions as if it were a past time-sphere TO when other TOs are related to it (see section 15). The use of past time-sphere tenses in (158,a–c) contrasts with the use of present time-sphere tenses in (159,a–b). I suggest that the use of these tenses follows from the fact that the relevant situation-TO is now not related to the situation-TO of the higher clause but rather to the implicit

Instead of the past perfect we then use the conditional perfect, which explicitly signals irrealis and is therefore capable of cancelling the factuality implicature:
(ii) *He died three days before he *had finished* the symphony. (Fillmore 1975:35)
(iii) Max died two months before he *would have retired.* (Heinämäki 1978:63)
(iv) We paid the gas bill a short time before the company *would have cut* our supply. (Heinämäki 1978:60)

post-present TO. That is, in (159,a) the situation of saving money is no longer represented as anterior to the loss (as it is in (158,b)), but rather as anterior to the implicit post-present TO to which the loss is also anterior. Since, in the Future Perspective System, the idea 'anterior to a post-present TO' is expressed by the future perfect, and since a restrictive relative clause can in principle use either the Present Perspective System or the Future Perspective System (see above), we may therefore expect the future perfect to be possible too in (159,a). This expectation is actually borne out:

(160) In six months' time we will perhaps have lost everything we *will have saved* in that period.

The analysis proposed here also accounts for (159,b), where the present tense is used to represent John's coming back as simultaneous with the implicit TO prior to which I will have finished my homework. Note that, if the speaker wanted to represent John's coming back as simultaneous with my finishing my homework, he would have to use the past tense (as in (158,a)):[106]

(161) I will have finished my homework precisely when John *came* back.

Similar remarks can be made in connection with the expression of posteriority:

(162) (a) He will have promised me a hundred times that he *would help* me.
(b) He will have promised me a hundred times that he *will help* me.

In both cases the promises are anterior to an implicit post-present TO (which might be indicated by an adverbial like *by the end of this term*). In (162,a) the situation of helping me is represented as posterior to the promises and therefore as possibly anterior to the implicit TO. For this reason the sentence can be followed by something like (163,a). In (162,b) the helping is represented as posterior to the implicit TO. For this reason the sentence can be completed with (163,b), but not with (163,a):

(163) (a) ... but I'm sure he won't have helped me yet.
(b) ... but I still doubt if he will really help me.

106. There are, of course, more adequate ways of expressing the temporal relations implied here, i.e. of expressing the idea 'At some time in the future it will appear that I finished my homework precisely when John came back'. This explains why sentences like (161) may seem hard to interpret (and hence unacceptable) when not presented in a suitable context.

In the preceding paragraphs we have discussed nonfactual *before*-clauses and some uses of the future perfect in order to prove and illustrate the existence of an implicit TO in the temporal structure of some tense forms. I will now point out a couple of other instances. Consider:

(164) When Mary had broken her leg, John had left the country.

This sentence does not represent one situation as anterior to another, but represents two situations as anterior to an implicit TO. (This is an example of what we have called indirect binding.) What is expressed is that there was a time at which it was the case both that Mary had broken her leg and that John had left the country. The time in question remains implicit because no situation is located at it, whereas it is not indicated adverbially either. There is also an implicit TO in sentences like the following:

(165) I was going to leave for London tomorrow (but the strike makes it impossible).

Here the situation is represented as posterior to a TO which remains unidentified. (The TO in question may be indicated by an adverbial, as in

(166) Yesterday I was still giving my paper this evening. (Huddleston 1977:732)

If such a sentence contains only one time adverbial indicating past time, it is ambiguous, because the adverbial may indicate either the implicit TO or the time of the situation. Thus, Huddleston (1977:730) points out that *He was leaving yesterday* can mean either 'It was planned/scheduled that yesterday he should leave' or 'Yesterday the plan/schedule was that he should leave at some unspecified future time.')[107]

25 ADVERBIAL TIME CLAUSES

Adverbial time clauses deserve special consideration because the rules governing the use of their verb forms are not particularly transparent. On a superficial consideration, time clauses raise all kinds of problems. However, most of these problems disappear when it is understood that the temporal conjunctions in English involve reference to an implicit TO.

All temporal conjunctions can be paraphrased by means of a prepositional phrase with the word *time*: *when* is equivalent to 'at the time when/that', *after* means 'after the time that', etc. This semantic observation accords with the fact that most temporal conjunctions appear to

107. However, this ambiguity arises only if the time adverb is not put in initial position. The sentence *Yesterday he was leaving* yields only the reading 'Yesterday the plan was that he should leave.'

have developed from prepositional phrases.[108] Thus, in Old English *after* was not yet used on its own as a conjunction. Instead a prepositional phrase of the form 'after then that' (realized as *after þam þat/þe* or something similar) was used (see Visser 1970:868; Mitchell 1987:346ff.). In Middle English these phrases were still used, though the phrase *after tyme* (which literally means 'after the time that') was also occasionally used (see *the Middle English Dictionary*, Part A.2. p. 137). In later times these prepositional phrases were reduced to *after þat*, which finally developed into *after*. Similarly, the conjunction *before* has developed from a phrase of the form 'before the time that' (variously realized in Old English as *toforan þam timan þe, foran to þam timan þe* and *toforan þam þe* – see Mitchell 1987:379).[109]

Apart from the diachronic evidence, the prepositional origin of the temporal conjunctions still appears from the fact that, like prepositional phrases, temporal clauses can be used as postmodifiers:

(167) (a) The years after that were difficult.
(b) The years after that had happened were difficult.
(168) (a) He felt very nervous during the days before the examination.
(b) He felt very nervous during the days before the examination took place.

The hypothesis that the temporal conjunctions have retained the meanings of the phrases from which they have developed (and thus involve reference to an implicit TO)[110] accounts for the seemingly deviant use of the tenses in temporal clauses (where conjunctions expressing anteriority (*after*) or posteriority (*before, until, by*) are often followed by a verb form expressing simultaneity instead of by a verb form expressing anteriority or posteriority). In this section I will attempt to show this for time clauses introduced by the three basic conjunctions, viz. *when, after* and *before.*

25.1 *When*-clauses

Consider the following sentences:

(169) (a) It happened when the police *were* there.
(b) It happened when the police *had left.*
(c) It happened when the police *were going to intervene.*

108. A notable exception is *when*, which was used as a question word or indefinite adverb ('at some time') in Old English and only later developed its use as a conjunction. (see Mitchell 1987:402)
109. However, these phrases were very unusual in Old English, which almost exclusively used *ær þam þae, ær þam* or *ær.*
110. Clark (1973:50), Ritchie (1980:209) and Harris (1982:154–5) also analyse temporal conjunctions as reduced prepositional phrases.

(170) (a) It will happen when the police *are* there.
(b) It will happen when the police *have left.*
(c) It will happen when the police *are going to intervene.*

In each of these sentences the *when*-clause expresses a temporal relation in the domain established by the head clause. In each case *when* is equivalent to 'at a time when', so that the following are good paraphrases of (169)–(170):

(169′) (a) It happened at the time when (it was the case that) the police were there.
(b) It happened at the time when (it was the case that) the police had left.
(c) It happened at the time when (it was the case that) the police were going to intervene.
(170′) (a) It will happen at the time when (it is the case that) the police are there.
(b) It will happen at the time when (it is the case that) the police have left.
(c) It will happen at the time when (it is the case that) the police are going to intervene.

These paraphrases make clear that the *when*-clauses locate their situations at, after or before the implicit time that is simultaneous with the head clause situation.

At first sight it may seem that the assumption of an implicit TO is not quite necessary in order to account for sentences like (169,a–c) and (170,a–c). However, this necessity becomes clearer when one considers examples like the following:

(171) (a) John had left when I *arrived.*
(b) John will have left when I *arrive.*

If we consider (171,a) in isolation we may feel tempted to say that the *when*-clause here establishes the central TO of a past domain and that the head clause situation is represented as anterior to this. However, when we compare (171,a) with (171,b), we see that this analysis is dubious. In (171,b) we cannot say that the *when*-clause establishes the central TO of the post-present domain. If it did, an absolute future tense form (*will arrive*) would have to be used. In reality we cannot use the future tense. We have to use the present tense (which expresses simultaneity) instead. This means that we must assume that the central TO of the post-present domain remains implicit (in the sense that it is not a situation-TO) and that the two situations are temporally related to it: John's leaving is represented as anterior to it, my arrival is represented as simultaneous with it. Since (171,a) is the past time-sphere counterpart of (171,b), it should probably be analysed in a similar way: *had left* represents John's

leaving as anterior to the central TO and *arrived* represents my arrival as simultaneous with this TO, but the TO in question remains implicit.

Further evidence in favour of this analysis is provided by sentences like the following:

(172) (a) When the fire brigade *had finished* evacuating the orphans, the fire had destroyed most of the building.
(b) When John *has finished* his homework, the other children will have left the house.

(173) (a) When the police *were going to ask* him questions, the suspect had disappeared.
(b) When the police *are* finally *going to question* John, he will have disappeared.

In each of these sentences the head clause situation is represented as anterior to a TO. This TO is not the situation-TO of the *when*-clause, for the latter is represented as anterior to the TO in question in (172,a–b) and as posterior to it in (173,a–b). The relevant TO is therefore an implicit TO. (Note that all these sentences sound better if we substitute *by the time that* for *when*. This is not surprising, for this phrase makes the temporal structure that is realized in the sentences more explicit: *by* expresses that the head clause situation is anterior to the implicit TO, and *the time* refers explicitly to the TO that remains implicit when we use *when*.)

Another piece of evidence supporting the claim that *when*-clauses relate their situations to an implicit TO instead of establishing a domain can be derived from indirect speech constructions. Consider:

(174) (a) Bill believed that it *was* five o'clock.
(b) John said that Bill had believed that it *was* five o'clock.

In (174,a) the verb referring to Bill's belief is the verb of the head clause and is therefore an absolute tense form (*believed*). In (174,b) the same verb appears in a *that*-clause and is therefore a relative tense form (*had believed*). In this case syntactic subordination thus goes hand in hand with temporal subordination. Since the relative tense used to represent a situation as anterior to a past TO is different from the absolute tense used to establish a past domain, this syntactic and temporal subordination entails an adaptation of the tense form from past to past perfect. This kind of adaptation is known in the literature as 'backshifting' or 'sequence of tenses'. However, the tense of the *that*-clause of (174,a) is not backshifted when we report (174,a) in the form of (174,b). The reason is that both in (174,a) and in (174,b) the *that*-clause situation is represented as simultaneous with the situation of its matrix. In both cases a relative past tense form (*was*) is used for this. The conclusion to be drawn from (174,a–b) is therefore that only absolute past tense forms can

undergo backshifting in indirect speech. This means that past tense forms that cannot be backshifted must be instances of relative tense.[111]

Consider now:

(175) (a) Bill was in London when Jenny was in Paris.
(b) I explained that Bill had been in London when Jenny was in Paris.

Examples like these show that the tense form of a direct speech *when*-clause is not normally backshifted in indirect speech.[112] This provides us with a strong piece of evidence for the claim that an adverbial *when*-clause does not establish its own domain, i.e. that it relates its situation to an implicit TO.[113]

It should be noted, however, that there is one type of *when*-clause that forms an exception to this rule. In chapter 3 we will discuss so-called 'narrative *when*-clauses', i.e. *when*-clauses which do not specify the time when the head clause situation holds but which 'push forward the action'. This type of *when*-clause (which we find in examples like *I was sitting quietly in the kitchen when suddenly a stranger entered the room*) shifts the domain (witness the fact that there is 'backshifting' in indirect speech: *I explained that I had been sitting quietly in the kitchen when suddenly a stranger had entered the room*) – see chapter 3.

25.2 *Before*-clauses

Before-clauses in principle allow two possibilities: temporal subordination and shifting of the domain. The latter possibility is realized in sentences like *Bill left before John arrived.* There is evidence that the *before*-clause here uses an absolute preterit, i.e. a preterit establishing its own domain. One piece of evidence is that *arrived* represents John's arrival as factual. That is, John's arrival is represented, not as something

111. Note that I am not saying that absolute past tense forms *must* be backshifted in indirect speech. As will be explained in chapter 4, the reporting speaker may have the *that*-clause establish its own domain, provided that the temporal relations are clear from the context (as in *Bill said that he washed the car earlier that day*). The point I am making is that an absolute preterit occurring in an unembedded direct speech clause *can* always be backshifted if that clause is embedded under a reporting verb in the past tense, whereas a relative tense form cannot normally be backshifted.

112. I am adding the proviso *normally* because there are types of subclause in which English allows indirect binding. Perhaps some speakers do accept sentences of the kind *?Bill said that he had been unhappy when he had been young.* However, it is clear that the use of the preterit is strongly preferred in the *when*-clauses of such examples.

113. Perhaps it is also worth noting that, if the *when*-clause situation is to be represented as counterfactual, it uses the conditional perfect:
(i) (I would have told her . . .) But when I *would have spoken* of this, I saw that she was looking into the future. (Jespersen 1931:309)
(ii) Maddeningly, just when I *would have welcomed* full information on the books he was reading, he started to mention nothing beyond authors and titles, often in a shortened form. (K. Amis, *The green man*, Frogmore: Panther, p. 97)

that was expected (at some past TO) to take place later, but rather as a past fact.[114] This means that the speaker relates John's arrival directly to t_0 and therefore uses *arrived* as an absolute preterit.[115] The second piece of evidence concerns the choice of tense in indirect speech. Both preterits in *Bill left before John arrived* can be 'backshifted' in indirect speech: *Mary said that Bill had left before John had arrived.* This proves that both preterits are absolute tense forms (see above).

The possibility of shifting the domain is systematically made use of in what we could call 'narrative *before*-clauses'. Like narrative *when*-clauses, these are clauses that push forward the action instead of functioning as time adverbials (specifying the time of the head clause situation). An example is: *Instead of entering the house at once he waited some time before he pushed the key into the lock.* (The indirect speech test confirms that the *before*-clause shifts the domain: *He said that instead of entering the house at once he had waited for some time before he had pushed the key into the lock.*)

As a rule, *before*-clauses that refer to the post-present do not allow this possibility of shifting the domain. Sentences of the type **Bill will leave before I will arrive* are ungrammatical. When the head clause establishes a post-present domain, the *before*-clause situation must normally be temporally subordinated. The reason for this is probably the following. As we will see below, the 'intensional domain' (opaque context) created by an intensional verb (verb of propositional attitude) constitutes a temporal domain, so that *before*-clauses that are to be interpreted as

114. The former possibility is realized in sentences like *Bill wanted to leave before I arrived* (see below).

115. Another argument supporting this analysis can be derived from Dutch. Unlike English, Dutch can use the present perfect to represent a situation as holding at a particular past time:

(i) (a) Ik heb Jan gisteren gezien. ('I have seen John yesterday')
(b) Jan is vertrokken voordat ik ben aangekomen. ('John has left before I have arrived')

However, the Dutch present perfect can replace the past tense only when the latter is an absolute tense. It cannot normally express simultaneity (see Janssen 1990:32, where it is pointed out that the present perfect cannot be used after *toen* ('when'), except in 'very particular cases'). Compare:

(ii) (a) Hij is weggegaan toen hij wist dat ik boos was. ('He has gone away when he knew that I was cross')
(b) *Hij is weggegaan toen hij heeft geweten dat ik boos ben geweest. ('He has gone away when he has known that I have been cross')

The fact that a Dutch present perfect cannot normally correspond to an English relative past tense is also clear from indirect speech:

(iii) (a) Jan gelooft dat de erfgenaam een meisje is. ('John believes that the heir is a girl')
(b) Betty zei dat Jan geloofde dat de erfgenaam een meisje *was/*is geweest.* ('Betty said that John believed that the heir was/*has been a girl')

Since the Dutch present perfect cannot normally be used as a relative tense, we must conclude that the present perfect in the *before*-clause of (i,b) is an absolute tense form, i.e. a form which shifts the domain. This corroborates our claim that the preterit in its English counterpart should be analysed in the same way.

opaque must be temporally subordinated. Now it is well-known that 'the future modality by itself produces opacity in its scope' (Givòn 1973:101). This means that a temporal domain established by a future tense form is also an intensional domain, and this accounts for the fact that *before*-clauses depending on a head clause in the future tense cannot shift the domain.

Though this rule appears to be a very strict one, occasional exceptions can be found:[116]

(176) (a) Tobacco, it is said, is gradually being ousted by sweets. We fancy, however, that it will be some little time before it *will be* a common sight to see men about town walking down Pall Mall with a sugar stick in the mouth instead of a cigarette. (Kruisinga 1931:494)

(b) John will see that film before you do/*will*. (Kittredge 1969:15)

A type of shift of domain that is more productively possible is the one that is to be observed in the following example:

(177) Tina said that she would leave before Sol *returns*. (Comrie 1986:296)

Here the *before*-clause establishes a present domain because the speaker wants to express that he treats the (post-present) return of Sol as a present fact. (The use of *returned* would express no more than that Tina expected Sol to return; the speaker's own opinion on the subject would not be expressed.)[117]

Apart from the (heavily restricted) possibility of shifting the domain, *before*-clauses can evince temporal subordination. In that case the *before*-clause situation (which is posterior to that of the binding clause) is not directly related to the latter (by the use of a verb form expressing posteriority). Instead, it is temporally related to an implicit TO, which is itself represented as posterior to the binding clause situation by the use

116. Harris (1982:268) accepts examples like *She will phone before/after he will leave*. Most speakers, however, find such sentences ungrammatical.

117. The form *returns* in the *before*-clause is an example of a present tense which actually refers to a post-present situation. It thus realizes a shift of perspective from the post-present to the present. This is exactly the same shift as is observed in independent clauses like *Sol returns tomorrow* or *The train leaves in an hour*. It is therefore not surprising that verbs that do not allow the shift in head clauses cannot be found in the present tense in *before*-clauses either:

(i) (a) *It *rains* tomorrow.
(b) *Tina said she would leave before it *rains*.
(ii) (a) *Tina *is* unhappy soon.
(b) *Tina said she would leave before she *is* unhappy.

(Incidentally, this proves that the present tense in *Tina said she would leave before Sol returns* establishes a domain and does not express simultaneity with some implicit TO in a post-present domain.)

of *before*. In other words, *before* is equivalent to 'before a/the time when'. The *before*-clause situation is then usually represented as simultaneous with this implicit TO. However, when the speaker wants to refer to the resultant state produced by the situation rather than to the situation itself, the relation expressed will be that of anteriority. For example:

(178) (a) Bill wanted to go home before it *was* night. (*would be)
(b) Bill wanted to go home before night *had fallen*. (*would have fallen)

(179) (a) Bill would have gone home before it *was* night. (*would be)
(b) Bill would have gone home before night *had fallen*. (*would have fallen)

(180) (a) Bill will want to go home before it *is* night. (*will be)
(b) Bill will want to go home before night *has fallen*. (*will have fallen)

(181) (a) Bill will have gone home before it *is* night. (*will be)
(b) Bill will have gone home before night *has fallen*. (*will have fallen)

As is clear from these examples, a (temporally subordinated) *before*-clause does not relate its situation directly to that of the head clause (in terms of posteriority). Rather it relates it to an implicit TO, which is itself posterior to the head clause situation.[118] This posteriority relation is expressed by *before*, which is therefore paraphrasable as 'before a/the time when'. In the (a) sentences the relation expressed is that of simultaneity; in the (b) sentences it is anteriority. In the latter case the reference is not so much to the situation itself as to the resultant state which it has produced.

The hypothesis that the *before*-clauses relate their situations to an implicit TO is also supported by two further observations. One is that the past tense forms in the *before*-clauses of the (a) sentences of (178)–(181) cannot be backshifted in indirect speech: *Mary said that Bill had wanted to go home before it was/*had been night*. As noted above, this proves that the past tense forms in question are relative tense forms. The second observation is that the verb forms in (179,a–b) and (181,a–b) remain unaffected when we substitute *by the time that* for *before*. (This is not so easily possible in (178,a–b) and (180,a–b) because *by the time that* strongly prefers the use of a perfect form in the head clause.)

(179′) (a) Bill would have gone home by the time it *was* night. (*would be)
(b) Bill would have gone home by the time night *had fallen*. (*would have fallen)

118. In Dutch either system is possible: *Bill wou vertrekken voordat Mary* ***vertrok/zou vertrekken*** ('Bill wanted to leave before Mary left/would leave').

In the phrase *by the time that* the NP *the time* makes explicit the TO which remains implicit when we use the conjunction *before*. The verb form that follows then clearly relates its situation to *the time*, not to the head clause situation. The fact that *before*-clauses use exactly the same verb forms therefore suggests that the same temporal relations are expressed.

One of the consequences of the temporal subordination process is that the *before*-clause situation is not represented as a fact. As noted before, a *before*-clause can represent a situation as factual by the use of an absolute preterit or present tense, as in *Bill left before I did* or *Bill said he would leave before I do*. However, when there is temporal subordination, the interpretation is necessarily in terms of (relative) nonfactuality. That is, since the tense used is not an absolute tense, the speaker does not treat the *before*-clause situation as a fact. Moreover, since the implicit TO is posterior to the binding clause situation, the situation that is located at the implicit TO is necessarily a situation which has not yet actualized at the time of the binding clause situation. Thus, both (178,a) and (178,b) express that Bill wanted to go home at a time when it was not yet night.

Nonfactual *before*-clauses often form part of an intensional (opaque) context. In fact, it appears that an intensional context functions as a temporal domain: any clause that is to be interpreted as belonging to the 'intensional domain' must be incorporated into the temporal domain containing the clause establishing the opaque context. In clauses that shift the domain, the situation is directly related to t_0, and hence to the point of view of the speaker. Such clauses must therefore be interpreted transparently.

The fact that a *before*-clause can receive a *de dicto* (opaque) interpretation only when it does not establish its own domain entails that a *before*-clause that is to be interpreted as opaque must as a rule be temporally subordinated. This is the case, for example, when its matrix expresses posteriority in a past domain (often in examples of indirect speech or RST). For instance:

(182) (a) Bill intended to notify the police before it *was* too late.
(b) Bill intended to notify the police before a murder *had been committed.*
(c) Tim ordered me to leave before John (*had*) *arrived.*
(d) John said he would leave before Mary (*had*) *arrived.*
(e) He would leave before Mary (*had*) *arrived.*

As before, there is an implicit TO. The preterit is used for simultaneity, the past perfect for anteriority. (In the latter case the speaker refers to the state resulting from the situation rather than to the situation itself.)

When the clause containing the *before*-clause forms part of an opaque context but does not represent its own situation as posterior to the

situation of its own head clause, the *before*-clause may or may not shift the domain:

(183) John said Bill had left before Mary *arrived.*

If John's actual words were *Bill left before Mary arrived*, the *before*-clause of (183) is interpreted as opaque. In that case it belongs to the temporal domain that corresponds with the opaque context. However, if John said something different (e.g. he said *Bill left before Joan left* and I, who know that Joan left precisely when Mary arrived, report this in the form of (183)), then I am reponsible for the truth of the *before*-clause. In other words, the *before*-clause then does not form part of the intensional domain. This means that it is not incorporated into the temporal domain established by *said* but establishes its own domain.

The fact that nonfactual *before*-clauses often occur in intensional contexts does not mean that they only appear there. In the following examples too the *before*-clause situation is represented as not yet realized (i.e. as nonfactual) at the time of the head clause situation:

(184) (a) Bill left before it *was* night.
(b) Bill left before night *had fallen.*
(185) (a) Bill will leave before it *is* night.
(b) Bill will leave before night *has fallen.*

Here again the mechanism is that a situation or its resultant state is located at an implicit TO which is posterior to the situation of the head clause. The tense form expressing simultaneity is used to refer to the situation itself; the tense form expressing anteriority is used to refer to its resultant state.[119]

I would like to conclude this discussion of *before*-clauses by drawing attention to the curious use of *before* in sentences like the following:[120]

(186) (a) It was the following holidays before they fought again. (D. Lessing, *This was the old chief's country*, London: Triad Grafton, p. 365)
(b) It was 7 p.m. before he arrived in London.
(c) But it was dawn before he slept again. (J. Caldwell, *Great lion of God*, London: Collins, p. 128)

At first sight these sentences would seem to be similar to the following:

(187) It was three weeks before Bill arrived.

119. As noted in footnote 105, we use the conditional perfect rather than the past perfect to express that the *before*-clause situation (which was still nonfactual at the time of the head clause situation) in fact never actualized:
(i) Max died two months before he *would have retired.* (Heinämäki 1978:63)

120. This type of sentence was brought to my attention by Shigeki Seki.

However, there is a difference. In (187) *three weeks* indicates the timespan between the past TO and the *before*-clause situation (i.e. three weeks elapsed before Bill arrived). In (186,a–c), by contrast, *the following holidays*, *7 p.m.* and *dawn* indicate the time at which the *before*-clause situation actualized. That is, as far as objective time is concerned, the way in which the situations are located in time in (186,a–c) is not different from the way this is done in (188,a–c):

(188) (a) ?It was the following holidays when they fought again.
(b) It was 7 p.m. when he arrived in London.
(c) It was dawn when he slept again.

However, we do not interpret (186,a–c) as identical with (188,a–c). We rather interpret them as equivalents of (189,a–c):

(189) (a) It was not until the following holidays that they fought again.
(b) It was not until 7 p.m. that he arrived in London.
(c) It was not until dawn that he slept again.

Whereas (188,a–c) just locate the *before*-clause situation at the time indicated, (186,a–c) and (189,a–c) do more than this: they also stress both the existence of an interval of time between the relevant TO and the *before*-clause situation and the nonactualization of this situation during the interval in question. (The use of *by the time that* instead of *before* in (186,a–c) would have the same effect.) In (186,a–c) the function of *before* is thus to bring out a posteriority ('later') relation which is expressed by no other constituent in the sentence.[121]

25.3 *After*-clauses

Diachronically speaking, the conjunction *after* is a reduction of something like 'after the time that'. (The *Oxford English Dictionary* gives some Old English examples showing this phrasal structure.) We may therefore expect that *after* still involves reference to an implicit TO. As a matter of fact, this expectation is borne out by examples like the following:

(190) Was that before or after the police *were going to intervene*?

This sentence is interpreted as 'Was that before or after the time when (it was the case that) the police were going to intervene?'. This means that the *after*-clause situation (the intervention) is located as posterior to an implicit TO which is itself located before or after the head clause situation-TO.

121. If the relationship is that of anteriority, *since* is used:
(i) Come on, you've learnt your part. – Yes, I'm just turning it over. You see, *it's last December since I played it.* (Geluykens 1984:95)

Needless to say, the grammaticality of examples like (190) is decisive evidence that the time expressed by the NP of the original prepositional phrase still functions as implicit TO in the meaning of the present-day conjunction. If this were not the case, it would be quite inexplicable that a verb form expressing posteriority can be used in a clause introduced by *after* (whose basic meaning is anteriority).

It is possible for the speaker to represent the *after*-clause situation as simultaneous with the implicit TO:

(191) (a) If he doesn't hurry, the doctor will arrive only after the patient *is* dead.
(b) Did you call the doctor before or after the patient *was* dead?
(c) It was only after the president *was* dead that the documents were made public.

In (191,a) *is* obviously expresses simultaneity in the post-present domain. That the past tense forms in the time clauses of (191,b–c) also express simultaneity is confirmed by the indirect speech test (which relies on the observation that an absolute preterit is normally backshifted after an introductory verb in the past tense, whereas a relative one is normally retained – see above):

(192) (a) The police asked if it was before or after the patient *was* dead that she had called the doctor.
(b) The police declared that it was only after the president *was* dead that the documents had been made public.

Apart from expressing posteriority or simultaneity, the verb of the *after*-clause can also represent its situation as anterior to the implicit TO:

(193) We know that the man came to live in London three months after he *had lived* in New York for two years. But we don't know where he lived during those three months.

This sentence is interpreted as '. . . the man came to live in London three months after the time when (it was the case that) he had lived in New York for two years'.

However, it is difficult to find examples in which the past perfect in the *after*-clause expresses anteriority to the implicit TO. In most cases the *after*-clause is interpreted as relating its situation directly to the head clause situation:

(194) John arrived after the bomb *had exploded.*

It would seem that this sentence is not interpreted as 'John arrived after the time after the explosion of the bomb', but rather as 'John arrived after the explosion of the bomb'. In other words, there seems to be no real difference of temporal interpretation between (194) and (195,a–b):

(195) (a) John arrived *after* the bomb *exploded.*
(b) John arrived *when* the bomb *had exploded.*

This suggests that the anteriority relation expressed by *had exploded* in (194) is actually the same as that expressed by *after*, i.e. that there is only one anteriority relation and that this relation is expressed twice: once by the choice of *after* and once by the use of the past perfect. As a matter of fact, there are a couple of further observations that support this analysis:

1 As is clear from (195,a–b), the speaker can also choose to express the anteriority relation only once, without this having any effect on the temporal interpretation.

2 In those cases where the past perfect does represent the *after*-clause situation as anterior to an implicit TO which is itself represented as anterior to the head clause situation by means of *after*, the speaker must have a reason for expressing this double relation. If he just wants to express that situation A followed situation B, there is no need for him to express this as 'A after implicit TO after B'. The simpler representation 'A after B' will do. It is only in cases like (193) that two relations must be expressed, because there is an adverbial pertaining to each of them: in (193) *three months* measures the distance between the head clause situation and the implicit TO, whereas *for two years* measures the time between the beginning of the *after*-clause situation and the implicit TO (which is the time when the *after*-clause situation comes to an end).[122]

3 The claim that (194) does not realize the schema 'A after implicit TO after B' accords with the observation that we cannot insert two measure phrases into the sentence:

(196) (a) *John arrived long after the bomb had just exploded.
(b) *John arrived three minutes after the bomb had just exploded.
(c) *John arrived three minutes after the bomb had exploded five minutes before.

If (194) expressed a double anteriority relation, there would be no reason why we could not add a measure phrase to each of them. (Note that in (196,c) the phrases *three minutes* and *five minutes before* are felt to contradict each other. This is because we apply both of them to the same time interval, viz. the timespan between John's arrival and the explosion. It is also worth noting that the insertion of the two phrases does not bring out the 'A after TO after B' reading. It just results in an ungrammatical sentence.)

122. The anteriority relation expressed by a perfect form may be of the type 'before and up to' – see chapter 7.

4 The hypothesis that I am arguing is that the use of the pluperfect in (194) (*John arrived after the bomb had exploded*) is simply induced by the use of *after*. Using *after*, the speaker has an anteriority relation in mind, and this leads him to use the past perfect. A final argument in favour of this analysis is that exactly the same thing appears to be possible in *before*-clauses – even though the possibility is a very marginal one in English. As noted above, English *before*-clauses that are temporally subordinated as a rule use a verb form expressing simultaneity with an implicit TO. However, in many European languages, the speaker may choose either a form expressing simultaneity or a form expressing posteriority. For example, a Dutch speaker can say either *Hij zou vertrekken vooraleer Jan vertrok* ('He would leave before John left') or *Hij zou vertrekken vooraleer Jan zou vertrekken* ('He would leave before John would leave'). Even in English, occasional examples can be found where *before* is followed by a conditional tense:

(197) They were passing away the few minutes before Pamela would be ready to join them in the dining-room above. (Jespersen 1931:320–1)

In this example the reason for the use of the conditional is probably that the speaker wants to represent the *before*-clause situation as part of an expectation rather than as a past fact. (Perhaps Pamela did not come to the dining-room at all, so that the expectation remained unfulfilled.) If the speaker had used a preterit, this could be understood as an absolute preterit, i.e. as a form representing the situation as a past fact. However this may be, the use of the conditional with *before* is similar to the use of the pluperfect with *after*. In both cases the use of a form expressing simultaneity with the implicit TO is shunned in favour of the use of a tense copying the relation that is expressed by the conjunction. (The term 'conjunction copying' would therefore seem to be a suitable label for this phenomenon.)[123]

Another question that merits going into (and which has already been touched upon above) is that of the nature of the past tense in the *after*-clause of sentences such as the following:

(198) John arrived after the bomb *exploded*.

Theoretically, there are two possibilities. Either *exploded* shifts the domain or it is a relative preterit which expresses simultaneity with the implicit TO. When dealing with *before*-clauses, we noticed the same two possibilities, and we pointed out two tests that are in principle capable of

123. Conjunction copying is similar to the phenomenon that many people use *doesn't* instead of *don't* in sentences like *He is one of those linguists that don't like formalization*. The verb of the relative clause here adopts the singular number of the words at the beginning of the sentence (*he*, *one*).

distinguishing between them. The first had to do with the meaning (factual or nonfactual) of the *before*-clause: if the *before*-clause represents its situation as a past fact (as in *John arrived before the bomb exploded*), it must establish a past domain; otherwise it expresses a relation in an already established domain. Unfortunately, this criterion is not applicable to *after*-clauses: since whatever precedes a factual situation must itself be factual, all *after*-clauses that depend on a head clause in the past tense represent their situations as factual (in the sense of 'realized at the time of the head clause situation'), even if they do not shift the domain. Thus, not only (198) but also *John arrived after the bomb had exploded* represents the explosion as factual.

The second test we used in connection with *before*-clauses concerned the possibility of backshifting the tense in indirect speech: an absolute past tense can always be backshifted (if the reporting clause is in the past tense), whereas a relative preterit is retained. Unfortunately, this criterion is not helpful either if we apply it to *after*-clauses, because the tense of an *after*-clause can copy the conjunction. That is, a past perfect in an *after*-clause in indirect speech need not be the result of 'backshifting'. It may also be an instance of conjunction copying. This means that the indirect speech version may use a past perfect even if the direct speech sentence uses a relative preterit in the *after*-clause.

Let us see if *after*-clauses depending on a verb in the future tense can make us any wiser:

(199) (a) John will arrive after the bomb *explodes/has exploded.*
(b) *John will arrive after the bomb *will explode/will have exploded.*

(200) (a) You will inherit that money only after your uncle *is* dead.
(b) *You will inherit that money only after your uncle *will be* dead.

These examples make clear that an *after*-clause can express simultaneity in a post-present domain but cannot establish its own post-present domain.

Should we conclude from this that the past tense form *exploded* in (198) (*John arrived after the bomb exploded*) is a relative tense form expressing simultaneity? I think we must be cautious not to jump to conclusions. There is a reason for doubting whether (198) must necessarily be comparable with (199,a) and (200,a). As noted above, an intensional domain constitutes a temporal domain, so that a time clause that is to be interpreted as opaque must be temporally subordinated. Since 'the future modality by itself produces opacity in its scope' (Givòn 1973:101), the post-present temporal domains established by the future tense in the head clauses of (199,a) and (200,a) are also intensional domains, and this could be an explanation for the fact that the *after*-clause must be temporally subordinated (as appears from the

ungrammaticality of (199,b) and (200,b)). The situation is different, however, in (198), since the past tense does not create an opaque environment.

If this is correct, the obligatory use of a relative tense form in an *after*-clause depending on a post-present head clause is no decisive evidence for assuming that the preterit in the *after*-clause of (198) (*John arrived after the bomb exploded*) is a relative tense form. On the other hand, it does seem reasonable to conclude that the *after*-clause uses a relative past tense in (201,a–b), which are the indirect speech versions of (199,a) and (200,a):

(201) (a) Mary said that John would arrived after the bomb *exploded.* (*would explode)
(b) The solicitor told John that he would inherit the money only after his uncle *was* dead. (*would be)

Since a truthful report respects the temporal relations expressed in the original utterance, it seems logical to assume that what is a relative tense form in direct speech is also a relative tense form in indirect speech, the only difference being that the tense form in question is adapted to the past domain in which the situation is now located. (See also chapter 4.)

Further evidence for the hypothesis that a preterit in an *after*-clause is (or, at least, can be) a relative tense form is to be derived from sentences like the following:

(202) Bill imagined that John would be fired. He also imagined that he too would be dismissed after John *was* fired.

As noted above, the verb *imagine* creates an intensional domain, and an intensional domain functions as a temporal domain. Since the *after*-clause is interpreted opaquely, it must therefore form part of the domain established by the head clause.

Up to now I have adduced several pieces of evidence that make clear that *after*-clauses can express a temporal relation with an implicit TO. No evidence has been advanced yet that suggests that *after*-clauses can also shift the domain (at least in the past time-sphere). In fact, as noted above, it appears difficult to find conclusive evidence confirming the existence of this possibility. Still, there are a couple of observations that are suggestive. One is that in Dutch the present perfect can be used in an *after*-clause depending on a head clause in the present perfect:

(203) Jan is vertrokken nadat Piet is toegekomen, niet voordien.
('John has left after Peter has arrived, not before')
(= John left after Peter arrived, not before)

As pointed out in footnote 115, Dutch can use the present perfect instead of an absolute preterit, but not normally instead of a relative past tense

form. Given the strong parallelism between the use of tenses in English *after*-clauses and that in Dutch *nadat*-clauses, this observation can be taken as an indication that *exploded* in (198) (*John arrived after the bomb exploded*) could be an absolute preterit form. A second observation corroborating this conclusion is the parallelism between (198) and *The bomb exploded before John arrived.* As noted above, there is strong evidence that both preterits of the latter sentence can be analysed as absolute tense forms. The construction is used if the speaker treats both situations as factual and merely wants to state the temporal order in which they occurred. In my opinion, if a speaker can do this by means of a sentence of the form 'X before Y', it must also be possible for him to do it by means of a sentence of the form 'Y after X'. That is, it is most unlikely that *after*-clauses should be incapable of shifting the domain if *before*-clauses are capable of doing so.

In sum, there are various pieces of evidence that an *after*-clause can express a domain-internal temporal relation with an implicit TO. No conclusive evidence has been found that a preterit in an *after*-clause can shift the domain, but it seems unlikely that this possibility should not be available, and there is certainly no evidence to the contrary.

26 CONCLUSION

26.1 In this chapter I have developed the apparatus that is necessary for interpreting the many observations that offer themselves in connection with tense and for making the correct generalizations on the basis of these observations. I have shown that all observations can be categorized if we make the following assumptions:

1 English treats time as divided into two time-spheres, a present time-sphere including t_0 and a past time-sphere excluding t_0.
2 t_0 is usually the time of utterance; exceptionally it is the time of decodification.
3 t_0 divides the present time-sphere into three sectors, so that there are four absolute sectors in all.
4 English has four absolute tenses, one for each of the four absolute sectors.
5 The four absolute tenses create domains, which can be expanded by the incorporation (temporal subordination) of other situations.
6 The past time-sphere has its own set of relative tenses to effect temporal subordination.
7 In the present time-sphere, temporal subordination often involves a shift of temporal perspective to another sector.
8 In some cases tense forms expressing simultaneity are used in a 'sloppy' way.

9 Temporal subordination may involve either direct or indirect binding.
10 Not every new clause is temporally subordinated. In some cases the speaker shifts the domain, or establishes a simultaneous domain, or re-establishes the existing domain.
11 Some tenses can be used in a special way. There are numerous examples, which can be analysed either as a shift of temporal perspective or as a result of a shift of temporal focus.
12 An adverbial time clause usually relates its situation to an implicit TO rather than to the situation-TO of its head clause.

26.2 The complexity of the English tense system is partly due to the fact that the speaker can often choose between two possibilities.[124] For example:

1 Choice of t_0: the speaker can select either the time of utterance or the time of decodification.
2 Choice of time-sphere: e.g. a situation that took place before t_0 can be represented as lying in the present time-sphere (where it is located in the pre-present sector) (e.g. *He has done it*) or as lying in the past time-sphere (e.g. *He did it*).
3 Choice of sector: e.g. a situation that has started before t_0 and is still holding at t_0 can often be located either in the pre-present sector (e.g. *I have been waiting for you*) or in the present sector (e.g. *I am waiting for you*).
4 Choice of temporal domain: e.g. when two situations belong to the past time-sphere, the speaker may represent them as lying in the same domain (e.g. *He left and would never come back*) or as lying in two different domains (e.g. *He left and never came back*).
5 Choice of temporal relation: e.g. when two situations follow each other in time, the first may sometimes be represented either as anterior to the other (e.g. *When the bell had rung we left the room*) or as (sloppily) simultaneous with it (e.g. *When the bell rang we left the room*).
6 Choice of temporal perspective: e.g. the speaker may or may not use the historical present in clauses that form part of a past narrative.
7 Choice of temporal focus: e.g. the speaker may use either *He went to Florence, which lay in a wide valley* or *He went to Florence, which lies in a wide valley*, depending on whether or not he wishes to represent

124. It should have been clear throughout this chapter that the use of the tenses has less to do with the temporal location of a situation in real time than with the subjective way in which the temporal location of the situation is viewed and expressed by the speaker. (see Lloyd (1984:139): 'Tenses reflect sequences of events as experienced by speakers, and may not reflect real-time relations. Since they depend on subjective estimates of experiential time, tenses may be used for other expressive purposes.')

the situation as it was experienced by the relevant participant in the situation (and hence at the time of the situation).

8 Choice between the Present Perspective System and the Future Perspective System: e.g. the speaker can say either *The newspapers will publish everything that is said tomorrow* or *The newspapers will publish everything that will be said tomorrow.*

9 Choice between direct and indirect binding: e.g. the speaker may say either *I had been reading while I was waiting* or (though less usually) *I had been reading while I had been waiting.*

26.3 If our description of the system of tenses is correct, we can formulate the following claims in connection with the relation between tense and time:

1 Those tenses that are used as absolute tenses are basically used to express the temporal location of the situation relative to t_0. As far as these tenses are concerned, there is therefore a direct relation between tense and time (as it is viewed by language users). English speakers divide time into two time-spheres and four absolute sectors, and there is an absolute tense corresponding with each of these sectors.

2 Apart from this basic use, the absolute tenses are also used to represent a situation as if it belonged to a particular sector. In this use (in which there is a shift of temporal perspective to a sector that is different from the one to which the situation actually belongs) there is only a metaphorical relation between tense and time.

3 Relative tenses express temporal relations, not absolute location in time.

26.4 It goes without saying that there are all kinds of restrictions on the possibilities and choices mentioned in sections 26.1 and 26.2. The most important of these will be investigated in chapter 3.

26.5 Since this chapter is little more than a rough sketch of the English tense system, it is worth testing the apparatus which we have developed by applying it to some specific areas of English grammar. This will be done in chapter 4, which will be devoted to a scrutiny of the tense system in indirect speech and in conditional clauses.

3 Tense in discourse 2: restrictions and possibilities

It is clear from chapter 2 that the complexity of the way the English tense system is used is due not only to the richness of the tense system itself (i.e. the number of available tenses) but also to the fact that the speaker often has the freedom of choice between several possibilities. Different options are available as regards the t_0, the time-sphere, the absolute sector, the temporal domain, the domain-internal temporal relation, the temporal perspective, the temporal focus, the choice of binding TO, and the system to locate a situation in a posterior sector. However, this freedom of choice may be severely restricted by the context, by the kind of clause in which the verb form is to be used, by the nature of the situation referred to or by the kind of verb that is used. In this chapter we will investigate some of these possibilities and restrictions.

1 THE CHOICE AND INTERPRETATION OF THE TEMPORAL DOMAIN

1.1 Unembedded clauses

When two situations are located within the same sector by means of unembedded clauses, these situations may or may not be represented as belonging to the same temporal domain. As we have seen, sentences like the following illustrate these two possibilities:

(1) (a) He left and *would* never *come* back.
(b) He left and never *came* back.

In (1,a) the situation in the second clause is incorporated into the domain established by the first clause: the second situation is represented as posterior to the first. In (1,b) we notice a shift of temporal domain: the second clause establishes a new domain. This means that there is no longer any temporal relation expressed between the second situation and the first.[1] If we still interpret the second situation as holding after the

1. See Adelaar and Lo Cascio (1986:260): 'the reciprocal order between states of affairs of

first, this is only because of our pragmatic knowledge of the world: we know that one can only come back after one has left. In some examples similar to (1,b) the pragmatically inferred temporal relation may be different. The most likely interpretation of *Mary was tall and John was small* is not in terms of posteriority (sequence) but rather in terms of simultaneity.

As we have seen, shifting the domain is only one of three possibilities when a domain is established. The other two possibilities are that of establishing a simultaneous domain (as in *John, who **will be** ill, will be in bed*) and that of re-establishing the domain (as in *John will be sad because Mary **will have left***). In the former case the tense is an absolute tense, in the latter it is an absolute–relative one.

1.1.1 Introduction

1.1.1.1 We have observed that a shift of temporal domain within the same sector is a kind of tense simplification (Couper-Kuhlen (1987:25) speaks of 'neutralization'). Instead of using a complex relative tense expressing a domain-internal relation (e.g. the conditional tense, or the past perfect) the speaker uses a simpler absolute tense, which creates a new domain. It goes without saying that this will normally be done only if the pragmatics of the sentence and its context are such that no confusion can arise concerning the location of the situations relative to each other or if these relations are deemed irrelevant (see Grice's (1975) Maxims of Quantity and Relation (Relevance)). It is therefore not surprising that in sentences where the different clauses establish different domains within the same absolute sector the precise temporal relation is very often indicated by a time adverbial.

1.1.1.2 An interesting question is whether a clause with a progressive verb form can be interpreted as establishing a domain. At first sight this might seem unlikely, because a progressive form refers to the middle part of a situation (see Comrie 1976:24) and therefore represents the situation as in progress at some TO. One could feel tempted to say that the TO in question, which must be identified in the context, is the time of the situation referred to in the preceding clause.[2] If that is correct, the

a text is not given by deictic [i.e. absolute] tenses. . . . A deictic tense does not express any temporal relation relative to the time of the preceding state of affairs of the same text. . . . The movement of the action in a narrative text is given by connective temporal adverbs or is stated with the help of our knowledge of the world.'

2. See König (1980:289): 'in contrast to the simple form of the verb, the progressive form does not introduce a temporal context but depends for its interpretation on a temporal context independently established'.

progressive verb form represents the situation as simultaneous with (i.e. as being in progress at) a TO in an already established domain, thus incorporating the situation into the domain in question.

However, one should not jump to this conclusion. Examples can be found where the TO with which the progressive situation is simultaneous is not given in the preceding context but is established in the progressive clause itself by means of an adverbial. For example:

(2) (a) They left the village at 8 o'clock. An hour later they *were walking* through the forest.
(b) A moment later, like an evil spirit put to flight, I *was stumbling* away down the stairs. (Bronzwaer 1970:69)

In these examples the progressive preterit does not represent its situation as simultaneous with the situation of the first clause. It must therefore be analysed as shifting the domain.

Once this possibility is recognized, there is no *a priori* reason to reject the view that any progressive preterit can establish its own domain. Even when the progressive situation is interpreted as simultaneous with the situation of the preceding clause, as in (3,a–b), we do not have to analyse the preterit as a relative tense: we can also analyse it as establishing a simultaneous domain.

(3) (a) Bill was in the study. He *was working.*
(b) The boys were singing and the girls *were dancing.*

In (3,b) the two situations are interpreted as simultaneous with (i.e. as being in progress at) the same TO, and hence as simultaneous with each other. (This interpretation is not obligatory in *The boys sang and the girls danced.*) However, in terms of our theory this need not mean that the progressive forms are relative tense forms. We can also say that the second clause establishes a domain that is simultaneous with the previous one. This analysis is not implausible, given the fact that the first clause of (3,b), which definitely establishes a domain, is also progressive. Moreover, when the reference is to the post-present, we see that the Future Perspective System must be used:

(4) (a) John will arrive next Sunday at 4.30. Bill *will be waiting* for him on the platform. (!is waiting).
(b) The boys will be singing and the girls *will be dancing.*

In examples like these the future tense in the second clause establishes a domain, but this domain is naturally interpreted as simultaneous with that of the first clause, unless there is an adverbial indicating otherwise, as in

(5) John will arrive at 4.30. An hour later he *will be sitting* at our table.

Further support for the view that the progressive in (3,a) establishes its own domain is offered by the indirect speech test which was referred to in section 25 of chapter 2: (3,a) will normally be reported as *The maid said that John had been in the study and that he had been working.* As noted above, only absolute preterits are normally backshifted in indirect speech.

In sum, there is good evidence that a progressive form in an unembedded clause creates its own domain rather than being a relative tense form.

1.1.2 The temporal interpretation of newly established domains

When there is a sequence of unembedded clauses, it sometimes happens that each clause establishes its own domain and that the temporal relation between the relevant situations is neither indicated by an adverbial nor recoverable from contextual information or pragmatic knowledge. In that case there is a conversational implicature regulating the temporal interpretation of the situations in question. It stipulates that the sequence will be interpreted in accordance with the following principle (see Dowty 1986; Hinrichs 1986):

(6) *Principle of unmarked temporal interpretation:*
In a sequence of unembedded clauses where each clause establishes its own domain (within the same absolute sector) and where there is no adverbial, contextual or pragmatic indication of temporal order, the unmarked temporal interpretation is as follows:
(a) when the situations are represented as bounded, they are interpreted as following each other in the order in which they are reported;
(b) when the situations are represented as unbounded, they are interpreted as simultaneous with each other;
(c) when one situation is represented as bounded and the other as unbounded, the bounded situation is interpreted as temporally included in the unbounded one.

The difference between 'bounded' and 'unbounded' situations is well known in the linguistic literature.[3] A situation is bounded if it is represented as reaching a terminal point, otherwise it is unbounded. Thus, (7,a–c) are bounded sentences (i.e. sentences that represent a

3. The term 'bounded' is due to Bloomfield (1933:205).

situation as bounded), whereas (8,a–f) are unbounded (i.e. do not represent a situation as bounded)[4]:

(7) (a) John drank five glasses of whisky before dinner.
(b) Betty made a drawing on the floor.
(c) The athlete ran 10 kilometres.

(8) (a) John was drinking whisky (when I saw him).
(b) Betty was making a drawing on the floor.
(c) The athlete ran/was running.
(d) The athlete was running the 10 kilometres.
(e) John was in the kitchen.
(f) Betty knew about our affair.

Sentences (7,a–c) are bounded because they each represent a situation as involving a terminal point which has to be reached if the situation is to be what it is claimed to be. If John drank fewer than five glasses of whisky, (7,a) is not a true representation of the situation. Similarly, if the athlete did not cover the full distance of 10 kilometres, (7,c) does not count as a truthful account, though the unbounded sentences (8,c–d) do.

The bounded/unbounded distinction cuts across the classification proposed by Vendler (1967:99–107), which looks as follows: (1) situations are 'accomplishments' if they are 'processes going on in time' which 'proceed toward a terminus which is logically necessary to their being what they are' (e.g. running a mile, drawing a circle); (2) processes going on in time in a homogeneous way, i.e. without proceeding towards such a necessary terminus are 'activities' (e.g. pushing a cart, running); (3) situations that 'occur at a single moment' (e.g. reaching the top, dying, spotting a plane) are 'achievements'; (4) durative situations that are not 'processes going on in time', i.e. do not 'consist of successive phases following one another in time', are 'states' (e.g. knowing something, loving someone); states do not happen, they just exist, without change, in

4. As we will see, boundedness and unboundedness are basically features of situations, not sentences. However, situations are not inherently bounded or unbounded: they are represented as (un)bounded by the sentences that describe them. Thus, both *John drank whisky* and *John drank six glasses of whisky* can refer to exactly the same situation, but in the former case the situation is (represented as) unbounded, whereas in the latter it is not. The conclusion must therefore be that boundedness and unboundedness are characteristics of situations assigned to them by sentences (i.e. linguistic representations of situations). Since both elements (the situation represented and the sentence representing it) are essential to the concept, we may adopt the convention of speaking of '(un)bounded situation' as well as of '(un)bounded sentence'. One should keep in mind, however, that these terms are really abbreviations of 'situation represented as (un)bounded by a sentence' and 'sentence representing a situation as (un)bounded', respectively.

The fact that we apply the notion (un)bounded to sentences and not to other linguistic expressions (such as verbs or verb phrases) is due to the fact that virtually any constituent of a sentence can in principle make clear that the situation is viewed as (un)bounded. (For a detailed study of the factors that render a sentence (un)bounded, see Declerck (1979b).)

time. These definitions, and the additional tests suggested by Vendler, make clear that states and activities are unbounded, whereas accomplishments and achievements are bounded.[5]

The distinction bounded vs. unbounded is also referred to as 'perfective' vs. 'imperfective' (Comrie 1981b:66), 'conclusive' vs. 'nonconclusive' (e.g. Quirk *et al.* 1985:201) and by various other names. It is also often identified with Garey's (1957) distinction between 'telic' and 'atelic' sentences – see e.g. Allen (1966:198), Andersson (1972:48), Dahl (1981), Declerck (1979b), Brinton (1988). However, it may be argued that there is a difference of meaning between bounded/unbounded and telic/atelic. The former distinction hinges on the criterion of 'reaching a terminal point', the latter on the criterion of having a 'natural end-point'. That is, a telic expression involves reference to a non-arbitrary point of completion towards which the action tends to proceed and beyond which it cannot continue. There are several pieces of evidence that the set of bounded expressions and the set of telic expressions need not coincide. To begin with, achievements are by their very nature bounded, but they may be atelic. (Couper-Kuhlen (1987:16) argues that achievements are always atelic. However, there appear to be two types of achievements: those that only realize a 'punctual nucleus' (e.g. *knock*, *flash*) and those that only realize a 'culmination point' (e.g. *reach*, *discover*) – see Mommer (1986:75). The former are atelic, the latter are telic.) Secondly, accomplishments are both bounded and telic when the verb is nonprogressive (e.g. *John ran a mile*), but not when the verb is progressive: sentences like *John is running a mile* are clearly unbounded (since progressive sentences by definition do not represent a situation as terminating – see below), but they are still telic, i.e. they still represent the situation as tending towards a non-arbitrary point of completion. Thirdly, activities and states are by definition atelic, but they may be bounded, e.g. by the use of a duration adverbial like *for three hours* or by the temporal relations expressed in the sentence (as in *John went home after he* ***had pushed*** *the cart* or *Will you wipe your feet on the mat when you* ***have been*** *in the garden?*). For all these reasons I will not equate (un)boundedness with (a)telicity.

The following are some well-known tests that can help us to make the bounded/unbounded distinction:

1 A sentence whose verb has progressive meaning is always unbounded. This is inherent in the meaning of progressive aspect, since progressive aspect means that the speaker just refers to the middle of a situation, and

5. According to Schopf (1984:236–8), both accomplishments and achievements involve three stages: an initial phase, a process phase and a final phase. A situation is represented as an accomplishment when the three phases are referred to as a single whole. It is represented as an achievement when only the final phase is asserted, whereas the other two are presupposed. In both cases the assertion of the final stage (terminal point) renders the situation bounded.

disregards its beginning and (possible) end (see Comrie 1976:3; Smith 1986:100). A progressive sentence thus by definition fails to represent a situation as terminating (bounded).[6]

It follows that a bounded sentence becomes unbounded when it is put in the progressive form,[7] and that, in consequence, the truth of such a progressive sentence does not entail the truth of its nonprogressive (bounded) counterpart. From the truth of *John was writing a book* we cannot infer the truth of *John wrote a book.* On the other hand, this kind of inference is possible if the nonprogressive sentence is unbounded: it does follow from *John was drinking tea* that *John drank tea,* since both sentences represent the situation as unbounded.

2 Unbounded sentences are typically compatible with 'non-inclusive' duration adverbials (i.e. adverbials which answer the question *For how long?*), while bounded sentences are typically compatible with 'inclusive' duration adverbials – Gabbay and Moravcsik (1980:78) speak of 'containers' – i.e. adverbials which answer the question *Within what time?*:

(9) (a) John was running for hours. (unbounded)
    (b) *John was running in an hour.
(10) (a) John ran a mile in an hour. (bounded)
    (b) John ran a mile for hours. (ungrammatical on a single-event reading; grammatical on the repetitive interpretation 'John spent hours repeatedly running a mile' – a reading which is again unbounded)

3 Unbounded situations resemble stuffs (indicated by mass nouns) in that they are 'homogeneous' (i.e. they are 'the same all the way through' (Galton 1984:154)) and consequently 'dissective' (i.e. an unbounded situation can be broken down into smaller subsituations of the same

6. There are many different opinions as to what constitutes the core meaning of the progressive form, but most linguists agree that the idea of 'incompleteness' is one of the basic features of progressive aspect. (See especially Hill 1958; Allen 1966.) A progressive verb form refers to an internal portion of the situation, without making explicit reference to its beginning or end. A nonprogressive verb form, in contrast, represents a situation as 'complete', i.e. it 'presents the totality of the situation referred to . . . without reference to its internal temporal constituency: the whole of the situation is presented as a single unanalysable whole, with beginning, middle, and end rolled into one; no attempt is made to divide this situation up into the various individual phases that make up the action' (Comrie 1976:3). (Note that the idea of '(in)completeness' as defined here has nothing to do with the question whether the situation is finished or not in real time; all that matters is how the speaker conceptualizes the situation.)
7. Translating this into Vendler's (1967) terminology, we might say that the use of the progressive turns accomplishments and achievements into activities. Several linguists explicitly state that they accept this formulation. Others, however, are reluctant to do so because in their opinion the most important semantic feature of an accomplishment is that it is telic, i.e. that it is a situation which 'proceeds towards a climax, or natural terminal point' (Lyons 1977:711). As noted above, this feature is not absent in the progressive.

nature). This means that, if we omit the duration adverbial (if any) from a sentence representing a situation as unbounded, the same sentence can also be used to refer to smaller parts of that situation.[8] For example, if John ran in the woods from two o'clock to four, then unbounded sentences like *John ran* or *John was running* are true representations not only of what John did between two and four, but also of what he did between two and three, and of what he did between two-thirty and a quarter to three, etc. The converse is also true: if we sum up several unbounded situations of that kind, the result is a longer unbounded situation of the same kind. Thus, unbounded situations are also 'additive'.

Bounded situations, by contrast, are 'heterogeneous', 'unitary' and 'non-additive'. Like objects indicated by count nouns, they are not 'the same all the way through' and they cannot be broken down into smaller entities that can be indicated with the same name (description). That is, a sentence that is used to represent a situation as bounded cannot also be used to refer to a section of that situation. In the situation outlined above we can say *John ran 10 kilometres* (*between two o'clock and four*) but the same sentence (*John ran 10 kilometres*) cannot be used as a true representation of any subpart of the overall situation. Moreover, if we sum up several past situations of John running 10 kilometres, the sum-total is not itself a situation which we can refer to by means of the sentence *John ran 10 kilometres.*

Now that the terms 'bounded' and 'unbounded' have been defined, we can return to the principle of unmarked temporal interpretation given in (6). However, before having a closer look at the claims made in (6), it is worth stressing the following point in order to rule out any misunderstanding. Principle (6) is claimed to hold only for sequences of clauses that satisfy the following two requirements: (1) each clause in the sequence establishes its own domain within one and the same absolute

8. Vendler (1967:101) makes too strong a claim in connection with the observation that 'activities' 'go on in time in a homogeneous way': according to him, homogeneity means that 'any part of the process is of the same nature as the whole'. In my opinion, this is an overstatement, because it does not necessarily follow from *That afternoon John was running* that John was actually running at every moment of the afternoon in question. The above sentence remains true if there were occasional intervals at which John was not actually running. Moreover, Taylor (1977) has pointed out that even if we consider uninterrupted situations only, it is not always the case that a sentence that represents a situation as unbounded can also refer to minimal subparts of this. Thus, the unbounded situation referred to in *John chuckled for a couple of seconds* is not homogeneous in Vendler's sense, 'since any sounds emitted in a microsecond during a period of chuckling (at the normal rate) hardly constitute chuckling themselves' but fall 'within some period of chuckling though themselves occupying a time too short to constitute such a period' (Taylor 1977:212). However, even durative situations like chuckling, which are heterogeneous in this (micro-)sense, are homogeneous in the (macro-)sense of our definition: if *John was chuckling* is true of a certain time interval, then there are at least some subintervals of which the sentence is also true. Bounded situations, by contrast, are heterogeneous in both senses of the word.

sector, and (2) the temporal order of the situations is neither indicated by an adverbial nor recoverable from contextual information or pragmatic knowledge. In order to ensure that we are dealing with domain-establishing clauses, I will illustrate each claim not only with examples in the past tense but also with examples that refer to the future. (As we have seen, the difference between absolute and relative tense forms may not always be clear when we refer to the past, since a preterit can be either a relative or an absolute tense form. The difference is, however, clear when the reference is to the future, because absolute and relative tense forms there belong to the Future Perspective System and the Present Perspective System, respectively. This means that we can conclude that the past tense clauses shift the domain if the corresponding clauses referring to the post-present use the future tense.)

Principle (6) involves the following claims:

1 The unmarked interpretation of a sequence of bounded clauses (each of which establishes its own domain) is in terms of what we may call 'iconic sequencing': the situations are interpreted as following each other in the order in which they are reported.[9] For example:

(11) (a) John went to the door and knocked three times. The door opened slowly.
(b) I will go to the door and (I will) knock three times. The door will open slowly.

In sentences like these the situations will be interpreted as following each other directly, unless such an interpretation is pragmatically impossible or unlikely.[10] This is the case in examples like

(12) (a) He moved to London and found a job there as a teacher. He courted a girl called Mary and married her.
(b) Perhaps he will move to London and find a job there as a teacher. And perhaps he will court a girl and marry her.

In sentences like these, the situations need not follow each other directly. There may be long intervals of time between them.

The tendency to interpret a sequence of bounded clauses in terms of succession has often been noted in the linguistic literature.[11] Still, it is a

9. As noted by Dowty (1986:38), some 'lexical stative verbs' (e.g. *stand*, *sit*, *realize*) often yield an 'inceptive' interpretation. In that case the sentence is interpreted as an achievement, i.e. as bounded. (It follows that its situation is not a background situation but one that moves the action forward.) For example:
(i) John entered the president's office. The president *realized* why he had come. (Dowty 1986:38)
10. In other words, the fact that the immediate succession reading is favoured is due to a conversational implicature, which can be cancelled. (I do not agree with Partee's (1984) claim that such a reading is obligatory when the two clauses express 'events'.)
11. To mention only a few examples: Ryle (1954:118) already noted that *She took arsenic and fell ill* does not have the same meaning as *She fell ill and took arsenic.* Labov's

tendency, not a strict rule. When the clauses have different subjects and there is no contextual or pragmatic reason for adopting the succession interpretation, any of the three temporal readings is in principle available. A sentence like *Bill studied chemistry and John studied mathematics* tells us nothing about the temporal order in which these two situations occurred.[12] The sentence is therefore true in each of the following cases: (1) Bill studied before John did; (2) John studied before Bill did; (3) they studied at (approximately) the same time.[13]

We may wonder, then, why it is that a sequence of bounded clauses is sometimes interpreted in terms of succession, as in (13,a–b), and sometimes allows any temporal interpretation, as in (14,a–b):

(13) (a) The man *went* to the door, *opened* it and *shouted* something.
(b) The man *will go* to the door, (he *will*) *open* it and (he *will*) *shout* something.

(14) (a) Did any of them marry? – Yes, Bill *married* Tina, and John *married* some Italian girl.
(b) Will any of them marry? – Yes, Bill *will marry* Tina, and John *will marry* some Italian girl.

There appear to be two factors that are conducive to a succession interpretation. The first is pragmatic knowledge. We know that going to the door must take place prior to opening it. The second is the kind of text or discourse that the sequence of bounded clauses is part of. If this is a story that progresses in time, then each bounded clause that is added will normally be interpreted as carrying the action forward. That is, each new situation will be interpreted as following the situation of the

(1972) rule of 'temporal juncture' refers to the same phenomenon. Partee's (1984:254) claim that 'the introduction of new event sentences moves the reference time forward' will, in our theory, be interpreted as meaning that, when a new bounded clause is produced and its situation-TO establishes a new domain, this domain is interpreted as following the previously established domain. (However, Partee's formulation is much too strong, since (1) not all event clauses are bounded, (2) not all bounded clauses establish their own domain (they may also be temporally subordinated), (3) not all domain-establishing bounded clauses are interpreted in terms of succession (in some cases the domain is interpreted as simultaneous with the existing unbounded domain – see below), and (4) the succession reading is anyhow only the unmarked interpretation, and may be overruled by contextual or pragmatic factors – see below).

The claim that the unmarked way of reporting bounded situations is to mention them chronologically is also in keeping with the so-called 'order of mention' principle which Clark and Clark (1977:358) have formulated on the basis of psychological experiments. (The experiments in question showed that situations that are reported in complex sentences are more accurately recalled if the sentences preserve the temporal order of the situations.)

12. See Richards (1982:86), who observes that the sentence *Sam changed the tyre and Bill filled the tank* is 'ambiguous' between the interpretations 'at the same time' and 'at different times'.
13. The two situations may partly overlap. (Vet and Molendijk (1986:143) note this in connection with the sequence *Jean went to his room. Pierre followed him: 'e_2 starts somewhere in the middle of e_1; at any rate after the beginning and before the end of e_1.'*)

previous clause.[14] Sentences (13,a–b) appear to be part of such a story with a linear temporal structure. The answers in (14,a–b) are not, and therefore need not be interpreted in terms of succession.

2 The second claim made in principle (6) is that in a sequence of (unembedded) unbounded clauses the unmarked interpretation is for the situations to be simultaneous.[15]

(15) (a) John stood by the window. Mary was in the kitchen.
(b) The man seemed reliable. He wore expensive clothes.
(c) John carried the suitcase. Bill pushed the cart.
(d) He enjoyed and admired the sonnets of Petrarch. (Leech 1971:9)

(16) (a) John will stand by the window. Mary will be in the kitchen.
(b) (In ten years' time) the man will seem reliable. He will wear expensive clothes.
(c) John will carry the suitcase. Bill will push the cart.
(d) (When he is older) he will enjoy and admire the sonnets of Petrarch.

(It was noted in chapter 2 that in cases like these we do not really have a shift of domain but the establishment of a simultaneous domain.)[16]

14. Blakemore (1987:114–21) argues that this follows from Grice's Principle of Relation (Relevance).
15. Remember that we are dealing with unembedded domain- establishing clauses that do not contain an adverbial indicating the temporal order of the situations.
16. At first sight one may wonder whether the preterits in the second clauses of (15,a–d) are not relative past tense forms expressing simultaneity rather than absolute tense forms establishing (simultaneous) domains. However, there are two pieces of evidence confirming that this is not the case. First, consider the post-present counterparts of (15,a–d), viz. (16,a–d). Each of these shows that the second clause requires an absolute tense form if it is to refer to the future. Using a relative tense form expressing simultaneity (i.e. a present tense) is not possible. Since (15,a–d) are quite similar to (16,a–d), this suggests that the past tense forms in the second clauses of (15,a–d) are absolute preterits. The second piece of evidence confirming this conclusion concerns the behaviour of the relevant tense forms in indirect speech. As noted above, an absolute past tense may be 'backshifted' in indirect speech, but a relative preterit is normally retained. Consider:
(i) (a) I was very unhappy when I stayed in Paris.
(b) Bill said he had been very unhappy when he stayed in Paris.
Whereas in (i,a) the situation of being unhappy is located in the past of t_0 (so that the preterit is used), it is represented as anterior to the reporting time (*said*) in (i,b) (hence the 'backshift' to the past perfect). The tense of the *when*-clause, by contrast, remains the same in (i,b) because it expresses the same relation as in (i,a): both in (i,a) and in (i,b) *stayed* is a relative past tense expressing simultaneity in a past domain. The behaviour of past tenses in reported speech thus presents us with a powerful test to distinguish between absolute and relative preterits: only absolute preterits can be 'backshifted' (which, however, does not mean that they always have to be – see chapter 4). When we apply this test to (15,a–d), we see that it confirms the view that the past tenses are absolute tenses, since all of them are naturally 'backshifted':
(ii) (a) The witness explained that John had stood by the window and that Mary had been in the kitchen.
(b) Bill said that the man had seemed reliable, as he had worn expensive clothes.

As we have seen, unbounded situations are represented as homogeneous and without reference to an initial or terminal point. For this reason, unbounded situations may in principle go on indefinitely. This explains why the two unbounded situations referred to in each of the examples in (15) and (16) are not normally interpreted as preceding or following each other. There is simply no reason why one of them should be interpreted as beginning only after, or as terminating before, the other situation. This is true even if the clauses form part of a story. Unbounded clauses (not containing a time adverbial) do not normally carry the action forward. They have a 'backgrounding' rather than a 'foregrounding' function.[17]

Progressive sentences form an exception to this rule. They are by definition unbounded, but they have a foregrounding function when the TO at which the situation is represented as in progress is later than the situation-TO of the preceding clause. This is the case, for example, when the progressive clause contains an adverbial like *soon* or *an hour later* which causes a shift of domain interpretation (as in *They left the house at five; Soon/an hour later they were walking through the forest* – see above).

At first sight it would seem that unbounded clauses that are interpreted inchoatively form a second exception. Such clauses naturally carry the action forward because they imply that the situation in question did not obtain earlier. This is the case, for example, in the second clauses of (17,a–b). (In the latter example the inceptive reading is induced by the presence of *suddenly* – see Dowty (1986:50)):[18]

(c) They said that John had carried the suitcase and that Bill had pushed the cart.

(d) Jenny told me that she had enjoyed and had admired the sonnets of Petrarch.

17. The terms 'foreground' and 'background' have received slightly different meanings in the linguistic literature (see Couper-Kuhlen 1989b:10). I will use them in accordance with the following definition, which seems to be the most widely accepted: 'The foreground of a narrative text is defined as linguistic material which charts the progress of a narrative through time, while the background is durative and descriptive material which serves to embellish and elaborate upon the foreground.' (Ehrlich 1987:363) In other words, foregrounded situations are those that move the narrative on in time.

    Our statement that unbounded sentences usually have a backgrounding function accords with the fact that the perfective/imperfective distinction has often been associated with the foreground and the background of narrative texts, respectively (see e.g. Weinrich 1964, Forsyth 1970:9, Labov 1972, Hopper 1979, 1982:16, Dry 1981:233–40, 1983:21, Aristar and Dry 1982:3, Givón 1984:288–9, Eisterhold 1986:126, Thompson 1987:443).

18. Dowty (1986:50) correctly points out that states that are interpreted inchoatively form an exception to Partee's (1984:262) rule that 'state-expressing sentences cannot by themselves move the action forward'. Hatav (1989:487) points out that there is a second exception, viz. when the duration of the state 'is restricted by overt linguistic marking, for example, adverbials such as 'for three hours''. (As we will see in chapter 6, an adverbial that specifies the precise duration of a situation renders this situation bounded. A sentence like *John was here for two hours* is thus bounded and can therefore have a foregrounding function.)

(17) (a) I switched off the light as I left. It *was* pitch-dark in the room.
(b) We were sitting in the dining-room. Suddenly there *was* a loud noise in the street.

However, states that are interpreted inchoatively are not quite as unbounded as other states: they are viewed as unbounded to the right but as bounded to the left. That is, they are conceptualized as beginning but not as ending. This distinguishes them from other states, which are unbounded on both sides. So we could argue that the claim that the default interpretation of unbounded clauses is in terms of simultaneity actually holds only for clauses that are fully unbounded. The nonsimultaneous interpretation of inchoative situations is then in accordance with the rule.

3 The third claim made in (6) is that when one of the two situations is bounded, whereas the other is unbounded, the unmarked interpretation is that the bounded situation is included in the unbounded one. (This is a special kind of simultaneity relation.)[19]

(18) (a) Mary was in the drawing-room. Suddenly Bill came in.
(b) John went out of the room. He was feeling very tired.
(19) (a) Mary will be in the drawing-room. Suddenly Bill will come in.
(b) John will go out of the room. He will be feeling very tired.

The reason why the unbounded clauses are here interpreted as simultaneous with the bounded ones is again that they represent backgrounded information.

Clauses that are interpreted inceptively again behave differently. In each of the following examples the relevant clause (which is bounded to the left but unbounded to the right) pushes the action forward because its situation is not interpreted as holding at the time of the situation referred to in the preceding clause:

(20) (a) There was a loud explosion. Suddenly all kinds of things *were flying* through the air.
(b) Away in front, the engine whistled. Trees, hills and road slid sideways and *were gone.* (Dry 1981:234)
(c) This time she was pushed out of the frightening fifth dimension with a sudden immediate jerk. There she *was,* herself again, standing with Calvin beside her. (ibid.)
(d) John asked where the children were. Mary *looked* anxiously out of the window. (ibid.)

19. This 'framing effect', as it is called by Jespersen (1931:180), is particularly clear when the unbounded situation is described in the progressive form. (As we have seen, progressive sentences are by definition unbounded.)

(e) The president began the interview in a coldly official manner, much as Mary had expected. But the next thing she *knew*, the president was offering her the ambassador post. (Dowty 1986:55)

(f) Jameson entered the room, shut the door carefully, and switched off the light. It *was* pitch dark around him, because the Venetian blinds were closed. (Hinrichs 1981:66)

It should be stressed that the above principles of temporal interpretation (summarized in (6)) stipulate only what is the 'normal' or 'unmarked' interpretation of sequences of unembedded clauses. As pointed out by Stegu (1983:322–8) and Molendijk (1984), the unmarked interpretation may easily be ruled out by pragmatic factors. This is clear from sentences like the following:

(21) (a) John was in the kitchen. Suddenly Bill came in.
(b) John was in the kitchen. Suddenly he went out into the yard.

In both cases we have a combination of an unbounded clause and a bounded one. But whereas (21,a) yields the unmarked inclusion interpretation, (21,b) can only be interpreted in terms of sequence. Similarly, in (20,b–d) and (20,f) it is because of pragmatic considerations that the simultaneity interpretation is ruled out in favour of the reading that the unbounded situation begins only after the bounded situation referred to in the preceding clause is completed. And in the following sequences (in which both sentences are bounded) the second sentence is interpreted as referring to the same action as the first, so that there is an interpretation in terms of simultaneity rather than in terms of temporal progression:

(22) (a) Yes, *I fired the gun. I shot the old devil through the head.* It was legitimate self-defence. I had no choice.
(b) Sheila had a party last Friday and Sam got drunk. (Partee 1984:245)[20]

Up to now we have identified three possible types of temporal relation in sequences consisting of two unembedded domain-establishing clauses (A and B): (1) Situation B follows situation A; (2) The two situations are simultaneous; (3) Situation B includes situation A, or vice versa. However, there are still other (marked) possibilities:

20. The most normal interpretation of this example is that the time of getting drunk is the time of the party. This and other observations induce Partee (1973) to conclude that tenses are anaphoric, i.e. that, like anaphoric pronouns, they have antecedents. My own analysis of (22,b) is that both sentences establish their own domains, and that it is due to an implicature that these two domains are interpreted as simultaneous (see section 18 of chapter 2). This analysis is quite incompatible with the view that the tense forms in (22,b) are anaphoric.

1 The situation of the second clause may be anterior to that of the first clause:

(23) (a) The grand-vizier died at the age of 88. He *was* a very wise man who *gave* much satisfaction. (Vet and Molendijk 1986:145)

(b) The grand-vizier will die at the age of 88. He *will have been* a very wise man who *will have given* much satisfaction.

(c) It was eventually arranged that for the present Babun should live in the house while he put up another house for himself and Punchi Menika. Silindu took no part in the discussion. After Karlinahami intervened, he *became* silent. (Nerbonne 1986:93)[21]

This kind of shift of domain is likely to occur only if the corresponding unmarked interpretation is pragmatically impossible or unlikely. For this reason it is relatively rare. To represent a situation as anterior to the situation of the previous clause we normally have to use a relative tense (as in *John could not take a taxi. He had gambled away all his money*) or add an adverbial signalling anteriority.

2 In sequences consisting of more than two unembedded clauses, we may find a mixture of the above three possibilities:

(24) (a) We made a long journey through Italy. We *visited* Rome and Venice, and afterwards we *went* to Naples.

(b) We will make a long journey through Italy. We *will visit* Rome and Venice, and afterwards we *will go* to Naples.

In each of these examples the two clauses in the second sentence establish their own domains (with *afterwards* indicating the temporal order between them), but both of these domains are interpreted as simultaneous with some part of the domain established in the first sentence. This is what typically happens when a clause referring to a situation consisting of various subsituations is followed by several clauses describing these subsituations.

3. In sequences of the above-mentioned type the order of the subsituations is often irrelevant:

(25) (a) In that year several of John's relatives had an accident. Betty *fell* down the stairs, Bill *drove* into a lorry, and Aunt Mildred *got* knocked over by a motorcycle.

(b) Next year several of John's relatives will have an accident. Betty *will fall* down the stairs, Bill *will drive* into a lorry, and Aunt Mildred *will get* knocked over by a motorcycle.

21. Nerbonne adds the comment: 'The passage clearly means that Silindu took no part in the discussion which *followed* Karlinahami's interruption and his becoming silent.'

Here the various domains established by the clauses following the introductory sentence are presented as an unordered set. This is also the case in examples like the following:

(26) (a) What did you do this morning? – Betty *did* the washing-up, I *fixed* the car, and John *planted* the tulips.
(b) What will you do this morning? – Betty *will do* the washing-up, I *will fix* the car, and John *will plant* the tulips.

In the above paragraphs we have identified all the kinds of relation that are logically possible between situations that are not bound to each other by the tense forms, i.e. which are expressed by absolute tense forms. In order to pick out the possibility that is intended by the speaker the hearer will be guided by (1) his pragmatic knowledge of the world, (2) the information provided by temporal adverbs (if any), and (3) the nature of the text or discourse containing the sequence, which may or may not be a story with a linear temporal structure.[22]

There is one final point to be made. The examples that have been adduced to illustrate the principle of unmarked temporal interpretation of independent domains have all been in the preterit or in the future tense. There have been no examples involving the present perfect. The reason is that pre-present domains are not normally interpreted as temporally related to each other. As we will see in chapter 7, it is typical of the present perfect that it is interpreted as relating the situation to t_0, not to any other TO. For this reason the situations referred to in sequences like the following are not normally interpreted as temporally ordered with respect to each other:[23]

(27) (a) I have visited Greece and I have been to Italy.
(b) John has opened the door. Bill has been in the kitchen.
(c) *I've read, I've listened to the radio, I've watched TV* – but I haven't enjoyed anything so much as just sitting and doing nothing. (Allen 1966:203)

22. The observation that, if there are no time adverbials, the hearer will often have to rely on pragmatic considerations poses a serious problem for any semantic theory that is based exclusively on the truth value of sentences. It is therefore not surprising that Rohrer (1986:88) considers a sentence like *The doctor has ascertained that Rummenigge has sprained his ankle* (in which there is a shift of domain) as a problem: 'If we evaluate (8) [the above example] with respect to the speech point, then the sentence would also be true in a situation where Rummenigge sprained his ankle after the doctor noticed that he sprained his ankle. An interpretation which is clearly absurd.'
23. Brinkman (1885:719) already noted that the present perfect does not relate the individual events reported to one another as sequential or simultaneous. Joos (1964:145) observes that the English perfect 'cannot be used for narration'. See also Duskovà (1974:70), Schopf (1987:218), Matthews (1987).

In such sequences the different clauses just express that there has been an occasion before t_0 when this particular situation held. The precise temporal location of these occasions is treated as irrelevant, and the hearer does not, therefore, assume the situations to be temporally related in a particular way. Apparently all that matters is that the situations have held at some time before t_0.[24] For lack of a temporal interpretation of the domains, the principle of unmarked temporal interpretation of independent domains (i.e. principle (6)) does not apply.[25]

1.2 The choice of temporal domain in subclauses

1.2.1 Restrictions

Like unembedded clauses, subclauses can establish a domain of their own. In some cases this procedure is quite normal, in others it is ruled out. The following appear to be the major principles.

1.2.1.1 Sometimes the shift of domain is triggered by the fact that the subclause involves an adverbial which relates the time of the situation to t_0 rather than to a TO in the current domain. Compare:

(28) (a) He will announce tomorrow that the next meeting *will take* place a week later.
(b) He will announce tomorrow that the next meeting *will take* place next week.

(29) (a) The butler reported yesterday that the duke *had vanished* a couple of days before.
(b) The butler reported yesterday that the duke *vanished* last week.

In the (a) sentences the subclause situation is incorporated in the domain established by the head clause. This accords with the fact that the adverbial (*a week later, a couple of days before*) computes the time of the situation from the central TO of the domain in question. In the (b) sentences, in contrast, the subclause establishes a new domain because the adverbial (*next week, last week*) again relates the time of the situation to t_0. In examples like these the use of the tenses is thus in keeping with

24. This is in keeping with Haegeman's (1989:310) observation that (i) 'strongly suggests that the one event causes the other', whereas (ii) 'could be taken as a mere list':
(i) He didn't finish his Ph.D. and he lost his job.
(ii) He hasn't finished his Ph.D. and he has lost his job.

25. However, occasional examples can be found where pragmatic and contextual factors impose a sequential interpretation:
(i) I have opened the door and looked inside.
In such sentences the two situations are felt to be part of a single occasion in the pre-present. (The fact that there is gapping (deletion of *have*) in (i), but not in (27,a–c), may have something to do with this.)

the temporal meaning of the adverbial. (However, this is not necessarily the case. Some people accept the use of an (absolute) past tense in combination with a (relative) time adverbial like *a couple of days before*.[26] And combinations of a relative tense with an adverbial that is an absolute deictic are quite common, e.g. *He said he would do it yesterday.*)

1.2.1.2 If the subclause does not contain an adverbial triggering a shift of domain, the shift generally remains possible, provided contextual information or pragmatic knowledge guides the hearer to the correct interpretation concerning the temporal relation between the domain in question and the current domain. Consider:

(30) (a) John was hurt because Bill (*had*) *hit* him.
(b) Though Mary *gave/had given* him the answer, John remained dissatisfied.
(c) John, who (*had*) *heard* the news from Mary, did not believe it.
(31) John will be hurt because Bill *has/will have hit* him.

In these sentences the subclauses may or may not establish their own domains (by using the preterits *hit*, *gave* and *heard* rather than the past perfect forms or by using the future perfect *will have hit* rather than the present perfect) because the only plausible interpretation anyhow seems to be that of anteriority. In other sentences, however, this interpretation does not enforce itself, so that a shift of domain cannot take place without causing either ambiguity or a breakdown of communication. Compare:

(32) (a) John was angry because Mary *had been* in London.
(b) John was angry because Mary *was* in London.
(33) (a) Though Mary *had lived* in the village she did not know the Smiths.
(b) Though Mary *lived* in the village she did not know the Smiths.

In the (b) examples the subclause situations will normally be interpreted as simultaneous with the head clause situations. To represent them as

26. The following example is from Costa (1972:44):
(i) Smith told the plumber that he *dropped* his car-keys down the plug-hole the day before.
According to Costa, the use of a 'shifted' adverbial form with an apparently unshifted verb form here means that the 'Sequence of Tenses' rule has applied, but that *had* has subsequently been deleted. In my opinion, however, different conclusions should be drawn from examples like (i). One is that the behaviour of tenses and that of adverbials need not always run parallel to each other. Another is that a shift of domain is only possible when there is some indication of how the situation referred to in the matrix and those referred to in the subclause are temporally related to each other. In (i) it is precisely the adverbial that serves an such an indication.

anterior we have to incorporate them in the head clause domains by using the past perfect, as in the (a) examples.

Adverbial time clauses also conform to this rule: they allow a shift of domain only if this does not blur the temporal relations. This means that the shift is in principle possible with conjunctions like *after* and *before*, which express a temporal relationship independently of the verb form used, but not with *when*.[27] Thus, there is no difference of temporal interpretation between (34,a) and (34,b), but (35,a) is not interpreted as synonymous with (35,b):

(34) (a) Bill arrived after John *left*.
(b) Bill arrived after John *had left*.
(35) (a) Bill arrived when John *left*.
(b) Bill arrived when John *had left*.

1.2.1.3 Progressive subclauses do not normally shift the domain (within the same sector), unless there is a context or adverbial indicating otherwise. Compare:

(36) (a) John went to the Hilton because Mary *was staying* there.
(b) John went to the Hilton because Mary *was staying* there the week before.

In (36,a) *was staying* binds the subclause situation to the head clause situation, thus yielding the unmarked simultaneity interpretation. In (36,b) this interpretation is excluded by the adverbial *the week before*. The past tense must therefore be analysed as shifting the domain.

Another case in which the progressive subclause does establish a domain of its own is when the tense form belongs to the Future Perspective System:

(37) (a) John will go to the Hilton tomorrow because Mary *will be staying* there.
(b) John will go to the Hilton tomorrow because Mary *will be staying* there next week.

As is clear from these examples, the progressive is again naturally interpreted in terms of simultaneity, except when there is a time adverbial indicating otherwise.

1.2.1.4 As noted in chapter 2, the possibility of shifting the domain depends on the kind of subclause. Some subclauses (e.g. concessive clauses, relative clauses, *because*-clauses, locative clauses, etc.) easily allow the shift because their semantics is such that they need not

27. However, there is a special use of *when*-clauses in which absolute tense forms are used. This use (for which I will use the label 'narrative *when*-clauses') will be discussed in section 2 below.

represent their situations as temporally dependent on the head clause situation. Other subclauses (e.g. conditional clauses, time clauses) allow the shift only in exceptional cases.[28] Even within the same type of subclause there may be possibilities and restrictions. In chapter 4 we will show this at great length for complement clauses and conditional clauses.

1.2.1.5 Temporal clauses are also subject to a special restriction. The restriction is that the time clause and its matrix must not establish domains in different absolute sectors. This is clear from the following:

(38) (a) *Today John has washed before he goes/will go to bed.
(b) *John will leave tomorrow after he arrived yesterday.[29]
(c) *He has left through the window before I am (now) opening the door.
(d) *Bill left before Susan comes.

One of the consequences of this restriction is that a head clause which establishes a present domain (i.e. refers to t_0) cannot be accompanied by a time clause referring to another absolute sector.[30]

(39) (a) *Alice is curtsying as soon as she has seen the Queen. (Smith 1975:83)
(b) ?I am (now) writing a letter before I go to bed.[31]
(c) *I am (now) washing after I (have) played football.

28. It is interesting to note the semantic effect that the choice of system has in clauses introduced by *so that*:
(i) You left the door open, so that the prisoner escaped.
(ii) You left the door open, so that the prisoner would escape.
In (i) the *so that*-clause shifts the domain and is interpreted as a result clause. In (ii), the *so that*-clause is temporally subordinated to the head clause and is interpreted as a subclause of purpose (goal). This semantic difference follows logically from the difference of temporal focus: in (ii), where the subclause situation is related (as posterior) to the matrix situation, the temporal focus resides with the past situation, which was performed in order to achieve a certain effect; in (i), where the subclause situation is directly related to t_0, the focus is with the speaker, who ascertains what was the effect (consequence) of the head clause situation.
29. Of course, it is possible to say *John will leave soon after he (has) arrived*, but this is no counterexample to the claim that we are making because the use of *(has) arrived* here results from a shift of temporal perspective (see chapter 2). That is, *(has) arrived* is a relative tense form expressing anteriority in the post-present domain, not an absolute tense form shifting the domain.
30. The restriction does not hold if the use of the present tense results from a shift of temporal perspective, e.g. when it is a historical present. Thus, a sentence like *When he has stopped beating me he puts me in jail* is impeccable as an instance of the historical present. The reason is that the time clause and the head clause are interpreted as belonging to the same temporal domain (and hence to the same sector).
31. Joost Buysschaert has pointed out to me that the sentence *I am writing this letter before I go to bed* seems all right if it forms a part of the letter itself. And according to Bernard Comrie the sentence *I'm just washing the dishes before we leave for the station* is fine. This indicates that the restriction is less strict when the time clause is a *before*-clause.

(d) *John is eating soup before Bill will be eating bread. (Janda 1976)

At first sight there would seem to be a couple of exceptions to this rule. The first concerns habitual sentences. The following are impeccable:

(40) (a) I usually wash after I have played football.
(b) The children go home as soon as the bell has rung.

However, habitual sentences such as these are no exception to the rule if we assume that they locate a temporally complex situation exclusively in the present sector. That is, (40,a) refers to a situation consisting of a number of subsituations, each of which has the structure 'x after y', but the overall situation is located exclusively in the present sector: (40,a) does not relate the habit as a whole to some pre-present situation; it just expresses that the habit holds at t_0. (I will return to the question of how to analyse habitual sentences in chapter 6.)[32]

The second seeming exception concerns clauses introduced by *now that*:

(41) (a) Now that John has left, Betty is enjoying herself.
(b) Now that Bill will not be here to help us tomorrow, we are doing our utmost to finish the work today.

However, these sentences turn out to be no exceptions either, because clauses introduced by *now that* differ from normal time clauses in that they do not locate a situation relative to another: unlike *after*, *before* and *while*, *now that* does not inherently indicate a particular temporal relation. (For this reason *now that* can be followed by any tense.) In fact, *now that* means 'As it is/was/etc. now the case that. . . .' and functions as a conjunction of reason rather than as a conjunction of time. (This explains the use of the Future Perspective System in (41,b), where a true time clause would require the Present Perspective System.)

The rule that a head clause situation that is located at t_0 cannot be temporally related to another absolute sector even holds if the temporal relation in question is expressed by an adverb or prepositional phrase. Sentences like the following are ungrammatical:

(42) (a) *I am now washing afterwards/after the match.
(b) *I am now working before eight o'clock.

Here again it would seem that exceptions to the rule are possible. Sentences like *I am now working before eight o'clock* may be acceptable

32. Smith (1976a) also observes that the present tense is incompatible with time indications that do not refer to the present, except in habitual sentences. Her explanation for the fact that habituals form an exception is that 'habituals do not specify a particular time at all' (p.13) – a claim which I cannot subscribe to (see chapter 6).

in special contexts. For example, suppose Bill has uttered the accusation that I never work before eight o'clock. A couple of days later I am already working when he comes in at 7.30. In that case I can say *Today I am working before eight o'clock.* However, the reason why this is possible is that I am no longer using the sentence to locate a situation (my working) relative to a given time (eight o'clock). Rather, I am using the sentence to assert that a particular situation (viz. the situation of my working at eight o'clock) is now holding. In other words, the element 'at eight o'clock' is now part of the structure of the situation itself. (This is also clear from the fact that *at eight o'clock* cannot be preposed.) In similar contexts sentences like (38,a) and (39,b) may also be impeccable. (Similarly, the sentence *John has washed before he goes to bed* may be judged grammatical on the reading 'the situation of John washing before going to bed has occurred at least once in the past'.)[33]

1.2.2 The identification of the temporal domain

If the subclause shifts the domain, its verb form does not indicate the precise temporal relation that holds between its situation and the head clause situation. In that case this temporal relation may be indicated by an adverbial (as in (36,b)) or may be clear from pragmatic factors (as in (43)).

(43) Slocum chose to follow the route that Magellan took. (Diver 1963:167)

33. There is still another kind of exception to the rule that the present tense is incompatible with an adverbial containing *after*. Osselton (1981:465) draws attention to the following example:

(i) After the 1974 Elections, electoral reform is gaining an unchallengeable place on the political agenda.

Osselton agrees with the suggestion of his readers that the reason for this use of *after* with the present tense (instead of *since* with the present perfect) is that 'there is a non-temporal element here – not only *post hoc* but *propter hoc*: as a result of the 1974 election there is an impetus towards electoral reform. The *since* construction followed by the expanded perfect was perhaps thought to be too exclusively temporal in import.' That this explanation is probably correct also appears from the fact that the head clause supporting this kind of *after*-clause can use an (absolute) present perfect form (which is not possible when *after* is purely temporal):

(ii) After Parliament's decision permanently to abolish the death penalty for murder, the Home Secretary has asked the Criminal Law Revision Committee to review all the law on offences against the person, including homicide. (*The Times*, 6 March 1970)

Another type of sentence in which *after* can collocate with a present or present perfect is the following:

(iii) Tom has come back to England (*or:* is back in England) after living (*or:* having lived) in Nigeria for many years.

Here again the *after*-clause does not appear to be purely temporal: it does not indicate the time of the head clause situation. (For this reason (iii) could not be used in answer to a question beginning with *when*.)

If there is no adverbial or pragmatic indication, the unmarked interpretation of temporal order will only be partly in accordance with the principle that we have observed in connection with unembedded clauses (viz. the principle formulated in (6):

1.2.2.1 If both the head clause and the subclause shifting the domain are bounded, the two situations tend to be interpreted as holding in the order in which they are reported only if the subclause is one of the clauses that carry the action forward (i.e. that have a foregrounding function). This may be the case in examples like the following:

(44) (a) John hit Bill, who hit him.
(b) John, who hit Bill, was hit by him.
(c) Bill hit John, who hit him.

There is a certain tendency here to interpret the situations as occurring in the order mentioned, particularly if these sentences are couched in a context consisting of clauses that carry the action forward. However, subclauses do not often carry the action forward.[34] The order of the clauses, therefore, often fails to reflect the temporal order of the events. Thus, (45,a) may be substituted for either (45,b) or (45,c):

(45) (a) Bill was killed on the same spot where John had his accident.
(b) Bill was killed on the same spot where John had had his accident.
(c) Bill had been killed on the same sport where John had his accident.

Similarly, the following (a) and (b) sentences do not suggest different temporal interpretations:

(46) (a) The man came to my office who won the first prize yesterday.
(b) The man who won the first prize yesterday came to my office.
(47) (a) The man who killed the rat bought us a beer.
(b) The man who bought us a beer killed the rat.

In each of these examples the temporal order of the two situations is simply unspecified.

34. Some linguists (e.g. Labov and Waletzky 1967; Labov 1972:362; Partee 1984;256; Reinhart 1984) claim that syntactically embedded clauses never carry the action forward but always function as 'background'. I think that this claim is an overstatement. Though it does hold for the majority of subclauses, there certainly exist subclauses which have a foregrounding function. (One type of them, so-called 'narrative *when*-clauses', will be discussed in section 2). (See also Thompson (1987), who argues that about 10 per cent of the embedded clauses in her material are foregrounding, i.e. describe a temporally sequenced situation, and Couper-Kuhlen (1989b:12).)

1.2.2.2 When the two clauses are unbounded, the unmarked interpretation is that of simultaneity:

(48) (a) John carried the suitcase because Bill pushed the cart.
(b) Tom, who lived in London, knew the way to the Strand.
(c) I didn't see the girls that wore red hats.
(49) (a) John will carry the suitcase because Bill will push the cart.
(b) Tom, who will then live in London, will know the way to the Strand.
(c) You won't see the girls that will wear red hats.

Repetitive and habitual sentences that are unbounded are interpreted similarly: the unmarked interpretation is in terms of simultaneity:

(50) (a) After a while John disliked his wife because she *read* his letters.
(b) You will dislike your future wife because she *will read* your letters.

If the repetitive or habitual situation is to be interpreted as anterior to the matrix situation, there is usually an adverbial indication of anteriority:

(51) John was angry because Mary *read* his letters the day before.

1.2.2.3 When one clause is unbounded while the other is bounded, the unmarked interpretation is that the bounded situation is included in the unbounded one. This is the case irrespective of whether the unbounded clause is the head clause or the subclause and irrespective of the surface order of the clauses:

(52) (a) Bill, who *seemed to know* everything about the stranger, *told* us who he was.
(b) Bill, who *told* us who the stranger was, *seemed to know* everything about him.
(53) (a) Bill, who *will* soon *know* everything about the stranger, *will tell* us who he is.
(b) Bill, who *will tell* us who the stranger is, *will appear* to know everything about him.

1.3 Conclusion

We have reached the following conclusions concerning the possibility of shifting the domain (in the same sector):

1 In unembedded clauses the domain may be shifted freely as long as this does not obscure the temporal order of the situations. If a shift occurs, the temporal order of the situations (which is no longer indicated by the verb form) is usually indicated by an adverbial or is recoverable

from contextual information or pragmatic factors. If this is not the case, the temporal interpretation will depend on the bounded or unbounded nature of the clauses. The unmarked interpretation is that of iconic sequencing if the two clauses are bounded, simultaneity if they are unbounded, and inclusion if one clause is bounded while the other is not.

2 Clauses that are syntactically dependent are preferably also interpreted as semantically dependent. That is, subclauses usually have a backgrounding function. This means that subclause situations are mostly incorporated into already existing temporal domains. However, some types of subclause may also shift the domain. In that case the temporal relation between the relevant situations is mostly apparent from adverbial, contextual or pragmatic information. If this is not the case, the principles of temporal interpretation are partly the same as in unembedded clauses: when both clauses are unbounded, they are preferably interpreted in terms of simultaneity; when one is bounded and the other is not, the unmarked interpretation is in terms of inclusion; however, when both clauses are bounded, the iconic sequencing interpretation only suggests itself in the exceptional cases where the subclause is interpreted as pushing the action forward (i.e. as providing foregrounded information).[35]

3 Some syntactically dependent clauses are more semantically (i.e. temporally) dependent on their matrix than others. The former will not so easily shift the domain (i.e. make use of absolute tenses).[36] For example, the semantic dependency is very great in adverbial time clauses. These allow the shift of domain only exceptionally (e.g. after the conjunction *before*). By contrast, nonrestrictive relative clauses have a maximal semantic independency and allow the domain to be shifted under the same conditions as obtain for independent clauses.

4 Situations that are temporally subordinated are normally interpreted as background. As a rule, only sentences that shift the domain push the action forward. (However, the converse is not true: not all sentences

35. According to Enç (1987: 635), sentences like *You knew that I was upset about the results* and *I heard that Sally was in London* are 'ambiguous' because the past tense in the subclause may locate its situation at a time that is before, at or after the time of the subclause situation. I would like to make three remarks in connection with this claim. First, the 'ambiguity' arises only if the past tense is interpreted as an absolute tense, not if it is a relative tense form. Secondly, if the past tense is read as an absolute tense form creating its own domain, the temporal relation between the two situations is simply unspecified. This means that we do not really have to do with ambiguity but rather with vagueness. Thirdly, even if the temporal relation is unspecified, so that situation B may in principle lie before, at, or after situation A, there is always one possibility that suggests itself as the preferred (unmarked) interpretation.
36. This is in keeping with Givón's (1984:315) general principle concerning the coding of TAM (= Tense, Aspect, Modality) categories in subordinate clauses: 'The more dependent the SUB-clause is semantically/pragmatically on the MAIN-clause, the less likely are independently-expressed TAM markers to appear in the SUB-clause.'

establishing a domain have a foregrounding function. For example, nonrestrictive relative clauses are mostly background. Moreover, unbounded clauses never have a foregrounding function, unless they are interpreted inchoatively.)

What has not been pointed out yet is that time clauses may form an exception to this rule. In the following pieces of discourse, the time clause each time pushes the action forward in spite of the fact that it is temporally subordinated to the head clause:

(54) (a) I softly opened the door and went inside. *After I had drawn the curtains*, I switched on the light and started looking for the safe.
(b) I went to see a solicitor. *When I had told him everything*, he told me what I should do.
(c) You will have to wait a little in this waiting room. *When the doctor is free*, he will come and find you here.

The temporal clauses here do not shift the domain in the technical sense of the term. That is, they do not establish the central TO of a new domain. However, it is clear that in each of these examples the time clause does mark the beginning of a new domain. This is because the time clause precedes its binding clause and thus happens to be the first clause to be mentioned of the clauses that make up the domain. It follows that the time clause is felt to push the action forward, even though it is a temporally subordinated clause.

2 THE CHOICE OF BINDING TO

When a situation is incorporated into a domain, it is temporally related to a 'binding TO'. This binding TO may be either the central TO of the domain or an 'ancillary' TO (see section 11 of chapter 2), but the speaker is not always free to choose it.[37]

37. What we call the binding TO has sometimes been called the 'reference point' or 'reference time'. Thus, when we say that the past perfect represents a situation as anterior to a past binding TO, this statement is comparable to Reichenbach's (1947) assertion that the past perfect implies that the 'event' lies before a past 'reference point'. Now, the linguistic literature contains a host of works dealing with the question of how 'reference times' are identified in texts (discourses). See e.g. Smith (1978a, 1978b), Kamp (1981), Hinrichs (1981), Kamp and Rohrer (1983), Partee (1984), Rigter (1986), Houweling (1986), Lo Cascio (1986), Lo Cascio and Rohrer (1986), Adelaar and Lo Cascio (1986), Couper-Kuhlen (1987), Hamann (1987), etc. However, in most of these works the term 'reference time' is not used in the sense of our 'binding TO', but rather in the sense of 'time with respect to which a clause is interpreted'. Thus, Partee (1984:257) states that, in a sentence consisting of a main clause and a *when*-clause, 'the main clause ... is interpreted with respect to a reference time descriptively characterized by the subordinate clause'. Since, as far as the tenses are concerned, it is mostly the *when*-clause that is bound by the main clause, and not vice versa, it is clear that the term 'reference time' is here used in a sense that is totally

2.1 Binding rules for subclauses

When the 'bound situation' (i.e. the situation whose TO is temporally related to the binding TO) is referred to in a subclause, the unmarked choice is what we have called 'direct binding'. Direct binding means that the binding TO is the TO of the situation referred to in the matrix clause which immediately dominates the subclause in question.[38] This is the case for each of the subclause situations in the following example:

(55) John explained that he *had told* Mary that Bill *had said* that she *was* ugly.

However, there are numerous cases in which a subclause situation is bound indirectly, i.e. is bound by a TO which is not the situation-TO of its matrix. The following are some typical examples.[39]

2.1.1 As we have seen, some types of subclause need not represent their situation as temporally dependent on the situation of their matrix. If the situation-TO of that matrix is itself a bound TO, such subclauses may (or, sometimes, must) represent their situation-TO as bound, not by the situation-TO of their matrix, but by the TO which also binds the situation-TO of the matrix. For example:

(56) John knew that Mary had said something which *had shocked* Bill.

In (56), the head clause situation (*knew*) is the binding TO for both subclauses. Though the relative clause is embedded in the *that*-clause, its situation is not related (as simultaneous) to the *that*-clause situation, but

different from our notion of 'binding TO'. This difference is also apparent from the fact that most of these linguists also speak of a reference time in connection with clauses where the domain is shifted or re-established, i.e. in cases where there is no binding TO. Obviously, the notion 'reference time' is a hybrid concept, and I will therefore abstain from using it. (I will return to this question in chapter 6.)

38. The claim that a subclause situation-TO is preferably bound by the situation-TO of its matrix accords with similar claims made by Smith (1978b:62), Hornstein (1981:140), Lo Cascio and Rohrer (1986:236).

39. Lo Cascio and Rohrer (1986:238) also point out that sometimes (but 'not very often') 'the verbal tense of the complement clause is not evaluated with respect to its matrix but with respect to a clause which is higher in the hierarchy'. They suggest the following as an example:

(i) I think he will be angry because you will not be waiting for him at the station.

Lo Cascio and Rohrer point out that the situation denoted by *you will not be waiting for him* is not posterior to the situation of its matrix (*he will be angry*) but to that of the head clause. In my opinion, however, (i) is not an example of indirect binding, because the relevant situation is not bound at all: its TO is not dependent on another TO in the same domain, but is the central TO of a new domain. That is, the *because*-clause does not incorporate a situation in the post-present domain established by the first subclause, but rather establishes its own (simultaneous) domain. It is for this reason that the *because*-clause uses the future tense (the FPS) rather than the present tense (PPS) (see chapter 2).

is related directly (as anterior) to the head clause situation.[40] The following is a similar example:

(57) John had been in bed during the preceding week because he *had been* ill.

Here both situation-TOs are bound by the same (implicit) TO. (This kind of binding is not obligatory for the *because*-clause, which could also use the relative tense form *was*.)

It is interesting to note that even *while*-clauses occasionally use this kind of indirect binding. Next to the (more normal) sentence (58,a), we can also use (58,b) (see footnote 57 of chapter 2):

(58) (a) We had been waiting while they *were fetching* their coats.
(b) We had been waiting while they *had been fetching* their coats.

2.1.2 A situation expressed by a subclause which is conjoined to a subclause which is also marked by relative tense may be bound by the situation-TO of the latter clause:

(59) (a) John said that he had met Mary in London and that she *had looked* (*or:* looked) quite all right then.
(b) An hour ago a witness told the police that the girl who was found killed last night had missed the last train home on Tuesday night and *would try* to get home by hitch-hiking.

When the past perfect *had looked* is used in (59,a), the situation is bound to (i.e. represented as anterior to) the matrix situation (*said*); when *looked* is used, the situation is bound to (i.e. represented as simultaneous with) the situation-TO of the first conjunct (*had met*). In (59,b) it is clear from contextual information that *would try* expresses posteriority with respect to the situation of the first conjunct (*had missed*) and not with respect to that of the matrix (*told*).[41]

Since the unmarked possibility is for a subclause to be bound by its matrix, the other possibility is not always available. For example, when the two conjoined subclauses using relative tense are nonrestrictive

40. Adelaar and Lo Cascio (1986:255) also point out that an 'anaphoric' (i.e. relative) tense may be bound not only by the anaphoric tense immediately preceding it but also by the deictic tense which is the head of the domain.

41. Examples like these force us to reject Adelaar and Lo Cascio's (1986) claim that 'anaphoric tenses cannot express temporal connection between coordinate clauses' (p. 264), i.e. that 'tenses express a temporal relation only with respect to temporal entities, functioning as evaluation times, which belong to a higher position in the (syntactic and semantic) hierarchy' (p. 269). As a result we also have to reject the more general claim that 'a time interval can function as evaluative time for a deictic or anaphoric tense if it c-commands but does not govern the node immediately dominating the tense in question' (p. 270).

relative clauses, both of them are normally bound by the TO of the head clause:

(60) This piece of information was given by John, who had had a conversation with the manager and *had heard* a couple of interesting things.

It is only when we add an adverbial representing the situations of the conjuncts as simultaneous with each other that the past tense can be substituted for the past perfect:

(61) This piece of information was given by John, who had had a conversation with the manager and (*had*) *heard* a couple of interesting things *in the course of that conversation.*

2.1.3 Time clauses are often bound by their matrix (e.g. *We arrived after John had left*). However, there are cases in which the time clause seems to bind the matrix:

(62) John *had left* before we arrived.

However, it was shown in section 25 of chapter 2 that *arrived* is not an absolute tense form in this example: it expresses simultaneity with an implicit TO. (This is in keeping with the fact that *arrived* is not 'backshifted' so as to become *had arrived* in indirect speech. As noted above (see e.g. footnote 16), a relative past tense form is not 'backshifted' because it expresses the same temporal relation in indirect speech as it does in direct speech.) The conclusion must therefore be that in (62) neither clause binds the other. Both relate their situation to an implicit TO.

Similar conclusions hold for sentences involving a *when*-clause, which may make use of two different ways of locating the two situations:

(63) (a) We arrived when John *had left.*
(b) John *had left* when we arrived.
(64) (a) We *were* in the kitchen when John left.
(b) John left when we *were* in the kitchen.

Only in the (a) sentences is the preterit (*arrived/were*) an absolute tense form. In the (b) sentences, where it appears in the *when*-clause, the preterit is a relative tense form which expresses simultaneity with an implicit TO (see section 25 of chapter 2).

However, there is a special type of *when*-clause which does use an absolute tense and whose situation-TO therefore does function as binding TO, not as bound TO. It is exemplified by the following:

(65) (a) I was sitting quietly in the drawing-room when suddenly John *came in.*

(b) Hardly had we arrived at our hotel when Betty *started* saying that she wanted to go home.
(c) We had just come to the swimming pool when it *started* raining. (Heinämäki 1978:27)
(d) My grandmother was standing on her own, watching a dance, when a beautiful young woman ... suddenly *abandoned* her partner. (Edgren 1971:178)

When-clauses like these differ from 'normal' *when*-clauses in that they do not indicate the time at which the head clause situation is located,[42] but carry the action forward. Since it has become customary to use Labov's (1972) term 'narrative clause' to refer to a clause that carries a foregrounded situation ahead in time, I will refer to such *when*-clauses as 'narrative *when*-clauses'.[43] Their special function entails that they behave differently from 'normal' *when*-clauses in several respects:

1 A narrative *when*-clause depending on a head clause with an absolute tense form shifts the domain and therefore also makes use of an absolute tense form. A normal *when*-clause, by contrast, expresses a domain-internal relation and therefore uses a relative tense form. Compare:

(66) (a) We were sitting in the kitchen when Bill *was reciting* that poem.
(b) We were sitting in the kitchen when all of a sudden Bill *recited* that poem.

Only (66,b) can be interpreted as involving a narrative *when*-clause. No such interpretation is possible for (66,a) because the progressive form *was reciting* cannot be interpreted as shifting the domain. (As noted above, progressive forms very seldom shift the domain in embedded clauses. They can only be interpreted as doing so if there is a time adverbial or context enforcing this interpretation.)

42. 'Normal' *when*-clauses have the same function as other time adverbials (*at five o'clock*, etc.). This appears clear when we compare *John left when Mary left* with *John left at five o'clock*. Both sentences can be used in answer to *When did John leave?* By contrast, sentence (65,a) cannot be used in answer to *When were you sitting in the drawing-room?*
43. This is also the term used by Couper-Kuhlen (1987, 1989a). In fact, the label 'sentence-final narrative *when*-clauses' would be a more adequate label, for I have pointed out above that 'normal' (i.e. temporally subordinated) *when*-clauses (and other temporal clauses) can also move the action forward if they precede the head clause, as in:
(i) People began to leave. When the room *was* empty, the janitors came in. (Partee 1984:262)
(ii) We had dinner. After/as soon as/when the table *had been cleared*, we had a rubber of bridge.
Our discussion of narrative *when*-clauses will only concern sentence-final narrative *when*-clauses.

The claim that the *when*-clause involves a relative preterit in (66,a) and an absolute one in (66,b) is confirmed by the fact that the following are the normal indirect speech versions of (66,a–b):

(67) (a) They said that they had been sitting in the kitchen when Bill *was reciting* that poem.
(b) They said that they had been sitting in the kitchen when all of a sudden Bill *had recited* that poem.

As noted above, an absolute preterit is normally 'backshifted' in indirect speech (after a reporting verb in the past tense), whereas a relative tense form is normally retained.

The observation that a narrative *when*-clause that depends on an unembedded clause shifts the domain explains why such a *when*-clause can only use an absolute tense form. That is, tenses that can never be used as absolute tenses (e.g. the past perfect, conditional, conditional perfect) cannot occur in such narrative *when*-clauses. (Note that we are talking here of narrative *when*-clauses that depend on an unembedded head clause, not of those that depend on a matrix which is temporally subordinated, as in (67,b).) And when the reference is to the post-present, the system to be used is the (absolute) Future Perspective System, not the (relative) Present Perspective System:

(68) I am afraid that the following will happen. We will be sitting quietly in the drawing-room when suddenly Bill *will come in* (**comes in*) and *will tell* (**tells*) us that. . .[44]

2 As is clear from (67,b), a narrative *when*-clause depending on a matrix with relative tense will make use of the same tense as its matrix. In other words, such a narrative *when*-clause is bound indirectly (i.e. is bound by the TO which also binds its matrix). A normal *when*-clause, by contrast, is bound directly (i.e. is bound by its own matrix), as in (67,a).

3 Because narrative *when*-clauses like (66,b) shift the domain, they can be paraphrased by means of *and then*. . . Normal *when*-clauses do not allow such a paraphrase. Compare:

(69) (a) We were sitting in the kitchen, and then all of a sudden Bill recited that poem.
(b) John was in the kitchen, and then I left.

Example (69,a) is a reasonably acceptable paraphrase of (66,b), but (69,b) does not have the same meaning as *John was in the kitchen when I left.*

44. The present tense forms are to be read as relative tense forms, not as historical presents. (When they are read as historical presents, sentence (68) is all right – see below.)

4 Narrative *when*-clauses allow the use of the historical present:

(70) I was quietly sitting in the sitting-room with my wife when suddenly this chap *comes* in and *threatens* us with a revolver.

As is well known, the use of the historical present is a strong device for foregrounding a situation (see Fleischman 1985; Wald 1987:486).

5 Narrative *when*-clauses in fact do not answer the question *when*? Thus, (66,b) is not felt to be an answer to the question *When were we sitting in the kichen?* (In fact, it is the head clause that expresses the temporal setting for the *when*-clause rather than the other way around.)

6 Because they do not specify the time of the head clause situation, narrative *when*-clauses do not function as time adverbials. One consequence of this is that they cannot be preposed:

(71) (a) I was quietly sitting in the drawing-room when suddenly the telephone rang.
(b) *When suddenly the telephone rang I was sitting quietly in the drawing-room.

Normal *when*-clauses are time adverbials and can therefore be preposed:

(72) (a) John was in the kitchen when I left.
(b) When I left John was in the kitchen.

7 Unlike time adverbials, narrative *when*-clauses cannot become the focus of a cleft sentence:

(73) (a) It was when I left (not when Mary left) that John was in the kitchen.
(b) ?It was when suddenly the telephone rang that I was sitting in the drawing-room.

If (73,b) is judged acceptable, its *when*-clause is interpreted as a normal *when*-clause.

8 Because normal *when*-clauses are time adverbials, they can be replaced by other time adverbials. Thus, the *when*-clause in (72,a–b) can be replaced by a clause introduced by *at the time that*. Narrative *when*-clauses do not allow such a substitution:

(74) (a) John was in the kitchen at the time that I left.
(b) We were sitting in the kitchen at the time that Bill all of a sudden recited that poem.

Example (74,b) is no paraphrase of (66,b).

9 When two narrative *when*-clauses are conjoined (with or without conjunction reduction), they may refer to different time intervals (like unembedded clauses). This is clear from (68) and (70). When normal

when-clauses (using the same tense) are conjoined (without repetition of *when*), they refer to the same interval:

(75) I was sitting in the kitchen when Tom came in and Mary left.

A narrative *when*-clause whose verb is a phrasal verb (e.g. *come in*) allows the adverbial particle to be preposed. A normal *when*-clause does not allow this. Thus, the following sentence can only be analysed as involving a narrative *when*-clause (see Morgan 1975:445):

(76) I was sitting in the kitchen when in came this man who I had never seen before in my life.

2.1.4 A fourth case of indirect binding is when the situation-TO of a relative clause binds the situation-TO of the matrix (rather than the other way round). This is possible only if the relative clause precedes the matrix. Compare:

(77) (a) A boy who arrived yesterday *had been mugged* in the underground the day before.
(b) A boy *had been mugged* who arrived yesterday.
(78) (a) The copy that was destroyed today *had* only *been made* a couple of hours earlier.
(b) A copy *had been made* a couple of hours earlier which was destroyed today.

In (77,a) the situation-TO of *had been mugged* is bound by that of *arrived*. The same thing is not possible in (77,b), where the only possible interpretation is that the mugging took place before some (unspecified) time other than the arrival. The past tense *arrived* then establishes a shift of domain to some other time interval in the past. The same difference in interpretation is also to be observed in (78,a–b).

2.1.5 It is not impossible for sentences involving a complement clause to exhibit the same binding conditions as the ones observed in (77)–(78):

(79) That the president left the country on 4 July *would* not *be made* public until two days later.

However, examples of this kind are rare (though perfectly grammatical). The same idea will more normally be expressed in the following form:[45]

45. Smith (1981:235) claims that the sentence
(i) That the treaty expired on Thursday had been announced the day before.
illustrates the possibility of a head clause situation being 'anchored' to a time expressed in a preceding complement clause. In my opinion, however, *expired* does not denote the TO to which *had been announced* is anterior. I think that *expired* is here similar to *would expire*, i.e. represents its situation as posterior to some (unspecified) TO (which is the same as the one to which the head clause situation (*had announced*) is anterior).

(80) That the president (had) left the country on 4 July was not made public until two days later.

2.1.6 There is also indirect binding when a noun clause relates a situation to a TO which is not given in the complex sentence of which the noun clause forms part but in the context preceding it. The following examples illustrate this possibility:

(81) (a) I am sure the president knew everything about the money that was given to the Contras. Of course, he denies everything now, but I am sure he *had been told* of the scheme beforehand and *had approved* of it.

(b) Albertine arrived at the hotel towards midnight. Marcel was not yet asleep, but he did not feel like seeing her. When, the next morning, Albertine wanted to know why he had not opened the door, he said that he *was* tired and that he *had taken* a sleeping pill which *was going to work* within five minutes. (translated from Rohrer (1986:90))

In (81,a) there is a shift from a past domain (*knew*, *was given*) to the present (*denies*, *am sure*), and then back to the past domain. That is, the binding TO of the past perfect situations (*had been told*, *had approved*) is the time when the money was given, i.e. a TO which is part, not of the domain referred to by the immediately preceding clause, but of the domain referred to before that. Similarly, the verb forms of the last three clauses of (81,b) express relations in a domain which is different from the domain to which the situation of the matrix clause (*said*) belongs. (The domain in question is established by the very first sentence.)

2.1.7 A relative clause that is embedded into a clause with a relative tense form (and belongs to the same domain as the latter) is normally bound by the TO that also binds the latter:

(82) (a) John, who *had been* ill, had been in bed during the preceding week.

In other words, the use of *expired* has resulted from a shift of temporal perspective: the past tense is used to refer to a situation which is in fact posterior. The same use is to be observed in the sentence *They told me on Wednesday that the treaty expired on Thursday*. Neither in this example nor in (i) does *expired* denote the central TO of the domain. (It should also be noted that if Smith's claim that the head clause of (i) is temporally bound by the complement clause were correct, the only possible interpretation would be that the announcement took place the day before the expiration of the treaty. However, *the day before* in (i) may be the day before any past time. In (ii), for example, it is likely to be the day before the situation of the first sentence:

(ii) It was Sunday. That the treaty expired on Thursday had been announced the day before.)

(b) The boy that *had been* in hospital for three weeks had had a car accident.
(c) The man *had lived* in the town where Monroe had been born.

In (82,a) both clauses are represented as anterior to the same implicit TO; the temporal relation between them is thus not expressed by the tense forms, but the two situations are pragmatically interpreted as simultaneous with each other. The same thing is true in (82,b), but now the relative clause situation is pragmatically interpreted as holding later than the head clause situation. (This, of course, is the clearest proof that the former is not bound by the latter.) In (82,c) both situations are again bound by one and the same implicit TO; the temporal relation between the two situations is not expressed (either of them may have preceded the other, or they may have been simultaneous).

However, in indirect speech the relative clause can be interpreted as bound by its matrix:

(83) (a) Tom said that he had visited the town that *had been destroyed* by the earthquake.
(b) Tom said that he had spoken to the man who *would be sent* to Mars.

Since indirect speech creates an opaque context (see Zwicky 1971, 1974; Hall Partee 1973:415; Coulmas 1986:3–4; Rohrer 1986:91–2), all NPs in a stretch of indirect speech yield both a *de dicto* (opaque) and a *de re* (transparent) interpretation. This affects the interpretation of the relative clauses, since these are parts of NPs (noun phrases). Thus, both in (83,a) and in (83,b) the relative clause provides information which either belonged to Tom's original utterance (the *de dicto* reading) or is added by the present speaker (the *de re* reading). In the former case the situation-TO of the relative clause is bound by that of its matrix, in the latter it is bound by the situation-TO of the reporting clause *Tom said.*

In sum, there are at least seven exceptions to the principle that the TO of a subclause situation is bound by the TO of the matrix situation. One type of exception concerns the fact that the binding TO may not be established in the complex sentence at all, but rather in the context preceding it. The other exceptions that we have noted concern either the possibility that the binding TO is not to be found in the matrix but in some other clause, or the possibility that it is the matrix TO that is bound, while the binding TO is established in a subclause.

2.2 Binding rules for unembedded clauses

When an unembedded clause (i.e. a simple sentence or co-ordinate clause) represents its situation as bound, the binding TO is usually given

in the preceding clause (as in *John was angry. He had just missed the bus*) or in the next clause (as in *He had lost his pen and could not do the exercise*).[46] Either of these possibilities realizes what we have called 'direct binding' of an unembedded clause (see chapter 2). However, unembedded clauses are occasionally bound indirectly. In that case one or more clauses separate the relevant bound clause from the clause containing the binding TO. For example:

(84) (a) I heard John shout that he would leave at once and never set foot in the house again. Apparently Mary *had said* something which had greatly upset him.
(b) At eight o'clock Pierre left his office. He had worked hard. He *felt* good. (Molendijk 1984:130)

In (84,a), the binding TO of *had said* is the situation-TO of the very first clause (*heard*). (The relative clause (*had upset*) is also bound by this TO.) Similarly, the binding TO of the situation in the last sentence of (84,b) is the situation-TO of the first sentence.

There are some general principles determining the ways in which an unembedded clause can be bound by a TO in the context preceding it:

2.2.1 An unembedded clause that follows a complex sentence consisting of a matrix and a subclause may in principle be bound by either of these clauses:

(85) John told me that he (had) met Susan in the park. An hour before he *had met* Bill there too.

Two interpretations are possible here: John met Bill in the park an hour before he met Susan or an hour before he told me about his encounter with Susan. Self-evidently, contextual or pragmatic factors can exclude one of the two interpretations:

(86) John told me yesterday that he met Susan in the park a week ago. Two days later he *would meet* her again in the library.

Because *two days later* can here only mean 'two days after the first meeting', the situation-TO indicated by *would meet* can only be interpreted as bound by the situation-TO of the *that*-clause.

46. Lo Cascio (1986) claims that the preceding clause must not have an 'anaphoric' (relative) tense if it is to bind the tense in the next independent clause. This claim is not correct, witness the following example:
(i) John told me last week that he had met Mary in London. Moreover, he had already met her the week before, in Cambridge.
In this example the TO binding the situation expressed by *had met* may be the situation-TO of the preceding *that*-clause.

2.2.2 When two unembedded clauses are conjoined and the first of them has relative tense, the second clause is not bound by the situation-TO of the first but by the TO which also binds the first:

(87) John looked tired. He had worked hard and *had taken* insufficient rest. (*took)

Although the situation of taking insufficient rest is simultaneous with the situation of working hard, we cannot use the past tense (*took*) to express this simultaneity. Instead, we have to express anteriority again, so that in fact no simultaneity is expressed between the situations of the co-ordinate clauses.

The following example is similar to (87), except for the fact that the bound conjuncts are not interpreted as simultaneous:

(88) While we were having dinner, a policeman brought in the boy that had disappeared the day before. He *had left* the group while they were climbing the hill and *had returned* to the village. He *had stolen* some apples from an orchard and *had spent* the night in a shed. In the morning he *had been detected* by the farmer and *handed over* to the police.

In this example the successive past perfect clauses, which are all bound by the same TO, report a number of situations that follow each other in time.[47]

2.2.3 The above two principles concern the possibility of an unembedded clause being bound, not by the situation-TO of the preceding clause, but by the TO binding the situation-TO of the preceding clause. In this type of indirect binding there is continuity of the binding TO: a number of clauses are bound by the same TO. (This is not the case when there is direct binding, because in that case there is a shift of binding TO: each clause is bound by the situation-TO of its immediate predecessor.) There exists another type of indirect binding, in which the binding TO does not remain constant. For example:

(89) In the evening the police brought in John, who had disappeared the day before. He *had sneaked* away while we were climbing the hill. He *had gone* back to the village and *had spent* the night with one of the local girls. **This will not surprise you. You know John is always sleeping with some girl or other.** In the morning he *had*

47. Note that the linear order of these past perfect clauses reflects the order in which the situations have taken place. These clauses are thus 'narrative clauses' in the sense of Labov and Waletzky (1967) and Labov (1972): they express foregrounded information, i.e. describe situations that carry the action forward.

gone to the police station and *told* them he had got lost in the woods and wanted to rejoin the group. So the police *had brought* him back to us.

As in (88), all the past perfects in italic are bound by the same TO. The only difference is that at a certain point the binding line is interrupted by two clauses (the ones in bold type) whose temporal reference is totally different. However, this interruption does not break the binding line, which is simply resumed after the clauses in bold. The reason why this is possible is that the past perfect clauses express the foregrounded situations (see footnote 47), whereas the clauses in bold express backgrounded information. The switch of TO in the clauses in bold thus does not affect the continuity of the binding TO in the foregrounded clauses.

3 THE CHOICE BETWEEN THE PRESENT PERSPECTIVE SYSTEM AND THE FUTURE PERSPECTIVE SYSTEM

As we have seen, the PPS is the unmarked system to express relations within a post-present domain. In this system, the binding TO to which the situation that is incorporated into the domain is related behaves as if it were t_0, so that the tense that is used is one of the four absolute tenses (present, preterit, present perfect or future tense). For example:

(90) (a) He will say that he does not believe what you *are telling* him.
(b) He will say that he *left* today.
(c) He will say that he *has never seen* you in his life.
(d) He will promise that he *will do* it.

The FPS comprises the (absolute) future tense as well as a couple of absolute–relative tenses, viz. the future perfect and *will be going to*. The future tense establishes a domain (which may or may not be simultaneous with one referred to in the preceding context); the other forms both establish (or re-establish) a post-present domain and indicate a relation in it (anteriority in the case of the future perfect, posteriority in the case of *will be going to*). For example:

(91) (a) Everybody *will disagree* with you if you *make* that suggestion. They *will say* that they *have* always *disapproved* of that kind of action. Moreover, they *will have heard* from John that you yourself *are* not quite certain that this *is* the best solution.
(b) Watch the sky when you arrive in Memphis. Perhaps it *will be going to rain*, and then you'd better take a taxi instead of walking to John's house.

The first clause in (91,a) uses the future tense (*will disagree*) to establish a post-present domain. The second clause uses a PPS form (*make*) to

incorporate a situation into the domain and relate it as simultaneous to the central TO. The clause *they will say* then again establishes a domain, which is simultaneous with the previously established one. The present perfect form *have disapproved* (which is a PPS form) expresses anteriority within this domain. The future perfect *will have heard* both re-establishes the current domain (through the use of *will*) and expresses anteriority within it (through the use of *have*). The present tense forms *are certain* and *is* express simultaneity within the domain.

As pointed out in chapter 2, the FPS will be used whenever the speaker deems it necessary to explicitly represent the domain he is referring to as a post-present domain. This is the case whenever there is no close temporal relation between the clause containing the relevant verb form and the clause(s) in the context referring to a post-present domain. Thus we will use the FPS rather than the PPS in independent clauses (e.g. *I will remember your name*) and in subclauses that establish their own time reference (e.g. nonrestrictive relative clauses: *I will ask John, who will know the answer tonight*). Subclauses whose situation is represented as temporally connected to that of the matrix (e.g. adverbial time clauses) use the PPS.

Most types of adverbial clauses may or may not be temporally dependent on their head clause and may therefore use either system. A good example of this is provided by the following sentences (see Tregidgo 1979:194–5):

(92) (a) Don't come and see me, because *I'll be* in hospital.
(b) Don't come and see me, because *I'm* in hospital.
(c) Don't come and see me because I *am* in hospital.
(d) Don't come and see me because I *will be* in hospital.

In (92,a) and (92,b) there is a break (signalled by a comma) between the two clauses. It follows that the *because*-clauses are interpreted as temporally independent of the head clause: both the future tense (*'ll be*) and the present tense (*'m*) are interpreted as absolute tense forms, referring to the future (relative to t_0) and the present, respectively. In (92,c–d), on the other hand, there is a close connection between head clause and subclause. The *because*-clause is therefore interpreted as temporally dependent on the head clause. That is, the present tense (*am*) in (92,c) expresses simultaneity within the post-present domain established by the head clause, whereas the future tense (*will be*) in (92,d) expresses posteriority in the same domain. Neither of these tense forms is interpreted as an absolute tense form. The correct interpretation of (92,c–d) is therefore as follows:

(92) (c′) If you come and see me, the reason for doing so should not be that I am in hospital (at the time of your visit).

(d′) If you come and see me, the reason for doing so should not be that I will be in hospital (at some time in the future of your visit).

Noun clauses usually use the PPS rather than the FPS. Hence the difference between (93,a) and (93,b):

(93) (a) Bill will explain to you how we *do* it.
(b) Bill will explain to you how we *will do* it.

In (93,a) the present tense *do* is either an absolute tense (referring to the present) or a relative tense (representing the noun clause situation as simultaneous with the head clause situation). In the latter case the present tense is part of the PPS and locates the situation in the post-present domain established by the head clause. In (93,b) the future tense *will do* can again be interpreted as an absolute or relative tense form. As an absolute tense form it establishes a post-present domain which may, but need not, be simultaneous with that established by the head clause. (On this interpretation the speaker refers both to a future way of doing something and to a future explanation of this, but since he relates both of these to t_0 he does not express a temporal relation between them.) The second reading of (93,b) is that on which the future tense form *will do* is a relative tense form belonging to the PPS. On this interpretation the noun clause situation is incorporated into the post-present domain established by the head clause and is represented as posterior to the head clause situation.

In restrictive relative clauses both the PPS and the FPS are commonly found:

(94) (a) The newspapers will print everything that *is said*.
(b) The newspapers will print everything that *will be said*.

Sentence (94,a) is ambiguous because *is said* can be interpreted either as an absolute tense form (referring to t_0) or as a relative tense form referring to the future (i.e. expressing simultaneity in the post-present domain established by the head clause).[48] In order to avoid this ambiguity the speaker may switch to the FPS, as in (94,b), which explicitly locates the subclause situation in the future of t_0. This double possibility is also illustrated by the following sentences:

(95) (a) If you will not be generous with your child, he will hate you for the hard time you *will have given* him. (Gabbay 1976:170)[49]

48. As we saw in chapter 2, it is often 'sloppy' simultaneity that is expressed by the PPS. In (94,a) it is pragmatically clear that the saying must precede the printing, yet the verb form selected (*is said*) is the form used for expressing simultaneity.
49. Gabbay (1976:170) makes the following (erroneous) claim in connection with (95,a–b): 'the future perfect is used to indicate that the "hard time" is in the past of "hate" but

(b) If you will not be generous to your child, he will hate you for the hard time you *gave/have given* him.

The relative clause uses the FPS in (95,a) and the PPS in (95,b). (In the latter case the time indicated by *will hate* behaves as if it were t_0; what lies before it can therefore be indicated by either the preterit or the present perfect.)[50]

Conditional clauses as a rule use the PPS, but the FPS is exceptionally possible. I will investigate this at great length in chapter 4. In that chapter I will also go more fully into complement clauses, which use either the PPS or the FPS (see above).

still in the future of "now". While ordinary past takes us back to the past of "now".' (In reality both the future perfect in (95,a) and the past tense in (95,b) just represent the situation as anterior to a post-present TO and leave its location with respect to t_0 unexpressed. It follows that in both cases the situation may in principle lie before, at or after t_0.)

50. The difference between the present perfect and the preterit will be explored in chapter 7.

4 Tense in discourse 3: two test cases

In chapter 2 we have developed a theory of tense which renders it possible to interpret and classify observations of all possible kinds. In chapter 3 we have gone further into some of the general principles that govern the choice of tenses. We have formulated rules in connection with the choice of domain, the choice of binding TO and the choice of perspective system to refer to the future. In this chapter we will scrutinize two areas of English grammar to which the choice of tense form is pertinent, and which can serve as test cases for the theory as a whole. The first is indirect speech, the second concerns *if*-clauses. Both of them are known as particularly troublesome areas because of the questions they raise in connection with the use of the tenses. The recent linguistic literature abounds with articles that aim to account either for the use of the tenses in indirect speech or for the occasional use of *will* in *if*-clauses. As before, I will argue that most of the problems have stemmed from the fact that all kinds of observations have been interpreted wrongly, for lack of a coherent descriptive theory of tense. In the theory argued in chapter 2 these observations neatly fall into place, so that seemingly problematic cases turn out not to be problematic at all.

1 TENSE IN INDIRECT SPEECH

Indirect speech sentences are typical instances of structures in which tense subordination is very frequent, but not always obligatory. Compare:

(1) (a) Last week John imagined his employer *would/*will fire* him.
    (b) Last week John told me his employer *would/will fire* him.
(2) (a) One day I will tell your wife that you *have treated/*are treating* me like this today.
    (b) One day you will regret that you *have treated/are treating* me like this today.

Tense subordination (i.e. incorporation of the complement clause situation into the head clause domain) appears to be obligatory in (1,a)

and (2,a). In (1,b) and (2,b), on the other hand, it is optional: the subclauses may also shift the domain.

One of the purposes of this section is to discuss the nature of restrictions such as the ones that are operative in (1,a) and (2,a). However, before doing so we will have a look at the various analyses of tense in indirect speech which are available in the linguistic literature.

1.1 Theories of tense in indirect speech

The phenomenon of tense subordination in indirect speech complement clauses is widely known under the name of 'Sequence of Tenses' (SoT). Other (though less familiar) labels are 'tense harmony' (Higgins 1976) and 'tense copying' (Noonan 1985:88). Although there is widespread agreement about the existence of SoT, there is no consensus of opinion as to the nature of the phenomenon. The competing views appear to be the following:

1 The most widespread view no doubt is that SoT is a formal (syntactic) rule which operates mechanically when the tense of the head clause is past (i.e. is one of the tenses typical of the past time-sphere). This view is inherent in Jespersen's (1924:290–9) well-know 'quotative analysis', according to which a complement clause after a verb of saying or thinking originates as a direct quote (e.g. '*I am late*') which then undergoes 'backshifting' when it is syntactically subordinated to a verb in the past tense (e.g. *John said/regretted he was late*).

The linguists advocating this theory fall into two groups. Some (e.g. Comrie 1986) claim that backshifting is a purely formal operation (triggered by the past tense of the reporting clause), which is not semantically motivated. Others hold that backshifting is part of the general deictic reorientation that takes place in indirect speech. Their view is that the syntactic subordinating operation entails that all the coreference relations (expressed by pronouns, tenses and deictic indications of time and place) which in direct quotes are defined in relation to t_0 are redefined in relation to the speech act expressed by the clause under which the quote is embedded.

Since the quotative analysis is in essence transformational, it was readily adopted in early transformational grammar (see e.g. Ross 1967; Bach 1968; Kiparsky and Kiparsky 1970; Vasudeva 1971:101; McCawley 1973). In later years its status as a transformational rule was defended by some (e.g. Wunderlich 1979:297) and challenged by others (e.g. Banfield 1973; Ejerhed Braroe 1974:5ff.; Dali 1981:3), but the view that SoT is a formal rule (whether semantically motivated or not) remained widespread (see e.g. Quirk *et al.* 1972:786ff.; Cygan 1972:6; Ladusaw 1977; Coulmas 1986:16). The most outspoken defence of this analysis in recent years is Comrie (1986). According to him, SoT is a

purely formal rule without semantic motivation. Henceforth I will refer to this analysis as the 'formal SoT hypothesis'.

2 An alternative analysis is what we can call the 'relative time hypothesis'. According to this hypothesis the tense form in the complement clause depends on (1) the tense (past or nonpast) of the head clause and (2) the temporal relation that is expressed between the complement clause and the head clause. If the head clause is in a past tense, the complement clause will use the preterit for simultaneity, the past perfect for anteriority, etc. According to this theory, the reason for the use of the preterit form *was* in *John said that he was ill* is that the speaker wants to represent the situation of John being ill as simultaneous with (i.e. as holding at the time of) the head clause situation. It follows that SoT is seen as a semantic (rather than syntactic) phenomenon.

Proponents of this relative time hypothesis are Allen (1966), Ultan (1978:117), Tregidgo (1979), Wekker (1980), Huddleston (1984:151), Schopf (1984:285), Dahl (1987:500–1).

3 A third analysis is based on the claim that tenses in complement clauses are of exactly the same kind as in head clauses, i.e. they are deictic (absolute) tenses. On this analysis the reason why *was* is used in *John said he was ill* is simply that the illness is to be located in the past. This 'absolute deixis hypothesis' is not very widespread, but it has been propounded by Brecht (1974:500–2), Riddle (1978) and Heny (1982:124):

> Viewed in this way, the SofT rule is a convention whereby embedded tenses are all defined from the point of view of the speaker. Accordingly, the embedded past tense in the sentences (19) below must be interpreted as signaling that the action or state in the complement is past, even though the action or state is either eternal or relevant at the time of the utterance. This latter meaning can only be explicitly signaled if the tense of the embedded verb is put into the present . . .
>
> (19) (a) How did you know what my name *was*?
> (b) Who told you he *was* coming?
> (c) How did Columbus find out that the world was round?
> (Brecht 1974:500)

Heny (1982:124) formulates it as follows:

> In a nutshell, tense will always be independent, with a deictic component – and in embedded propositions will simply receive a standard interpretation.

4 As should be clear from chapters 2 and 3, our own view is that the tense used in a complement clause may be either a relative tense

(incorporating the situation into the domain established by the head clause) or an absolute tense (creating a new domain). In some cases the speaker has to make use of the former possibility, in others he has the choice of the two options. Which possibility is realized depends on restrictions and factors that have been dealt with in chapter 3. For our present purpose it suffices to point out the following general principle concerning the speaker's choice of relative or absolute tense forms:

(3) If the head clause situation is not located at t_0, the use of a relative tense in the complement clause is the unmarked (default) choice. This means that in that type of sentence the complement clause can always use a relative tense, whereas there are restrictions on the use of an absolute tense form (which shifts the domain). For example, a shift of domain within the same absolute sector is allowed only if the temporal order of the situations (which is not expressed by the absolute tense forms) is clear from a temporal adverb, the context or from the hearer's pragmatic knowledge of the world.

When evaluating the different analyses,[1] we can to a certain extent refer to Comrie (1986), who argues effectively that the absolute deixis hypothesis (which holds that indirect speech complement clauses make use of absolute tense forms) is untenable because (unlike Comrie's own formal SoT hypothesis) it does not make the correct predictions in each of the following cases:

1 The absolute deixis hypothesis predicts that the indirect speech version of (4,a) is (4,b), and not (4,c):

(4) (a) Diana will say, 'I am dancing.'
(b) Diana will say that she will be dancing.
(c) Diana will say that she is dancing.

Comrie's formal SoT hypothesis (according to which the tense of the original utterance is retained in the *that*-clause if the tense of the verb of reporting is nonpast) accounts for the fact that the correct reported version of (4,a) is not (4,b) but (4,c). The hypothesis we have been arguing also accounts for this fact, because it states that the *that*-clause in (4,c) represents the situation of dancing as simultaneous with the head clause situation. As we have seen, the relative tense used to express simultaneity in a post-present domain is the present tense.

1. Leuschner (1972) argues for yet another analysis, which will be disregarded here because it is quite incompatible with the whole theoretical framework we have developed. (Leuschner bases his analysis on Weinrich's (1964, 1970) idiosyncratic theory of tense, which hinges on the claim that there is no basic relation between tense and time.)

2 The second argument concerns sentences like the following:

(5) (a) In 2010, Ebenezer will say, 'I got tenure in 2000.'
(b) In 2010, Ebenezer will say that he got tenure in 2000.
(c) *In 2010, Ebenezer will say that he will get tenure in 2000.

Since the year 2000 is future with respect to the moment of speech, the absolute deixis hypothesis predicts that Ebenezer's statement should be reported in the form of (5,c). However, the correct indirect version of (5,a) is not (5,c) but (5,b). Comrie's formal SoT hypothesis is in keeping with this, as it states that a past tense should be retained after a verb of reporting in a non-past tense. Our own hypothesis also accounts for the data, because, as we have seen, it provides for the possibility that the expression of temporal relations in a domain established by a future tense involves a shift of temporal perspective to the past time-sphere. That is, the past tense is used to represent a situation as being past relative to the central situation of the future domain.

3 Consider:

(6) (a) Tomorrow, Frances will say, 'I was absent yesterday.'
(b) Tomorrow, Frances will say that she was absent today.
(c) *Tomorrow, Frances will say that she is absent today.

The absolute deixis hypothesis wrongly predicts that (6,b) should be ungrammatical (since an absolute past tense cannot refer to present time) and that (6,c) should be the correct indirect speech version of (6,a). The formal SoT hypothesis, on the other hand, correctly predicts the grammaticality of (6,b) and the ungrammaticality of (6,c). Our own hypothesis does so too, because it provides for the use of the past tense to represent a situation as past with respect to the central situation of a future domain. This is the case in (6,b), where *today* indicates a time that is past with respect to the time indicated by *tomorrow*.

It should be noted, however, that sentences like (6,c) are not always ungrammatical. The use of the present tense in the subclause is occasionally possible, witness the following examples:

(7) (a) One day John will regret that he *is treating* me like this. (Dowty 1982:50)
(b) They will report tomorrow that Harry *is transmitting*. (Ejerhed Braroe 1974:47)

In these examples, the *that*-clauses can be interpreted as referring to present (rather than future) time. In that case they are the indirect representations of a thought or utterance which, in direct speech, would not be phrased in the present tense, but rather in the preterit. That is, (7,b) (on the relevant interpretation) is the indirect version of (8,a), not of (8,b):

(8) (a) They will report tomorrow: 'Henry was transmitting (yesterday).'
(b) They will report tomorrow: 'Henry is transmitting.'

Similarly, the thought which is reported in (7,a) and which is the cause of the subject's regret is 'I treated (*or:* was treating) her badly', and not 'I am treating her badly.'

Sentences like (7,a–b) are subject to very severe restrictions: the substitution of the present tense for the preterit is only exceptionally possible (see section 1.2 below). Even so, there are examples (such as (7,a–b)) that are grammatical, and this grammaticality should be accounted for by an adequate theory of indirect speech. And now we see that Comrie's formal SoT hypothesis is no longer up to the mark: since it holds that 'the tense of the original utterance is retained' 'if the tense of the verb of reporting is non-past' (Comrie 1986:279), it cannot account for the fact that a thought or utterance in the past tense becomes a present tense statement in indirect speech. The theory we have been arguing, however, does provide for the grammaticality of (7,a–b). It holds that the tense in an indirect speech complement clause may be either a relative tense (expressing a domain-internal relation) or an absolute tense (shifting the domain), and that the latter possibility is marked, and therefore subject to restrictions. This theory predicts that the normal (unmarked) reported version of (8,a) is *They will report tomorrow that Henry was transmitting today* (in which *was transmitting* is a relative tense), but it also provides for the possibility that the *that*-clause shifts the domain from the future to the present by using the present tense (as absolute tense). Moreover, it correctly predicts that this shift of domain is subject to restrictions, since it is the marked possibility.

In sum, this is a clear example of a case in which our hypothesis proves to be superior to Comrie's formal SoT analysis.

4 The fourth argument advanced by Comrie (1986) against the absolute deixis hypothesis concerns sentences like the following:

(9) (a) In 1970, Graham said, 'I will get tenure in 1980.'
(b) In 1970, Graham said that he would get tenure in 1980.
(c) *In 1970, Graham said that he got tenure in 1980.

The absolute deixis hypothesis incorrectly predicts that (9,c) should be grammatical (since the subclause refers to a time that is past with respect to the moment of speech). The formal SoT analysis rules out (9,c), as it states that *will* must be backshifted to *would* after a verb of reporting in the past tense. Our own hypothesis also predicts the grammaticality of (9,b), as it states that the tense in a complement clause is normally a relative tense, and the relative tense needed here (to express posteriority in a past domain) is, indeed, the conditional tense. (Note that, if (9,c) had been grammatical, we would have accounted for that in terms of a shift

of domain. However, we have seen that shifting the domain is a marked possibility, which is subject to (often severe) restrictions, and there is apparently such a restriction forbidding the shift of domain in sentences like (9,c) (see section 1.2 below).)

5 The final piece of evidence adduced by Comrie against the absolute deixis hypothesis concerns examples like the following:

(10) (a) Yesterday, Henrietta said, 'I will be absent tomorrow.'
(b) Yesterday, Henrietta said that she would be absent today.
(c) *Yesterday, Henrietta said that she is absent today.

The absolute deixis hypothesis incorrectly predicts that (10,c) should be grammatical ('since a time point that is subsequent to a time point prior to the present moment could happen to coincide with the present moment' (p. 282)). Comrie's hypothesis (which requires backshifting after *said*) accounts for the grammaticality of (10,b) and the ungrammaticality of (10,c). Our hypothesis does so too. It accounts for the use of the conditional tense in (10,b) (since the conditional is the tense to be used to represent a situation as posterior to the central situation of the past time-sphere domain). And it accounts for the ungrammaticality of (10,c) because it holds that the use of an absolute tense (shifting the domain) in an indirect speech complement clause is a marked possibility which is often blocked. (The restriction in question will be discussed in section 1.2 below.)

The above discussion can be summarized as follows. There are five sets of data which the absolute deixis hypothesis cannot account for. All these data are consistent with both Comrie's hypothesis and ours. However, the discussion has revealed that there are other data (viz. sentences like (7,a–b)) which are accounted for by our hypothesis, but not by Comrie's formal SoT analysis.

Apart from the above arguments there is other evidence against the absolute time hypothesis. For example, the absolute time hypothesis fails to account for the fact that *used to* can replace the preterit in the head clause of (11,a) but not in the indirect speech complement clause of (12,a):

(11) (a) When I was young, I often went to the seaside.
(b) When I was young, I used to go to the seaside.
(12) (a) Bill said that he often went to the seaside.
(b) Bill said that he used to go to the seaside.

Whereas (11,a) and (11,b) are equivalent in that both locate my going to the seaside in the past sector, (12,a) is not equivalent to (12,b): (12,a) locates the subclause situation at the time of the head clause situation, while (12,b) locates it before that time (i.e. *used to go* is roughly

equivalent to *had often gone*). The absolute time hypothesis, which treats *went* as an absolute preterit in both (11,a) and (12,a), cannot account for this. Our hypothesis, on the other hand, can trace the observed difference in replaceability of *went* by *used to go* to the fact that *went* is an absolute tense form in (11,a) and a relative one in (12,a). Apparently *used to* can take the place of an absolute preterit, but not that of a preterit expressing simultaneity. This is also clear from sentences like the following:

(13) (a) My wife often came to see me while I was in hospital.
(b) *My wife often came to see me while I used to be in hospital.

The reason why *used to* can replace an absolute preterit only is that it represents a habitual situation as anterior to a TO. In unembedded clauses this TO is t_0, so that there is no difference between *I used to go* and *I usually went* (since an absolute preterit also locates its situation before t_0). However, when *used to* appears in a subclause that is temporally subordinated, the TO in question is the situation-TO of the head clause. In that case *used to* differs from the preterit in that it locates its situation before the TO, whereas the preterit represents it as simultaneous with the TO. (It follows that we use the past perfect, not the preterit, when we replace *used to* by a tense form in a temporally subordinated clause:

(14) (a) Bill said that he used to believe in ghosts when he was a kid.
(b) Bill said that he had believed in ghosts when he was a kid.)

In sum, the observations in connection with *used to* confirm the distinction we make between absolute and relative past tense forms – a distinction which is incompatible with the formal SoT analysis (and which is therefore explicitly denied by Comrie, see below).

Let us now look at the evidence which Comrie (1986) adduces against the 'relative time hypothesis'. (This evidence should also refute our hypothesis, since this is basically a relative time hypothesis, supplemented with the claim that the use of absolute tense is sometimes possible too.) Since the relative time hypothesis is more widely accepted than the absolute deixis hypothesis, we might expect Comrie to devote even more space to refuting the former than he does to refuting the latter. However, this is by no means the case. Comrie treats the relative time hypothesis as if it were hardly worth considering, and adduces only two arguments against it. The first of them runs as follows: 'Russian does use relative time reference; English differs from Russian, therefore English cannot use relative time reference' (p. 278). In my opinion, this argument is circular. It asserts that Russian has relative tense and that English differs from Russian in this respect. However, the latter claim is exactly what has to be proved. The second argument hinges on the example *Andrew said that he was sick* (*although he now claims to be better*) and runs as follows:

> One possible interpretation of the indirect speech, and the one that we shall be concerned with here, is that Andrew's actual words were *I am sick*, i.e. that the sickness in question is simultaneous with Andrew's utterance. If tense were conditioned by relative time reference, however, the past tense would be inappropriate in these circumstances, or at least the present (simultaneity with a contextually given reference point) should also be possible, but in fact the present is not possible on this interpretation (in particular, where Andrew's real or putative sickness does not extend to the present). Thus tense in indirect speech in English is not conditioned by relative time reference.
>
> (p. 278)

There are two (explicit or implicit) claims underlying this argument: (1) English has no tense that can express simultaneity with a past TO; (2) if the English tense system did provide for the possibility of expressing simultaneity with a past TO, it would necessarily be the present tense that was used for this. Claim (1) is made explicitly elsewhere in the article. After pointing out (on p. 272) that the preterit, the present tense and the future tense express absolute time in independent clauses, Comrie goes on to claim that the same thing is true in embedded clauses:[2]

> It is important to note that these characterizations of the meanings of these three tenses apply just as much in subordinate clauses as in main clauses, as in [15]:
>
> [15] While Ian was singing, Jennifer was dancing.
>
> In [15], the past tenses in both clauses are determined by past time reference relative to the here-and-now, i.e. are absolute deictic expressions; the fact that the actions of singing and dancing were simultaneous (relative deixis) is not expressed grammatically in English.

It is clear that we cannot subscribe to this view, which is asserted dogmatically, without evidence supporting it. The claim that the preterit cannot be used as a relative tense leads to the conclusion that English has tenses that can be used to represent a situation as anterior or posterior to a past TO (viz. the pluperfect and the conditional tense) but has no tense that can be used to represent a situation as simultaneous with a past TO. This conclusion is one that I cannot accept. As noted in chapter 2, simultaneity is the unmarked of the three relations (i.e. it is the default category), and (as is well known in markedness theory) the existence of

2. I have substituted my own reference numbers for Comrie's in this quotation, and in the ones that follow.

marked possibilities requires the existence of an unmarked possibility.[3] Moreover, the claim that the past, present and future tenses can only be used as absolute tenses is flatly contradicted by examples like *Jennifer will be dancing when Ian is singing*, where the present tense (*is*) cannot be interpreted as referring to the moment of speech and can only be taken to be a relative tense expressing simultaneity.

Once it is accepted that English can use tense forms to express simultaneity with a past or future TO, the second claim underlying Comrie's argument (viz. that only the present tense can express simultaneity) is automatically disproved, for it is simply contradicted by the facts. One cannot accept that English can use a tense to express simultaneity with a past TO without recognizing that the tense in question is the preterit.

In sum, Comrie's arguments against a relative time hypothesis do not appear to be valid.[4] This takes us back to the conclusion arrived at above, viz. that so far we have pointed out one set of examples that are inconsistent with the formal SoT analysis, but none that are inconsistent with the hypothesis we have been arguing.

Comrie (1986:283) also considers some 'apparent problems' for the formal SoT hypothesis. Let us examine whether these data are also problems for our hypothesis.

1 A formal SoT rule stipulating that backshifting of tenses is obligatory after a verb of reporting in the past tense wrongly predicts that sentences like the following should be ungrammatical:

(16) (a) Yesterday, John said that he *will arrive* tomorrow.
(b) Bill said that he *is* ill.

Comrie solves this problem by adding a restriction to his SoT rule. The rule now stipulates that backshifting after a verb of reporting in the past must take place, except 'if the content of the indirect speech has

3. Note that Comrie denies not only that English has no tense whose meaning (or basic meaning) it is to express simultaneity with a past TO but also that English has no tense which *can* be used to express simultaneity with a past TO, i.e. that this relation simply cannot be expressed by a tense form. I can accept the former claim. Since the preterit is not formally marked for simultaneity (whereas the past perfect and the conditional are marked for anteriority and posteriority, respectively), it seems correct to say that English has no tense whose (basic) meaning is 'simultaneity with a past TO'. However, it is something quite different to claim that English has no tense which *can* be used to express this relation. I hope to have shown conclusively in chapter 2 that the preterit can be used in this way (see also chapter 7). Similarly, I agree that 'simultaneity with a post-present TO' is not the basic meaning of the present tense, but I do not see how one can deny that one can use this (and only this) tense to express this meaning (see *Jennifer will be dancing while Ian is singing*).
4. In Comrie (1985), which also contains a section on indirect speech, we also find most of the above arguments against the absolute deixis hypothesis and in favour of the formal SoT hypothesis. The relative time hypothesis is not considered at all.

continuing applicability'. In that case 'the backshifting is optional' (p. 285).

Comrie also points out that some of these cases of continuing applicability cannot be handled by the absolute deixis hypothesis. For example:

(17) *Yesterday, Linda said she *will arrive* tomorrow, but she immediately changed her mind.

Comrie's argument runs as follows:

> Absolute deixis is incapable of distinguishing between grammatical examples like [16,b] and ungrammatical examples like [17], because in both sentences the verb in the future has future time reference relative to the here-and-now of the reporter, and should therefore be grammatical. But whereas in [16,b] the content of the original utterance still has continuing applicability, in [17] the final clause makes it clear that the content of the original utterance does not have continuing applicability. What is crucial in the examples discussed in this section is thus not the time reference of the content of the original utterance, but rather its continuing applicability, and examples like [17] show that these are not necessarily the same.
>
> (p. 286)

It is clear that this criticism applies to the absolute deixis hypothesis but not to the one we have been arguing. Our hypothesis explains the use of present time-sphere tenses in the *that*-clauses of (16,a–b) as resulting from a shift of domain. It also says that a shift of domain is a marked possibility (the unmarked alternative being the use of relative tenses), and is therefore subject to restrictions (see also below). In the case of (16,a–b) and (17) the restriction is that the statement must have continuing applicability. (By relating the subclause situation to his own here-and-now the speaker makes clear that he believes in the truth of the subclause. Of course he cannot do this – at least not without violating Grice's Maxim of Quality – if he has (and reports) information to the contrary.) Our hypothesis is thus perfectly in keeping with Comrie's remarks. Moreover, it is superior to Comrie's analysis in that it accounts for the use of present time-sphere tenses in the *that*-clauses of (16,a–b) without having to introduce an additional (and rather *ad hoc*) stipulation. Furthermore, it should be pointed out that Comrie's solution (according to which backshifting is optional if the original utterance has continuing applicability) fails to account for the fact that it is sometimes impossible to use the present tense, in spite of the fact that there is continuing applicability:

(18) (a) This is John's wife. – Yes, I THOUGHT he *was* married.
(b) I knew you *liked* her.

As pointed out by Oakeshott-Taylor (1984:4), we cannot substitute *is married* for *was married* in the reply sentence of (18,a) (which is to be read with the nuclear accent on *thought*). Yet it is obvious that the speaker takes the truth of the *that*-clause for granted. The same is true of (18,b), where the use of *like* would be 'thoroughly unacceptable' (Huddleston 1984:152). Examples like these make clear that there are restrictions on the use of the present tense in indirect speech clauses depending on a matrix in the past tense. This is in accordance with our theory, which treats the use of the present tense as a shift of domain, i.e. as a marked possibility,[5] which is naturally subject to restrictions.[6] Comrie's theory, however, does not provide for such restrictions.

2 The second of Comrie's 'apparent problems for the SoT hypothesis' arises from examples like the following, which should be interpreted as uttered on Wednesday:

(19) (a) On Friday, Oswald will say, 'I arrived on Thursday.'
(b) On Friday, Oswald will say that he *arrived* on Thursday.
(c) ?On Friday, Oswald will say that he *arrived* tomorrow.

The absolute deixis hypothesis wrongly predicts that (19,b) and (19,c) are ungrammatical. The formal SoT analysis predicts that they should both be grammatical, and therefore also meets with a problem, for it turns out to be the case that many speakers find (19,c) ungrammatical. Comrie's solution to this problem is that, for those speakers who reject (19,c), a collocation restriction should be formulated which says that a sentence is ungrammatical if it combines a tense that has past time reference with a 'time adverbial whose meaning incorporates future time reference' (p. 288). (Note that *on Thursday* in (19,b) 'does not have future time reference as part of its meaning' (p. 288). It is only because of the context that it is here interpreted as referring to the future.) Comrie concludes: 'If this collocation restriction account is correct, then the ungrammaticality of (19,c) (for some speakers) is accounted for by a principle independent of the sequence of tenses rule, and the validity of this rule is unaffected.' (p. 288). Since this statement is also applicable to our hypothesis – we will also have to accept the existence of some kind

5. The claim that the use of a relative tense form is the unmarked possibility is also in keeping with Poutsma's (1926:213) remark that, even when the present tense could be used in the *that*-clause, it is often the preterit that is used, 'owing to the fact that the shifting from the preterite in the reporting sentence to the present in the reported statement requires some effort of the mind'.
6. The restrictions in question appear to be rather intricate. Kiparsky and Kiparsky (1970) and Costa (1972) claim that the determining factor is the (non)factivity of the matrix verb. McGilvray (1974:32–3) and Riddle (1978:46–51) show that this is not correct. My own views on the subject are set out in section 1.2 below.

of restriction[7] – the conclusion would seem to be that cases like (19,c) provide evidence neither for nor against one of the competing hypotheses.

3 Comrie stresses that his SoT rule 'is completely independent of the meaning of the tense forms involved, it is a purely formal operation' (p. 290). It follows that sentences like (20,b), in which *arrived* is not backshifted to *had arrived* in spite of the past tense in the reporting clause, need to be accounted for.

(20) (a) Yesterday, Wendy said, 'I arrived yesterday.'
(b) Yesterday, Wendy said that she *arrived* the day before yesterday.
(c) Yesterday, Wendy said that she *had arrived* the day before yesterday.

Before going into Comrie's solution, I would like to point out that the fact that both (20,b) and (20,c) are grammatical is no problem for our hypothesis: (20,c) illustrates the normal rule that the past perfect is used when a situation that is incorporated into a past time-sphere domain is represented as anterior to another situation in that domain; (20,b) is an instance of a shift of domain: the *that*-clause situation is not incorporated into the existing domain but is represented as establishing a domain of its own (by the use of the preterit as absolute tense).

According to Comrie, (20,b–c) can be accounted for if we accept the following two principles:

a. 'Where the verb in the original utterance is in a past tense already, it remains unchanged after a main clause verb in the past tense in indirect speech'. (pp. 290–1)
b. 'In English, any event in the past can be referred to by the past. If that event is located prior to some contextually established reference point in the past, then the past may be replaced by the pluperfect'. (p. 292)

According to Comrie, principle a explains the grammaticality of (20,b), while principle b accounts for that of (20,c).

Several things should be noted here:

7. Comrie's formulation of the restriction may actually be too general. Harris (1982:266) offers the following example involving *tomorrow*:
(i) A person who registered as late as tomorrow will still be eligible to vote on Tuesday.
Notice also:
(ii) On Friday, Oswald told me that he would be abroad tomorrow.
(iii) I joined tomorrow. (from J. Conrad, *Youth* – mentioned in Rauh 1988:35)
These sentences too combine a tense 'part of whose meaning is past time reference' with 'a time adverbial whose meaning incorporates future time reference' (Comrie 1986:288).

a. Accepting principle a boils down to building an exception into the SoT rule. Our hypothesis, in contrast, accounts for (20,b–c) without any additional apparatus.
b. Principle b involves reference to a temporal relation. This means that Comrie is moving away here from a pure formal SoT hypothesis (which hinges on the claim that SoT happens independently of temporal relations) and is trespassing on the ground of the relative time hypothesis. (As a matter of fact, this is the second time that Comrie has recurred to semantics, for the notion of continuing applicability is also a semantic notion.)
c. Principle a is contradicted by the observation that *I loved her* cannot be reported as *He said he loved her*, unless there is a context that excludes interpreting *loved* as a relative tense form expressing simultaneity. (Since Comrie's principles are of a purely formal nature, they operate independently of the context. Principle a therefore wrongly predicts that *I loved her* can *always* be reported as *He said he loved her.*) (Note that our theory naturally accounts for the fact that this kind of report is not always possible, as it holds that a shift of domain is only admissible if it does not obscure the temporal relations.)
d. The two principles do not account for the restrictions that are to be observed in some of the following examples:

(21) (a) I spoke to her when she came home.
(b) He said he had spoken to her when she came/had come home.
(c) *He said he spoke to her when she had come home. (ungrammatical as a report of (21,a))

(22) (a) I spent some time with John, who felt lonely.
(b) She said that she had spent some time with John, who felt/had felt lonely.
(c) *She said she spent some time with John, who had felt lonely. (ungrammatical as a report of (22,a))

(23) (a) I was sad while I was alone.
(b) She said she had been sad while she was/had been alone.
(c) *She said she was sad while she had been alone.

(24) (a) John said: 'I told Betty that I was feeling ill.'
(b) John said that he had told Betty that he was feeling/*had been feeling ill.

(24) (a) I left the money where it was.
(b) He said he had left the money where it was/*had been.

In our theory the use of the past tense in the most deeply embedded subclauses of the (b) sentences (i.e. in the subclauses that have final position) follows naturally from the fact that these subclauses represent their situations as simultaneous with a past binding TO. The

fact that the past perfect is sometimes also possible follows from the fact that in some types of subclause indirect binding is possible too (see chapters 2 and 3). In Comrie's theory, principle b will have to account for the past perfects in the sentence-final clauses of (21,b), (22,b) and (23,b): we will have to say that the past perfect can be used in the third clause because the situation referred to is anterior to the situation of the first clause, whose verb is in the past tense (*said*). However, this explanation fails to account for the fact that the past perfect is not possible in the third clause of (24,b) and (25,b), where the same anteriority relation exists between the third clause and the first. (I grant, however, that this problem would disappear if Comrie could supplement his analysis of the pluperfect with a set of restrictions on its use. My own explanation (in terms of indirect binding) also crucially refers to restrictions: as we have seen, indirect binding is a restricted option, which is only available if it does not obscure the temporal relations.)

Another problem for Comrie's theory concerns the (c) sentences of (21)–(23). According to Comrie the past tense of the original report can always be retained (which is what happens in the first subclause of each (c) sentence) or can under certain conditions be backshifted to a past perfect (which is what happens in each of the final clauses of the (c) sentences). The condition under which the past perfect can be used is that there is some contextually given past reference point to which the relevant situation is anterior. This condition is satisfied in the (c) sentences, since the situation of the final clause is each time anterior to that of the first clause. Still, the (c) sentences are no grammatical reports of the (a) sentences. There are two ways in which this observation could be interpreted. One could say that the ungrammaticality results from the use of the past perfect in the final clause. If this is correct, the problem could perhaps be obviated by the formulation of a suitable restriction on the use of the pluperfect. (However, I think that without the framework developed in chapter 2, the restriction could at best be formulated. It would not be accounted for.) The alternative way of defining the problem is to say that the ungrammaticality results from the use of the preterit in the second clause. (The (c) sentences become grammatical if the pluperfect is used in that clause.) In that case the ungrammaticality of the (c) sentences is a real problem for Comrie's analysis, as this hinges on the claim that a past tense can always be retained.

In our theory the observation that the final clause of the (c) sentences cannot be in the past perfect is explained from the fact that this clause is of a type that must be bound by its matrix and that the relation to be expressed is that of simultaneity. (Using a past perfect would be an instance of indirect binding, but this is not possible here because the second clause is not bound: it shifts the domain. The

second and third clauses thus belong to a domain that is different from the one established by the first clause. Self-evidently this means that the first clause cannot bind the third.)

e. Last but not least, Comrie's principles a and b seem to imply that, in pairs like (20,b–c), where the *that*-clause refers to a situation that is anterior to the head clause situation, the preterit is the unmarked choice in the *that*-clause, whereas the use of the past perfect is a marked alternative. (Though Comrie does not state this himself, I think it follows from the formulation of claims a and b: compare the categorical 'remains unchanged' in a with 'the past may be replaced by the pluperfect' in b. At any rate, Comrie claims that the past tense can always be retained, and this claim will turn out to be incorrect.)

In our theory, it is the use of the past tense (which shifts the domain) that is the marked possibility. The unmarked alternative is the use of the past perfect (which incorporates the *that*-clause situation into the existing domain). Now, it is not difficult to prove that it is indeed the past tense that is the marked choice, i.e. the choice that is subject to restrictions. When a direct speech statement in the past tense is embedded under a verb of reporting in the past tense, the 'backshift' from the absolute preterit to the pluperfect is always possible. It is not subject to any restrictions whatsoever. The use of an absolute past tense, however, requires that the anteriority relation between the *that*-clause situation and the head clause situation should be clear from an adverbial, the context or the hearer's knowledge of the world (see above). If that is not the case, the preterit will make for a sentence that is unacceptable because it violates the generally accepted principles of conversation. Thus a sentence like *John said he had been ill*, if presented out of context, cannot be replaced by *John said he was ill* without violation of Grice's (1975) Maxim of Quantity (which requires that the speaker should make his contribution sufficiently informative to guide the hearer unequivocally to the intended interpretation).

Examples of domain shifting in temporal clauses confirm this conclusion. (I feel entitled to include time clauses in the discussion because Comrie (1986:292) does so to: he refers to the tense vacillation in *After he ate/had eaten breakfast, Xenophon put on his armour* as evidence for his principle b.) Like complement clauses, time clauses allow a shift of domain (entailing that the situation is no longer related to any other situation) only if the temporal relation between the situations of head clause and time clause are recoverable in some other way. In the above example, the relation is recoverable because it is explicitly expressed by *after*. But the relation is no longer recoverable if we replace *after* by a temporally vague conjunction such as *when*. In consequence, the Maxim of Quantity forbids replacing

When he had eaten breakfast, Xenophon put on his armour by *When he ate breakfast, Xenophon put on his armour.*

We must conclude that Comrie's treatment of sentences like (20,b) is not satisfactory. This is another case, then, where the formal SoT hypothesis is deficient, whereas the analysis argued here neatly accounts for the data.

Let us now consider some data which are not mentioned in Comrie (1986) and which provide further evidence against the formal SoT hypothesis.

1 Examples can be found of indirect speech sentences whose complement clauses use the present perfect although the head clauses are in the past tense:

(26) (a) The Secretary of Labor stated the other day that in the past couple of months there *have* only *been* 200 lost man days. (Ota 1963:115)
(b) Not too long ago ... a gentleman who'd been on foreign service mentioned that Latin America *has* not *had* its New Deal. (ibid.)
(c) The prime minister stated in Parliament yesterday that Britain's economy *has looked up* considerably during the past two months.

In our theory such sentences are explained in terms of a shift of domain: the head clause establishes a domain in the past time-sphere, whereas the subclause establishes one in the present time-sphere. As usual, the shift is semantically motivated. In this case the motivation is that the embedded statement concerns a period which started in the past and reaches up to the present. As noted above, the present perfect is the normal tense to refer to such a period.

Sentences like (26,a–c) run counter to the formal SoT hypothesis, because they exhibit no backshifting in spite of the fact that the verb of reporting is in the past tense. To salvage the SoT rule one would have to formulate still another *ad hoc* restriction providing for this kind of exception.[8] Needless to say, this would erode the rule even further.

8. Comrie might try to counter this argument by saying that in his theory the present perfect expresses current relevance, and that cases like (26,a–c) are therefore covered by the stipulation that was added to the SoT rule in order to account for the use of nonpast tenses when the situation has 'continuing applicability'. However, I will argue in chapter 7 that current relevance is no more than an implicature of the use of the present perfect, and that there are therefore many cases in which the present perfect is used without triggering a current relevance interpretation. This is the case, for example, in (i). However, the present perfect of (i) can be retained in indirect speech, witness (ii). The use of the present perfect in (ii) is therefore not covered by the SoT rule as it is formulated by Comrie.
(i) This bike has been mine, it has been my little brother's, and now it is used by my grandchild.
(ii) I was just telling my friend that the bike has been mine, that it has been my little brother's, and that it is now used by my grandchild.

2 We now come to a piece of evidence that is really detrimental to the formal SoT hypothesis. As noted in chapter 2, domain-internal relations within a domain established by a so-called 'indefinite' perfect are normally expressed by the system of tenses typical of the past time-sphere. That is, once the present perfect has established the domain, there is a shift of temporal perspective to the past time-sphere. The following sentences exemplify this:

(27) (a) He's been here once or twice, while his wife *was* on holiday.
(b) Have you ever spoken to her after she *had had* one of her fits?

The same shift of temporal perspective is to be observed in indirect speech sentences:[9]

(28) (a) I have never said that you *were* stupid.
(b) (He has met her several times but) he has never told his wife that he *had met* her.
(c) (He has met her many times and) he has always promised her that he *would* not *say* anything to her husband.
(d) I've always said that eight *was* a perfect number. (Leuschner 1977:98)
(e) So far British Embassy sources have said that Britain *was* opposed to an exchange of prisoners ... (Leuschner 1977:103)

When we reconstruct the direct speech statements that are reported in (28,a–c), we see that they are in the present, present perfect and future tense, respectively. This means that, in the terminology of the formal SoT hypothesis, (28,a–c) are instances of backshifting after a verb in the present perfect. Such sentences constitute an insurmountable problem for Comrie's formal SoT hypothesis, as they are incompatible with its most basic claim, viz. the claim that backshifting is 'a purely formal operation' (1986:290) which is triggered by the past tense of the verb of reporting in the head clause and which consequently cannot occur if the tense of the verb of reporting is non-past. Since sentences like (28,a–c) have a reporting verb in the present perfect, they present a case of backshifting which simply falls outside the scope of the formal SoT rule. (The present perfect is a nonpast tense. Comrie himself treats it this way on p. 290 of

9. We can also find indirect speech sentences that use a present perfect in the complement clause after a present perfect in the head clause:
(i) I have always maintained that I have never said that.
In this sentence the speaker does not use a relative tense incorporating the subclause situation into the head clause domain. Instead, he uses an absolute tense which shifts the domain, i.e. which creates a new pre-present domain. The semantic motivation for this shift is that the speaker wants the complement clause to refer to a time interval reaching up to the moment of speech (see chapter 7).

his article.)[10] This thrusts to the heart of the SoT hypothesis. This time the hypothesis can no longer be salvaged by the formulation of an *ad hoc* restriction on its applicability.

3 Another observation that leads to the same conclusion is that 'backshifting' can take place in complement clauses depending on a nonfinite verb form:

(29) (a) Bill seems to have told Mary: 'I feel depressed.'
(b) Bill seems to have told Mary that he felt depressed.
(30) (a) Bill regrets telling Mary: 'I feel depressed.'
(b) Bill regrets telling Mary that he felt depressed.

As has been stressed in chapters 1 and 2, nonfinite clauses are untensed. A nonfinite verb form just expresses a temporal relation. The relation in question may happen to be a relation with t_0, but it is not inherent in the meaning of a nonfinite verb form that it relates its situation (directly or indirectly) to t_0. This means that the use of the past tense form *felt* in (29,b) and (30,b) is not provoked by the use of a past tense form in the matrix.

It is important to see that, even if it were claimed that the perfect infinitive was a past tense form, the use of *felt* in (30,b) would still be unaccounted for, as the *that*-clause of (30,b) depends on a present infinitive. Clearly, the use of *felt* cannot be triggered by the form of the matrix verb. It can only be triggered by the temporal meaning of this verb, i.e. by the fact that the situation of telling is interpreted as anterior to that of the head clause. (This is the case not only in (29,b), where anteriority is formally marked by the use of *have*, but also in (30,b), where the anteriority relation is not marked on the infinitive (which is a 'sloppy simultaneity' form) because it follows from the meaning of *regret*.) This, then, is another telling argument supporting the claim that SoT is a semantic phenomenon, not a formal rule.

4 Up to now I have only adduced arguments that are based on observations. Obviously, these arguments in themselves are amply sufficient to demonstrate that the formal SoT hypothesis is untenable. However, it is worth pointing out that this analysis also raises all kinds of theoretical problems.

a. As has often been pointed out, a quotative analysis of indirect speech (i.e. an analysis which derives indirect speech sentences from direct speech ones) meets with the problem that some verbs (e.g. *realize*, *regret*) can be followed by an indirect speech (or indirect thought) clause but not by a direct quote.

10. The relevant passage is where Comrie defines 'a past tense' as 'one of the tenses given in the righthand column of (95)'. The column in question lists the preterit, the pluperfect, the conditional and the conditional perfect.

b. Comrie himself argues that the 'adaptation' of deictics other than tense (i.e. pronouns and adverbs of time or place) in indirect speech is not the result of a formal rule. The deictics in question are related either to the here-and-now of the reporting speaker (as in ***He** told **me** that **he** would do it **tomorrow***) or to the time of the head clause situation (as is the case for *the next day* in *He said he would do it **the next day***). Given this fact, it is hard to imagine that the verbal deictics should be analysed in a totally different way. Our hypothesis, which holds that verb forms in indirect speech are either absolute deictics (relating the situation to the here-and- now of the reporting speaker) or relative ones (relating the situation to the time of the head clause situation), treats verbal deictics in exactly the same way as the pronominal and adverbial ones.[11]

c. Comrie's analysis of indirect speech is not applicable to so-called 'free indirect speech', i.e. to the mode of discourse that we have labelled 'Represented Speech and Thought' (RST) (see chapter 2) because in RST there is no reporting clause in the past tense that can be claimed to trigger backshifting. (In order to account for the occurrence of SoT in RST Comrie would have to assume that an RST sentence is the result of ellipting the reporting clause of a direct speech sentence. However, this solution is not without problems. To mention one only: how is the hearer to know whether the verb of the deleted clause is *thought* or *believed, said to himself, realized, noticed*, etc.? That is, this kind of deletion would not satisfy the well-known recoverability requirement.) Our theory, however, does not need to posit a deleted hyperclause in order to account for the 'backshifted' tenses in RST. As noted in chapter 2 (sections 22 and 23), the speaker may assume the temporal standpoint of a participant in a past situation in order to suggest that the situation is expressed from his point of view. This means that he relates the situations to a particular past time (the time when the relevant character said, thought or observed something) rather than to his own here-and-now (i.e. t_0). The past time-sphere tenses in RST are thus explained as tenses that are bound by the past time that the speaker is focussing on. Our analysis therefore does not need to consider RST sentences as derived (through deletion of the reporting clause) from indirect speech sentences (which, on this hypothesis, are themselves derived from sequences involving a direct quote).

d. Comrie argues that he can account for the use of tenses in indirect speech without referring to the semantics of the temporal relations. He repeatedly makes statements of the following kind:

11. According to Ejerhed Braroe (1974:75–8), one cannot accept the quotative analysis without assuming that not only the tense forms but also the time adverbs shift. However, as she points out, this raises all kinds of problems.

> One feature of tense backshifting that takes place in indirect speech after a main verb in the past tense is that (apart from the stipulation concerning continuing applicability) it is completely independent of the meaning of the tense forms involved, it is a purely formal operation.
>
> (Comrie 1986:289–90)

However, statements like this raise intriguing theoretical questions. How can we reconcile this view (that the use of the tenses in indirect speech bears no relation to the meaning of the tense forms in question) with Comrie's definition of tense as 'the grammaticalization of location in time' (Comrie 1985:1)? The claim that backshifting is a 'purely formal' operation, which is 'independent of the meaning of the tense form involved', suggests that *was* in *John said he was ill* has the same meaning as *am* in *I am ill*. If this is not the case, Comrie's SoT rule is not a purely formal rule, as it has semantic import. But of course *was* is semantically different from *am*. Comrie himself admits as much when he formulates the 'collocation restriction' which leads to the reduced acceptability of *?On Friday, Oswald will say that he arrived tomorrow* as follows:

> what is wrong with [the above example] is the apparent direct conflict between the past tense, part of whose meaning is past time reference, and a time adverbial whose meaning incorporates future time reference. (Comrie 1986:288)

If past time reference is part of the meaning of *arrived* in the above example, then that element of meaning must have been introduced by the backshifting operation. But if so, we cannot maintain the fiction that the operation is a purely formal one. Moreover, once this is clear we must ask the question whether the element of past meaning results from the fact that the form of the verb is changed from present tense to preterit or whether it is the other way round (i.e. the choice of verb form is determined by the meaning to be expressed). In my opinion, there is little doubt that the latter hypothesis is the correct one. If this were not the case, the 'backshifting' rule would be without any motivation whatsoever.

If one reads the article carefully, one easily sees that Comrie does not succeed in eliminating semantics from his allegedly 'purely formal' rule. Each of the three additions to the original formulation of the rule crucially refers to the meaning of the tense forms. Moreover, even the first formulation encroaches on semantics, and Comrie himself points this out when he writes:

> the sequence of tenses rule . . . says either (for non-past main clause verb) that the tense of the original utterance is retained, or (for past tense main clause verb) that a transposition is to be applied to the

> tense of the original utterance. In either case, the tense of the original utterance is crucial.... Thus, there is a very real sense in which relative deixis, relative to the speech situation of the original utterance, is part of tense in indirect speech in English.
>
> (Comrie 1986:283)

However, if one accepts this, there is absolutely no reason left to assume the existence of a formal backshifting rule: given that tenses in indirect speech (at least, the 'backshifted' ones) express relative deixis, and given the fact that the relative tenses to be used for this if the binding time belongs to a past domain are past time-sphere tenses, the use of past time-sphere tenses in indirect speech after a head clause in a past tense is automatically accounted for.

e. One of the questions raised by the formal SoT hypothesis is why it is completely impossible to backshift the past perfect. Comrie's (1986:291) answer to this is that there is no verb form **had had seen.* (In fact **had have seen* would have been a more likely candidate.) However, if backshifting were a rule that was operative in English, it is rather unlikely that the form **had have seen* would not have existed. After all, this is not a very complex form. It is no more complex than *would have seen* or *will have seen* and even less complex than *was going to have seen, will have been going to see*, etc. This suggests that the non-existence of **had have seen* fails to be explained by the suggestion that this is too complex a form to be acceptable. Moreover, as noted by Wekker (1987:461), some speakers occasionally do use forms like *had have seen* in conditionals,[12] yet they never use this form as the backshifted version of a pluperfect in indirect speech. Furthermore, we observe that *would see* is not backshifted either in indirect speech. Yet, we note the existence of the forms *had been going to see* and *would have seen*. One of these would surely do if English needed a backshifted version of *would see*. The conclusion is clear: if the past perfect and the conditional tense are not backshifted, it is not for lack of suitable verb forms. It is simply because there is no such thing as a formal rule of backshifting in English.[13] The truth is that the past perfect and the conditional express a temporal relation in a past domain and that they do so irrespective of the location of the binding TO in the domain. This means that the reason why the past perfect is not backshifted when we report *I had already left when Mary*

12. The following example quoted by Wekker (1987:461) is from a corpus examined by Jenny Cheshire:
    (i) If I hadn't have gone to the hospital when I did, I wouldn't have got there at all.
13. This is not to say that the adaptation of tense forms that may result from a deictic reorientation in indirect speech (as when *I am ill* is reported as *He said he was ill*) cannot be called backshifting. The point I am making is that backshifting does not exist as a separate rule that operates in indirect speech only.

arrived as *Bill said he had already left when Mary arrived* is that in both sentences Bill's leaving is expressed as anterior to Mary's arrival. And a similar remark can be made in connection with the conditional tense: *I thought I would meet her here* is reported as *He said he had thought he would meet her there* without there being any need for 'backshifting' *would meet* because the temporal relation remains the same.

f. According to Comrie, one of the advantages of the formal SoT hypothesis is that it offers a unified account of the data and is therefore more economical than an analysis which is based on two or more principles:

> Sequence of tenses ... provides a homogeneous solution to the general problem of tense in indirect speech in English, covering all examples with a single set of principles. Of course, it would surely be possible to take absolute deixis in those instances where it happens to make the same predictions as sequence of tenses, then take some other principle (perhaps relative deixis) in some of the cases where absolute deixis makes incorrect predictions, plus perhaps some other principle where even this combination fails to make the correct prediction, and come up with a composite account which would have different principles or sets of principles operating in different sets of cases. This simply illustrates a commonplace of science, namely that given a particular hypothesis which covers a certain range of data, it is always possible to come up with some more complex theory that covers the same range of data. The beauty of the sequence of tenses rule is that it covers the whole range of data with just a single simple set of principles.
>
> (Comrie 1986:282–3)

At first sight Comrie's formal SoT analysis might seem to be simpler than our analysis (which holds that indirect speech complement clauses use either relative or absolute tense forms). However, it soon becomes clear that this is not true. The above quotation refers to the preliminary version of the SoT rule, i.e. to the version which covers only the 'backshifted' tense forms. Our analysis also needs only one principle to account for these, since the cases in which there is 'backshifting' coincide with the cases which we analyse in terms of temporal subordination. However, immediately after the above quotation Comrie has to add a couple of *ad hoc* stipulations to his rule in order to accommodate the various cases in which there is no 'backshifting'. In our theory all these cases are analysed as shifts of domain. This analysis, which works with two principles only, is therefore more economical than Comrie's, which works with a basic rule and a number of exceptions. Moreover, our analysis is applicable

not only to indirect speech subclauses but to subclauses generally. Comrie, by contrast, treats the 'backshift' of tenses in indirect speech complement clauses as an isolated phenomenon, unrelated to the behaviour of tenses in other environments. This not only makes his analysis less general and hence less economical than ours but also appears unjustified, since we notice a striking parallelism between what happens in indirect speech and what happens in other clauses (both embedded and unembedded). Compare:

(31) (a) The man said he *had heard* a shot in the street.
(b) The man phoned the police. He *had heard* a shot in the street.
(c) The man, who *had heard* a shot in the street, phoned the police.
(d) The man phoned the police because he *had heard* a shot in the street.
(e) The man reported the fact that he *had heard* a shot in the street to the police.

In the theory I have argued, the use of the past perfect receives the same explanation in each of these examples: the past perfect is used because the situation referred to is incorporated into the past time-sphere domain established by the head clause or preceding context, and because it is represented as anterior to the central situation of that domain. (The first element of the explanation accounts for the use of a past tense (the *past* perfect), the second for the use of a tense signalling anteriority (the past *perfect*).) Our hypothesis therefore makes a much larger generalization than the formal SoT hypothesis could ever achieve: it accounts for the fact that 'tense harmony' takes place not only in complement clauses after verbs of saying or thinking (where it is called SoT) but also in other types of noun clause (e.g. *It was clear that he* ***was lying***; *There was a chance that she* ***would die***; *The idea that she* ***was*** *perhaps dead had not occurred to me*), and in fact in any kind of clause that allows temporal subordination (e.g. *Last week he made a promise which he* ***would*** *not* ***keep***; *I got the job because I* ***knew*** *the manager well*; *They needed someone who* ***had studied*** *languages*).[14] It also accounts for the fact that tense harmony may be 'violated' both in complement clauses and in other types of clause (e.g. *Last week he made a promise which he*

14. The following show how tense harmony works in larger discourses:
(i) John has a very good memory, which he *has* always *thought* of as a great advantage. He *stores* up the things his friends *say* to him and if he *does* not understand them, he *goes* back to them for clarification after he *has turned* them over in his mind.
(ii) John had a very good memory, which he *had* always *thought* of as a great advantage. He *stored* up the things his friends *said* to him and if he *did* not *understand* them, he *went* back to them for clarification after he *had turned* them over in his mind.

will *never* ***keep***): tense harmony disappears whenever the speaker prefers to shift the domain.

Another generalization achieved by our theory is that it also accounts for the use of verb forms in nonfinite clauses. Consider the following examples (from Costa 1972:49):

(32) (a) Marmaduke believed himself to be of royal blood.
(b) *Marmaduke believed himself to have been of royal blood.

According to Costa, such examples show that SoT must not apply to infinitival complements, and the reason for this is that 'it is not the complement verb itself but the *tense* of the complement which is subject to SoT' (p. 49): since nonfinite clauses are untensed, SoT cannot apply to them. I would suggest a slightly different explanation. In infinitive clauses there is no special set of past sector tense forms to express temporal relations. All infinitive clauses use the 'present infinitive' form for simultaneity and posteriority, and the 'perfect infinitive' form for anteriority, irrespective of the tense of the head clause. Now the ungrammaticality of (32,b) is simply due to the fact that the wrong temporal relation is expressed: the belief *I am of royal blood* (which expresses simultaneity with the time that the belief is held) must be reported as *Marmaduke believed himself to be of royal blood* (where the infinitive also expresses simultaneity). If the belief is about a situation that is located before the time of believing, then it can be reported with the help of a perfect infinitive (e.g. *John believed himself to have been wronged by Bill*). (This is not possible in (32,b) because the belief in question concerns a situation which cannot be over at the time of believing.)

Note that, in a certain sense, there is SoT in nonfinite clauses. It should be clear from the above discussion that SoT should not simply be equalled with backshifting. Rather it should be equalled with temporal subordination. That is, there is SoT whenever a subclause situation is incorporated into the domain including the head clause situation. If the domain in question is a past domain, as in *John said he was ill*, this operation involves backshifting, because the complement clause uses a past sector tense for the expression of its temporal relation, whereas the corresponding direct quote uses a present time-sphere tense. But this does not mean that SoT (incorporation into the domain) should be identified with backshifting. Contrary to what is claimed by Costa, I therefore hold that there is SoT in (32,a), although there is no backshifting: there is SoT in the sense that the subclause situation is incorporated into the head clause domain.

Our conclusion from the above discussion of the various analyses of tense in indirect speech can be as follows. We subscribe to Comrie's (1986)

claim that there are data which cannot be accounted for by the absolute deixis hypothesis. We also agree that a 'relative time hypothesis' which holds that the tenses in indirect speech complement clauses must invariably be relative tenses is untenable. However, contrary to Comrie (1986), we have shown that the formal SoT hypothesis is not adequate either. Several arguments have been adduced to this effect, the most important of which are the following:

1 There are too many data that are incompatible with the basic claims of the formal SoT hypothesis, and which can only be treated as exceptions.
2 The formal SoT hypothesis does not cover cases of backshifting after a reporting verb in a non-past tense.
3 Neither does it account for the possibility of backshifting after a tenseless verb form.
4 The formal SoT hypothesis does not cover cases in which there is no backshifting in spite of the fact that the reporting verb is in a past tense. (I am referring here to the fact that tense forms expressing a temporal relation in a past domain are not backshifted in indirect speech.)
5 We have mentioned no fewer than six objections of a theoretical nature that can be made against the formal SoT hypothesis. One of the most serious is that it misses an important generalization because it treats tense in indirect speech complement clauses as totally unrelated to tense in other environments.

The theory we have argued, according to which complement clauses may use either relative tenses or absolute tenses (the former possibility being the unmarked one), is subject to none of these objections. It accounts naturally for all the possibilities that manifest themselves, both in indirect speech and in other contexts.

1.2. Restrictions on the use of absolute tenses in complement clauses

Perhaps the most relevant conclusion from the preceding section is that in our theory SoT need not be formulated as an independent rule. SoT is simply the phenomenon that a subclause situation is preferably incorporated into the domain of the head clause situation, rather than establishing a new domain of its own.[15] As we have seen, this option is

15. This mechanism also works in the complement clauses of languages that are said not to have a (formal) SoT rule. However, in such languages the system of tenses used to express temporal relations in a past domain is the same as is used to locate situations relative to t_0. Modern Hebrew is a case in point: it uses the present, past and future tenses to express simultaneity, anteriority and posteriority with respect to any TO, whether it is t_0 itself or lies in the past or future of t_0 (see Cole 1974). Russian is another well-known example of a language using this system. (However, literary

the unmarked choice in subclauses (and is therefore always possible). If the speaker decides not to incorporate the situation (in complement clauses: not to apply SoT) but to shift the domain instead, he must have a reason for doing so. In complement clauses there appear to be two such reasons: the tendency towards tense simplification and the fact that the speaker sometimes wishes to represent a past situation as still relevant at t_0. Let us have a closer look at these two possibilities.

1.2.1 Tense simplification

As noted in chapter 2, it is sometimes possible for the speaker to use a preterit instead of a past perfect or a conditional tense:

(33) (a) I went to bed after John *had left.*
(b) I went to bed after John *left.*
(34) (a) The boy left for the front. His fiancée *would be* dead when he returned.
(b) The boy left for the front. His fiancée *was* dead when he returned.

Such a shift of domain (within the past time-sphere) is probably due to the tendency to replace a tense that has a complex temporal structure by a simpler tense wherever possible. As we have seen, this operation is a marked choice, which is permissible only if it does not obscure the temporal relations between the relevant situations.

Of the two types of shift exemplified by (33,b) and (34,b) only the first appears to be possible in indirect speech complement clauses. This is clear from the following examples:

(35) (a) Yesterday, Wendy said: 'I *arrived* yesterday.' (Comrie 1986:291)
(b) Yesterday, Wendy said that she *had arrived* the day before yesterday. (ibid.)
(c) Yesterday, Wendy said that she *arrived* the day before yesterday. (ibid.)
(36) (a) Two days ago Betty said: 'I *will throw* a party tomorrow night.'
(b) Two days ago Betty said that she *would throw* a party last night.
(c) *Two days ago Betty said that she *threw* a party last night.
(37) (a) When she left he felt sad because he knew she *would* never *come* back.

Russian also allows the use of absolute tenses (shifting the domain) in complement clauses – see Brecht (1974:496).)

(b) *When she left he felt sad because he knew she never *came* back.

The ungrammaticality of (36,c) and (37,b) reveals that complement clauses are subject to a restriction that does not affect unembedded clauses like (34,b). The restriction concerns shifting the domain instead of expressing posteriority in the current past domain. In my opinion, the fact that the restriction affects complement clauses but not unembedded clauses is due to the fact that the latter anyhow represent situations as actualized (factual): both (34,a) and (34,b) state that the boy's fiancée was dead when he returned. This is not the case in (36)–(37). Here the past tense represents the situation as factual, but the conditional tense does not. The use of the conditional tense entails that the complement clause does not refer to a fact but expresses what the subject of the head clause was thinking in connection with the future. The subclause situation is represented as an expectation, i.e. as something hypothetical rather than factual. This kind of representation requires the expression of posteriority as a domain-internal relation. That is, the contents of the mental activity referred to by the head clause verb must be located in the same domain as the mental activity itself. A shift of domain is therefore excluded.[16]

The same explanation accounts for the fact (observed above) that (38,b), not (38,c) is the correct reported version of (38,a):

(38) (a) Yesterday, Henrietta said, 'I *will be* absent tomorrow.' (Comrie 1986:282)
(b) Yesterday, Henrietta said that she *would be* absent today. (ibid.)
(c) *Yesterday, Henrietta said that she *is* absent today. (ibid.)

As is clear from (38,a), the content of the clause to be reported is something that Henrietta expected to happen in the future. Sentence (38,b) does justice to this content, but (38,c) does not. The latter sentence no longer expresses Henrietta's expectation but represents the situation from the point of view of the reporting speaker, who treats it as a present fact.

16. This explanation accords with what Heny (1982:126) argues in connection with the example
(i) Sam believed that he *would be* in Boston. (*was)
According to Heny, the use of the preterit is excluded in such sentences because 'the object of belief must be relativized to the same time as the belief sentence itself or to some interval intersecting with that time. ... It would clearly do violence to the fact that Sam's belief was about the future – a future relative to that past time at which it was held – if we used the deictic past to fix the evaluation of the embedded proposition, rather than the available *would*.' In other words: 'a report about an attitude must not do violence to the structure of the attitude in question' (ibid.). See also Hüllen (1989:619).

It is worth noting that the above explanation of the ungrammaticality of (36,c), (37,b) and (38,c) is exactly the same as that which was offered in section 25 of chapter 2 to account for the difference between (39,a) and (39,b):

(39) (a) John left before Bill arrived.
(b) John intended to leave before Bill arrived.

We have pointed out that there is both semantic and syntactic evidence that *arrived* is an absolute preterit in (39,a) and a relative one in (39,b), and we have traced back this difference to the fact that *intend* creates an intensional domain. Intensional domains behave as temporal domains: every situation that forms part of what is to be interpreted intensionally must be located in the temporal domain established by the intensional verb. This is logical, since the use of an absolute preterit means that the speaker relates the situation to his own here-and-now (t_0) and (since he locates it in the past) represents it as factual. Needless to say, this excludes an intensional interpretation. (It may be interesting to note that in Dutch the conditional tense can replace the preterit in the time clause of (39,b), but not in that of (39,a). This is in keeping with what we have argued, for the use of the conditional is an instance of indirect binding. As it goes without saying that only situations that are conceived of as bound can be represented as indirectly (rather than directly) bound, it is predictable that the conditional cannot replace the absolute preterit of the time clause of (the Dutch equivalent of) (39,a).)

There is another observation that can be accounted for along the same lines. When discussing Comrie's backshifting rule I have pointed out that this rule incorrectly predicts that backshifting is never obligatory when the relevant clause has continuing applicability. In examples like (40,a–b) it is impossible to substitute the preterit for the present tense in the complement clause, in spite of the fact that the content of the clause does have continuing applicability.

(40) (a) This is John's wife. – Yes, I THOUGHT he *was* married.
(b) I knew you *liked* her.[17]

17. Huddleston (1984:152) notes that, although (i) is ungrammatical, (ii) is all right:
(i) *I knew you like her.
(ii) Did you know that Jill's in hospital again?
The reason for this difference is obvious. In (i) the *that*-clause belongs to the intensional domain created by *I knew* and must therefore be temporally subordinated. In (ii), on the other hand, the speaker represents the contents of the *that*-clause as something that is known by him but not necessarily by the hearer. The *that*-clause therefore need not be represented as belonging to the intensional (and temporal) domain created by *did you know*.

Here again the explanation is that temporal subordination is obligatory because the relevant clause belongs to an intensional domain.[18]

1.2.2 Present relevance of a past situation

Sometimes the reason why the speaker does not apply SoT (temporal subordination) to the complement clause is that he wants to express that the situation is somehow still relevant at t_0. In that case he locates the situation in one of the sectors defined in relation to t_0 instead of relating it to the past TO of the matrix. We can distinguish the following possibilities:

1 The speaker will locate the situation in the present sector (provided the shift of domain is allowed – see above) if he wishes to represent it as a situation that is, or may be, still valid at t_0.[19] According to Riddle (1978), this use of the present tense has the following 'implications' (I am using the term in a nontechnical sense):

a. The present tense suggests that the speaker believes in the truth of the statement.[20] Thus, (41) suggests that the speaker believes Bill's statement that he has a house in New York:

(41) Bill told me yesterday that he *has* a house in New York.

My own opinion is that Riddle's claim is actually too strong. The present tense may suggest that the speaker tends to believe the statement, but I do not think that it implies more than that.[21] It should also be noted that the use of the past tense in (41) would not necessarily

18. There may be exceptions to the rule that *that*-clauses that belong to an intensional domain must be temporally subordinated:
(i) Bill said/knew/believed that his wife was born in London.
(ii) Bill said/?believed that the Eiffel Tower stands in Paris.
However, it is never possible to use an absolute tense to report a past expectation concerning the future:
(iii) Tom said/knew/believed that Betty would help him. (*helped)

19. This minimally requires that the situation 'lasts long enough to be present both for the original speaker and for the reporter' (Huddleston 1969:794). This explains the ungrammaticality of **On coming in he told me it's five o'clock.*

20. Self-evidently, this can only be implied in indirect statements, not in indirect questions. There is no implication of truth in examples like the following:
(i) Back in the living-room, we asked Signor Moravia how he *goes* about his work. (Leuschner 1972:85)

21. The suggestion in question can easily be cancelled by the context:
(i) Harry told me that the world *is* flat, but of course I did not believe him.
Sentences like these make clear that, contrary to what is often asserted, the use of the present tense does not necessarily mean that the speaker 'is committing him/-herself personally' (Randriamasimanana 1987:195). This point is also made by Huddleston (1969:794–5), who notes that (what we call) a shift of domain to the present 'does not imply that the relevant proposition is true. But this is not to deny that, where the reporter has a choice . . ., he is less likely to orient the tenses with respect to his own speech act if he disagrees with the original speaker about the truth of the proposition.'

indicate that the speaker believed that Bill's statement was false. It would just mean that the speaker remained totally noncommittal about the truth of the statement.[22] (It follows that, contrary to what is sometimes claimed, the past tense can be used even if the complement clause expresses an 'eternal truth'. Using the present tense is never obligatory:[23]

(42) The Ancient Greeks did not know yet that the earth *is/was* round.)

b. In appropriate contexts, the present tense implies that 'the referent of the subject of the embedding verb has some current involvement in the state of affairs described in the complement' (Riddle 1978:11).[24] This current involvement may be:

i. Involvement with the subject of the embedded clause. For example, (40) suggests that the person in question is still Jack's room-mate:

(43) Jack told me that his room-mate *has* blue eyes.

ii. Involvement with the place referred to in the complement clause. This means that the 'place has some personal significance for that person at the time of the speech act that goes beyond mere awareness of its objective characteristics' (Riddle 1978:20). Compare:

(44) (a) I didn't mind the fact that Urbana *was/?is* so far from Chicago.

(b) At first, I didn't mind the fact that Urbana *was/is* so far from Chicago, but now the trip is getting to be a strain.

According to Riddle, the oddity of *is* in (44,a) 'is due to the fact that the present tense in the *that*-clause implies current involvement on

22. In some contexts, where the introductory verb itself makes clear that the speaker realizes that the indirect statement is false, the present tense cannot be used:
(i) This is John's fiancée. – Oh, that's a surprise! Somebody told me he *was/*is* married.

23. Jespersen (1931:152) states that in some cases the use of 'backshifted' tense forms 'is not required logically, but is due simply to mental inertia: the speaker's mind is moving in the past, and he does not stop to consider whether each dependent statement refers to one or the other time, but simply goes on speaking in the tense adapted to the leading idea. In many combinations it requires a certain effort to use the present tense, even if something is stated as universally true at all times or as referring to the present moment in contrast to the time of speaking.' Needless to say, this statement is quite consonant with our claim that temporal subordination is the default choice.

24. What Riddle calls presence or absence of 'current involvement' corresponds with what Blackstone (1962:17) calls 'immediacy or remoteness of interest'. (Blackstone offers an excellent discussion of the influence of this factor on the choice of (present or past) tense.)

the part of the subject in the distance between Urbana and Chicago, but the use of the past tense for the higher verb implies exactly the opposite' (p. 21).

When there is current involvement, the present tense can be used even if the truth of the statement is explicitly denied (Riddle 1978:27):

(45) Fred said that his girlfriend *was/is* stubborn, but I think it's the other way round.

c. In suitable contexts, the present tense suggests that 'the state of affairs being described is unresolved', i.e. 'that the state of affairs described by the sentence holds at the time of the speech act but that further development or change in status is expectable. It follows from this claim that the use of the present tense is incompatible with the description of a situation whose outcome is indicated contextually' (Riddle 1978:29). For example:

(46) (a) It was alleged in the newspapers recently that the mayor *was/is* the head of a big swindling racket. This is really going to shake up the union elections if it doesn't get settled soon.

(b) It was alleged in the newspapers recently that the mayor *was/??is* the head of a big swindling racket, but then they found evidence that she was framed.

According to Riddle, it is because the present tense may have one or more of these three 'implications' that the speaker may wish to use it in the complement clause. The choice of the present tense is therefore a marked choice. The speaker selects the present tense only if he wishes to convey one or more of these implications. Otherwise he chooses the past tense. Riddle (1978:46) traces this back to Grice's (1975:45) Maxim of Quantity:[25] 'if the use of the present tense in a particular sentence would convey implicatures which are superfluous to the intended communication, the speaker may choose the past tense as a strategy to avoid conveying these implicatures'.

It is worth pointing out that this conclusion accords perfectly with the analysis we have proposed. In our theory the use of the present tense (after a matrix in one of the past tenses) represents a shift of domain. A shift of domain is a marked choice (the unmarked choice being temporal subordination), which is made only in order to achieve a particular effect. The effect in question is to suggest one of the 'implications' referred to above.[26]

25. The relevant stipulation is the one saying that we should not make our contribution more informative than is required.

26. It is thus the use of the present tense, not the use of the preterit that conveys special shades of meaning. I therefore subscribe to the following claim made by Haberland (1986:252–3):

2 The shift of domain need not always be a shift to the present sector (t_0). In some cases it is a shift to the post-present. The speaker makes use of this possibility if he wishes to affirm his belief that the situation will take place after t_0:

(47) Someone said that John *would/will do* it tomorrow.

The use of *would* here implies that the speaker is just reporting what was said; *will* suggests that the speaker believes in the post-present actualization of the complement clause situation (or at least sees this actualization as a real possibility).

This use of *will* in noun clauses is quite similar to the use of *will* in restrictive relative clauses depending on a past tense matrix (e.g. *He bought something which he will need next week*). In both cases there is a shift of domain from the past sector to the post-present sector.

3 The speaker can also shift the domain from the past time-sphere to the pre-present:

(48) (a) The Secretary of Labor stated the other day that in the past couple of months there *have* only *been* 200 lost man days. (Ota 1963:115)
(b) Not too long ago ... a gentleman who'd been on foreign service mentioned that Latin America *has* not *had* its New Deal. (ibid.)
(c) Your father was telling me that you *have succeeded* in isolating the germ of that fever epidemic that's broken out at Lyon. (Allen 1966:80)

It should be noted that in such examples the present perfect situation is no longer represented as anterior to the matrix situation, because there is a shift of domain, which means that the present perfect is used as an absolute tense (representing the situation as anterior to t_0). Since the shift in question is also a shift from the original speaker's point of view to that of the reporting speaker, it does not normally occur after verbs like *believe*, *suppose*, *think*, etc. As we have seen, these verbs prohibit any shift of domain that would do violence to the structure of the belief in question. The following sentences are therefore peculiar:

> The use of the past tense in reports of a present tense model is sometimes described as expressing a certain reservation on the side of the speaker. ... I do not think that this is a quite fitting description. It is not the use of the past tense as such that expresses the reservation. It is the narrow scope reading of the present tense of the model that, by way of a Gricean implicature ..., indicates that the speaker does not want to go into a discussion of how far he or she subscribes to the claims of the speaker of the model. Reporting the model with the narrowest reading possible minimizes the report speaker's commitment. So if the speaker does not want to commit her- or himself to the truth of what the original speaker said, he or she has to use a past tense for a present tense of the model, but this being the unmarked case, the past tense in itself does not express any reservation on behalf of the speaker.

(49) (a) ?Sam believed that Bill *has been* here.
(b) ?Bill thought that Latin America *has had* its New Deal.

4 In the previous sections we have spoken about the possibility of shifting the domain from the past time-sphere to one of the present time-sphere sectors. In each case the motivation is that the speaker wants to relate the subclause situation to t_0 rather than to the past TO of the head clause. Similar shifts of domain appear to be possible within the present time-sphere too. Consider:

(50) Perhaps someone will tell the police that one of the cars *has disappeared.*

Sentence (50) is ambiguous, because the present perfect can be interpreted as locating the situation either before t_0 or before the situation-TO of the head clause. In the former case the present perfect represents a shift of domain to the pre-present; in the latter case it expresses anteriority within the post-present domain established by the head clause.

We might expect a similar kind of ambiguity in sentences like the following:

(51) Tomorrow I will tell your father that Bill *refuses* to pay us.

We might expect that *refuses* can refer either to t_0 (the shift of domain interpretation) or to the same time as *will tell* (the temporal subordination interpretation). Actually, the former interpretation is not available. That is, in (51) the time of refusing can be simultaneous with the telling, or it can be a time interval including both t_0 and the time of the telling, but it cannot be simultaneous with t_0 and anterior to the time of the telling. The latter idea requires the use of the present perfect (*has refused*), which is the normal tense to express anteriority in a post-present domain. There is thus no possibility of shifting the domain from the post-present to the present. However, this restriction is not absolute. If either of the two clauses contains lexical material that makes clear that the subclause situation can only be anterior to the head clause situation, then the present tense can be substituted for the present perfect:

(52) (a) You will soon forget that I *am* now *helping* you.
(b) One day you will remember what I *am* now *showing* to you.
(c) One day you will regret that you *are treating* me like this. (Dowty 1982:50)[27]

27. Dowty (1982:50–1) notes the difference between the grammaticality of this sentence and the ungrammaticality of (i), but offers no explanation:
(i) Smith will claim on the witness stand that he was/*is in Mexico.

The grammaticality of these examples contrasts sharply with the ungrammaticality of the present tense in the following:

(53) When you are dead, I will tell everybody that you *have treated*/**are treating* me badly.

Clearly, the use of the present tense to refer to a situation that holds at t_0, but no longer holds at the post-present TO, is possible only if there is some indication revealing these temporal relations. This is the case, for example, if the complement clause contains a time adverbial referring to t_0, or if the head clause verb is one that has the feature anteriority as part of its meaning (e.g. *regret*, *remember*, *forget*).

1.3 Further remarks

1.3.1 The theory of SoT argued here would seem to predict that tense harmony will more easily be violated in subclauses that are temporally less dependent on their matrix and therefore more easily allow the domain to be shifted or re-established. When dealing with the phenomenon of re-established domains in chapter 2, we noted that subclauses of reason, manner, comparison, result, concession and place are of this type. It should therefore be expected that SoT is easily violated in such clauses, and this expectation appears to be borne out. Ota (1963:111–13) reports that in the corpus he has investigated SoT is fairly strictly observed in noun clauses, time clauses, conditional clauses and restrictive relative clauses, but is often violated in subclauses indicating reason, manner, comparison and result. His comment is that the latter types of clause 'have more unmatched cases because what these clauses indicate can be semantically independent of the time indicated by the verb of the main clause', so that 'the choice of tense forms . . . is made in reference to the moment of speaking rather than in reference to the tense form of the verb of the main clause'.[28]

1.3.2 It should be noted, finally, that in relative clauses the speaker's choice between observing SoT and shifting the domain may have additional semantic effects. Consider:

(54) (a) Bill said he was referring to the girl who *lived* in London.
(b) Bill said he was referring to the girl who *lives* in London.

As in the case of noun clauses, the use of the present tense (*lives*) implies that the situation still holds at t_0, whereas the use of the preterit does not. From this it follows that the shift of domain also has another semantic effect: the statement that the girl lives in London at t_0 must be traced

28. Gelhaus (1974:121–2) reaches similar conclusions in connection with German.

back to the reporting speaker (i.e. the speaker uttering (54,b) at t_0).[29] In other words, the relative clause in (54,b) has a transparent (*de re*) interpretation, whereas the relative clause in (54,a) (which implies no more than that the girl lived in London at the time of the head clause situation (*said*)) has an opaque (*de dicto*) interpretation. (See also section 2.1.7 of chapter 3.)

2 THE PRESENT PERSPECTIVE SYSTEM VERSUS THE FUTURE PERSPECTIVE SYSTEM IN CONDITIONAL CLAUSES

2.1 Introductory remarks

As a second test case for the theory developed in chapter 2, we will now investigate the use of the tenses in conditional clauses referring to the post-present sector.

It is well known that conditional sentences[30] referring to the future can involve the future tense (*will*) in the head clause, but not normally in the conditional clause:

(55) If it rains/*will rain tomorrow, the game will be cancelled.

In the theory developed in chapter 2, this means that the head clause makes use of the FPS, whereas the *if*-clause uses the PPS and, moreover, treats the *if*-clause situation as (sloppily) simultaneous with the head clause situation.

Exceptions to the rule that *if*-clauses do not use the FPS, though possible, are relatively infrequent, and are therefore usually disregarded in grammatical handbooks. In the linguistic literature, however, the use of 'pure future' (nonmodal) *will* in *if*-clauses has received ample attention (see e.g. Poutsma 1926; Jespersen 1931; Palmer 1974; Quirk *et al.* 1972:781; Tregidgo 1974, 1979, 1980; Wekker 1976, 1980; Comrie 1982; Haegeman 1983a; Haegeman and Wekker 1984; Jacobsson 1984; Nieuwint 1986). I myself have tackled the subject in an earlier publication (Declerck 1984a). Though I still believe that the analysis presented there is basically correct, I think it can be much improved in the light of the theory that has been developed in chapter 2.[31] I will

29. This does not mean that (54,b) cannot be a report of Bill's statement *I am referring to the girl who lives in London*. It just means that the reporting speaker locates the situation of the girl living in London at t_0, thus assuming responsibility for an assertion for which Bill assumes responsibility in the original utterance.
30. I will use the term 'conditional sentence' to refer to a complex sentence consisting of a head clause and a conditional subclause.
31. My treatment of the subject in Declerck (1984a) differs from the other treatments in that I have not tried to reduce all the possible cases of *if* + *will* to a single type. Instead, I have argued that there are no fewer than nine different types of *if*-clauses in which (nonmodal) *will* appears, and that an adequate theory will have to account for all of these.

therefore reconsider all the observations and reformulate the conclusions in terms of that theory.

As we have seen, there are two systems available to the speaker for locating a situation in the post-present sector. The FPS uses the future tense or the future perfect to establish or re-establish a post-present domain, whereas the PPS incorporates a situation into an already established post-present domain. According to this theory, we may expect that the use of (nonmodal)[32] *will* in *if*-clauses may be due to two reasons: the speaker may use *will* either to represent the *if*-clause as establishing its own post-present domain or to express posteriority in a post-present domain. (In the former case *will* is a FPS form; in the latter it is a PPS form.) It is the aim of this section to prove that this expectation is borne out by the facts.

Let us first concentrate on conditional sentences of the type exemplified by (55), which use the PPS in the *if*-clause. (Henceforth I will refer to such conditionals as 'standard conditionals' because they are much more frequent than conditionals that involve the future tense in the *if*-clause.) Conditionals of this type have the following characteristics:

1 They express an 'open' condition, i.e. a condition which may or may not be fulfilled in the future, but whose fulfilment is seen as a real possibility. Thus, in

(56) If you refuse that deal, the papers will publish it.

the speaker clearly reckons with the possibility that the addressee may refuse the deal in question.

Not all conditionals express an open condition. Conditions may also be 'closed', 'hypothetical' or 'counterfactual'. A closed condition is one whose fulfilment is taken for granted, i.e. the fulfilment is treated as a present fact,[33] as in

(57) (a) If this is the famous Mona Lisa, I am not impressed by it.
(b) A: (*child in kitchen shouting to mother upstairs*) Mummy, the kettle is boiling.

32. In this discussion we will only consider the 'pure future' use of *will*. As is well known, *will* can also be used as a modal auxiliary (expressing some kind of volition or willingness). In this meaning it can easily appear in *if*-clauses (e.g. *If you will lay the table, I will wash the dishes*).

33. As a matter of fact, the condition expressed in a closed (or 'factual') conditional is not treated as 100 per cent factual: there is still a trace of evidential doubt. This is clear from (57,b), where the mother can say *If the kettle is boiling* . . . only because she has no first-hand knowledge (she only has the child's words to rely on). She could not use the same conditional if she were in the kitchen herself and could see with her own eyes that the kettle was boiling (see Akatsuka (1986:341ff.); where it is argued that 'the prototypical meaning of "if *p*" is the speaker's uncertainty/uncontrollability of *p*'). So, even in a closed conditional the *if*-clause retains at least a trace of its basic meaning of supposition. (This is also the case in (57,a), where the speaker standing before the Mona Lisa apparently finds it difficult to believe that he is actually looking at the famous painting.)

B: (*mother*) If the kettle is boiling you must take it off the fire.

A hypothetical (or 'theoretical') condition is one that represents the possibility of fulfilment as rather remote (i.e. as purely theoretical). For example:

(58) If you parked/were to park your car there, it would be towed away.

Finally, a condition is counterfactual ('rejected') if it is represented as unfulfilled in the past, present or future:

(59) (a) You would not say that if you were older.
(b) If you had parked your car here, it would have been towed away.
(c) If you had come tomorrow instead of today, you would have found my wife at home.

2 Standard conditionals express what I will call a 'situation-condition',[34] not a 'truth-condition'. A truth-condition is one that can be paraphrased as *if it is true that...* or *if it is the case that...*, as in

(60) If this is love, I don't want it. (Tregidgo 1980:188)

Sentences like these may be called truth-conditionals because the head clause typically refers to a conclusion that is conditional on the truth of the assumption made in the *if*-clause. However, we speak of a situation-conditional when the fulfilment of the condition expressed in the *if*-clause consists in the actualization of a situation (i.e. the happening of an event, the performance of an action, the occurrence of a process or the existence of a state). Thus, the following are different examples of (open) situation-conditionals:

(61) (a) If that window breaks, you will be held responsible.
(b) If you say that in public, the papers will publish it.
(c) If the town continues to change its appearance that rapidly, you will hardly recognize it when you come back in ten years' time.
(d) If our next child is a girl, we shall call her Mary.

As pointed out by Tregidgo (1980:188), a present tense in a situation-conditional refers to the future, whereas 'in a truth-condition a present tense ... is an independent present, referring to present time and not future'.

34. Tregidgo (1980:187) uses the term 'event-condition'. However, since the 'event' in question may also be an action, process or state (see below), it is better to stick to our terminology and speak of 'situation-condition'.

3 The logical relation expressed in standard conditionals is 'if p, then q'. This means that the head clause situation is represented as conditional on the *if*-clause situation. In other words, the head clause is not established as true independently of the condition expressed. (Note that this does not mean that the head clause situation can only hold if the condition is fulfilled. If I say *If the weather is good, John will go to the seaside*, and it turns out to be the case that the weather is not good but John goes to the seaside all the same, the sentence is not falsified. All that matters is that the speaker, when uttering the sentence, represents John's going to the seaside as conditional on the weather.)[35]

4 In English, standard conditionals make use of the PPS in the *if*-clause. This means that the *if*-clause is always temporally subordinated. The reason for this is the close logical relation between the two clauses: the *if*-clause provides the hypothetical basis for the assertion made in the head clause. The two clauses thus form one 'hypothetical domain', which (like the intensional domains discussed above) behaves as a temporal domain. This means that the English tense system treats the expression of the logical relation between the two clauses as more important than the expression of the post-present location of the *if*-clause situation (which is anyhow recoverable in other ways). (If the latter were judged the more important, the FPS would be used in both clauses. However, this would mean that the two situations were located in different domains, so that the close relation between them would be left unexpressed.)[36]

5 The fact that the PPS is used in the *if*-clause means that the *if*-clause situation is temporally subordinated to the head clause situation. In most cases the relation expressed is that of simultaneity. However, as noted in chapter 2, this simultaneity relation is often of the 'sloppy' kind. In an example like

(62) If you say that in public, the papers will print it.

the *if*-clause situation is not simultaneous with the head clause situation in the strict sense of the term. Since the head clause situation

35. It is true, however, that the sentence is normally interpreted as meaning that *John will go to the seaside only if the weather is good.* According to Eilfort (1987:56) this interpretation is due to Grice's (1975) Maxim of Relevance, which requires that 'in a conditional utterance, all of the relevant conditions are given. Thus, in the case of a single antecedent, there is an implicature that the antecedent is both a necessary and a sufficient condition on the consequent.' According to James (1986:458–9) the idea of 'sufficient condition' follows from the Maxims of Quantity and Quality. However this may be, it is clear that some kind of implicature is involved, since the idea 'if and only if' can be cancelled, e.g. by the addition of *even*. (For other discussions of the subject, see Geis (1971:562) and Geis and Zwicky (1971), where this is treated as a case of 'invited inference', and de Cornulier (1983), who speaks of 'the presumption of exhaustivity'.)

36. In many European languages (e.g. French, Dutch, German) the speaker may express either idea and can therefore choose between the FPS and the PPS.

is represented as conditional on the possible future actualization of the *if*-clause situation, the actualization of the *if*-clause situation can only be temporally prior to that of the head clause situation. Strictly speaking, the relation is therefore one of anteriority. In spite of this, the relative tense used in the *if*-clause is the present tense. And this is usually the case in standard conditionals. The present perfect is normally only used when the speaker wants to express explicitly that the *if*-clause situation is anterior to the head clause situation, as in *If you have not left by then . . .*[37] Otherwise the temporal relation between p and q is considered to be (sloppy) simultaneity.[38] In fact this does not seem unjustified. Since the *if*-clause expresses the possible future actualization of one situation, while the head clause refers to the actualization of another situation logically dependent on this, both situations must necessarily belong to the post-present sector.[39]

37. As a matter of fact, it is only relational adverbials like *by 5 o'clock* that often provoke the use of a verb form expressing anteriority (e.g. *I will punish you if you haven't finished by 5 o'clock*). Even when the two clauses contain adverbials referring to different times, they may still be related to each other as if they were simultaneous (e.g. *If John doesn't do/hasn't done it today, I will do it myself tomorrow*). This means that the speaker will represent the two situations as simultaneous if he is just thinking of them as two post-present situations that are logically connected with each other. It is only when he wants to draw explicit attention to the fact that one situation is over when the other takes place that he selects a verb form expressing anteriority.
38. What I am arguing accords perfectly with what Tregidgo (1979:195) writes in connection with the sentence *You will put your coat on and you will feel warmer*:
    > This clearly indicates two closely-related events, the first leading directly to the second. If we subordinate the first clause to the second we also subordinate the first tense to the second:
    >
    > *After/When/If you put your coat on, you will feel warmer.*
    >
    > The present tense of *put* here does not indicate present time, but the *same* time as that of feeling warmer, or more exactly a time immediately preceding it and leading up to it. The use of 'will put' as a non-modal future is not possible here because, if subordinate, it would indicate that putting on the coat is later than feeling warmer; and if not subordinate, it would no longer relate the two future events at all.
39. This claim would seem to be contradicted by the fact that some conditional sentences with an *if*-clause referring to the future use the present tense or present perfect in the head clause:
    (i) (a) If you help me do the washing-up, I *promise* I will give you a fiver.
        (b) If John doesn't win this race, I *don't think* he will ever compete again.
    (ii) (a) If you don't give me the money, you'*re* a dead man.
        (b) If he doesn't move his queen now, he'*s lost* the game.

    In spite of appearances, I do not think that these sentences should be treated as distinct from standard conditionals. The examples in (i) can be analysed as derived from (iii,a–b) by fronting of the *if*-clause:
    (iii) (a) I promise I will give you a fiver if you help me do the washing-up.
        (b) I don't think John will ever compete again if he doesn't win this race.

    As noted by Postal (1970:145), G. Lakoff (1972), Bell (1974:131), Geis (1975:133), etc., this kind of preposing of an adverbial element to the front of a higher clause is not uncommon with several types of adverbial, provided the verb of the higher clause is of the right type. The phenomenon is further illustrated by examples like
    (iv) (a) Last night I think Sam smoked pot.
        (b) When/he can get it cheap, I think Sam smokes pot.
        (c) At midnight I expect him to be here.
        (d) Tomorrow I expect to have time to read this.

Moreover, there is an implicature that the lapse of time between them is the shortest possible timespan that is allowed by the adverbials (if any) and by the pragmatics of the situations and the context.

In sum, the use of the PPS in *if*-clauses is subject to several conditions, which must be satisfied simultaneously:

1 The condition expressed in the *if*-clause must be an open condition.[40] This means (a) that it is a condition which is not yet fulfilled at t_0, and (b) that its future fulfilment is conceived as a real possibility.
2 The condition must be a situation-condition. This means that it must not concern the (present) truth of a statement (as in *If that man is a professor, I am Napoleon*) but rather the (future) actualization of a situation.
3 The logical structure of the sentence must be of the type 'if p, then q'. This means that there must be a close logical relation between the two clauses: the head clause situation must not be asserted as holding independently of the *if*-clause situation, but must be represented as conditional on it.

As to (ii,a–b), these are examples of a shift of temporal perspective (see section 20 of chapter 2): the post-present domain established by the head clause is treated as if it were a present or pre-present one. In this way the actualization of the head clause situation under the given condition is guaranteed in the speaker's here-and-now: the potential future situation is treated as if it were a present factual one. In (ii,a) this gives more force to the threat expressed in the head clause.

There is one type of conditional, however, in which the head clause situation does not lie in the post-present, although the *if*-clause situation does:

(v) If our visitors don't turn up, we'*ve* just *done* a lot of work for nothing.

This sentence should be compared with (vi):

(vi) If the visitors don't turn up, we *will have done* a lot of work for nothing.

This is a standard conditional whose meaning is something like 'If the visitors don't turn up, it will be the case that we have done a lot of work for nothing.' As this paraphrase reveals, what is conditional here on the possible future actualization of the p-situation is a (future) conclusion concerning a situation anterior to the conclusion (and possibly anterior to t_0). In chapter 2 we have noted that this idea of 'future evaluation of an anterior situation' is expressed by the future perfect. In that case there is a 'shift of temporal focus' to some post-present TO. However, if the work has already been done at t_0, the conclusion can also be presented as a conclusion that holds at t_0 (rather than at some post-present TO). In that case the speaker may use either the preterit or the present perfect to refer to the anterior situation. This is what happens in (v). However, it is important to see that what is conditional on the post-present *if*-clause situation is not the pre-present head clause situation but the post-present evaluation of this.

40. Closed conditionals like our above *If this is the Mona Lisa, I am not impressed by it* do not use the PPS in the *if*-clause. That is, they do not use the present tense as a relative tense (expressing simultaneity in a post-present domain) but as an absolute tense (referring to t_0). The fact that the other absolute tenses can also easily be found in this type of *if*-clause confirms this analysis:
    (i) If this *was painted* by Da Vinci, I am not impressed by it.
    (ii) If the museum *has paid* a million pounds for this, it still does not mean that everybody is impressed by it.
    (iii) If the museum *will/is going to pay* a million pounds for this painting, it still does not mean that everybody is impressed by it.

4 The speaker must feel there to be a temporal relation between p and q which is sufficiently close to warrant its representation as a relation of (sloppy) simultaneity.[41]

2.2 Types of conditionals that use the FPS

If the above claims are correct, all the cases in which (nonmodal) *will* is found in a conditional clause should be explainable as violations of one or more of these conditions. In order to see whether this is correct I will now discuss the various types of *if* + *will* clause that were distinguished in Declerck (1984a) and which cover all the cases that have been noted in the linguistic literature.[42]

2.2.1 Type I

Type I is the type that has been most extensively discussed in the linguistic literature. It consists of such sentences as the following:

(63) (a) Now if all the dresses *will* be finished by about next Monday, why don't you bring them yourself? (Jespersen 1931:400)
(b) If he *will* go to China next year, then we should publish his book now. (Glendinning-Johnson 1975:76)
(c) If he *won't* arrive before nine, there's no point in ordering dinner for him. (Quirk *et al.* 1972:781)
(d) If the lava *will* come down as far as this, we must evacuate these houses immediately. (Close 1975:256)

Sentences like these do not satisfy the requirements that standard conditionals must come up to. One difference is that they represent p as a 'closed' condition rather than as an open one. That is, the future actualization of the situation described in the *if*-clause is represented as something that is taken for granted rather than as a mere possibility. The implication is clearly that the speaker knows (or at least assumes) that the

41. See Tregidgo (1980:187): in standard conditionals 'the two time-references are closely linked, that of the condition leaning, so to speak, on that of the consequence, and being necessarily simultaneous with it or immediately preceding it. ... In these circumstances, the *if*-clause cannot contain an independent future tense.'

42. The subcategorization into nine types that will now be presented and the discussion of these types differ in a number of respects from the discussion in Declerck (1984a). Some of the changes are due to remarks made by P.S. Tregidgo in a letter he sent me in reaction to this article. Through this letter I also got acquainted with Tregidgo (1979) and Tregidgo (1980) – two brief articles which had escaped my attention and which accord very well with the theory I argued in 1984. I am thankful to Mr Tregidgo for his letter and apologize for overlooking his articles (which I now realize were among the best in the linguistic literature on the subject).

Recently, my 1984 article has received some attention from Nieuwint (1986). Unfortunately, Nieuwint's summary of the article does not do justice to its contents, and the subsequent criticism is therefore rather unfair. I leave it to the reader to judge this for himself.

condition is fulfilled, i.e. that the situation expressed in the *if*-clause will actualize. The situation referred to in p is seen as something factual, even though it lies in the future.[43] For this reason the *if* of the *if*-clause can be paraphrased as 'if it is the case that', 'if it is a fact that', 'assuming it is true that'.[44] (As is clear from the latter paraphrase, this type of conditional does not imply that the speaker commits himself to the truth of the *if*-clause. What is implied is no more than that, for the sake of the argument, the speaker entertains the assumption that the prediction made in the *if*-clause is correct.)[45]

Apart from the fact that the *if*-clauses express a closed condition, (63,a–d) also differ from standard conditionals in that the condition expressed is always a truth-condition rather than a situation-condition (see the above paraphrases), and in that the logical and temporal relations between p and q that are typical of standard conditionals are absent. That is, whereas in a standard conditional like *If she comes I will be happy* we have 'if p, then q' in the sense that the possible future actualization of q depends directly on the possible future actualization of p, the 'condition' for q in (63,a–d) is the expected truth of the prediction that the p-situation will actualize, not the actualization itself. (Note that the actualization of the p-situation may even follow that of the q-situation, as in (63,b) or (63,d).) Moreover, what depends on the expectation that the p-situation will actualize is not the actualization of the q-situation but

43. Leech (1971:59–60) argues that the use of the present tense in standard conditionals like *If she comes I will be happy* follows from the fact that the situation referred to 'is not a prediction but a fact that is taken as given': 'a conditional sentence . . . has the structure "If *X* is a fact, then I predict *Y*."' Hence in the *if*-clause, it is appropriate to use the Present Tense, with its assumption of factual certainty, rather than the predictive auxiliary *will*.' It should be clear now that this explanation (which treats the present tense in the *if*-clause of a standard conditional as an absolute tense) is quite unacceptable. A standard conditional does not represent a situation as a fact. (If it did, a sentence like *If he comes, which is rather doubtful, I will ask him for an explanation* would be semantically anomalous, which it is not.) Moreover, in those cases in which the *if*-clause does treat the future as a fact, it is the future tense that is used in the *if*-clause, not the present tense (see (63,a–d)).
44. The observation made by Palmer (1979:118) that examples like (63,a–d) are seldom found because 'it would be much more usual to use *be going to* in such circumstances' is in keeping with this. As noted by Leech (1971:54–7), *be going to* is the normal form to express 'future fulfilment of the present', i.e. to represent a future event as depending on a present intention or cause. Now, if there is subjective certainty on the part of the speaker about the future occurrence of p, this can only mean that the speaker knows or assumes that someone has the intention to perform p (Leech's 'future of present intention') or that there are perceptible signs in the present of what is going to happen in the future ('future of present cause'). In either case there is the expectation that p will happen because all necessary conditions for its occurrence have been met. Since *be going to* is the natural auxiliary to express this idea, it can easily be substituted for *will* in the *if*-clauses of (63,a–d) and may actually be more frequent than the latter in such sentences.
45. As noted by Comrie (1982:148), this type of conditional requires that the content of the *if*-clause be contextually given. In most cases this means that the prediction in question has been made in the preceding context. The speaker then expresses what in his opinion follows from the assumption that the prediction is correct.

a decision or wish (i.e. a cognitive act) concerning the actualization of the q-situation. There is thus no direct cause–effect relation between two future actualizations, but a more deductive relationship between two present cognitive acts (viz. the expectation that the p-situation will hold and the conclusion that the action constituting the q-situation should or should not be performed). It is because of this deductive nature (the conditional having the logical structure 'p, hence q') that the *if*-clause must necessarily precede the head clause – a restriction which does not hold for standard conditionals. Another consequence is that it is not possible to substitute *provided* for *if* in (63,a–d). Rather, the meaning of *if* comes very close to that of *as* or *since*. The *if*-clause expresses no real condition but the reason why the speaker has reached a particular conclusion.

In sum, none of the above mentioned four conditions for incorporating p into the domain established by q (i.e. for the use of the PPS) is fulfilled. The speaker therefore has to use the FPS in the *if*-clause. As in head clauses, this use normally involves *will*,[46] but other means of expressing futurity (e.g. *be going to* or the present tense) may also be used if particular semantic conditions are satisfied. (Roughly speaking, *be going to* represents the future as having its roots in, or as otherwise predictable from, the present (see footnote 44); the present tense represents the future as somehow predetermined by the present.) Thus, there is a perfect parallelism between the verb forms in the independent clauses of (64,a–d) and those in the *if*-clauses of (65,a–d):[47]

(64) (a) The decision will/is going to be taken tomorrow.
(b) It is going to/*will rain.
(c) The delegation arrives/is arriving tonight.
(d) All America is going to become middle-class.

(65) (a) If the decision *will/is going to* be taken tomorrow, we will read all about it in tomorrow's evening papers.
(b) If it *is going to/*will* rain, we had better take our umbrellas.
(c) If the delegation *arrives/is arriving* tonight, we must see that we are ready to receive them.

46. Our claim that the use of *will* in *if*-clauses follows from the fact that the FPS is used, which locates a situation as posterior to t_0, accords with the following claim by Wekker (1976:2): 'I claim that this use of the non-modal future in the *if*-clause always implies marked future reference with the time of the utterance as reference-point.' (However, this general claim leads Wekker to reduce all instances of *if* + *will* clauses to one single type. I do not agree with this, as there are clear differences between the nine types that will be identified below.)

47. Jacobsson (1984:139) objects to the theory that the use of *will* and *be going to* is the same in *if*-clauses as it is in independent clauses. He points out that, although both **It rains* and **It will rain* are unacceptable in isolation, the former can occur in an *if*-clause, while the latter cannot. Our answer to this is that neither form can be used in an *if*-clause of type I, since neither form is possible in the FPS. However, *it rains* can be used in the *if*-clause of a standard conditional (e.g. *If it rains, we'll stay at home tomorrow*), where the present tense form follows from the use of the PPS.

(d) If all America *is going to* become middle-class, will not all America become Republican? (Jacobsson 1984:132)

To close off the discussion of conditionals of type I, we can point out that the observation that the time of the occurrence of the p-situation is not related to that of the q-situation in this type is borne out by the fact that q need not even refer to the future at all (whereas p always does):

(66) (a) If Claude will be here tomorrow, there's no need to call him now. (Close 1980:107)
(b) If you will see him tomorrow, why are you phoning him now?
(c) If, as you say, this cloth will shrink when it is washed, it can't be genuine dralon.
(d) A: Who will be given the job?
B: I think it will be George.
A: George? But... Tell me, *if it will be George, how come I have not seen his application?*

In these examples the conclusion arrived at does not concern some post-present situation. In (66,a–b) it concerns a present course of action; in (66,c) it is a conclusion concerning the truth of q (which is a gnomic statement expressed by a 'universal' present); in (66,d) it is a question concerning the pre-present. It is clear, then, that, unlike standard conditionals, type I conditionals do not relate p temporally to q.

As a matter of fact, this property of type I conditionals is wholly predictable from the fact that type I conditionals are truth-conditionals, i.e. conditionals in which the head clause expresses a conclusion which is conditional on the truth of what is said in the *if*-clause. It is a general characteristic of truth-conditionals that both clauses make use of absolute tenses, so that all kinds of combinations of tenses are possible. For example:

(67) (a) If John is in London, Bill is/must be there too. (present + present)
(b) If Oswald didn't shoot Kennedy, then someone else did. (Braine 1979:36) (past + past)
(c) If John will be in London tomorrow, Bill will be there too. (future + future)
(d) If John is in London, Bill will soon be in London too. (present + future)
(e) If John has lived in London for five years, he knows the way to Trafalgar Square. (present perfect + present)
(f) If John was in London yesterday, Bill will soon go to London too. (past + future)
(g) If Bill is alive, then Mary didn't kill him. (present + past)

(h) If the lights are not on, you haven't pressed the button hard enough. (present + present perfect)
(i) If John will go to London tomorrow, then (surely) Fred sent him an invitation. (future + past)
(j) If adultery won't necessarily land me in Hell then it makes it awfully tempting to indulge now and pay later. (Jacobsson 1984:136) (future + present)
(k) If John will have finished his book by next month, he has worked much faster than I had expected him to do. (future perfect + present perfect)

etc.

Any such conditional having absolute future time reference in the *if*-clause is a type I conditional.[48]

2.2.2 Type II

The second type of conditional in which *will* can appear in the protasis involves an *if*-clause which does not stipulate a condition for the occurrence of q but expresses a felicity condition for uttering the head clause. That is, the *if*-clause is of the type of conditional clause that does 'no more than indicate the conditions under which the main clause has any relevance or news value' (Palmer 1974:143). The following sentences illustrate this kind of *if*-clause. Those in (69) involve *will* in the protasis and therefore exemplify a second type of 'exception' to the rule that pure future *will* cannot occur in *if*-clauses:

(68) (a) If you want to know, I've sold my house.
(b) If you're hungry, there are some biscuits on the table.
(c) If you want to go out, don't forget it's snowing.
(d) If you need me, I'll be at Finster's house. (Eilfort 1987:56)

48. If both clauses use the future tense, both establish a post-present domain of their own (since the future tense is an absolute tense in both cases). In that case no temporal relation is expressed between the two domains, which means that the sentence leaves vague whether the head clause domain is anterior to, posterior to, or simultaneous with the *if*-clause domain. In some cases this information is clear from contextual or pragmatic elements (as in (i,a–b), in some cases it is not (as in (ii)):
(i) (a) If the dress will be ready tomorrow, I'll pay for it straightaway. (Tregidgo 1979:196)
(b) If the dress will be ready tomorrow, I'll be able to wear it tomorrow night. (ibid.)
(ii) If he'll be left destitute, I'll change my will. (Palmer 1974:149)
As noted by Palmer, (ii) can mean either 'I will change my will if he will be left destitute as a result of that change' or 'I will change my will if in its present form I shall leave him destitute.' In our theory this follows simply from the fact that the sentence offers no clue as to whether the head clause domain is later in time than the *if*-clause domain or vice versa.

(e) If you're interested, Gerald Ford's real name is Leslie L. King, Jr. (ibid.)

(69) (a) If you *won't* be with us for the late news, goodnight. (Haegeman and Wekker 1984:53)[49]

(b) If it'*ll* make you feel any better, we know now that it wasn't your fault.

(c) If it *will* not be too much trouble, please give this letter to the captain when you see him next month.

In examples like these the *if*-clause states a condition for the relevance of uttering the head clause. Following Haegeman (1984) we will call them 'utterance-conditionals'.[50]

Utterance-conditionals involving *will* in the protasis differ from type I conditionals in at least two respects: First, they express a condition that is not closed, as in type I, but open (in the sense that the speaker treats the actualization of the p-situation as a future possibility, not as a present fact); second, type II conditionals are not deductive in the sense that type I conditionals are. For this reason it is hardly possible to insert *then* before the head clause. (Since *then* signals either a cause–effect relation or a deductive relationship, its insertion is possible both in type I and in standard conditionals like *If she comes I'll be happy.*)

Utterance conditionals like (69,a–c) use the FPS in the *if*-clause because they differ from standard conditionals in three respects. First, they express a truth-condition rather than a situation-condition. Thus, (69,a) can be paraphrased as 'If it is true/the case that you will not be with us for the late news, goodnight.' Second, they do not express the logical formula 'if p, then q'. That is, they do not represent the actualization of the p-situation as a condition for the actualization of the q-situation (see the impossibility of adding *then* to the head clause). Because of this they have several characteristics that are not shared by standard conditionals: (1) they cannot be reformulated in the form of '(*Let*) p *and* q' (whereas standard conditionals can – see Haiman (1983) for some discussion); (2) they do not have the 'if and only if' implicature that is typical of standard conditionals (see footnote 35);[51] and (3) they

49. Haegeman and Wekker (1984:53) add the following comment to this example: 'Goodnight' only 'counts' for those viewers who will have gone to bed by the time the announcer presents the late news. In other words, only those listeners should process the utterance. For other listeners (those who will be watching the late news) the utterance 'Goodnight' is obviously not relevant, hence the instruction is 'do not process'.

50. For more recent discussions of this type of conditional, see Davies (1979) (who calls them 'telling conditionals'); Haegeman and Wekker (1984) (who speak of 'peripheral conditionals'); de Cornulier (1985); and Eilfort (1987). (The latter defines an utterance-conditional as a conditional in which the *if*-clause 'seems to give felicity conditions on the speech act expressed by the consequent' (p. 57).)

51. See Eilfort (1987:56).

cannot (like standard conditionals) occur in the form of a counterfactual conditional.[52] The following sentences illustrate point 1:[53]

(70) (a) If John *wins* the race, Harry will be green with envy.
(b) Let John win the race and Harry will be green with envy.
(69) (b) If it'*ll* make you feel any better, we know now that it wasn't your fault.
(b′) ?!Let it make you feel any better and we know now that it wasn't your fault.

A third general difference between (69,a–c) and standard conditionals is that, in (69,a–c), there is no temporal relation expressed between the two situations.[54]

The fact that utterance-conditionals like (69,a–c) thus differ from standard conditionals entails that the usual conditions for incorporating the p-situation into the temporal domain established by the head clause are not satisfied. It is not surprising, therefore, that the tense forms used to refer to the future in the *if*-clauses of (69,a–c) belong to the FPS.[55]

It should be noted, however, that, in spite of the above remarks, some type II conditionals use the PPS to refer to the future in the *if*-clause:

(71) (a) If you ever *come* into a lot of money, I know a good way of investing it for you.
(b) If you *get* hungry tonight, there is some food in the refrigerator.
(c) If I *don't see* you before you leave, have a nice vacation! (Eilford 1987:13)

52. See Eilfort (1987:62–3): 'While the assertion of a central [= standard] conditional is the assertion of a proposition with conditions on the proposition, the assertion of a peripheral conditional [= utterance-conditional] is an assertion of a proposition with conditions on the assertion. . . . There are no peripheral counterfactuals because there are no counterfactual speech acts, that is, acts that take place in another possible world.'
53. If the head clause is a question, there is the further difference that the answer still implies the condition in the case of the standard conditional, whereas it does not in the case of the utterance-conditional. Compare:
(i) If we get into trouble, who is going to help us? – John.
(ii) If you're so smart, who is the oldest boy in this room? – John.
In (i), *John* is elliptic for 'John is going to help us if we get into trouble'; in (ii) *John* is elliptic for 'John is the oldest boy', not for 'John is the oldest boy if I am so smart' (see Eilfort 1987:58).
54. Along with these semantic differences between standard conditionals and utterance-conditionals there are some syntactic differences due to the fact that utterance-conditionals are less 'central' (more 'peripheral') than standard conditionals. For example, we can say (i) but not (ii):
(i) You should invite her to tea if you see her again, if you like her so much.
(ii) ??You should invite her to tea, if you like her so much, if you see her again.
For a fuller discussion, see Haegeman and Wekker (1984:48–9).
55. This is in keeping with Haegeman's (1984) claim that the 'temporal and modal interpretation' of utterance-conditionals 'is similar to that of independent main clauses: the present tense, for instance, refers to the speaker's present'.

On the other hand, the present tense can hardly be substituted for *will* in (69,a–c), nor in the following:

(72) (a) If you *will* be alone on Christmas Day, let us know now. (Close 1980:104)
(b) If you *will* not be in receipt of a scholarship or Award or if the Award *will* be inadequate to meet full fees and expenses of your course and your maintenance, please state how you propose to meet those fees and expenses. (Haegeman 1983a:153)
(c) Will you eat today? If you *will*, then please spare something for an old person overseas who will be 'lucky' to have even one scanty meal. (ibid.)

When we compare (71,a–c) with (72,a–c) we see that the only difference between them concerns the extent to which the condition is felt to be open. In (71,a–c) the condition is open with respect to both the speaker and the hearer: neither of them knows whether the p-situation will actualize or not. In (72,a–c) the condition is open with respect to the speaker but closed with respect to the addressee. Thus, in (72,a) the speaker does not know whether the addressee will be alone on Christmas Day, but he assumes the adressee knows. Similarly, it may or may not be the case that what is expressed in the *if*-clause of (72,b) applies to the person to whom the sentence is addressed at a particular time, but it is clear that the person in question is assumed to know this for himself. And, of course, the same is also true of (72,c).

It appears, then, that the distribution of the PPS and the FPS in the *if*-clauses of utterance-conditionals ultimately depends on whether the condition (which is anyhow open as far as the speaker is concerned) is represented as open or closed with respect to the hearer. In the former case we get one step closer to a standard conditional, and this appears to be sufficient for using the PPS, at least in sentences like (71,a–c), where the *if*-clause and the head clause together form a close-knit syntactic structure. If this is not the case, i.e. if the *if*-clause is relatively free or is added as an afterthought, the FPS is again possible:

(73) (a) The next king – if there *will* be a next king – will have no power whatever.
(b) The factory – if it *will* ever be built – will provide work for 150 people.
(c) It has become clear . . . that my aspiration to the Bar must give way to matters of practicality. Therefore, I shall qualify initially as a Solicitor, and then I will be in a position to consider higher things. Always supposing that, at that stage, I *will* still wish to. (Haegeman 1983a:160)

2.2.3 Type III

The third type is illustrated by examples like the following:

(74) (a) If it'*ll* make you feel any better, I'll take it back. (Tregidgo 1974:105)
(b) If it *will* make any difference, I'll gladly lend you some money. (Quirk *et al.* 1972:781)
(c) I will come if it *will* be (of) any use to you. (Jespersen 1931:400)
(d) If it'*ll* be of any help, I'll come along. (Palmer 1974:148)
(e) I don't want to call on Mrs Fustle, but I'll see her if it *will* do any good. (Allen 1966:179)
(f) I will lend you the money you ask for if you *will* use it to pay your debts, not if you *will* use it to buy whisky.
(g) I will give her a loan only if that *will* solve all her problems.
(h) British firms should not hesitate to tie up with American ones if this *will* help them keep up with the most advanced techniques. (Jacobsson 1984:136)

Sentence (74,a) can be paraphrased as 'I'll take it back provided that will have the effect of making you feel better.' The other sentences of (74) yield similar readings. Thus, (74,b) means something like 'I'll gladly lend you some money if my doing so will make any difference.' Another (equally revealing) paraphrase is 'I'll gladly lend you some money if it will make any difference if I do so.'

Palmer (1974) and Tregidgo (1974) attempt to account for such examples by suggesting that *will* is the grammatical means of expressing the so-called 'after-future' in English. That is, the use of *will* here follows from the fact that the condition p does not precede its consequence q but actually follows it in time.[56] Wekker (1976:72) rejects this explanation, saying that 'it is logically impossible for a consequence to be earlier in time than the condition which leads to it'. Instead he suggests an explanation which is essentially the same as we have argued for our type I: *will* is used because 'the future is especially marked, in the sense that the future event is regarded as very likely or inevitable' (p. 71); the *if*-clause may therefore be considered 'an elliptical form of "If you think that . . ." or "if it is the case that . . ." ' (p. 72). However, it is clear that an explanation which does not distinguish between type I and type III is not adequate. There is a difference of meaning between the two types. In conditionals of type I (e.g. *If he will be abroad for the next few weeks, there is no point in sending him an invitation for the party*) the *if*-clause represents the actualization of the p-situation as a present fact rather than as a future possibility, so that the condition is closed and *if* cannot be

56. In a later article Tregidgo (1979) admits that this analysis is not correct. As noted above, the alternative theory which he proposes is very similar to the one I am arguing.

replaced by *provided.* In contrast, the *if*-clause of a type III conditional does not represent p as a fact; it represents it as a possibility. The condition is therefore open and *provided* can be substituted for *if*:

(75) (a) I will come provided it *will* be any use to you.
(b) I will subscribe ten pounds, provided that 20 others *will* do the same. (Close 1980:106)

A further difference between the two types is that the use of *be going to* (which, as we have seen, is quite natural in type I) leads to unacceptable results in type III:[57]

(76) *I will come provided it *is going to* be of any use to you.

This means that, unlike type I, type III involves an *if*-clause which really expresses a condition for q. It is inherent in the meaning of *be going to* that this form is only used if all the conditions necessary for the occurrence of the situation have been met. As we have seen, this is the case in type I, where p is seen as a closed condition, so that its consequence q is also regarded as certain (inevitable). However, this is not the case in type III. The q clause of (75,a) does not express the unconditional promise that I will come. Rather, it says that I will come provided that I am given some reason for assuming that my coming will be of some help. The sentence is therefore paraphrasable as 'Tell me that it will help you if I come, and I will come' – a kind of paraphrase that is not applicable to type I examples at all.

Another difference between type I and type III lies in the relation between p and q. As we have seen, the relation expressed in type I is between the present assumption that the p-situation will take place and the present conclusion that q is therefore necessary or true. In type III the head clause also expresses a present conclusion (which is either a decision, intention or promise to perform the q-situation (as in (74,a–g)) or a conclusion as to the advisability of performing it (as in (74,h)), but the *if*-clause serves a different purpose: it no longer represents the factual grounds on which the conclusion has been drawn, but expresses a (yet to be fulfilled) condition for the validity of the conclusion. Thus, (74,d) can be paraphrased as 'I promise to come along, provided I get the assurance that my coming along will be of some help.' Whereas in type I the conclusion is based on the subjective certainty that the p-situation will actualize, in type III the conclusion is a provisional one and depends on whether or not there will be any guarantee that the p-situation will actualize. It is because of this that (74,a–g) suggest that the speaker invites the hearer (or some other person) to confirm that p will happen

57. Sentence (76) is unacceptable for those speakers who use *be going to* only in the senses described above (i.e. the future event is dependent on a present intention or is predictable from present causes). Speakers who do not distinguish between *will* and *be going to* may of course accept (76).

and that this confirmation will lead him to execute his decision (intention, promise) to perform q.[58]

What the above remarks amount to is that, whereas the logical structure of type I conditionals is something like 'p, hence q', that of type III conditionals is 'if [if q, then p], then q'.[59] In fact, every one of the examples in (74) can be reformulated in such a way that its syntactic structure reflects this formula:

(74) (a′) If it will make you feel any better if I take it back, I will take it back.
(b′) If it will make any difference if I lend you some money, I will lend you some money.
(c′) If it will be any use to you if I come, I will come.

etc.

It follows that *if*-clauses of type III have a double time reference. The actualization of the p-situation is clearly represented as lying in the future of t_0; it even follows the actualization of the q-situation (hence the idea of 'after-future').[60] Since it is logically impossible for a condition (cause) to follow its consequence (effect), the condition for executing the (present) decision expressed in the head clause cannot be the actualization of the p-situation. Rather, it is the confirmation that the p-situation will actualize consequent on the actualization of the q-situation. The speaker expects this confirmation to be given now. As noted by Palmer (1974:149), the *if*-clause is therefore paraphrasable as 'provided it now seems to be the case that p will happen'. Since the *if*-clause thus invites the hearer to confirm that the p-situation will take place, the meaning of type III sentences comes close to something like 'Will p happen? If so, I promise to do q.' This is in keeping with Close's (1975:60–1) feeling that (77,a) is paraphrasable as (77,b):

(77) (a) I can lend you five pounds, if that will help.
(b) I can lend you five pounds. Will that help?

58. The following example is a particularly good realization of the formula 'I promise to do X if you can give me the assurance that X will (not) entail Y':
(i) Later, as he showed Trevor Roper various samples of his Hitler papers, Heidemann said: 'I'll produce anything, if you just *won't* put me through the mincing machine.' (Jacobsson 1984:137)

59. The claim that type III conditionals realize this logical structure accords with Comrie's (1982:151) observation that this type of conditional 'requires a causal link between apodosis and protasis'. (Thus *If it will rain, I'll take my umbrella* is acceptable as a type III conditional only if it is assumed that my taking an umbrella will provoke rain.)

60. The fact that the occurrence of p can follow the occurrence of q accords with our theory. If (as we say) both p and q establish a post-present domain of their own, then both of them are related to t_0 but not to each other. That is, all that the tenses express is that both occurrences lie in the future of t_0. How they are ordered relative to each other is left unspecified.

and with Jespersen's (1931:400) statement that the *if*-clause in *I will come if it will be of any use to you* is 'a shade politer than *if it is. . .*'. The latter claim (though rejected by e.g. Wekker (1976:71)) is quite correct: the use of *will* in the *if*-clause suggests that the hearer is invited to confirm or deny that my visit would be helpful to him; it therefore suggests that I leave it to him to decide whether he wants me to come or not. The use of *will* is therefore felt to be politer than the use of the present tense (*if it is any use to you*), since in the latter case the hearer is not involved in the making of the decision at all.

The fact that a type III conditional realizes the formula 'if [if q, then p], then q' rather than the formula typical of standard conditionals (viz. 'if p, then q') accounts for the use of the FPS in the *if*-clause. As we have seen, the reason why standard conditionals use the PPS is that there is a close logical and temporal relation between the two situations: the p-situation leads up to the q-situation and is (sloppily) simultaneous with it. In type III conditionals this close logical and temporal relation is absent. The head clause refers to a present conclusion[61] concerning the future actualization of the q-situation (and hence uses the FPS), whereas the *if*-clause establishes a post-present domain consisting of a matrix situation (p), which constitutes the central TO of the domain, and an (implicit or overt) *if*-clause (q), which is logically and temporally related to this central TO. It is therefore predictable from our theory that the *if*-clause of a type III conditional uses the FPS: in order to establish a post-present domain we have to use the FPS, not the PPS.

2.2.4 Type IV

A typical example of this type is:

(78) Try to find out whether or not John will attend the meeting. *If he* ***will*** *attend it, we must warn Mary as soon as possible.*

This sentence should be compared with the following:

(79) Have you heard? John will attend the meeting! – Good heavens! *If John* ***will*** *attend the meeting, we must warn Mary as soon as possible.*

The italicized conditional in (79) is a clear example of type I: the *if*-clause expresses a closed condition (i.e. it represents the future actualization of the p-situation as a fact), while the head clause refers to a future course of action which is seen as a natural consequence of this fact. The

61. The *Longman Dictionary of Contemporary English* (p. 557) states that (i) is paraphrasable as (ii). It adds the comment that 'though the condition expressed by *if* is in the future, the whole sentence speaks of a truth in the present'.
(i) If it will help, I'll lend you 50 pounds.
(ii) If it is true now that 50 pounds will help, I'll lend you 50 pounds.

conditional sentence in (78) differs from this in only one respect: whereas (79) treats the truth of p as a present fact, (78) treats it as something that may turn out to be a fact in the future. That is, (78) can be paraphrased as follows:

(78′) If (at some future time t_i) it becomes an established fact that John will attend the meeting (at some time t_j), then we must warn Mary as soon as possible (i.e. at some time later than t_i but earlier than t_j).

As in type I, the actualization of the p-situation (John's attending the meeting) is posterior to the actualization of the q-situation (Mary's being warned), whereas it is anterior to it in the corresponding standard conditional *If John attends the meeting, let me know at once.* For this reason p does not allow the PPS (which would represent p as temporally dependent on q and thus produce the standard conditional) in (78). The information that the *if*-clause situation lies in the future of t_0 can therefore only be expressed by the FPS.

The same remarks also apply to the following examples, which further illustrate type IV. (Note that in each of these examples the conditional is embedded in a context which makes clear that the condition expressed is not closed but open.)

(80) (a) I'll write down how much petrol I'll need to make the journey. You will be given exactly the same amount when you start. *If, say, 500 gallons* ***will*** *be sufficient for me, then I don't see why it should not be sufficient for you.*

(b) . . .a new society is coming. The issue is, what will it be like? And the answer is not yet written, for it will be made by men. It still could be positive, and, *if this* ***will*** *be the case, then most of the decadences described here are happy omens.* (Jacobsson 1984:137)

(c) It is not certain yet whether the firm can be saved by financial support. At any rate, the union leaders have already declared that, *if it* ***will*** *help the firm to solve some of its problems, the government must give as much support as it can.*

2.2.5 Type V

Examples of this type are:

(81) (a) I'll come down to your office after one o'clock, if it *will* suit you. (Poutsma 1926:190)

(b) And you'll go to Freeman Founders to dine with him, won't you? – Yes, if it *will* please you. (ibid.)

(c) I should like to try farming, Dad; if it *won't* cost too much. (ibid.)

(d) Well, if you *won't* think it rude, Harold, I shall toddle in. (Jacobsson 1984:137)
(e) I'll leave the mule here, if you *won't* mind. (ibid.)

This type of conditional has much in common with type III (i.e. with conditionals like (82,a–b) which realize the logical structure 'if [if q, then p], then q').

(82) (a) British firms should not hesitate to tie up with American ones if this *will* help them keep up with the most advanced techniques. (Jacobsson 1984:136)
(b) I will give her a loan only if that *will* solve all her problems.

Like (82,a–b), (81,a–e) contain a head clause expressing a present conclusion or decision whose validity is made dependent on a certain condition. However, the kind of condition is different. In (82,a–b) the condition is that the speaker gets the assurance that, if the q-situation (referred to in the head clause) is actualized, this will entail the p-situation (expressed in the *if*-clause). The actualization of the q-situation is thus logically and temporally prior to that of the p-situation. In (81,a–e) this is not the case. Here the condition on which the validity of the present conclusion expressed in the head clause depends is that the speaker gets the assurance that the p-situation will actualize together with (i.e. at the same time as) the actualization of the q-situation. Thus, *I will come to your place tomorrow, if you won't mind* expresses that the speaker will go to the addressee's house the next day if the adressee will not mind the speaker's doing so. The crucial point here is that the situation of not minding is not located at t_0 – *if you don't mind* would be used for this – but at the time of the future visit. Thus, in type V conditionals the p-situation coincides with the q-situation, whereas it is posterior to the latter in type III conditionals. This means that the FPS in the conditional clause establishes a future domain which is simultaneous with that established in the head clause. In fact, since the *if*-clause typically follows the head clause in this type, we can say that the future tense in the *if*-clause re-establishes the post-present domain of the head clause (see chapter 2). This re- establishment is necessary because the use of the present tense in this type of *if*-clause would represent the p-situation as simultaneous with t_0 (as in *I will leave the mule here, if you don't mind*).

It is interesting to note that there are different reasons why the speaker may wish to have his *if*-clause refer to the future. Apart from the fact that future time reference may be necessary because of the situation referred to (as in (81,c)), *will* is apparently often used to achieve an effect of tentativeness. The *if*-clause (which is often added as an afterthought after the speaker has stated his decision (intention etc.)) has the function of indirectly asking the hearer whether the arrangement is all right for him

(as is provisionally assumed by the speaker). Thus, in *I'll come to your house at 5 o'clock, if that suits you*, the appended *if*-clause practically has the value of a question: 'Does that suit you?' (As a matter of fact we could easily put a question mark after the *if*-clause.) The use of the future tense (*if that will suit you*) enhances the tentativeness (politeness, tactfulness) of this *if*-clause question: by using *will* the speaker suggests that the hearer needn't commit himself now, but may defer doing so until later. This tentativeness can still be heightened by the use of hypothetical *would*, as in (83,a–b):

(83) (a) Tomorrow afternoon perhaps – about three o'clock, if that *would* suit her. (Poutsma 1926:193)
(b) 'You must bring Irene to the works', my father said. 'That is, if she'*d* be interested?' (Haegeman 1983a:161)
(c) Let's have some tea. And then if it *would* amuse you, I'll show you some of my pictures. (Haegeman and Wekker 1984:46)

Using a conditional form, the speaker renders the issue still more uncertain, thus leaving the other person plenty of room for a polite refusal.[62]

2.2.6 Type VI

In chapter 2 we have seen that, in subclauses that can be interpreted as temporally independent of their matrix, the FPS can be preferred to the PPS in order to avoid ambiguity. Consider, for example:

(84) (a) The teacher will punish everybody who *looks* untidy.
(b) The teacher will punish everybody who *will* look untidy.

As explained in section 17 of chapter 2, (84,a) is ambiguous because (for lack of disambiguating adverbial or contextual elements) the present tense in the relative clause can be interpreted either as referring to t_0 or as expressing simultaneity with the post-present situation referred to in the head clause. If the speaker intends the latter interpretation, he may therefore prefer to use the unambiguous (84,b).

This use of the FPS instead of the PPS is also possible in conditional clauses. Compare:

(85) (a) If anyone *is* untidy, the teacher will punish him.
(b) If anyone *will* be untidy, the teacher will punish him.

Again, the *if*-clause of (85,a) is ambiguous between present and post-present time reference. (In the former case the condition is a truth-

62. The possibility of using (nonvolitional) *would* in the *if*-clause of course confirms the difference between type V and standard conditionals.

condition, in the latter it is a situation-condition.) The speaker may therefore prefer to use (85,b). Similar examples of the use of *will* after *if* are:

(86) (a) I'd like – tonight – to drop in and see you. At ten? If you *will* be alone. (Jacobsson 1984:136)
(b) Will you really take me out to the bar of the crazies? – Sure. If you *won't* be scared. (ibid.)
(c) It might be fun to go to the exhibition of great Victorian paintings at the Royal Academy this morning. . . . Afterwards we could go on as planned to the Courtauld Institute, if that *won't* be a surfeit of painting. (Jacobsson 1984:136–7)
(d) If John *will* be ill, he will have caught the measles. (Jacobsson 1984:137)
(e) We have strong faith that a rally to the defence of the Act will be a feature of next year's politics, if the Tory Party *will* have the courage to come into the open and declare war upon it. (Jacobsson 1984:137–8)
(f) It is important, of course, to pay attention to what you eat – particularly if your weight problem *will* involve a lengthy period of dieting. (Haegeman 1983a:153)

It is not surprising that all these examples involve a stative verb in the *if*-clause. The simple present of a dynamic verb form cannot normally be used to refer to a (nonrepetitive and non-gnomic) situation that is holding at t_0, except in performative sentences and some other marked sentence types. It follows that the simple present of a dynamic verb in an *if*-clause accompanied by a head clause in the future tense can only be interpreted as referring to the future. In that case there is no reason for using the FPS.

The above conclusion does not hold for dynamic verb forms in the progressive: the present progressive can be used both to refer to t_0 and to express simultaneity in a post-present domain. Examples where the FPS is used for clarity are therefore conceivable:

(87) It is important, of course, to take medical advice – particularly if you *will* be following a strict diet.

It is also worth pointing out that this use of *will* to refer to the post-present is matched by the use of *would* to express posteriority in a past domain:

(88) If they could be sure that Miranda would not return to the flat, then the flat would do. . . . Certainly London hotels were either squalid or expensive, and *John was loath to pay for a room if the flat* ***would*** *be free.* (Jacobsson 1984:140)

2.2.7 Type VII

In Type VI the speaker uses the future tense (which re-establishes the post-present domain) to make clear that the *if*-clause situation lies in the post-present and not in the present. Type VII differs from this in that the *if*-clause situation not only lies in the post-present, but is actually posterior to the head clause situation. That is, the *if*-clause situation is not represented as posterior to t_0 but as posterior to the post-present head clause situation. In order to do so the speaker has to use a form which expresses posteriority in the PPS, viz. *will* or *is going to*:

(89) (a) Where are the children? – They're playing in the park. – *Will they come in if there* ***is going*** *to be a storm?* – I suppose so. They are old enough to know what they should do.
(b) I'll ring you up if I'*m going to* be late for dinner. (Jacobsson 1984:132)
(c) In order to see whether the baby will be a boy or a girl, put the pregnant woman in the middle of a flock of wrens. The baby will be a boy if they fly away in terror, because white boys stalk and torture and kill them. *If it* ***will*** *be a girl, they will cluster around singing*, because girls grind grain and scatter some for the wrens. (adapted from an example (cited below) from Jacobsson 1984:140)

Sentence (89,a) can be paraphrased as 'Will they come in if there are signs of an impending storm?' This means that the head clause situation establishes the central TO of a post-present domain and that the *if*-clause represents its situation as posterior to that TO. In doing so, the *if*-clause uses *be going to* because the posteriority expressed is of the type usually expressed by this auxiliary: the *if*-clause situation is represented as already perceptible (i.e. as having its initiation) at the time of the head clause situation (see footnote 44). The *if*-clause situation is thus not represented as posterior to t_0 – (89,a) does not suggest that there are perceptible signs of an impending storm at t_0 – but as posterior to the head clause situation (which establishes the central TO of the domain). *Be going to* is therefore used as a PPS form here, not as a FPS form.

Similar remarks can be made in connection with (89,b–c). The former can be paraphrased as 'I'll ring you up if (at the time of ringing you up) it is clear that I'm going to be late for dinner.' Similarly, in (89,c) the *if*-clause situation (*will be a girl*) is represented as posterior to the head clause situation (*will cluster around singing*). The verb form used in the *if*-clause is thus not a FPS form; it is a PPS form expressing posteriority in the post-present domain.

As might be expected, this use of *will/is going to* in a post-present domain is matched by the use of *would/was going to* in a past one:

(90) (a) The children would come in if there *was going to* be a storm.
(b) He would ring me up if he *was going to* be late for dinner.
(c) Put the pregnant women in the middle of a flock of wrens. The baby would be a boy if they flew away in terror, because white boys stalked and tortured and killed them. *If it* ***would*** *be a girl, they'd cluster around singing*, because girls ground grain and scattered some for the wrens. (Jacobsson 1984:140)

2.2.8 Type VIII

This is a minor type, consisting of *if*-clauses that are used independently of a head clause:

(91) Hang it all! *If that idiot* ***won't*** *be there as well!* Who the hell sent him an invitation?

The reason for using *will* is obvious here: as there is no head clause establishing a post-present domain, the *if*-clause must establish such a domain itself and must therefore recur to the FPS.[63]

2.2.9 Type IX

Examples of this type involve *if*-clauses which are themselves interpreted as containing an embedded *if*-clause:

(92) (a) If there *will/is going to* be trouble if we attend the meeting, we'd better stay at home.
(b) If you *will* be unhappy if I leave, then I promise you I will not leave.
(c) This system of subsidies will be maintained if the farmers *will* suffer considerable losses if it is abolished.
(d) We will not use these abbreviations if it *will* result in confusion (if we use them).

The rule that we do not use the PPS in an *if*-clause which is itself interpreted as the matrix of an embedded *if*-clause forms part of a more general principle which holds for conditional sentences in general: if an *if*-clause itself contains or implies a conditional clause, then it must make use of the verb form that is normally used in an apodosis (i.e. in the head clause of a conditional) (see also Poutsma 1926:190, Curme 1931:6). The

63. A more productive type of independent *if*-clause is that which is used as a polite imperative:
(i) After this brief introduction we can now enter the museum. *If you will follow me through this door.*
However, the *will* used in this type of *if*-clause is the modal auxiliary (expressing willingness), not the auxiliary of the future tense.

sentences in (92) show this for conditionals with an open condition. The following are examples involving a hypothetical or counterfactual condition:

(93) (a) Each hypothetical statement made by this primary use of *if* is acceptable if the antecedent clause if made or accepted, *would* in the circumstances be a good reason for accepting the consequent statement. (Strawson 1952:150)
(b) A behind-the-scenes struggle is now going on over a possible compromise with Mr Heath, on the basis of a backbencher's amendment which would enable price maintenance to continue for certain goods 'if the public as purchasers, consumers, or users of any goods' *would* be harmed if it were abolished. (Haegeman 1983a:158)
(c) If JOHN *would* have succeeded (had he tried), why shouldn't I succeed?

In many cases the embedded condition is implied rather than overtly expressed:

(94) (a) If a loan *would* have saved him, why didn't you give him one? (= if it would have saved him if you had given him a loan. . .)
(b) After *'d rather* we have to use an infinitive clause rather than a finite subclause with a modal past tense if the subject of the finite subclause *would* be the same as that of *'d rather.*
(c) If a table of random numbers *would* have helped John set up his experimental design, one would have helped Sam too. (Glendinning-Johnson 1975:83)
(d) I should not take this medicine if it *would* upset you (if I took it). (Poutsma 1926:193)
(e) The father of Ancient Rome . . . exercised the right of life and death over a new-born baby. If the living *would* suffer as a result of its birth . . . it was not allowed to live. (Jacobsson 1984:140)

It should be noted that the use of *would* (*have*) in the *if*-clause expresses a meaning which a subjunctive or 'modal past' could not express. This is clear when we compare the following:

(95) (a) I wouldn't do that if it *were* upsetting to you.
(b) I wouldn't do that if it *would* be upsetting to you (if I did it).

In (95,a) the *if*-clause has a straightforward counterfactual meaning: p is simply not true. In (95,b) it has the 'rejected possibility' meaning that is typical of head clauses with hypothetical *would*: p would be possible

under certain conditions, but these conditions do not appear to be fulfilled. A similar semantic difference is to be observed in (96,a–b):

(96) (a) If John *had succeeded*, we would have succeeded as well.
(b) If John *would* have succeeded (had he tried), we would have succeeded as well.

In (96,a) the *if*-clause merely rejects the idea that John succeeded; in (96,b) it expresses that John would possibly have succeeded, but that he did not, because some necessary condition was not fulfilled. In other words, (96,a) suggests that John tried but did not succeed, while (96,b) suggests that he did not try, though he might have succeeded if he had. The two sentences of (96) thus involve a condition that is counterfactual for a different reason: whereas (96,a) implies that the possibility of John succeeding did not materialize when it was put to the test, (96b) implies that the possibility did not materialize because it was not put to the test.[64]

An interesting fact about type IX conditionals is that they do not share the restrictions on the possible combinations of tense forms which we observe in 'canonical' conditionals.[65] In canonical conditionals there is a correlation between a past tense in the *if*-clause and *would* in the head clause and between a past perfect and *would have*. In type IX there are no such restrictions. This appears from (93)–(94) and from examples like the following:

(97) If JOHN *would* have broken the record (if he had not fallen)
(a) WE will certainly break it.

64. Exactly the same observations can be made in connection with adverbial time clauses. Like *if*-clauses, these normally forbid the use of *would (have)*, but these forms become possible when there is an (overt or implicit) *if*-clause embedded in them:
(i) (I would have told her . . .) But when I *would* have spoken of this, I saw that she was looking into the future. (Jespersen 1931:309)
(ii) They were passing away the few minutes before Pamela and her aunt *would* be ready to join them in the dining-room above. (Jespersen 1931:320–1)
(iii) Maddeningly, just when I *would* have welcomed full information on the books he was reading, he started to mention nothing beyond authors and titles, often in a shortened form. (K. Amis, *The green man*, Frogmore: Panther, p. 97)
Sentence (i) can be paraphrased as 'At the moment that I would have spoken of this (if nothing had intervened) I saw that . . .' *Would have spoken* thus expresses a possibility that is rejected because it depends on some unfulfilled condition. Using *had spoken* could not express this. Similarly, (ii) can be paraphrased as 'They were waiting the few minutes before Pamela and her aunt would be ready to join them (if everything passed off according to schedule) . . .' *Would be ready* thus expresses a conditioned possibility, whereas *were ready* would be a statement of fact (and would therefore imply that everything did pass off according to schedule). Sentence (iii), finally, is quite similar to (i).

65. By 'canonical conditionals' I mean conditionals of one of the following three types:
(i) If she comes I will be happy.
(ii) If she came I would be happy.
(iii) If she had come I would have been happy.

(b) maybe WE would be able to break it.
(c) WE could have broken it too.

(98) If JOHN *would* be able to break the record (if he really tried),
(a) WE can certainly break it.
(b) WE would be able to break it too.
(c) WE could have broken it a long time ago.

(99) But if loss of the final consonant *would* result in confusion between the present and past forms of the verb, it is retained. (Jacobsson 1984:142)

This means that the combination of the *if*-clause and the embedded (or implied) *if*-clause is a self-contained whole and that the usual close relation between the *if*-clause and the head clause is absent here. Put in other words: the situations described in the *if*-clause and the embedded *if*-clause belong to the same domain, but the head clause situation does not form part of it. Since the domain in question is thus not established by the head clause, the *if*-clause must establish it by itself. This explains why the *if*-clause must contain a tense form from the FPS (i.e. a form which is used for establishing a domain) rather than a tense form which merely expresses a domain-internal relation.

Another thing worth noting is that type IX overlaps most of the other types of conditional that we have considered. Sentences like (92,a–b) can be interpreted as belonging to type I. Thus (92,a) allows the paraphrase 'If it is the case that there will be trouble if we attend the meeting, then we'd better stay at home'. It also becomes clear now that type I may also involve *would* or *would have* in the *if*-clause:

(100) (a) If only 400 weapons *would* destroy the Soviets, what would 17,000 weapons do to the world? (Jacobsson 1984:141)
(b) If John *would* have succeeded if he had tried, you must have a try at once. You are much better than him.
(c) If, as you say, that man *would* be able to open our safe if he had the right instruments, it is high time we bought another, more sophisticated one.

As in other examples of type I, the speaker assumes that p is true.[66]

An example of a conditional that is both type IX and type II (i.e. the type in which the *if*-clause is an utterance-conditional) is the following:

(101) If it *would* comfort your mother's heart to know that your daughter will be in good company, I think I can give you that comfort. (Jespersen 1931:302)

66. Jacobsson (1984:140–1) points out that the paraphrase 'If it is true that . . .' fits the only example that he has found of an *if*-clause itself embedding another conditional clause, viz. (i):
(i) Yes, said Stella Armstrong, if he *wouldn't* be happy unless he had wiped Humphrey and Kate off his obligations, he had better do so.

The conditional clause embedded in the *if*-clause here assumes the form of an infinitival clause (*to know* is equivalent to *if you knew*). There's nothing unusual about this, as *if*-clauses often have the value of a subject clause or object clause (as in *It would be nice if you could win*) and vice versa.

It is also easy to find examples of conditionals that belong both to type IX and to type III. In fact, most of the examples of type III that we have given imply a condition in the form of the pro-form *it* or *that*, which has the value of a subject or object clause. Thus, the sentence *If it* ***will*** *solve your problems, I will buy the house* is equivalent to 'If it will solve your problems if I buy the house, I will buy it.' Even if there is no pro-form in the *if*-clause, a condition may be implied, as in (102,a). Because of this, the *if*-clause may easily involve *would*, witness (102,b):

(102) (a) If money *will* solve your problems, I'll give you a loan. (= if it will solve your problems if I give you some money . . .)
(b) If money *would* solve your problems, I would give you a loan.

Type IV conditionals may also imply an embedded *if*-clause, provided the (superordinate) *if*-clause contains a pro-form for the *if*-clause in question. This is the case in one of the examples that have been given:

(80) (c) It is not certain yet whether the firm can be saved by financial support. At any rate, the union leaders have already declared that, *if it* ***will*** *help the firm to solve some of its problems, the government must give as much support as it can.*

Type V conditionals systematically imply a covert *if*-clause:

(103) I'll pay you back tomorrow, if that *will* suit you. (= . . . if it will suit you if I pay you back tomorrow)

Once again it is the pro-form *that* which entails the conditional meaning. Examples with *would* are also easily found:

(104) (a) Tomorrow afternoon perhaps – about three o'clock, if that *would* suit her. (Poutsma 1926:193)
(b) I'd like to have some tea please, if that *would* be possible.
(c) Have you come to stay? – Well, I'd like to, if it *wouldn't* be an awful bore to you. (Jacobsson 1984:141)

Even counterfactual conditionals are possible here:

(105) I could have pretended to have forgotten, if it *would* have pleased you better. (Jespersen 1931:308) (= . . . if it would have pleased you better if I had . . .)

Some examples of type VI conditionals also allow interpreting a pronoun as a covert *if*-clause:

(87) (c) It might be fun to go to the exhibition of great Victorian paintings at the Royal Academy this morning. . . . Afterwards we could go on as planned to the Courtauld Institute, if that *won't* be a surfeit of painting. (Jacobsson 1984:136–7) (= if it won't be a surfeit of painting if we do all that)

In an example like this, it is also perfectly possible to use *would* in the *if*-clause.

In type VII, on the other hand, the *if*-clause does not itself embed an implicit conditional clause:

(89) (a) Will the children come in if there *is going to* be a storm?
(b) I'll ring you up if *I'm going to* be late for dinner. (Jacobsson 1984:132)

The reason why there cannot be an (overt or implicit) conditional clause embedded in the *if*-clause of this type is that the *if*-clause is now temporally subordinated: its situation is represented as posterior to the post-present head clause situation. In this respect type VII differs from types I–VI: in each of the latter types the *if*-clause establishes its own post-present domain, which can be further expanded by the addition of an *if*-clause. In other words, in types I–VI the *if*-clause behaves as if it were a head clause, and can therefore support its own *if*-clause. This is not the case in type VII, where the *if*-clause does not establish a domain but uses a PPS form expressing posteriority.

The observation that types I–VI overlap type IX raises the question whether type IX should really be considered a type of its own.[67] To be able to answer this question in the affirmative we should be able to adduce examples that do not overlap any of the other types. As a matter of fact, I have found no such examples. This suggests that type IX is not really a type of its own. This conclusion is corroborated by the following observation. If type IX were a type of its own, i.e. if the grammaticality of *will* in the *if*-clause followed automatically from the presence of an embedded *if*-clause, then we would expect that it should be possible to insert an *if*-clause into the *if*-clause of a standard conditional, provided we substitute *will* for the present tense in the latter. In practice this expectation is not borne out:

(106) (a) If she is willing to see me, I will be happy.
(b) *If she will be willing to see me if she has time, I will be happy.

In fact, we can build examples in which the present tense, but not the future tense, is grammatical:

67. The insertion of an *if*-clause is presumably also possible in type VIII. However, because of the extremely low frequency of this type no attested examples could be found.

(107) (a) Even if she is only willing to see me if I insist, I'll be happy.
(b) *Even if she will only be willing to see me if I insist, I'll be happy.

These observations show that it is not the embedding of an *if*-clause into an *if*-clause that renders the use of *will* in the latter possible. Rather it is the other way round: when an *if*-clause establishes its own domain, it is capable of supporting another *if*-clause.

If the above conclusion is correct, we should not consider type IX as a type of its own. That is, the examples adduced to illustrate type IX in fact belong to one of the types I–VI.

2.3 Conclusion

We have identified a number of different types of sentences in which the use of *will* (or *be going to*)[68] to express future time in the *if*-clause is quite grammatical.[69] In each case the conditional sentence differs from a standard conditional in at least one respect:

1 Each of the first six types differs from a standard conditional in that the *if*-clause does not refer to a possible future actualization of the p-situation which, if it takes place, will lead to the actualization of the q-situation. Because of this, there is no close (conditional and temporal) relation between p and q. Instead of being incorporated into the domain established by the head clause, the *if*-clause therefore establishes a post-present domain of its own.[70] It follows that the system used is not the PPS but the FPS.

68. Some speakers (especially in the USA) find many of the examples more acceptable with *be going to* than with *will*. As noted before, this does not affect our theory, for *be going to* can fulfil the same functions as *will*, i.e. it can either establish a post-present domain (in which case it is a FPS form) or express posteriority in a post-present domain (in which case it is a PPS form).
69. The fact that eight or nine types of *if* + *will* clauses have been identified (no more, no fewer) of course does not mean that this classification is an absolutely necessary one. It may be that there are other possible types that have escaped my attention. In fact, it does not make any difference to the basic analysis – the generalization explaining the use of the future tense – how many types there are, or whether a particular sentence, on a particular reading, belongs to one particular type rather than another. Still, the classification is useful because it enables us to distinguish between the idiosyncratic features of such *if*-clauses (which do not by themselves occasion the use of *will*) and the common element which is responsible for the use of the FPS in all of them. This distinction is important, for it is only by setting off the constant factors from the other (more incidental) characteristics that we can escape making the mistake which has been made in previous treatments of the subject, viz. the mistake of reducing all cases of *if* + *will* to one single type and hence of explaining the use of *will* from one or more of the incidental characteristics.
70. Tregidgo (1979) comes to the same conclusion when he states that the use of *will* means that the tense of the *if*-clause is not subordinate to that of the head clause but is independent of it. A similar statement is made by Wekker (1980:100–1): in *if*-clauses in the present tense 'the speaker is oriented towards a point in the future', whereas in *if*-clauses in the future tense 'his point of orientation is the present'

2 Type VII does use the PPS, but the temporal relation expressed is not simultaneity but rather posteriority. For that reason *will* or *be going to* has to be used.
3 In type VIII the conditional clause is used as unembedded clause and must therefore make use of the FPS.

Because the conditional clause of types I–VI uses a verb form typical of unembedded clauses, it can support another *if*-clause.

5 The structure of tense 1: previous analyses

In the first part of this book we have explored the principles that determine the selection of tenses in clauses, both embedded and unembedded. We have spoken about the use of absolute and relative tenses, about temporal subordination and shifting the domain, about direct and indirect binding, about shifting the temporal perspective, sloppy simultaneity, temporal focus, etc. In doing so, we have explored the potentialities of the English tense system, asking such questions as: Which tenses can be used in simple clauses, compound sentences, complex sentences, text and discourse? What principles govern the selection of the tenses in these structures and environments? What restrictions and conditions are there on the application of the general principles? Needless to say, the answers that have been given to these questions are sketchy and incomplete, but if they are on the right track, they constitute the basic principles of what I have called a 'descriptive theory'. Such a theory is a prerequisite to building 'higher order' theories of tense, for it enables us to classify and interpret the data in a coherent way.

I will now turn to another aspect of the theory of tense, viz. the analysis of the temporal schemata that are realized by the tenses. Irrespective of how they are used in discourse, each tense has its own temporal structure, which consists of a number of primitives (temporal zero-point, time of the situation, etc.) and temporal relations holding between them. It is my aim, in the second part of this book, to reconstruct these temporal schemata for each of the eight tenses considered in the first part. In this chapter I will begin this investigation by considering two of the analyses that are available in the linguistic literature: the extremely influential theory of Reichenbach (1947) and the recent proposals by Comrie (1985). In chapter 6 I will then develop a theory of my own, which borrows some basic insights from both Reichenbach and Comrie, but departs from them wherever these analyses seem to be unsatisfactory. In chapter 7, finally, the theory argued will be applied to each of the eight tenses.

Let us begin by looking at Reichenbach (1947) and Comrie (1985).

1 REICHENBACH (1947)

1.1 Reichenbach's analysis

Reichenbach (1947:288–98) has developed a system in which there are three elements (primitives) involved in the description of the tenses: a speech act, an 'event' and a 'reference point'. The way in which these three are interrelated is illustrated with the help of the past perfect:

> Let us call the time point of the token the *point of speech.* ... From a sentence like 'Peter had gone' we see that the time order expressed in the tense does not concern one event, but two events, whose positions are determined with respect to the point of speech. We shall call these time points the *point of the event* and the *point of reference.* In the example the point of the event is the time when Peter went; the point of reference is a time between this point and the point of speech. In an individual sentence like the one given it is not clear which time point is used as the point of reference. This determination is rather given by the context of speech.
>
> (Reichenbach 1947:288)

In this theory, the meaning of the past perfect involves reference to three different points, located in a particular order on the time line:



(The symbols E, R and S stand, respectively, for 'point of the event', 'point of reference' and 'point of speech'.)

Reichenbach then goes on to claim that these three points are relevant to every one of the tenses, not just to tenses such as the past perfect or future perfect:

> In some tenses, two of the three points are simultaneous. Thus, in the simple past, the point of the event and the point of reference are simultaneous, and both are before the point of speech. ... This distinguishes the simple past from the present perfect. In the statement 'I have seen Charles' the event is also before the point of speech, but it is referred to a point simultaneous with the point of speech; i.e. the points of speech and reference coincide. ... We see that we need three points even for the distinction of tenses which, in a superficial consideration, seem to concern only two time points. The difficulties which grammar books have in explaining the meanings of the different tenses originate from the fact that they do not recognize the three-place structure of the time determination given in the tenses.
>
> (Reichenbach 1947:289–90)

The three points of time can be ordered in no fewer than thirteen different ways. These thirteen configurations are represented as follows (with dashes representing an interval of time and commas representing simultaneity):

	Reichenbach's label	*Traditional label*
E–R–S	Anterior past	Past perfect
E,R–S	Simple past	Past tense *or* preterit
R–E–S	Posterior past	Conditional tense
R–S,E	id.	id.
R–S–E	id.	id.
E–S,R	Anterior present	Present perfect
S,R,E	Simple present	Present tense
S,R–E	Posterior present	Future tense
S–R,E	Simple future	id.
S–R–E	Posterior future	id.
S–E–R	Anterior future	Future perfect
S,E–R	id.	id.
E–S–R	id.	id.

(In Reichenbach's terminology, the words 'past', 'present' and 'future' refer to the position of R relative to S. The terms 'anterior', 'simple' and 'posterior' indicate the position of E relative to R, the word 'simple' being used for the coincidence of R and E.)

According to Reichenbach, these schemata also accommodate the use of progressive verb forms. Progressive tense forms (which Reichenbach refers to as 'extended tenses') indicate explicitly that 'the event covers a certain stretch of time' (p. 290). Their temporal schemata therefore differ from those of the corresponding nonprogressive tenses only in that E is not a point on the time line but a longer timespan.

1.2 Objections to Reichenbach's system

Although Reichenbach's system has been extremely influential, a number of critical remarks can be made in connection with it:

1.2.1 It has often been pointed out that speaking of the '*point* of reference' and '*point* of event' is infelicitous, since both reference time and event time are usually longer than a moment (point): a term like 'interval' or 'time' would therefore be a better choice.

1.2.2 It has been pointed out (e.g. by McCoard (1978)) that it is a major weakness of Reichenbach's system that it generates more possibilities than are actually to be found in natural language. For example, Reichenbach provides for three different future perfect tenses (corresponding to the configurations 'S–E–R', 'S,E–R' and 'E–S–R'), but no

language appears to have three different future perfect tenses. The same overcapacity is observed in connection with the three tenses corresponding to the English future, and in connection with the three tenses corresponding with the English conditional. McCoard (1978:91) puts it as follows:

> The problem is that Reichenbach's model makes it appear to be an accident that English has no set of *grammatical* forms to express just these distinctions – nor does any other language that comes to mind. The set of point-orderings implies a systematic richness which outstrips the actual resources of natural languages.

1.2.3 Comrie (1981a:26) points out that, if there is only one tense to realize three different configurations, this tense will have to be three ways ambiguous (rather than vague). However, there is no evidence whatever for claiming that the future perfect, the future and the conditional tense are ambiguous in English, nor in any other language. Consider, for example, the following sentence:

(1) I don't know if my assistant has finished the report yet, but I can assure you that he *will have finished* it before the end of the week, for he has promised to do so and I know that he always keeps his promises.

Obviously, there are three situations in which the clause *he will have finished it before the end of the week* will turn out to be true: either the report has already been finished, or the assistant is finishing it right now, or he will do so between the time of uttering (1) and the time indicated by *the end of the week*. However, this does not mean that the clause is three ways ambiguous. The clause just does not contain any information as to the exact time that the report is finished. That is, the clause is vague in this respect, not ambiguous. (This vagueness is similar to that observed in sentences like *I saw John in the distance*. This sentence does not contain any precise information as to whether I saw John's face or not, but that does not mean that the sentence is ambiguous between the readings 'I saw John, including his face' and 'I saw John's figure, but not his face.' Clearly, this is a matter of vagueness, not of ambiguity.)[1]

1. We see this conclusion confirmed when we apply G. Lakoff's (1970) ambiguity test to the sentence:
   (i) (a) John will have finished my report before the end of the week, and Bill will have done so too.
   (b) John will have finished my report before the end of the week, and so will Bill.
   Both these sentences may be true irrespective of whether either report is finished before, at, or after t_0. Thus, one of the cases in which they are true is that in which John has already finished his report at t_0 and Bill still has to finish his. This proves that *will have finished* is vague with respect to the relation between the time of the finishing and t_0. If it were ambiguous, the reading selected in the first conjunct of (i,a) and (i,b) – say, the reading 'before t_0' – would also have to be selected in the second conjunct.

It should be noted that this problem cannot be solved in the Reichenbachian framework, because it is inherent in the system itself. In fact, the problem will crop up in any system that describes tenses in terms of configurations of time points on the time line, for placing time points on the time line inevitably means that each point is located relative to each of the other points. In such a system the temporal relation between two points cannot possibly remain vague.

1.2.4 The above problem arises not only in connection with such tenses as the future perfect and the conditional but also in connection with the preterit and the future tense. Consider the following:

(2) Where's John? – *He was in the kitchen five minutes ago*, so I suppose he's still there.

Reichenbach analyses *He was in the kitchen five minutes ago* as realizing the pattern E,R–S, in which both E and R are anterior to S. However, it is not correct to say that this sentence asserts that E is anterior to S: it does not exclude the possibility that the 'event' of John being in the kitchen still continues at S. (In fact, the context in which it is couched in (2) invites precisely this interpretation.) This means that, though it is correct that R (established by *five minutes ago*) lies before S and that there is some kind of simultaneity relation between E and R, the precise relation between E and S is not expressed in the sentence: the 'event' may or may not continue into the present. This vagueness cannot be handled by a system that incorporates the claim that 'in the simple past, the point of the event and the point of reference are simultaneous, and *both* are before the point of speech' (Reichenbach 1947:289 – my emphasis).

The same problem arises in connection with the future tense. Reichenbach's analysis of this tense locates E after S, thus making the claim that the future tense cannot be used to refer to an event that already holds at S. However, this claim is false, as appears from a sentence like *In five minutes' time John will still be here.* Like the preterit, the future tense fails to specify the precise relation between E and S, and therefore defies an analysis which puts S and E in a particular order on the time line.

1.2.5 Prior (1967:13) and many others in his wake have pointed out that Reichenbach's scheme is too simple in that it provides for only one reference point, while at least two points of reference are necessary to account for tenses that are more complicated than the past perfect or future perfect, as in *The others would have left by then* or *I shall have been going to see John.* The introduction of a second reference point also seems required to account for clauses with a dual time orientation, such as *We are now leaving next Saturday* (see Fenn 1987:155).

One might say that this objection carries little weight, since there is no reason why we should not allow the possibility of a second reference point in the Reichenbachian system. However, the introduction of a second reference point vitiates the 'generative' aspect of Reichenbach's theory which is explained in the following passage:

> If we wish to systematize the possible tenses we can proceed as follows. We choose the point of speech as the starting point; relative to it the point of reference can be in the past, at the same time, or in the future. This furnishes three possibilities. Next we consider the point of the event; it can be before, simultaneous with, or after the reference point. We thus arrive at 3 . 3 = 9 possible forms, which we call *fundamental forms.* Further differences of form result only when the position of the event relative to the point of speech is considered; this position, however, is usually irrelevant. Thus the form *S–E–R* can be distinguished from the form *S,E–R*; with respect to relations between *S* and *R* on the one hand and between *R* and *E* on the other hand, however, these two forms do not differ, and we therefore regard them as representing the same fundamental form. Consequently, we need not deal with all the 13 possible forms and may restrict ourselves to the 9 fundamental forms.
>
> (Reichenbach 1947:296)

The importance of the generative aspect of the system is also clear from the following passage from Hornstein (1981:130). (Note that Hornstein modifies Reichenbach's theory on two points: (1) Hornstein does consider it relevant whether or not two times are 'associated'. Thus, he claims that S,E–R (in which S and E are associated) does differ in a relevant way from S–E–R. (2) Hornstein claims that the linear order of the primitives is important, even if they are associated. He thus treats S,E–R as different from E,S–R. In this way the total number of possible configurations in Hornstein's theory is higher than that in Reichenbach's, viz. 24.)

> In a Reichenbachian theory, tense is a configuration of *S R E* elements structured by linearity and associativity. The total number of possible combinations yielded by these two operations is twenty-four. Thus, a Reichenbachian system, one can say, provides a fairly limited number of possible tenses for natural languages to choose from. Each tense need not appear in any one specific language, rather the Reichenbachian theory of tense universals delimits the range of tenses a language learner can choose from in constructing the particular tense grammar of its language. In specific, on this account there are no tenses like the past past past . . . past perfect in any natural language for no permutation of the elements *S R E* yields a configuration which can be so interpreted.

It is clear that the view that there are as many possible tenses as there are possible combinations of the primitives S, E and R becomes untenable when a fourth primitive R_2 is added. If three primitives yield a limited number of possible combinations (thirteen in Reichenbach's theory, twenty-four in Hornstein's), four primitives no longer do so (seventy-five possible combinations in Reichenbach's theory, and well over a hundred in Hornstein's). This vitiates the theory, for it seems preposterous to assume that the number of possible tenses can be so high.[2] Moreover, since Reichenbach claims that all the tenses realize a particular combination of *all* the primitives, the introduction of R_2 as a fourth primitive means that a second R is introduced in the structure of tenses that do not appear to involve more than one R. For example, the structure of the present tense would have to be S,R,R,E rather than S,R,E. (As a matter of fact, even the presence of one R in the structure of the present tense has often been challenged – see e.g. Katz 1972; Ultan 1978; Barense 1980; Bertinetto 1982; Comrie 1985.)[3]

1.2.6 The observation that some tenses involve more than one reference time leads Prior (1967:13) to formulate another point of criticism:

> once this possibility is seen, it becomes unnecessary and misleading to make such a sharp distinction between the point or points of reference and the point of speech; the point of speech is just the *first* point of reference. . . . This makes pastness and futurity *always* relative to some point of reference – maybe the first one (i.e. the point of speech) or maybe some other.

One of the implications of this remark is that Reichenbach's analyses in which S and R coincide become suspect. As McCoard (1978:92) puts it: 'In what sense could we speak of "simultaneity" of the point of speech with some other reference point, except as an artefact of description?'

1.2.7 According to Reichenbach (1947:294), temporal adverbials refer to R, not to E:

> When a time determination is added, such as is given by words like 'now' or 'yesterday', or by a nonreflexive symbol like 'November 7, 1944', it is referred, not to the event, but to the reference point of the

2. If we follow Reichenbach in discarding the position of E relative to S as irrelevant, the number of 'fundamental forms' is somewhat lower than seventy-five, but it is still unacceptably high.
3. Vikner (1985) actually argues that the present tense should be analysed as '*S,R,R,E*'. More in general, Vikner holds that there are two R's in the structure of any tense. The argumentation, however, is very weak. It mainly hinges on the observation that introducing two R's is necessary for analysing the conditional perfect, and is therefore required if we wish to analyse all tenses in a 'parallel' way.

> sentence. We say, 'I met him yesterday'; that the word 'yesterday' refers here to the event obtains only because the points of reference and of event coincide. When we say, 'I had met him yesterday', what was yesterday is the reference point, and the meeting may have occurred the day before yesterday. We shall speak, therefore, of the *positional use of the reference point*; the reference point is used here as the carrier of the time position. Such usage, at least, is followed by the English language.

This theory is hard pressed by the simple observation that *I had met him yesterday* is actually ambiguous between the reading referred to above (according to which the meeting occurred before yesterday) and the interpretation on which *yesterday* indicates the time of the encounter. The latter interpretation is brought to the fore in (3):

(3) This morning mother wanted me to go and see grandfather, but I told her it was not necessary because *I had met him yesterday.*

In this sentence *yesterday* refers to the time of the encounter and does not establish the reference point that lies between the event point and the speech point (since the structure of the past perfect in Reichenbach's theory is E–R–S). In (3) the reference point in question is referred to by *this morning.*

There are two ways to accommodate this observation. The first is to give up the claim that temporal adverbials can only refer to R and accept that they can refer to either R or E. (This solution is adopted by Hornstein (1975, 1977), Smith (1978b), Comrie (1985).[4]) The second way out is to assume that in a past perfect sentence like *I had met him yesterday* there are two reference points, one of which is simultaneous with the event point, while the other lies between the event point and the point of speech. (This solution is adopted by Declerck (1986a).)

1.2.8 There is a second way in which Reichenbach's claim that time adverbials refer to R proves incompatible with his own analysis of the past perfect. Consider a sentence like the following:

(4) I heard yesterday that John *had been* in London the day before.

If the past perfect in the *that*-clause is to be analysed as 'E–R–S', then R must be the time indicated by *yesterday*, not the time indicated by *the day before*, for E (the time of being in London) is anterior to the former time, not to the latter. But which time is then established by *the day before*? Either we will have to give up the principle that time adverbials

4. As noted by Salkie (1989:7), this view leads to difficulties if one assumes there to be an R in the structure of the present perfect (as Hornstein and Dinsmore do), for the present perfect cannot collocate with an adverbial specifying E:**I have seen him yesterday.*

refer to R and not to E and say that *the day before* indicates the time of E, or we will have to say that the structure of the past perfect contains two Rs (that is, E is simultaneous with one R ('the day before') and anterior to another ('yesterday')). This is exactly the same conclusion as we reached in connection with (3).

Dahl (1985:30) offers a similar piece of evidence. He draws attention to a sentence like

(5) When I arrived, Peter had tried to phone me twice during the preceding week.

and comments: 'R here must be the time when I arrived, and E – or rather the E's – are the time points when Peter tried to phone me. But we see that there is nothing in Reichenbach's scheme that corresponds to the time referred to by *during the preceding week*. The function of that phrase could be said to be to specify a *temporal frame* for the E points, that is, a time period within which they are located.'

What this criticism amounts to is that Reichenbach's claim that temporal adverbials are accommodated in his structures (since they must be taken to refer to R) is not correct. Some types of temporal specifiers appear to refer to times that do not correspond to any of the times ('points') in the Reichenbachian system.

1.2.9 A final point of criticism[5] is that Reichenbach's system, which provides for the presence of S in the structure of every tense, does not account for the use of relative tense. Consider, for example, the following:

(6) John told us a minute ago that Bill *was* in the kitchen.

I have argued in chapter 2 that the relative past tense form *was* just represents the *that*-clause situation as simultaneous with the head clause situation (which acts as binding TO, i.e. as Reichenbach's R). It does not express any temporal relation between the *that*-clause situation and t_0 (speech time). This means that *was* does not realize the full schema 'E,R–S'. It expresses no more than 'E,R'. The relation between R and S is indicated in the head clause.

5. I will disregard such criticism of Reichenbach's system as is irrelevant to English. The following is an example:

> One addition that must be made to the system is made necessary by those languages, such as the Bantu language Luganda and the Australian language Yandruwandha, to cite but two among many, that have different grammatical forms for different degrees of remoteness of past and future, e.g. different tenses for Recent past and Remote past. The relation of *precedence* is thus not solely qualitative, but also quantitative; Reichenbach's system can easily be modified to permit specification of the *quantity of precedence*, i.e. precedence becomes a vector, with direction and magnitude.'
>
> (Comrie 1981a:25) (see also Botne 1981:108–10)

This problem can only be solved if we give up one of Reichenbach's basic claims, viz. that all three elements E, R and S are necessarily present in the structure of every tense. We will have to say that the relative preterit only expresses the relation between E and R.[6]

2 COMRIE (1985)

2.1 Comrie's analysis

Some of the above points of criticism (viz. points 2, 3 and 5) are explicitly made in Comrie (1981a, 1985) and have led Comrie (1985) to propose a theory of tense that is different from Reichenbach's. In this theory, all that we need for representing the three 'absolute' tenses (present, preterit and future)[7] is two time elements (viz. the time of speech (S), and the time of the event (E)) and three relations (viz. simultaneity, anteriority and posteriority):

present tense	E simul S
past tense	E before S
future tense	E after S

For the representation of the other tenses one more primitive is necessary, viz. the reference point (R):

pluperfect	E before R before S
future perfect	E before R after S
conditional	E after R before S

For more complicated instances of location in time several reference times can be provided for. For example:

conditional perfect	E before R_i after R_j before S

The above system differs from Reichenbach's in the following respects:

6. Comrie (1981a:25–6) makes the same point, but wrongly assumes that tense forms that only express the relation between E and R are to be found in some foreign languages only:

   > Many languages have what is traditionally called 'relative tense', i.e. tenses which do not include any overt relation to *S*: in Classical Arabic, for instance, the so-called Imperfect indicates overlap between *E* and *R*, but does not relate either of these to *S*. The temporal relation between *R* and *S* may of course be given by the context, and in the absence of any other indication *R* is taken to overlap *S*. Again, a relatively minor modification to Reichenbach's system is called for: specification of *S* is now optional, rather than obligatory.

   For a discussion of some examples of relative tense in African languages, see Wald (1973), Welmers (1973), Hyman (1980), Contini-Morava (1983).
7. In my theory there is a fourth absolute tense, viz. the present perfect. According to Comrie, however, the present perfect realizes the same temporal schema as the preterit (see below).

2.1.1 Comrie rejects Reichenbach's principle that 'we need three points even for the distinction of tenses which, in a superficial consideration, seem to concern only two time points' (Reichenbach 1947:290). According to Comrie, there is no R in the schemata of the 'absolute' tenses.[8]

This restriction on the occurrence of R comes down to saying that there are no schemata in which there is an R that coincides with S or E. This principle solves the problem of overcapacity that we have observed in connection with Reichenbach's theory. If we disregard the possibility of a second R, the number of possible structures resulting from combining the time elements and the available relations is no more than seven in Comrie's system:

E simul S	present tense
E before S	past tense
E after S	future tense
E before R before S	past perfect
E before R after S	future perfect
E after R before S	conditional
E after R after S	future tense

This system, in which there are six English tenses corresponding to seven possible schemata, seems more satisfactory than Reichenbach's system, in which there are no fewer than thirteen possible schemata for only six tenses. On the other hand, once one allows for the introduction of a second R (and this is inevitable for the analysis of the conditional perfect), the number of theoretically possible combinations increases enormously, so that Comrie's theory ultimately encounters the same problem of overcapacity as Reichenbach's. (It is true, however, that Comrie does not propose his theory as a 'generative' one. He does not claim that there are as many 'possible tenses' as there are possible combinations.)

2.1.2 Unlike Reichenbach, Comrie does not place S, R and E on the time line. In doing so, he avoids stating relations which are not actually expressed by the English tenses. For example, we have seen that in Reichenbach's theory there are three different configurations corresponding to the future perfect (since E can either follow or precede S, or coincide with it) and that this is unsatisfactory because a future perfect is not ambiguous between three readings but just vague as to the position of E relative to S. Comrie's analysis of the future perfect does not meet with this problem, since it places E before S without making any claim

8. Some people (e.g. Katz 1972; Ultan 1978; Barense 1980) had already made this claim in connection with the present tense. Bäuerle (1979:49) makes it in connection with the two simple tenses he recognizes, viz. the present and the preterit.

as to the position of E relative to S. In this respect Comrie's theory is superior to Reichenbach's.

2.1.3 The claim that there is no R in the temporal structure of the 'absolute' tenses renders it impossible for Comrie to offer separate analyses for the past tense and the present perfect: he is forced to claim that 'in terms of location in time, . . . the perfect is not distinct from the past' (p. 78). Comrie thus rejects the Reichenbachian analysis on which the preterit involves a past point of reference, whereas the present perfect involves a reference point that coincides with the point of speech. According to Comrie, both tenses just locate a situation as prior to the present moment and do not involve a reference point at all. To account for the obvious difference that we feel there to be between the two tenses, Comrie has recourse to the notion of 'perfect aspect': the present perfect implies 'current relevance', the past tense does not. According to Comrie, this warrants his disregarding the present perfect in his treatment of the tenses: 'however perfect differs from past, it is not in terms of time location' (p. 78).

2.2 Objections to Comrie's theory

2.2.1 Before formulating objections proper, we can point out that Comrie's theory is a theory of verb tense only, not a general theory of time reference. This means that Comrie does not take into account the role that adverbials and context play in the way that situations are located in time. Although this is a deliberate choice, which does not seem unwarranted in an introductory book on tense, it does constitute a major limitation. In my opinion, it is better not to deal with tense in isolation but to build a richer system which is also capable of handling the temporal specification effected by adverbials and context.[9]

2.2.2 The view that there is no reference time in the 'absolute' tenses makes it necessary for Comrie to reject Reichenbach's (intuitively attractive) view that the present perfect locates a situation in time

9. One of the objections that have been raised in connection with Reichenbach's theory is that it cannot deal with sentences like *I found out that John had been in London during the preceding week* because it provides for no element corresponding to the time referred to by *during the preceding week*. This remark self-evidently also applies to Comrie's theory. When we look at the sentence in question, we ascertain that, apart from the moment of speech, it refers to three different times: the time of my finding out, the time when John was in London and the time referred to by *during the preceding week*. Comrie's analysis of the past perfect ('E before R before S') provides for only two time-points besides S. Since R is obviously the time of finding out, and E is the time of John's being in London, Comrie's theory fails to provide for the time referred to by *during the preceding week* (which is a time relative to which E is located).

relative to a reference point which coincides with the present moment. The basic argument runs as follows:

> If one were to provide an analysis of the present perfect analogous to that of the pluperfect and the future perfect, then one would say that the reference point for the perfect is simultaneous with the present moment, rather than being before the present moment (as for the pluperfect) or after the present moment (as for the future perfect). The situation in question would then be located in time prior to this reference point. In terms of location in time, however, this would give precisely the same result as the past, which also locates a situation as prior to the present moment. Thus, however perfect differs from past, it is not in terms of time location.

This basic argument is then supplemented with cross-linguistic and diachronic observations, which, according to Comrie, show that there is no absolute parallelism between the present perfect and the other perfect tenses in natural language.

Several critical remarks can be made here:[10]

1 Comrie's basic argument seems to me to be fallacious. It purports to examine what would be the consequences if we assumed that there was a reference point in the structure of the present perfect and the past tense and leads to the conclusion that 'in terms of location in time ... this would give the same result' for the two tenses. But this conclusion is false. As pointed out by Comrie himself, the result in the case of the present perfect would be an analysis in which R coincides with S and E is anterior to R. In the case of the past tense, however, the result would have to be different. One would have to say that R is anterior to S and that E is located as simultaneous with R. Thus, in *John was here at some time yesterday* one would have to say that the event time (i.e. the time that John was here) is located relative to the reference time indicated by *yesterday* and that this R is anterior to S. There is no way in which one can analyse this sentence as involving an R that is simultaneous with S. The conclusion, then, is that Comrie's argument provides no evidence for the claim that present perfect and past tense do not differ in terms of time location. Rather, the outcome of the discussion is that there is every reason to assume that the two tenses do differ in their temporal schemata, and this in its turn leads to the conclusion that Comrie's abolition of R in the descriptions of the 'absolute' tenses is unwarranted.

10. Salkie (1989), which appeared after I had finished the first draft of this book, explicitly discusses and rejects Comrie's arguments. Especially the cross-linguistic and diachronic evidence is discussed at great length and rejected as inadequate. Since there is no point in reduplicating this refutation here, I will restrict myself to my own objections to the basic argument and refer to Salkie's article for the rebuttal of the cross-linguistic and diachronic arguments, as well as for Salkie's objections to the basic argument.

2 An observation supporting the above conclusion is that the past tense and the present perfect differ in the ways they collocate with temporal adverbials. A great number of time adverbials (e.g. *yesterday, two days ago, in 1983, then, at the time*, etc.) can combine with a past tense but not with a present perfect. Conversely, adverbs like *just* and *latterly* require the present perfect (at least in British English). It is not clear to me how one can account for this different behaviour if one assumes that there is no difference in the temporal structures expressed by the two tenses. (After all, adverbs like *yesterday* or *latterly* express time, not aspect. So the difference between past and present perfect cannot be purely aspectual.)[11]

3 If it were correct that the two tenses differed only in aspect (presence or absence of 'perfect aspect', i.e. of 'current relevance'), we would expect this to be reflected in the distribution of the two tenses. That is, we would expect that the present perfect should be used (or should at least be possible) whenever there is an idea of current relevance. However, this expectation is not borne out:

(7) I know what Tom is like. *I* (**have*) *spent my holidays with him two years ago.*

There can be no doubt that there is current relevance here: if I know what Tom is like it is *because* I spent my holidays with him. Still, we have to use the preterit, apparently because the situation is located at a time which wholly precedes the present moment. (Note that the perfect would have to be used if we substituted *for the last two years* for *two years ago.*) This means that the primary factor determining the use of the present perfect and the preterit is not the presence or absence of the idea of current relevance, but rather the way in which the situation is located in time.

4 The claim that 'current relevance' is the basic meaning of the present perfect is not uncontroversial. Examples can be found in which there seems to be no clear sense of current relevance:

11. According to Comrie (1985:78), the observation that we cannot say **I have arrived yesterday* supports his claim that the present perfect is basically different from the pluperfect, seeing that the pluperfect can collocate with an adverbial specifying the temporal location of E (e.g. *I had arrived yesterday*). However, it will be shown in chapter 7 that the incompatibility between the perfect and adverbials like *yesterday* follows from the fact that the use of the present perfect (which locates the situation in the pre-present sector, i.e. 'before NOW') implies that the speaker is thinking of NOW, whereas the adverbials in question refer to the past time-sphere (THEN). It is not possible for an English speaker to be thinking of NOW and THEN at the same time. In the case of the past perfect there is no such incompatibility, because the past perfect locates the situation in the past time-sphere. The time adverbial facts observed by Comrie thus follow naturally from the temporal meanings of the two tenses.

(8) These forks have been ours, they have been my cousin's, and now they belong to you. (Hirtle 1975:5)

I will discuss this point at some length in chapter 7. (I will argue there that current relevance is an implicature which follows from the basic temporal meaning of the present perfect.)

5 The claim that the present perfect does not have the same temporal structure as the preterit is further corroborated by the observation that in sentences like (8,a–b) the past reference point necessary to account for the past perfect can be established by the past tense, but not by the present perfect:

(9) (a) Tom (*has) arrived after I had left. (So, he can't have seen me.)
(b) I had left before Tom (*has) arrived. (So, he can't have seen me.)

Obviously, the use of *has arrived* should be grammatical if the present perfect located a situation in time in exactly the same way as the preterit does. (Note that the sentence added within brackets makes it clear that there is current relevance. This condition for using the present perfect is therefore satisfied.)

6 This conclusion is also in keeping with the fact that a stretch of discourse in the historical present uses the present perfect, not the preterit, to refer to a situation that is anterior to one described in the present tense:

(10) All of a sudden I get a letter that I'*ve been* accepted. (Wolfson 1982:5)

As noted by Wolfson (1982:4), the historical present uses the present tense and the present perfect where normal narrative uses the preterit and the past perfect, respectively. This suggests that the present perfect and the preterit belong to two different systems, i.e. represent different ways of locating situations in time. (As shown in chapter 2, the crucial difference is that the present perfect locates the situation in the present time-sphere, whereas the preterit locates it in the past time-sphere.)

7 If the present perfect had the same temporal meaning as the past tense, it would trigger 'Sequence of Tenses' in indirect speech in the same way as the past tense does. However, it is well known that this is not necessarily the case. (As argued in chapters 2 and 4, it is only when the present perfect is an 'indefinite perfect' that the process of temporal subordination involves the use of past time-sphere tense forms.) It is therefore not surprising that, in his 1986 article on indirect speech, Comrie treats the present perfect as one of the nonpast tenses (p. 290).

8 As we have seen in chapter 2, there are cases in which the past tense is used instead of the present tense because the speaker wishes to focus on some past situation. For example, when someone has just answered a telephone call from a client, he may be asked the question *What did that man want?* The use of the preterit in this question means that the speaker places the temporal focus on the time of the telephone call, which is now over. If the present perfect located situations in time in the same way as the preterit does, we should expect that the present perfect could be used similarly. However, this expectation is not borne out. In the given context it would be quite inappropriate to ask the question *What has that man wanted?*

9 Similar remarks can be made about the metaphorical uses of the preterit. As we saw in chapter 2, the speaker can shift the temporal perspective from the present to the past in order to represent the contents of his utterance as tentative (e.g. *I wanted to show you something*), hypothetical (e.g. *I would appreciate if you left now*) or unreal (e.g. *If only I knew his name!*). The present perfect has no similar uses.[12]

10 Another serious objection to Comrie's treatment of the present perfect is that in examples like

(11) I have not visited him since my car has been out of order.

it is clearly not true that the situations described by the present perfect forms are located entirely before the present moment. In (11), each clause represents a situation as covering a timespan which reaches up to (and includes) the moment of speech. For this reason we cannot substitute past tense forms for the present perfect forms in this sentence, for the past tense involves reference to a time which does not last up to the present moment. In other words, the representation 'E before S' which Comrie argues for the past tense does not fit the present perfect forms in (11), for Comrie states explicitly (on p. 122) that 'before' is to be interpreted as 'wholly before'.

11 As noted before, Comrie claims that his analysis is supported by the fact that, cross-linguistically and diachronically, there is no absolute parallelism between the present perfect and the other perfect tenses. Comrie adduces a number of observations in support of this claim. These various observations and the ensuing generalizations are examined in great detail by Salkie (1989), who comes to the following conclusions:

12. This, I would claim, is due to the fact that the present perfect does not locate its situation in the past time-sphere, whereas the preterit does. Since it is only the past sector that is felt to be temporally remote from t_0, only past tenses can be used metaphorically to express nontemporal remoteness.

a. Most of the generalizations lack a sound empirical basis. (For example, Comrie claims that Swahili has a present perfect but no pluperfect. On close inspection both claims turn out to be dubious. The same is true, according to Salkie, of similar claims which Comrie makes in connection with Luganda, Portuguese, French and German.)
b. Even where the data are unassailable, they do not warrant the conclusions that Comrie draws from them. (For example, even if Swahili has no pluperfect, this does not support the claim that the present perfect and the pluperfect are basically different: 'There are languages which have a past tense but no future tense. . . . This fact does not, however, lead to the conclusion that past and future tenses must be treated in radically different ways. Languages have gaps in their paradigms for many different reasons: such gaps are not normally given the kind of theoretical significance that Comrie assigns to the lack of the pluperfect' (Salkie 1989:13–14). (I would add to this that, more generally, there is no a priori reason why data from other languages or from the past should warrant conclusions concerning the present perfect in present-day English.)
c. As a matter of fact, the cross-linguistic data support rather than refute the view that the various perfect tenses should be analysed in a parallel way.[13]

2.2.3 The general conclusion from the above arguments is that the present perfect cannot be assigned the same temporal structure as the preterit. However, the assumption that the two tenses do have the same temporal structure is essential to Comrie's theory: if the temporal structures are not the same, one can no longer uphold the claim that there is no reference time involved in them. Clearly, if the temporal schema realized by *I have done it* is not the same as that realized by *I did it*, the difference can only be due to the different location or nature of the reference time, for the relation between the situation and the moment of speaking is the same in both cases: E wholly precedes S.

There are several other pieces of evidence that confirm the conclusion that even the 'absolute' tenses cannot be analysed without reference to R:

1 McGilvray (1974:37) notes that sentences like *John had seen Mary* or *John will have seen Mary* 'when uttered initially (at the start of a conversation) and independently of a special context are odd. . . . They would normally require a specification of a reference point by means of

13. Dahl, who originally subscribed to Comrie's position (see Dahl 1985:114), has dissociated himself from it in more recent work (Dahl 1987:498):

> If a language has a perfect, and in addition past and/or future tenses which can cooccur with the perfect, the combinations of these forms will be used as a pluperfect and a future perfect, respectively. In my material, there seem to be no counter-examples to this generalisation.

temporal adverbials (adsententials?) or discourse context.' McGilvray goes on to point out that the same thing is true of past tense sentences like *The farmer killed the duckling.* In fact, an innumerable number of linguists have made this claim in connection with the preterit (see e.g. McCawley 1973; Partee 1973;[14] Fillmore 1975:39; Kratzer 1978:69; Tregidgo 1979:191; Tichý 1980:344; Bäuerle and von Stechow 1980:397; Bach 1980:30).[15] It goes without saying that this observation can only be accounted for if one assumes that there is a reference time in the structure of the preterit.[16]

2 When dealing with the 'relative' tenses, Comrie repeatedly notes that time adverbials may have the function of establishing a reference point. For example:

> Often, the reference point is given by a time adverbial. . . . In the next example, . . . for the text to make sense *at ten o'clock* must be interpreted as the reference point prior to which John had left: *Mary came to visit John at ten o'clock; but John had already left at ten o'clock.* (pp. 65–6)

However, if it is true that *at ten o'clock* establishes a past reference point and that the past perfect *had already left* represents a situation as prior to this reference point, it seems logical to hold the view that *at ten o'clock*

14. It is this observation that induced Partee (1973) to reject the analysis of the preterit which was up to then generally accepted in (Priorian) tense logic. According to that analysis a sentence like *I didn't turn off the stove* is true if there is some time in the past at which I didn't turn off the stove. (In technical terms: the proposition 'I didn't turn off the stove' is true at time t in world w iff there is a time t' which precedes the time of utterance and 'I do not turn off the stove' is true at time t' in world w.) Partee correctly observed that the sentence means more than this: it means that in the context of utterance there is a particular past time which the hearer is expected to be able to identify, and that 'I do not turn off the stove' is true at that past time. (see also Dowty 1982:37; Richards 1982:75; Heny 1982:133; Barwise and Perry 1983:35).
15. The importance of the reference time also appears from the fact that when small children tell a story, they tend to begin every new sentence with a word like *then.* The following are authentic German and French examples noted by Kratzer (1978:83):
    (i) *Da* hat er gelesen. *Da* hat er geschlafen. Und *da* hat die Zeitung gebrannt und *da* ist er aufgewacht und *da* hat er ne Bierflasche oder ne Weinflasche und *da* hat er's ausgemacht. *Da* hat er den Hund genommen und *da* hat der Hund en Würstchen gekriegt.
    (ii) *Alors* c'est un petit lapin et un canard qui se promènent dans les champs. *Alors* ils arrêtent et ils voient une ruche. *Alors* ils doivent se penser qu'il y a du miel. *Alors* ils y vont mais il y a des abeilles. *Alors*, eux, ils sont perdus parce que les abeilles les piquent . . .

    Kratzer (1978:84) adds the comment: 'In den Erzählungen der beiden Kinder wird mit der ersten Äusserung von "da", beziehungsweise "alors", eine Betrachtzeit aus dem Blauen heraus eingeführt, und die folgenden "da"s und "alors"s beziehen sich jeweils auf eine Zeit, die kurz nach der soeben betrachteten Zeit liegt.'
16. Perhaps Comrie would say that the time which needs to be clear from an adverbial or from the context is E, not R. However, this would immediately raise the question why E needs specifying in the past tense, but not in the past perfect, future perfect, etc. (In *John had already left* it is not the time of the event but the reference time that must be identifiable if the sentence is to be fully interpretable.)

also serves as a reference point for the location in time effected by the past tense, i.e. that *came* represents a situation as simultaneous with this past reference point. Such an analysis would have the additional advantage of explaining why the preterit, and not the present perfect, can collocate with adverbials like *at ten o'clock*, *yesterday*, etc., i.e. with adverbials that establish a past reference point. (The claim that the present perfect locates a situation relative to a reference point which coincides (at least partly) with the point of speech would automatically explain why this tense cannot collocate with such adverbials.)

3 The assumption that the temporal structure of the preterit involves the relation 'E simul R' also proves necessary if we want to explain the difference in meaning between e.g. *At five o'clock I went into the church* and *At five o'clock the procession was going into the church.* The difference between the nonprogressive and the progressive past tense here concerns the way in which the situation is located relative to the reference point: in the former sentence the situation and the reference point are strictly simultaneous (commensurate); in the latter sentence the progressive represents the reference point as being included in the longer timespan through which the situation lasts. (The consequences this has for the definition of *simul* will be discussed below.)

4 Another argument is provided by sentences like *The balloon burst when they were high up in the sky.* If we assume that the only times involved in the use of the past tense are the time of the situation and the time of utterance, we get into problems, for neither of these times can be the time specified by the temporal clause. (The temporal clause cannot define the time of the situation, since the latter is punctual whereas the clause refers to a timespan.) There is no problem, however, if we assume that the time of the situation is located relative to (i.e. as simultaneous with) some time of reference, for the temporal clause can then be taken to specify the time of reference.[17]

5 Another argument to the same effect can be found in Comrie's own treatment of nonfinite clauses:

17. Dahl (1985:24) also disregards this necessity of taking three different times into account. He notes that one of the 'properties typical to tense categories' is that 'they semantically depend on the relation between the time that "is talked of" in the sentence and the time of the speech act'. As an example he gives the sentence *It was raining yesterday*, noting that this sentence 'concerns an interval of time which wholly precedes the point of speech'. However, I would say that there are two times 'talked of' in this sentence: the time during which it was raining and the time indicated by *yesterday*. In this example these two times may be commensurate, but in many sentences they are not. In *I met John yesterday* the past tense relates the time of my meeting John to the time indicated by *yesterday*, which is in its turn related to the time of the speech act.

> With the English non-finite verb forms, it seems in general clear that they have basically relative time reference, i.e. time reference defined relative to some deictic centre established by the context, so that the primary interpretation of *those sitting on the benches were asked to leave* is as 'those who were (at that time) sitting on the benches were asked to leave'.
>
> (pp. 21–2)

If we assume that the use of a past tense (*were asked*) creates a past reference point relative to which the situation expressed by the reduced relative clause is located,[18] then we must assume that exactly the same thing happens when we use a full relative clause (*who were sitting*) instead of the reduced version. That is, the past tense *were sitting* must also be taken to represent a situation as simultaneous with a past point of reference.

It is interesting to note that Comrie himself refers to this piece of evidence. However, his intention in doing so is to claim that it is not a valid argument. The examples he considers are the following (p. 57):

(12) (a) The passengers awaiting flight 26 proceeded to departure gate 5.
(b) The passengers who were awaiting flight 26 proceeded to departure gate 5.

According to Comrie, *awaiting* has relative time reference in (12,a), but the same thing is not true of *were awaiting* in (12,b). That is, Comrie claims that, unlike the former, the latter does not express simultaneity:

> the version with the finite verb merely locates the time reference of the awaiting as being in the (absolute) past, without equating it with any other point of time in the past. It is thus possible, though pragmatically less likely, that the past time points at which *were awaiting* and *proceeded* are located in this sentence are different time points in the past.
>
> (p. 57)

However, the fact that *were awaiting* and *proceeded* need not refer to the same time is no argument against the view that *were awaiting* is a relative tense form. On the contrary, it is precisely typical of relative tense forms that they need not represent the situation as temporally related to the head clause situation. In fact, Comrie himself points this out: 'for relative tenses all that is required is the identification of a reference point, the range of potential reference points being in principle all those compatible with the given context' (p. 58). Thus, the reference point (TO) with which

18. This assumption is also made by Vlach (1973), who introduces the operator THEN to account for sentences like *John was once going to cite everyone (then) driving too fast.*

the awaiting in (12,a) is simultaneous may be either the TO of the head clause or t_0 or any other TO established in the context. Comrie illustrates the latter possibility as follows:

> Suppose someone asked about the passengers who had been awaiting various flights yesterday, in order to find out what gate they had proceeded to on their flight of the previous day. In this context, the partial answer *the passengers awaiting flight 26 proceeded to gate 5* would receive as its only coherent interpretation *the passengers who were awaiting flight 26 yesterday proceeded to gate 5 the day before.* (p. 58)

It is clear, however, that (12,b) would receive exactly the same interpretation in this context. This proves that the time reference of *were awaiting* can also be related to several possible reference points.[19] The conclusion must therefore be that both *awaiting* in (12,a) and *were awaiting* in (12,b) are relative tense forms, which locate a situation relative to a reference time.

6 According to Comrie, the preterit can be represented simply as 'E before S'. In this representation, E is the 'time point or interval which is occupied by the situation to be located in time' (p. 122). The relation 'before' is defined as 'wholly (*or*: properly) before', i.e. 'on the time line an interval X is before an interval Y (X before Y) if and only if each time point within X is to the left of each point within Y' (pp. 122–3). Given this specification, the representation 'E before S' can only mean that the situation referred to is located exclusively in the past, i.e. is no longer continuing at the moment of speech. However, although a great many linguists have stated that this is, indeed, an aspect of meaning of the past tense,[20] the inference that the situation no longer holds at the time of speech is actually no more than an invited inference (implicature).[21]

19. The only interpretation which (12,b) does not share with (12,a) is the one on which t_0 is interpreted as the reference point for the relative tense form. The participle *awaiting* allows this interpretation, but *were awaiting* does not, because the past tense can serve as a relative tense in a past domain only.
20. See e.g. Leech 1971:9 ('the happening takes place before the present moment ... the past tense applies only to completed happenings'); Quirk *et al.* 1985:183 ('The event/state must have taken place in the past, with a gap between its completion and the present moment'); Eckersley and Eckersley 1960:160 ('The Simple Past (or Preterite) Tense is used ... to express an action wholly completed at some point, or during some period, in the past'); Scheurweghs 1959:323 ('The past tense refers to something in the past that has no longer any connection with the present'); Canavan 1983:36 ('the action was completed before the now segment began'); Dekeyser *et al.* 1979:64; Saurer 1981:67; Botne 1981:102; Quirk 1986:68; Brinton 1988:15, etc.
21. As usual, Jespersen (1931:60) hits the nail on the head when he states that the preterit 'refers to some time in the past without telling anything about its connexion with the present moment'. Similar remarks are also made by Oversteegen (1986:208) and Riddle (1978:77–81), who offers a number of arguments to prove that the notion 'no longer true at the time of the speech act' is not entailed, but conversationally implicated, by the past tense.

Interestingly, Comrie himself notes this in another section of his book:

> It should also be noted that use of the past tense only locates the situation in the past, without saying anything about whether that situation continues to the present or into the future, although there is often a conversational implicature that it does not continue to or beyond the present. . . . Thus, English *John was eating his lunch* (*when I looked into his room*) . . . says nothing about whether the situation still continues at the present moment or not.
>
> (pp. 41–2)

The observation made here by Comrie can only be accounted for if we assume that what is located completely before the point of speech is not E (the time of the situation) but some reference time. The past tense can then be taken to represent the situation as holding at this reference time. In other words, the structure of the past tense is 'E simul R before S'.[22] (Note that no direct relation is specified between E and S.) The fact that *John was eating his lunch* does not exclude the possibility that John is still eating at S can then be explained in one of the following ways:

a. One possibility would be to define the relationship of simultaneity in terms of either coincidence or overlapping. (This definition differs from Comrie's, who defines *simul* in terms of coincidence only: *'X simul Y* means that each time point in X is also in Y and vice versa.' (p. 123).) According to this definition E may either coincide with R completely (as in *At 5 o'clock we heard a shot* or *He was here from two o'clock until five*) or be shorter than R (as in *He left yesterday*) or be longer than R (as in *At two o'clock he was reading a book*). In the latter case part of the situation may precede the past reference time or follow it (including the possibility that the situation continues into the present or post-present).
b. There is an alternative way of accounting for this possibility. Instead of changing the definition of *simul* we can redefine the notion E. If we interpret E as meaning 'the time of that part of the situation that is referred to in the sentence' (rather than 'the time of the whole situation'),[23] then we can define 'E simul R' as expressing simultaneity in the strictest sense (i.e. coincidence). For example, we can say that the sentence *At five o'clock John was here* expresses no more than that

22. Mugler (1988:41) also argues that the semantics of the preterit involve these two relations.
23. In the linguistic literature the Reichenbachian notion 'event time' is usually interpreted as 'the time of the whole situation'. See, for example, Saurer 1981:66 ('If the event in question is a walk by John . . . the event time corresponding to it is the time beginning with the moment at which John starts walking and ending with the moment at which he stops walking.'); Berezovsky 1978:7 ('the interval of time that an occurrence, as described by the verb, takes place'); Hirtle 1975:18 ('the time contained within an event', i.e. 'the event's duration'); Bertinetto 1982:90 ('the interval of time corresponding to the given event'); Rigter 1980a:415; McGilvray 1974:38; Vet 1980:25, etc.

at R (five o'clock) a certain situation ('John was here') was holding. This means that the sentence does not necessarily refer to the whole interval of time that John was here but just to that interval that was (strictly) simultaneous with R. Whether or not that interval is part of a larger situation (in this case: whether or not John was here at other intervals besides R) is information that is not given in the sentence: the sentence is vague on this point. The fact that *At five o'clock John was here* does not exclude the possibility that John is still here at S is thus accounted for, while the relation *simul* in 'E simul R' can be defined in terms of coincidence.

The question which of these solutions should be accepted need not be answered here.[24] The point of the present argument is that Comrie's representation of the past tense as 'E before S' should be given up in favour of 'E simul R before S'.[25]

To be fair to Comrie, I must admit that I am not quite sure that the above discussion does full justice to his views. The point is that it is not clear whether Comrie interprets E as the time of the full situation or as that part of it that is referred to by the sentence. I have assumed that Comrie assumes the former interpretation, but the statements he makes concerning E are not really explicit on this point. On p. 122 Comrie defines E as the 'moment of event', adding that this 'moment' can be 'a point or an interval of time longer than a point' and that the term 'event' is here meant to refer to 'situations in general'. E thus stands for the 'time of the situation' and is said to be the 'time point or interval which is occupied by the situation to be located in time' (p. 122). This formulation suggests that E is the point or interval of time that is taken up by the situation on the time line (i.e. the time through which the situation lasts). However, on other occasions Comrie uses a different kind of formulation. On p. 122 he says that E is 'the time at which the situation is located'. Does this mean that the concept 'time of the situation' should be interpreted as denoting the time (indicated by a time adverbial or by the context) with which the situation is said to be simultaneous? If so, Comrie's representation of the past tense ('E before S') is an abbreviation of 'event at time t before S', which is in fact equivalent to the analysis that I have been arguing, viz. 'E simul R before

24. In Declerck (1986a) I accepted the first solution. However, in chapter 6 I will argue that solution 2 is better.

25. Exactly the same thing can be argued for the future tense. In Comrie's representation 'E after S', *after* is to be interpreted as 'completely after' (p. 123), yet the sentence *When you enter the kitchen there will be a loaf lying on the table* does not exclude the possibility that the loaf is already lying there, nor that it has been lying there for some time. This means that the future tense does not locate the time of the situation of the loaf lying on the table after the point of speech, but rather relates the situation to some time of reference (established here by the *when*-clause) which lies completely in the future relative to the moment of speech. The relation between the time of the situation and speech time is left unexpressed.

S'. At other places too, Comrie's formulations are confusing. For example, when he states that 'the meaning of the past tense is . . . location in time prior to the present moment' (p. 41), it is not clear what exactly it is that is located in the past. Is it the situation itself, or is the situation located at a time which is itself located prior to the moment of speech? Comrie's statements sometimes suggest the former interpretation (e.g. 'past tense means location of the situation prior to the present moment' (p. 36)), sometimes the latter (e.g. on p. 43 he states that the future tense can be defined 'as locating a situation at a time subsequent to the present moment').

In my above rejection of Comrie's analysis of the preterit ('E before S') I have assumed that Comrie interprets E as the time of the full situation. In my opinion, only this interpretation is compatible with the formula 'E before S'. If one agrees that what is located (wholly) before S is not the situation itself but the time at which the situation is located, then one must accept a formula like 'E simul R before S'.

In sum, if Comrie interprets E as the time of the full situation, then the formula 'E (wholly) before S' is not correct because the preterit does not imply that E is over at S. If, on the other hand, Comrie interprets E as that part of the situation that is referred to by the sentence, i.e. as that part of the situation that is located in time, then a theory that aims at dealing with time reference generally rather than just with tense should accept the formula 'E at R before S'. (As noted above, Comrie does not aim to provide a full theory of time reference. If what Comrie has in mind is the latter interpretation of E, my conclusion is probably no real objection to his theory (which deals with tense only). However, the point I have made cannot be ignored by a more full-fledged theory of time reference.)

7 The arguments that we have offered up to now in support of the claim that we cannot do without a reference time in the analysis of the 'absolute' tenses have all been arguments concerning the preterit or the future tense. Let us now see if the claim can also be argued in connection with the present tense.

As noted before, Comrie's representation of the present tense is 'E simul S'. At various places in the book (e.g. pp. 2, 36, 37) he states explicitly that S is a 'point in time', i.e. that S is not a timespan. (On p. 37 he discusses a couple of cases in which 'there is literal coincidence between the time location of a situation and the present moment', but concludes that 'a more characteristic use of the present tense is in referring to situations which occupy a much longer period of time than the present moment, but which nonetheless include the present moment within them'.) Now, the relation 'simul' is defined as follows: '*X simul Y* means that each time point in X is also in Y and vice versa. *Simul* is, of course, a symmetrical relation, i.e. *X simul Y* is equivalent to *Y simul X*.'

(p. 123). If we combine this definition of 'simul' with the claim that S is punctual (nondurative), then the representation 'E simul S' which Comrie proposes for the present tense must imply that E (the time of the situation) must not be a timespan, i.e. that only punctual situations can be referred to by the present tense. This is, of course, incompatible with Comrie's own observation that the present tense can be 'used to speak of states and processes which hold at the present moment, but which began before the present moment and may well continue beyond the present moment, as in *the Eiffel Tower stands in Paris* and *the author is working on chapter two*' (p. 37).

One might object that this argument no longer holds if one defines E (the time of the situation) as 'the time of that part of the situation that is referred to in the sentence' rather than as 'the time of the full situation'. For, if one accepts this definition, the formula 'E simul S' predicts that a sentence like *John is in London* makes a statement about just that part of the situation that is contemporaneous with the time of speech – a prediction which might seem to be correct. However, this is only seemingly so. It suffices to add a time adverbial to the sentence to see that the analysis 'E simul S' is still deficient:

(13) John is in London today.

This sentence is not interpreted as an assertion about just that part of the situation that coincides with the moment of speech. It is clearly an assertion about a longer time interval. That is, the correct temporal interpretation of (13) is that the time that John is in London wholly or partially coincides with the time indicated by *today*, and that this time itself includes the moment of speech.[26] The temporal structure realized in (13) must therefore involve the three primitives E, S and R (where R is the time referred to by *today*).[27]

8 Last but not least, it should be pointed out that Comrie (1985) treats the present, past and future tenses exclusively as absolute tenses. However, we saw in chapter 2 that each of them can also be used as a relative tense, i.e. as a tense that relates a situation to some reference

26. Again, it should be pointed out that Comrie does not seek to fully account for the temporal interpretation of a sentence but only considers the contribution made by tense (while disregarding the role of time adverbials). It may therefore be argued that the point I am making is not really an objection to Comrie's theory. However, there can be no doubt that the point in question cannot be ignored by a full theory of time reference, and I personally have serious doubts about a theory of tense that cannot be fitted into a larger theory of time reference.

27. This is a simplification. In the next chapter it will be argued that a sentence like *John is in London today* actually involves reference to four primitives, as we must make a distinction between the full timespan indicated by *today* and that subinterval of it (possibly the entire timespan) during which the situation of John being in London holds. However, it would be premature to emphasize this distinction at this point. What I am trying to make clear in this chapter is that Comrie's elimination of any primitives besides E and S from the structures of the absolute tenses is untenable.

time other than t_0. Comrie's analyses, which each time link the situation directly to t_0, are of course totally incompatible with this relative use. Obviously, we need a reference time in the temporal structure of any relative tense. And since numerous arguments have been advanced to show that there is also a reference time in the structure of the absolute tenses, we can conclude that the temporal structure realized by the preterit, present tense or future tense is essentially the same in the relative use of the tense as it is in the absolute use. Thus, the preterit has a structure which involves the relations 'E simul R' and 'R before t_0'. When it is used as absolute tense, R is indicated or implied in the clause itself.[28] When it is used as relative tense (as in *I read a book while I **was waiting***), R is the time to which the situation is temporally subordinated (i.e. the binding TO). In that case only the relation 'E simul R' is realized in the clause involving the relative past tense;[29] the relation 'R before t_0' is realized in the clause containing the binding TO.

3 CONCLUSION

The purpose of this chapter has been to establish a couple of basic principles which must form part of any theory that aims at analysing the temporal structures underlying the tenses. These principles have been derived both from the theories of Reichenbach (1947) and Comrie (1985) and from the criticism that has been levelled at these theories. In the next chapter I will refine the principles further and supplement them with others. In this way we will gradually arrive at a coherent framework for analysing the temporal schemata expressed by the tenses.

28. That is, the clause either contains an adverbial referring to R or it takes an R from the context (see Partee 1984:265).
29. This naturally explains why a relative past tense always expresses simultaneity, not anteriority.

6 The structure of tense 2: general principles

1 PRIMITIVES AND RELATIONS

The discussion of Reichenbach (1947) and Comrie (1985) has revealed a couple of basic facts and principles, which our theory will have to accommodate.[1]

1.1 The primitives

Both Reichenbach and Comrie work with three primitives: moment of speech, time of the situation (event time) and reference time. Some of the problems noted above suggest that a larger set of primitives is required. The theory developed here will make use of the following:

1.1.1 We first of all need something corresponding to Reichenbach's 'event time'. I will speak of 'the time of the situation'. In doing so I will continue to use the term *situation* as a cover-term for anything that can be expressed by the predicate of a sentence. That is, a situation may be a state, an act, an activity, a process or an event.[2] The abbreviation TS will be used for 'time of the situation'. Since the term 'situation' is a

1. The above criticism of Reichenbach (1947) and Comrie (1985) has induced me to attempt to develop a theory that is not subject to these objections. A first version of such a theory was written in 1985 and published in 1986. This first version (Declerck 1986a) differs from the one I will argue presently in a couple of respects. The most important difference concerns the way in which the notions 'time of the situation' and 'simul' are defined. Another difference is that the new version will also try to accommodate the use of progressive forms, which was disregarded in Declerck (1986a). For the rest, however, the two theories have much in common, so that it would be rather uneconomical if I expounded them separately. For that reason I prefer to set out the theory as I see it today and indicate where necessary how it distinguishes itself from the earlier version.
2. That Reichenbach's term 'event' is not a felicitous term for what we call a 'situation' has often been remarked in the linguistic literature. For example, Barense (1980:20) points out that 'many verb phrases are not associated with any events'; she therefore replaces 'time of the event' by 'predicate satisfaction time'. Lo Cascio (1986:194) replaces 'event' with 'state of affairs'. De Groot (1983:73) chooses the same term as cover-term. I prefer to use 'situation' because this term has become increasingly widespread since it was used by Comrie (1976).

cover-term for any kind of state or 'happening', whether it is durative or punctual (nondurative), TS may in principle be either a time point or a longer timespan. In both cases we may say that TS is a 'time interval'.

1.1.2 Since I will be using the term *situation* instead of Reichenbach's and Comrie's *event*, it would be confusing to go on using the symbol S for 'speech time'. I will therefore replace the symbol S by the initials TU, which stand for 'time of utterance'.[3]

1.1.3 As noted in chapter 2, the time that serves as primary 'deictic centre' for the speaker is usually TU, but it may also be the time of decodification. We therefore need the notion of 'temporal zero-point' (t_0) to cover these two possibilities.

1.1.4 The term 'reference point' or 'reference time' also deserves some comment. What is striking in Reichenbach (1947), Comrie (1985) and most other treatments of tense is that the term covers more than one aspect of meaning. The reference time is usually defined as 'the point of perspective from which an event is viewed' (Guenthner 1977:83) or as the time 'with reference to which we ... locate situations in time' (Comrie 1985:14). But the reference time is also a time which can be 'established' (Comrie 1985:14) by an adverbial or by the context.[4] The former statement means that the reference time is the 'time pointed from', the latter represents it as a 'time pointed to'. In other words, 'reference time' will mean 'time referred from' in one context and 'time referred to' in another.[5] This may be confusing, and I will therefore give

3 The term 'time of utterance' should be interpreted in the broad sense of 'coding time' (which may also be the time of writing a letter, printing a journal, etc.). Some (logically oriented) linguists also use terms like 'evaluation time' (de Vuyst 1983:168, 171; Guenthner 1977:6; Heny 1982:112) or 'context of evaluation' (Bach 1981:66), but such terms are rather confusing, as they are more often used in the sense of Reichenbach's 'reference time' (see Smith 1983:275 ('the time of evaluation of a sentence is Reference Time in Reichenbach's terminology'); Lo Cascio 1986:192 ('Reference time is the time in relation to which an event E is true'); Vet 1981:159, 1983:199; Hinrichs 1983:173, etc.). Barense (1980) uses still another term, viz. 'base time'.

4 Both elements of meaning also figure in Reichenbach's definition (or rather illustration) of 'reference point'.

5 Sometimes one of these meanings is reflected in the term that is used to indicate 'reference time'. For example, Dieling and Kempter (1983:31) clearly have the notion 'time referred to' in mind when they speak of 'besprochene Zeit' ('time spoken of'). Similarly, Smith (1977b:569) defines 'reference time' as 'the time indicated in a sentence', while Rigter (1986:101–2) replaces the term *point of reference* by *focal episode* (i.e. 'that interval ... on which the speaker's attention is focused', i.e. 'that the speaker is talking about').

When R is defined as 'the time at which the situation is relevant' (Papp 1985) or as 'the time at which the truth of the sentence is evaluated', it is not always clear which of the elements 'time referred from' and 'time referred to' predominates.

It is also worth repeating that what we have called a binding TO is not necessarily a reference time according to these definitions. For instance, in *I was there when he arrived*

up using the term in favour of the unambiguous labels 'time of orientation' (henceforth: TO) and 'time (that is) established' (henceforth: TE). A TE is a time that is established by an adverbial or by the context. A TO is a time to which a situation or another TO is related: whenever we use a tense to refer to a situation, this situation is located relative to (i.e. as simultaneous with) a TO. This TO may itself be related to some other TO, etc. For example, the future perfect *will have arrived* in *John will have arrived by that time* represents the situation of John's arrival as simultaneous with a TO which is anterior to another TO, which is itself posterior to t_0. The temporal schema realized by the future perfect therefore involves the relations 'TS simul TO_3', 'TO_3 before TO_2' and 'TO_2 after TO_1' (where TO_1 is t_0). (The attentive reader will have noted that the first of these three relations was not explicitly referred to when we discussed the future perfect in chapter 2. The reason was that the relation 'TS simul TO' is realized by any tense form, so that it was not necessary to draw explicit attention to it when the different uses of the tenses were explored. However, the relation was not quite ignored, for we did introduce the notion 'situation-TO': this concept implicitly refers to the relation 'TS simul TO', since a situation-TO is nothing other than a TO with which a situation (more precisely: a TS) is simultaneous.)

This clear distinction between 'time of orientation' (TO) and 'established time' (TE)[6] does not mean that the two notions are mutually exclusive. In a sentence like *At 5 p.m. yesterday John had already left*, the time which is referred to by the time adverbial *at 5 p.m.* (and which is therefore a TE) serves as a TO for the use of the past perfect: the past perfect expresses that the situation-TO was anterior to the TO 'at 5 p.m.'. On the other hand, the TE established by *yesterday* does not function as a TO: the sentence does not express that John left before yesterday. Rather, the function of *yesterday* is to establish a TE which acts as a 'frame' (see Dahl 1985:30) for (i.e. includes) the TE established by *at 5 p.m.* This means that a TE may or may not function as a TO.

The reverse is also true. Although the TO at which a tense locates a situation (i.e. the TO with which TS is simultaneous, in other words, the situation-TO) is mostly either specified by an adverbial or recoverable from the preceding context, there are exceptional cases in which it is not established in either of these ways. A typical example is offered by narrative texts that begin *in medias res*. If the first sentence of such a text contains no time adverbial, the situation-TO is not specified. In that case it is the nature of the tense itself that tells us how the TO in question is located relative to t_0. Another kind of discourse in which the situation-TO is not specified is illustrated by the following sequence:

the head clause specifies the binding time for the subclause, but the subclause indicates the evaluation time for the truth of the head clause (see Enc 1987:634).

6 The notion TE corresponds with what Bertinetto (1982) and Rohrer (1986) call 'localization' and 'temporal location time', respectively.

(1) Steve Davis has now collected twenty-seven points. *He needed those points badly*, for White had made a break of fifty-six.

The TO with which the situation denoted by *needed* is simultaneous (and to which the situation denoted by *had made* is anterior) is not established by the preceding sentence, nor by an adverbial. Still, the verb form *needed* clearly implies that the speaker has a TO in mind which lies in the past with respect to t_0.

In cases like the above ones the verb form implies the existence of a TO and situates it with respect to t_0. Still, such a TO is not 'established' (in the sense of 'identified'), and I will therefore not consider it to be a TE. This means that not all TOs are TEs.[7] Since, as we have seen, the converse is also true (i.e. a TE may or may not function as a TO), it may be handy in some contexts to have different symbols for TEs that are TOs and for TEs that are not. I will therefore introduce the symbol TOE for the former class. The abbreviation TOE will be used when we wish to express either that the TE we are talking of also functions as a TO, or that the TO we are talking of is a TE, i.e. is established by the context or by an adverbial.

1.1.5 The final primitive that I will introduce is the 'basic time of orientation' ('basic TO' or 'TO_1'). The basic TO is that TO in the structure of a tense from which the temporal relations expressed by the tense begin to be computed. The unmarked situation is for t_0 to function as TO_1, but in some cases TO_1 is some other time interval. This becomes clear when we compare the following sentences:

(2) (a) John had left before his wife arrived.
(b) (Do not leave too early, or) the newspapers will write that you had left before your wife arrived.

7 Like Bertinetto (1982:88–9), Richards (1982:75) and Heny (1982:134), I do not agree with Smith's (1977b:570, 1978b:80, 1981:215) claim that a sentence in the past tense is semantically incomplete and hence not fully interpretable when there is no time adverb establishing a reference time. In my opinion, sentences like the following are quite all right:

(i) (a) We know that *John lived in Boston for some time and was quite a respectable citizen there.* But we do not know when that was, nor where he went afterwards.
(b) Is Bill in the house? – No, *he went away.*
(c) What became of your sisters? – Oh, *Jane married a sailor, Sue bought a gold mine, and Marjorie joined the air-force.* (Heny 1982:134)

The italicized sequences are fully interpretable, although there is no overt time adverbial, nor a context establishing a particular time reference.

Moreover, it should be noted that not all temporal adverbials really identify a specific time interval. According to Ritchie (1979:97), 'time clauses serve to locate events or situations *relative to one another*, not with respect to some absolute time-line.' This certainly seems true of examples like (ii), which leaves the times of both situations referred to indefinite:

(ii) When I lived in London, the rents were high.

In both (2,a) and (2,b) the past tense *arrived* locates a situation at a TO which is past with respect to TO_1. In (2,a) this basic TO_1 is t_0; in (2,b) it is a time interval that is posterior to t_0 but which behaves as if it were t_0 (see chapter 2). Since it seems reasonable to say that the past tense realizes the same temporal schema (viz. 'TS simul TO_2 before TO_1')[8] in both cases, we must conclude that the temporal relations expressed by a tense form need not always start from t_0: the 'basic TO' (TO_1) can also be some other time interval. (Of course, this other time interval will ultimately also be related to t_0, but this temporal relation is not expressed by the tense form itself. Thus the temporal schema realized by *arrived* in (2,b) contains only the relations 'TS simul TO_2' and 'TO_2 before TO_1'. The relation 'TO_1 after t_0' is not established by *arrived* but by the context in which it is used.) The same thing can be argued in connection with the form *had left*. In both (2,a) and (2,b) the past perfect realizes the schema 'TS simul TO_3 before TO_2 before TO_1'. In (2,a), TO_1 is t_0. In (2,b) TO_1 is a time interval that is posterior to t_0. This posteriority relation is not expressed by the past perfect itself.

In sum, we will use symbols as indicated in (3):

(3)

t_0	temporal zero-point (time of utterance or time of decodification)
TU	time of utterance (coding time)
TS	time of the situation
TE	time established (by a time adverbial or by the context)
TO	time of orientation
TOE	time that is both TO and TE
TO_1	basic TO (the TO that is the starting point for the temporal relations expressed by a tense form)

1.2 Relations

Apart from the above primitives (kinds of times) we need a number of relations. Both Reichenbach and Comrie work with three relations only: simultaneity, anteriority and posteriority. I will argue below that this set is not sufficient. (We also need the inclusion relation.) Moreover, I will stress that it is very important to define these relations (especially simultaneity) very accurately. This is something I will do shortly.

1.3 Time adverbials

One of the remarks we have made in connection with both Reichenbach's theory and Comrie's is that they do not offer a full theory of time reference, so that their temporal schemata do not provide for the

8 This, of course, is an abbreviated way of saying that the (absolute) past tense involves two relations in its structure: 'TS simultaneous with TO_2' and 'TO_2 anterior to TO_1'.

analysis of elements other than tense. Thus, neither theory is suficiently rich to offer a full temporal analysis of sentences like *When I arrived, Peter had tried to phone me twice during the preceding week*, because neither theory provides for an element corresponding to the time referred to by *during the preceding week*. When we look at the sentence in question, we ascertain that, apart from the moment of speech (t_0), it refers to three different times: the time of my arrival, the times when Peter tried to phone me and the time referred to by *the preceding week*. Reichenbach's and Comrie's analyses of the past perfect ('E–R–S' and 'E before R before S') provide for only two time-points besides S. Since R is obviously the time of my arrival, and since E would seem to be the times when Peter tried to phone me, Reichenbach's and Comrie's theories fail to provide for the time referred to by *during the preceding week*, i.e. the time relative to which the events of John trying to phone me (which make up E) are located.

The exclusion of R from the 'absolute' tenses in Comrie's theory entails similar problems even in connection with very simple sentences such as *Today my lecture is about tense and time*. Comrie's analysis of the present tense ('E simul S') contains no element corresponding to *today*, for S (TU) is by definition punctual (nondurative), and the time of the situation, however it is defined, is certainly not the same time interval as that denoted by *today*.

In my opinion, it is more interesting to build a full-fledged theory of time reference than to restrict oneself to the role played by tense. What is more, I do not believe one should even try to build a theory of tense in isolation. A theory of tense must form part of a full theory of time reference, and its value can only be judged with reference to the latter. In my opinion, a theory of tense can only be correct if it is consonant with, and can be fitted into, a full theory of how situations are located in time.

Comrie defines tense as the grammaticalization of location in time. However, it seems obvious that locating a situation in time is not something which tenses do quite on their own, without the co-operation of time adverbials or context.[9] As we have seen, it is the time adverbials and the context that define (at least some of) the TOs in the structures of the tenses. For this reason Reichenbach has tried to assign to the adverbials and to the context a role in his analyses. He states explicitly that 'reference points' are established by temporal adverbials or by the (linguistic or nonlinguistic) context. Moreover, he claims that a temporal adverbial always denotes such a reference point. Comrie (1985) does not provide for time adverbials in his temporal schemata, but he does claim that a time adverbial refers either to a reference point (if there is one) or

9 As we saw in chapter 2, the present tense and the preterit have various uses involving a shift of temporal perspective (e.g. the historical present, the present referring to the future, the modal past). In such uses the real time reference of the tense must be clear from adverbial, contextual or pragmatic information.

to the time of the situation. In sum, both Reichenbach and Comrie touch upon the role played by time adverbials, but they ultimately disregard them because they choose to build a theory of tense rather than a more general theory of time reference.

In what follows I will assume that locating a situation in time is a task that is performed not by the tense of the verb form alone but by the sentence as a whole (and this, moreover, with the help of the context).[10] This means that the temporal schemata that I will propose will have to be such that they contain sufficient elements for every temporal adverbial in the sentence to correspond with a particular element in the schema.

1.4 Time of orientation

Our discussion of Comrie (1985) in chapter 5 has led to three further conclusions in connection with the notion TO:

1 Prior (1967) has stressed that an adequate theory of tense must be such that no sharp distinction is made between 'speech time' and 'reference time', since TU is just the first 'reference time' in the structure of the tenses. Our theory will be in accordance with this remark, as it treats t_0 as a TO: we say that, if the temporal relations start from t_0, t_0 is the basic TO (i.e. TO_1) in the schema. (However, we depart from Prior when we state that the basic TO need not always be t_0 – see above.)

2 When we say that a situation 'is located at' a certain time, what we mean is that it is simultaneous with that time. That is, any situation that is referred to in a tensed sentence is represented as simultaneous with some TO. This means that we cannot accept formulas such as Comrie's 'E before S' (for the past tense), in which the situation is directly related to t_0. Rather, we must represent the structure of the past tense as 'TS simul TO_2 before TO_1'. This general principle – that any tensed clause represents its situation (more precisely: its TS) as simultaneous with some TO – is in keeping with McCoard's (1978:92) claim that 'every event is simultaneous with some time; to happen means to become present at some time.' (See also Prior 1967:15;[11] Rigter 1980a:414; Bartsch 1988/9:146.)

10 See Dowty (1979:323): 'tenses in English are primarily parasitic on time adverbials ... and cannot be properly understood without an understanding of their interaction with time adverbials'. See also Bäuerle (1977) and Fenn (1987:9), who stress that tenses and adverbs have a complementary function.

11 Prior bases his tense logic on the claim that *I was eating my breakfast* means 'It was the case that I am eating my breakfast' and that *I will be eating my breakfast* means 'It will be the case that I am eating my breakfast.' He comments:

> The construction I am sketching embodies the truth behind Augustine's suggestion of the 'secret place' where past and future times 'are', and his insistence that wherever they are, they are not there as past or future, but as present. The past is not the present but it *is* the past present, and the future is not present but it *is* the future present. (Prior 1967:8)

As a matter of fact, we have already taken account of this claim in chapters 2–4, where we worked with the notion of 'situation-TO'. In chapter 2 (section 4) I defined the situation-TO as 'the TO with which the situation is represented as simultaneous'. (I also briefly referred to the relationship between this notion and the concepts 'time of the situation' and 'full situation'. I will return to this relationship shortly.)

3 It follows from this that any tense, whether it is used as relative or as absolute tense, has a structure that involves at least one TO (viz. the situation-TO) next to TO_1. This is another requirement that our analysis will have to satisfy.

1.5 Chains of relations

We have also observed that tenses must not be analysed as configurations of times on the time line (as in Reichenbach's theory), because a tensed sentence expresses relations between some times, but not necessarily between all the times involved in the temporal structure. That is, the temporal structure of a tensed sentence is like a chain of relations: in the same way as each link in a chain is only connected with the immediately preceding link and the immediately following one, each time in the temporal structure is related to some other time(s) but not to all of them. Thus, the conditional tense in *John said that he would do it* expresses that the situation-TO of the *that*-clause (the time of his doing it) is posterior to the situation-TO of the head clause, and that this TO is itself anterior to t_0, but it does not express a direct relation between the situation-TO of the *that*-clause and t_0 (i.e. it does not locate the time of 'doing it' relative to t_0). This vagueness (as to whether John has already done it, is actually doing it right now, or is still to do it) cannot be expressed in a system (such as Reichenbach's) which places all the relevant times on the same line. In such a system each point occupies a position relative to every other point. That is why, as we have seen, Reichenbach's system generates three different representations for the conditional tense, thus (wrongly) claiming that the conditional tense is three ways ambiguous, rather than just vague with respect to the relation between its situation-TO and t_0.

1.6 Further preliminaries

There are three other issues that need settling before we can start reconstructing the schemata underlying the different tenses:

1 How do we define the notion TS?
2 What does it mean to say that two time intervals are simultaneous?

3 If the TE established by an adverbial is not a TO in the structure of the tense of the sentence, what is then the relation between this TE and the TO(s) in that structure?

These three questions are closely related to each other. In fact, the different answers that can be given to the first question will lead to different answers to the other two questions as well. It is therefore impossible to treat these questions in isolation.

Let us begin by considering the first question. What do we mean when we speak of the 'time of the situation' (TS)? Reichenbach just uses the term ('time of the event') without defining it in any way. Comrie (1985:122) says that TS ('event time') is the 'time point or interval which is occupied by the situation to be located in time' (p. 122). As noted before, this formulation suggests that Comrie uses the term in the way that Reichenbach's notion 'event time' has mostly been interpreted in the linguistic literature: the time of the event is 'the interval of time that an occurrence, as described by the verb, takes place' (Berezovsky 1978:7). That is, 'if the event in question is a walk by John ... the event time corresponding to it is the time beginning with the moment at which John starts walking and ending with the moment at which he stops walking' (Saurer 1981:66).[12] According to this definition, TS in both (4,a) and (4,b) is the stretch of the time line occupied by the situation of the book being in the kitchen (i.e. the length of time of which the statement *The book is in the kitchen* is true).

(4) (a) A moment ago the book was in the kitchen.
(b) When you come home *the book will be in the kitchen.*

However, when discussing Comrie (1985), I have pointed out that this definition of TS is incompatible with the analyses 'E before S' and 'E after S' which Comrie proposes for the past and future tenses, respectively. If TS is the timespan occupied by the situation, and if 'before/after' is defined as 'wholly before/after' (see Comrie 1985:122–3), then the analyses 'TS before t_0' and 'TS after t_0' wrongly predict that (4,a–b) exclude the possibility that the situations also hold at t_0 (i.e. that TS includes t_0).

In our theory, which holds that TS is by definition simultaneous with a TO, there appear to be two *a priori* ways to avoid this problem. If we share the view that TS is the full time occupied by the situation – let us provisionally use the term 'situation span' (SS) to refer to this – then we can change the definition of 'simul' in the relation 'TS simul TO'. We can say that 'X simul Y' means no more than that X and Y have at least one time point in common. The statement that a situation is simultaneous with a TO is then true even if the situation is a longer time interval than

12 For similar definitions see e.g. McGilvray (1974:38); Woisetschlaeger (1976:94); Rigter (1980a:415); Vet (1980:25); Bertinetto (1982:90).

the TO in question. Since the structure we assign to the past tense is 'TS simul TO_2 before TO_1', this definition of 'simul' would seem to account for the fact that (4,a) does not exclude the possibility that the situation continues up to (and even beyond) TO_1 (= t_0): the situation that is simultaneous with TO_2 (which corresponds with the time indicated by *a moment ago*) may be much longer than TO_2, and since it is TO_2 (rather than the time of the situation itself) that is represented as anterior to t_0, there is nothing that prevents the situation from continuing up to t_0.

Another reason why this treatment of 'simul' in terms of either coincidence or overlapping would seem to be satisfactory is that it creates the possibility of incorporating the time adverbials in the schemata of the tenses. If we assume (with Reichenbach) that a time adverbial indicates TO (which, in our terminology, is therefore a TOE), then every one of the following cases would seem to be accounted for:

(5) (a) At that moment the gun went off.
(b) I was in the house from seven o'clock until lunch.
(c) I left yesterday.
(d) At that moment John was sleeping.

These four sentences would seem to illustrate four different kinds of 'simul' relation. In (5,a) and (5,b), SS (the timespan occupied by the situation) and TOE (the time interval identified by the adverbial) coincide, the only difference being that both are punctual in (5,a), whereas both are durative in (5,b). In (5,c), the situation span (SS) is an interval of time that is included in the longer timespan TOE. In (5,d), it is the other way round: TOE is shorter than the length of time taken up by the situation and is included in it. In the latter case the sentence does not exclude the possibility that the situation is still holding at t_0.

In sum, it would seem that several data are accommodated by a theory which holds that the definition of the relation 'TS simul TO' involves the following claims: (1) 'X simul Y' means that X and Y have at least one time point in common; (2) we are talking here of the situation as a whole, i.e. TS (the time of the situation) is SS (the timespan occupied by the situation).

However, in spite of these apparent advantages, this theory is untenable.[13] This should be clear from the following pieces of evidence:

1 There is a problem concerning the present tense. In the above theory, the structure of the present tense will be said to be 'SS simul TO simul t_0'. Thus the sentence *Today my lecture is about tense and time* will be analysed as meaning that SS (the time taken up by my lecture) is simultaneous with a TOE (the time indicated by *today*) in the sense that SS is a subinterval of TOE, while TOE is itself simultaneous with t_0 in

13 I myself argued for this theory in Declerck (1986a). I have now come to the conclusion that it should be given up in favour of an alternative analysis (to be presented below).

the sense that TOE includes t_0. In this case none of the three times coincides with either of the other two, but this is in keeping with the way we have (provisionally) defined 'simul' (viz. in terms of either coincidence or overlapping). However, the above analysis raises an insurmountable problem. If 'simul' is defined in this way, then 'SS simul TO simul t_0' does not entail 'SS simul t_0'. This means that, on this analysis, a sentence in the present tense need not represent a situation as holding at t_0. Thus the sentence *Today my lecture is about tense and time* should be appropriate if it were uttered at, say, 5 p.m., while the lecture referred to was actually given in the morning of the same day. (In this case the structure 'SS simul TO simul t_0' would be realized, since TO is *today*, which (on the present definition of 'simul') is simultaneous both with SS (a time interval in the morning) and t_0 (a time interval in the afternoon).) However, it does not appear to be the case that the sentence *Today my lecture is about tense and time* is appropriate if the lecture is in the morning, while t_0 is at 5 p.m. In a situation like this I can only use the preterit or the present perfect. Clearly, a sentence in the present tense necessarily implies that the situation coincides with or includes t_0. However, on the present definition of 'simul', the representation 'SS simul TO simul t_0' wrongly predicts that this is not necessary.

2 There is another problem in connection with the present perfect and the preterit. In Declerck (1986a) I argued that the present perfect differs from the preterit in that the latter implies that the TO with which SS is simultaneous lies wholly before t_0, whereas the present perfect implies that this TO starts before t_0 and reaches up to it. This introduction of a second kind of 'before' relation (with the meaning 'before and up to') seemed to account for the various uses of the present perfect:

(6) (a) John has been here since five o'clock.
(b) I have done it this morning.
(c) I have lived in Paris (but now I am living in Brussels).

In (6,a), the TO which reaches to t_0 is established by *since five o'clock*; the relation 'SS simul TO' is one of coincidence. In (6,b), a similar TO is indicated by *this morning*; SS overlaps only a section of TO that does not include t_0. In (6,c), the TO is not specified in the sentence itself, but the use of the present perfect implies its existence, and it is pragmatically interpreted as being the lifetime of the speaker; SS again does not reach up to t_0.

Satisfactory as the above analyses may seem, there is a problem in connection with sentences like (7,a–b):

(7) (a) I have done it today.
(b) I did it today.

Both these sentences are grammatical. As to (7,a), this grammaticality is predicted by the current theory: we can say that *today* indicates the TOE that lasts up to t_0 and that SS is simultaneous with TOE in the sense that it coincides with a section of TOE that lies wholly before t_0. However, the grammaticality of (7,b) is not predictable. According to the analysis of the past tense presented above (viz. 'SS simul TO before t_0', in which 'before' means 'wholly before'), it should be impossible to use the past tense if TOE is established by an adverb referring to a timespan that includes t_0. Yet, sentence (7,b) is impeccable. So there must be something wrong with this analysis.

The observation that (7,b) is grammatical in spite of the fact that it combines the use of a past tense with an adverb referring to a stretch of time that does not properly precede t_0 can be interpreted in two ways. We might say that, since *today* establishes a TOE, the representation of the past tense cannot involve the formula 'TO before t_0'. However, this conclusion is quite untenable. If there is one thing that seems to be beyond question, it is the fact that the past tense expresses this kind of 'before' relation. In consequence, the only possible conclusion is that the TE specified by *today* in (7,b) does not establish the TO with which SS is simultaneous. Whereas TE reaches up to t_0, TO does not. In other words, the time interval indicated by *today* is a TE but no TOE (i.e. it is a TE which does not function as TO). This means that we will have to enrich our representation of (7,b) so that it involves reference to this TE. (More specifically: in the case of (7,b) we will have to supplement the relations 'TS simul TO' and 'TO before t_0' with the relation 'TE (properly) includes TO' – see below.)

The conclusion that *today* does not indicate TO in (7,b) is intuitively satisfactory. We do, indeed, interpret (7,b) as meaning that the situation is located at a time which is part of today but which wholly precedes t_0. The time in question is not identified in the sentence, but it is the TO with which the situation is represented as simultaneous.[14]

The above change in the interpretation of TO provides an answer to the question how it is possible to use the past tense in combination with adverbials like *today*, but it does not eliminate a second problem that is raised by sentences like (7,b). Even if we accept that SS is simultaneous with an (unidentified) TO that is anterior to t_0, our current definition of 'simul' entails that any sentence in the past tense must allow the possibility that SS reaches up to and even beyond t_0. (Since 'SS simul

14 Lo Cascio (1986:205) remarks the same thing in connection with the past perfect: in the sentence *I said yesterday that he had called up* 'the event time of the telephone call is stated as prior to the time which binds the event "say" and not to the time "yesterday"'. This confirms our above conclusion that a time adverbial establishes a TE rather than a TO. In what follows we will argue that the relation between TEs and TOs is one of inclusion. That is, a TO may be properly included in a TE (as is the case in the above example) or can be commensurate with TE (as, for example, when TE is punctual).

TO' holds as soon as SS and TO have one time point in common, SS can be a much longer time interval than TO, so that it should always be possible for SS (provided it is durative and not punctual) to reach up to t_0.) However, this is not the case in (7,b): in spite of the fact that the situation of 'my doing it' is a durative situation, (7,b) clearly implies that SS lies wholly before t_0.

The reason why SS cannot include t_0 in (7,b) or (8,a), whereas it can in (8,b), is that the former sentences represent their situations as bounded, whereas (8,b) represents its situation as unbounded:

(8) (a) Five minutes ago John ate a biscuit.
(b) Five minutes ago John was in the kitchen.

The notions 'bounded' and 'unbounded' have been discussed at some length in chapter 3. Since these notions will also prove of crucial importance in the present chapter, it may be useful to summarize the main points of that discussion:

a. Boundedness and unboundedness are basically features of situations, not sentences. However, situations are not inherently bounded or unbounded: they are represented as (un)bounded by the sentences that describe them. Thus, both *John drank whisky* and *John drank six glasses of whisky* can refer to exactly the same situation, but in the former case the situation is (represented as) unbounded, whereas in the latter it is not. This means that boundedness and unboundedness are characteristics of situations assigned to them by sentences (i.e. linguistic representations of situations). Since both elements (the situation represented and the sentence representing it) are essential to the concept, we have adopted the convention of speaking of '(un)bounded situation' as well as of '(un)bounded sentence'. One should keep in mind, however, that these terms are really abbreviations of 'situation represented as (un)bounded by a sentence' and 'sentence representing a situation as (un)bounded', respectively.
b. A situation is bounded if it is represented as reaching a terminal point. If it is not represented as terminating, it is unbounded. Thus, (9,a–c) are bounded sentences, whereas (10,a–f) are unbounded:

(9) (a) John drank five glasses of whisky before dinner.
(b) Betty made a drawing on the floor.
(c) The athlete ran 10 kilometres.
(10) (a) John was drinking whisky (when I saw him).
(b) Betty was making a drawing on the floor.
(c) The athlete ran/was running.
(d) The athlete was running the 10 kilometres.
(e) John was in the kitchen.
(f) Betty knew about our affair.

Sentences (9,a–c) are bounded because they each represent a situation as involving a terminal point which has to be reached if the situation is to be what it is claimed to be. If John drank fewer than five glasses of whisky, (9,a) is not a true representation of the situation. Similarly, if the athlete did not cover the full distance of 10 kilometres, (9,c) does not count as a truthful account, though the unbounded sentences (10,c–d) do.

c. Situations that are 'accomplishments' or 'achievements' in Vendler's (1967) classification are bounded. 'Activities' and 'states' are unbounded.[15]

d. Progressive sentences always represent situations as unbounded. Since progressive aspect means that the speaker just refers to the middle of a situation, and disregards its beginning and (possible) end, a progressive sentence by definition fails to represent a situation as terminating (bounded).[16]

It follows that a bounded sentence becomes unbounded when it is put in the progressive form,[17] and that, in consequence, the truth of

15. In other words,'states and activities are imperfective, while accomplishments are perfective' (Gustafsson 1980:444. See also Recktenwald 1975:33; Zydatiss 1976; Smith 1983:278, 1986). (It should be noted, however, that activities and states are only in principle unbounded. We will see below that they can be represented as bounded by the use of a specific duration adverbial like *for two hours*. And another point to be made is that an accomplishment becomes unbounded when it is referred to in a progressive sentence. (However, in that case it is debatable whether we can still speak of an accomplishment – see footnote 17).)

16. As noted in chapter 3, 'incompleteness' (imperfectivity) is an inherent characteristic of progressive aspect. A progressive verb form refers to an internal portion of the situation, without making explicit reference to its beginning or end. A nonprogressive verb form, in contrast, has perfective meaning, i.e. it represents a situation as 'complete': it 'presents the totality of the situation referred to . . . without reference to its internal temporal constituency: the whole of the situation is presented as a single unanalysable whole, with beginning, middle, and end rolled into one; no attempt is made to divide this situation up into the various individual phases that make up the action' (Comrie 1976:3. See also King 1983:126–46; Smith 1986:100–1). (Note that the idea of '(in)completeness' as defined here has nothing to do with the question whether the situation is finished or not in real time; all that matters is how the speaker conceptualizes the situation.)

One linguist who has argued against the view that the progressive picks out a subinterval (viz. the middle) of a larger situation is Vlach (1981). According to Vlach, this analysis is not applicable to the sentence *Is someone sitting there?* if uttered in a cinema with reference to a seat where nobody is actually sitting at the time of the utterance. However, Hinrichs (1983:178) points out that a sentence like this presents no counterevidence if we assume that speakers observe 'structural constraints' (as the term is used in 'Situation Semantics' – see e.g. Barwise and Perry 1983): 'there exists a conventional constraint which says that a chair has been taken in a movie theatre, if someone has claimed it and has possibly left his coat on the chair, even though he might not actually sit in the chair at any given time' (Hinrichs 1983:178).

17. Translating this into Vendler's (1967) terminology, we might say that the use of the progressive turns accomplishments and achievements into activities. Several linguists (e.g. Heinämäki 1978:7; Bennett 1981; Edgren 1985:82; de Vuyst 1985:161) explicitly state that they accept this formulation. However, some linguists (e.g. Dowty 1979:147–8; Fenn 1987:43) are reluctant to do so because in their opinion the most

such a progressive sentence does not entail the truth of its nonprogressive (bounded) counterpart. From the truth of *John was writing a book* we cannot infer anything concerning the truth of *John wrote a book.* On the other hand, this kind of inference is possible if the nonprogressive sentence is unbounded: it does follow from *John was drinking tea* that *John drank tea*, since both sentences represent the situation as unbounded.

e. Unbounded sentences are typically compatible with 'non-inclusive' duration adverbials (i.e. adverbials which answer the question *For how long?*), while bounded sentences are typically compatible with 'inclusive' duration adverbials, i.e. adverbials that measure the distance between the beginning and the end of the situation (e.g. *in fifteen minutes*):

(11) (a) John was running for a long time.
(b) *John was running in five minutes.[18]
(c) John ran a mile in five minutes.
(d) *John ran a mile for a long time.[19]

f. Unbounded situations resemble stuffs (indicated by mass nouns) in that they are 'homogeneous' (i.e. they are 'the same all the way through' (Galton 1984:154)) and consequently 'dissective' (i.e. an unbounded situation can be broken down into smaller subsituations of the same nature).[20] This means that, if we omit the duration adverbial

important semantic feature of an accomplishment is that it is telic, i.e. that it is a situation which 'proceeds towards a climax, or natural terminal point' (Lyons 1977:711). This feature would still seem to be present in the progressive. In fact, it may be argued that a sentence like *John was drawing a circle* is both telic (i.e. an accomplishment) and unbounded (see Zydatiss 1976:227, 1978:354). (This may seem paradoxical, since telicity inherently implies reference to a terminal point, whereas unboundedness means that the situation is not represented as reaching a terminal point. For two different solutions to this 'imperfective paradox' see Dowty (1977) and Declerck (1979c).) In what follows I will distinguish carefully between the two notions: I will consider as telic those expressions that represent situations as having a natural end-point (point of completion) and as bounded those expressions that assert that a (possibly arbitrary) terminal point is reached. (For further arguments for keeping the two notions apart, see section 1 of chapter 3.)

18. The asterisk is meant to indicate that the sentence *John was running in five minutes* is ungrammatical on the reading 'Five minutes elapsed between the beginning and the end of John's running'. (The sentence is acceptable on the reading 'It lasted five minutes before John started running', see van Voorst 1987:266.)
19. The asterisk is meant to indicate that the sentence *John ran a mile for a long time* is ungrammatical on the single event reading. (It is grammatical on the interpretation 'For a long time John repeatedly ran a mile' because this is again an unbounded reading.) Notice also that some bounded sentences (viz. those that imply that the situation produces a result state when the terminal point is reached) are compatible with a *for*-adverbial, but that in that case the adverbial measures the length of the result state, not that of the situation itself (e.g. *John opened the door for a couple of minutes*).
20. Bennett and Partee (1978:17) call this characteristic the 'subinterval property'. Carlson (1981:48) speaks of 'partitivity', while Hatav (1989:512) calls it 'the distributivity property'. Cooper (1986:27) dubs it 'temporal ill-foundedness'.

(if any) from a sentence representing a situation as unbounded, the same sentence can also be used to refer to smaller parts of that situation. For example, if John ran in the woods from two o'clock to four, then unbounded sentences like *John ran* or *John was running* are true representations not only of what John did between two and four, but also of what he did between two and three, and of what he did between two-thirty and a quarter to three, etc. The converse is also true: if we add together several unbounded situations of the same kind, the result is a longer unbounded situation of the same kind. Thus, unbounded situations are also 'additive'.

In contrast, bounded situations are 'heterogeneous', 'unitary' and 'non-additive'. Like objects indicated by count nouns, they are not 'the same all the way through' and they cannot be broken down into smaller entities that can be indicated with the same name (description). That is, a sentence that is used to represent a situation as bounded cannot also be used to refer to a section of that situation. In the situation outlined above we can say *John ran 10 kilometres (between two o'clock and four)* but the same sentence (*John ran 10 kilometres*) cannot be used as a true representation of any subpart of the overall situation. Moreover, if we add together several past situations of John running 10 kilometres, the sum-total is not itself a situation which we can refer to by means of the sentence *John ran 10 kilometres.*

Returning to our main topic, we can say that it is this difference between bounded and unbounded situations that accounts for the fact that some sentences in the past tense allow the possibility that the situation continues into the present, while others exclude this possibility. Let us again have a look at some examples:

(12) (a) Bill wrote a poem today.
(b) Five minutes ago Bill ate an apple.

(13) (a) Bill was writing a poem today.
(b) Five minutes ago Bill was in the kitchen.
(c) Bill was at his office today.

Neither (12,a) nor (12,b) allows the possibility that the situation referred to still continues at t_0. The reason is that both sentences represent their situation as bounded. As we have seen, a bounded situation is always 'unitary', i.e. the sentence representing it can only refer to the situation as a whole (including the terminal point). It follows that if the sentence locates the situation at some time in the past, the whole situation is located in the past. Bounded sentences in the past tense therefore exclude the possibility that the situation may continue up to t_0.

Sentences (13,a–c), by contrast, are unbounded. Since unbounded situations are 'dissective', these sentences can refer to subsituations. For

example, *Bill was at his office* can refer either to the whole length of time that Bill was at his office or to any subinterval of this.[21] In a sentence like (13,c) both interpretations are in principle possible. The latter (i.e. the reading on which the sentence refers to a subinterval) is compatible with the possibility that Bill is still at his office at t_0. In that case the speaker represents a subsituation (of the situation of Bill being at his office) as simultaneous with some past time interval (the situation-TO). (The fact that the speaker uses *today* does not alter the fact that the interval in question is past, for the time interval which the speaker selects as situation-TO is not TE (the whole interval indicated by *today*) but some (pragmatically defined) past part of it (presumably the time that he himself was at the office).) In sum, an unbounded sentence can refer to any subinterval of the largest interval that the sentence is true of.[22] It follows that, in such a sentence in the past tense, the time of the situation which is located as simultaneous with some past (situation-)TO may be such a subinterval, in which case there is nothing to prevent some other subinterval from including t_0.[23]

It should be noted that unbounded sentences become bounded if we add an adverbial to them which specifies the duration of the situation (see Allen 1966:51; Vasudeva 1971:140; Dowty 1979:61; Mommer 1986:24; Mittwoch 1988:210):

(14) Today Bill was at his office for four hours.

This sentence clearly represents the situation as bounded, i.e. as heterogeneous: the predication *Bill was at his office for four hours* applies to the situation as a whole and cannot be used to describe portions of it.[24]

21. See Schopf (1987:180): 'states can be true at moments (points) of time, in as much as we can relate a state to a moment of time as encompassing this moment'. Rigter (1980b:217) puts it as follows: 'Although in LTM (Long Term Memory) a state x may be specified as obtaining for a period of time Sx, only a relevant segment of Sx is activated in the working-memory when x is referred to [in a tensed sentence].'
22. The phrase *any subinterval* is a slight overstatement. It is clear that an unbounded sentence can only refer to subsituations that 'fully instantiate' (Schachter 1981:398) the situation of which they form part. That is, the subsituation in question must be long enough to be recognizable as a (subsituation of a) situation of a particular kind. (See also footnote 8 of chapter 3.)
23. This explains the difference of meaning between the following sentences:
    (i) How about inviting John? – *I considered inviting him.*
    (ii) How about inviting John? – *I was considering inviting him.*
    As noted by Legault (1976:99), (i) 'can suggest that speaker B is flatly ignoring or denying A's suggestion' while (ii) 'at least permits the polite fiction that B is entertaining A's suggestion and is open to persuasion or, at least, to further consideration of the matter'. The reason is that speaker B represents the situation (of considering A's suggestion) as both past and bounded in (i), but as past and unbounded in (ii). Only in the latter case is it possible for the full situation to extend beyond TS and into the present.
24. The claim that (14) is bounded might seem to be in contradiction with the claim (made above) that unbounded sentences are typically compatible with non-inclusive duration adverbials (e.g. *for two hours, all day long*), whereas bounded sentences are not. However, there is no contradiction. Though it is true that (14) is bounded, this

(For this reason it is incompatible with a punctual indication of time like *at five o'clock*.)[25] Since the sentence can only refer to the interval corresponding to the duration that is specified, the 'situation referred to in the sentence' is actually also the 'full realized situation', i.e. TS is SS (the time of the full situation). Since the past tense in (14) represents the situation as simultaneous with a TO which is anterior to t_0, (14) therefore implies that the full situation lies before t_0, i.e. that Bill is not at his office any more at t_0.

The ensuing generalization is that, in sentences in the past tense, it is possible for the full situation to include t_0 only if TS need not be interpreted as the time of the full situation. This possibility is not available when the sentence represents the situation as bounded, because in that case the sentence predicates something of the complete (full) situation.[26]

boundedness results exclusively from the presence of the adverbial *for four hours*. If the adverbial is dropped, what remains is an unbounded sentence (*Today Bill was at his office*). The claim that unbounded sentences are typically compatible with non-inclusive duration adverbials thus does not mean that sentences containing such an adverbial are automatically unbounded. What it means is that non-inclusive duration adverbials can always be added to unbounded sentences and never to bounded ones. However, if the adverbial that is added indicates a specific length of time (as in (i,a–b)) the result is that the originally unbounded sentence becomes bounded. If the timespan indicated is not specific (as in (ii,a–b)) the addition of the adverbial does not affect the unbounded nature of the original sentence.

(i) (a) I was in the house for four hours.
(b) All day long I was bothered by telephone calls.
(ii) (a) I was in the house for some time.
(b) For many years Britain ruled the seas.

25. The observation that a duration adverbial indicating a specific length of time renders its sentence bounded also explains why such an adverbial is incompatible with progressive aspect (see Leech 1969:150; Mittwoch 1988:224ff.):
(i) It rained/*was raining for two hours.
The progressive (which renders the sentence unbounded) conflicts with the durational adverbial (which renders the sentence bounded). In other words, *for two hours* (which measures the length of the situation as a whole) is not compatible with a verb form which refers only to the middle of the situation. (However, there is no such incompatibility when the sentence is in the present perfect because the adverbial then specifies the duration of the situation up to t_0 rather than the duration of the full situation: *It has been raining for two hours (now)*. There is no incompatibility either if the duration of the situation (and hence the terminal point) is fixed before the situation begins. Thus, a sentence like *John is/was working for two hours* is impeccable if it was arranged beforehand how long John was going to work.)

26. Although a situation represented by a past tense can in principle continue at t_0 provided the above condition is fulfilled, a past tense sentence will normally suggest that the situation does not include t_0, unless there is an indication to the contrary. That is, the preterit carries the implicature of 'discontinued situation'. According to Comrie (1985:41–2), the origin of this implicature is Grice's (1975:45) Maxim of Relation (Relevance), in that, 'other things being equal, statements about the present moment are more relevant than those about other times, so that use of a form explicitly locating a situation in the past suggests that that situation does not hold at the present, otherwise the present tense would be used'. As is predictable, the implicature can be cancelled by the context or by other pragmatic considerations. As noted by de Vuyst (1983:168), the sentence *Harry knew English* does not so easily suggest that Harry does

The observation that TS may be a subinterval of the largest interval that the sentence is true of obliges us to give up our former definition of TS. We can no longer say that TS is 'the time of the situation referred to' in the sense of 'the time that is occupied by the situation on the time line' (i.e. what we have called the 'situation span' (SS)). We must now accept a more restrictive definition, which says that the notion 'situation referred to' means no more than 'that part of the situation that is talked of in the sentence'. Thus, in *Five minutes ago John was eating in the kitchen* we can no longer take TS to be the stretch of time occupied by the full situation of John eating in the kitchen. Rather we must say that TS is the time taken up by just that portion of the situation that is talked of in the sentence, i.e. the time of that portion of the situation that is said to have held at the past TOE. In other words, irrespective of how long John was actually eating in the kitchen, TS is just that length of time that coincides with the TOE established by *five minutes ago.*

The fact that TS is just that interval of the situation that coincides with TOE means that we also have to revise our definition of 'simul'. (In fact, we had already reached this conclusion on different grounds.) Instead of defining 'simul' in terms of either coincidence or overlapping, we must define it in terms of coincidence only. 'TS simul TO' means that both time intervals coincide (i.e. are commensurate), so that 'TS simul TO' is reversible to 'TO simul TS' and is in fact equivalent to 'TS = TO'. In other words, when we utter a tensed sentence like *Five minutes ago John was eating in the kitchen* we do not make a statement about SS (the full situation span, i.e. the time interval that John was actually eating in the kitchen). What we do is state a (strict) simultaneity relation between a TO (here corresponding with the TE indicated by *five minutes ago*) and some part (proper or otherwise) of the situation in question. This is what we call 'locating a situation at a past TO'. The statement we make concerns only that part of the situation that is commensurate with the TO. It is this time interval that we now call 'the time of the situation' (TS). If the situation is unbounded, TS may be a proper subpart of SS (the time interval occupied by the full situation). Since the sentence only makes a statement about TS, the time intervals contained in SS but not in TS are not referred to: these portions of the situation are not located in time. This explains why the sentence *Five minutes ago John was eating in the kitchen* leaves vague whether John is still eating in the kitchen and

not know English any more, because we know that knowing a language is not a statewhich (barring special circumstances) comes to an end abruptly. But *Mary swam in the river* strongly suggests that she does not swim in the river any more. (According to de Vuyst, this is even the only possible interpretation. However, it is sufficient to couch the sentence in a suitable context to see that this is not correct: *When she was young, Mary swam in the river, and perhaps she still does.*)

whether he was already eating in the kitchen before the time indicated by *five minutes ago*.[27]

An inevitable consequence of this definition is that we must give up another basic principle argued in Declerck (1986a), viz. the claim that, in a sentence with a time adverbial, the adverbial always establishes the TO (or one of the TOs if there are several). This principle (first formulated by Reichenbach (1947)) cannot be upheld because it is clear that, in a sentence like *Bill left yesterday*, the TS (the time of Bill's leaving) cannot be simultaneous (in the newly defined sense) with the interval (TE) established by *yesterday*. The TO with which TS is commensurate must be some unspecified subinterval of this TE. This not only confirms a conclusion arrived at before, viz. that not any TE is a TO and that not any TO is a TOE, but also shows that the bounded/unbounded distinction is also relevant to time adverbials.

Like situations, the TEs identified by time adverbials may be bounded or unbounded. A TE is bounded if the adverbial refers to its boundaries (beginning and end) rather than merely indicating duration.[28] Adverbials like *from two o'clock to/until four* explicitly represent a TE as bounded. In *I have been here since midday* the TE is also represented as bounded because the beginning of the time interval is explicitly referred to while the use of the present perfect implies that t_0 is the ending-point. Something similar happens in *I will be here until five o'clock*, where the end point is indicated and the beginning of the interval is taken to be t_0 or some other point established by the context. I will say that adverbials like *from two o'clock to four*, *since midday*, *until five o'clock*, etc., are 'bounded adverbials', i.e. time adverbials that represent an interval of time as bounded. It is typical of such adverbials that they are 'unitary', i.e. they can only indicate the interval as a whole.

Adverbials like *yesterday*, *in 1983*, etc., refer to an interval without drawing attention to its boundaries. For that reason they represent an interval as unbounded and can therefore be used both to refer to the interval as a whole and to refer to subintervals of this. In other words, they are 'dissective'. A sentence like *John was here yesterday* may mean either that John was here all the time yesterday or that he was here for some time in the course of yesterday. The analogy with situations is clear: like unbounded situations, unbounded TEs are homogeneous ('the same

27. These conclusions are in keeping with Rigter's (1980b:217, 1980d) observation that state sentences need not refer to a state as a whole, witness the following example:
    (i) Pete's car isn't blue at all! *It is red. It has been red* as long as I can remember. *It was red* when he bought it.
    Rigter points out that each of the italicized clauses here picks out a particular portion of the state as 'focal episode', i.e. as the time segment on which the speaker focusses his attention. This notion of focal episode seems to be the same as our notion of situation-TO, i.e. the TO with which TS is represented as simultaneous.
28. Reference to one of the boundaries is sufficient if the other is recoverable from the context – see below.

all the way through') and dissective. In the same way as sentences representing situations as unbounded have been called 'unbounded sentences', adverbials that represent TEs as unbounded will be called 'unbounded adverbials'.

The distinction between bounded and unbounded TEs renders it possible to maintain that TS is simultaneous with TO (in the sense that TS and TO are commensurate, i.e. TS = TO) in sentences like *Bill left yesterday*. We will say that TO is not the time interval indicated by *yesterday* but some (unspecified) subinterval of this. This is possible because the TE established by *yesterday* is an unbounded TE, every subinterval of which can also be referred to as being 'yesterday'. A similar analysis can be argued for *Bill left today*. Here the TE established by the unbounded adverbial includes t_0, but this is irrelevant to the use of the tense because the TO that is commensurate with TS is some unspecified subinterval of TE which lies entirely in the past. This explains why the adverb *today* is not incompatible with the past tense: since *today* establishes TE, not TO, the temporal structure typical of the past tense (viz. 'TS simul TO_2 before TO_1') is realized.

It is important to see that we now have all the elements necessary for a correct understanding of the statement that a tense 'locates a situation in time'. Locating a situation in time means stating a relationship of simultaneity (coincidence) between TS and some TO. Since TS may be a proper subinterval of the time of the 'full situation', and since TO may be a proper subinterval of TE (i.e. of the full time interval referred to by the time adverbial), the time of the full situation need not be simultaneous with the full time interval indicated by the adverbial. Thus, when we say that the sentence *John was here yesterday* locates the situation of John's being here at the time indicated by *yesterday* we say no more than that the statement 'John is here' was true at some time (possibly all the time) yesterday.

At the beginning of this section I referred to three questions that needed settling: How do we define 'simul'? How do we define TS? And what is the relation between TE and TO? We now know the answers to these questions. The relation 'simul' is one of coincidence: 'TS simul TO' means that TS and TO are commensurate. This definition is in keeping with the fact that TS has been defined as just that portion of the 'fully actualized' situation that is located in time (i.e. that is represented as simultaneous with TO). And the relation between TE (if any) and TO is one of inclusion. This means that TE either properly includes TO or coincides with it.

1.7 Temporal vagueness

The fact that not only situations but also adverbially established intervals may be either bounded or unbounded entails that, even in sentences with

a time adverbial, we often have no precise information about the length of time taken up by TS (and hence by TO_{sit},[29] since TS and TO_{sit} are commensurate), nor about the precise location of TO_{sit} (and TS) on the time line. This vagueness is greatest when both TS and TE are unbounded, as in *John was here yesterday*. Since *John was here* is an unbounded expression, TS may be either the whole time that John was here or any subinterval of this. Similarly, since *yesterday* is an unbounded adverbial, TO_{sit} may be the whole interval indicated by the adverb (in which case TO_{sit} = TE) or any subinterval of TE. The combination of *John was here* and *yesterday* thus gives rise to several possible interpretations:

1 TS may be the time of the full situation, i.e. the whole interval of John's being here. In that case there are two possibilities:
   a. TS (and hence TO_{sit}) may be commensurate with TE. (In that case John was here all day yesterday and it was not the case that he was here at any time interval that was not part of yesterday.)
   b. TS (and hence TO_{sit}) may be commensurate with only part of yesterday. (In that case John arrived and left yesterday, but he did not stay for the whole day.)
2 TS may be a (proper) subinterval of the time of John's being here. Here again there are two possibilities:
   a. TS (and hence TO_{sit}) may be commensurate with TE. (In that case John's arrival is situated before yesterday or/and his departure is situated after yesterday. The possibility exists that he is still here at t_0.)
   b. TS (and hence TO_{sit}) may be commensurate with only part of yesterday. (In this case it is left unspecified both what is the length of the timespan taken up by John's being here and what is the precise relation of this timespan to the time line. The sentence actually gives no information besides the information that some part of the situation of John's being here was simultaneous with some part of TE (yesterday). The possibility that John is still here is therefore left open.)

It should be stressed that these different possibilities are a question of vagueness, not of ambiguity. It is not true that the sentence *John was here yesterday* is four-ways ambiguous. (If it were, the two clauses of *John was here yesterday, and so was Bill* would have to be interpreted in terms of the same possibility (see G. Lakoff 1970), but this is not the case. The sentence is perfectly appropriate if, say, John was here before midday, while Bill was here in the evening.)

29. From now on I will be using 'TO_{sit}' as an abbreviation of 'situation-TO' (i.e. the TO with which TS is represented as simultaneous).

There are fewer possibilities if the situation is bounded. If TE is bounded too, the only possible interpretation is that TS and TE are commensurate. Examples of this are *John wrote two letters between two o'clock and three* and *The doctor has seen three patients since 5 p.m.* If TE is unbounded (while, as before, the situation is bounded), as in *John wrote two letters this morning*, TS is commensurate either with TE or with some part of it. (That is, it may or may not have taken John all morning to write the two letters.)

The final logical possibility is that the situation is unbounded while TE is bounded (e.g. *John was here from two o'clock till five, Bill has been working since lunch*).[30] In that case TE must also be commensurate with TO_{sit}, and there are two theoretical possibilities as to the nature of TS. The first is that TS is the time of the whole situation. In that case the boundaries referred to by the bounded adverbial are also the boundaries of the full situation (since in this case TS = TO_{sit} = TE). The second is that TS is only a subinterval of the timespan covered by the full situation. This interpretation is less obvious than the former, as it will mostly run counter to Grice's (1975) Maxim of Quantity. For example, if John was here all day yesterday, it is misleading (though perfectly true) to say *John was here from two o'clock till five*, especially if this is meant to be the answer to the question *When was John here?* However, there is no such problem if TE (*from two o'clock till five*) is not the point of the question (as, for example, when we use the sentence in answer to *Where was John from two o'clock till five?* or *Who was here from two o'clock till five?*) In that case the possibility that TS is only a portion of the time of the whole situation is quite open (i.e. the sentence does not exclude that John's being here extends beyond TE and even up to t_0).

Up to now we have not considered the possibility of TS or/and TE being punctual (nondurative). It is clear, however, that time intervals consisting of a single time point behave like bounded intervals. This means that we have the following possibilities:

1 If both TS and TE are punctual (as in *John came in at 5:13 p.m.*), the only possible interpretation is that TE also functions as TO_{sit} and coincides with TS (i.e. TE = TO_{sit} = TS).[31]

30. As noted by Dowty (1975:585) such sentences are odd when 'the property predicated is not easily construed as temporary':
    (i) !The Washington Monument was heavy in 1934.
    (ii) !The man who answered the phone was Harvey's brother last week.
    In my opinion this oddity is due to a violation of Grice's Maxim of Relation (Relevance): since statements about the present are more relevant than statements about the past (see footnote 26), a situation that still holds at the time of speech should not be located exclusively in the past (unless there is some good reason for doing so).
31. The relation of simultaneity (coincidence) between TO_{sit} and TS may be of the 'sloppy' kind referred to in section 16 of chapter 2. This is often the case when TE is established by a *when*-clause:
    (i) When the shot was fired, the girl began to scream.

2. If TS is punctual, and TE is durative, TO_{sit} (which must be punctual too, since it coincides with TS) is some unspecified point of TE. In that case the durative TE is necessarily unbounded. (Thus, we can say *John arrived yesterday* but not (barring a repetitive (hence durative) reading) *!John arrived from 2 o'clock till four.*)
3. If TS is durative whereas TE is punctual, there are two possibilities, according to whether the situation is bounded or unbounded:

   a. If the situation is unbounded (e.g. *John was here at five o'clock*), TS is normally some unspecified punctual subinterval of the situation. However, with some stative verbs (e.g. *know*) it is impossible to interpret TS as punctual (see Bach 1981:73); in that case the punctual TE is interpreted as the beginning of a period with which TS is commensurate. For example: *At that moment I finally knew what he meant.*
   b. If the situation is bounded (e.g. *At five o'clock I wrote a couple of letters*) there should normally be no acceptable interpretation, for it is logically impossible for a durative bounded situation to be commensurate with a nondurative TE. In practice, however, such sentences are acceptable in English because they are conventionally interpreted as meaning that it is the initial point of TO_{sit} that is simultaneous with the punctual TE. That is, the above example is interpreted as 'At five o'clock I started writing a couple of letters.'[32] A similar example is *They will go to the station at seven o'clock.* This example should be compared with *They will come to the station at seven o'clock*, where it is the final, not the initial, point of TO_{sit} that is interpreted as simultaneous with the punctual TE. This means that when a bounded durative situation is located as simultaneous with a punctual TE, we will look for a suitable interpretation by assuming that it is either the initial or (less usually) the final point of TO_{sit} that actually coincides with TE.[33] The kind of situation referred to (e.g. going vs. coming) will guide us towards one of these interpretations.

(ii) When the bell rang, the headmaster came out.

The normal interpretation here is not that TE and TS are perfectly simultaneous, but rather that TS immediately follows TE. (In the above sentences this interpretation concurs with the fact that we tend to interpret the situation expressed in the *when*-clause as being the cause of (and hence as being prior to) the situation referred to in the head clause.)

32. This paraphrase is actually inaccurate in that it asserts only the beginning of the situation, while *At five o'clock I wrote a couple of letters* asserts the entire situation (see Comrie 1985:30). A more accurate paraphrase is therefore 'I wrote a couple of letters and I started doing so at five o'clock.'
33. This mechanism is similar to what we observe in sentences like (i)–(ii), where a verb phrase denoting a punctual situation collocates with a duration adverbial:
    (i) Tomorrow I will leave the country for five days.
    (ii) Jill has come for a couple of hours.

2 PROGRESSIVE SENTENCES

As far as I can see, the above observations provide us with everything that we need to analyse the temporal structure of a tense such as the preterit. Even verb forms expressing progressive aspect require no additional apparatus, for situations represented by progressive forms can be analysed in exactly the same way as other unbounded situations. (As we have seen, a progressive sentence necessarily represents a situation as unbounded,[34] because it refers to the middle of the situation while disregarding its beginning and (possible) end.)[35] This means that we have the following theoretical possibilities:

2.1 When TE is punctual and hence simultaneous with TO_{sit}, TS is some unspecified punctual interval of the 'middle of the situation' (henceforth: MidS).[36] An example of this is *John was sleeping when I arrived.* (As has often been observed, the progressive in such a sentence refers to a situation which functions as a 'frame' or 'background' for some other situation or time interval, in the sense that the progressive situation extends beyond the boundaries of the latter. This observation is, of course, accounted for in our theory: we say that a punctual TOE

Since TS is punctual, the duration adverbial cannot be interpreted as specifying the length of TS. The adverbial is therefore interpreted as specifying the duration of the state that is produced by the situation. Thus, sentence (i) is interpreted as meaning that I will leave the country and stay away for five days, and (ii) is interpreted as meaning that Jill has come and stayed for a couple of hours (or has come with the intention of staying for a couple of hours).

34. Of course this statement holds only for *be . . . -ing* forms that really express progressive aspect (reference to the middle of the situation), not for progressive forms that are used, for example, to express no more than the intended or expected future actualization of a situation. Such sentences may be bounded, as appears from
    (i) Tom is in great shape and has a good chance of winning the contest. *Insiders expect he'll be running the 100 metres in 10.04 seconds.*
35. The fact that progressive situations are unbounded, like states, has induced some people to claim that they *are* states. Thus, Rohrer (1986:96) states that 'if the main verb of a sentence contains the feature value +PROG, then the sentence denotes a state'. The same claim is made by Vendler (1967:114); Steedman (1977:222); Bach (1981:68); Vlach (1981:274); Galton (1984:24); Dowty (1986:44); Mittwoch (1988:234–5); Salkie (1989:8–10); and Parsons (1989:222). However, de Vuyst (1983:71) points out that, although 'progressives and state descriptions behave alike' one cannot say that 'progressives *are* state-descriptions' because, unlike states, progressive situations are dynamic, i.e. imply movement (see also Ota 1963:59; Kaluża 1972:16; Saurer 1981:166; Smith 1982:175ff.; Edgren 1985:74; de Vuyst 1985:161). Another difference is that progressive situations need not be continuous, i.e. allow gaps or interruptions (as is clear from e.g. *I am writing a book*), whereas states are by definition continuous (see Gabbay and Moravcsik 1980:63). Furthermore, states can be bounded by a duration adverbial (e.g. *John was here for three hours*), whereas this is not normally possible for progressive situations (**John was eating for three hours*) (see Hatav 1989:501).
36. See Langacker (1978:862): 'Since an imperfective event is construed as an on-going affair, without regard to beginning or end, to verify its existence we need sample only one point in time.'

coincides with one of the time points that make up the middle of the situation.)[37]

2.2 When TE is durative and bounded (in which case it is again simultaneous with TO_{sit}), TS (MidS) must also be a durative interval (since TO_{sit} = TS). However, since the term 'middle of the situation' applies to any subinterval of the situation that lies after its starting-point and before its ending-point, TS may be either a long or a short subinterval of the situation. Thus, the sentence *From two o'clock till four John was asleep* is true if John slept from one o'clock until six, and if he slept from one-thirty till nine, etc. All that is expressed in the sentence is that TE (= TO_{sit}) is commensurate with a TS which is 'the middle of the situation'.

It should be noted that it does not necessarily follow from this that the situation whose middle is the MidS referred to in the sentence must have other intervals in the actual world. It is typical of progressive aspect that the speaker using a progressive form does not commit himself to any claim which does not concern the middle of the situation. More specifically, he does not guarantee the eventual actualization of what follows MidS, viz. the terminal point. Consider, for example, the following sentences:

(15) (a) Between three o'clock and four Bill read a book.
(b) Between three o'clock and four Bill was reading a book.

In sentence (15,a) the situation of Bill reading a book is represented as bounded (complete) and as commensurate with the interval denoted by the bounded adverbial. If the speaker does not wish to represent the situation as bounded, he must use (15,b). In doing so he just refers to the middle of the situation and says nothing about its terminal point. In fact, it is not impossible that the situation never got beyond this MidS, i.e.

37. This 'framing effect', as it is called by Jespersen (1931:180) and Leech (1971:17), is not an automatic consequence of the use of the progressive, and neither is it to be found with the progressive only. In fact, the framing effect appears whenever TO_{sit} (= TS) is felt to be a proper subinterval of the time of the full (complete) situation. This is possible whenever the situation is unbounded, i.e. not only in progressive sentences like (i,a) but also in nonprogressive stative sentences like (i,b–c):
(i) (a) I was sleeping when he arrived.
(b) I was asleep when he arrived.
(c) I was in my bed when he arrived.
Conversely, progressive sentences do not have the framing effect if TO_{sit} is not felt to be shorter than the full situation. This is the case in examples like the following:
(ii) (a) All through dinner they were talking of nothing but the match. (Charleston 1955:274)
(b) They were watching television while we were working.
(c) They have been watching television since five o'clock.
(d) We learn from the scientists that our globe is continually shrinking. (Vasudeva 1971:134).
(e) I was knitting for two hours this morning. (Charleston 1955:275)

never reached the (potential) terminal point. That is, (15,b) remains a truthful sentence if John never resumed reading the book after four o'clock, and so never finished it. In that case MidS may actually be the whole 'actualized situation' (provided John had never been reading the book before the time indicated by *three o'clock*).

2.3 We have seen that when TE is unbounded (as when the adverbial is *yesterday*), TO_{sit} may in principle either coincide with TE or be a subinterval of TE. This general rule also applies when the sentence is progressive, i.e. when TS is the time of MidS. For example, the sentence *John was eavesdropping at your door yesterday* is true both if John was eavesdropping all day and if he was eavesdropping only for some time in the course of the day.[38] (Here again we may note that MidS may actually be the only part of the situation that is actualized. The sentence is therefore true even if John was not eavesdropping at my door at any other time.)

A special case of unbounded TE is that in which the time adverbial is itself a clause in the progressive form, as in

(16) (a) Mary was sleeping while I was working.
(b) Mary was sleeping when I was working.

These sentences express no more than that Mary's sleeping and my working were simultaneous at a certain time (TO_{sit}). Since the full situation of Mary's sleeping may extend beyond TS (and hence beyond TO_{sit}, since TO_{sit} and TS are by definition commensurate), and since TO_{sit} may be a proper subinterval of TE, which, moreover, may be a proper subinterval of the full situation of my working, (16,a–b) give us no other information about the temporal relation between the two full situations. All that they assert is that there was some time interval in the past during which both situations were holding. (However, even though

38. Gabbay and Moravcsik (1980:81) observe the same thing in connection with the example *He was working on his paper yesterday*. However, they trace back the two interpretations to an (alleged) ambiguity of *yesterday*: they consider *yesterday* a 'durational modifier' on the interpretation that the situation lasted throughout the day, and a 'referential time modifier' (like *at five o'clock*) on the other interpretation. Levinson (1983:74) makes a similar claim when he says that words like *yesterday* 'have two kinds of referent: they can either refer to the entire span itself . . . or to a point within the relevant span'. The analysis I have presented does not need to treat adverbials like *yesterday* as ambiguous. In fact, I do not think there can be any question of ambiguity, for two reasons. First, since the sentence *John was here yesterday* is true if John was here at no matter what time yesterday, this view of ambiguity would come down to saying that *yesterday* must be ambiguous between an infinite number of interpretations. Second, the possibility of interpreting *yesterday* as 'at some time yesterday' arises only when the adverb is used in a sentence, i.e. when a TS is located as simultaneous with it; when used in isolation, or as an NP (as in *Yesterday was Sunday*), *yesterday* only yields the interpretation on which the day as a whole is referred to.

when and *while* do not assert that the head clause and the time clause are coextensive, they do conversationally implicate this – see Heinämäki (1978:105–6); Ritchie (1979:94).)

2.4 When the progressive sentence contains no time adverbial establishing a TE, a TO_{sit} must normally be recoverable from the linguistic or nonlinguistic context,[39] and, as always, TS (the time of MidS) is interpreted as commensurate with this TO_{sit}. If there is no TO_{sit} recoverable from the linguistic or nonlinguistic context, the progressive sentence is not fully interpretable.[40] Still, in that case the tense itself gives us the minimal information that, whatever the timespan occupied on the time line by TO_{sit}, TO_{sit} lies before t_0 (in the case of the past tense), or after t_0 (in the case of the future tense), or before some TO that itself follows t_0 (in the case of the future perfect), etc. In other words, the progressive necessarily implies the existence of a TO_{sit}, because the sense of incompleteness[41] which it conveys 'necessarily means incompleteness with reference to a point in time' (Hill 1958:209; see also Charleston 1955:272; Zydatiss 1976:248; Rohrer 1979:73; Johnson 1981:152–4; Moravcsik 1982:99; Legenhausen 1985:119; Dorfmüller-Karpusa 1985:45, 1988:140; Mommer 1986:246; Mugler 1988:122).[42]

39. If the TO_{sit} is given in the linguistic context, this does not necessarily mean (as is sometimes claimed) that it is identified in the preceding or superordinate clause. Molendijk (1984:130–1) makes clear that we sometimes have to go back several clauses before we find the TO_{sit}, and that we need to appeal to our knowledge of the world in order to decide in which of the clauses in the context it is to be found. Compare:
    (i) John opened the door and went out. He blinked, as the light *was blinding* his eyes.
    (ii) John began to beat his daughter. After a while his sister intervened and scolded him. She said he *was behaving* like a degenerate father.

    See also chapter 3.
40. Houweling (1986:162) observes the same thing when he notes that sentences that express progressive aspect 'cannot stand on their own'. (The fact that the question *What are you doing?* can be used on its own is not, as Houweling assumes, an exception: this sentence is fully interpretable because the use of the present tense makes clear that it is t_0 that serves as TO_{sit} here.)
41. Though the term 'incompleteness' is often used (see e.g. Hill 1958:209; Leech 1971:16; Palmer 1974:79) to express that the progressive represents a situation as imperfective, i.e. as in progress at some TO, it is actually an infelicitous term when applied to atelic situations (which do not imply that the situation tends towards a natural terminal point at which it is completed). Fenn (1987:35) rightly wonders: 'In what sense ... can statements such as *John was standing outside the pub*, or *The bride was wearing white* be said to be "incomplete"?'
42. It is the presence of this implicit TO_{sit} that is exploited in sentences like
    (i) (a) Was that before or after you were waiting for John?
        (b) I was very tired after I had been studying so long.

    In such examples it is the implied TO_{sit} which is the time X in the relation 'before/after X'. (Note that similar sentences are often hardly acceptable (see Allen 1966:77; Kittredge 1969:17; Berezovsky 1978:61–2; Smith 1982:175–2; Hamann 1989:79):
    (ii) ??Bill went home before/after he was eating.

The observation that TS must be commensurate with TO_{sit} when the sentence is progressive accords with the fact that when a progressive clause expresses simultaneity in a temporal domain, the simultaneity in question can never be of the sloppy kind. Compare:

(17) (a) If he comes, I'll tell him the news.
(b) If he is coming, I'll tell him the news.

As noted in chapter 2, the *if*-clause of (17,a) represents its situation as simultaneous with the head clause situation. This is sloppy simultaneity, since the two situations actually follow each other closely. (In technical terms: the TO_{sit} of the *if*-clause is represented as being (commensurate with) the TO_{sit} of the head clause, in spite of the fact that these are not really simultaneous.) A similar interpretation is not possible in (17,b). Here the use of the progressive entails that a time that is not strictly simultaneous (i.e. commensurate) with the TS of the *if*-clause cannot function as (i.e. be commensurate with) the TO_{sit} of the *if*-clause. Thus, the TO_{sit} of the head clause, which is clearly later than the TS of the *if*-clause, cannot be interpreted as being (commensurate with) the TO_{sit} of the *if*-clause. It follows that some other time (e.g. t_0) is interpreted as being (commensurate with) the TO_{sit} of the *if*-clause (i.e. as being the time during which the coming is in progress).

3 REPETITIVE AND HABITUAL SENTENCES

Up to now we have not paid any explicit attention to sentences that yield a repetitive or habitual interpretation. It is important, however, to show that such sentences can also be analysed in terms of the theory that has been developed so far.

Let us use the label 'repetitive sentence' for a sentence that represents a situation as repetitive (iterative), i.e. as consisting of repeated occurrences of the same situation. These different occurrences form subsituations which are either identical or similar to each other. They are identical when all the subsituations involve exactly the same participants, as in (18,a), and similar when they involve participants of the same type, as in (18,b):

(18) (a) John kicked the ball for hours.
(b) John drank a cup of coffee every hour.

The reason is that the progressive leaves vague the times of the beginning and end of the situation, while the interval X in 'before/after X' is preferably a well-defined bounded interval, i.e. an interval whose starting point and end are specified.)

The inevitable presence of an (explicit or implicit) TO_{sit} is also the reason why a progressive usually suggests that the situation is 'particular' (Leech 1971) or 'definite' (Sweet 1891:103; Diver 1963; Penhalluriack 1981:245), i.e. nongeneric: '*I am enjoying the seaside* would be spoken when the speaker is actually at the seaside; this is not necessarily true of *I enjoy the seaside*' (Leech 1971:16).

In (18,a) the repetitive situation consists of subsituations whose participants are John and a particular (each time the same) ball; in (18,b) it consists of subsituations whose participants are John and an each time different cup of coffee.[43]

A repetitive sentence (or the situation represented by it) is 'bounded-repetitive' or 'unbounded-repetitive' according as there is or is not an indication specifying or otherwise bounding the number of subsituations. Thus, (18,a–b) and (19,a) are unbounded-repetitive, whereas (19,b–d) are bounded-repetitive:

(19) (a) Mary has knitted this kind of sweater for years.
(b) John kicked the ball three times.
(c) Ted five times drank a cup of coffee.
(d) Ted drank five cups of coffee.

Repetitive sentences present no problems for our analysis, since repetitive situations can be treated exactly like nonrepetitive (semelfactive) situations:

1 If the sentence is bounded-repetitive, the 'situation referred to in the sentence' is the complete situation (consisting of a bounded number of subsituations) and TS is the time interval taken up by that situation on the time line. A sentence like (19,b) can therefore be analysed in exactly the same way as *John kicked the ball* (*once*).
2 If the sentence is unbounded-repetitive, TS may be either the situation as a whole or a subinterval of this. These are the same two possibilities as we have observed in connection with nonrepetitive unbounded sentences. (It must, however, be noted that repetitive unbounded sentences can only refer to 'relevant' subintervals, i.e. to subintervals that are sufficiently long to be repetitive themselves, i.e. to refer to more than one subsituation. Self-evidently, if this were not the case, the sentence would not be repetitive.)

Unbounded-repetitive sentences can be put in the progressive form. The effect, as usual, is that the reference is no longer to the full situation but to its middle (MidS):

(20) (a) When I spotted him, *John was furiously kicking the ball with his left foot.*
(b) John is drinking cups of coffee.
(c) Mary has been knitting this kind of sweater for years.

43. Note that the pluralization of the situations is also different in these two sentences: in (18,a) there is reference to 'more than one event on the same occasion' (Leech 1969:125), while (18,b) refers to 'one event on more than one occasion'. However, we can disregard this difference, which is immaterial to the present discussion. In both cases we can say that the reference is to a repetitive situation, i.e. to a situation consisting of a number of subsituations of the same kind.

Since progressive aspect always renders a sentence unbounded, bounded-repetitive sentences cannot normally be put in the progressive without losing their bounded-repetitive character. A sentence like *John was kicking the ball thirty times* is no longer bounded, but refers to an unbounded (hence dissective and homogeneous) situation.[44]

In the above examples the speaker locates a repetitive situation as simultaneous with a particular TO_{sit} without referring explicitly to the individual subsituations that make up the repetitive situation. This is different, however, in a sentence like

(21) John always had a cup of tea at 4 p.m.

Here it is not only the repetitive situation as a whole that is located in time but also each of the subsituations: each of the subsituations is located at a TO_{sit} of its own (4 p.m. of an each time different day). The TS of the sentence as a whole is an interval comprising the TSs of all the subsituations, and the TO_{sit} with which this overall TS is simultaneous similarly comprises the TO_{sit}s of the different subsituations. In sum, the overall situation consists of a repetition of subsituations of John having a cup of tea at four, all of which lie wholly before t_0.

An example similar to (21) is

(22) John was always sleeping when I came in.

This sentence differs from (21) only in that each subsituation (of sleeping) is now represented as a MidS. The repetitive situation consists of an indefinite number of subsituations of the type 'John was sleeping when I came in'.

If a dynamic situation is repeated an indefinite number of times over an extended period and is therefore viewed as a characteristic (rather than incidental) feature of that period, the sentence representing the situation in this way will be said to have a 'habitual' meaning (see Comrie 1976:27–8). For example:

(23) (a) For many years I have eaten an egg at breakfast.
(b) John drives an Austin.
(c) John beats his wife.

As pointed out by Comrie (1985:39), 'sentences with habitual aspectual meaning refer not to a sequence of situations recurring at intervals, but rather to a habit, a characteristic situation that holds at all times'. In other words, the 'situation referred to' in a habitual sentence is a state,

44. Sentences like *John was kicking the ball thirty times* may seem odd, but this is only because they require a specific kind of setting. If John had announced in advance that he was going to kick the ball thirty times and the speaker heard this, then there is nothing unusual about the speaker saying something like *John was kicking the ball thirty times when the headmaster came in.*

not a dynamic situation.[45] This means that a habitual sentence (like any sentence referring to a state) is by definition unbounded and hence dissective and homogeneous.[46] This explains why we can use a sentence like *John beats his wife* even when John is not actually beating his wife at t_0. What we assert in this sentence is that it is characteristic of John that he beats his wife now and then. That is, it is a characteristic that we can predicate of John at any time, irrespective of whether or not he is actually beating his wife when we utter the sentence.

Habitual sentences do not normally appear in the progressive form, except in two cases. The first is when each of the subsituations is viewed as in progress at some TO. This is the case in examples like *Bill was always working on his dissertation when I visited him.* The second case is somewhat special. Since habitual sentences are necessarily unbounded, we do not need to put them in the progressive form to refer to MidS: the nonprogressive form can refer to any part of the whole situation. On the other hand, since a state may be indefinitely long, it is certainly useful to have a means of representing a state as temporary, i.e. as being of restricted (rather than permanent or indefinite) duration. As a matter of fact, the progressive form is used to this effect in habitual sentences:[47]

(24) (a) John has been beating his wife recently.
(b) In those days John was driving a Ford.

45. See Rigter (1980b:223): 'sentences like *Delmer walks to school* ... are not statements about events, but statements about conceived world structure. ... Accordingly, generic sentences behave in all respects as structures referring to states.' The view that habitual sentences describe states is also put forward by Vendler (1967:108), Hirtle (1967:50), Leech (1969:138, 1971:5), Vasudeva (1971:155), Hirtle (1975:95), Lyons (1977:716), Steedman (1977:222), Berezovsky (1978:30), O'Grady (1979:145), Rigter (1980d:150), Vet (1980:15), Bach (1981:76), Partee (1984:270), Fenn (1987:63–4), Mittwoch (1988:209), etc. The only linguist who explicitly rejects this view is Brinton (1987). According to her, habits are similar to states, but differ from them in that they are agentive, dynamic, and iterative (whereas states are non-agentive, nondynamic, and continuous). In my opinion, this view is debatable, but I do not think it is necessary to discuss it here. Even if habits should be different from states in these respects, this difference is immaterial to the analysis that will be argued. For our present purpose, all that matters is that habits, like states, are unbounded, so that they can be represented as holding at a particular time, even if the full habitual situation also extends before and/or after the time interval in question.

46. The observation that a habitual sentence is necessarily unbounded accords with the fact that (bounded) sentences like the following are felt to be repetitive, but not habitual:
(i) (a) In two years' time I have eaten 712 eggs in the mornings.
(b) John has beaten his wife at least 100 times in the last year.
Compare also:
(ii) (a) They went to the movies three times a week.
(b) They went to the movies three times that week.
As noted by Smith (1978b:70), (ii,a) is a habitual sentence, but (ii,b) is not, because the number of subsituations in (ii,b) is specified. (This specification renders (ii,b) bounded.)

47. See Leech (1969) and Dowty (1977) for a discussion of the temporary connotation of progressive sentences.

In the above paragraphs it has been pointed out that a repetitive sentence may or may not have habitual meaning, i.e. may or may not represent the repetitive situation as characteristic of some extended period of time. If it does (e.g. *My dog chases cars*), the sentence expresses a state. Such a sentence is thus positively marked for the features [repetition] and [state], whereas repetitive sentences that do not yield a habitual (characterizing) reading (e.g. *The boy kicked the ball three times*) have the features [+ repetition], [− state].

Sentences like *My dog chases cars* are normally included in the class of so-called 'generic' sentences.[48] Generic sentences can be defined as sentences that predicate some typical characteristic of either a kind (as in *Beavers build dams*) or an individual. The characteristic in question need not consist in the performance of some habitual activity (as in the above examples): in sentences like *Beavers know how to build dams* or *I can speak French* it is a continuous state. This means that generic sentences are always [+ state], but may be either [+ repetitive] or [− repetitive].

Generic sentences may or may not be 'omnitemporal', i.e. make a statement that is valid at all times (an 'eternal truth'). An example like *The sun rises in the east* may be assigned the feature [+ omnitemporal], but sentences like *My dog chases cars* or *Dinosaurs had long tails* are [− omnitemporal]. Whether or not a generic sentence has the feature [+ omnitemporal] often depends on whether the referent of the subject NP (or of another relevant participant NP) is subject to temporal restrictions. This is clearly the case in the above examples: the sentence *My dog chases cars* can only be true as long as my dog is alive and capable of performing the activity in question; and in *Dinosaurs had long tails*, where the characteristic is predicated of a species, the use of the past tense is consonant with the fact that the species in question has ceased to exist.[49]

48. Lawler (1972, 1973a, 1973b) has made a lot of interesting observations in connection with generic sentences of this kind. The question why certain sentences are interpreted generically while others are not has been dealt with by Declerck (1986b, 1987c, 1989, 1991).

49. An omnitemporal sentence makes a claim which holds for all times, and is therefore incompatible with the preterit, present perfect or future tense, unless there is a suitable adverbial making clear that the truth of the statement is not restricted to the period indicated. Compare:

    (i) (a) !Two and two made four.
        (b) !Two and two has made four.
        (c) !Two and two will make four.
    (ii) (a) At that time two and two already made four.
        (b) Two and two has always made four.
        (c) Two and two will always make four.

    Generic sentences that are not omnitemporal can be used freely in the preterit, present perfect or future tense:

    (iii) (a) The Egyptians built splendid pyramids.
        (b) Until now this kind of dog has been easy to train.
        (c) John will soon be able to speak French.

Some sentences are [+ state] and [+ omnitemporal], but are not generic, because they do not predicate a typical characteristic of the referent of the subject NP. The sentence *Two plus two is four* is an example. This means that we can distinguish the following four possibilities in connection with what we could call 'gnomic' sentences (i.e. sentences making an assertion that is 'general' in some way):

1 [+ state, + repetition, + omnitemporal][50]

(25) (a) The sun rises in the east.
(b) Any body that is not subject to external forces maintains a constant velocity.
(c) Like poles of a magnet repel each other.

2 [+ state, + repetition, – omnitemporal]

(26) (a) My dog chases cars.
(b) Bob works in a factory.

3 [+ state, – repetition, + omnitemporal]

(27) (a) Two and two is four.
(b) The beaver builds dams.
(c) Hydrogen is the lightest element.

4 [+ state, – repetition, – omnitemporal]

(28) (a) John can speak English.
(b) Bill has a good memory.

What all of these have in common is that they represent a situation as a state. The present tense represents this state as holding at t_0. This means that gnomic sentences can be analysed in terms of temporal schemata that are quite similar to those assigned to non-gnomic states (as in *John's hair is untidy*).[51] It is therefore unnecessary to posit a special kind of

50. As noted at the beginning of this section, we call a sentence repetitive both if it refers to a number of identical subsituations (involving exactly the same participants) and if it refers to a number of similar subsituations (involving different participants). This means that a sentence has the feature [+ repetition] not only if the VP implies repetition but also if the plurality of one or more of the participant NPs entails reference to several 'cases' (in the sense of Lewis 1975). Thus, *People come to pray here once in their lives* has the feature [+ repetition] no less than *That man regularly comes to pray here.*

51. As noted in footnote 45, Brinton (1987) rejects the view that habits are states. One of the arguments she adduces is that 'habits do not occur comfortably with *right now*, whereas states do' (p. 201), witness the following examples:
(i) (a) ?They eat dinner right now.
(b) ?They take their holidays right now.
(ii) (a) He is happy right now.
(b) She is at work right now.
At first sight this observation would seem to refute the claim that habits should be treated as unbounded situations that are located at t_0. (Thus, Brinton concludes from (i)–(ii) that 'the activities constituting a habit are not normally thought of as actually

'generic' or 'universal' tense in order to account for them.[52]

As has often been noted, gnomic (general) sentences that involve the feature [+ repetition] do not imply that one of the subsituations is actually holding or occurring at the relevant TO. For the sentence *My dog chases cars* to be true it is not necessary that my dog is actually chasing a car at t_0. In this respect repetitive gnomic sentences resemble non-gnomic progressive sentences (e.g. *John is writing a novel*), which also refer to a situation that is marked by internal gaps, and which remain true even if t_0 happens to be included in such a gap. This similarity with progressive sentences means that the possibility that no subsituation is actually holding at t_0 (or at some other TO) should not be interpreted as meaning that repetitive gnomic sentences are not 'anchored' to a TO. (The latter claim is not unfamiliar in the linguistic literature, see e.g. Botne 1981:106, Steube 1980:113.)

The claim that gnomic sentences are not anchored to a TO has also been made on different grounds. According to some linguists, a sentence that is omnitemporal (i.e. valid at all times) is also 'timeless' (i.e. not located in time). Thus, Bache (1985:107) claims that 'all-time' expressions do not locate situations in time relative to the temporal context of the speech situation', and Barense (1980:20) similarly states that gnomic sentences are not used 'to make a deictic reference to the time of speech' (see also Sweet 1891:104; Crystal 1966:22; Lyons 1977:681; Ultan 1978:87; Canavan 1983:35; Schopf 1984:253, 1985:86; Papp 1985:100;

occurring at the present moment, whereas states do exist at the present moment' (p. 201).) However, it appears that many states are also incompatible with *right now*:

(iii) (a) ?John knows English right now.
(b) ?Fred has long legs just now.

As a matter of fact, we have already noted (see footnote 30) that states which are not normally conceived as temporary cannot comfortably be located at a relatively short time interval (see *?John knew English this morning, ?Fred had long legs at 5 o'clock*). It follows that the above observations actually support the view that habits are states, for the habits referred to in (i,a–b) are also represented as permanent situations. (As noted above, temporary habits are expressed by progressive sentences.)

The fact that permanent states (including habits) cannot normally be located at a punctual or short time may also seem to be in contradiction with the claim that a sentence describing a state in the simple present tense locates that situation at t_0 (i.e. represents it as holding at t_0). However, this apparent problem will disappear when, in chapter 7, we refine this analysis of the present tense. It will be shown there that the time at which the situation is located (i.e. TO_{sit}) need not coincide with t_0. In most cases TO_{sit} is a much longer timespan which includes t_0. In the case of habitual sentences like *John's dog chases cars*, TO_{sit} is interpreted as commensurate with the time of the habit as a whole.

52. Comrie (1985:40) offers the same analysis:

> A sentence like *cows eat grass* is claimed to refer only to the present moment, the interpretation of this as a universal truth being on the basis of structural and extralinguistic factors beyond the meaning of the present tense. . . . There can be no universal tense, i.e. a tense that is used for truths that hold at all time.

This view is also in keeping with the fact that Dahl (1985) reports that he has found no evidence (in the 64 languages examined) for a tense/mood/aspect category restricted to generics.

Bache 1986:8).[53] Such a view inevitably leads to the conclusion that some present tense sentences locate a situation in time, while others do not. In my opinion, this abandonment of a unitary treatment of the present tense is quite unnecessary. As we will see in chapter 7, the present tense basically implies no more than that t_0 is included in the situation-TO (which coincides with TS). Since this is clearly the case when the reference is to an 'all-time' state, there is no reason for claiming that such sentences resist the analysis of the present tense.[54]

The general conclusion from this section is that repetitive and gnomic sentences present no problem for the analysis we have argued. Repetitive dynamic situations are located in time in exactly the same way as nonrepetitive dynamic situations, while (repetitive or nonrepetitive) gnomic situations are states that are located in time in just the same way as other states.

4 TIME ADVERBIALS

We have seen that a TE may or may not function as a TO. That is, a TO (for example, the TO_{sit} that is simultaneous with TS) may coincide either with TE as a whole or with a subinterval of TE. Thus, *John was here yesterday* leaves vague whether John was here all day yesterday or only during part of the day. (In technical terms: TO_{sit} (= TS) may coincide with TE as a whole or with a subinterval of TE.) This means that the relation between TO_{sit} and TE is not describable in terms of the 'simul' relation as it has been defined above (viz. 'X simul Y' means 'X = Y'). The relation between TE and TO_{sit} is one of inclusion: TE includes TO_{sit} in the sense that TO_{sit} is either a subinterval of TE (in which case we can speak of 'proper inclusion') or coincides with TE.

In some sentences there is more than one time adverbial, and hence more than one TE. In that case the same relationship of inclusion will also hold between the different TEs. Thus, in *John was born at three o'clock in the morning on Christmas Day, 1967* there are four TEs, which are related in terms of inclusion: the TE denoted by *at three o'clock* (which is punctual and must therefore be the time that functions as (i.e. is simultaneous with) TO_{sit}) is properly included in the TE established by *in the morning*, which is itself a proper subinterval of the TE set by *on Christmas Day*, which is in its turn properly included in the TE referred

53. The view that omnitemporal sentences are timeless (nontemporal) has even induced some linguists (e.g. Anderson 1973a:481ff.; McGilvray 1974:31; Canavan 1983:32ff.) to claim that such sentences are 'tenseless'. As noted by Lyons (1977:679), this confusion of omnitemporality with tenselessness is unfortunate: in English, 'there are no tenseless declarative or interrogative system-sentences'. (See also McCawley 1973 ('tense is an obligatory category in English'), Levinson 1983:77.)

54. Jespersen (1931:17) already stated that it is wrong to call omnitemporal sentences timeless, and that the reason why such sentences use the present tense is simply because they 'are valid now'. Matthiesen (1983:400–1) argues the same point at some length.

to by *1967*. This appears to be a general rule in connection with the use of time adverbials: there may be several time adverbials establishing different TEs, but since there can only be one TO_{sit}, these different TEs must be related to each other in such a way that all of them ultimately contribute to establishing the same time, viz. the time (TO_{sit}) at which the situation is said to hold.[55] This means that one TE will be included in another,[56] and that TO_{sit} either coincides with the shortest TE (as in the above example) or is an unspecified subinterval of it (as in *He left early in the morning on Christmas Day, 1967*).

It should be noted that these conclusions remain unchanged when the different adverbials are not contiguous:[57]

(29) Yesterday John left at five o'clock in the morning.

I consider such sentences to be the result of 'preposing' or 'fronting' of an adverbial – an operation which applies freely to time adverbials if the following two conditions are satisfied:

1 The adverbial in question must constitute given information. Thus, (29) could be used in answer to *When did John leave yesterday?* but not in answer to *When did John leave?*.
2 If there are several time adverbials, only the one with the widest scope (i.e. the one indicating the longest interval) can be fronted. Unlike sentence (29), the following sentences do not satisfy this condition and are therefore ungrammatical:

(30) (a) *At five o'clock in the morning John left yesterday
(b) *In the morning John left at five o'clock yesterday.

Apart from piling up time adverbials which identify the same time with different degrees of preciseness,[58] we sometimes also use time adverbials

55. Of course I am disregarding here sentences like *John got up at ten o'clock every day last week*, which are interpreted repetitively and whose time adverbials therefore collaborate in establishing a series of times rather than a single TE.
56. This is not the case when the second adverbial is added as a kind of appositive to the first. For example:
(i) John came at 5 p.m., before Mary left. (Kittredge 1969:93)
However, in this case too, the two adverbials specify the same time interval.
57. Huddleston (1969:779) makes the same point. In connection with the example *Last week she went to London on Tuesday* he notes: '*On Tuesday* here narrows the specification expressed in *last week*. In such cases we should probably regard the two (surface) specifiers as forming a single deep constituent.' Similar remarks are made by Bertinetto (1982:94) and Verkuyl and Le Loux-Schuringa (1985:242).
58. Some linguists (e.g. Hornstein 1977; Dieling and Kempter 1983:15) attempt to solve the problem of discontinuous time adverbials by assuming that the fronted adverbial indicates Reichenbach's R while the other(s) establish(es) E. This, of course, has the unfortunate consequence that one must abandon the Reichenbachian principle that time adverbials can only refer to R. Moreover, the solution would not seem to apply so easily to *Yesterday John left in the afternoon* as it does to *Yesterday John left at five o'clock*: whereas *at five o'clock* denotes a point of time, and can therefore be taken to denote the time of the punctual situation of John leaving, the same is not true of *in the afternoon*, which refers to a longer timespan.

which are 'complex' in the sense that they refer to several time intervals and state temporal relations between them. The following are some examples:

(31) (a) John left the day before yesterday.
(b) John left a fortnight ago from yesterday.
(c) I did it the day before the last Sunday before Christmas.

Time adverbials like these would seem to realize temporal structures that are very similar to those realized by the tenses. For example, one might say that *the day before yesterday* and *a fortnight ago from yesterday* realize the structure 'TE_2 before TE_1 before t_0'. This is very similar to the schema of the past perfect, which, apart from the relation 'TS simul TO_{sit}', consists of the structure 'TO_{sit} before TO_2 before TO_1' (in which TO_1 is usually t_0). However, it is the preterit and not the past perfect that is used in (31,a–b).[59] This means that the complex time adverbials establish only one time relevant to the use of the tenses: in both sentences they indicate the TE in which the TO_{sit} is properly included. More generally speaking, we can say that, as far as the use of the tenses is concerned, complex adverbials that refer to several time intervals and state temporal relations between them establish only one TE. This principle has also been observed by Smith (1978b:48), who states that 'complex time adverbials are single units in temporal interpretation, even though they need not occur as a unit in surface structure'.[60] See also Dowty (1982:39).

As far as I can see, the above conclusion also holds for sentences like the following:

(32) (a) (From) yesterday, John left a week ago. (Hornstein 1977:532)
(b) (From) tomorrow, John will leave in a week. (ibid.)

According to Hornstein, who holds the view that 'presentential adverbs' always indicate R (reference time) in the Reichenbachian system, the initial time adverbial in such sentences establishes R, while the adverbial that has end position establishes E (event time). (Smith (1978b:52) makes

59. We can, of course, use the past perfect with these complex adverbials, but in that case we have a different meaning. While *John left the day before yesterday* expresses that the situation of John leaving held at a TO_{sit} that preceded t_0 (and whose approximate position on the time line is indicated by *the day before yesterday*), the sentence *John had left the day before yesterday* expresses that the situation held at a TO_{sit} which was anterior to some other TO_2 which itself preceded t_0. (As we will see below, the complex time adverbial can indicate either the TE that includes TO_{sit} or the TE including TO_2.)

60. However, Smith's analysis differs from ours in that she applies the term 'reference time' to one of the constituents of such a complex adverbial: in her (Reichenbachian) analysis, the adverbial (*some time*) *before noon* in *He left* (*some time*) *before noon* establishes the 'event time', whereas *noon* indicates the 'reference time' (see Smith 1978a:77, 1978b:59). (This analysis is also accepted by Lo Cascio (1981).) In our theory, the adverbial as a whole establishes the TE which includes TO_{sit}. The internal structure of the adverbial is irrelevant to the temporal schema realized by the tense.

the same claim.) My own analysis of (32,a–b) is different.[61] In my opinion, the two adverbials again co-operate to specify a single TE, as is clear from the fact that there is no semantic difference between (32,a–b) on the one hand and (33,a–b) or (34,a–b) on the other:

(33) (a) John left a week ago (from) yesterday.
(b) John will leave in a week from tomorrow.
John will leave tomorrow in a week. (Hornstein 1975:82)
(34) (a) John left a week before yesterday.
(b) John will leave a week after tomorrow.

The only difference between (32,a–b) and (33,a–b) on the one hand and (34,a–b) on the other is that they exemplify two different ways of relating the two times that are involved in the internal structure of the complex time indication, more specifically, two different ways of computing the distance between these times. In (32,a–b) and (33,a–b) the speaker expresses that the location of the time to be established is computed from some other time (hence the use of the preposition *from*); in (34,a–b) he expresses that the time to be established is prior or posterior to some other time (hence the use of *before/after*). In both cases he also indicates the distance between the two times. Different though these two types of time indication may be, they ultimately serve to establish the same time: the (continuous or discontinuous) complex time adverbials are different lexicalizations of the same idea.

Another type of time adverbial that can be treated on a par with the type *the day before yesterday* is the kind exemplified in

(35) (a) He left before the war.
(b) He wrote the letter after breakfast.

The adverbials *before the war* and *after breakfast* have the same meaning as *at some time before the war* and *at some time after breakfast*.[62] Since

61. Hornstein's analysis of (*From*) *yesterday, John left a week ago* (according to which (*from*) *yesterday* establishes R and *a week ago* indicates E) leads him to conclude that the representation of *John left a week ago* (*from*) *yesterday* is 'E–R–S' (p. 532). However, this conclusion is incompatible with the tense of the sentence, since 'E–R–S' is the schema for the past perfect (in both Reichenbach's theory and Hornstein's).

62. At least, the semantic meaning of *after breakfast* is the same as that of *at a time after breakfast* (see also Leech 1969:111; Fillmore 1975:35; Heinamaki 1978:126; Partee 1984:267; Schopf 1984:356). However, the two phrases may have different elements of pragmatic meaning in addition to this. For example, the mere fact that we say *He wrote the letter after breakfast*, and not *He wrote the letter before lunch*, suggests that the letter was written immediately or soon after breakfast (at any rate that the situation was closer in time to breakfast than to lunch). The overt use of the indefinite phrase *at a/some time* cancels this implicature (see Konig 1974:560). An aspect of pragmatic meaning that is certainly lost when we replace *after/before* by *at a time after/before* is the universal interpretation which is assigned to *after/before* in certain contexts, as in
(i) Before the invention of printing all texts had to be copied by hand.
In sentences like this, *before* is pragmatically interpreted as 'all the time before' (see Kroch 1972:261.; Heinämäki 1978:107ff.). Since the indefinite quantifiers *a* and *some*

the latter adverbials belong to the type *the day before yesterday*, it seems warranted to conclude that the former belong to this type as well.[63] (Perhaps we can even consider *before/after the war* as a reduction from something like *at (a time) before/after the war*. This would be in keeping with the fact that some prepositions can appear overtly before *before/after*: *I stayed until after the game was over, I haven't seen him since before he left for America.*)[64]

Adverbial clauses introduced by conjunctions like *before* and *after* can be analysed in the same way.[65] Thus, in

(36) After he had left I suddenly remembered where I had seen him before.

the adverbial *after he had left* means 'at some time after he had left'.[66] This adverbial indicates the time interval which functions as TE for the head clause, i.e. the time interval which includes the TO_{sit} of the head clause (i.e. the time when I remembered).

The above conclusion (that complex adverbials like *the day before yesterday*, *a week ago from yesterday* or *before the war* establish only one time interval relevant to the use of the tenses) does not hold for what I

conversationally implicate 'not all' (see Horn 1972; Declerck 1987a), this universal interpretation is no longer available when we replace *before* with *at a/some time before*.

63. As pointed out in footnote 60, this conclusion is at variance with Smith (1978a:77, 1978b:59). However, what Smith argues in these articles does not accord with what she notes elsewhere in connection with the sentence *Sam will announce before midnight that Sue left three hours earlier*:

> The sentence can only be interpreted to mean that the point of reference for *three hours earlier* is *before midnight*, not *midnight*. (Smith 1977a:160)

This observation supports our analysis, according to which *before midnight* does not refer to midnight as a time interval that is relevant to the use of the tenses.

Our analysis also accords with Bertinetto (1982), who argues at some length that phrases like *before midday* 'must be assigned a single function for the semantic interpretation' of the sentences in which they appear (p. 73). That is, 'it is wrong to take the X of *before* X as an R.' (p. 85). Partee (1984:267), Vet (1984:195), Schopf (1984:356), Harkness (1987:83) and Hamann (1987:44) also adopt this point of view.

64. The mistaken view that, in an adverbial of the form 'after/before X', it is X that sets the TO, rather than the adverbial as a whole, can lead to strange conclusions. Thus, it induces Allan (1986:330) to challenge the widely accepted view (also argued above) that a verb form with progressive meaning typically refers to the middle of a situation: 'it is by no means certain that the English progressive invariably describes an event that has commenced at the point of orientation'. According to Allan, this is clear from an example like *He was writing after I entered*, which can mean that the situation of writing began after the speaker entered. However, the error is obvious: the TO corresponding with (i.e. included in) the TE established by the adverbial *after I entered* is not the time of my entering but a time after that. And this TO does coincide with some middle part of the situation described in the head clause.

65. The observation that temporal clauses belong to the same syntactic and semantic category as temporal adverbs is a familiar one – see e.g. Schopf (1984:356–7), Harkness (1987:85–6), Hamann (1989:39).

66. Wollmann (1985:142–3) argues that *after*, *before*, *since* and *until* have the following meanings: *after* = AT (SOME t (AFTER (X)); *before* = AT (SOME t (BEFORE (X)); *since* = AT (ALL t (AFTER (X)); *until* = AT (ALL t (BEFORE (X)).

will call 'relative time adverbials', i.e. adverbials of the type *the day before*. Unlike the former, the latter require (or at least prefer) the use of the past perfect: we say *John had left the day before* rather than *John left the day before*.[67] The reason why the past perfect is the normal tense here is that a relative time adverbial not only establishes a TE but also relates it to a time which functions as a TO in the preceding context and which is also one of the relevant TOs in the temporal structure of the tense form of the sentence in which the relative time adverbial is used. That is, *the day before* means 'the day before the TO at which the TS of the previous clause is located'.[68] Since this TO is necessarily a past TO – if it were t_0 we would have to use *yesterday* instead of *the day before* – the situation which is said to hold on the day before this TO is referred to by means of the past perfect. Relative time adverbials are thus different from complex adverbials.[69] Although the latter involve reference to several intervals, there is only one interval that is relevant to the use of the tenses. On the other hand, when a relative time adverbial is used, there are two such intervals. Moreover, the relation expressed by a relative time adverbial (in this case: 'before') is one of the relations that make up the temporal structure of the tense. This is not true of (any of) the relation(s) expressed by a complex adverbial.[70]

67. Some speakers appear to accept the use of the preterit with adverbials like *the day before*. As pointed out in chapter 2, we can account for this in terms of a shift of domain: instead of representing the situation as anterior to the relevant past TO, the speaker represents the situation as establishing the central TO of a new past time-sphere domain.
68. In other words, *the day before* means 'the day before the TO_{sit} of the previous clause'. This is clear from an example like the following:
    (i) Last week John came to see me. He had had an accident the day before.
    Here *the day before* means 'the day before he came to see me' (i.e. the day before the TO_{sit} of the first clause), not 'the day before last week'.
69. This difference also appears from examples like the following:
    (i) (a) When the doctor arrived, John had died the day before.
    (b) *When the doctor arrived, John had died before Christmas.
    (c) *When the doctor arrived, John had died the day before Christmas.
    There is a clear difference in acceptability between (i,a) and (i,b–c). This must be due to the difference between the relative adverbial *the day before* and the complex adverbial (*the day*) *before Christmas*.
70. It is interesting to note that the adverb *earlier* may or may not behave as a relative time adverbial. That is, we can say either (i,a) or (i,b), and either (ii, a) or (ii,b):
    (i) (a) John had left earlier.
    (b) John left earlier.
    (ii) (a) John will have left earlier.
    (b) John will leave earlier.
    In the (a) sentences, *earlier* is interpreted as 'at some time earlier than t', and t behaves as a TO that is relevant to the choice of the tense. In the (b) sentences, t is not felt to indicate a relevant TO because the idea 'at some time earlier than t' is interpreted as establishing a single TOE (in the same way as *the day before yesterday* establishes only one TO). (In other words, the perfect tense forms in the (a) sentences are relative tense forms which relate a situation to a binding TO (t) in a given domain, whereas the tenses in the (b) sentences are absolute tenses establishing a new domain.)

The claim that the implied TO in adverbials like *the day before* is a TO that is relevant to the use of the tense is confirmed by the fact that it may appear overtly:[71]

(37) On Tuesday, John had left three days before.[72]

The separation of *on Tuesday* from *three days before* is relevant: as noted above, an adverbial of the type *three days before Tuesday* refers to only one relevant time, and we therefore use the preterit to refer to a situation that is simultaneous with it. However, putting *on Tuesday* at the beginning means that 'Tuesday' is a relevant time too, i.e. that it has the status of a full TO in the temporal schema of the tensed sentence (see Smith 1978b:52). This means that the two relevant past TOs of the structure of the past perfect are referred to, so that this is the tense that must now be selected.[73] An example that is similar, but expresses future-in-the-past instead of past-in-the-past, is the following:

(38) (John said that) yesterday at noon Bill was going to leave three hours later. (Gabbay and Rohrer 1978:109)

We can end this discussion of complex and relative adverbials by pointing out a couple of exceptional cases in which there are two time adverbials that do not constitute a single complex adverbial. One such case is illustrated by the following sentences:

(39) (a) Yesterday you were coming tomorrow. (Huddleston 1969: 781)
(b) A week ago, John would have left yesterday. (Pankhurst 1980:128)[74]

In chapter 2 (section 24) it was noted that sentences like *He was coming here himself tomorrow* involve an 'implicit TO'. Sentence (39,a) is quite similar, but now the TO in question is indicated by *yesterday*. One might

71. According to Smith (1978b:52), sentences like this are not grammatical for all speakers.
72. According to Smith (1977a:158, 1978b:84), some (American) English speakers also accept *ago* instead of *before* in examples like this. In fact, *ago* becomes the normal form when the initial time adverb refers to the future while the second (and hence the complex adverbial as a whole) refers to a time which is anterior to t_0:
(i) Tomorrow, John will have left three days ago.
However, we again use *before* if the time established by the complex time indication as a whole lies in the future with respect to t_0:
(ii) In two weeks' time, John will have left three days before.
As noted by Fillmore (1976:97), *earlier* is also possible in such sentences.
73. The past perfect also collocates with two adverbials specifying related times in examples like
(i) When I visited John, he had been sick since Thursday. (Dowty 1979:345)
Here one adverbial (*since Thursday*) indicates a period leading up to some TO while the other (the *when*-clause) specifies the TO in question.
74. According to Pankhurst, (39,b) 'could mean something like: "Judging by the way he was acting a week ago, John would have left at the earliest opportunity, namely yesterday, but he is still here, so things must have changed".'

argue that this sentence results from the reduction of something like 'Yesterday you said that you were coming tomorrow.' At any rate it is clear that the two adverbials each perform a job of their own. (For this reason we cannot replace the two of them by a single *then*.) This is in keeping with the fact that their positions are rather fixed: the adverbial referring to the TO that is not simultaneous with TS must be in initial position, whereas the adverbial indicating the TE which includes the TO_{sit} must come at the end.

A second type of sentence in which there are two time adverbials that do not co-operate in specifying a single time interval is exemplified by the following:

(40) (a) When I met John, he had often got up at five o'clock.
(b) Up to now I have seldom left at five o'clock.

In these examples, one time adverbial (viz. *at five o'clock*) is related to the subsituations of the repetitive situation, whereas the other either specifies another TO (*when I met John*) or indicates a timespan leading up to such a TO (*up to now*). I will not further examine such sentences here. I will consider them in detail in the sections on the present perfect and the past perfect in chapter 7.

A third exception concerns sentences involving the adverb *now*:

(41) (a) Now we will have no money at the end of the month. (Huddleston 1969:789)
(b) He would now leave the country before the crime was detected.
(c) Now he would have left the country before the end of the day.
(d) Now he will have left the country when the day is over.

In each of these sentences *now* co-occurs with another time adverbial without creating any conflict. The reason is that *now* is different from other temporal adverbs, such as *yesterday*, *then*, *tomorrow*, etc. In none of the sentences (41,a–d) can *now* be interpreted as specifying the TE that includes TO_{sit}. Instead, *now* is interpreted as the reduction of a kind of hypersentence: 'now it is/was the case that . . .'.[75] This use of *now* serves to indicate the time that receives the 'temporal focus' (see chapter 2, section 22).[76] As noted by Banfield (1982:158), sentences like *Now he felt*

75. Vet (1984:194) suggests that *now* is the reduction of a *now that*-clause expressing the fulfilment of a condition: *Now he will be more careful* means something like 'Now that this has happened (e.g. now that he has been punished) he will be more careful.' In my opinion, however, *Now he will be more careful* means no more than that it is now the case that the person referred to will be more careful. The fact that this state of affairs logically presupposes the fulfilment of some condition is not something that is expressed in the sentence itself. It is therefore unwarranted to analyse *now* as elliptical for a *now that*-clause.

76. As appears from (41,b–d), the temporal focus can only lie with the central TO of the domain, not with an 'ancillary' (noncentral) TO. Thus, in (41,b–c) *now* indicates the central TO of the past domain, not the time of leaving.

very tired 'recount events as represented from within a consciousness', and *now* indicates the time of the act of consciousness. That is, the 'NOW' of the sentence is the time when the relevant person (the 'SELF') experiences the situation. The fact that this use of *now* is different from that of normal time adverbs entails that we can find instances in which this *now* co-occurs with a time adverbial specifying the TE including TO_{sit}.[77]

Apart from the above three exceptions, there are some other cases in which two time adverbials co-occur without establishing a single time interval. However, the description of these cases presupposes more information about the structure of some individual tenses than has been furnished in this chapter. I will therefore defer going into these cases until the past perfect is discussed in chapter 7.

5 CONCLUSION

In this chapter we have identified the structural elements that constitute the temporal schemata realized by the tenses. The following are the most important claims that have been made:

1 The temporal schema of a tense minimally involves the elements TO_1 (basic TO), TO_{sit} (situation-TO) and TS (time of the situation).
2 TO_1 may or may not be t_0. (In chapter 7 we will see that TO_1 is t_0 when the tense is used as an absolute tense form, but not necessarily when it is used as a relative tense form.)
3 TS (the time of the situation) is the time interval taken up by that part of the situation that is being referred to (and located in time) by the clause involving the relevant tense form.
4 If the situation is bounded, TS is the time of the full situation. If it is unbounded, TS may be either the time of the full situation or a subinterval of this.
5 TS is by definition represented as simultaneous with a TO. We have referred to this TO as TO_{sit}. The simultaneity relation in question is one of coincidence: TS and TO_{sit} are commensurate.
6 TO_{sit} is related directly or indirectly to t_0. (For example, the future perfect represents TO_{sit} as anterior to a TO which is itself posterior to t_0.)
7 If there is a (simple or complex) time adverbial, it indicates a time interval which we refer to as TE. If there are several time adverbials, they normally co-operate in establishing a single TE.

77. When used in this way it is virtually impossible for *now* to occur at the end of the sentence. Its normal position is at the beginning, or at least before the main verb form. The reason for this is that *now* anaphorically indicates the time that receives the temporal focus and therefore functions as theme (topic) of the sentence. As is well-known, the theme of a sentence is preferably put at the beginning (see e.g. Declerck 1984b).

8 If there is a duration adverbial, it measures the length of the full situation. If this measurement is a specific indication (e.g. *for two hours*), it renders the situation bounded. In that case the time of the full situation coincides with TS.
9 The relation between TE and TO_{sit} is one of inclusion: TE either properly includes TO_{sit} or coincides with it. (Since TS = TO_{sit}, the same inclusion relation holds between TE and TS.)
10 Non-gnomic (i.e. dynamic) repetitive sentences are analysed in the same way as other dynamic sentences.
11 Progressive sentences and gnomic sentences are analysed in the same way as other unbounded sentences.

As we have seen, the term 'reference time' is used in more than one way in the linguistic literature:

1 When reference time means 'the time referred to' by an adverbial,[78] it corresponds with our TE.
2 When it is a 'time referred from', it corresponds to one of the TOs in our system:

   a. The R in Reichenbach's analysis of the past perfect (E–R–S) is in our system the TO to which TO_{sit} is anterior and which is itself anterior to TO_1.
   b. The R in Reichenbach's analysis of the past tense (E,R–S) corresponds with our TO_{sit}.

3. Allen (1966:167) uses the term 'point of reference' in the sense of our 'binding TO', i.e. the time to which the TO_{sit} of a temporally subordinated clause is related by the use of a relative tense. If the temporally subordinated clause is also syntactically subordinated, this binding TO is the TO_{sit} of the matrix clause. Otherwise it is either the TO_{sit} of a clause in the context, or a TE (as in *At five o'clock John had already left*) or (but seldom) a time that remains unspecified (i.e. what we have called an 'implicit TO').

The term 'evaluation time' (i.e. time in relation to which the situation is true) is also ambiguous:

1 Some people use it in the sense of our TO_1 (which is t_0 in the absolute uses of the tenses) (see footnote 3).
2 Those who do not use 'evaluation time' in the above way use it either in the sense of 'time referred to' or in the sense of 'time referred from':

   a. Some claim that a time adverbial indicates the evaluation time. In that case 'evaluation time' corresponds with our TE.

78. Bäuerle (1979) and Fabricius-Hansen (1986, 1987) call this time 'die Betrachtzeit' (the 'time looked at').

b. Otherwise it corresponds with TO_{sit} or with a TO intermediate between TO_{sit} and TO_1.

Finally, when the time adverbial is a clause, it functions as TE for the head clause in one of the following ways:

1 When the TO_{sit}s of the two clauses are interpreted as (strictly) simultaneous (as in *John was reading while he was waiting*), the TO_{sit} of the time clause establishes the TE of the head clause and this TE is interpreted as commensurate with the TO_{sit} of the head clause.
2 When the TO_{sit} of the head clause is interpreted as properly including the TO_{sit} of the time clause (as in *John was sleeping when I left*), the TO_{sit} of the time clause functions as TE for the head clause and the TO_{sit} of the head clause is commensurate with this TE. The inclusion interpretation then stems from the fact that the TS of the head clause is not felt to be the time of the full situation. (In the above example this follows from the use of the progressive.)
3 When the TO_{sit} of the head clause is interpreted as properly included in that of the time clause (as in *I left when/while John was sleeping*), the TO_{sit} of the time clause functions as TE for the head clause and the TO_{sit} of the head clause is interpreted as properly included in this TE.
4 When it is clear from the conjunction and/or the tense forms used that the TO_{sit} of the time clause is anterior to that of the head clause (as in *John arrived after/when I had left*), the time indicated by the time clause (and which serves as TE for the head clause) is the time following the TO_{sit} of the time clause (in the above example: the time following my departure). (The right bound of this timespan is not specified, but the timespan will normally be interpreted as a pragmatically restricted span.)
5 When the conjunction and/or the tenses locate the TO_{sit} of the time clause as posterior to that of the head clause (as in *John had arrived before/when I left*), the time indicated by the time clause (and which serves as TE for the head clause) is the (pragmatically restricted) timespan preceding the TO_{sit} of the time clause.

7 The structure of tense 3: analysing the tenses

1 THE PRETERIT

1.1 In chapter 6 we have mostly used examples in the past tense to illustrate the basic principles that were being introduced. In this way the typical characteristics of the past tense have already been referred to, so that it is not necessary to go into them again at great length here.

The most typical feature of the past tense is that a situation is located at a time (TO_{sit}) which is related to TO_1 in such a way that TO_{sit} lies in the past time-sphere, whereas TO_1 lies in the present time-sphere. TO_{sit} must therefore lie wholly before TO_1.[1] (Remember that TO_1 is the 'basic TO', which is either t_0 or another TO defined in the context. The latter possibility is realized in *He will say that he was not at home tonight*, where the TO_1 in the structure of the past tense form *was* is the TO_{sit} of the head clause.) The temporal schema of the preterit thus consists of the following relations:

1 TS simul TO_{sit}
2 TO_{sit} before TO_1
3 TO_{sit} belongs to the past time-sphere
4 TO_1 belongs to the present time-sphere

This can be represented as follows:

1 if TS and TO_{sit} are punctual: Figure 7.1.
2 if TS and TO_{sit} are durative: Figure 7.2.

1. Perhaps the best illustration of the fact that TO_{sit} must lie wholly before t_0 is to be found in sentences like the following:
   (i) Right now, Mary put out the lights in the hall.
   (ii) John went out just now.
   In spite of the adverbials *right now* and *just now* (which can refer to the present, as in *Don't disturb me; I'm having an important telephone conversation right/just now*), these sentences do not refer to the time of utterance, but to a point of time just before TU.

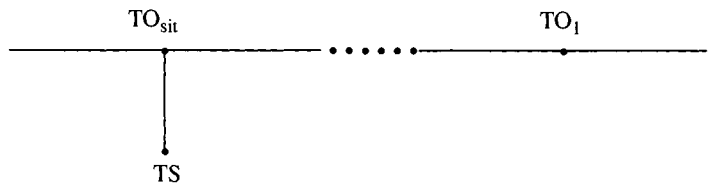


Figure 7.1

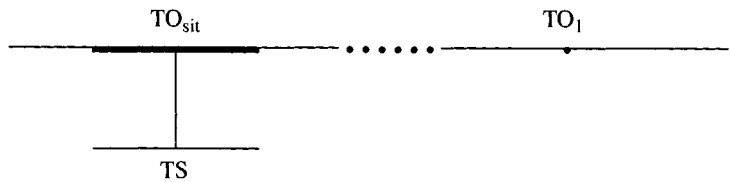


Figure 7.2

(The conventions used in the figures are the following. To represent the present and past time-spheres on the time line, the time line is divided into two parts, separated by a dotted line. What is before the dotted line is the past time-sphere; the part after the dotted line (which is the part including t_0) is the present time-sphere. The use of the dotted line is meant to express the fact that there is felt to be a break between the past time-sphere and the present one. A vertical line linking two elements indicates that the elements in question are simultaneous. A punctual time interval is represented by a dot, a durative interval by a bold line.)

1.2 The analysis presented here covers the various ways in which the past tense can be used:

1 The past tense can be used in an absolute way, as in *John was ill yesterday*. In that case TO_1 is t_0.

2 The past tense can be used as a relative tense (expressing the domain-internal relation of simultaneity). For example:

(1) John said he *was* ill.

The past tense form *was* here represents the situation of John being ill as simultaneous with the TO_{sit} of the head clause. This TO_{sit} lies before TO_1 (which is t_0 here).

3 The use of the preterit may be the result of a shift of temporal perspective (see chapter 2):

a. Situations that hold at t_0 are sometimes located in the past time-sphere (e.g. *I wanted to ask you a few questions*). In that case the

temporal structure realized is the same as in the normal use of the preterit, only there is a metaphorical relation between this temporal structure and real time.

b. When we relate a situation to a post-present TO, this TO behaves as if it were t_0. A situation lying before the TO can therefore be referred to by the preterit, as in *Tomorrow John will say that he* ***was*** *ill tonight.* In this use the preterit realizes the same structure as in its normal absolute use, except for the fact that TO_1 is not t_0 but a time posterior to t_0. The latter fact entails (i) that the past time-sphere in which the situation is located is not the 'real' past time-sphere (which is defined relative to t_0) and (ii) that the past tense behaves as if it were an absolute tense form, whereas in fact it serves to express a temporal relation in a post-present domain.

c. A case of shift of temporal perspective in which the past tense is used as relative tense (expressing simultaneity) is when a pre-present domain is expanded as if it were a past domain:

(2) He has never admitted *that he was not feeling well.*

In this example the situation-TO of the italicized clause is represented as simultaneous with the situation-TO of its matrix. The fact that the preterit is used for this means that the pre-present matrix situation is treated as if it belonged to the past time-sphere.

1.3 When the past tense is used as a relative tense, it expresses simultaneity. This simultaneity relation can be defined in the same way as the 'simul' relation that holds between TS and TO_{sit}, viz. as a relation of coincidence ($TS = TO_{sit}$). This is clear from examples like the following:

(3) John said that Bill was in the kitchen.

In this example, the absolute tense form in the head clause involves the relations 'matrix-TS simul matrix-TO_{sit}' and 'matrix-TO_{sit} before TO_1' (where TO_1 is t_0 and the matrix-TO_{sit} belongs to the past time-sphere). The relative tense in the subclause expresses the relations 'subclause-TS simul subclause-TO_{sit}' and 'subclause-TO_{sit} simul matrix-TO_{sit}'. If we define 'simul' in terms of coincidence, it follows from this that (3) should also imply 'subclause-TS simul matrix-TS'. As a matter of fact, (3) does imply this. To see this, one must remember that TS (the time of the situation) has been defined as the time interval occupied by that part of the situation about which a claim is made in the clause, which means that TS need not be the time of the full actualized situation if the latter is not bounded. What is claimed in sentence (3) is that John said that Bill was

in the kitchen *at that very moment*, i.e. at the time when he said this.[2] In other words, (3) represents the subclause-TS as strictly simultaneous with the matrix-TS. In this respect (3) is similar to *At five o'clock Bill was in the kitchen*, which represents TS as strictly simultaneous with the time indicated by *at five o'clock* (see chapter 6).

1.4 Sentences in the past tense mostly involve a time adverbial establishing a TE. If this is not the case, TE is nearly always recoverable from the preceding context. As we have seen, TE is related to TO_{sit} in terms of inclusion: TE includes TO_{sit}. This inclusion can be either 'proper inclusion' (i.e. TO_{sit} overlaps only a portion of TE) or coincidence (TO_{sit} = TE). Since both TO_{sit} and TE may be punctual or durative, this leads to the following possibilities:

1 TO_{sit} and TE are both punctual (e.g. *John left at two-thirty*). In that case TO_{sit} and TE must coincide. Figure 7.3 represents this possibility.

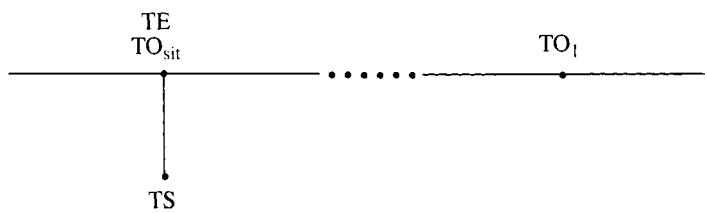


Figure 7.3

2 TO_{sit} is punctual, TE is durative. In that case TE may or may not include TO_t, but TO_{sit} must precede TO_t. Examples are *John left yesterday* (Fig. 7.4) and *John left today* (Fig. 7.5). (In both cases TO_t is t_0.)

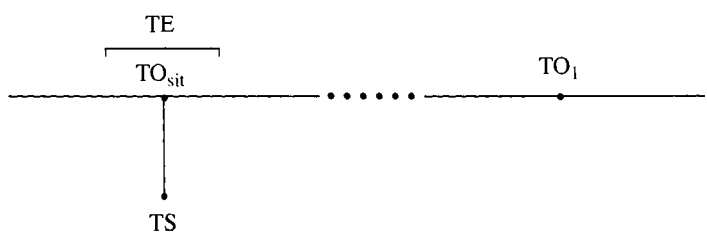


Figure 7.4

2. I am concentrating here on the most natural interpretation of *John said that Bill was in the kitchen*, viz. the reading on which *was* is a relative tense form expressing simultaneity. In chapter 2 we saw that in sentences like this the preterit in the *that*-clause can also be an absolute tense form, shifting the domain. (In that case *was* receives the same interpretation as *had been*.) This possibility will be disregarded here.

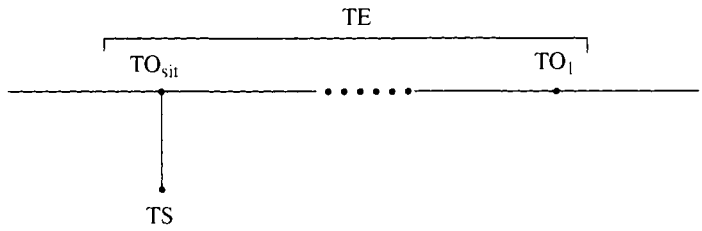


Figure 7.5

3 TO$_{sit}$ and TE are both durative. In that case there are several possible structures because there are different possibilities as to the relation between TO$_{sit}$ and TE, and as to the location of TE relative to TO$_1$:

a. TO$_{sit}$ may coincide with TE or be a proper subinterval of TE.
b. If TO$_{sit}$ is a proper subinterval of TE, TE may or may not include TO$_1$.

This gives us the following possibilities:[3]

a. TO$_{sit}$ = TE (e.g. *John was here all day yesterday*) (Fig. 7.6)

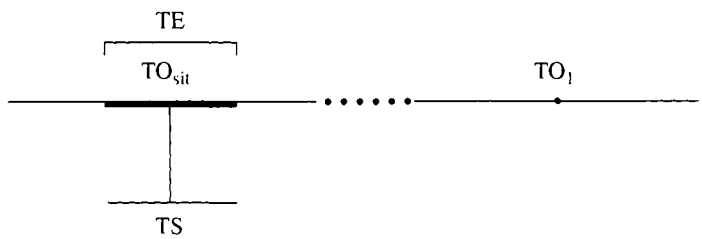


Figure 7.6

b. TE properly includes TO$_{sit}$ and TE properly includes TO$_1$. (e.g. *John was here today*) (Fig. 7.7)

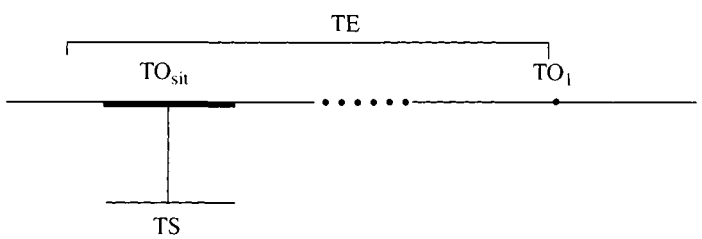


Figure 7.7

3. For the sake of simplicity I am only giving here examples in which TO$_1$ is t_0. However, examples in which TO$_1$ is a different time are available too. The following sentences illustrate this possibility for each of the three cases pointed out in the text:
   (i) Tomorrow he will say that John *was* here all day today.
   (ii) If he discovers this he will realize that John *was* in his house that very same day.
   (iii) I will tell him that John *was* here for some time today.

c. TE properly includes TO_{sit} and TE does not include TO_1 (e.g. *John was here for some time yesterday*) (Fig. 7.8).

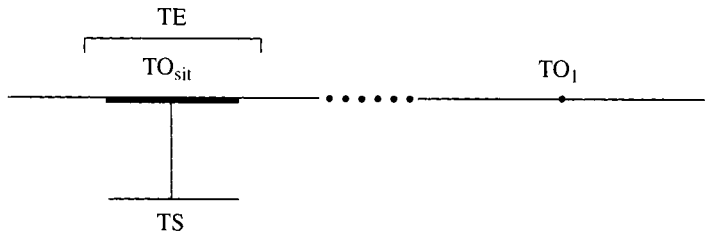


Figure 7.8

1.5 As we have seen, TS is that portion of the situation that is located in time by the sentence and may or may not be the time interval corresponding with the full actualized situation. If the situation is represented as bounded (as in *I wrote four letters this morning*) the time of the full situation coincides with TS. (In other words, there is no portion of the actualized situation that is not referred to (i.e. not located in time) by the sentence.) However, if the situation is unbounded, TS may be a proper subinterval of the time of the full situation. For instance, *At the time of the war John was in London* does not exclude the possibility that John was also in London before or/and after the war. Since a sentence never involves reference to any subintervals of the full situation besides TS, we do not need to provide for such subintervals in our schemata. However, if we wish to indicate explicitly that TS is only a proper subinterval of the full situation, we can make use of a notation in which the point or dash representing TS lies on a dotted line. In that case the dotted portions of the line represent the sections of the full situation that are not referred to in the sentence. This kind of notation may be useful to indicate the difference between progressive and nonprogressive sentences. For example:

1 Example: *John read a book yesterday.* (Fig. 7.9).

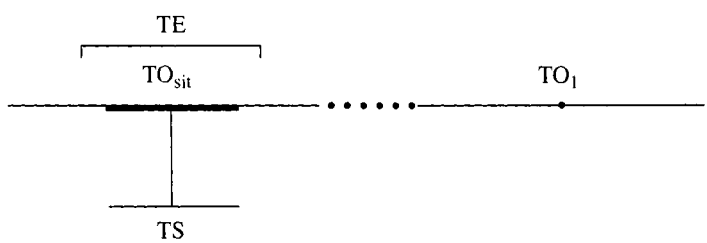


Figure 7.9

2 Example: *John was reading a book yesterday.* (Fig. 7.10).
3 Example: *At three o'clock John was reading a book.* (Fig. 7.11).

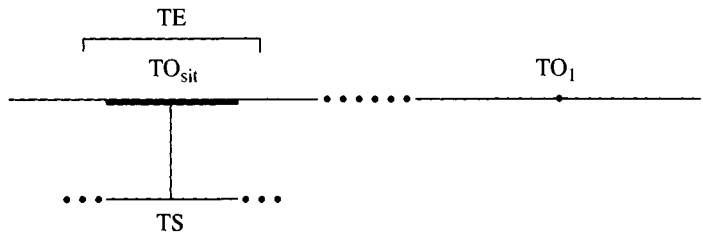


Figure 7.10

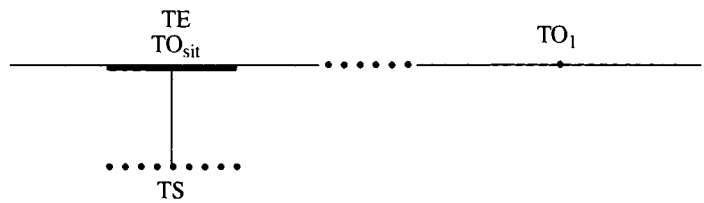


Figure 7.11

1.6 Another point worth noting is that, since TS need not be the full actualized situation and since our analysis does not state a direct relation between TS and TO_1, we have provided for the possibility (already referred to above) that the full situation continues up to TO_1. For example, *John was in the kitchen five minutes ago* does not exclude the possibility that John is still in the kitchen. On the other hand, the fact that TS is located as simultaneous with a time which lies wholly before TO_1 (rather than with TO_1 itself) will often suggest that the situation does not continue up to TO_1.[4] This is especially the case when it is the time adverbial that conveys the new information in the sentence. For example:

(4) When was John in London? – John was in London yesterday.

In this example *John was in London yesterday* implicates that John is no longer in London now, since the speaker would be violating Grice's Maxim of Quantity if this were not the case. That is, if John is still in London, the reply *John was in London yesterday* is not untrue, but it is deceiving because it is insufficiently informative: while the speaker asking the question expects specification of the full relevant timespan, the reply actually specifies only part of it.

On the other hand, there is no such clear implicature if the point of the question is the nature of the situation and not its location in time. For example, if the sentence *John was in London yesterday* is part of a

4. Comrie (1985:41) notes that 'there is often a conversational implicature' that the situation 'does not continue to or beyond the present'. We have already referred to this implicature in footnote 26 of chapter 6.

conversation about yesterday, it does not so clearly invite the hearer to infer that John's whereabouts are different today.

The above conclusions also hold for past tense sentences that do not contain an adverbial denoting past time. For example, Quirk *et al.* (1985:176) make the following remarks in connection with the sentence *Albatrosses were large birds*: (1) the speaker 'does not commit himself to the continuation of the past state of affairs it describes into the present'; (2) it follows that the sentence is a statement 'suggesting that the albatross, like the dodo, is extinct'; (3) nevertheless, the sentence 'does not exclude the possibility of such a continuation. It is possible to assert, without inconsistency: *Albatrosses were, are, and always will be large birds*'.[5] These three remarks accord perfectly with three claims that are inherent in the theory we have argued: (1) the preterit expresses no more than that the time of the situation that is talked of in the sentence (and which may or may not be part of a larger 'full' situation) coincides with some past TO_{sit}; (2) since there must be some point in using the preterit rather than the present tense (especially in a sentence like this, where the past TO_{sit} is not identified), the use of the preterit carries the conversational implicature that the situation does not continue into the present; (3) like any conversational implicature, this suggestion can be cancelled by a statement to the contrary.

1.7 According to many linguists, the preterit locates a situation at some 'definite' time in the past. (See e.g. Allen 1966:155–7; Leech 1969:143; Schopf 1985:60; 1987:198; Comrie 1985:41; Papp 1985:105.) Quirk *et al.* (1985:183–4) formulate the claim as follows:

> The speaker or writer must have in mind a definite time at which the event/state took place. . . . It is not necessary, however, for the past tense to be accompanied by an overt indicator of time. All that is required is that the speaker should be able to count on the hearer's assumption that he has a specific time in mind. In this respect, the past tense meaning of DEFINITE PAST time is an equivalent, in the verb phrase, of the definite article in the noun phrase. Just as with the definite article . . ., so with the verb phrase, an element of definite meaning may be recoverable from knowledge of (a) the immediate or local situation; (b) the larger situation or 'general knowledge'; (c) what has been said earlier in the same sentence or text; or (d) what comes later on in the same sentence or text. . . . Definiteness by virtue of the

5. It is a rather surprising that elsewhere in their book Quirk *et al.* (1985) deny the possibility that the situation can continue into the present: they claim that it is one of the 'features of meaning' of the past tense that 'the event/state must have taken place in the past, with a gap between its completion and the present moment' (p. 183). As we have seen, this claim holds for bounded situations only, not for unbounded ones, and cannot, therefore, be an inherent part of the meaning of the preterit.

larger situation (or general knowledge . . .) may be invoked to explain the use of the simple past in historical or biographical statements which have specific people, places, or objects as their topics:

(6) Byron *died* in Greece.
(7) I have a friend who *was* at school with Kissinger.
(8) This picture *was painted* by the owner's grandfather.
(9) Rome *was* not *built* in a day. (a proverb)

It is a matter of general knowledge that Byron is a historical personage (and therefore that he must have died at some time or another). The past tense in (6) presupposes such common ground between speaker and hearer: it is as if the speaker had said: 'We all know that Byron died at some time or other; well, *when* he died, he died in Greece'. Similarly, (7) presupposes that most people spend a particular period at school, (8) presupposes that a picture one sees on the wall must have been painted at some time or another; and (9) presupposes that all cities (including Rome, the 'Eternal City') have to be built at some time or other.

In my opinion, this parallel between the past tense and definite noun phrases is problematic. When the term 'definiteness' is applied to noun phrases, it is defined as follows: the speaker uses a definite NP as a signal to the hearer that the latter can somehow identify the referent, i.e. that the referent is recoverable from the situation of speaking, from the linguistic context, from the hearer's general knowledge of the world, etc. (see Hawkins 1978; Declerck 1987a, 1987b). Thus, while both *I have found a handkerchief* and *I have found the handkerchief* refer to the finding of a particular (specific) handkerchief, only the latter sentence implies that the hearer is expected to be able to identify the handkerchief in question.[6] If we define the notion 'definite time interval' in a similar way, we will have to say that a time interval is definite if it is identifiable to the hearer. It is clear, however, that the preterit does not necessarily imply reference to a time interval that is definite in this sense. The above mentioned example *Byron died in Greece* does not imply that the time of Byron's death is identifiable to the hearer. Similarly, it may be true that *This picture was painted by the owner's grandfather* 'presupposes that a picture one sees on the wall must have been painted at some time or another' (Quirk *et al.* 1985:184), but the preterit does not imply that the hearer is expected to be able to identify this time. Heny (1982:133–4) makes the same point when he states that 'there are in fact many . . .

6. This need not mean that the hearer is expected to be able to pick out the handkerchief from a set; it is sufficient that he is familiar with the description *the handkerchief* and knows that it applies to a unique object, even if he cannot pick it out from a set. (That is, *the handkerchief* presupposes identifiability in the sense that it is either 'referential' or 'attributive' in the sense of Donnellan (1966). In both cases *the handkerchief* implies the existence of a particular handkerchief, the ability to refer to which is shared by the speaker and hearer.)

sentences which are hard to reconcile with any account of the past tense which requires that it refer directly to some definite interval'. For example, we can say *Brutus killed Caesar* 'without any indication, presumed or otherwise, of when we take this to have happened'.

Another argument against the claim that 'the past tense refers to a DEFINITE time in the past' (Leech and Svartvik 1975:65) is that the past tense is not incompatible with indefinite adverbials of time:

(5) (a) It happened at some time before the war.
(b) All this was a long time ago.
(c) The expected disaster never occurred. (Penhalluriack 1981: 260)

Because of problems like these,[7] some people have given up the claim that the preterit refers to a definite time in favour of the (weaker) claim that it refers to a specific time. For example:

> If I say, 'Jackie was biting Molly', my statement refers to a specific time in the past.
>
> When we say that my use of *Jackie was biting Molly* refers to some particular past time, we do not suggest that I could accurately identify or describe that time, say by offering some definite description. ... Rather, it is just that the meaning of *Jackie was biting Molly* requires that I be speaking about some particular past time, one with which I am somehow connected. Thus if Jackie has bitten Molly on many occasions, the statement can only be about one of them. This contrasts with my saying, 'Jackie has bitten Molly,' which doesn't describe any particular past biting of Molly by Jackie. Since the past is determined by the time of discourse, the referential use of tense also exploits the discourse situation.
>
> (Barwise and Perry 1983:35)

This statement is not quite correct either. If there has been one instance in the past of Jackie biting Molly, then it is not impossible to use the present perfect to refer to this specific occasion (as in *Jackie has bitten Molly before, you know* or *Jackie has bitten Molly once, not twice*). Similarly, *I have just met your brother* refers to a meeting which took place at a specific time in the past. So, the claim that reference to a specific time in the past is implied by the preterit, but not by the present perfect, is not correct. Moreover, even the past tense does not always appear to refer to a specific time. This is clear from examples like the following (some of which have already been noted above):

(6) (a) We know that *John lived in Boston for some time and was quite a respectable citizen there.* But we do not know when that was, nor where he went afterwards.

7. See also McCoard (1978:76–87), Mommer (1986:108–10), Mugler (1988:26–30).

(b) Is Bill in the house? – No, *he went away.*
(c) What became of your sisters? – Oh, *Jane married a sailor, Sue bought a gold mine, and Marjorie joined the air-force.* (Heny 1982:134)
(d) 'I see,' Beesley said sniffing, *'I didn't know that before.'* (K. Amis, *Lucky Jim*)
(e) *She never was a great artist.* (Leech 1969:143)
(f) He is no longer the player *that he was.*
(g) A London Council *which agreed to put £100,000 in a local firm,* is to consider having its own 'enterprise board' to award loans to local industry. (Fenn 1987:165)
(h) What do you know about Abraham Lincoln? – *He was the first president of the USA.*

In these examples the preterit forms locate situations in the past, but they do not appear to locate them at specific time intervals. Of course, if a situation has held in the past, then it must have held at some specific time – this is true irrespective of whether we use a present perfect or a preterit to describe the situation – but it does not appear to be the case that the preterit necessarily implies that the speaker is thinking of, or referring to, this time.[8] Clearly, the above examples illustrate that there are cases in which the speaker need not have a definite or specific time in mind and in which it is therefore irrelevant for the hearer to try and identify such a time.

The above conclusion does not run counter to the observation that for a full interpretation of a past tense utterance the hearer often has to reconstruct a definite time. (This is clearly true of Partee's (1973:602) example *I didn't turn off the stove*, which is uttered by a speaker who is 'half-way down the turnpike'.) The point is simply that such utterances are meant as statements about a state of affairs that holds at a specific time and place, and that the hearer needs to reconstruct this time and place if he is to grasp the relevance of the remark. However, the fact that knowledge of time and place is necessary for a full semantico-pragmatic interpretation of an utterance in the past tense need not mean that reference to a definite time (and place?) is inherent in the semantics of the past tense. It just means that the hearer has to melt the semantics of the sentence with the pragmatics of its context. That is, he needs to scan the information that is pragmatically available to him in order to find a suitable time and place in relation to which he can interpret the sentence.

8. Fenn (1987:166) makes the same point:
> It is a clear and unproblematical rule in English that reference *expressis verbis* to a past time makes the past tense obligatory. There are many cases, however, in which the past tense is required although there is no such reference in the utterance, nor indeed any evidence that the speaker's attention is in any way focused on a particular time.

Besides, there are cases (illustrated by (6,a-h)) where even this is not required.

What, then, is the true semantic difference between the preterit and the present perfect? In my opinion, the answer to this question is simply that the preterit locates a situation in the past time-sphere whereas the present perfect locates it in the present time-sphere. Since the present time-sphere includes t_0, whereas the past time-sphere lies wholly before t_0, these different ways of locating a situation correspond with different ways of conceptualizing the situation. The choice of locating the situation in the one time-sphere rather than in the other is not an arbitrary choice. It depends on a variety of considerations:

1 A past situation that is located in the present time-sphere is located in the pre-present sector, i.e. in a sector which reaches up to t_0. This kind of location is therefore natural in the following cases:

a. when there is an adverbial which, apart from referring to the past, also involves direct or indirect reference to t_0, i.e. an adverbial which refers to a timespan that reaches up to t_0 (e.g. *since 1950, today*) or almost reaches up to it (e.g. *just, this minute, lately*).
b. when the past situation continues into the present (e.g. *I have been waiting for her for a long time*).
c. when the speaker wishes to represent the past situation as part of the structure of the world that holds at t_0. (This is the familiar notion of 'current relevance'.) For example, the speaker may be thinking of a present state of affairs that has resulted from the past situation (the so-called 'resultative perfect'). Or he may represent the past situation as something which is still part of the relevant person's present knowledge and experience (the so-called 'perfect of experience', as in *Have you ever tasted lobster?*). In fact, there are many ways in which the speaker may consider a past situation as forming part of the world structure holding at t_0. We can refer here to well-known examples like *Einstein has visited Princeton* (where the speaker is not concerned with what Einstein did but, for example, with the many famous people that have visited Princeton up to now, i.e. with the idea 'famous visitors of Princeton' which is part of the present structure of the world). Similarly, the use of the present perfect in *Shakespeare has written wonderful plays* shows that the speaker is not concerned with Shakespeare's deeds but rather with their effect on the present world structure.

2 Since the past time-sphere does not include t_0, the preterit will be the normal tense whenever the speaker wishes to refer to a past situation without referring directly or indirectly to t_0. The following cases can be distinguished:

a. The clearest case is when there is an adverbial referring to a (definite or indefinite) time interval that does not reach up to t_0 (e.g. *two/some weeks ago*). If there is such an adverbial, the situation is explicitly located in the past time-sphere. The reason is that, if the speaker concentrates on a time interval which lies wholly before t_0, he cannot simultaneously concentrate on the structure of the world that holds at t_0.[9]

b. The past time interval that the speaker is concentrating on need not be indicated in the past tense clause itself. It may also be established by the context, or the situation of speaking. In this connection we must mention the well-known fact that the preterit is often used when we go on talking about a situation which has been introduced by means of a present perfect sentence: *I've seen John today. I met him in the park.* This shift from a present perfect to a preterit is usually explained as a shift from an indefinite tense form to a definite one – a shift which is similar to the shift from an indefinite NP to a definite one in sequences like *There was a man in the park. . . .The man told me that . . .*[10] In my opinion, a few remarks should be made here. First, the parallelism with NPs is only superficial, as the switch from the present perfect to the preterit is not always obligatory (e.g. *I have already met John and I have told him the news*), whereas the shift is obligatory where noun phrases are concerned: once we have spoken of *a man* we cannot go on using this indefinite NP to refer to him. Secondly, granted that there is a shift from indefinite time to definite time in examples like the above one (*I've seen John today. I met him in the park*), this does not warrant the generalization that definite time reference is inherent in the preterit. (In fact, we have already pointed out that this generalization is not correct.) The reason why there is a shift from the present perfect to the preterit in such examples is not so much that there is a shift from indefinite to definite time reference, but that there is a shift in what the speaker is concerned with. In *I have seen John today* the speaker concentrates on the fact that a situation has occurred which is relevant to the structure of the world at t_0. In *I met him in the park* he concentrates on the situation itself (more specifically, on the question of where it took place) and therefore locates the situation in the past time-sphere. This is not to

9. The adverb *just* refers to an interval which does not include t_0 but lies very close to it. For American English speakers the fact that t_0 is not included tends to be decisive: in clauses with *just* they normally use the preterit. British English speakers, in contrast, prefer the present perfect. They apparently rely on the consideration that what has just happened is likely to affect the present world structure. (The same observation can be made in connection with *lately*, *recently*, etc.)

10. See Barense (1980:46): 'Definite articles are commonly used to refer back to indefinite nominals which establish the frame of reference. In a parallel way, past tense constructions commonly presuppose a framework of time reference already established by a preceding perfect tense construction.'

say, however, that he could not have gone on thinking of the structure of the world that holds at t_0. In a sequence like *I have gone to a priest and I have told him about my troubles* this is exactly what the speaker is doing, and he therefore continues to use the present perfect in spite of the fact that there is again a shift from indefinite to definite reference.

So, what is crucial to the use of the tenses in such examples is whether the speaker wishes to locate the situation in the present time-sphere or in the past one. The question of whether the reference (including the temporal reference) is definite or not is only relevant in so far as the presence of an overt indication of definite past time is incompatible with the location of a situation in the present time-sphere. (As said before, the speaker cannot be concerned with the structure of the world at t_0 if he concentrates on the time of a past situation.)

The above remarks are in keeping with the descriptive framework which was developed in chapter 2. When the speaker has used an indefinite perfect to locate a situation in the pre-present sector, he can go on talking about the past in one of the following ways. First, he can incorporate the next situation he describes into the domain established by the present perfect. As noted in chapter 2, he will use a past time-sphere tense to do so (e.g. *He has often said that he* ***was*** *unhappy*). This switch to the past time-sphere is logical, since the situation is not seen as related to t_0 but as related to a TO_{sit} which lies before t_0. Second, the speaker can relate the situation to t_0 by locating it in the pre-present time-sphere (e.g. *I have seen him and* ***have told*** *him the news*). In that case there is a shift of domain (from one pre-present domain to another). Third, the speaker may wish to represent the situation as unrelated to t_0 and as not temporally dependent on the present perfect situation. In that case he will have the new situation establish its own domain in the past time-sphere (e.g. *I have met John today and he* ***told*** *me the news*).

c. When there is no adverbial or context referring to some (definite or indefinite) past time, the speaker is free to locate the situation in either of the time-spheres. His choice simply depends on whether or not he is concerned with the structure of the world at t_0. In some cases both tenses are therefore possible:

(7) (a) What became of your sisters? – Oh, Jane *married* a sailor, Sue *bought* a gold mine, and Marjorie *joined* the air-force. (Heny 1982:134)

(b) What has become of your sisters? – Oh, Jane *has married* a sailor, Sue *has bought* a gold mine, and Marjorie *has joined* the air-force.

Unlike (7,a), (7,b) is concerned with the structure of the world at t_0. It therefore suggests that Jane is presently married to a sailor, that Sue is the owner of a gold mine, and that Marjorie is serving in the air-force. The sentences of (7,a) do not suggest this, because they locate the situations in the past time-sphere.

In many cases the pragmatics of the sentence are such that only one of the two tenses is suitable. Thus, we say *Byron died in Greece* and not **Byron has died in Greece* because the fact that the speaker is concerned with the question where Byron died entails that he is not concerned with the present world structure. But we may say *Byron has written several interesting books* because the books in question form part of the present world structure and this is what the speaker is now concerned with. In a sentence like *She's not so active as she was* we cannot escape using the preterit because an explicit contrast is established between the present world structure and a past one. (Note that there is no reference to definite time in the subclause. In this sentence the reason for using the preterit (i.e. for locating the situation in the past time-sphere) is that the reference is to a world structure which is no longer holding at t_0. The question of (in)definite time reference is irrelevant to this.)

In sum, the following claims can be made about the preterit:

1 The essential difference between the present perfect and the past tense is not that they refer to indefinite and definite past time, respectively. The difference is that they locate situations in different time-spheres.

2 When the speaker locates the situation in the pre-present sector, this means that he is concerned with the structure of the world at t_0 rather than with some aspect (when?, where?, how?, why?, etc.) of the situation itself. This means that the present perfect will be used (a) if the situation still continues at t_0, (b) if the speaker is concerned with the fact that the situation has held and has thus become part of the structure of the world rather than with another aspect of the situation, and (c) if there is some other reason for the speaker to view the past situation as part of the world structure at t_0. (In none of these cases is the speaker concerned with the past time when the situation held. Reference to this time is therefore incompatible with the present perfect.)

3 The speaker will use the past tense if he does not want to represent the situation as part of the structure of the world at t_0.

a. One typical case is if the speaker wishes to establish a contrast between the situation and the present structure of the world, as in (8,a–b):

(8) (a) He is not so active as he *was*.
(b) I *didn't know* you were a teacher.

The latter sentence means something like 'I didn't know you were a teacher until you told me so (or: until I found out) just now' (see Erades 1975:2, Fenn 1987:194). A contrast is thus established between the past and the present. The same thing happens in sentences like the following:

(9) (a) Charlie, you are a good-looking boy. I *had* no idea you were so handsome. (Fenn 1987:186)
(b) I never *thought* I'd see the day when you'd be in custody. (ibid.)

Incidentally, it may be noted that the reference to past time is completely indefinite here.

b. The speaker is not thinking of t_0 either if he focusses exclusively on the past situation. Two possibilities can be distinguished here:
i. Sometimes the speaker is only concerned with one of the aspects of the past situation (e.g. with the question of when, why, where, how, etc. it took place, or who was involved in the situation).[11] For example:

(10) (a) I can't remember where I *bought* that vase.
(b) It *was* you who *phoned* my wife this morning, *wasn't* it? (Fenn 1987:173)
(c) Just tell me how you *did* it.

In cases like these the speaker focusses on the past situation and not on the structure of the world at t_0. This is also the case in sentences which provide further details concerning a previously mentioned situation or which explain its past origin:

(11) (a) Why do you say that? – I have been making enquiries. It *was* not difficult. The whole island community is in an uproar. (Fenn 1987:177)
(b) He's not with us any more. – You mean he *resigned*? – No, he *was thrown* down an elevator shaft in Goodge Street. (Fenn 1987:175)

ii. Sometimes the speaker just wants to inform the hearer of the fact that the situation took place, but without representing it as part of the structure of the world at t_0. For example:

(12) (a) You know, I nearly *did take* you into my confidence. (Fenn 1987:195)
(b) The expectation of life of a heroin addict is about 2 years. A doctor friend of mine *was telling* me. (ibid.)

11. Fenn (1987:168) calls this 'occurrence focus'.

(c) Much more recently, the same woman, still married to the same husband, *saw* Sean Connery again – this time he was arriving at Pinewood to make his first film in Britain for 6 years. (ibid.)

(d) Is John here? – No, he *went* away.

In such examples too, the time of the past situation remains completely indefinite.

4 When we make a statement about a particular aspect of a past situation, as in (10,a–c) and (11,a–b), we must be sure that the situation in question is identifiable to the hearer. That is, the hearer must dispose of the essential information that makes it possible for him to understand the speaker's reference to the situation. This information may be available to him in several ways:

a. The hearer will of course know which situation is being talked about if it has already been mentioned. This is the case, for example, in *I have tasted lobster once, but I* ***did****n't like it.* We notice a familiar pattern here: the present perfect is used to introduce the situation (i.e. to say that it has held before t_0), the preterit is used when we subsequently focus on another aspect of the situation.

b. When a situation that is mentioned for the first time is located in the past time-sphere, it can be identifiable to the hearer in one of the following ways:

   i. In some cases the situation is uniquely identifiable for the simple reason that it is unique: there is no other situation that can be described in the same terms. Sentences like *Byron died in Greece* or *The Hindenburg exploded* refer to situations of this type. For a full interpretation of such sentences it is required that the hearer be familiar with the referents of the relevant NPs. It is not necessary that the time of the situation be known to him.

   ii. The situation is also identifiable if it is given in the extralinguistic context (i.e. in the immediate situation of speaking). For example, 'if I see you with a broken arm, then I can ask *how did you break your arm?* with the non-Perfect form, since I am taking as already given the fact that you broke your arm, and am eliciting further information about the incident' (Comrie 1976:55). (In this case too, the precise time location of the past situation is irrelevant.)

   iii. When the situation that is referred to for the first time is one that may in principle have held at different times (e.g. *John locked the door*), the hearer can know which particular instance of the situation is being referred to only if he knows on which occasion (i.e. at which time) the situation in question held. (This is why the sentence *John locked the door*, when uttered *in vacuo*, automati-

cally begs the question 'when?'.)[12] In this case identifiability thus depends on information concerning the time of the past situation. This information may be given by a (definite or indefinite) time adverbial in the speaker's sentence itself (e.g. *three days ago, a long time ago*). If there is no such adverbial, the necessary temporal information must be available from the linguistic or extralinguistic context in which the sentence is used or from the knowledge that is shared by the speaker and the hearer (as in *I saw your wife at the races*).[13]

It is clear from these principles that the following two claims, which are very often made in connection with the preterit, are incorrect generalizations:[14]

1 It is often claimed that 'a simple Past tense sentence would remain uninterpretable if a specific reference time were not supplied either co-textually or contextually' (Schopf 1987:186). This claim is correct only for the cases mentioned in 4.b.iii. It is incorrect as regards all other types of past tense sentences (e.g. *He is not so active as he was, Byron died in Greece,* etc.)
2 The claim that it is inherent in the past tense that the reference is to a definite time in the past is *a fortiori* incorrect. The preterit refers to a definite (i.e. identifiable) time only in those cases when identifying the time of the situation is necessary to know which situation is being referred to.[15]

I will return to the question of how past situations are located in time when dealing with the present perfect (section 3).

12. For the same reason, a question like *Did John lock the door?* cannot be answered with *Yes* or *No* if the time of the situation is unknown (see Bach 1980:31), unless the question is understood as equivalent to *Has John locked the door?* (which is possible in American English). Richards' (1987:357) observation that *Mandy did not phone* is not equivalent to *Mandy never phoned* relates to the same phenomenon: unlike the latter sentence, *Mandy did not phone* asserts that Mandy did not phone at some specific time in the past.
13. There is, of course, a similarity here between the use of the preterit and the use of a definite NP. If used in isolation, a sentence like *Did you put the cat out?* presupposes some shared knowledge. To fully understand this sentence, the hearer must check his memory stock, not only for a suitable referent for the NP *the cat* but also for a suitable occasion (time) for the situation to have taken place.
14. Fenn (1987:198) reaches the same conclusion.
15. According to Schopf (1987), the past tense has the function of a 'search instruction'. It tells the hearer to 'look out for a co-textually or contextually given specific time in the past ... and relate the untensed or temporally unspecified proposition to it as simultaneous' (p. 186). This formulation is reminiscent of the way in which Hawkins (1978) defines definite reference: by using a definite NP the speaker tells the hearer to find a 'shared set' of objects and to 'locate' the referent in it. Schopf's analysis of the past tense is therefore nothing but a reformulation of the familiar theory that the past tense refers to a definite (i.e. identifiable) past time.

2 THE PRESENT TENSE

According to Comrie (1985), the structure of the present tense involves only one relation, viz. 'E simul S'. However, in chapter 6 we have formulated the general principle that every tense represents TS as coinciding with a TO_{sit} which is (directly or indirectly) related to t_0. It would seem, then, that according to this principle we must represent the structure of the present tense as 'TS simul TO_{sit} simul t_0'. However, this representation appears to be unacceptable. Since we have defined 'simul' in terms of coincidence, and since t_0 is by definition punctual,[16] the representation 'TS simul TO_{sit} simul t_0' would entail that the TS referred to in a sentence in the present tense would necessarily be punctual. This, of course, runs counter to our intuition, which tells us that sentences like *John is in London today*, *I am not feeling well these days* and *The Eiffel Tower stands in Paris* are statements about situations that occupy more than a single point on the time line. (In other words, *John is in London today* states a temporal relationship between John's being in London and today, not between John's being in London and t_0.)

One might try to obviate this problem by pointing out that the situations referred to in the above two sentences are states, that states are by definition unbounded, that an unbounded sentence refers to a TS which is only a portion of the full actualized situation, that consequently a sentence like *The Eiffel Tower stands in Paris* (even though it only refers to a punctual subinterval of the full situation) implies the existence of a longer full situation, and that this explains why the sentence tends to be interpreted as being a statement about a durative situation. However, this explanation fails when the sentence involves a bounded durative time adverbial. Consider:

(13) Today John is in London from eight o'clock this morning until seven tonight.

According to the above theory, this sentence states that the situation of John being in London holds at t_0 and suggests (because being in London is an unbounded situation) that John is actually also in London for some time before and after t_0. However, this theory is deficient, because this is not what the sentence means. The sentence clearly states that John's being in London is simultaneous with the time interval indicated by the bounded time adverbial *today from eight o'clock this morning until seven tonight.* That is, TS is represented as durative rather than punctual.

This conclusion is confirmed by sentences like the following:

16. Salkie (1989:15) suggests that 'in English, the present moment is regarded as a point in time without duration, while in French and German the present is regarded as an interval of time'. This would account for the fact that in French and German, but not in English, the simple (nonprogressive) present can be used to describe an ongoing nonstative situation. It would also explain the existence of the progressive in English and its absence in French and German.

(14) I am looking after the children while my wife is away.

The TE established by the *while*-clause is durative and is represented as simultaneous with TS. TS must therefore also be durative.

How can we accommodate these observations? One of the principles formulated in chapter 6 is that, although TO_{sit} may generally either coincide with TE or be a proper subinterval of TE, TO_{sit} always coincides with TE if the latter is established by a bounded time adverbial. Since TS coincides with TO_{sit}, this means that a bounded TE ultimately specifies the length of TS. (For example, *John was here from two to four* states a relationship of simultaneity between the situation of John being here and the timespan indicated by the time adverbial.) If we apply this principle (according to which TE = TO_{sit} = TS if TE is bounded) to sentence (13), the conclusion is that, since TE is bounded and durative, TS must be durative too. This conclusion concurs exactly with the interpretation we have assigned to the sentence. A similar conclusion can be reached in connection with (14): because of the fact that *while* inherently indicates both duration and simultaneity, not only TE but also TO_{sit} and TS must be durative.

It follows that the temporal structure of the present tense must be such that it involves the relation 'TS simul TO_{sit}', but not the relation 'TO_{sit} simul t_0'. Since t_0 is punctual, while TO_{sit} may be durative, the relation must be one of inclusion: TO_{sit} includes t_0 in the sense that t_0 either coincides with TO_{sit} (if the latter is punctual too) or is properly contained in TO_{sit}. The latter possibility is illustrated by (13) and (14); an example of the former is:

(15) I hear a shot!

There is a further piece of evidence corroborating this conclusion. As has already been pointed out, states (including habits) that are conceived as permanent rather than temporary cannot be located at a punctual time interval. Sentences like the following are therefore unacceptable, except in a context where it makes sense to represent the situations in question as temporary:

(16) (a) !At five o'clock John knew English.
(b) !Whales were mammals when I moored the boat.
(c) !My father has ten toes right now.

On the other hand, sentences like *John knows English* or *My father has ten toes* are impeccable. This suggests that such sentences do not locate the situation at t_0, but at a time that is much longer than t_0. In other words, the relation between TO_{sit} (the time at which the situation is located) and t_0 must be that of inclusion rather than coincidence. In fact, since there is no time adverbial specifying a TE including TO_{sit}, and since there is no duration adverbial restricting the duration of TS (and

hence of TO_{sit}), TO_{sit} will be interpreted maximally, i.e. as commensurate with the time of the full situation.

The above arguments suffice to conclude that the present tense (in its absolute use) expresses two relations, viz. 'TS = TO_{sit}' and 'TO_{sit} includes t_0' (where inclusion, as usual, means either coincidence or proper inclusion). Figure 7.12 and Figure 7.13, which represent the structures of (13) and (15), respectively, illustrate this.

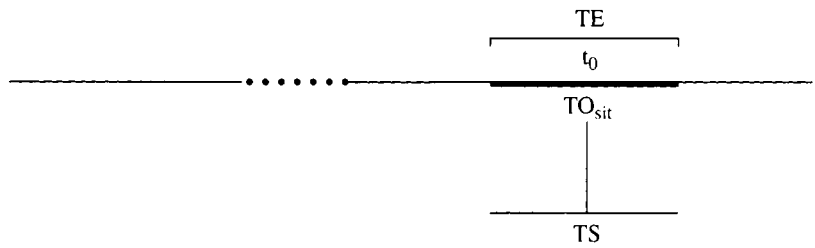


Figure 7.12

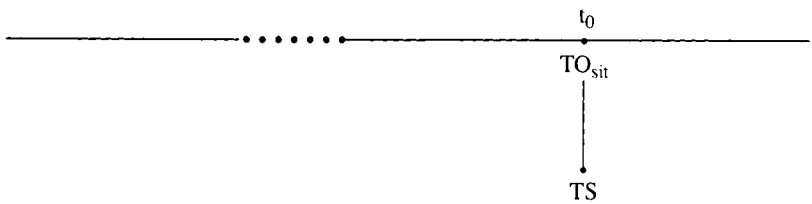


Figure 7.13

It should be noted that Figure 7.12 represents TO_{sit} (and hence TS) as commensurate with TE. Sentence (13) is, indeed, interpreted as meaning that the situation of John being in London holds for the full timespan indicated by the time adverbial. As noted above, this follows naturally from the fact that the time adverbial is a bounded one: in that case TO_{sit} (and hence TS) is always commensurate with TE. When the time adverbial is not a bounded one (as in *John is in London today*) the interpretation on which TE (today) and TS coincide is also predominant, but it is in fact not the only possible one. This is clear from sentences like the following:

(17) Mist and clouds usually render it impossible to see the sun rise from the sea. But *today we can see it* because the weather conditions are quite exceptional.

In this example, TS (the time of seeing the sunrise) coincides with a TO_{sit} which is not commensurate with TE (today) but is a proper subinterval of TE. The conclusion is that the relationship between TO_{sit} and TE in the present tense is the same as in the other tenses. That is, TE includes TO_{sit}, either in the sense that TO_{sit} is a proper subinterval of TE or in

the sense that TO_{sit} is commensurate with TE. However, there seems to be a conversational implicature that the latter interpretation prevails if it is not excluded (or rendered improbable) by the context or by pragmatic considerations. Thus, the normal interpretation of *John is in London these days* is that he is in London throughout the period indicated by *these days*.

Figure 7.13 represents the structure not only of sentences like *I hear a shot* (where the situation is punctual in the strict sense of the word) but also of sentences whose situation is conceptualized as beginning and ending at the time of speech. This kind of conceptualization is typically found in performative sentences (e.g. *I wish you a merry Christmas*), commentaries, demonstrations, exclamatory sentences of the type *Here they come*, etc.

Apart from the temporal schemata represented by Figures 7.12–7.13, sentences in the present tense can also have the structures represented by Figures 7.14–7.15. Figure 7.14 represents the temporal structure of a

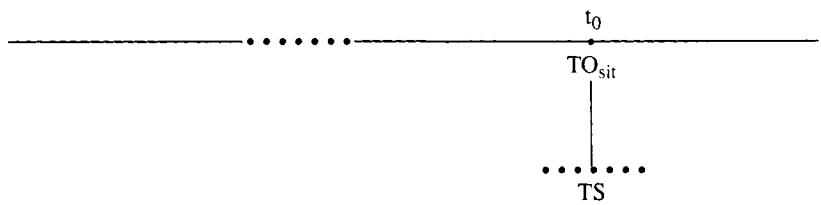


Figure 7.14

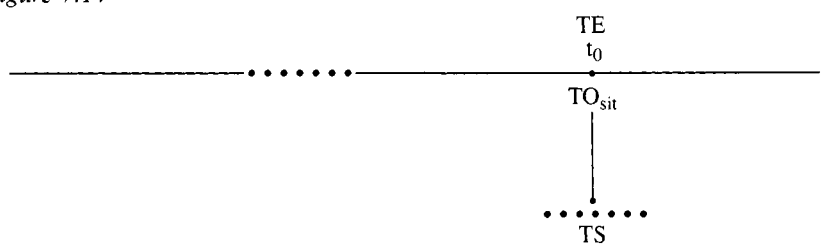


Figure 7.15

sentence like *John is sleeping*, in which there is no TE established by an adverbial and the situation is unbounded. Figure 7.15 represents the structure of an unbounded sentence containing a punctual time adverbial referring to t_0, e.g. *John is sleeping at this moment.*

In chapter 6 it was pointed out as a problem for the theory argued in Declerck (1986a) (more specifically, for the definition of 'simul' proposed there) that we cannot truthfully say *John is here today* if John is not actually here at the moment of speaking. That is, the sentence is not true if John's being here is simultaneous with some part of TE which does not include t_0. If John was here from 8 a.m. till noon, I can report this at 2 p.m. by means of *John has been here today*, but not by means of *John is*

here today. This presents a problem if one defines 'simul' in terms of either coincidence or partial overlapping, for 'TS simul TO_{sit} simul t_0' then does not guarantee that t_0 is included in TS (since TS can be simultaneous with a subinterval of TO_{sit} that does not include t_0). The theory I am now arguing does not encounter this problem, as it states explicitly that t_0 is always included in TO_{sit}, and hence in TS (which is commensurate with TO_{sit}).

As noted in chapter 6, gnomic sentences (i.e. generic sentences, including the 'habitual' ones of the type *Mary works in a factory*, and omnitemporal sentences like *Two and two is four*) present no problem for the analysis of the present tense: such sentences represent world structure states, and will therefore be analysed in the same way as non-gnomic sentences representing states, such as *John is in the house*. The only differences between the two are (1) that a gnomic state is generally much longer, even indefinitely long when it is omnitemporal, and (2) that the state expressed in a gnomic sentence that implies repetition (e.g. *My dog chases cars*) involves a multiplicity of dynamic and possibly noncontiguous subsituations. However, neither of these differences alters the fact that a gnomic situation is conceived as a state, i.e. is an unbounded (hence homogeneous and dissective) situation which does not imply change. Gnomic situations can therefore be analysed in terms of the same type of temporal schema as non-gnomic states.

In chapter 6 we also noted that sentences which are repetitive without being gnomic (e.g. *John is kicking the ball*) are analysed in exactly the same way as nonrepetitive non-gnomic situations. The only difference is that the situation now consists of a number of subsituations. This means that TS (the time of the situation as a whole) comprises the TSs of several subsituations, and that the over-all TO_{sit} comprises the several TO_{sit}s that are simultaneous with these sub-TSs. What is worth noting in this connection is that, when the sentence is progressive, the sub-TSs may be the times of either the full situations or of the middles of these situations. The former possibility is realized in *John is kicking the ball*, where each subsituation is one complete occurrence of the full situation of John kicking the ball. The latter possibility is realized in the head clause of a sentence like the following:

(18) John is always sleeping when I come home.

Here the subsituations in the overall situation are not represented as full (complete) situations but as subsituations in progress. The reference is each time to the middle of the subsituation: it is each time a MidS which is simultaneous with one of the sub-TO_{sit}s.

It is clear from this that in a repetitive sentence the structure of the present tense is, as it were, reduplicated: each sub-TS is simultaneous with a sub-TO_{sit}, and all sub-TSs together form a TS which is simultaneous with a TO_{sit} that is the sum-total of all the sub-TO_{sit}s.

Figure 7.16, which represents the structure of the present tense in the head clause of (18), illustrates this.

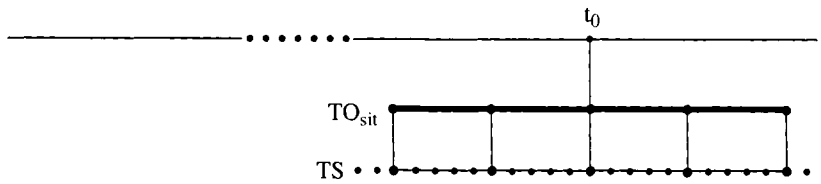


Figure 7.16

The description of the present tense which has been given here involves the relation 'TO_{sit} includes t_0'. This formulation self-evidently only applies to the present tense in its absolute use. If it is used as a relative tense, as in *I will be sad when you are away*, the relevant relation is 'TO_{sit} includes TO_1', where TO_1 is a time different from t_0. For the rest, the structure of the present tense remains the same. (More specifically, the relation 'TS simul TO_{sit}' is unaffected by the question whether the present tense is used as absolute or relative tense.) However, the fact that TO_1 is not t_0 entails that TO_1 may be durative (whereas t_0 is by definition punctual). This is, for example, the case in the *that*-clause of (19), whose structure is represented by Figure 7.17.

(19) It will have been clear for days that John is the culprit.

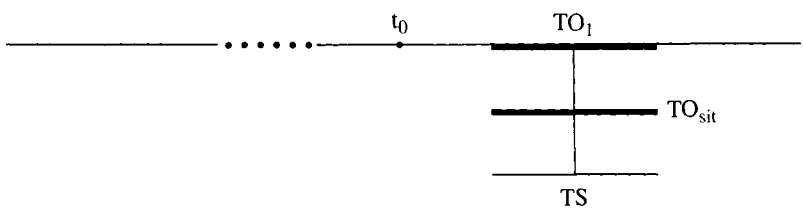


Figure 7.17

A final thing to be noted in connection with the present tense is that it cannot normally be used to refer to a durative bounded situation whose TO_{sit} includes t_0. This is clear from examples like the following:

(20) (a) John is in London.
(b) John is writing a letter.
(c) I declare this meeting closed.
(d) John writes a letter.

The first two sentences both represent their situation as unbounded and as including t_0. Sentence (20,c) represents its situation as punctual and as coinciding with t_0. However, sentence (20,d) requires a special kind of interpretation. It is acceptable only in dramatic speech (e.g. as a stage direction or as a historical present). Otherwise it cannot be used to

represent the writing of the letter as including t_0. This follows from the fact that the nonprogressive verb phrase *writes a letter* represents the situation as bounded. Apparently it is impossible to represent a situation at the same time as bounded and as including t_0. I suggest that the reason is the following. There are only three logical possibilities for a durative situation to include t_0: either the situation begins at t_0, or it is in progress at t_0, or it terminates at t_0. The first two possibilities are incompatible with the use of the simple present tense because they entail that the terminal point of the situation follows t_0. If this is the case, the present tense can only represent the situation as still in progress at t_0. It must therefore be a progressive present tense, which represents the situation as unbounded. The third possibility (viz. that t_0 coincides with the terminal point of the situation) is also incompatible with the present tense, because in that case the time of the situation constitutes a TE-up-to-t_0, which necessitates the use of the present perfect.[17]

3 THE PRESENT PERFECT

As pointed out above, Comrie (1981a, 1985) discards the present perfect from his treatment of the tenses because, in his opinion, the present perfect does not differ from the past tense in terms of time location: both tenses locate a situation as prior to TU. The difference between the two is claimed to be one of aspect only: the present perfect implies 'current relevance', the past tense does not. This position is roughly the same as that defended by traditional grammarians like Jespersen (1924:269) and Poutsma (1926:209), who also disregard the present perfect in the temporal schemata they propose for the tenses. In fact, there is quite a tradition of linguists taking this view. Even McCoard (1978:17), who explicitly rejects the 'current relevance' theory, adopts the same position as far as the meaning of the perfect as a tense is concerned: 'We shall claim in this work that the preterit and the present perfect are not, in fact, distinct in terms of tense or sequence: their distinctiveness lies elsewhere'.

Reichenbach (1947) takes a different view. In his opinion, the temporal schema realized by the present perfect is different from that of the preterit. The difference is that the present perfect involves a 'reference point' that coincides with the 'speech point' (E–R,S), whereas the past

17. Smith (1976b:574–5) suggests a different explanation. According to her, the reason why sentences like *Stanley builds a house* are ungrammatical (except on a dramatic interpretation) is that such sentences are perfective: 'Perfective aspect implies completion and, therefore, more than one point in time; Present sentences may not specify more than one point in time.' This explanation (if I interpret it correctly) hinges on the claim that the present tense requires that TO_{sit} should not be a longer time interval than t_0 – a claim which I have rejected at the beginning of this section.

tense involves a reference point that coincides with the past 'event point' (E,R–S).

In this book I am defending a position similar to Reichenbach's. I will argue that the basic difference between the two tenses lies in their temporal schemata. The crucial distinction is that the past tense locates a situation in the past time-sphere, whereas the present perfect locates it in the present time-sphere. The various differences of use and interpretation that can be observed between the two tenses naturally follow from this. There is no need to assume the existence of such a linguistic category as 'perfect aspect'.

The reasons why I do not subscribe to the view that the present perfect and the preterit do not differ in terms of time location are the following:[18]

1 There are time adverbials that collocate exclusively with one of the two tenses. For example, *yesterday* requires the past tense, *since yesterday* can only be used with the present perfect. This suggests that the two tenses differ in the way they locate situations in time.[19]

2 If the only difference between the two tenses were one of aspect (i.e. the presence or absence of 'current relevance' or some similar notion), then the following claims would have to be true:

Claim A: It is an (a) sufficient and (b) necessary condition for the use of the present perfect that the speaker wishes to represent a past situation as having current relevance.

Claim B: If the idea of current relevance is to be expressed, the speaker cannot use the preterit. (This follows from claim A,a.)

However, both of these claims are wrong, as appears from the following sentences:

(21) A: How do you know that Mr Benson has a difficult character?
B: I know what Mr Benson is like because *I worked for him a couple of years ago.*

(22) A: Have you always lived here?
B: No, *I have lived in London too,* but that was a long time ago.

The italicized sentence in (21,B) is in the preterit, in spite of the fact that the past situation is explicitly represented as having current relevance. This vitiates claim (A,a) and claim (B). Claim (A,b) is refuted by the fact that in (22,B) the use of the present perfect is not accompanied by any suggestion of current relevance.[20]

18. Some of these arguments have already been given in chapter 5. These are only briefly repeated here.
19. I admit that this by itself need not be a decisive argument. One could argue that the adverbials in question are sensitive to the aspectual distinction perfect/imperfect.
20. The observation that not all present perfect sentences imply current relevance is not new. See e.g. Diver (1963), Penhalluriack (1981:45–51).

3 Unlike the preterit, the present perfect can represent a situation as continuing from the past into the present (e.g. *I have been living here for three years*). Note that an unbounded sentence in the preterit does not exclude that the full situation continues into the present (see above). However, in examples like the above one the perfect (together with the adverbial) does more than this: whereas *John was waiting in the kitchen for half an hour* leaves vague whether or not John is still waiting in the kitchen, *John has been waiting in the kitchen for half an hour* asserts that John is still waiting in the kitchen. This is a basic difference, which must have to do with the ways the two tenses locate situations in time.

4 In section 22 of chapter 2 we saw that the past tense can be used to represent a situation as it was experienced or perceived in the past by some participant in the situation. The present perfect can never be used in this way. That is, if the speaker shifts the temporal focus from the present to the past, he uses the preterit, not the present perfect. This is clear from examples like the following:

(23) (a) I saw a film in which a young boy *insults/insulted/*has insulted* the Queen.
(b) I have seen a film in which a young boy *insults/*insulted/*has insulted* the Queen.

In (23,a) the speaker can use either *insults* or *insulted*. This means that he may or may not shift the temporal focus to the time when he saw the film. The present perfect *has insulted* cannot be used to effect this shift.[21] In (23,b) only *insults* can be used. The present perfect in the head clause does not allow a shift of temporal focus to the past time of seeing the film.

This incompatibility between the present perfect and a past temporal focus also appears from the way we interpret sentences with *now*:

(24) (a) Now he had finally managed to be completely independent.
(b) Now he has finally managed to be completely independent.

It should also be pointed out that some linguists who do claim that current relevance is the basic feature of meaning of the present perfect broaden the sense of 'current relevance' to such an extent that the notion applies to virtually any sentence. Thus, Palmer (1974:50) defines current relevance as meaning 'that in some way or other (not necessarily in its results) the action is relevant to something observable at the present'. Penhalluriack (1981:51) rightly points out that 'the extremely vague sense of the expression that we are reduced to is objectionable on the grounds that the idea that "the event is not irrelevant to the moment of speaking" is applicable to virtually any verb form. For it is fundamental to communication that one does not, as a rule, introduce irrelevant material into one's conversation. To do so would be to baffle one's hearer.' (see Grice's (1975:45) Maxim of Relation (Relevance), which stipulates that a co-operative speaker must 'be relevant').

21 *Has insulted* is, of course, grammatical if it is meant to express that the situation has already taken place at the point where the film story begins. This reading can be disregarded here.

It was noted in chapter 2 that in sentences like these *now* refers to the time that receives the temporal focus. In (24,a) this time belongs to the past time-sphere. *Now* is therefore paraphrasable as 'now it was the case that . . .'. In (24,b), in contrast, the temporal focus is not on a past time interval, but on the present. This is clear from the fact that *now* is now equivalent to 'now it is the case that . . .' and not to 'now it has been/was the case that . . .'. All this shows that the present perfect does not locate a past situation in time in the same way as the preterit does.

5 As we saw in section 15 of chapter 2, only past time-sphere tenses can be used to incorporate a situation into a past domain. The present perfect cannot be used for this purpose.

6 The preterit can be used as a relative tense expressing simultaneity. The present perfect has no such use.

7 In sentences like *Tom had left when I arrived* the past tense form (*arrived*) indicates a TO which plays a role in the structure of the past perfect: it indicates TO_2 in the structure 'TO_{sit} before TO_2 before TO_1' (where TO_1 is t_0 in the above example). This TO_2 cannot be established by the present perfect: **Tom had left when I have arrived.*

8 As a rule, a stretch of discourse in the historical present uses the present perfect, not the preterit, to refer to a situation that is prior to one described in the present tense:

(25) All of a sudden I get a letter that I'*ve been* accepted. (Wolfson 1982:5)

Historical present uses the present and the present perfect where normal narrative uses the preterit and the pluperfect, respectively. This suggests that the present perfect and the past tense belong to two different systems, i.e. differ in the way they locate situations in time. (The crucial difference, of course, is that the present perfect is a present time-sphere tense whereas the preterit is a past time-sphere tense.)

9 The past tense can be used to express 'future in the past' (e.g. *He was leaving the next day*). The present perfect cannot be used in this way.

The above arguments, I think, suffice to say that the present perfect and the preterit do not realize the same temporal schema. As I have already dealt with the temporal structure of the past tense, I will now proceed to examine the structure of the present perfect.

Let us begin by investigating the meaning of the present perfect in sentences that do not involve time adverbials:

(26) (a) John has lived in Paris for some time.
(b) John has lived in Paris.
(c) John has been living in Paris.

Sentence (26,a) is ambiguous between two readings. The first, which is often called the 'continuative' interpretation, is that John went to live in Paris at some time in the past and is still living there. The second, which is called the 'indefinite' ('existential', 'indefinite past') interpretation (and which is less salient when the sentence is used in isolation) is that John lived in Paris at some (unspecified) time in the past, but is not living there any more now.[22] Sentence (26,b) yields only the latter interpretation (unless there is a context excluding it – see below). The reason apparently is that the continuative interpretation of the present perfect requires reference to a period that reaches up to t_0.[23] That is, the continuative interpretation is possible (1) when there is a time adverbial like *so far*, *since two o'clock*, etc., (2) when there is a duration adverbial specifying a timespan which can be interpreted as lasting up to t_0 (e.g. *for three years*), (3) when the sentence is negative (e.g. *He has not been helpful at all*),[24] (4) when the context enforces the continuative interpretation (e.g. *John has lived in Paris, and perhaps he still does*),[25] and (5) when the verb is in the progressive form. (In the latter case the usual interpretation is that MidS (i.e. the middle of the situation referred to) reaches up to t_0 and may extend into the future – e.g. *I have been waiting for Tom.*) It is interesting to note that only possibilities (1) and (4) actually involve explicit reference to t_0. In the other cases the reference is to a timespan which can, but in theory need not, include t_0.[26] Still, present perfect sentences of these types primarily suggest a continuative interpretation. It is apparently an implicature of the use of the present perfect that, if duration is asserted by the use of an adverbial or by a progressive form, the continuative interpretation prevails as long as it is

22. As noted by Needham (1976:313), this use of the present perfect involves 'existential quantification over moments earlier than the present'.
23. This explains the unacceptability of (grammatical) sentences like *Paris has stood on the Seine* (Leech 1969:153) or *Horses have been mammals* (Lees 1961:160). Without a time adverbial such sentences are interpreted as involving an indefinite perfect. However, the implication that the situation no longer holds clashes with our knowledge of the world.
24. As noted by Dubois (1972:10–11), Feigenbaum (1981:401) and Vermant (1983:61–3), negative present perfect sentences (in which the negation has wide scope) are usually interpreted continuatively. That is, the fact that a particular situation has not held is interpreted as a state of affairs which reaches up to t_0. (There are, however, occasional exceptions. The following example was noted by Vermant (1983:62):
    (i) (*said with reference to someone who has just left the room*) I hope I haven't driven him off.)
25. This means that it is only by implicature that *John has lived in Paris* (when used in isolation) is automatically assigned the indefinite interpretation. (See also footnote 28.)
26. Thus, the duration adverbial in *John has attended Oxford University for eight years* indicates a period which may or may not include the present moment. It is also worth noting that in both cases the period may or may not consist of contiguous time intervals. That is, the sentence is true if John attended Oxford from 1976 to 1980 and from 1982 to 1986, or if he attended Oxford from 1976 to 1980 and has attended it again for the last four years. (Note that the latter interpretation is again indefinite, not continuative, because a continuative interpretation of the present perfect requires the situation-TO to be an uninterrupted period up to the present.)

not prevented or rendered unlikely by the pragmatics of the sentence or its context.[27] Thus, *John has lived in Atlanta since 1982* strongly suggests a continuative interpretation, yet the indefinite reading is enforced if we add *and in Moscow, Paris, Rome and a lot of other places* (see Canavan 1985:114). (The observation that we often put *ever* before *since* also illustrates the fact that sentences with *since* (without *ever*) just implicate a continuative reading. We need to add *ever* if we want to assert this reading.)[28]

Examples like (26,a–c) show that, in terms of time location, the present perfect allows two possibilities: on the indefinite interpretation the perfect refers to a situation that lies wholly before t_0, on the continuative interpretation it refers to a situation that is still continuing at t_0. These two possibilities correspond to a bounded and an unbounded reading, respectively. On the indefinite interpretation the situation has reached its terminal point before t_0. It is therefore conceptualized as bounded. On the continuative reading it is viewed as still in progress at t_0, i.e. as unbounded. Bounded sentences (e.g. *I have written two letters*) do not, therefore, yield the continuative reading.[29]

The indefinite interpretation is often accompanied by the idea that the past situation has yielded a result that is relevant to the present. This is the case, for example, in

(27) (a) You can come in. *I've opened the door.*
(b) *I HAVE eaten lobster*, but I did not like it.

In (27,a) the message conveyed by *I've opened the door* is really 'The door is open'; in (27,b) *I have eaten lobster* suggests 'I know what lobster tastes like; eating lobster is part of the experience that I have acquired in my lifetime.' This resultative meaning has induced some linguists to speak of a 'resultative perfect' next to (or even instead of) the indefinite and the continuative perfect.[30] Thus, Joos (1964:138–44) claims that

27. As we will see shortly, one of the cases in which a continuative interpretation is excluded is when the sentence represents the situation as bounded.
28. Dowty (1979:343) notes that *John has lived in Boston for four years* is in principle ambiguous between a continuative and an indefinite reading, but that the continuative reading is preferred out of context. He too suggests that this may be the result of conversational principles: since the preterit (*John lived in Boston for four years*) is available to assert unambiguously that the situation is over at t_0, the fact that the speaker does not choose the preterit leads the hearer to conclude that the situation is not yet over at t_0.

    On the other hand, when the present perfect is used without a marker of duration, as in *I have lived here*, the unmarked interpretation is the indefinite one. This is again due to Gricean principles: if the speaker 'were still living there, he should have expressed the situation as *I live here*. When there is *no apparent reason* for him to involve a specifically past-to-present reference, he would not be expected to use the perfect' (McCoard 1978:48).
29. This point is also made by Bauer (1970) and Brinton (1987:197).
30. McCawley (1973:263) claims that the present perfect is ambiguous between four readings, which he calls 'universal', 'existential', 'stative' and 'hot news'. The

resultativeness is the essential feature of the present perfect, and that the idea of anteriority is at best a connotation.[31] However, it is doubtful that resultativeness is an inherent part of the meaning of the present perfect. The following observations appear to contradict this:

1 Examples can be found in which there is no clear sense of resultativeness:

(28) (a) I *have* never *heard* such nonsense before.
(b) These forks *have been* ours, they *have been* my cousin's, and now they belong to you. (Hirtle 1975:5)
(c) Some people *have been* reluctant to believe this, but everybody certainly believes it now.

2 It often depends on the context which particular resultative interpretation is assigned to the present perfect. According to the context in which it is used, a sentence like *They've fallen into the river* suggests various resultative interpretations, such as 'Please help them', 'That's why they are soaking wet', 'That's how they got pneumonia', 'That's why they are late', etc. None of these propositions is predictable from the semantics of the sentence: they depend on the context. This means that at best one could argue that the present perfect functions as an instruction for the hearer to interpret the situation referred to as having some result, while leaving it to him to identify the result in question by taking into account the (linguistic and extralinguistic) context. However, it seems difficult to accept that this is the most essential element of meaning of the present perfect, which distinguishes it from the preterit. (Note that the preterit too can suggest a resultative reading. As Bryan (1936:373–6) says, *It rained hard all last night* will lead us to infer that the earth is wet this morning, but nobody would say that this is because of the past tense.) In my opinion, the above definition of resultativeness is so weak that an analysis which treats resultativeness (if it is present at all) as following from the temporal meaning of the perfect is much more plausible than an analysis which treats resultativeness as the core meaning of the perfect.

3 If resultativeness were an inherent part of the meaning of a perfect that is interpreted as resultative, it would not be possible to couch such a perfect in a context which explicitly denies resultativeness. Thus, the sentence *I have closed the door* could not be followed by *but now it is open*

'universal' and 'existential' readings correspond with what I label 'continuative' and 'indefinite', respectively. By 'stative' McCawley means 'resultative'. The 'hot news' interpretation is illustrated by *Malcolm X has just been assassinated.* In my opinion, the 'existential', 'stative' and 'hot news' readings are different interpretations allowed by the temporal meaning of the indefinite present perfect in particular contexts. I do not subscribe to McCawley's claim that these readings are a question of semantic ambiguity. (See also footnote 41.)

31. The same claims are also made by Hajicová, Panenová and Sgall (1973:206–7).

again without contradiction.[32] In practice, however, such a sequence is not contradictory at all. (See also Guenthner 1977:96.)

The above arguments make clear that it is unwarranted to speak of a 'resultative perfect' next to the 'continuative perfect' and the 'indefinite perfect'.[33] Resultativeness is not an inherent part of the meaning of the present perfect:[34] we can say no more than that the present perfect allows a resultative interpretation when such an interpretation is invited by the lexical meaning of the verb and/or by the context. This conclusion is not surprising: whereas the labels 'indefinite' and 'continuative' refer to different ways of locating a situation in time, the label 'resultative' indicates a totally different kind of phenomenon. (The phenomenon in question is sometimes called aspectual, sometimes pragmatic. However this may be, it certainly is not a question of location in time.) I will have more to say about this subject below.

Returning to the topic of how the present perfect locates a situation in time, we can point out a couple of differences between the present perfect and the past tense. As we have seen, the structure of the preterit typically involves the following relations:

1 TS simul TO_{sit}
2 TO_{sit} before TO_1 (TO_1 is usually t_0)
3 TO_{sit} belongs to the past time-sphere
4 TO_1 belongs to the present time-sphere

If there is a time adverbial indicating a TE, the preterit implies that TE must include TO_{sit} (i.e. TE must either include TO_{sit} properly or be

32. Compare this sentence with the unacceptable *!I have killed him, but he is not dead.* This sentence is really contradictory, because the idea of a resultant state ('be dead') is inherent in the meaning of the verb *kill*, and is therefore necessarily implied when the verb is used in a perfective construction.
33. For ease of reference I will use the label 'indefinite/continuative perfect' to refer to a perfect in a sentence that yields an indefinite or continuative interpretation, respectively. (I am aware of the fact that the interpretation in question is not due solely to the tense form.)
34. The view that there is such a thing as a resultative perfect was already expressed by Sweet (1891:98). It was widely accepted by the leading traditional grammarians, viz. Poutsma (1926:256), Jespersen (1931:47) and Kruisinga (1931:390ff.). The first linguist who explicitly rejected this view was Bryan (1936:369):

> I believe that any idea there is of result or consequences is not implied in the perfect tense form but derives from the meaning or character of the verb, or from the statement as a whole. That is, to assign to the perfect tense form itself a resultative function means a failure in analysis.

In recent linguistics we find proponents of both points of view. The view that resultativeness (current relevance) is an inherent part of the meaning of the perfect or of some type of perfect is upheld by e.g. Twaddell (1963), Joos (1964), Palmer (1965:73–5), Strang (1968:162), Leech (1969, 1971), McCawley (1973), Hirtle (1975), Comrie (1976:52), Feigenbaum (1981), Moravcsik (1982:101), Papp (1985:106), Fenn (1987). Among the linguists that reject this view we note Sørensen (1964:79–82), Bauer (1970), Zydatiss (1976:158, 1978), McCoard (1978), Rigter (1980d:150), Canavan (1983:39), King (1983), Rauh (1985:14ff.), Mugler (1988:77ff.), etc.

commensurate with it). TE may also include t_0, as long as TO_{sit} wholly precedes t_0 (e.g. *It happened today*).

The description of the present perfect also involves the relations (1) and (4), but differs from the description of the preterit in the following respects:

1 TO_{sit} belongs to the present time-sphere (more specifically: to the pre-present sector).

2 TO_{sit} (and hence TS) may either lie entirely before t_0 (as in *I have visited Paris four times*) or reach up to t_0 (as in *I have been here for two hours*).

3 There must be no element in the context that is incompatible with the interpretation that TO_{sit} lies in the pre-present sector. For example, adverbials that can only refer to the past time-sphere cannot co-occur with the present perfect (see ***I have done it yesterday**, **I have learned that at school*).[35]

4 If there is no TE established by an adverbial (or by the context) and TO_{sit} does not reach up to t_0, the only difference between the preterit and the present perfect is that the former locates TO_{sit} in the past time-sphere, whereas the latter locates it in the present time-sphere (pre-present sector). Which of these two options is selected depends upon what is uppermost in the speaker's mind. If the speaker is somehow thinking of t_0 (NOW), he will locate the situation in the 'before-NOW' (i.e. in the pre-present sector).[36] If he is not thinking of NOW, he will locate it THEN, i.e. in the past time-sphere. This is the case not only when he is thinking of the time when the situation took place, but also if he sees the past situation as contrasting with the present (e.g. *He is not so active as he **was**, I **did**n't **know** you were married*) and if he focusses on a particular aspect of the situation itself (see section 1 of this chapter).[37]

35. In the latter example the place adverbial *at school* involves temporal reference to a time that is completely over at t_0. (Place adverbials often imply temporal reference. As a matter of fact, Geis (1975) argues that such adverbials are actually 'residues of *while*-clauses' (p. 7). According to him, *at school* is derived from something like 'during the time while I was at school'.)
36. See Rigter (1980d:146), who argues that the basic meaning of the present perfect is that it locates the situation before the 'focal episode' (i.e. the time focussed on), which in this case is t_0. Similarly, Papp (1985:103) writes that the present perfect 'describes the present in an indirect way, i.e. the time of the action/state that we name precedes present, the time that we describe. (Present fact, pre-present state/action.)'
37. For example, the question *Is John in the house?* may be answered by either *No, he left* or (more commonly) by *No, he has left*. In the former case the speaker focusses on the situation itself, which belongs to the past; in the latter case he focusses on the present state of affairs that has been produced by the past situation. Similarly, the difference between *I lost my umbrella* and *I have lost my umbrella* is that in the former case the speaker is concerned with aspects of the situation itself (especially with the place and time at which it might have occurred), whereas in the latter case it is the resulting state of affairs that is uppermost in his mind (see below).

The statement that the speaker using a present perfect is thinking of NOW should be interpreted as meaning that the context in which he intends his utterance to be interpreted consists of, or at least contains, one or more propositions in the present tense (see N. Smith 1981). Thus, the sentence *I have opened the door* is appropriately used (because relevant) in answer to *Can we get in?* but it is inappropriate as a reply to *What happened?* In general, the fact that the present perfect locates the situation in the present time-sphere entails that 'it indicates that the propositions relevant to the interpretation of the sentence include some present tense ones' (N. Smith 1981:260).

As noted before, this leads to the conclusion that there are two typical uses of the present perfect: the 'indefinite past' use and the 'continuative use'. In the former use, TO_{sit} is represented as wholly preceding t_0, but its precise location in time is left indefinite because the speaker is not concerned with this question. In the continuative use, TO_{sit} (and hence TS) reaches up to t_0. What the two uses have in common is the presence of an explicit or implicit TE which reaches up to t_0.[38] This is most evident in the continuative use: since TE by definition includes TO_{sit} (i.e. TE either includes TO_{sit} properly or is contemporaneous with it), TE must reach up to t_0 if TO_{sit} does. A time adverbial accompanying a continuative present perfect will therefore always indicate a timespan leading up to (and including) t_0. When the perfect is of the indefinite past type (e.g. *I have done that before*), the TE established by the time adverbial is also a timespan continuing into the present. TO_{sit} (= TS) is then interpreted as being some unspecified subinterval of TE which lies wholly before t_0. If there is no explicit TE established by an adverbial or by the context, there is an implicit one, viz. the pre-present sector itself. Again, this is most obvious if the meaning is continuative, for 'continuation up to the present' necessarily means continuation through a period leading up to the present. As far as the indefinite perfect is concerned, the very fact that the speaker chooses the present perfect and not the preterit means that he conceptualizes the situation as lying, not in the past time-sphere, but in the 'before now' part of the present time-sphere.[39] The implicit TE-up-to-t_0 into which a sentence like *John has sold his house* locates its situation is thus the pre-present sector as a whole. Since this TE exists only in the mind of the speaker, it may be conceived as a timespan of any length. In some cases it is conceived as the whole of mankind's history (as in *There have always been wars* or *Skeletons buried in a contracted position have been found with knees*

38. This timespan continuing up to t_0 is sometimes referred to as an 'extended now' (see e.g. Bennett and Partee 1978; McCoard 1978; Saurer 1981:126; Dowty 1982:27).
39. As we saw in section **1**, this is because the speaker is primarily thinking of NOW when he is referring to the past situation. If this is not the case (for whatever reason), the situation is located in the past time-sphere (i.e. it is located THEN rather than 'before NOW').

drawn up under the chin (Scheurweghs 1959:326)); in other cases it is the speaker's lifetime (e.g. *I have been in Italy once*); mostly, however, it is a much shorter timespan (e.g. *I have opened the door*). In fact, because of the Gricean Maxims,[40] the implicit TE-up-to-now (pre-present sector) is as a rule conceived and interpreted as the shortest possible timespan that is in keeping with the pragmatics of the sentence and its context.[41] For example, in the sentence *Have you had breakfast yet?* the implicit TE is 'today'; an answer like *Yes, I did three weeks ago* would therefore be quite inappropriate. (This answer would imply that breakfast is not eaten every day but only once in three weeks.) In *John has already visited Rome*, the TE will maximally be the timespan corresponding with John's lifetime,[42] but it may also be a much shorter period, e.g. the timespan occupied by that particular journey to Italy which John is making at this very moment.[43]

It follows from the above observations that the present perfect is compatible with some types of time adverbial only:

1 The type that is probably most frequent with the present perfect consists of time adverbials which establish a timespan that reaches up to the present moment. Such an adverbial may be a bounded adverbial which explicitly refers to a timespan including t_0 (e.g. *until now, so far, from 1978 till now*), or a bounded adverbial which refers to a timespan

40. The Maxim involved is probably that of Relation (Relevance). As noted before, it is a fact that (other things being equal) statements about the present are appreciated as more relevant than statements about the past. For this reason the implicit pre-present TE will be interpreted as lying as close to t_0 as possible.
41. This is the reason why the indefinite perfect often suggests that the situation occurred in the recent past (see Quirk *et al.* 1985:193) and why it often yields what McCawley (1973) calls a 'hot news' interpretation (e.g. *Malcolm X has been assassinated*). (McCawley actually treats the 'hot news' use of the perfect as different from the 'existential' (indefinite) use. In my opinion, this distinction is unwarranted. The 'hot news' interpretation is an indefinite interpretation. It arises automatically if two pragmatic conditions are fulfilled, i.e. if (1) what is said is new information (news) to the hearer, and (2) there is nothing to prevent the interpretation that the implicit TE-up-to-TO_1 (and hence the distance between TO_{sit} and TO_1) is very short. (If this TE is very short, what is said is not only news but also hot news.))
42. If John happens to be dead, the timespan corresponding with his lifetime does not reach up to t_0, so that the indefinite perfect is no longer appropriate. This has often been observed in connection with the low degree of acceptability (under neutral sentence intonation) of examples like *Newton has believed in God* or *Einstein has visited Princeton*. (See also below.)
43. If there is no such adverb as *already*, the latter possibility is the one that normally suggests itself. But the interpretation that the TE is a whole lifetime can still be elicited by putting a heavy accent on *have*. Compare:
    (i) I have been to FRANCE.
    (ii) I HAVE been to France.
    Sentence (i), which shows the neutral intonation pattern (i.e. with the nuclear accent on the last open class item expressing new information), suggests a recent visit to France. Sentence (ii) suggests an indefinite interpretation: I have been to France (at least once) in my lifetime. (See also Crystal 1966:8.)

which is interpreted as leading up to a TO which, if the present perfect is used, is t_0 (e.g. *since then, from 1984 onwards*), or an adverbial which establishes a well-defined[44] period including t_0 (e.g. *today, this week, this year*),[45] or an adverbial which refers to a period whose beginning lies indefinitely far in the past and which, if used in combination with a present perfect, is interpreted as reaching up to t_0 (e.g. *before, ever, never, already, always, in the past*).[46]

2 A second class of adverbials that is compatible with the present perfect consists of adverbs like *recently, this minute, just*, etc. These differ from the previous ones in that they do not refer directly to a TE including t_0. Rather, they indicate (in a vague way) the position of TO_{sit} relative to t_0 within an unspecified timespan reaching up to t_0. Thus, *recently* informs us that the distance between TO_{sit} and t_0 is relatively short; *just* and *this minute* tell us that this distance is even shorter.[47] In both cases the

44. This kind of TE is, of course, definite: since t_0 is a definite time (see Allen 1966:164), any interval of a well-defined length that includes t_0 is a definite interval. However, this is not in contradiction with the fact that we call a present perfect accompanying such an adverbial an 'indefinite perfect'. What is indefinite in this use of the perfect is the location of TO_{sit} in the TE-up-to-now, not the TE itself. (McCoard (1978:79ff.) wrongly applies the label 'indefinite' to the TE when he claims that the possibility of using the perfect with definite adverbials like *today* refutes the theory that the perfect expresses 'indefinite past'.)
45. Adverbials like these can also co-occur with the preterit, as in *I saw Mary today*. (This observation leads Fenn (1987:203) to reject the theory that the present perfect differs from the preterit in that it locates the situation in a timespan reaching up to t_0. According to Fenn, this theory wrongly claims that *today* indicates a period which includes t_0 if the present perfect is used, but which does not include it when the adverb collocates with the preterit. However, it will be shown below that this is only seemingly a problem. As we have seen in chapter 6, adverbials like *today, this year*, etc. are unbounded and can therefore be used to refer to subintervals of the day, year, etc. in question. This means that the speaker using such an adverbial may be thinking of a subinterval which lies wholly before t_0. That is, even though *today* always establishes a TE which includes t_0, TS (and hence TO_{sit}) may coincide with a portion of TE which lies wholly before t_0. If the speaker is not concerned with t_0, the situation can then be conceptualized as belonging to the past time-sphere. The fact that *today* can collocate with the preterit is therefore not incompatible with the view that *today* always indicates a timespan which includes t_0.)
46. Of course, these adverbs can also be used in a context which makes clear that t_0 is not included in the reference. In that case we have to use one of the past tenses:
    (i) When I was young I always/never *told* my parents the truth.
    (ii) I thought I *had seen* him somewhere before.
    In sentences like the following, the preterit is used because of the contrast that is established between the present and the past:
    (iii) I never *thought* of that before. (Blackstone 1962:89)
    (iv) 'I see,' Beesley said, sniffing. 'I *didn't know* that before.' (ibid.)
    Clearly, these sentences imply a contrast: 'but I am thinking of it now', 'but I do know it now'.
47. *Just* is basically a restrictive adverb. In this case it restricts the timespan lying between TO_{sit} and t_0: it indicates that this timespan is very short. It should be noted, however, that shortness is a subjective and pragmatically determined notion here: the timespan indicated by *just* is much shorter in *I have just opened the door* than in *The summer has just come to an end* (see Mommer 1986:193). (For more on the semantics of *just*, see Bourdin 1982; Dominicy 1983:325; Lee 1987.)

present perfect is used because the speaker is thinking of the relative position of TO_{sit} within a timespan up to t_0.[48]

3 The present perfect is also compatible with adverbials that refer to t_0, as in *I have now done half of the work* or *At present I have finished the first two chapters of the book*. In that case the interpretation is that the situation has occurred within (or over) a period which reaches up to t_0. (In other words, what is specified by the adverbial is the right bound (t_0) of an implicit period.)

4 Even adverbials referring to a timespan which lies wholly before t_0 may be compatible with the present perfect:

(29) (a) I think I've met him some time or other.
(b) You don't believe *I've ever left at five o'clock*? I can assure you, *I HAVE left at five o'clock.*

The use of an adverbial excluding t_0 (*some time or other, at five o'clock*) does not rule out the present perfect because the times indicated are clearly conceived as intervals within a period-up-to-now. (In both sentences this period is likely to be the speaker's lifetime.) What is uppermost in the speaker's mind is not the time when the situation held but the fact that it has held at a certain time in a period up to t_0. To see this better, compare:

(30) (a) John HAS left the house at five o'clock.
(b) John left the house at five o'clock.

In (30,b) *at five o'clock* is a definite time indication because it is interpreted in relation to a particular time that is identifiable from the context: the adverbial is of the 'anchored' (Smith 1978b) or 'deictic' (Barense 1980) type. In (30,a), in contrast, *at five o'clock* is not a definite indication of time because it is an 'unanchored' or 'nondeictic' adverbial, i.e. it is not interpreted in relation to any particular day referred to in the context. This different interpretation accords with the fact that *at five o'clock* is felt to identify a TO_{sit} in (30,b), but not in (30,a). Whereas (30,b) states that five o'clock was the time of the situation of John leaving the house, (30,a) states that the situation of John leaving the house at five o'clock has held in the past (i.e. within an unspecified period up to now). That is, the situation that is located in time in (30,b) is John's leaving the house, whereas in (30,a) it is the situation of John leaving the house at five o'clock. In (30,a) the element 'at five o'clock' thus belongs to the

48. This, at least, is true of British English. American English speakers apparently interpret these adverbs as establishing a TE which does not include t_0 and therefore use the past tense.

description of the situation that is being located in time,[49] rather than indicating the TO_{sit} at which the situation is located.[50]

This semantic difference between (30,a) and (30,b) has a syntactic correlate in the fact that *at five o'clock* is an element 'inside the VP' in (30,a) whereas it is an element 'outside the VP' in (30,b) (unless it is the only constituent there that expresses new information).[51] This appears from the fact that *at five o'clock* can be fronted in (30,b) but not in (30,a):

(31) (a) *At five o'clock John HAS left the house.
(b) At five o'clock John left the house.

As is well known, adverbials that are inside the VP cannot be fronted.

As is predictable from the above remarks, 'anchored' past time adverbials (i.e. adverbials that indicate a past interval which is interpreted relative to some definite TO) normally rule out this use of the perfect:

(32) (a) *I have got up yesterday.
(b) *I have met John in 1978.

The reason why the present perfect is ungrammatical here is that the anchored adverbials establish reference to an interval which lies before t_0 and which is not interpretable as lying in a period-up-to-t_0.

It may be noted, finally, that present perfect sentences of the kind exemplified by (30,a) often involve a repetitive time adverbial:

(33) (a) John has often gone swimming at six o'clock in the morning.
(b) Many a time, passengers have been terrified when their plane began to lose height.
(c) I have occasionally left before Tom (did).

However, the sense of repetition does not come exclusively from the adverbial. Even if the latter is dropped (in which case *have* normally receives a primary accent), there is a sense of (at least potential) repetition. The following sentences suggest paraphrases like 'It has happened on occasion that . . .' or 'It has happened at least once that . . .':

49. In other words, the time adverbial forms 'one constituent' (Hamann 1989:40) with *has left the house*. It 'should be understood as providing more information about the temporal profile of the event under discussion' (Harkness 1987:105).
50. Since (30,a) expresses no more than that the situation of John leaving the house at five o'clock has held in the past (i.e. was simultaneous with at least one unidentified TO_{sit} in a period-up-to-now), the sentence can easily be followed by another sentence specifying the TO_{sit} (or TO_{sit}s) in question:
(i) A: John has left the house at five o'clock before, you know.
B: I don't believe you. Tell me when that was.
A: He left at five o'clock on 4 May, 1985, and he did last Sunday too.
51. The distinction between adverbials inside the VP and adverbials outside the VP has often been made in transformational grammar – see e.g. Verkuyl (1972), Gruber (1976). It corresponds to the difference between 'obligatory' ('necessary') and 'optional' adverbials.

(34) (a) John HAS gone swimming at six o'clock in the morning.
(b) Passengers HAVE been terrified when their plane began to lose height.
(c) I HAVE left before Tom (did).

I suggest that it is precisely the lack of information concerning the temporal location of TO_{sit} that produces this potentially repetitive reading. Sentences like (34,a–c) express no more than that, in a period up to now, there have been one of more instances of a situation of the type 'swimming at 6 p.m.' or 'leaving before Tom'. No information is given concerning the precise temporal location of these instances, nor about their frequency.

5 Apart from the above type of sentence, in which the past time adverbial forms part of the description of the situation reported, we must note the fact that adverbials like *at five o'clock, yesterday,* etc. can co-occur with a present perfect if the idea of a TE-up-to-t_0 is also explicitly present. This is the case in examples like the following:

(35) George *has proved* difficult last month, Tuesday, and yesterday so far. (Dinsmore 1981:476)
(36) (a) Prince Henry has decided to travel to Tokyo by the overland route. Twice already he *has visited* Japan, in 1898 and 1900. (Erades 1975:138)
(b) 'We appreciate your help exceedingly, Captain Newman', said the Personage. 'Now we *have* already *discussed* this affair at some length last night . . .'. (Erades 1975:137)
(c) I *have tried* to call him yesterday, last night, and today, but with no success. (Dowty 1979:373)
(37) (a) A: How has he been occupying himself this week?
B: Well, he's *played* golf on Tuesday, *ridden* horseback on Wednesday, and *rested* on Thursday. That's all I know about so far. (McCoard 1978:104)
(b) A: Have you ever seen *Macbeth* on the stage?
B: Yes, I'*ve seen* it ages ago, when I was a child. (Quirk *et al.* 1985:195)
(38) (a) The motor *has relegated* the cabriolet to the coach-house long ago. (Erades 1975:138)
(b) There *have been* more deaths in Northern Ireland yesterday. (Huddleston 1976:342)

The above examples illustrate four different possibilities concerning the TE-up-to-t_0. In (35) the TE is indicated explicitly by *so far.* Since this adverb occurs in the sentence itself, the present perfect must be used, in spite of the other time adverbials. In (36,a–c) the presence of an implicit TE-up-to-now is suggested either by *already* or by the enumeration of TEs (the first of which lies in the past, while the last includes t_0). In

(37,a–b) we see that speaker B adopts the TE-up-to-t_0 from the question that he is answering. The 'up to now' frame of the question persists in the answer and provokes the use of the perfect in spite of the presence of the (anchored) past time adverbial(s).[52] In (38,a–b), finally, the TE-up-to-now is not established by an adverbial. It is an implicit TE, suggested by the pragmatics of the sentence and its context.[53]

In the preceding paragraphs I have argued that the difference between the preterit and the present perfect is that these tenses locate situations in the past time-sphere and the pre-present sector, respectively. The distribution of time adverbials over the two tenses appears to be in keeping with this. In the following paragraphs attention is drawn to some additional observations confirming the above conclusions:

1 A question beginning with *when* will normally be in the past tense rather than in the present perfect. We say *When did the accident happen?* and not **When has the accident happened?* (See also Berezovsky 1978:80; Fenn 1987:170.) This is in keeping with the general principle, since a speaker asking a *when*-question is normally concerned with THEN, not with NOW. That is, when a speaker asks a *when*-question (concerning a past situation) he is seeking information concerning the temporal location of TO_{sit} in the past time-sphere. (Note that the fact that he uses *when* implies that the speaker expects that TO_{sit} does not include t_0. If he expected that TO_{sit} reached up to t_0, the speaker would use *since when?* or *for how long?*. The use of the preterit with *when* and of the perfect with *since when* is therefore predictable.)

However, occasional examples can be found of questions where *when* means 'at what time(s) in a period-up-to-now'. In that case the present perfect is predictably possible, as in the following example (from A. Christie, *4.50 from Paddington*, London: Fontana, p. 104):

(39) 'Me, I do not like the police!' She scowled at them, without camouflaging her dislike of the visit. 'Always, if they can, they make me embarrassments.'
'No, no, Madame, you must not say that,' said Dessin. . . . *'When have I ever caused you embarrassments?'*

2 According to Comrie (1981a:30), the use of the perfect with *long since* (as in *I have long since given up smoking*) is a 'potential counterexample' to the rule that time adverbials accompanying a present perfect cannot

52. Quirk *et al.* (1985:195) consider such examples as 'performance errors, induced by B's copying the form of A's question'. I suggest that it is not so much the form as the TE of the question that is copied. The statement that doing so is an error seems highly questionable.
53. Examples like these are very rare. McCoard (1978:157) lists some others, which have been noted at different places in the linguistic literature. He adds that, personally, he would star all of them as anomalous.

exclude t_0 from their reference. However, there is no reason for treating *long since* as an exception. The adverb *since* refers to a TE which starts at a certain time in the past (contextually identified) and reaches up to some other time. If this other time is t_0, all the conditions for using the present perfect are satisfied: there is a TE leading up to t_0, and TO_{sit} is an unspecified subinterval of this TE. The addition of *long* just means that this unidentified TO_{sit} within TE does not lie close to t_0. (That is, *long* measures the distance between TO_{sit} and t_0 in the same way as adverbials like *just, recently, this minute* do.)

3 An interesting observation in connection with adverbials that establish a TE-up-to-t_0 is that some of them are compatible with the present perfect only (as in (40,a)), while others can collocate with both the perfect and the preterit (as in (40,b)). This is something our analysis will have to account for.

(40) (a) So far I have seen him twice. (*saw)
(b) I have seen/saw him several times this week.

The difference between these two sentences follows from the different nature of the adverbials. As we have seen in chapter 6, time adverbials may be bounded or unbounded. This also applies to time adverbials that establish a TE including t_0: some of them (e.g. *up to now, so far, since 1985*) are bounded, others (e.g. *today, this week, this year*) are unbounded. Some (e.g. *between 1985 and now*) can even be used either way.

A time adverbial is bounded if it explicitly represents an interval (TE) as leading up to a terminal point. As we saw in chapter 6, bounded adverbials are 'heterogeneous' and hence 'unitary': they can only refer to the timespan as a whole, not to any subinterval of this. Because of this, the inclusion relation between a TE established by a bounded adverbial and TO_{sit} is never that of proper inclusion: it is always the case that TE coincides with TO_{sit}. It follows that bounded adverbials referring to a timespan whose terminal point is t_0 cannot co-occur with the past tense. Since TE here coincides with TO_{sit}, and since TE includes t_0, TO_{sit} must also include t_0. This rules out the preterit, which necessarily locates TO_{sit} wholly before t_0. Hence the obligatory use of the present perfect in (40,a) and in similar sentences like the following:[54]

(41) (a) Up to now seven people have died in such accidents.
(b) Nobody has spoken to us since that incident.

54. This explanation accords with the observation that sentences with such adverbials cannot be used in answer to questions beginning with *when*. Thus, (41,a) provides no answer to the question *When did seven people die in such accidents?* Such a question would ask for the specification of the past TO_{sit}, but adverbials like *so far* or *up to now* are no such specifications. (They specify TE, not TO_{sit}.) So, the speaker of (41,a–b) is not concerned with the question when the situation happened, but with the question 'how many times has it happened?' or 'has it happened at all?'

(Note that TO_{sit} includes t_0 not only in (41,b), where the perfect refers to a state that continues into the present,[55] but also in (41,a). In (41,a) there is reference to a repetitive situation consisting of seven subsituations. Though the individual subsituations do not include t_0, the repetitive situation as a whole does.)[56]

Unlike bounded adverbials, unbounded adverbials (e.g. *this year, today*) refer to a timespan which is 'homogeneous' (the same all the way through) and 'dissective' (i.e. the adverbial can refer not only to the interval as a whole but also to subintervals of it). This means that the inclusion relation between the TE established by an adverbial like *this year* and TO_{sit} may now be that of proper inclusion. It follows that the fact that TE includes t_0 does not entail that TO_{sit} must also include t_0. And if TO_{sit} does not include t_0, both the preterit and the present perfect are predictably possible. The choice of tense simply depends on whether the speaker wishes to locate the situation in the past time-sphere or in the pre-present sector. Compare:

(42) (a) Seven accidents have happened this year.
(b) The accident happened this year.

In (42,a) the speaker is concerned with the number of accidents that happened in a period up to the moment of speech. This number is a feature of the structure of the world at t_0. In (42,b), in contrast, he is concerned with the question when a particular accident happened: unlike a sentence like (42,a), this sentence could serve as an answer to *When did the accident happen?*

Some time adverbials allow both a bounded and an unbounded interpretation. Compare:

(43) (a) Between five o'clock and six John was running in the park.
(b) Between five o'clock and six John ran 18 kilometres.

In (43,a) the adverbial (and the sentence as a whole) is unbounded; in (43,b) it is bounded (see also Declerck (1979b:770–2)). Adverbials like these can collocate both with the preterit and with the present perfect:

(44) (a) Seven of them have died between 1985 and now.
(b) Seven of them died between 1985 and now.
(c) John retired somewhere between 1985 and now. (*has retired)

55. As pointed out by Dowty (1986:44), the situation expressed in such a negative sentence is always a state.
56. I claim that this is the case even if the (potentially) repetitive situation turns out to contain only one subsituation, as in *So far one person has died.* As we have seen, it is typical of repetitive situations that there may be gaps between the subsituations. This is clearly the case in this example, where TO_{sit} consists partly of gaps and partly of the subsituation referred to. (Note also that, if the situation were not conceived as repetitive (pluralized), it would be normal to use *a* instead of *one*.)

In (44,a) the adverbial *between 1985 and now* yields the bounded (heterogeneous, unitary) interpretation. The speaker considers the timespan indicated as a whole and indicates how many people died within it. Since the timespan includes t_0, he uses the present perfect in doing so. In (44,b) the speaker is not concerned with the number of people that died within the stated period, but rather with the (complex) past situation itself. Although the adverbial *between 1985 and now* still establishes a TE which reaches up to t_0, TO_{sit} is not thought of as a time interval that includes t_0. Instead, it is conceived as an interval that belongs to the past time-sphere, and is therefore referred to by means of the past tense. This is possible because the adverbial allows the unbounded interpretation (on which TO_{sit} is a proper subpart of TE).

Sentence (44,c) can be explained in the same way. Here the speaker is concerned with the question when it was that John retired. He does not specify that time (TO_{sit}) precisely but just tells us that it lies somewhere in the timespan (TE) between 1985 and t_0. In doing so he can make use of the adverbial *between 1985 and now* because this need not refer to TE as a whole but can also be used to indicate the subinterval (TO_{sit}) at which TS is located (and which lies wholly before t_0). Since the speaker is not concerned with NOW but with THEN (more specifically, with the time when John retired), he uses the preterit to locate the situation in the past time-sphere.

4 In sentences like *He did it* and *He has done it* there is no time adverbial, but there is an implicit time reference of the kind we have been examining. When using *He did it* the speaker is not concerned with a timespan reaching up to t_0, but rather with the situation itself, which held at some past time (TO_{sit}). This sentence will therefore normally be used in a context in which the time in question has been referred to, and if this is not the case, it means that the speaker is thinking of the situation itself rather than of a period up to now.[57] The use of *He has done it*, in contrast, means that the speaker is not thinking of the past TO_{sit}, but is concerned with the present world structure, one characteristic of which is that the action referred to has taken place.[58]

57. As noted in the section on the past tense, this focus on the situation itself presupposes that the situation be identifiable to the hearer.
58. This theory is consistent with Inoue's (1975:54) claim that the basic function of the present perfect is 'to explain the state of affairs of the thing which is being talked about. It is an explanatory and/or illustrative expression. The activity, event or state which is described in the present perfect explains, provides an illustrative example, or gives a reason for the thing being as it is at the time of the speech act.' To illustrate this claim Inoue adduces examples like the following:
    (i) What's new with Kissinger? – *Kissinger has met with President Sadat of Egypt.*
    (ii) Mrs. Gandhi's door is always open. *John Chancellor of NBC has met with her, Walter Cronkite of CBS has met with her . . .*
    (iii) How are you? – Oh, I am feeling rather low. *My dog has died.* You know how much I loved him.

Summing up what has been argued so far, we can say that the present perfect differs from the preterit in the following ways:

1 The preterit locates TO_{sit} in the past time-sphere; the present perfect locates it in the pre-present sector of the present time-sphere.
2 Because of this, the preterit necessarily represents TO_{sit} as lying entirely before t_0. The present perfect, on the other hand, is compatible both with a TO_{sit} that lies entirely before t_0 and with a TO_{sit} that reaches up to (and includes) t_0.
3 Time adverbials that establish a TE which does not reach up to t_0 can collocate with the preterit only, except when the time interval indicated is an unanchored time which forms part of the structure of the situation. The preterit is also possible when TE is an unbounded time interval including t_0. (In that case TO_{sit} coincides with a portion of TE which lies before t_0.) The present perfect, on the other hand, allows only time adverbials that establish a TE-up-to-t_0 or are compatible with such a TE. It cannot be used with adverbials referring to an anchored past time.
4 When there are no time adverbials, the speaker's choice of the past time-sphere or the pre-present sector will depend on whether he is concerned with aspects of the past situation itself or with the structure of the world at t_0.

The fact that it is relevant whether or not TO_{sit} or TE reaches up to t_0 means that we will have to distinguish between two types of anteriority relation. For the description of the past tense we need an anteriority relation of the type 'X before Y' in which *before* means 'wholly before', i.e. X does not have any time point in common with Y. For the description of the present perfect we also need a relation of the type 'X leading up to Y', i.e. X is a time interval whose final point coincides with Y.

The temporal structure of the present perfect, then, consists of the following elements and relations:

(45) (a) TS simul TO_{sit}
(b) TO_{sit} lies either wholly before TO_1 or before-and-up-to TO_1
(c) TO_1 is either t_0 or a TO behaving as if it were t_0 (as in *He will say that he has done it*)
(d) TO_{sit} belongs to the pre-present sector.

In (i) 'the speaker is describing Kissinger in terms of his diplomatic activities. The speaker's focus is not in the event of him meeting with Sadat, but rather in the state of Kissinger's diplomatic activities' (p. 55). In (ii) 'the sentences in the present perfect are given as illustrative examples of her being accessible to reporters'. In (iii) 'Speaker B says that her psychological state is not good, and she gives the recent death of her beloved dog as a reason for it' (p. 58).

This can be represented schematically as in Figure 7.18 (e.g. *I have eaten lobster once (but I didn't like it)*) and Figure 7.19 (e.g. *John has been sitting here since Midday*).[59]

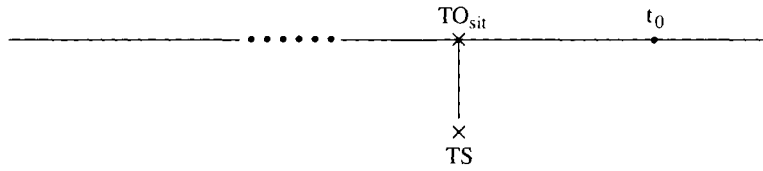


Figure 7.18 Example: *I have eaten lobster once*

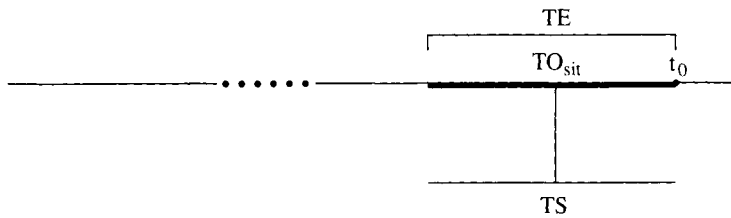


Figure 7.19 Example: *John has been here since midday*

I will close this section on the present perfect with a few considerations pertaining to some issues that are relevant to the use and meaning of the tense in question.

1 It is sometimes argued that the present perfect has only one basic meaning, and that labels like 'indefinite' or 'continuative' relate to how that meaning interacts with the context in which the perfect is used.[60] What I have argued is not at variance with this. In footnote 33 I have stressed that I am using the labels 'indefinite perfect' and 'continuative perfect' only as abbreviations for 'perfect in a sentence that receives an indefinite/continuative interpretation'. This means that I agree that the perfect itself has no indefinite or continuative meaning. The perfect expresses no more than that the situation is located somewhere in a TE-up-to-t_0. Whether it yields an indefinite or a continuative interpretation depends on whether or not this 'somewhere' is interpreted as including t_0. However, even if it is true that the present perfect has only one basic meaning, it is clear that we cannot simply collapse the indefinite and the continuative readings. It does make a difference in terms of temporal structure (and this is what we are concerned with here) whether TO_{sit} lies wholly before t_0 or includes it. The difference between

59. Up to now I have represented situations by either a line or a dot, according to whether they were durative or punctual. For ease of representation I will now add the symbol x, which will be used to represent a situation without reference to whether it is durative or punctual.
60. See e.g. Zydatiss (1978).

the relations 'wholly before' and 'before and up to' is relevant to the way in which we interpret present perfect sentences. This is clear from the fact that we interpret a sentence like *John has lived in Paris for four years* as having two clearly differentiated readings. These two readings are not just a question of vagueness. It would not do to say that the sentence is vague as regards the precise location of TO_{sit} in the TE-up-to-now, so that it is true not only if John lived in Paris two years ago but also if he lived there three years ago, or five years ago, and also if he is still living there. Clearly, the latter interpretation is not on a par with the former ones. It is felt to be distinct from them. In other words, *John has lived in Paris for four years* is felt to be ambiguous (rather than just vague) between a continuative reading and an indefinite one (the latter in fact being a set of indefinite readings). (For this reason the two conjuncts of *John has lived in Paris for four years and so has Bill* yield the same reading. It is not possible to assign an indefinite interpretation to one of them and a continuative interpretation to the other.)

2 At the beginning of this section it was claimed that the basic difference between the present perfect and the preterit is to be found in their temporal schemata and not in the presence or absence of 'current relevance'. This does not mean, however, that the notion of current relevance is not pertinent to the study of the present perfect. I therefore propose having a closer look at it.

a. A first thing to be noted is that the notion of current relevance was initially introduced to account for the fact that the present perfect often yields a resultative interpretation. It was only afterwards that some linguists (e.g. Joos 1964) went a step further and claimed that current relevance was the basic aspect of meaning of the perfect. As noted above, this claim inevitably leads to a complete erosion of the term: faced with the observation that a great many perfect forms do not yield a resultative reading, Palmer (1974:50) ultimately defines current relevance as meaning 'that in some way or other (not necessarily in its results) the action is relevant to something observable at present'. Similarly, Dowty (1979) defines it as meaning that 'the event described has some relevance or other to the present context, the nature of which is to be inferred entirely from contextual factors'. Since Grice's (1975:45) Maxim of Relation (Relevance) stipulates that a co-operative speaker should only say things that are relevant, this watered-down notion of current relevance ultimately applies to any statement, whether it is in the present perfect or not.
b. There are situations that imply a certain result whenever they are represented as completed, irrespective of the tense that is used to describe them. For example, it is inherent in the situation of opening a door that the door in question is open immediately after the action has been performed. This implication of a resultant state has nothing

to do with the choice of tense: not only *I have opened the door* but also *I opened the door, I will open the door*, etc. imply the idea of the door being open right after the completion of the situation.[61] On the other hand, there are situations that do not automatically imply a resultant state. The situation of a door being open is an example of this. If such a situation is referred to (as in *The door has been open, The door was open, The door will be open*, etc.), there is no implication of any well-defined resultant state (see also Chafe 1970:172). Similarly, *I have heard/will hear/heard/*etc. *the truth* implies *I know/will know/knew/*etc. *the truth*, but there is no similar resultative implication in *I have known/will know/knew/*etc. *the truth*. This means that, if there is an idea of resultativeness, we should not trace it back to the choice of the tense alone: the nature of the situation and the particular words chosen to describe it are equally important. More specifically, the situations that automatically suggest a resultant state of a particular kind (if used in a bounded sentence) are those that are represented as telic, i.e. as tending towards a natural end-point (see Bauer 1970; Dahl 1973:19; Nehls 1978:99; Zydatiss 1976:156, 1978:339).[62] Atelic sentences can also yield a resultative reading (e.g. *John has been in the cellar* can suggest 'So he must be dirty/cold'), but this reading is always one that is suggested by the context or the pragmatics of the situation rather than being an automatic consequence of the fact that a natural terminal point is reached, as in the case of telic situations.

c. What does it mean, then, to say that *I have opened the door* implies current relevance, whereas *I opened the door* does not? The answer is that, as a result of the different temporal schemata underlying the two tenses, the resultant state which is entailed by the situation of opening

61. If such a verb is accompanied by a duration adverbial, the latter is normally interpreted as indicating the duration of the result state rather than that of the situation: *Bill opened the door for five minutes* (see also Mommer 1986:9).

62. There is one type of present perfect sentence that yields a resultative reading in spite of the fact that the situation referred to is neither bounded nor telic:
    (i) You've been fighting again!
    (ii) Look at her eyes. *She's been crying.*
    (iii) You're dirty! – Yes. *I've been digging the garden.*
    Sentences like these typically refer to a situation which is not continuing any more (in spite of the use of the progressive form) but whose effects are still visible at t_0. (In most cases the effects in question are unintended side-effects.) This use is sometimes called the 'explanatory–resultative' use of the present perfect, because the sentences in question are mostly used to offer an explanation, excuse or apology for some existing state of affairs.
    The result referred to in a sentence of this type is special in that it is not the natural result produced by the completion of a telic situation but a result (often an unintended side-effect) which is produced by a (telic or atelic) situation as soon as the latter is in progress. (Hence the use of the progressive form. The nonprogressive form would refer to a different kind of effect, viz. to the effect that is normally produced if the situation is completed. For example, while *I've been digging the garden* in (iii) suggests 'That is why I am dirty', *I've dug (up) the garden* would suggest 'The garden is dug up now; it will not need digging for some time.')

the door is located in different time-spheres. In *I opened the door* the action and the concomitant resultant state are located in the past time-sphere, i.e. in the time-sphere which does not contain t_0 and which is conceptualized as unrelated to t_0. In consequence, *I opened the door* tells us no more than that the situation (with its resultant state) held THEN. It tells us nothing about whether or not the resultant state still obtains NOW. That is, *I opened the door* does not imply current relevance in the sense of 'persistence of the resultant state into the present': it implies no more than that the resultant state obtained immediately after the completion of the past action. The situation is different when the present perfect is used. The present perfect locates a situation BEFORE NOW, i.e. it represents it as holding within or throughout a period leading up to t_0. This means that the speaker is not concerned with the THEN of some past situation, but with NOW. It follows that, if the reference is to a situation implying a resultant state, this state will be interpreted as still obtaining at t_0, unless this interpretation is contextually or pragmatically excluded.[63] As we have seen, the sentence *I have opened the door* (if couched in an appropriate context and pronounced with a suitable accentuation pattern, i.e. with the nuclear accent on *have*) can be interpreted as 'There has been (at least) one occasion in my lifetime when I opened the door'. In that case there will be no implication of current relevance (in the sense that the resultant state still holds at t_0) because the lapse of time between the action and t_0 may be very large. However, *I have opened the door* can also be interpreted differently. In that case the implicit TE within which the action is understood to have taken place is not my lifetime but is a much shorter period. As a matter of fact, it is (as we have argued above) the shortest possible interval that is allowed by the pragmatics of the sentence, the context and the situation of speaking. On this interpretation the action will therefore be taken to have occurred in the recent past, so that (failing an indication to the contrary) there is no reason to assume that the resultant state is not obtaining any more.[64]

63. This analysis is consonant with the following suggestion by Woisetschlaeger (1976:83): What if the appropriate notion were, instead of relevance, something more like *maintenance of the status quo*? The status quo involved here would be whatever state arises automatically from the narrated event or situation, where 'automatically' just means 'in the absence of any unusual, disturbing, or interfering circumstances'. Maintenance, the second part of our descriptive phrase, is not to be endowed with any agentive overtones; it is to be understood in the sense of 'persistence through time'.

64. Fenn (1987:125) adduces the following sentences in support of the claim that indefinite ('experiential') and resultative perfects constitute two different categories:
    (i) You've never cleaned the car yourself. You've always paid someone to do it. – That's not true. *I* ***have cleaned*** *it, you know*. Several times.
    (ii) ***I've cleaned*** *the car*. So it's ready for tomorrow.
    According to Fenn, we have an 'experiential' perfect in (i) and a resultative one in (ii). It is claimed that 'this difference is a difference not merely between two individual

What I am arguing, then, is that current relevance (resultativeness) is no more than a conversational implicature attached to the use of the present perfect (of some verbs) as a result of the fact that the present perfect locates the situation in the present (rather than the past) time-sphere and therefore implies that the speaker is concerned with NOW rather than with THEN.[65] The implicature arises only when it is not denied by the pragmatics of the linguistic or extralinguistic context. This means that, like any conversational implicature, it can be cancelled. For example, the implicature 'I know his name' which arises from *She has already told me his name* is easily cancellable, as in *She has already told me his name, but I have forgotten it.*[66] (However, this cancellation will be impossible when the resultant state is irreversible, as in *John has died.*)

The claim that the difference between the present perfect and the preterit lies in their different temporal schemata, while current relevance is just a conversational implicature accompanying the use of the perfect, explains the observations that were made at the beginning of this section: not every present perfect verb form suggests a resultative interpretation; even when there is a clear idea of resultativeness, the preterit must be used when there is an indication of definite past time (e.g. *I know what Paris is like. I* ***lived*** *there during the war*); some present perfect sentences (e.g. *They've fallen into the river*) may suggest different resultative interpretations, according to the context in which they are used. All these observations accord with the theory that resultativeness is an implicature. However, they become totally inexplicable on the hypothesis that resultativeness is an inherent part of the meaning of the present perfect.

Another piece of evidence supporting this conclusion is that the resultative interpretation suggested by an indefinite perfect of a telic verb disappears when the perfect is made progressive:

utterances, but between two identifiable classes of context'. In my opinion, we have an indefinite perfect in both cases: the situation of cleaning the car is located somewhere in the pre-present. The difference between (i) and (ii) is of a pragmatic nature: in (i) the implicature that the state produced by the telic situation still holds at t_0 is cancelled, because *I have cleaned it, you know* is interpreted as referring to occasions that lie in the more distant past (so that there have probably been other occasions of cleaning the car since). In (ii), in contrast, the implicature is not cancelled, because there is nothing in the context that contradicts the assumption that the distance between the cleaning and t_0 is very short. (If this distance is very short, the cleaning in question represents the latest occasion of cleaning the car, and the resultant state is likely to be still holding.)

65. Matthews (1987:135) also treats resultativeness as an implicature of present perfect sentences that refer to a 'conclusive' (telic) situation. See also Givón (1984:280–4) and Salkie (1989:6): 'current relevance should ... be seen as a natural inference from the basic temporal meaning of the perfect'.

66. We therefore have to reject Moravcsik's (1982:101) claim that an indefinite perfect sentence 'is true just in case the completion took place at a point before the point of evaluation *and the state still holds*' (my emphasis).

(46) (a) I have learned to type.
(b) I have been learning to type.

Example (46,a) suggests that I can type now. This is because the ability to type is the goal implied by the telic verb phrase *learn to type*, and because the perfective verb form implies that this goal was reached. In (46,b) the progressive (imperfective) verb form refers to a stage of the situation at which the goal is not yet attained. Since the perfect represents t_0 as included in this stage, the sentence not only fails to suggest that I can type but actually suggests that I cannot type yet (since I am still in the process of learning it). We can therefore conclude that the kind of resultative reading suggested by (46,a) is only suggested by present perfect forms that are at the same time telic, indefinite (not continuative) and perfective (nonprogressive). And even then, the resultative interpretation (i.e. the idea that the state produced by the situation still holds at t_0) arises only by way of an implicature, which may be cancelled by the context.[67] All this runs counter to the view that resultativeness is the criterial characteristic of the present perfect tense.

d. The above conclusion is not in contradiction with the observation that the present perfect is extremely common as a means of expressing a present state. Speakers frequently choose sentences because of their implicatures. This has often been pointed out in connection with sentences like *It's cold in here*, which are used because of the implicated message 'I would like you to close the window'. In the same way we often use a present perfect as an indirect way of drawing attention to the implicated present result. This explains why a child may say *I've finished my homework* as an indirect means of asking for permission to go out to play (see Palmer 1988:49) and why we can answer the question *Is John here?* with *No, he has left.*

e. It should be noted, finally, that the term 'current relevance' has occasionally been linked up with a notion that is different from resultativeness, viz. the idea that 'the person or thing referred to must be living or existing and thus related to the present' (Inoue 1975:22). This idea has often been expressed in the literature, though it has

67. The same is true of other readings which an indefinite perfect may suggest and which have sometimes been treated as separate meanings of the perfect (next to 'indefinite' and 'continuative'), e.g. the 'experiential' reading and the 'fulfilment' reading. These readings suggest themselves chiefly in unbounded sentences where the implicature of resultativeness is cancelled (for example, by the fact that the distance between TO_{sit} and t_0 is rather long). As noted by Matthews (1987:136), 'the choice seems to be pragmatically determined. Thus *work in a factory, smoke, drive a cab* will normally *sui generis* prefer an "experiential" reading, while *play chess, dance, sing* will *sui generis* prefer a "fulfilment" reading.'

seldom been formulated in terms of current relevance. It is often illustrated by means of the following sentences:[68]

(47) (a) Einstein has visited Princeton.
(b) Princeton has been visited by Einstein.

Chomsky (1971) claimed that (47,a) presupposes that Einstein is still alive and is therefore unacceptable (because the presupposition is false), whereas (47,b) presupposes that Princeton still exists (which it does) and is therefore impeccable. However, McCawley (1973) has pointed out that it is not necessarily the referent of the subject NP that must still be in existence. The requirement holds for whichever NP is the topic of discourse. Thus, (47,a) is as acceptable as (47,b) in a context whose topic is Princeton University. Inoue (1975:27) adds to this that (47,a) is also all right when 'people are talking about who, among the Nobel Prize winners, has visited Princeton' and when people are discussing 'German scholars and how many of them have come to the United States over the years'. In both cases the topic of discourse is not Einstein but something else, and the case of Einstein is just brought up as an example. In my opinion, the fact that (47,a) is only acceptable in one of these settings is consonant with the claim that a present perfect sentence requires TO_{sit} to be located in a TE-up-to-t_0. If the TE is not established by an adverbial it must be recoverable from the context. In that case it will maximally be the time of existence of the entity (person or thing) that is currently the topic of discourse. In other words, it is in the first place the spatio-temporal restrictions of the topic that determine the length of TE. The acceptability of sentences like (47,a) is thus directly related to the question of whether or not the topic NP is pragmatically compatible with the idea of a TE-up-to-now.

It should also be noted that if the topic NP does not pragmatically exclude the idea of a TE-up-to-now, some other NP in the sentence can do so. McCawley (1981:354) draws attention to examples like the following (also discussed by Leech (1969)):

(48) (a) Have you seen the Monet exhibition?
(b) Did you see the Monet exhibition?

For sentence (48,a) to be acceptable it is necessary not only that the referent of *you* should not be incapacitated from seeing the exhibition but also that the context should allow the possibility of the exhibition being seen (e.g. the exhibition must not be already closed). Our general conclusion, then, is that a present perfect sentence will be acceptable

68. Similar sentences are discussed by Curme (1931:360), Jespersen (1931:66–7), Chomsky (1971), Leech (1971:35), Anderson (1973b), McGilvray (1974:36–7), Palmer (1974:53), Inoue (1975:40ff.), Lyons (1977:716), McCoard (1978:62–3), Barense (1980:48), McCawley (1981:353), Vermant (1983:85–8), Fenn (1987:188–93).

only if the TE-up-to-now which is pragmatically required for its interpretation is not incompatible with other semantico-pragmatic elements in the sentence and its context.[69]

There is another interesting observation to be made in this connection. Even if there are no contextual or pragmatic factors excluding the idea of a TE-up-to-t_0, some present perfect sentences are unacceptable. Compare:

(49) (a) Someone has painted the garage.
(b) *Someone has built the garage.
(50) (a) Who has corrected this manuscript?
(b) *Who has written this manuscript?

In each of these sentences the referents of the two noun phrases may still be in existence, yet the (b) sentences are unacceptable, whereas the (a) sentences are not.[70] According to Inoue (1975:41ff.), the difference has to do with the fact that the verb in the (b) sentences is a 'verb of creation'. What distinguishes such verbs from the verbs of the (a) sentences is that they presuppose that the referent of the object NP does not exist prior to the activity and that they refer to a situation which is unique and not repeatable. According to Inoue, the latter characteristic explains the ungrammaticality of the (b) sentences.[71] The fact that the same sentences with an indefinite object NP are grammatical (see (51,a–b)) is taken as evidence for this, as the indefinite NP allows the situation to be interpreted as repeatable:

(51) (a) Someone has built a garage.
(b) Who has written a manuscript?

In my opinion, however, repeatability is not the decisive factor. The following sentences are impeccable although the situation is obviously unique and not repeatable:

69. What we argue is consistent with the following remark by Fenn (1987:189) (in which he rejects particular claims made by Jespersen (1931:66–7)):
It is certainly not true, in my opinion, that *Newton has explained the movements of the moon* is acceptable simply on the strength of the fact that his theories are considered to be valid today. If the theme of discourse were the movements of the moon, if, that is, they had been mentioned in discourse prior to this actual utterance, the perfect would be unacceptable because of the occurrence focus brought about by reference to the agent, Newton. By the same token *Newton explained the movements of the moon from the attraction of the earth* need by no means presuppose or imply that this theory had been abandoned: the past tense here could be motivated, in a given context, simply by the fact that the speaker chooses to focus on the circumstances surrounding the event *explain*.
70. B. Comrie has pointed out to me that sentences similar to (49,b) may be acceptable in certain contexts. According to him, *John has built this wall* is fine, e.g. if the speaker is showing a visitor the various improvements his friends have made to his house. I will suggest an explanation for this below.
71. McCawley (1973) had suggested the same explanation.

(52) (a) John has died. (see *John has been born)
(b) The letter has been destroyed.
(c) Somebody has broken the vase.

In my opinion, the ungrammaticality of sentences like **Someone has built the garage* and **Who has written this manuscript?* is due to the fact that a speaker using a proposition like 'Someone build the garage' or 'Someone write this manuscript' cannot be concerned with the structure of the world at t_0 and therefore does not conceptualize these situations as lying in an implicit TE-up-to-t_0. That he cannot do so follows from two factors: the nature of the situation and the definiteness of the object NP. (If one of these factors is missing, as in *John has painted the garage* or *John has built a garage*, the present perfect is not ungrammatical.) The nature of the situation is relevant because the restriction on the perfect holds only for propositions like 'John write this novel', 'Someone build the garage', 'Columbus discover America', etc. and not for propositions like 'John read this novel', 'John paint the garage', 'John visit America', etc. What is typical of the former type of propositions is that they express the coming into existence of the referent of the object NP. (In the case of 'Columbus discover America' this is, of course, a coming into existence in a sense which is different from the most literal sense.) The definiteness of the object NP is also relevant, as a definite NP implies the existence of a referent that is identifiable to the hearer. That is, a speaker using a definite NP considers the current existence of the referent as given. This entails that when he uses a proposition referring to the coming into existence of this referent he is not concerned with the fact that the referent has come into existence and is now part of the structure of the world. Instead he is concerned with giving new information, i.e. information concerning one of the aspects (when?, how?, who?, etc.) of the situation that led to the existence of the referent. Thus, in sentences like *Who built the garage?*, *John wrote the novel*, *Columbus discovered America*, etc. the focus of interest is on the agent of the action (i.e. the question 'Who did it?'). Clearly, the speaker using the proposition 'John write the novel' is not concerned with the current existence of the novel. The fact that he uses the definite NP *the novel* means that he presupposes this existence. What he is concerned with is necessarily an aspect of the (past) activity of writing the novel. He is thus concerned with THEN, not with NOW, and this explains why he cannot use the present perfect.[72]

72. After struggling to achieve this understanding of why sentences like **The garage has been built by John* are ungrammatical, I was pleased to find that Dinsmore (1981: 490–1) offers essentially the same explanation.

The situation is different when the speaker uses an indefinite NP. Since an indefinite NP is typically used in a sentence which introduces the referent for the first time, the proposition 'John write a novel' can be used to refer to the existence of a novel by John. In expressing this, the speaker is concerned with the structure of the world NOW. He therefore uses the present perfect, thus locating the writing in an implicit TE-up-to-NOW.[73]

As a further illustration of the above analysis, consider the following sentences:

(53) (a) Shakespeare has written impressive dramas. (R. Lakoff 1970:844)
(b) *Shakespeare has written these dramas.
(c) *Shakespeare has written these dramas in London.
(54) Rome was not built in a day. (*has been built)

In (53,a) the speaker is concerned with the fact that there exist impressive dramas written by Shakespeare which now form part of the world literature. In (53,b–c) the existence of the dramas referred to is taken for granted (see the definiteness of the NP); the speaker is no longer concerned with NOW but with an aspect of the situation that happened THEN, viz. with the question of who was the agent performing the action or where it was that the action was performed. As explained above, this focus on the situation itself requires the use of the preterit. Sentence (54) can be accounted for in a similar way: the focus is not on the current existence of Rome (which is taken for granted) but on the duration of the activity.

The above theory also accounts for the fact that sentences like *Who has painted the garage*? are grammatical in spite of the definite object NP (although **Who has built the garage?* is unacceptable). The use of the definite NP *the garage* again means that the speaker presupposes the existence of the garage, but this does not clash with the perfect because the situation referred to (painting the garage) is not the bringing into existence of the garage. The situation is one which took place at some time in the course of the 'lifetime' of the garage. Since the garage obviously still exists, this lifetime constitutes a TE-up-to-t_0, in which the situation can be located by means of the present perfect.

A final thing to be noted is that there may be exceptions to the principle that has been formulated. As noted in footnote 70, a

73. The (in)definiteness of the object NP is thus of crucial importance. If the NP is definite, it means that the referent's existence as part of the structure of the world has already been established. Any sentence concerning its creation is therefore a sentence about an aspect of the situation, not about the structure of the world at t_0. If the NP is indefinite, the existence of the object as part of the structure of the world has not yet been established. The present perfect can therefore be used to establish this.

sentence like *John has built this wall for me* is not unacceptable, e.g. if the speaker is showing the hearer the various improvements his friends have made to his house. In spite of the verb of creation and the definite object NP, the speaker is then thinking of NOW rather than THEN. (The thought that is uppermost in his mind is probably something like 'Look what John has done for me', 'How very nice of him.')

3 Habitual sentences present no problem for the analysis of the present perfect that we have been arguing. As pointed out in chapter 6, habitual situations are states, which are located in time in exactly the same way as states of any other kind. The temporal schema underlying habitual sentences like *John has beaten his wife for some time* and *Jim has sung in our choir for five years* is therefore the same as that of *John has been in Venice for some time.* On the continuative interpretation, both types of sentences imply that TO_{sit} (and hence TS) coincides with the implicit TE-up-to-t_0; on the indefinite interpretation, both types imply that TO_{sit} is some nonfinal subinterval of the implicit TE-up-to-t_0.

4 Present perfect sentences that are repetitive but not habitual may also be of either the continuative or the indefinite type. The continuative ones (e.g. *Bill has been kicking that ball since breakfast*) again present no problem at all, as they are analysed in exactly the same way as nonrepetitive continuative sentences (e.g. *John has been in the kitchen since breakfast*). The only difference between the two types is that, when the sentence is repetitive, TS is understood to be an interval which comprises the TSs of several subsituations.

As far as the indefinite interpretation of nonhabitual repetitive sentences is concerned, there are three possibilities. A sentence like

(55) John has kicked the ball a couple of times.

can be true in three cases:

a. The sentence is true if there have been several occasions in the past on which John kicked the ball once. In this case several TSs are located in the pre-present sector (as shown by Figure 7.20).

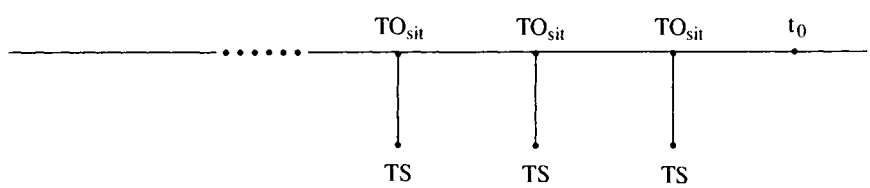


Figure 7.20

b. Sentence (55) is also true if there has been one occasion in the past on which John kicked the ball a couple of times. In this case there is

reference to one situation consisting of an indefinite number of subsituations (Figure 7.21).

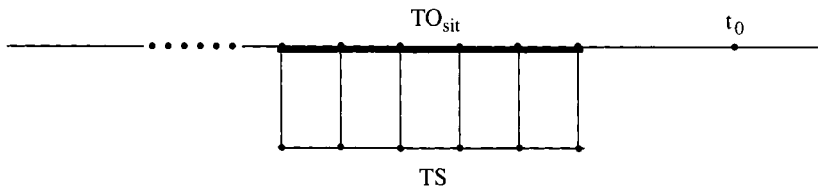


Figure 7.21

c. Sentence (55) is also true if there have been several occasions in the past on which John kicked the ball several times. In this case there is reference to several situations consisting of an indefinite number of subsituations (Figure 7.22).

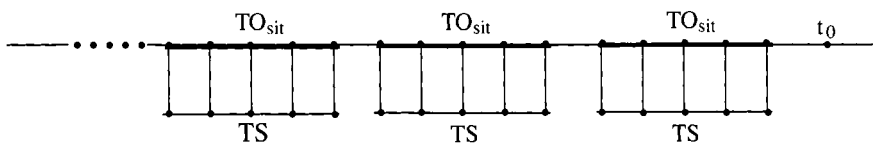


Figure 7.22

As noted before, an indefinite perfect can combine with an adverbial of time of the unanchored type which does not establish TE:

(56) John has arrived at five o'clock before, you know.

As usual, this sentence allows a repetitive interpretation: there have been (one or more) occasions in the past of John arriving at five o'clock. On this interpretation the time adverbial refers to the TO$_{sit}$s in the TE-up-to-now. What is especially noteworthy in connection with this is that each of these situation-TOs may itself behave as if it were t$_0$.[74] That is, it is possible to use a present perfect implying a TE-up-to-TO$_{sit}$ (instead of a TE-up-to-t$_0$) and representing another situation as holding within or throughout this TE. This is what happens in sentences like the following:

(57) (a) Every day, when I arrive at 12, Mary has been practising the piano since nine o'clock. (Rivière 1980:127)
(b) Every time I have seen John, he has been ill.

Sentence (57,a) says that there have been several occasions in the past when I arrived at twelve and that on each of these occasions the proposition 'Mary has been practising the piano since nine o'clock' was true. Similarly, (57,b) states that I have seen John several times in the

74. This is a case of shift of temporal perspective which we have not yet referred to.

past and that on each of these occasions the proposition 'John has been ill' was appropriate. This means that the strings *Mary has been practising the piano* and *John has been ill* have the same meanings in (57,a–b) as they normally have in isolation, except for the fact that the relevant TE is not a timespan reaching up to t_0 but a timespan reaching up to the times indicated by the time clauses (which in both sentences are interpreted as being indefinite times in a period up to the moment of speech). That (apart from this) the strings have their usual meanings appears from the fact that the string *John has been ill* is as ambiguous in (57,b) as it is in isolation: it yields either a continuative or an indefinite interpretation. That is, (57,b) means either 'Every time I have seen John he was ill and had been for some time' or 'Every time I have seen John he had been ill (but no longer was at that moment)' (see Huddleston 1969:4, Palmer 1988:69–70).

Sentences like (57,a–b) differ from other perfect sentences in that they involve an intermediate type of TO_1 in their temporal schema: whereas the present perfect normally represents TO_{sit} as an indefinite interval within or throughout a TE-up-to-t_0, the present perfect in (57,a–b) expresses that there are a number of TO_{sit}s, which are each indefinite intervals within or throughout a TE reaching up to an (each time different) TO_1, and that the latter TO_1s are themselves indefinite intervals in an implicit TE-up-to-t_0. Figure 7.23 and Figure 7.24 represent this for the continuative–repetitive reading of (57,b) and for the indefinite–repetitive reading, respectively:

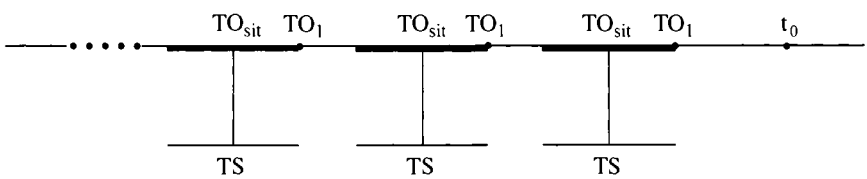


Figure 7.23 Represents the reading 'Every time I have seen John he was ill and had been for some time'

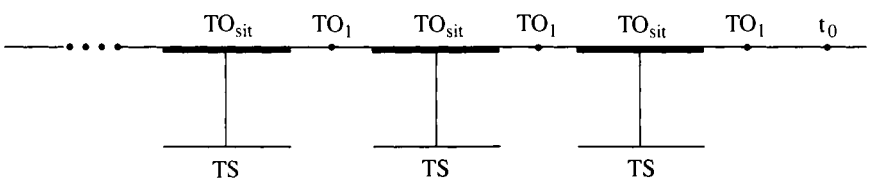


Figure 7.24 Represents the reading 'Every time I have seen John he had been ill (but no longer was at that moment)'

There are two further remarks to be made in this connection:

a. The present perfect cannot realize the above kind of temporal schema in a nonrepetitive sentence. The grammaticality of (57,b) contrasts

with the ungrammaticality of **When I have seen John, he has been ill* (in which *when* is to be read as 'on the particular occasion when'). The sentence that must be used instead is *When I saw John, he had been ill (for some time)*. The past perfect in this sentence will be assigned a structure that is not dissimilar to that of (57,b), i.e. with an additional TO intermediate between the situation-TO and t_0.

b. The use of an intermediate TO functioning as if it were t_0 is also exemplified by present perfect sentences of a different kind:

(58) He has been going to invite me for months (but he has not done so yet).

Here again the implication is that there have been several occasions in the past when a proposition was applicable which is normally applicable at the time of speech only (viz. the proposition 'He is going to invite me'). Again this use of the perfect is only possible on a repetitive reading ('For months he has been telling me that he was going to invite me').

5 In the preceding sections I have dealt with habitual and repetitive sentences. Generic and omnitemporal sentences can be drawn into the picture as well:

(59) (a) Two and two has always been four.
(b) Man has been afraid of wars throughout history.

Sentences like these must be interpreted continuatively and therefore require specification of a TE-up-to-t_0. They can be analysed in exactly the same way as other present perfect sentences that yield a continuative reading.

There also exist generic sentences in the present perfect that do not require explicit reference to a TE-up-to-t_0. The following are some examples:

(60) (a) A problem of this kind has been solved as soon as the necessary parameters have been found.
(b) A student has not studied the subject properly if he cannot answer this question.
(c) You have not really lived if you have never left your home village.

Sentences like these are generic because they express logical conditions and have a generic subject. A sentence with an indefinite generic subject NP refers to an indefinite number of 'cases' (see Lewis 1975) and hence an indefinite number of times. The analysis of such sentences is therefore similar to that of repetitive present perfect sentences like (57,a–b): like (57,a–b), (60,a–c) imply that there are an indefinite number of situations which each realize the temporal schema of the indefinite perfect, except

for the fact that the implicit TE is not a timespan leading up to t_0 but an interval leading up to some unspecified TO_1 within a timespan up to t_0. Thus, (60,a) is analysed in the same way as the repetitive sentence

(61) Every time, the problem has been solved as soon as the necessary parameters have been found.

The only difference between this sentence and (60,a) is that in (61) the idea of an indefinite number of cases (situations) arises from the repetitive meaning of *every time*, whereas in (60,a) it follows from the fact that the subject NP is generic.

6 It was pointed out in chapter 3 that when several situations are reported in the present perfect, they are not normally interpreted as related to each other (i.e. as sequential or simultaneous). On the other hand, situations described in the preterit are usually interpreted in this way. Compare:

(62) (a) I fell from a tree, I was run over by a bike, and I broke my left leg.
(b) I have fallen from a tree, I have been run over by a bike, and I have broken my left leg.

In both examples each clause establishes its own domain. In spite of this, sentence (62,a) suggests that the three situations followed each other in the order in which they are mentioned.[75] This is because of a particular pragmatic principle, which we have discussed in chapter 3. Sentence (62,b), on the other hand, does not suggest a sequential reading. It is interpreted as an account of the various accidents that I have had in my life, and which have not necessarily happened in the order in which they are reported here. In other words, the pre-present domains established by the indefinite present perfects in (62,b) are not interpreted as temporally related to each other. The reason for this should be clear from the analysis that has been assigned to the present perfect. An indefinite present perfect expresses no more than that the situation has held in the 'before NOW'. The precise temporal location of the situation is deemed irrelevant by the speaker, and hence also by the hearer. In consequence, examples like (62,b) will receive no temporal interpretation besides the claim that the situations have held in the pragmatically determined pre-present sector.[76]

75. In other words, the sentence suggests that the three situations form part of the same happening (occasion). This accords with the fact that the subject need not be repeated in the second and third clauses: *I fell from a tree, was run over by a bike, and broke my left leg.*

76. This explanation is confirmed by the fact that (62,b) becomes ungrammatical if we add adverbs that indicate the order of the situations (see Mommer 1986:227):
(i) (I have had several accidents in my life.) *First I have fallen from a tree, next I have been run over by a bike, and then I have broken my left leg.

4 THE PAST PERFECT

4.1 It has often been observed (see e.g. McCawley 1981:348)[77] that the pluperfect has two basic uses: one in which it is, as it were, the past of the preterit and one in which it is the past of the present perfect. Thus, there is a certain parallelism between (63,a) and (63,b) on the one hand, and between (64,a) and (64,b) on the other:

(63) (a) John has been here since yesterday.
(b) John had been there since the day before.
(64) (a) John left yesterday.
(b) John had left the day before.

This means that the temporal structure of either the past or the present perfect will also be part of the structure of the past perfect. As a matter of fact, the only difference between the past perfect and the former two tenses is that the part played by TO_1 (which is usually t_0) in the structures of the (absolute) preterit and present perfect is played by another TO (TO_2) in the structure of the pluperfect. This TO_2 is itself represented as (wholly) anterior to TO_1.

It follows that the temporal schema of a past perfect which is the 'past of a present perfect' involves the following elements and relations:

(65) (a) TS simul TO_{sit}.
(b) TO_{sit} either wholly precedes TO_2 or reaches up to it.
(c) TO_2 lies wholly before TO_1. (TO_1 is often t_0.)
(d) TO_2 belongs to the past time-sphere.
(e) There is an explicit or implicit TE-up-to-TO_2.
(f) This TE-up-to-TO_2 includes TO_{sit} (i.e. TO_{sit} either coincides with TE or is a proper subinterval of TE).

For example, the temporal structure of (63,b) (where the past perfect has continuative meaning) is represented by Figure 7.25. The structure of a sentence like *He had tasted lobster before* (which involves an indefinite past perfect) is represented by Figure 7.26.

The temporal structure of a past perfect which is the 'past of a preterit' consists of the following elements and relations:

(66) (a) TS simul TO_{sit}.
(b) TO_{sit} wholly before TO_2.

77. McCawley considers the following examples:
(i) When John married Sue, he had met Cynthia five years earlier.
(ii) When John married Sue, he had already read *War and Peace* three times.
McCawley notes that (i) is 'the past of a past' and (ii) 'the past of a present perfect', 'in the sense that the content of the main clause, if expressed at the time of John's marriage to Sue, would be expressed by a past tense (*John met Cynthia five years ago*) or a present perfect (*John has already read "War and Peace" three times*), respectively.'

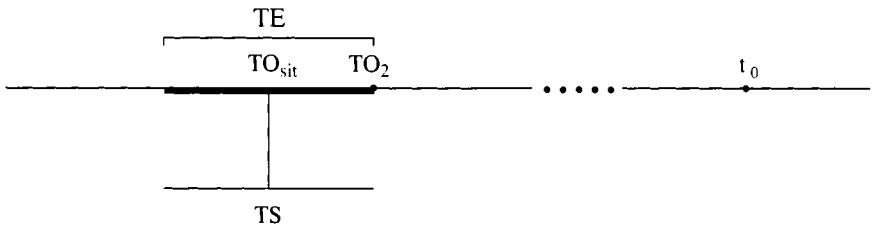


Figure 7.25

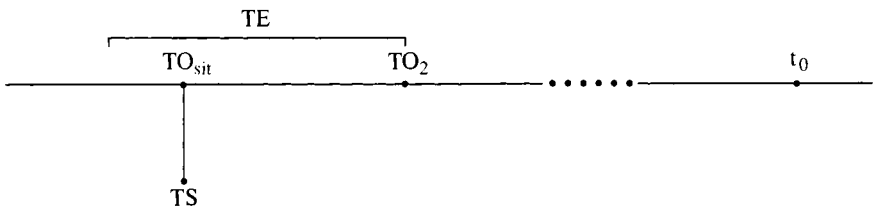


Figure 7.26

(c) TO_2 lies wholly before TO_1. (TO_1 is often t_0.)
(d) TO_2 belongs to the past time-sphere.

An example of such a temporal schema is Figure 7.27, which represents the structure of the past perfect in a sentence like *John had closed the door before he left.*

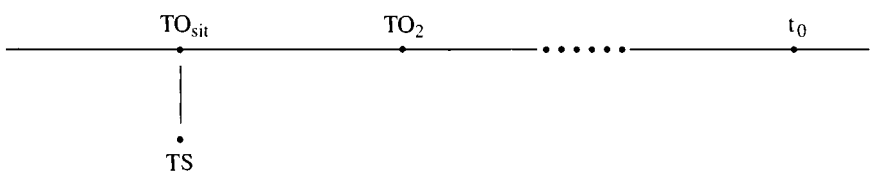


Figure 7.27

Apart from its two basic meanings (the 'past of a present perfect' and 'the past of a past'), the past perfect may also be the 'past of a past perfect'. This is the case in (67,a), which is the past report of (67,b):

(67) (a) John told me that the night before Bill had already informed him that he *had heard* from Susan that Peter *had left* before the others arrived.
(b) Yesterday night Bill had already informed me that he *had heard* from Susan that Peter *had left* before the others arrived.

As appears from such examples, a complex sentence may involve reference to quite a number of TOs lying between the TO_{sit} of the situation that is earliest in time and TO_1.

The observation that the past perfect may be, as it were, the past of a present perfect, the past of a preterit, or the past of a past perfect raises

the interesting question of whether the past perfect is three ways ambiguous. McCawley, who considers only the 'past of a past' and the 'past of a perfect' meanings, states that the past perfect is ambiguous between these meanings. Barense (1980:66) rejects this claim:

> McCawley has not shown that the past perfect is ambiguous. To accomplish this . . . he would have to offer a sentence containing a past perfect verb which has alternative paraphrases which are not paraphrases of each other. Then he would have to show that these alternative paraphrases are EXPLAINED by the fact that one is the past of a past and the other the past of a present perfect.

In my opinion, the question whether the past perfect is ambiguous in this way is a fallacious question, which would not suggest itself if the deceiving labels 'past of a preterit' and 'past of a present perfect' were not used. (These labels are deceiving because they are based on the 'formal SoT hypothesis' which we have rejected in chapter 4.) What we should say about the past perfect is that it always expresses anteriority in a past domain. As we have seen, anteriority may be either of the type 'before and up to' or of the type 'wholly before'. This distinction is one which may give rise to true ambiguity. A sentence like *John had lived in Paris for some time* is ambiguous between an indefinite and a continuative interpretation: John's living in Paris may or may not have lasted up to TO_2.[78] However, there is no other kind of ambiguity. The fact that there are two tenses that locate a situation before t_0 (viz. the preterit and the present perfect) and only one tense that locates a situation before a past TO does not entail that the latter is ambiguous. (The same point can be made in connection with the future tense, which is the only tense that locates a situation after t_0.) And the fact that the situation represented by the past perfect may be temporally related either to the central TO of the past domain (in which case we have a 'past of a past' reading) or to a TO which is itself anterior to the central TO (the 'past of a past perfect' reading) does not lead to ambiguity either. (In *He would say that he had seen her* the relevant situation is anterior to a TO which is itself posterior to the central TO of the past domain. Yet no one has ever suggested that the past perfect can be the 'past of a conditional tense'.)

4.2 If there is a time adverbial specifying a TE, the relation between TO_{sit} and TE is exactly the same as we have observed in connection with the preterit: TO_{sit} may coincide with TE or (if TE is not punctual) be a proper subinterval of TE. This gives us four possibilities, exemplified by the following sentences: (a) *John had left at three o'clock* (Fig. 7.28);

78. As noted in the section on the present perfect, this kind of ambiguity arises only when the situation is unbounded and there is no adverbial excluding either of the two readings.

(b) *John had left the day before* (Fig. 7.29); (c) *John had been there for a while the day before* (Fig. 7.30); (d) *The day before John had been there all day* (Fig. 7.31).

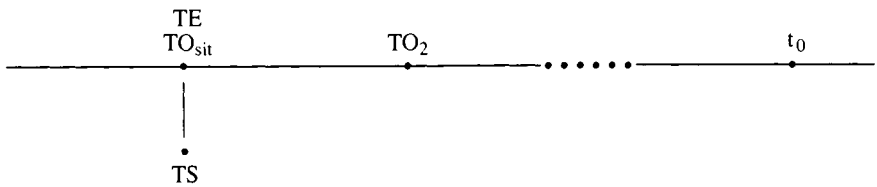


Figure 7.28 Example: *John had left at three o'clock*

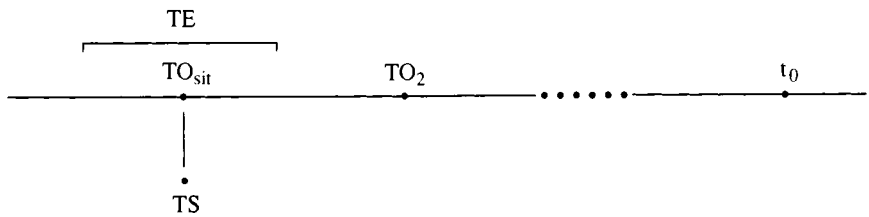


Figure 7.29 Example: *John had left the day before*

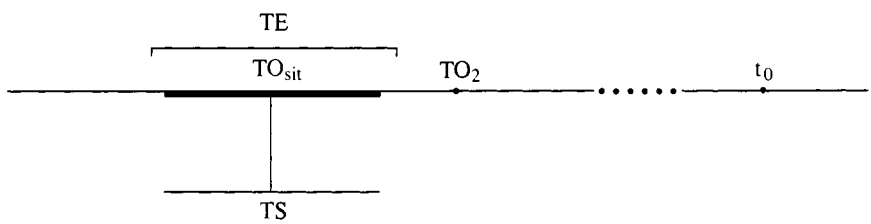


Figure 7.30 Example: *John had been there for a while the day before*

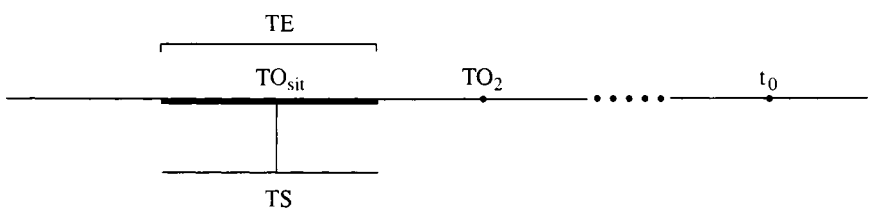


Figure 7.31 Example: *The day before John had been there all day*

In the above examples, TE is a time which wholly precedes TO_2 (and hence t_0). However, this is not necessarily the case. Since the relevant 'before' relation does not involve TE (as it is TO_{sit} and not TE that is located before TO_2), there is nothing that excludes the possibility that TE should also include TO_2, and even t_0. The former possibility is represented by Figure 7.32, which shows the temporal structure of *Yesterday John had already left when we arrived.* (TS is the time of John's

leaving; TE is yesterday; TO_2 is the time of our arrival.) Figure 7.33 represents the schema realized by *Today John had already left when we arrived.* (TE is today; TS and TO_2 are as in the previous example.)

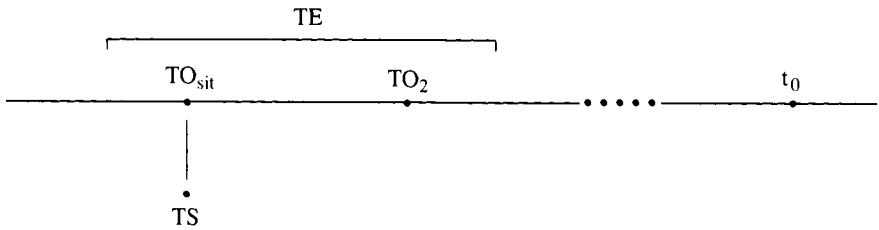


Figure 7.32 Example: *Yesterday John had already left when we arrived*

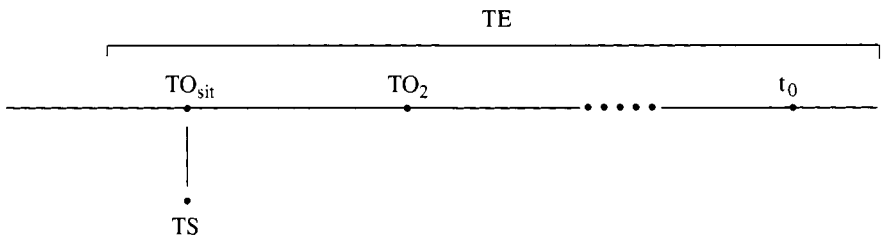


Figure 7.33 Example: *Today John had already left when we arrived*

The observation that a past TE such as that indicated by *yesterday* may include either TO_{sit} only or both TO_{sit} and TO_2 makes us wonder whether it is possible for TE to include TO_2 but not TO_{sit}. If this appears to be possible, it is probably a general principle that, when there is an adverbial establishing a TE in the past time-sphere, this TE may indicate (i.e. include) any TO or combination of TOs. (The latter possibility is, of course, not available when TE is punctual.) Actually, it does turn out to be the case that TE can include TO_2 without including TO_{sit}. This appears from an example like *The others had (already) left at five o'clock.* Here *at five o'clock* can indicate either TO_{sit} (the time when the others left) or TO_2 (the time prior to which the others left, i.e. the time at which the others had already left).[79] Either of these interpretations can be brought out by the context:

(68) (a) When I arrived late in the evening, John was the only one who was still there. *Everybody else had left at five o'clock.*
(b) I arrived at five o'clock, but it was too late. *Everybody had (already) left at five o'clock.*

79. When I say that an adverbial 'indicates' a TO, I really mean that the adverbial 'establishes a TE which includes' the TO in question. In what follows I will frequently use the former (sloppy) kind of formulation because it is simpler and less cumbersome than the latter.

As noted by Hornstein (1977:532) and Dinsmore (1982:225), the latter interpretation becomes predominant when the adverbial is put in initial position (*At five o'clock everybody had left*)[80] and is virtually obligatory when *already* is added.

The possibility for a time adverbial to refer to a TE which includes either TO_{sit} or TO_2 exists not only for 'unanchored' adverbials like *at five o'clock* but also for 'anchored' ones, such as *yesterday*. A sentence such as *Bill had already done it yesterday* can mean either 'Yesterday it was the case that Bill had already done it' or 'It was the case that Bill had already done it yesterday'. (The latter interpretation is invited by a context like the following:

(69) When I wanted to do it this morning, I found that *Bill had already done it yesterday.*)

Since we have seen that a durative TE such as the one indicated by *yesterday* can also include both TO_{sit} and TO_2 (see Fig. 7.32), we must conclude that sentences in the pluperfect which contain a durative adverbial indicating past time should in principle allow three different interpretations as to the time indicated by the adverbial.[81]

This does not mean, however, that the three possibilities are always equally available. As already noted, an adverbial that occupies front position or is accompanied by *already* is normally interpreted as

80. This may be part of a larger phenomenon. Dowty (1979:343–6) draws attention to the following examples:
    (i) (a) John was solving the puzzle in five minutes.
        (b) In five minutes, John was solving the puzzle.
    (ii) (a) John has lived in Boston for four years.
        (b) For four years, John has lived in Boston.
    (iii) (a) John has been in Chicago since 1971.
        (b) Since 1971, John has been in Chicago.
    Sentence (i,a) is ambiguous between the readings 'before five minutes had elapsed John was trying to solve the puzzle' and 'John was trying to solve the puzzle, for which five minutes' time had been allotted to him'. The latter reading (on which the adverbial specifies the duration of TS (= TO_{sit})) is no longer available in (i,b). Sentences (ii,a) and (iii,a) are both ambiguous between a continuative and a (less salient) indefinite reading. The latter is virtually excluded in (ii,b) and (iii,b). If we recall that adverbials like *for four years* and *since 1971* specify a TE which coincides with TO_{sit} when the interpretation is continuative and which properly includes TO_{sit} on the indefinite reading, then we see that it is each time the reading on which TE coincides with TO_{sit} that gets lost (or at least becomes relatively inaccessible) when the adverbial is preposed. This also appears to be the case in *At five o'clock John had left.* In all four cases the only available reading is that on which the preposed adverbial relates to a time other than TO_{sit}.
81. The fact that a time adverbial accompanying a past perfect may indicate either (what we call) TO_{sit} or (what we call) TO_2 has often been noted in the linguistic literature (see e.g. Smith 1975:77; Hornstein 1977:532, 1981:128; Heny 1982:140; Dinsmore 1982: 225). However, these linguists have not drawn attention to the fact that the time adverbial may often also indicate a timespan that comprises both TO_{sit} and TO_2.

indicating TO_2.[82] In other cases the context often guides us towards one of the possibilities (as in (69)). Moreover, not all kinds of time adverbial allow the two or three possibilities provided for by the general principle. The following restrictions should be pointed out:

1 As noted above, it is self-evident that punctual adverbials like *at five o'clock* cannot establish a TE including both TO_{sit} and TO_2, because such a TE is necessarily durative.

2 Bounded adverbials like *from two o'clock to four* cannot be interpreted as specifying different time intervals. This follows from the fact that bounded time adverbials entail that TE = TO_{sit} = TS (see chapter 6). This means that a bounded adverbial can only specify TO_{sit}, even if it is preposed (as in *From two o'clock to four John had been in the house*). (What is possible, of course, is that TO_{sit} reaches up to TO_2, as in *John had been in the house from two o'clock until then.* In that case too TE coincides with TO_{sit}.)

3 The same conclusion holds for time adverbials that signal simultaneity between TO_{sit} (= TS) and some other situation:

(70) John had been watching TV while his mother was away.

Since the *while*-clause indicates a TE which must be interpreted as simultaneous with the TO_{sit} (and hence with the TS) of the head clause, and since this TO_{sit} lies wholly before TO_2, TE cannot be interpreted as including TO_2. (In other words: the TS (= TO_{sit}) referred to by the temporal clause cannot include TO_2.)[83]

4 Time adverbials that do not indicate a time precisely tend to be related to TO_{sit} rather than to TO_2. For example:

(71) (a) John had left at approximately two o'clock.
(b) We had received the letter about noon.
(c) I had met her one day in August.
(d) I had been introduced to him at some party or other.

We will not normally interpret the adverbials as establishing TO_2 in such sentences (except, perhaps, when the time indication is an echo from the

82. Sharwood Smith (1982:68) notes that the tendency to interpret a preposed time adverbial as indicating TO_2 is especially strong when the sentence is negative or interrogative:
(i) At five, Gerald had not left the office.
(ii) At five, had Gerald left the office?
However, he also points out that contrastive stress on anything except the time adverbial can still reverse the reading in favour of the other interpretation:
(iii) At five, Gerald had not LEFT the office: he had in fact ARRIVED then.

83. This does not mean, of course, that the full situation of mother's being away cannot include TO_2. (Since the *while*-clause is unbounded, it may not refer to the full situation. If it does not, the full situation may extend beyond the TO_{sit} of the head clause and include TO_2.)

preceding context). Apparently TO_2 (i.e. the 'reference time' in the structure of the past perfect) is preferably better identifiable than by means of an adverbial indicating approximate time. This requirement is also clear from the fact that adverbials of the type '*after/before* + NP' are normally associated with TO_{sit} rather than with TO_2:

(72) (a) John had left after breakfast.
(b) We had received the letter before noon.

Adverbials like *after breakfast* and *before noon* are indefinite indications of time. As noted in chapter 6, they are interpreted as 'at some time after/before ...'. This means that they denote a timespan which is indefinitely long (though it may be pragmatically restricted) because only one of the two boundaries is specified, and that they offer no clue as to which particular part of the timespan is actually the time that is relevant to the interpretation of the sentence. (In other words, they are unbounded time adverbials, which leave vague the precise location of the relevant TO within TE). Such indefinite time indications will normally be associated with TO_{sit} rather than with TO_2. The reason is that TO_2 is the time to which TO_{sit} is temporally related. There is little point in 'anchoring' TO_{sit} to a TO if the temporal location of this TO is unclear.

5 Prepositional phrases with *by* can only indicate TO_2:

(73) John had left by five o'clock.

It appears, then, that the prepositions *by* and *before* show a kind of complementary distribution: *by five o'clock* can only indicate TO_2, *before five o'clock* can only indicate TO_{sit}.

6 Time adverbials that indicate a TE-up-to-TO_1 when they collocate with the present perfect (where TO_1 is usually t_0) indicate a TE-up-to-TO_2 when they are used with the past perfect. In that case the TE in question must also include TO_{sit}. This is clear from examples like the following:

(74) (a) He had been there before.
(b) She had not seen him since seven o'clock.

7 In connection with the present perfect we have seen that adverbials like *just, this minute, recently*, etc. refer in a very indefinite way to the position of TO_{sit} within the TE-up-to-TO_1. That is, they measure the distance between TO_{sit} and TO_1. These adverbials have a similar function when they collocate with the past perfect: they give us an indication of what is the length of time between TO_{sit} and TO_2. Since this is the only temporal information they can convey, they differ from adverbials

like *yesterday* in that they cannot be interpreted as specifying either TO_{sit} or TO_2 or a timespan including both.

8 In sentences like the following, the time adverbial can only be interpreted as referring to TO_2:

(75) (a) When I came back, Mary had written two letters.
(b) At that time the civil war had cost the lives of several hundred people.

Although the past perfect is interpreted here as expressing anteriority of the type 'before and up to', the time adverbial does not indicate the TE-up-to-TO_2. It can only be interpreted as indicating TO_2, i.e. as indicating the (perhaps provisional) end point of the TE. The reason is that the time indicated by the adverbial is too short to be the TE-up-to-TO_2 as a whole. The situations referred to in (75,a–b) are 'cumulative situations, which necessarily take a certain time (viz. the whole of the TE-up-to-TO_2.) In both sentences this time is necessarily longer (for pragmatic reasons) than the time indicated by the adverbial. The adverbial is therefore interpreted as referring to TO_2.[84] (This is consistent with the observation that an adverbial of this kind strongly prefers front position.)

9 A time adverbial which can in principle refer either to TO_{sit} or to TO_2 can only be interpreted as referring to TO_{sit} if the sentence is progressive:

(76) John had been reading a book at five o'clock.

As we have seen, a progressive sentence represents a situation as in progress at TO_{sit} and therefore depends for its temporal interpretation on the identification of TO_{sit}. It follows that, if TO_{sit} is not indicated by the context, the time adverbial is automatically interpreted as indicating TO_{sit}.

10 As we have seen in chapter 6, relative time adverbials like *the day before* have a double function: they both indicate TO_{sit} and temporally relate it to a TO (which, in the case of the past perfect, is TO_2). This means that they cannot be interpreted as indicating either TO_{sit} or TO_2:

(77) John had left the day before.

11 In examples like *Afterwards I was glad that I had left when I did* the *when*-clause can only be interpreted as indicating the TO_{sit} of the

84. In (i) the time indicated by the adverbial is long enough to be interpretable as TE-up-to-TO_2. This explains Huddleston's (1969:5) remark that (i) is ambiguous:
(i) In March John had read only two of the books.
The ambiguity results from the fact that *in March can* indicate either TO_2 or the TE-up-to-TO_2. (The former interpretation is the more salient because *in March* occupies the front position.)

that-clause. This follows logically from the fact that *did* is a pro-form for the VP of the *that*-clause and expresses simultaneity.

In sum, there are various kinds of restrictions on the general rule that a time adverbial accompanying a past perfect can in principle establish a TE which includes either TO_{sit} or TO_2 or both.[85] However, apart from these exceptions, the rule does appear to be valid. As a matter of fact, we will see below that the same principle is also at work in the structure of other tenses that involve one or more intermediate TOs between TO_{sit} and TO_1.[86]

4.3 There is another interesting observation to be made in connection with the adverbial specification of TOs. As pointed out by Bertinetto (1982:74–6), a sentence like (78,a) is ungrammatical, although the corresponding two sentence sequence (78,b) is fine:

(78) (a) *At 2 p.m. John had left before midday.
(b) It was already 2 p.m. John had left before midday.

The only difference between (78,a) and (78,b) is that the time which is TO_2 in the structure of the past perfect (viz. 2 p.m.) is indicated adverbially in the past perfect clause in (78,a), whereas in (78,b) it is identified in the preceding sentence. Why this should entail a difference

85. The above list of restrictions does not mention a constraint whose existence is claimed by Ejerhed Braroe (1974:135–6). The claim is that the possibility of associating the time adverbial with (what we call) TO_2 is lost when the clause is embedded under a nonfactive verb. According to Ejerhed Braroe, this should be clear from the following:
(i) John had cashed my check on the 25th.
(ii) When I talked to him, John regretted that he had cashed my check on the 25th.
(iii) When I talked to him, John believed that he had cashed my check on the 25th.
In (i), *on the 25th* can indicate either the day on which the check was cashed (TO_{sit}) or a day prior to which it was cashed (TO_2). According to Ejerhed Braroe, this is also the case in (ii), but not in (iii): in the latter sentence the adverbial can only refer to the day on which the check was cashed. Ejerhed Braroe claims that this difference between (ii) and (iii) is due to the fact that *regret* is factive, whereas *believe* is nonfactive.
I have not mentioned this putative restriction because I do not believe that the claim concerning (iii) is correct. In my opinion, (iii) yields exactly the same interpretations as (i) and (ii). The reading on which *on the 25th* indicates TO_2 can be brought out in the usual ways, i.e. by fronting the adverbial or by the addition of *already*:
(iv) When I talked to him, John believed that on the 25th he had cashed my check.
(v) When I talked to him, John believed that on the 25th he had already cashed my check.

86. The principle also holds for sentences like *He was leaving yesterday*. As shown in chapter 2 (section 24), such a sentence has an 'implicit TO', viz. the time at which the referent of the subject NP voiced his intention to leave. There are thus two TOs in (i), viz. the implicit TO and TO_{sit}. Our principle predicts that *yesterday* can indicate either (or both) of them. This prediction is borne out, for Huddleston (1977:730) points out that *He was leaving yesterday* is ambiguous between the readings 'It was planned/scheduled that yesterday he should leave' and 'Yesterday the plan/schedule was that he should leave at some unspecified future time.' (Huddleston neglects the possibility that the time indicated by *yesterday* encompasses both TOs.)

in grammaticality is not immediately clear.[87] After having a speculative look at a couple of tentative explanations, Bertinetto (1982:91–5) finally comes to the conclusion that the best he can say is that the ungrammaticality of (78,a) is due to a language-specific constraint: apparently English contains a rule which stipulates that 'any sentence is ungrammatical whenever it contains, within its own boundaries . . . an L and an explicit R' (p. 94). (Bertinetto's R ('reference time') corresponds with our TO_2, whereas L ('location') corresponds with our TE.) This rule not only accounts for the ungrammaticality of (78,a) but is also in keeping with a general principle that we have argued in chapter 6, viz. the principle that there must be only one time specification per clause. That is, if there are several time adverbials in one clause, they must normally co-operate in establishing reference to one single time interval (as in *I left at five o'clock yesterday*). However, it was pointed out in chapter 6 that there may be exceptions to this rule. It is therefore not surprising that Bertinetto's rule turns out to be too restrictive. It rules out not only sentences like (78,a) or (79,a–b), which are indeed ungrammatical,[88] but also examples such as (80,a–e), which appear to be grammatical:

(79) (a) *At midday, Bill had arrived at 10 a.m./before lunch.
(b) *When I arrived, Bill had arrived before lunch.

(80) (a) By the end of the day, two soldiers had been killed since the beginning of the manoeuvres.
(b) When I arrived, Peter had tried to phone me twice during the preceding week. (Dahl 1985:30)
(c) When we arrived, our host had died a couple of hours earlier/before.
(d) When I got acquainted with him, Bill had never yet said a prayer before eating.
(e) When he came to help us last week, John had got up at six o'clock.

87. Smith (1976a:5) also observes that a sentence like (78,a) is 'odd' and offers the easy explanation that this unacceptability is due to the fact that 'with two calendar times, useless information is given'. It is clear, however, that this explanation is untenable. Sentence (78,a) offers no more information than (78,b), which is impeccable.
The ungrammaticality of sentences like (78,a) is also noted by Rigter (1980a:27). His explanation is that, once the reference time has been specified, the event time can only be specified in terms of its relation to the reference time. However, this does not explain why (78,b) is so much better than (78,a).

88. I am disregarding the reading on which *At 2 p.m. John had left before midday* is short for 'At 2 p.m. X said that John had left before midday'. This reading is possible e.g. in a context where a policeman has repeatedly asked someone by what time John had left and has received different answers at different times. In such a context the policeman might say:
(i) At 2 p.m. John had left before midday, but at 3 p.m. he had left in the afternoon. Which was it?

It would appear from (80,a–e) that we must formulate a couple of restrictions on the rule that there must be only one time specification per clause (hence that it is not possible to specify both TO_2 and the TE including TO_{sit} in one and the same sentence). The exceptions appear to be the following:

1 There is no ungrammaticality if one adverbial specifies TO_2 whereas the other specifies a TE-up-to-TO_2. This is the case in (80,a), where *by the end of the day* establishes TO_2 and *since the beginning of the manoeuvres* indicates a timespan leading up to TO_2 (the end of the day). This also accounts for (80,b), where *when I arrived* specifies TO_2 and *during the preceding week* indicates a timespan leading up to TO_2.

2 The sentence is not ungrammatical either if the adverbial establishing TE is what we have called a 'relative time adverbial' (see chapter 6), i.e. if the adverbial not only specifies a time but also relates it to TO_2. This is the case in (80,c), where *when we arrived* specifies TO_2 and *a couple of hours earlier/before* indicates a time interval relative to TO_2.

3 The sentence may also be grammatical if one time adverbial refers to TO_{sit}, whereas the other specifies either TO_2 or a TE-up-to-TO_2. This is the case in (80,d) and in the following examples:

(81) (a) Up to then John had often got up at five o'clock.
(b) When I met John, he had often got up at five o'clock.

To understand the temporal structure of such sentences we should compare them with the corresponding present perfect sentences:

(82) (a) Up to now John has often got up at five o'clock.
(b) John has often got up at five o'clock.

As we saw in section 3, the adverbial *at five o'clock* does not establish TE (since TE is a period-up-to-TO_1) but belongs to the temporal structure of the situation itself. That is, (82,a–b) both refer to a repetitive situation consisting of a number of subsituations, each of which is located at five o'clock (of an each time different day). Exactly the same thing happens in (81,a–b): *at five o'clock* forms part of the description of the repetitive situation that is being located in time. And the same thing is true of (80,d). Here the head clause is not interpreted as being about the situation of Bill not having said a prayer yet and as locating that situation at the time identified by *before eating*. Rather, the sentence is about the situation of Bill saying a prayer before eating: it predicates of this situation that it had not held yet at TO_2. In other words, *before eating* does not establish a TE relative to which a situation is located; rather it is part of the situation that is being described.

In sum, it appears from (80,d), (81,a–b) and (82,a–b) that a sentence that locates a repetitive situation in a period up to some TO may involve

two time adverbials if one forms part of the situation described whereas the other specifies either the TO in question or the period leading up to it.

4 Finally, there is the grammaticality of (80,e) (repeated here), which is in strong contrast with the ungrammaticality of (83):

(80) (e) When he came to help us last week, John had got up at six o'clock.
(83) *At seven o'clock, John had got up at six o'clock.

I suggest that the difference between (80,e) and (83) is due to the different nature of the fronted adverbial: in (83) the adverbial is a very precise specification of TO_2; in (80,e) it is much vaguer. In the latter example, *when* means 'on the occasion when' rather than 'at the time when', which means that the *when*-clause is not a purely temporal adverbial. It is presumably this 'nontemporal' element in the interpretation of the adverbial that renders the sentence grammatical. For *when*-clauses whose primary function is to define 'cases' (in the sense of Lewis (1975)) and not to specify time can easily co-occur with a time adverbial establishing a TE:[89]

(84) (a) When they are not treated adequately, such patients may die within the hour.
(b) Children do not always wash at seven o'clock/after getting up when they have no parents to look after them.
(c) When John has to catch the early plane, he gets up at 4.30.

In such examples the *when*-clause does not refer to a specific time interval which is different from the time established by the other adverbial. Rather, it defines a number of 'cases', and each case is felt to cover a length of time which includes the time indicated by the other adverbial. This is also true of (80,e), where the particular case defined is associated with a particular day – *when* is interpreted as 'on the day when' – and the TE (six o'clock) is an interval in the course of that day.

The ensuing generalization seems to be that we cannot add two different time adverbials to a sentence (the one specifying a TO and the other specifying a TE including TO_{sit}), unless there is some kind of relation between the TO and the TE. This relation may be one of the following:

a. TE reaches up to TO. (This is our above exception 1.)
b. TE is established by a relational adverb which relates TE to TO (our above exception 2).
c. TO is felt to be included in TE (exception 4).

89. For a discussion of such 'atemporal' or 'restrictive' *when*-clauses see Carlson (1979), Farkas and Sugioka (1983), Declerck (1988b).

The sentences discussed under (3), in which the second time adverbial specifies a time that belongs to the structure of the repetitive situation itself, form no real exceptions, because the time adverbial does not specify a TE including TO_{sit}.

4.4 There is another observation that is accounted for by the temporal structure that we have assigned to the past perfect. Like the preterit, the past perfect expresses that TS lies wholly before TO_1, but it tells us nothing about the temporal relation between the 'full situation' and TO_1 (except if the situation is represented as bounded, in which case TS coincides with the full situation). Consider:

(85) (a) John had had breakfast when he came to me half an hour ago.
(b) John had not had breakfast yet when he came to me half an hour ago.

In (85,a) the head clause situation is represented as bounded. This means that the situation as a whole is referred to in the sentence (i.e. TS is the time of the full situation of John having breakfast). Since the past perfect locates TS (= TO_{sit}) relative to a TO_2 which lies wholly before t_0 (which functions as TO_1 here), the sentence excludes the possibility that the full situation includes t_0. In (85,b), on the other hand, the situation is represented as unbounded (since the situation of John not having had breakfast yet is a state, and states are by definition unbounded (unless they are bounded by the addition of a specific duration adverbial like *for two hours*)). This means that the full situation may extend beyond TS, so that we may expect the sentence to be compatible with the possibility that the full situation includes t_0. This is actually the case, for it is quite irrelevant to the truth and acceptability of (85,b) whether or not the situation of John not having had breakfast yet is still holding at t_0.

4.5 When the past perfect represents TO_{sit} as anterior to a past TO_2 established by a different clause, and when both TO_{sit} and TO_2 are adverbially specified, the relation of anteriority may or may not be also explicitly expressed by the choice of a relative adverbial indicating TO_{sit}. For example, we can use either (86,a) or (86,b) to refer to the same sequence of situations in the past:

(86) (a) John left on Tuesday after the others had arrived the day before.
(b) John left on Tuesday after the others had arrived on Monday.

In (86,a) the anteriority relation between the two situations is expressed not only by the past perfect *had arrived* but also by the relative adverbial

the day before. In (86,b) this is not the case: *on Monday* is not a relative adverbial. It represents a timespan as anchored (related) not to the time indicated by *on Tuesday* but directly to t_0.

As a matter of fact, not only the adverbial but also the tense form may be directly related to t_0. Next to (86,a–b) we can also say

(87) John left on Tuesday after the others arrived on Monday.

In this sentence neither the adverbials nor the tense forms are anchored to each other. Everything is anchored to t_0. In chapter 2 we have analysed this possibility in terms of a 'shift of domain'. We have pointed out that the shift is possible because the relation of anteriority between the two situations is anyhow expressed by the use of the conjunction *after* (and, in this particular example, by the use of *on Monday* versus *on Tuesday*).

4.6 To close off this section on the past perfect I will briefly consider a remark made by Tichý (1980:362–3). According to Tichý, 'it is often mistakenly maintained that the English pluperfect serves to indicate which of two past events came earlier than the other'. To show that this claim is mistaken, Tichý considers the following sentence:

(88) On 1 January 1977 Tom was drunk when he had finished breakfast.

According to Tichý, the past perfect here does not serve 'to indicate that the breakfast took place before the drunkenness'. In his opinion, the pluperfect does not have 'anything to do with the temporal relationship between Tom's getting drunk and Tom's finishing his breakfast' since (88) 'implies that the former event took place before the latter'. In my opinion, this observation is quite correct, but it does not warrant the conclusion that the past perfect does not indicate anteriority (i.e. does not indicate 'which of two past events came earlier than the other'). The flaw in Tichý's argument is that it assumes that the 'event' expressed in the head clause of (88) is Tom's *getting* drunk, whereas in fact it is Tom's *being* drunk. Since being drunk is a state (and hence by definition homogeneous and dissective), it may be represented as holding at a particular point of time even if the 'full' state occupies a much longer timespan (see chapter 6). Thus, what happens in (88) is that the state of Tom being drunk is located at a past time at which it was the case that Tom had already finished his breakfast. No other temporal relations are expressed. The fact that Tom's getting drunk (i.e. the beginning of the state) must have preceded the finishing of the breakfast has nothing to do with the semantic interpretation of (88). Tichý's claim that (88) proves

that the past perfect does not express anteriority is therefore without any foundation.[90]

5 THE FUTURE TENSE

If we wish to represent a situation as lying before t_0 we can choose either the preterit or the present perfect. However, if we wish to represent a situation as posterior to t_0, there is only one tense that we can use, viz. the future tense. It is therefore to be expected that the future tense can effect a double kind of location in time, i.e. that it can represent a situation either as lying wholly after t_0 or as holding from t_0 onwards. These two possibilities would run parallel to the two ways in which a situation can be located before t_0: 'before' can mean either 'wholly before' or 'before and up to'. As a matter of fact, the expectation that the future tense can effect this double kind of location in time is borne out by examples like the following:

(89) (a) I will do it next week.
(b) They will arrive any minute now.
(90) (a) From now on nobody will receive any more grants from us.
(b) I will use the notation 'X > Y' to represent the relation 'X after Y'.
(c) The term which I will henceforth use to refer to this phenomenon is . . .
(d) As from today I will give the orders here.
(e) We will stay here for another week.

In (89,a–b) the future tense locates the situation completely after t_0. In (90,a–e) the situation is located in a timespan which includes t_0 and stretches into the future. Thus, (90,a) can be said to someone who, at t_0, is applying for a grant. The sentence then serves to inform the applicant of the fact that no grant can be given to him. It is clear that in that case the situation referred to already holds at t_0. The same is true of (90,b), since the speaker is already using the particular kind of notation he is referring to. And similar remarks can be made in connection with (90,c–e).

90. Tichý also considers it as evidence for his claim that we can not only say (i,a) but also (i,b), and that we can say (ii) (where the pluperfect appears in the *before*-clause 'although the sentence implies that the event reported in the main clause took place *before* the finishing of the breakfast)':
(i) (a) He had opened the window before he got into bed.
(b) He opened the window before he got into bed.
(ii) On 1 January 1977 Tom was drunk before he had finished breakfast.
However, neither (i,b) nor (ii) supports Tichý's claim. In (i,b) there is a shift of domain, so that no anteriority is expressed, but this does not alter the fact that anteriority is expressed by the past perfect in (i,a). And the same is true in (ii), where the past perfect represents the situation as anterior to some implicit TO (see section 24 of chapter 2).

It should be noted that the future tense is not the only means we can use to express the relation 'TO_{sit} from t_0 onwards'. We can also make use of the present tense or the construction with *be going to*:

(91) (a) I am going to use the notation 'X > Y' to represent the relation 'X after Y'.
(b) As from today you are in charge. (*Longman Dictionary of Contemporary English*, p. 49)

As noted in chapter 2, I consider *be going to* as an alternative to *will*, which is especially used when the speaker wishes to represent the future situation as having its roots in the present.[91] The use of the present tense is an example of a shift of temporal perspective from the post-present to the present.

Although the future tense can in principle refer to a situation that starts at t_0, it is not often used in this way. In fact, the future tense carries the implicature that the situation is not yet holding at t_0, and it is only when this implicature is cancelled that the reading 'from t_0 onwards' is suggested. This is the case in (90,a–e), where the implicature is cancelled either by an adverbial or by the pragmatics of the situation of speaking. The implicature in question has the same origin as the one that we have observed in connection with the preterit (which implicates that the situation described is over at t_0): since, other things being equal, statements about the present are more relevant than statements about the past or future, a situation that holds at t_0 should not be located in the past or future sector, unless there is a good reason for doing so.

The fact that there are two kinds of posteriority relation expressed in (89,a–b) and (90,a–e) means that the future tense can realize two temporal structures: one involving the relations mentioned in (92,a) and one involving those given in (92,b):

(92) (a) TS simul TO_{sit}.
TO_{sit} after TO_1.
(b) TS simul TO_{sit}.
TO_{sit} from TO_1 onwards.

91. Fleischman (1982) investigates the use of the future tense and the *go*-future in various languages and comes to the following conclusion:
> Summing up our findings regarding these two futures, we conclude that along the tense axis future and go-future are currently equivalent. Along the relevant modal axes the two cover much, though not quite all, of the same territory. *This combination of factors militates strongly in favor of regarding the go-paradigm as a legitimate future-tense form.* But the go-future has retained, as an important overtone on its basic future meaning, a connection with the speaker's present which is lacking in the simple future and which translates into grammar as an aspect of prospection. (p. 97; the emphasis is mine.)

Fleischman's conclusion accords with Haegeman (1989), where it is argued 'that at the level of sentence meaning *be going to* and *shall/will* are equivalent, and that the difference between them is to be found in the constraints they impose on the processing in context of the utterance in which they occur' (p. 291)

In the absolute use of the future tense TO_1 is t_0. In the relative use (where the future tense just expresses posteriority) TO_1 is some other TO. Thus, in

(93) He will promise that he will do it.

it is the TO_{sit} of the head clause that functions as TO_1 for the structure of the future tense in the *that*-clause.

The two sets of relations expressed in (92,a–b) are represented by Figures 7.34 (e.g. *John will do it*) and 7.35 (e.g. *As from now I will give the orders here*).

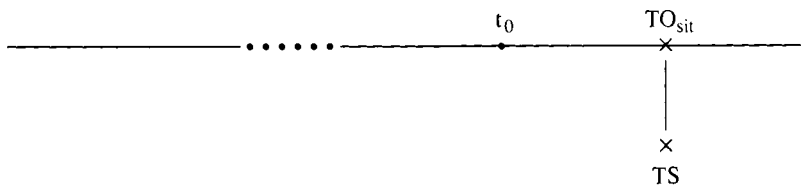


Figure 7.34 Example: *John will do it*

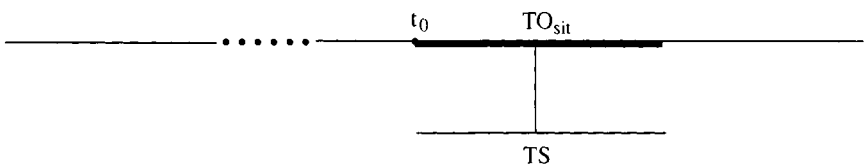


Figure 7.35 Example: *As from now I will give the orders here*

If the sentence contains a time adverbial, the temporal structure of the sentence will involve the following relations in addition to those specified in (92):

(94) (a) TE includes TO_{sit}. (As usual, this means that TE either includes TO_{sit} properly or is simultaneous with it.)
(b) TE lies in the future of TO_1. It may or may not include TO_1.

This yields the following possibilities:

1 In sentences like *John will arrive at five o'clock* the temporal structure is as follows:

(95) (TS = TO_{sit} = TE) wholly after t_0.

This is represented by Figure 7.36.

2 A sentence such as *I will be somewhere else all day tomorrow* also has the structure represented by (95), but TS, TO_{sit}, and TE are now durative (see Fig. 7.37).

3 The structure realized by *John will arrive tomorrow* consists of the following relations:

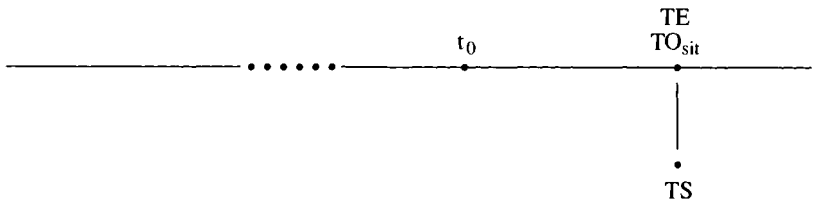


Figure 7.36

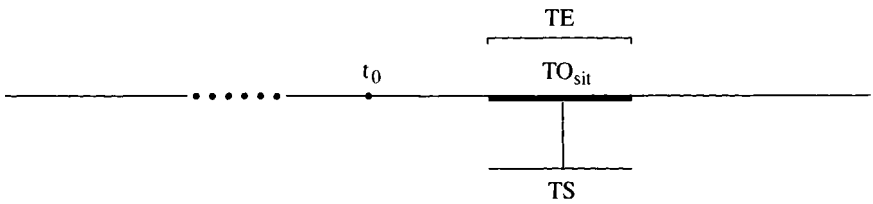


Figure 7.37

(96) (a) TS = TO_{sit}.
(b) TE properly includes TO_{sit}.
(c) TE wholly after t_0.
(d) TO_{sit} wholly after t_0.

This is represented by Figure 7.38.

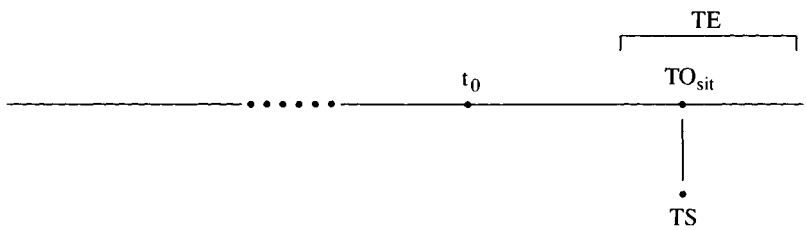


Figure 7.38

4 A sentence like *I will not be here for some time tomorrow* has the following structure (see Fig. 7.39):

(97) (a) TE properly includes TO_{sit}.
(b) TS simul TO_{sit}.
(c) TO_{sit} wholly after t_0.
(d) TE wholly after t_0.

5 The temporal structure of the sentence *I will not be here for some time today* also involves the relations specified in (97,a–c), but instead of (97,d) we now have the relation 'TE includes t_0' (Fig. 7.40).

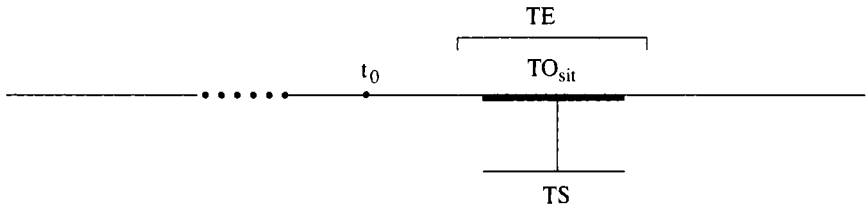


Figure 7.39

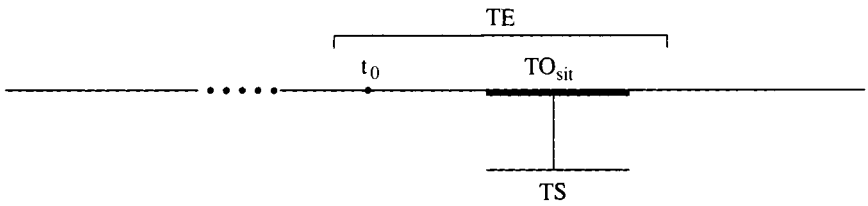


Figure 7.40

6 The sentence *John will arrive today* has the same temporal structure, but TO_{sit} and TS are now punctual (Fig. 7.41).

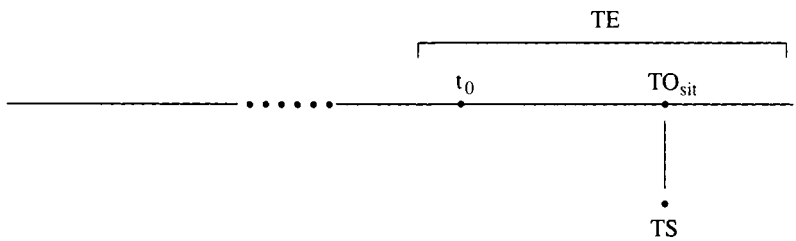


Figure 7.41

7 A sentence such as *I will henceforth use the term in this sense* realizes the following structure:

(98) (TS = TO_{sit} = TE) from t_0 onwards.

This is represented by Figure. 7.42.

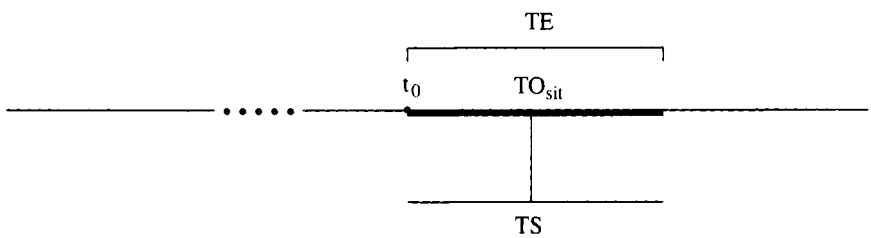


Figure 7.42

When we look at Figures 7.36–7.42, we ascertain that they are the mirror images of some of the structures we have argued to hold for either the present perfect or the preterit:

1 Figure 7.36 (which represents the temporal structure of *John will arrive at five o'clock*) is the mirror image of Figure 7.3 (*John arrived at five o'clock*).[92]
2 Figure 7.37 (*John will be here all day tomorrow*) is the mirror image of Figure 7.6 (*John was here all day yesterday*).
3 Figure 7.38 (*John will arrive tomorrow*) is the mirror image of Figure 7.4 (*John arrived yesterday*).
4 Figure 7.39 (*John will be somewhere else for some time tomorrow*) is the mirror image of Figure 7.8 (*John was somewhere else for some time yesterday*).
5 Figure 7.40 (*John will be somewhere else for some time today*) is the mirror image of Figure 7.7 (*John was somewhere else for some time today*).
6 Figure 7.41 (*John will arrive today*) is the mirror image of Figure 7.5 (*John arrived today*).
7 Figure 7.42 (*As from now I will give the orders here*) is the mirror image of Figure 7.19 (*Up to now I have given the orders here*).
8 We may add to this that the future tense may also be the mirror of an indefinite perfect. Compare *I have tasted every kind of fish once* (*in my life*) with *I will taste every kind of fish once* (*in my life*).

These parallels with the preterit and the present perfect show clearly that the future tense covers the two possibilities which, when the reference is to the past, are distributed over the preterit and the present perfect.

A final thing to be noted[93] is that the above structures all represent uses of the future tense as absolute tense. When the future tense is used as a relative tense, the only difference is that TO_1 is not t_0 but another TO (which necessarily lies in the future of t_0). This is the case, for example, in

(99) Bill will declare that *he will leave the country.*

Here *he will leave the country* realizes the structure '(TS = TO_{sit}) wholly after TO_1', where TO_1 is the TO_{sit} of the head clause, which lies wholly after t_0.

92. Of course, when speaking of 'mirror image' here I disregard the fact that the structure of the preterit involves reference to the two time-spheres, whereas the future tense involves reference to the present time-sphere only.
93. I do not think it is necessary to (once again) consider repetitive and gnomic sentences separately. It has been sufficiently demonstrated that these can be analysed in the same way as nonrepetitive and non-gnomic state sentences, respectively.

6 THE FUTURE PERFECT

6.1 The temporal structure expressed by the future perfect involves the following relations:

(100) (a) TS simul TO_{sit}.
(b) TO_{sit} anterior to TO_2 (where 'anterior to' means either 'wholly before' or 'before and up to').
(c) TO_2 after TO_1. (In the absolute use, TO_1 is t_0.)

An example of this is

(101) John will have left when we arrive.

whose temporal structure can (provisionally) be represented by Figure 7.43.

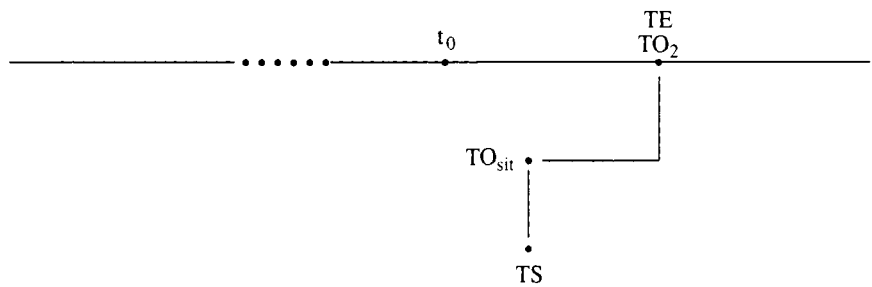


Figure 7.43

TE is the time established by the punctual adverbial *when we arrive.* This TE indicates TO_2. (In technical terms: TE includes TO_2; since TE is punctual, this means that TE = TO_2.) The time of John's departure (TS) is simultaneous with a time (TO_{sit}) which lies before TO_2.

It should be noted that, for the first time in our analyses of the tenses, we have not placed all the relevant TOs on the time line: TO_{sit} is related to TO_2 in terms of anteriority (the hooked line is meant to express this), but its relation with respect to t_0 is left unspecified. This, of course, is as it should be. We have argued in chapters 2 and 5 that the future perfect is vague with respect to this relation. The sentence *John will have arrived at your house before tonight* may be true not only if John is still to arrive at the house but also if (unknown to the speaker) John has already arrived there or arrives at the house precisely at t_0. As noted before, this does not mean that the sentence is three ways ambiguous. It just means that the sentence is vague on this point.[94] Figure 7.43 is in keeping with

94. Both Vikner (1985:88) and Comrie (1985:71–4) argue that, because of a conversational implicature, we normally interpret a situation referred to in the future perfect as following t_0. However, they make clear that this is not inherent in the meaning of the future perfect. (As noted in footnote 52 of chapter 2, one of the available arguments is that sentences like *If it rains tomorrow, we'll have worked in vain yesterday* may be perfectly acceptable in a suitable context.)

this, for it relates TO_{sit} only to TO_2 and not to t_0. This, of course, is a consequence of the fact that TO_{sit} is not placed on the time line. If it were placed there, it would have to precede, follow or overlap t_0, and so the schema would express one of the three theoretically possible relations between a durative TO_{sit} and t_0.

Another thing that should be noted is that, just as in the structure of the past perfect, the 'before' relation between TO_{sit} and TO_2 can be either the relation 'wholly before' or the relation 'before and up to'. Figure 7.43 (which represents the temporal structure of the future perfect in (101)) involves the relation 'wholly before'; the relation 'before and up to' is involved in

(102) (a) When he stops working he will have repaired all these cars.
(b) By tonight he will have been away for twenty-four hours.

Since the temporal structure of (102,a–b) differs from that of (101) only in that it involves a different type of 'before' relation, we will have to introduce different types of notation for these different types of 'before' relation in the temporal schemata of these sentences. Let us use a dotted hooked line to represent 'wholly before' and a hooked line with a bold horizontal part to represent 'before and up to'. The temporal structures of (101) and (102,a–b) are then represented by Figure 7.44 and Figure 7.45, respectively. Figure 7.46 can then be used as a general representation of the structure of the future perfect in which the type of 'before' relation is not made explicit. (Remember that time intervals are represented by the symbol x if we do not want to specify whether they are punctual or durative.)

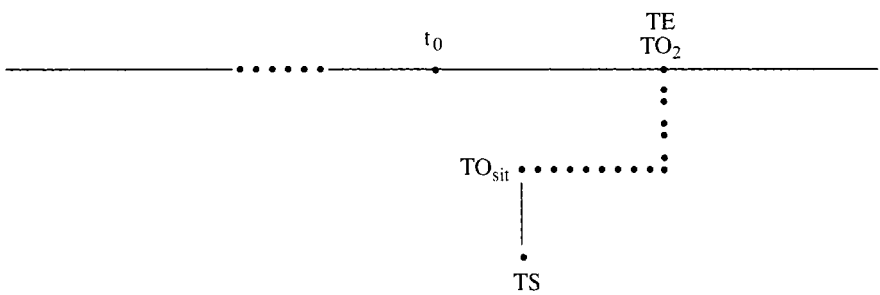


Figure 7.44

6.2 When dealing with the future tense we saw that the relation 'X after Y' can also be of two types: 'X wholly after Y' or 'X from Y onwards'. We may therefore wonder whether these two possibilities are also involved in the use of the future perfect. The answer to this question is negative. It is not possible to find examples of the future perfect in which 'TO_2 after t_0' can be interpreted as 'TO_2 from t_0 onwards'. If such

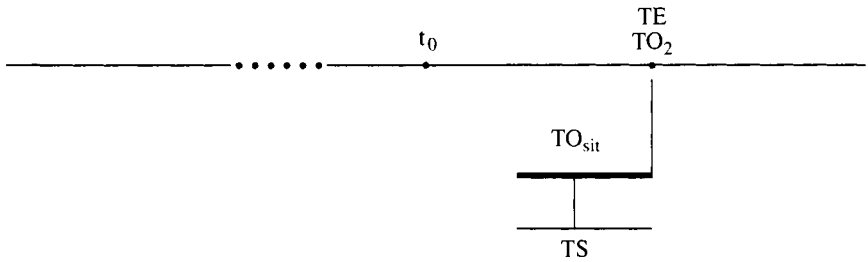


Figure 7.45

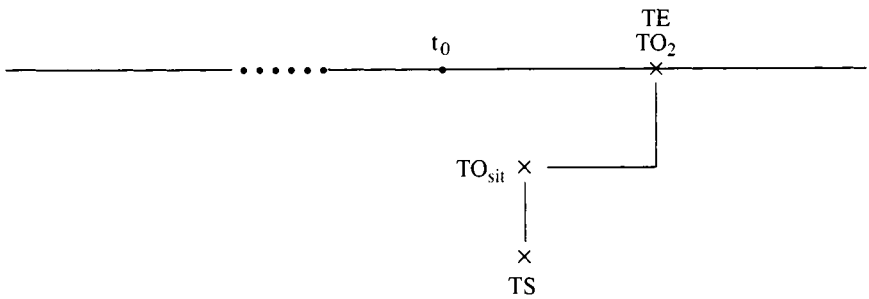


Figure 7.46

examples were possible, they would have to be of the kind illustrated by (103), which is ungrammatical (except perhaps as an elliptic version of *From now on we will assume/say that he has married Jill*).

(103) *From now on he will have married Jill.

It thus appears to be the case that the relation 'X from Y onwards' can hold only between TO_{sit} and TO_1.

6.3 Like the other tenses, the future perfect illustrates the principle that the TE established by a time adverbial can in principle be interpreted as referring to (i.e. as including) any TO or combination of TOs.[95] Consider the following sentences:

(104) (a) John will have left at five o'clock.
(b) John will have left that same day.
(c) John will have left today.

In (104,a) the time adverbial refers to a point of time which can be either TO_2 or TO_{sit}. This becomes clear when we use the sentence in a suitable context:

95. This principle and the exceptions to it were discussed in section 4 of this chapter.

(105) (a) Don't wait until it's five o'clock. I am telling you this because I know *John will have left at five o'clock.*
(b) You will be here until six o'clock, but *John will have left at five*, so you will be all on your own for a whole hour.

In (104,b) the time adverbial establishes a TE which (because of the use of *same*) is normally interpreted as involving both TO_{sit} and TO_2. In (104,c) TE may involve t_0 and TO_{sit} or it may involve t_0, TO_{sit} and TO_2. It is again the context that will guide us towards one of these interpretations:

(106) (a) Tomorrow we will hear what has happened. I expect it will be as follows: Mary will have left yesterday, and *John will have left today*, so Mary will have been the first to speak to Mother.
(b) Tonight we will hear what has happened. I expect it will be as follows: Mary will have left yesterday, and *John will have left today*, so Mary will have been the first to speak to Mother.

In (106,a) *today* indicates a timespan containing both t_0 and TO_{sit} (= TS), but not TO_2 (which is located as being 'tomorrow'). In (106,b), TO_2 is 'tonight', so that all three times are contained in the TE established by *today*.[96]

6.4 An interesting observation that can be made here is that a *when*-clause specifying TO_{sit} cannot be in the same tense as a *when*-clause specifying TO_2. Compare:

(107) (a) Bill will already have left when John *arrives.*
(b) (Bill will not be there at that moment.) He will have left when John *arrived.*

Both sentences yield only one interpretation: in (107,a) the *when*-clause specifies TO_2; in (107,b) it specifies TO_{sit}. The reason why different tenses are involved should be clear from the theory that has been argued in chapter 2. We have described the future perfect there as a 'absolute–relative' tense, i.e. as a tense which not only expresses anteriority in a post-present domain but also (re-)establishes the domain in question. Thus, in both (107,a) and (107,b) the future perfect represents a situation as anterior to the central TO of a post-present domain. (It is this TO that we have been referring to as TO_2.) In (107,a) the *when*-clause represents a situation as simultaneous with this TO_2. In accordance with the

96. An anchored time adverbial specifying a time interval that lies before t_0 can, of course, only be related to TO_{sit} (since this is the only TO in the structure of the future perfect that can precede t_0). The following sentence exemplifies this:
(i) If it rains tomorrow, we will have worked in vain yesterday. (Comrie 1985:73)

Present Perspective System (which is the normal system for expressing relations in a post-present domain) the present tense is used for this. In (107,b) the *when*-clause represents the situation of John's arrival as simultaneous with the situation of Bill's leaving. In order to do so it uses the preterit, which is the normal tense for expressing simultaneity in a past time-sphere domain. This means that the expression of anteriority effected by the future perfect has brought about a shift of temporal perspective to the past time-sphere, so that a past time-sphere tense is used to relate a situation to the situation described by the future perfect. This shift of temporal perspective is not surprising, for we have seen that the expression of anteriority in a post-present domain often involves such a shift of perspective to the past time-sphere.[97] This is, for example, the case in (108), which is a close paraphrase of (107,b):

(108) (Bill will not be there at that moment.) It will be the case that he *left* when John *arrived.*

7 THE CONDITIONAL TENSE

7.1 In a certain sense the conditional tense is the mirror image of the future perfect. Whereas the latter expresses 'past in the future', the conditional expresses 'future in the past'. An example is *would weather* in

(109) The faded red brick of the house had weathered many London storms and *would weather* many more. (Tottie and Övergaard 1984:151)

The temporal structure of this verb form involves the following relations:

(110) (a) TS simul TO_{sit}.
(b) TO_{sit} after TO_2.
(c) TO_2 before TO_1.
(d) TO_1 belongs to the present time-sphere (in the above example TO_1 is t_0).
(e) TO_2 belongs to the past time-sphere.

The relation 'TO_{sit} after TO_2' may be either 'TO_{sit} wholly after TO_2' or 'TO_{sit} from TO_2 onwards'. The following sentences illustrate these possibilities:

(111) (a) Later this play would be called Welcome to Our City. (Tottie and Övergaard 1984:156)
(b) From that time onwards he would really hate his mother.

97. Anteriority in a post-present domain is expressed in the same way as anteriority with respect to t_0. This means that the speaker can choose either the preterit (in which case there is a shift of perspective to the past time-sphere) or the present perfect (in which case there is a shift of perspective to the pre-present sector).

Figures 7.47 and 7.48 represent the temporal structures of (111,a) and (111,b), respectively. To represent the structure of the conditional tense

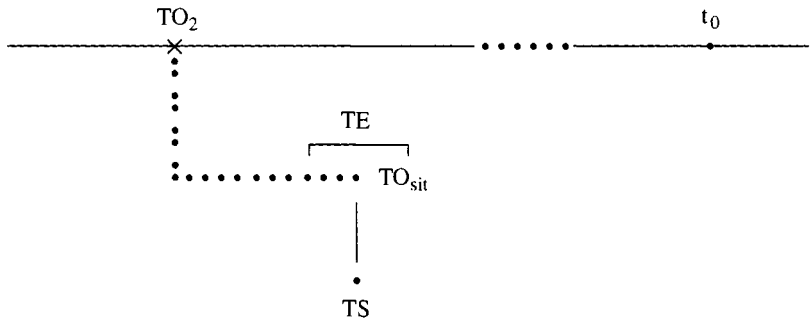


Figure 7.47

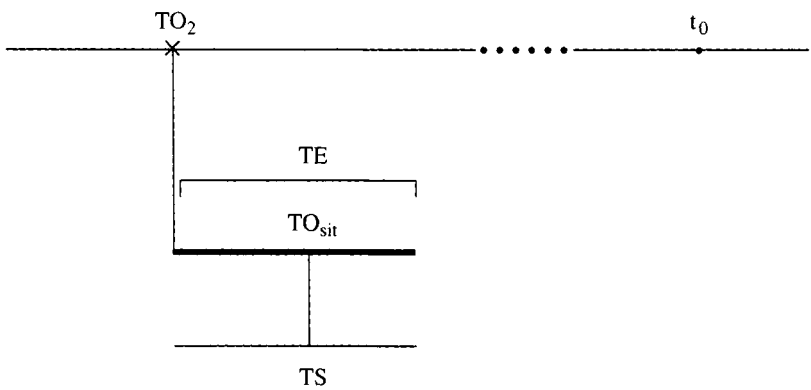


Figure 7.48

generally (i.e. without reference to TE and without making explicit which type of 'after' relation is involved, nor whether the TOs and TS are punctual or durative) we can use Figure 7.49.

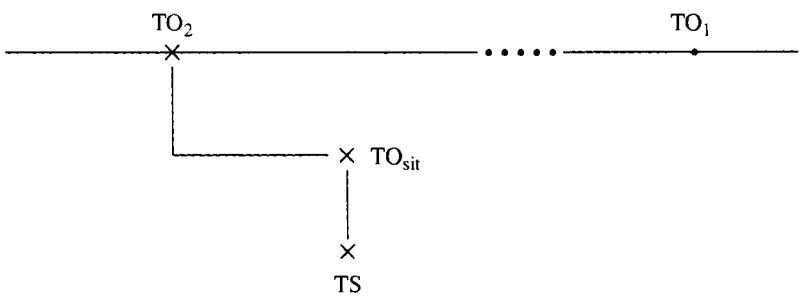


Figure 7.49

The fact that TO_{sit} is not situated on the time line serves the same purpose as we have observed in connection with the future perfect: since a sentence in the conditional tense leaves vague the temporal relation between TO_{sit} and TO_1, we must not place TO_{sit} (before, at, or after TO_1) on the time line. We can indicate no more than that TO_{sit} is located after TO_2.

7.2 According to context, a time adverbial establishing a past TE can again refer to the various TOs that belong to the past time-sphere. For example, *the next day* in

(112) He would leave the country the next day.

may refer to TO_{sit} (= TS), as in (113,a), or to TO_2, as in (113,b), or to both TO_{sit} and TO_2 (which is another possible interpretation of (113,b)).

(113) (a) I was never to see him again because *he would leave the country the next day.*
(b) He could not make up his mind. One day he would stay, *the next day he would leave the country.* I believe that in the end he moved to London.

In examples like (113,b), which express that, at a certain time in the past, the referent of the subject had the intention of doing something later, 'future in the past' reference is more normally expressed by *was going to* or by the progressive past tense (*was leaving*). This is, however, immaterial to the point that is being made here, viz. that the TE established by the time adverbial may refer to TO_2, or to both TO_{sit} and TO_2. (In (113,b) the latter interpretation means that the referent of *he* intended to leave on the very same day that he voiced this intention (the day in question being indicated by *the next day*).)

7.3 In examples like (113,b), TO_2 is an 'implicit TO' (see section 24 of chapter 2). It is not impossible for such an implicit TO_2 to be adverbially indicated (and hence to become an explicit TO_2), even when there is another adverbial referring to TO_{sit}:

(114) (a) Yesterday you were coming tomorrow. (Huddleston 1969: 781)
(b) (John said that) yesterday morning Bill was going to leave on the 25th.

(In such sentences we usually use the past progressive or *was going to* because the idea expressed is that of a past intention to perform a future action. *Would* is not normally used to express this kind of idea. However, the temporal relations are the same as those involved in the structure of the regular conditional tense.)

7.4 As noted in chapter 1, the use of *will* to predict a future situation combines reference to future time with a modal connotation (potentiality). The modality in question follows from the fact that a situation which lies in the future of t_0 is necessarily nonfactual at the time of speech. These modal overtones can also attach to the conditional tense, but only if the time of the situation follows t_0 or is left unspecified. If it is assumed that the situation (which is represented as posterior to a past TO_2) actualized before t_0, the conditional tense does not have the connotation of nonfactuality. This use of *would* to refer to a factual situation whose TO_{sit} lies between TO_2 and t_0 is not infrequent in historical narrative. The following are examples of this kind:

(115) (a) Twenty years later, Dick Whittington *would be* the richest man in London. (Leech 1971:48)
(b) Wharton *would* never *be deceived* about the world of her youth: it *would* always *be remembered* as a narrow, emotionally limited environment. (Tottie and Övergaard 1984: 160)

It has often been noted that this kind of *would* comes very close in meaning to 'was destined to' and that the same idea can also be expressed by *was to* (which is in fact strongly preferred in British English).[98] Even *was going to* can occasionally be found with this meaning:

(116) (a) Years were to pass before these plans came off the paper. (Tottie and Övergaard 1984:155)
(b) Through 1787 operations on both sides had been lackadaisical; those of 1788 *were going to prove* decisive, though many of their details are obscure. (Tottie and Övergaard 1984:160)

However, *was going to* is not often used in this sense. In fact, it appears that in unembedded clauses there is a kind of complementary distribution between *would* (or *was to*) and *was going to*: the former represents its situation as factual, the latter usually represents it as something which was expected or intended to happen later (without implying anything about whether it actually happened or not). For this reason, the following two sentences do not seem to be truth-conditionally equivalent:

(117) (a) The Queen would arrive three hours later.
(b) The Queen was going to arrive three hours later.

98. Tottie and Övergaard (1984:154) note that they found 19 examples of this kind of *would* in the Brown Corpus of Edited American English, and only one example in the Lancaster–Oslo/Bergen Corpus of British English. The former corpus counts thirty-six instances of *was/were to*, the latter forty-eight.

As pointed out by Haegeman (1989:313), 'in the event that the Queen did not arrive, the first sentence appears to be false, but the second is probably true'.

It should be noted, however, that there is no similar difference of meaning between *would* and *was going to* in embedded clauses. Neither of the following sentences represents the Queen's arrival as factual:

(118) (a) They said/thought that the Queen would arrive three hours later.
(b) They said/thought that the Queen was going to arrive three hours later.

These sentences are examples of reported speech/thought in which the complement clauses are temporally subordinated. As noted before, this means that the complement clauses receive a *de dicto* interpretation. For this reason the interpretation that the reporting speaker treats the complement clause situation as factual is excluded. It follows that we can trace back the factual interpretation of *would* in (117,a) to the absence of a superordinate clause creating an opaque context. Apart from Represented Speech and Thought (free indirect speech), where *would* is interpreted as in (118,a) rather than as in (117,a), the absence of a superordinate clause can only mean that the situation is represented as it is viewed by the speaker, i.e. as a factual past situation which is seen as posterior to another past situation. The fact that *was going to* in (118,b) is interpreted differently (at least normally – see (116,b)) follows from the fact that *be going to* usually suggests that the situation which is posterior to some TO has its roots (initiation) in some state of affairs which already holds at the TO in question. This means that the situation is viewed from a point of view associated with the TO in question rather than from the point of view of the speaker.

8 THE CONDITIONAL PERFECT

8.1 The conditional perfect is the most intricate of the traditional English tenses. It involves the following relations:

(119) (a) TS simul TO_{sit}.
(b) either: TO_{sit} wholly before TO_3.
or: TO_{sit} before-and-up-to TO_3.
(c) TO_3 after TO_2.
(d) TO_2 before TO_1.
(e) TO_1 belongs to the present time-sphere.
(f) TO_2 belongs to the past time-sphere.

The fact that there are two possible relations in (119,b) means that a conditional perfect may be either an 'indefinite' conditional perfect or a 'continuative' conditional perfect. An example of the former type is

(120) The others would have left by then.

whose structure is represented by Figure 7.50. An example of a continuative conditional perfect is (121).

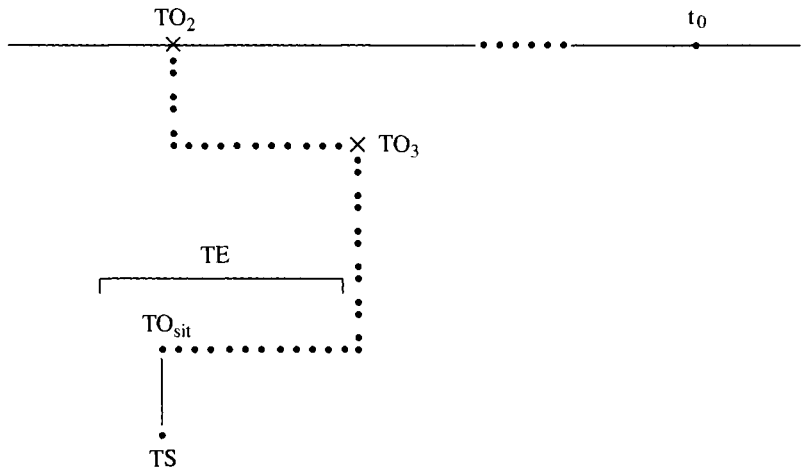


Figure 7.50

(121) On 7 July he would celebrate his sixtieth birthday in prison. *He would have been in prison for seven and a half years then.*

Here TO_3 is the time of the celebration. TO_{sit} (the time of his being in prison) leads up to (and includes) TO_3. The full temporal structure is as in Figure 7.51.

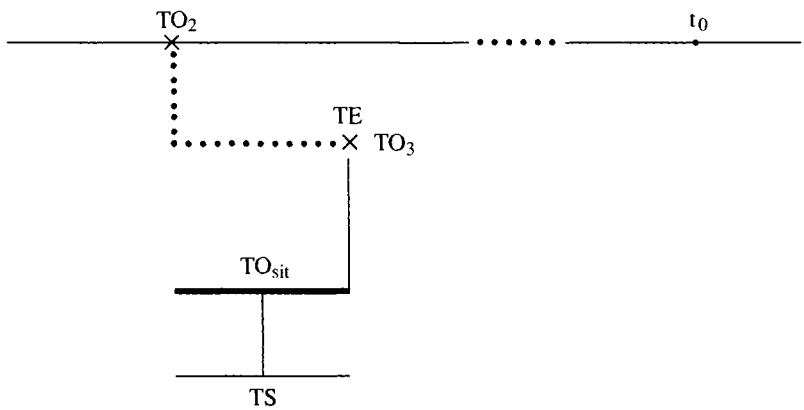


Figure 7.51

8.2 If there is a time adverbial establishing a past TO, it may again specify each of the past time-sphere TOs in the structure of the tense. That is, it may (at least in principle) indicate (i.e. include) either TO_{sit} or TO_3 or TO_2. Thus, in

(122) Mr Smith would have left the hotel at five o'clock.

at five o'clock can refer either to TO_{sit} (the time at which Mr Smith was expected to leave the hotel) or to TO_3 (the time by which Mr Smith would have left the hotel) or to TO_2 (the time when the opinion was expressed that Mr Smith would have left the hotel before a particular time referred to in the context). Each of these interpretations can be invited by a suitable context:

(123) (a) The plan was as follows. Mr Smith would go to the hotel at three o'clock and leave it at five. Bill would go to the hotel shortly after five. At that time Mr Smith would no longer be there, for *he would have left the hotel at five o'clock.* In this way both Mr Smith and Bill would be able to go to the hotel without being seen in each other's company.

(b) The plan was as follows. Mr Smith would leave the hotel at four o'clock. Bill would arrive there at five. In this way Bill would not be seen in Mr Smith's company, for *Mr Smith would (already) have left the hotel at five o'clock.*

(c) Judging from the statements that are issued every hour, the White House is not sure whether Mr Smith will have left the hotel before the Russian delegation arrives. At four o'clock the statement was that Mr Smith would not have left the hotel at that time. However, *at five o'clock he would have left the hotel.* Let us wait and see which of these conflicting statements will be confirmed at six o'clock.

When the past TE is durative, it can also be interpreted as including two of the three past time-sphere TOs (i.e. either TO_{sit} and TO_3 or TO_3 and TO_2) or all three of them. Thus the sentence

(124) Mr Smith would have left yesterday.

is true in each of the following cases:

1 Yesterday someone expressed the opinion that Mr Smith would have left before the end of the day. (In this case both TO_2 and TO_3 are included in the TE; the temporal location of TO_{sit} is left vague.)
2 Yesterday Mr Smith announced that he would have left before the end of the day. (Here all three TOs must be interpreted as included in the TE.)

3 The day before yesterday Mr Smith announced that he would have left before the end of the following day. (Here TE includes TO_3. It may also include TO_{sit}.)

8.3 It may be noted, finally, that a conditional perfect like *would have left* is the mirror image of a form like *will have been going to leave.* The structure of the former is something like (125,a), that of the latter something like (125,b):

(125) (a) TO_{sit} before TO_3 after TO_2 before TO_1.
(b) TO_{sit} after TO_3 before TO_2 after TO_1.

8 Conclusion

1 SUMMARY

1 The basic aim of this book has been double: to offer a full-fledged 'descriptive' theory of tense in discourse and to offer insight into the temporal schemata realized by the various tenses. However, I have not restricted myself to discussing tense. I have also taken into account the role played by time adverbials. In this way the theory I have developed is actually a full theory of time reference.

2 The descriptive theory has started from a number of basic observations:

a. Tense is the grammaticalization of the part played by the verb in the way situations are located in time.
b. Tensed clauses refer to situations that are directly or indirectly related to t_0.
c. Since this happens only in finite clauses, nonfinite clauses are untensed. Nonfinite verb forms do no more than express a (single) temporal relation with some TO available in the (linguistic or extralinguistic) context.
d. In the conception of English speakers there are two time-spheres in which situations can be located: a time-sphere containing the temporal zero-point and consisting of sectors that are contiguous with the present, and a time-sphere that is conceived as lying in the past and noncontiguous with the present.
e. Together with the past time-sphere, the three sectors of the present time-sphere (viz. the pre-present, the present and the post-present) form the four 'absolute sectors'.
f. There is no evidence that there are only two tenses in English. On the other hand, there is evidence that there are two sets of tenses: those that locate the situation in the past time-sphere and those that locate it in the present time-sphere.
g. There is no evidence that the present perfect and the future tense should not be considered as tenses. On the contrary, both locate

situations in time in a particular way and realize a particular temporal schema, so that there is every reason to consider them as tenses.

3 There are 'absolute' and 'relative' tenses. An absolute tense relates a situation directly to t_0, i.e. it establishes a temporal domain in one of the four absolute sectors. The four tenses that can be used in this way are the preterit, the present perfect, the present and the future. A relative tense expresses a relation inside a temporal domain. If the domain is located in the past time-sphere the tenses used for anteriority, simultaneity and posteriority are the past perfect, the preterit and the conditional tense, respectively.

4 A tense like the future perfect, which both establishes a domain and indicates a relation in it, is an 'absolute–relative' tense. The conditional perfect, by contrast, has no absolute time reference as part of its meaning. It expresses a double temporal relation inside a past domain.

5 The relative tenses that express relations in a past domain are also used when a pre-present domain that does not include t_0 is further expanded. This phenomenon, that a pre-present domain is expanded as if it were a past one, is an instance of a 'shift of temporal perspective'.

There is also a shift of temporal perspective when a post-present domain is expanded: when a situation is related to a TO (time of orientation) belonging to a post-present domain, this TO behaves as if it were t_0. This means that the tenses that are used to express relations in a post-present domain are actually the same as can be used as absolute tenses. For this reason we can call the tenses that express relations in a post-present domain 'pseudo-absolute' tenses.

6 Clauses may or may not be 'temporally subordinated'. We speak of temporal subordination (or 'temporal binding') when the clause uses a relative tense. When the speaker uses an absolute tense, he either shifts the domain or establishes a simultaneous domain or re-establishes the existing domain. In none of these cases is the situation referred to related to that of the preceding clause. Instead it is directly related to t_0. It follows that in a sequence of clauses with absolute tense forms the temporal relation between the situations is not expressed by the tense forms.

7 There are a couple of special uses of tenses:

a. In some cases a form expressing simultaneity is used to express what is really 'sloppy' simultaneity. This possibility is very frequently made use of in conditional clauses, which consequently use verb forms expressing simultaneity where other languages use (or can also use) verb forms expressing posteriority.
b. Sometimes a situation that is temporally subordinated is not temporally related to the situation of its matrix or to that of the preceding

clause. In that case it is bound 'indirectly' (whereas 'direct binding' is the default option).

c. Sometimes the speaker departs from the default choice of tense in order to place the 'temporal focus' on a particular past or post-present time.

d. The speaker may represent a past situation as if it were a present one, or a present situation as if it were a past one, etc. Such a shift of temporal perspective is always semantically motivated.

8 In adverbial time clauses that use a relative tense, the situation is mostly not related to the situation of the head clause but rather to an 'implicit TO' (implicit time of orientation), whose location relative to the head clause situation is signalled by the temporal conjunction. This explains why adverbial time clauses whose situation is posterior to that of the head clause use a verb form expressing simultaneity rather than a verb form expressing posteriority.

9 The speaker often has the freedom of choice between several possibilities. Different options are available to him as regards the t_0 (time of coding or decoding), the time-sphere, the absolute sector, the temporal domain, the domain-internal temporal relation, the temporal perspective, the temporal focus, the choice of binding TO, etc. However, these possibilities (at least the nondefault ones) are subject to restrictions. Some of these were investigated in chapter 3.

a. We have examined the restrictions on the possibility of shifting the domain in unembedded clauses and in embedded ones. We have also investigated how a new domain is interpreted temporally in relation to the previous domain.

b. We have discussed the rules that determine the choice of binding TO (which entails either direct or indirect binding).

c. We have gone into the principles underlying the choice of tense when the reference is to the post-present, i.e. the choice between the 'Future Perspective System', which consists of tenses that create post-present domains, versus that of the 'Present Perspective System', which consists of the 'pseudo-absolute' tenses that express relations within an already established post-present domain.

10 In chapter 4 we have scrutinized two controversial areas of English grammar to which the choice of tense is pertinent, and which are therefore good test cases for the descriptive theory developed in chapter 2.

First we have examined the use of tenses in indirect speech, more specifically the phenomenon of 'backshifting'. We have agreed with Comrie (1986) that there are data which cannot be accounted for by the 'absolute deixis hypothesis' (which holds that backshifted tense forms are absolute tense forms). We have also subscribed to Comrie's claim that a

'relative time hypothesis' which holds that the tenses in indirect speech complement clauses must invariably be relative tenses is untenable. However, contrary to Comrie (1986), we have shown that the 'formal SoT hypothesis' (which treats backshifting as a purely formal operation triggered by the past tense of the introductory verb) is not adequate either. Several arguments have been adduced to this effect, the most important of which are the following:

a. There are too many data that are incompatible with the basic claims of the formal SoT hypothesis, and which can only be treated as exceptions.
b. The formal SoT hypothesis does not cover cases of backshifting after a reporting verb in a non-past tense.
c. Neither does it account for the possibility of backshifting after a tenseless verb form.
d. The formal SoT hypothesis does not cover cases in which there is no backshifting in spite of the fact that the reporting verb is in a past tense. (I am referring here to the fact that tense forms expressing a temporal relation in a past domain are not backshifted in indirect speech.)
e. We have mentioned no fewer than six objections of a theoretical nature that can be made against the formal SoT hypothesis. One of the most serious is that it misses an important generalization because it treats tense in indirect speech complement clauses as totally unrelated to tense in other environments.

As an alternative to the three above-mentioned theories I have argued that complement clauses may use either relative tenses or absolute tenses (the former possibility being the default one). This analysis is subject to none of the objections that were raised against the other theories, and it accounts naturally for all the possibilities that manifest themselves, both in indirect speech and in other contexts.

The second test case for our descriptive theory has been the use of tense in conditional clauses that refer to the post-present. After noting that 'standard conditionals' (which are invariably of the type 'if p then q') use the present tense in the *if*-clause, a number of different types of sentence were identified in which the use of (nonmodal) *will* (or *be going to*) in the *if*-clause is quite grammatical. In each case the conditional sentence was shown to differ from a standard conditional in at least one respect:

a. The first six types differ from standard conditionals in that the *if*-clause does not refer to a possible future actualization of the p-situation which, if it takes place, will lead to the actualization of the q-situation. Because of this, there is no close (conditional and

temporal) relation between p and q. Instead of being incorporated into the domain established by the head clause, the *if*-clause therefore establishes a post-present domain of its own. It follows that the system used is not the Present Perspective System but the Future Perspective System. (One consequence of this is that the conditional clause, which uses a verb form typical of unembedded clauses, can support another *if*-clause.)

b. Type VII does use the Present Perspective System, but the temporal relation expressed is posteriority, not simultaneity. For that reason *will* or *be going to* has to be used.

c. In type VIII the conditional clause is used as unembedded clause and must therefore make use of the Future Perspective System.

11 Chapter 5 is the introductory chapter to the second part of the book, which investigates the structure of the tenses, i.e. the different temporal schemata which they realize. In this chapter we have discussed and criticized the analyses offered by Reichenbach (1947) and Comrie (1985) and have come to the conclusion that the following basic principles must form part of any theory that aims at analysing the temporal structures underlying the tenses:

a. Any situation is simultaneous with a TO. It follows that any tense has a structure that involves at least one TO next to the 'basic TO'. (The latter is the TO that is the 'origin' of the relations that form part of the structure of the tense. It is usually t_0.)

b. Tenses must not be analysed as configurations of times on the time line (as in Reichenbach's theory), because a tensed sentence expresses relations between some times, but not necessarily between all the times involved in the temporal structure. That is, the temporal structure of a tensed sentence is like a chain of relations: in the same way as each link in a chain is only connected with the immediately preceding link and the immediately following one, each time in the temporal structure is related to some other time(s) but not to all of them. Thus, the conditional tense in *John said that he would do it* expresses that the situation-TO of the *that*-clause (the time of his doing it) is posterior to the situation-TO of the head clause, and that this TO is itself anterior to t_0, but it does not express a direct relation between the situation-TO of the *that*-clause and t_0 (i.e. it does not locate the time of 'doing it' relative to t_0).

12 In chapter 6 we have identified the structural elements that constitute the temporal schemata realized by the tenses. The following are the most important claims that have been made:

a. The temporal schema of a tense minimally involves the elements TO_I (basic TO), TO_{sit} (situation-TO) and TS (time of the situation).

b. TO_1 may or may not be t_0. (In chapter 7 it has been shown that TO_1 is t_0 when the tense form is used as an absolute tense form, but not necessarily when it is used as a relative one.)
c. TS (the time of the situation) is the time interval taken up by that part of the situation that is being referred to (and located in time) by the clause involving the relevant tense form.
d. If the situation is bounded, TS is the time of the full situation. If it is unbounded, TS may be either the time of the full situation or a subinterval of this.
e. TS is by definition represented as simultaneous with a TO. We have referred to this TO as the 'situation-TO' (TO_{sit}). The simultaneity relation in question is one of coincidence: TS and TO_{sit} are commensurate.
f. TO_{sit} is related directly or indirectly to t_0. (For example, the future perfect represents TO_{sit} as anterior to a TO which is itself posterior to t_0.)
g. If there is a (simple or complex) time adverbial, it indicates a time interval which we refer to as TE. If there are several time adverbials, they cooperate in establishing a single TE.
h. If there is a duration adverbial, it measures the length of the full situation. If this measurement is a specific indication (e.g. *for two hours*), it renders the situation bounded. In that case the time of the full situation is also TS.
i. The relation between TE and TO_{sit} is one of inclusion: TE either properly includes TO_{sit} or coincides with it. (Since TS is commensurate with TO_{sit}, the same inclusion relation holds between TE and TS.)
j. Non-gnomic (i.e. dynamic) repetitive sentences are analysed in the same way as other dynamic sentences.
k. Progressive sentences and gnomic sentences are analysed in the same way as other unbounded sentences.

13 Chapter 7 has been devoted to the description of the temporal structures underlying the various tenses. In the course of this description some further points have been made, the most important of which are the following:

a. It is not correct to say that it is part of the meaning of the preterit that the speaker must have a definite or specific time in mind.
b. It is not correct to say that the preterit and the present perfect do not differ in the way they locate a situation in time. The preterit locates a situation in the past time-sphere, whereas the present perfect locates it in the pre-present sector of the present time-sphere. This basic difference explains the further differences that we observe between the two tenses, e.g. that the present perfect, but not the preterit, can be used to represent a past situation as continuing into the present.

c. Reference to a resultant state is not part of the meaning of the present perfect. Rather it is an implicature following from the 'indefinite' use of the perfect.
d. The TE (established time) indicated by a time adverbial can in principle be applied to (i.e. include) any of the TOs in the structure of a tense. There are, however, a number of restrictions on this principle.

2 FURTHER REMARKS

2.1 In this book I have dealt with the eight tenses that are traditionally distinguished in descriptive grammar. Apart from these, there are other tense forms for which no traditional label is available:

(1) (a) He *has been going to repair* that sink for several months, but it is still leaking.
(b) At the end of this week he *will have been going to repair* that sink for one year, but it is still leaking.
(c) John *had been going to repair* the sink for several months when his wife finally decided to call a plumber.
(d) For three weeks John *had been going to have repaired* the sink before the end of the week.
(e) By the end of next week it will be three weeks that John *has been going to have repaired* the sink before the end of the week.

Tense forms like these present no problem for the theory that has been argued. Those in (1,a) and (1,b) are similar to the future perfect in that they are absolute–relative tense forms: *has been going to repair* establishes a pre-present domain and expresses posteriority in it, whereas *will have been going to repair* establishes a post-present domain and expresses both anteriority and posteriority in it. The tense forms in (1,c–e) are similar to the conditional perfect, since they express more than one relation within the same domain. In order to analyse the use of these more complex tense forms no additional theoretical apparatus is required.

2.2 In the linguistic literature the temporal schemata underlying the tenses are sometimes treated as semantic structures that are linked up with different truth conditions. For example, Hornstein (1977) argues for a slightly modified Reichenbachian system and claims that the different tense representations have different truth conditions. I would like to make clear that, although I do present my temporal schemata as semantic structures, I do not think that different schemata necessarily mean different truth conditions. In fact, it is clear from various observations that this is not the case. For example, we have already noted that the use

of the preterit in unbounded sentences does not entail (though it may implicate) that the situation is no longer holding at the time of speech. When uttered at 11 p.m., the sentence *At 10 p.m. he was already sleeping* even suggests that the situation continues into the present. This sentence may therefore have the same truth conditions as *He has been sleeping for more than an hour*. Moreover, it does not follow from the truth of the sentence in the past tense that *He is sleeping* must be false.

Similarly, the sentence *You have just said something different* is not truth-conditionally distinct from *Just now you said something different*. It is not possible to assert one sentence while denying the other. And the same is true of the following pair:

(2) (a) Ah, there you are! I'm waiting for you.
(b) Ah, there you are! I've been waiting for you.

On the other hand, there are cases in which different tense forms do entail different truth conditions. Since a bounded sentence in the past tense (e.g. *Bill had two beers*) implies that the situation as a whole is located in the past (see chapter 6), such a sentence cannot have the same truth conditions as the corresponding sentence in the future tense (*Bill will have two beers*). Even within the same absolute sector, different tenses may make for different truth conditions. As noted by Haegeman (1989:314), *Mary left when John arrived* is not truth conditionally equivalent to *Mary had left when John arrived.*

As far as I can see, what is criterial for the truth conditions is the way the situation as a whole is located relative to a particular TO. Consider, for example, the following:

(3) (a) John walked five miles.
(b) John will walk five miles.
(c) John has walked five miles.

All these sentences are bounded (perfective). They therefore locate the situation as a whole in one of the absolute sectors. Because of this, (3,a) and (3,b) must have different truth conditions, since the one locates the situation as a whole before t_0, whereas the other locates it as a whole after t_0. On the other hand, (3,a) and (3,c) have the same truth conditions, since both represent the situation as terminating before t_0.

Consider now:

(4) (a) John is walking.
(b) John has been walking.
(c) John was walking.
(d) John will be walking.

All these sentences represent their situations as unbounded. This entails that none of them entails the falsity of any of the others. (Note that (4,c) implicates, but does not entail, that John is not walking any more at t_0.

Adding *already* to the sentence removes this implicature. Similarly, (4,d) implicates that John is not yet walking at t_0. The addition of *still* cancels this implicature.)

Even when we use relative tenses, this has no effect on the truth or falsity of the proposition 'situation x is holding at time y':

(5) (a) John had been walking for some time (when I met him two minutes ago).
(b) (In a minute John will say that) he has been walking for some time.
(c) (When I met John two minutes ago I realized that) he would be walking for some time.
(d) (When I met John two minutes ago I realized that) by the end of the day he would have been walking for a long time.

None of these sentences excludes the truth of any other sentence in (5), nor of the sentences (4,a–d). That is, none of the sentences in (4)–(5) entails that there is a (past, present or future) time y such that the proposition 'situation x is holding at time y' must be false. This means that different tenses do not entail different truth conditions if the situation is represented as unbounded. The reason is that in an unbounded sentence the 'full situation' may be much longer than TS (i.e. that part of the situation that is located in time) and may therefore extend indefinitely far into the past and future – see chapter 6. Unbounded sentences may thus in principle be true of any time. (It is true that their duration is usually pragmatically restricted, but this has nothing to do with the semantics of the tense that is used.)

Apart from unbounded sentences, there are a number of cases in which bounded sentences also have the same truth conditions in spite of the fact that they use different tenses. I have already noted that a bounded sentence in the past tense is not truth-conditionally distinct from the corresponding sentence in the indefinite perfect (see *John walked three miles* vs. *John has walked three miles*). Another case is when the tense is the conditional or the future perfect. Consider:

(6) (a) (Mary said that) John would write two poems.
(b) John will have written two poems before 8 p.m.

In sentences like these, the tense form does not locate the situation on the time line, so that the sentence is true irrespective of whether the situation of writing two poems actualized before t_0, is actualizing at t_0 or will actualize after t_0. The sentences are therefore not truth-conditionally distinct from any other sentence that locates the same situation after (in the case of *would write*) or before (in the case of *will have written*) the relevant TO. Thus, neither sentence excludes the truth of *John wrote two poems, John has written two poems, John is writing two poems* or *John will write two poems*. Sentence (6,a) does not even exclude the truth of

John had written two poems, provided that the TO to which the writing is now represented as anterior is later than the TO to which the writing is represented as posterior in (6,a).

In sum, there are many cases in which different tenses do not entail different truth conditions. This inevitably leads to the conclusion that a semantics that is based on truth conditions is ill-equipped to describe the different meanings of the tenses. In fact, this conclusion must have been obvious from the moment that a distinction was made between the past time-sphere and the pre-present sector. Situations that lie before t_0 can in principle be located in either of these. As stressed in chapter 7, the speaker's choice depends exclusively on how he conceptualizes the situation in relation to t_0. In other words, what counts is 'subjective time', not 'objective time'. Since only the latter is pertinent to truth conditions, it is hopeless to try and base the semantic analyses of the tenses on truth conditions.

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Index

absolute sector, definition of 19–20
absolute tense, definition of 19
absolute–relative tense 52–3, 57–8, 378
accomplishment 120–1, 262
achievement 120–1, 262
activity 120–1, 262
additive 264
Adelaar, M. 20, 117, 141, 143
adverb(ial) 16, 17–18, 23, 60–2, 94, 97, 109, 118–19, 127, 132–4, 147, 168, 176–7, 196, 229–31, 234, 236, 240–1, 251–5, 268–70, 284–93, 298–300, 327–38, 356, 358–68, 377–8, 381, 385; *see also* adverbial time clause; complex adverbial; relative time adverbial
adverbial time clause 43–4, 46, 54, 59, 63, 97–113, 134, 140, 154, 172, 185, 191, 217, 275, 288, 293–4; *see also after*-clause; *before*-clause; *when*-clause
after-clause 44, 107–13, 137, 172, 276, 288
Akatsuka, N. 193
Allan, K. 288
Allen, R.R. 25, 30, 38, 42, 47, 51, 53, 60, 67, 121–2, 131, 159, 189, 206, 265, 276, 293, 302, 330
ambiguity 24, 38, 40, 49–50, 97, 140, 155, 190, 212, 226, 230, 233, 270, 275, 293, 323–4, 340, 351, 356
anchored time adverbial 331–2, 338, 359, 378
ancillary time of orientation, definition of 21
Anderson, J. 284, 345
Andersson, S-G. 121
anteriority 19, 21–4, 26–7, 34–8, 43, 45, 47–8, 52–3, 58, 63, 68, 77, 81, 86, 88, 92, 94–7, 99–100, 104–6, 108–10, 130, 169, 171–2, 180–1, 190–1, 294, 338, 357, 367–9, 378–9
Aristar, A. 127
aspect 10–12, 121, 234, 236, 262, 273–84, 319–20; *see also* nonprogressive; progressive
atelic 121, 276, 341; *see also* bounded

Bach, E. 158, 240, 250, 272–3, 280, 312
Bache, C. 73, 283–4
background 72, 127–8, 138, 140–1, 153, 273
backshift 94, 101–2, 104, 108, 111, 157–91; *see also* sequence of tenses
Bally, C. 89
Banfield, A. 89–90, 158, 291
Barense, D.D. 229, 233, 249–50, 283, 307, 331, 345, 356
Bartsch, R. 255
Barwise, J. 240, 262, 304
basic time of orientation, definition of 252–3
Bauer, G. 324, 326, 341
Bäuerle, R. 233, 240, 255, 293
be going to 27, 33, 35, 45, 53–4, 153, 199–200, 207, 214–15, 370, 381–3, 386
before-clause 44, 93, 101–7, 110–11, 135, 276, 288; *see also* nonfactual
Bell, M. 196
Bennett, M. 88, 262–3, 328
Berezovsky, H.C. 31–2, 63, 86, 244,

Berezovsky, H.C.—*continued*
257, 276, 280, 334
Bertinetto, P.M. 229, 244, 251–2, 257, 285, 288, 363–4
binding time of orientation, definition of 20
Binnick, R. 53
Blackstone, B. 187, 330
Blakemore, D. 126
Bloomfield, L. 119
Bolinger, D. 74
Botne, R.D. 11, 67, 231, 243, 283
bounded 119–29, 138–41, 261–75, 278–80, 282, 292–3, 314–19, 324, 335–8, 341, 344, 360–1, 367
Bourdin, P. 330
Braine, M.S. 201
Brecht, R.D. 20, 159, 183
Brinkman, F. 131
Brinton, L. 90, 121, 243, 280, 282, 324
Bronzwaer, W.J.M. 72, 83, 118
Bryan, W. 325–6
Bull, W.E. 14, 16
Bybee, J.L. 88

Canavan, J.R. 10, 88, 243, 283–4, 324, 326
Carlson, G. 366
Carlson, L. 263
central time of orientation, definition of 21
Chafe, W.L. 341
Charleston, B.M. 17, 274, 276
children's games 77
Chomsky, N. 8, 345
Chung, S. 14, 16, 75
Clark, E.V. 125
Clark, H.H. 44, 98, 125
Close, R.A. 198, 201, 205, 207–8
closed condition 193, 197–8, 203, 205–7, 209–10
Cohn, D. 89
Cole, P. 182
complement clause 38–9, 45, 49, 68, 91–2, 142, 148–9, 155, 157–91, 383
complex time adverbial 286–90
Comrie, B. 5–6, 8, 10–12, 14–15, 19, 57–8, 69, 80, 86, 88–9, 103, 117, 121–2, 158, 160–84, 192, 199, 208, 223, 226, 229–30, 232–50, 253–5, 257, 262, 266, 272, 279, 283, 301–2, 311, 313, 319, 326, 334, 375, 378, 389–91
conditional clause 39, 42, 46, 54, 59, 77–8, 135, 178, 192–222; *see also* closed condition; counterfactual condition; hypothetical condition
conditional perfect tense 16–17, 27, 58–9, 64, 94, 101, 382–6
conditional tense 27, 47, 64, 110, 184–5, 212–30, 225–6, 232–3, 256, 379–82
conjunction copying 110–11
Contini-Morava, E. 232
continuative perfect 28–9, 323–4, 328, 339–40, 249, 351–2, 354, 356, 384
Cooper, R. 73, 263
Costa, R. 133, 168, 181
Coulmas, F. 89–90, 150, 158
counterfactual condition 77–8, 194, 204, 216–17, 219
Couper-Kuhlen, E. 117, 121, 127, 138, 141, 145
Crystal, D. 61, 69, 73, 283, 329
Curme, G.O. 19, 215, 345
current relevance 157, 167, 185–91, 234–7, 306–7, 319–20, 326, 340–9
Cygan, J. 10, 158

Dahl, Ö. 8, 10–11, 42, 69, 88–9, 121, 159, 231, 239, 241, 251, 283, 341, 364
Dali, F. 158
Davidsen-Nielsen, N. 13
Davies, E. 203
Declerck, R. 84, 94, 120–1, 192, 198, 230, 245, 249, 258–9, 263, 268, 281, 288, 292, 303, 316, 336, 366
de Cornulier, B. 195, 203
default interpretation 128; *see also* markedness
default tense 69–70, 73
de Groot, C. 249
deixis 7–8, 14, 90, 133, 158–68, 176, 178, 331
Dekeyser, X. 243
descriptive theory, definition of 1
de Vuyst, J. 250, 262, 266, 273
Dieling, K. 72, 250, 285
Dinsmore, J. 230, 333, 347, 359
direct binding, definition of 62–6

dissective 263–4, 268–9, 280, 317, 336, 368
distance *see* remoteness
Diver, W. 137, 277, 320
domain-internal relations 26–40
Dominicy, M. 330
Donnellan, K.S. 303
Dorfmüller-Karpusa, K. 67, 276
Dowty, D. 119, 124, 127, 129, 161, 190, 240, 255, 262–3, 265, 271, 273, 280, 286, 290, 324, 328, 333, 336, 340, 359
Dry, H. 90, 127–8
Dubois, B.L. 323
duration adverbial 121–3, 263, 265, 268, 293, 323, 341, 359, 385
durative 120, 250, 272, 274, 295–6, 298–9, 313–14, 318–19
Duskovà, L. 131
Dutch 44, 47, 68, 102, 104, 110, 112, 185, 195
dynamic situation 213, 273, 279, 284, 293

Eckersley, C.E. and J.M. 243
Edgren, E. 145, 262, 273
Ehrlich, S. 127
Eilfort, W.H. 195, 202, 203–4
Eisterhold, J.C. 127
Ejerhed Braroe, E. 44, 55, 66, 158, 161, 176, 363
Ejerhed, E. 38
Enç, M. 8, 19–20, 50, 140, 251
endpoint 119–21, 261–3, 264, 266, 268, 319, 335, 341
epistemic necessity 85
epistolary tenses 15
Erades, P.A. 78, 310, 333
erlebte Rede *see* Represented Speech and Thought
Er-Rayyan, M.R.H. 8
establish a (temporal) domain, definition of 21, 25, 45–62
established time (TE), definition of 251–2
evaluation time 85–6, 88, 196, 250, 293–4
existential perfect *see* indefinite perfect
expectation 67, 84, 110–11, 184, 186, 199–200, 382
experiential perfect *see* indefinite perfect

Fabricius-Hansen, C. 293
factual 67, 78, 88, 101, 105, 110, 184–5, 193, 196, 199, 382–3; *see also* counterfactual condition; nonfactual
Farkas D.F. 366
Fehr, B. 90
Feigenbaum, I. 323, 326
Fenn, P. 30, 32, 78, 227, 255, 262, 276, 280, 305, 310, 312, 326, 330, 334, 342, 345–6
Fillmore, C. 14–15, 95, 240, 287, 290
Fleischman, S. 7, 10, 35, 73, 79, 147, 370
focus *see* temporal focus
Fodor, J. Dean. 44
foreground 127, 138, 140–1, 145, 147, 153
Forsyth, J. 127
free indirect speech *see* Represented Speech and Thought
French 9–12, 44, 49, 63, 77, 86–8, 195, 239–40, 313
future perfect tense 16, 50–3, 57–8, 60, 86, 88, 95, 224–6, 232–3, 251, 375–9
Future Perspective System (FPS) 50–6, 64, 92, 118, 124, 134, 146, 153–6, 192–222
future tense 2, 9–11, 19, 33, 35, 39, 50–1, 56–7, 83–4, 89, 103, 192–222, 225, 227, 232–3, 369–74

Gabbay, D. 122, 155, 273, 275, 290
Galton, A. 122, 263, 273
Gardner, H. 73, 76
Garey, H.B. 121
Geis, M.L. 195, 196, 327
Gelhaus, H. 191
Geluykens, R. 107
generic 84, 280–3, 317, 352–3
German 44, 68, 191, 195, 239–40, 313
gerund 45
Givòn, T. 103, 111, 127, 140, 343
Glendinning-Johnson, J-C. 198, 216
gnomic 84, 282–4, 293, 317; *see also* generic
Godfrey, D.L. 69, 73
Grewendorf, G. 68
Grice, H.P. 47, 117, 172, 188, 195, 266, 271, 321, 340

Gruber, J.S. 332
Guenthner, F. 250, 326
Gustafsson, M. 262

Haberland, H. 188
habitual 136, 139, 164, 279–82, 317, 349
Haegeman, L. 35, 53, 132, 192, 203–5, 212–13, 216, 370, 383, 394
Haiman, J. 203
Hajicová, E. 24, 325
Hall Partee, B. 150
Halliday, M.A.K. 19, 90
Hamann, C. 141, 276, 288, 332
Harkness, J. 288, 332
Harris, Z. 74, 98, 103, 169
Hatav, G. 127, 263, 273
Hawkins, J.A. 303, 312
headlines 70
Heinämäki, O. 93, 95, 106, 145, 262, 276, 287
Heny, F. 8, 159, 184, 240, 250, 252, 303, 305, 308, 359
heterogeneous situation 123, 264–5
Higgins, F.R. 158
Hill, A.A. 122, 276
Hinrichs, E. 119, 129, 141, 250, 262
Hirtle, W.H. 55, 70, 74, 237, 244, 280, 325–6
historical present 25, 72–3, 75–6, 135, 147, 237, 322
homogeneous situation 120, 122–3, 127, 263, 280, 317
Hopper, P.J. 127
Horn, L. 288
Hornstein, N. 30, 57, 93, 142, 228, 230, 285–7, 359, 393
Houweling, F. 141, 276
Huddleston, R. 19, 42, 63, 68, 80–1, 97, 159, 168, 185–6, 285, 290–1, 333, 351, 362–3, 381
Hüllen, W. 184
Hutchinson, C. 79
Hyman, L. 232
hypothetical condition 77, 194, 216

imperfective 121, 127, 262, 276, 344; *see also* bounded
imperfective paradox 262
implicature 88, 93, 119, 124, 129, 173, 188, 195, 197, 203, 237, 243–4, 266, 276, 287, 301–2, 316, 323–4, 343–4, 370, 375
implicit time of orientation 92–7, 144, 276, 293, 363, 369, 381
inclusion 129, 140, 253, 269, 284, 293–4, 314–15, 335–6
indefinite perfect 28–33, 37, 174, 308, 323–4, 328, 339–40, 343–4, 348–54, 356, 359, 384
indirect binding, definition of 62–6, 142–53
indirect speech 4–5, 90–2, 100–2, 104, 108, 111–12, 150, 157–92, 237, 383
infinitive 34, 45, 175, 181, 219
Inoue, K. 337, 344–6
irrealis *see* unreality
iterative *see* repetitive

Jacobs, R.A. 8
Jacobsson, B. 192, 200–2, 206, 208, 210–11, 213–16, 218–20
Jakobson, R. 90
James, D. 78, 80
James, F. 195
Janda, A. 136
Janssen, T. 4, 74, 79, 102
Jenkins, L. 66
Jespersen, O. 8, 83, 85, 89, 101, 110, 128, 158, 187, 192, 198, 206, 209, 217–19, 243, 274, 284, 319, 326, 345–6
Johnson, M.M. 67, 276
Johnstone, B. 73
Joos, M. 8, 70, 74, 79, 131, 324, 326, 340
just 330

Kałuża, H. 273
Kamp, H. 20, 141
Katz, J. 229, 233
Kempter, F. 72, 250, 285
King, L.D. 19–20, 74, 77, 88, 262, 326
Kiparsky, C. 158, 168
Kiparsky, P. 72–3, 158, 168
Kittredge, R.I. 44, 103, 276, 285
König, E. 117, 287
Koops, A. 73–4
Kratzer, A. 240
Kress, G.R. 10, 79
Kroch, A. 287
Kruisinga, E. 87, 95, 103, 326

Kuryłowicz, J. 11

Labov, W. 124, 127, 138, 145, 152
Ladusaw, W. 5, 48, 158
Lakoff, G. 66, 196, 226, 270
Lakoff, R. 83–4, 348
Langacker, R. 74, 79, 273
Latin 10, 15
Latzel, S. 15, 68, 83
Lawler, J.M. 281
Lee, D. 330
Leech, G.N. 42, 53–4, 57, 71, 126, 199, 243, 266, 274, 276–8, 280, 287, 302, 304–5, 323, 326, 345, 382
Lees, R.B. 323
Legault, S.R. 88, 265
Legenhausen, L. 276
Le Loux-Schuringa, J.A. 285
Leuschner, B. 79, 160, 174, 186
Levinson, S.C. 275, 284
Lewis, D. 43, 282, 352, 366
Lewis, M. 16, 74–5, 67, 79, 83
Li, C.N. 90
Lloyd, P.M. 114
Lo Cascio, V. 15, 20, 117, 141, 142–3, 151, 249–50, 260, 286
Lorck, E. 89
Lutzeier, P.R. 93
Lyons, J. 7–11, 14–16, 70, 74, 79, 91, 122, 263, 280, 283–4, 345

McCawley, J. 8, 158, 240, 284, 324, 326, 329, 345–6, 354, 356
McCoard, R. 29, 64, 225–6, 229, 255, 304, 319, 324, 326, 328, 330, 333–4, 345
McGilvray, J.A. 168, 239, 244, 257, 284, 345
McKay, K.L. 73
Mårdhi, I. 70
markedness 14, 24, 42, 45, 48, 63, 71, 75–6, 81–3, 119, 123–4, 126, 128–32, 139–40, 142–3, 160, 162–3, 165–8, 172, 188, 252, 324
Martinet, A.V. 42
Matthews, R. 131, 343–4
Matthiesen, C. 11, 19–20, 53, 72, 74, 80, 284
Maxims of conversation 47, 75, 88, 117, 167, 172, 188, 195, 266, 271, 301, 320, 329, 340
Mitchell, B. 98
Mittwoch, A. 265–6, 273, 280
modal auxiliary 57, 85, 87–9, 193
modality 10–11, 76–9, 94, 382
Molendijk, A. 125, 129–30, 151, 276
moment of coding *see* temporal zero-point
Mommer, K.E. 121, 265, 276, 304, 330, 341, 353
Moravcsik, J.M. 122, 273, 275–6, 326, 343
Morgan, J.L. 148
Mugler, A. 244, 276, 304, 326

narrative 71–3, 117, 127, 131, 382
narrative *before*-clause 102
narrative past 71
narrative *when*-clause 101, 145–8
Needham, P. 323
Nehls, D. 68, 75–6, 79, 341
Nerbonne, J. 130
Nieuwint, P. 192, 198
nonfactual 77–9, 93–4, 105–6, 382; *see also* remoteness
nonfinite verb form 34, 175, 181, 241–2; *see also* gerund; infinitive; participle
nonprogressive 117, 121–2, 241, 262–3, 274, 300, 313, 319, 341, 344
nonrestrictive relative clause 39, 55, 58, 63, 140–1, 143, 154
Noonan, M. 158
noun clause *see* complement clause

Oakeshott-Taylor, J. 168
O'Grady, W.D. 280
omnitemporal 281–4, 317, 352; *see also* generic; gnomic
Osselton, N.E. 137
Ota, A. 8, 53, 173, 189, 191, 273
Övergaard, G. 379, 382
Oversteegen, L. 243

Palmer, F.R. 3, 12, 87, 192, 199, 202, 206, 208, 276, 321, 326, 340, 344–5, 351
Panenovà, J. 24, 325
Pankhurst, J.H. 290
Papp, N. 69, 250, 283, 302, 326–7
Parsons, T. 273
Partee, B. 60, 88, 124–5, 127, 129,

Partee, B. — *continued*
138, 141, 145, 240, 248, 263, 280, 287–8, 305, 328
participle 42, 243
past perfect tense 16, 26, 37, 58, 64–6, 77, 108–11, 178, 224–5, 230–1, 233, 237, 239, 286, 289–90, 352, 354–69
past sector, definition of 19–20
past tenses 16; *see also* preterit
past time-sphere, definition of 16
Penhalluriack, J. 277, 304, 320–1
perfective 121, 127, 262, 344; *see also* bounded
Perry, J. 240, 262, 304
pluperfect *see* past perfect tense
point of orientation *see* time of orientation
point of view 82–3, 89–92, 105, 176, 194, 189, 383
Postal, P. 196
posteriority 19, 23, 27, 33–9, 43–5, 49–50, 53–4, 67–8, 96–7, 103–8, 184, 214, 370
post-present sector, definition of 19–20
Poutsma, H. 17, 30–1, 63, 68, 168, 192, 210, 212, 215–16, 219, 319, 326
prediction 67, 87–9
pre-present sector, definition of 19–20
present perfect tense 16, 28–34, 37–8, 50–2, 68, 131, 173–4, 189–90, 224–5, 234–41, 268, 304–9, 319–56
Present Perspective System (PPS) 50–6, 64, 92, 124, 146, 153–6, 192–3, 195, 197, 204–5, 214, 379
present tense 3, 16, 32–4, 38–9, 43–4, 48, 50, 54, 66, 68–72, 74–6, 79–80, 83, 86, 99, 185–8, 190–1, 225, 229, 232–3, 237, 246–7, 258–9, 282, 284, 313–19
preterit 3–4, 16, 21, 24, 26–9, 32, 37, 50, 61, 64, 71, 77–80, 83–4, 86, 91, 100–2, 104, 108, 110–13, 118, 169–72, 185, 225, 227, 232–46, 248, 252, 259–60, 264–6, 281, 295–312, 319–22, 326, 333–8, 453–6
Prior, A.N. 227, 229, 255
progressive 67, 74, 117–19, 121–2, 127, 134, 213, 225, 241, 273–80, 300–1, 317, 319, 323, 341, 343–4, 362, 381
punctual 121, 247, 250, 254, 271–3, 295–6, 298, 314, 360

Quirk, R. 16, 68–70, 72, 121, 158, 192, 198, 206, 243, 302–3, 329, 333–4

Randriamasimanana, C. 73, 186
Rauh, G. 57, 74, 169, 326
rebinding, definition of 64
Recktenwald, R.P. 262
re-establish a (temporal) domain, definition of 52–9
reference point *see* reference time; time of orientation
reference time 141, 224–5, 227–30, 232, 234–5, 237, 239–44, 246–8, 250–1, 255, 293; *see also* time of orientation
Reichenbach, H. 6, 140, 223–34, 248–50, 253–8, 268, 319, 391
Reinhart, T. 138
relative clause 24, 39, 46, 53, 63, 77, 134, 148–50, 242; *see also* relative *when*-clause; restrictive relative clause; nonrestrictive relative clause
relative tense, definition of 20–1, 25
relative time adverbial 289, 362, 365–7
relative *when*-clause 55
remoteness 4, 76–80, 238
repetitive 29–30, 33, 121, 139, 277–84, 317, 332–3, 336, 349–53, 365
Represented Speech and Thought (RST) 89–92, 105, 383
Rescher, N. 7
restrictive relative clause 39, 55, 63, 155, 189, 191
result 104–6, 191, 340–4; *see also* resultative perfect
resultative perfect 306, 324–6, 340–4
Richards, B. 43, 48, 125, 240, 252, 312
Riddle, E.M. 81, 159, 168, 186–8, 243
Rigter, B. 33, 67, 141, 250, 268, 280, 326–7

Rigter, G.H. 54, 82, 244, 255, 257, 265, 268, 280, 364
Ritchie, G.D. 44, 98, 252, 276
Rivière, C. 350
Rodari, G. 78
Rohrer, C. 9, 17, 20, 131, 141, 142, 149–50, 251, 273, 276, 290
Rosenbaum, P.S. 8
Ross, J.R. 44, 56, 158
Rot, S. 70
Ruin, I. 90
Ryle, G. 124

Salkie, R. 49, 64, 230, 235, 238–9, 273, 313, 343
Sampson, G. 44, 56
Saurer, W. 68, 243–4, 257, 273, 328
Schachter, P. 265
Scheurweghs, G. 243, 329
Schiffrin, D. 73, 90
Schopf, A. 37, 121, 131, 159, 265, 283, 287–8, 302, 312
Schulze, R. 1
sector, definition of 19
Sennholz, K. 15
sequence 117, 119, 124–6, 129, 140, 353
Sequence of Tenses (SoT) 100, 157–91, 237
series 285; *see also* repetitive
Sgall, P. 20, 24, 325
Sharwood Smith, M. 360
shift of (temporal) domain, definition of 21–4, 46–9
shift of (temporal) perspective 24–5, 28–30, 36, 38–9, 41, 49–50, 66–76, 78, 174, 196, 238, 296–7, 350, 370, 379
simple form *see* nonprogressive
simultaneity 18–19, 26, 30, 32–4, 37–45, 54, 68, 77, 80–1, 83, 85–6, 98–100, 104–6, 108, 110–11, 118–19, 126, 128–30, 139–40, 165–6, 181, 195–6, 241, 244–8, 251, 255–60, 267–9, 277, 296–7, 313–14, 316–17, 360; *see also* simultaneous domain
simultaneous (temporal) domain, definition of 49–52
situation, definition of 14, 249
situation span (SS), definition of 257
situation-time of orientation (TO_{sit}), definition of 18, 251
sloppy simultaneity, definition of 41–5
Smith, C.S. 12, 27, 29, 31, 46, 74, 87, 95, 135–6, 141–2, 148, 230, 250, 252, 262, 273, 276, 280, 286, 288, 290, 319, 331, 359, 364
Smith, H.L. 8
Smith, N.V. 328
Sørensen, H.S. 13, 326
Spielman, R.W. 73, 76
state 120–1, 128, 262, 265, 273, 279–82, 314, 317
stative 213, 272, 274
Steedman, M.J. 273, 280
Steele, S. 79
Stegu, M. 129
Steinberg, G. 90
Steinmeyer, G. 87
Steube, A. 283
Strang, B. 326
Strawson, P.F. 216
Streng, A.H. 8
subclause of comparison 53
subordinate clause *see* individual subclauses
Sugioka, S. 366
Svartvik, J. 304
Sweet, H. 277, 283

Taylor, B. 123
Tedeschi, P.J. 78, 93
telic 121–2, 262, 341, 343–4; *see also* bounded
temporal clause *see* adverbial time clause
temporal domain, definition of 20
temporal focus 80–9
temporal perspective *see* shift of temporal perspective
temporal relations *see* domain-internal relations
temporal subordination, definition of 25
temporal zero-point (t_0), definition of 14–16
tense, definition of 7; *see also* absolute tense; relative tense; individual tenses
tense simplification 46, 117, 183–6
tenseless clause 7–8, 45, 175, 181, 284

tentativeness 78–9, 88, 211–12
terminal point *see* endpoint
that-clause *see* complement clause
Thomas, W. 73
Thompson, S.A. 127, 138
Thomson, A.J. 42
Tichý, P. 240, 368–9
Timberlake, A. 14, 16, 75
time clause *see* adverbial time clause
time line 16, 224, 227, 233, 375, 381
time of orientation (TO), definition of 17–18
time of the situation (TS), definition of 249–50, 257–67
time of utterance (TU) *see* temporal zero-point
time-sphere, definition of 16–17
timeless 71, 76, 283–4
TOE, definition of 252
Tottie, G. 379, 382
Trager, G.L. 8
Tregidgo, P.S. 25, 154, 159, 192, 194, 196, 198, 202, 206, 221, 240
truth conditions 82, 131, 240, 367, 382, 393–6
Twaddell, W.F. 8, 74, 79, 326

Ultan, R. 16, 159, 229, 233, 283
unbounded *see* bounded
unitary 264, 268, 335, 337
unreality 93–4, 101; *see also* counterfactual condition, nonfactuality.
Urquhart, A. 7

vagueness 43, 47–8, 140, 202, 226–7, 233, 245, 256, 267, 270, 276, 321, 361, 375, 381
van Gunsteren-Viersen, J. 89
van Voorst, J. 263
Vasudeva, H.N. 17, 19, 44, 158, 265, 274, 280
Vendler, Z. 120, 122–3, 262, 273, 280
verb of communication 69
Verkuyl, H.J. 285, 332
Vermant, S. 32, 323, 345
Vet, C. 125, 130, 244, 250, 257, 280, 288, 291
Vetter, D.C. 66
Vikner, S. 229, 375
Visser, F.T. 11, 98
Vlach, F. 242, 262, 273
von Stechow A. 240

Wachtel, T. 37, 93
Wald, B. 147, 232
Waletzky, J. 138, 152
Warnant, L. 78
Waugh, L. 74
Weinrich, H. 4, 70, 74, 127, 160
Wekker, H.C. 25, 44, 159, 178, 192, 200, 203–4, 206, 209, 212, 221
Welmers, W. 232
when-clause 43, 55, 65, 98–101, 144–8, 362, 366, 378–9; *see also* narrative *when*-clause; relative *when*-clause
Whorf, B.L. 16
will: expressing belief 87–9; expressing characteristic behaviour 87–9; expressing inherent capacity 87–9
will-deletion 44, 56, 66
Windfuhr, G.L. 74
Woisetschlaeger, E.F. 257, 342
Wolfson, N. 73–4, 76, 237, 322
Wollmann, A. 288
Wright, S. 73
Wunderlich, D. 19, 83, 158

Zwicky, A.M. 150, 195
Zydatiss, W. 67, 262–3, 276, 326, 339, 341